图纸履历 PLAN HISTORY

日期 DATE	标记 REV.NO	简 述 DESCRIPTION	设绘 DWN	审查 CHKD	审定 APPD
	O	供施工 FOR WORKING			

FIKAB NAVAL ARCHITECTS-MARINE CONSULTANTS

03257-2-115-01

174000 DWT 双壳散货船

174000 DWT DOUBLE SIDE SKIN BULK CARRIER

BOHAI SHIPBUILDING HEAVY INDUSTRY CO.,LTD.

详 细 设 计
DETAIL DESIGN

标记 Mark	数量 Qty	修改单号 Rev.Sh.No	签字 Sign	日期 Date

设计	Designed		
校对	Checked		
标检	Stan.Rev		
审核	Reviewed		
审定	Approved		

FINAL TRIM AND STABILITY BOOK

BH410-020-35A

本图适用于 2 号船
Suitable for No.2

重量 Weight		比例 Scale	
共 719 页 Total		第 1 页 Sheet	

渤海船舶重工有限责任公司
BOHAI SHIP BUILDING HEAVY INDUSTRY CO., LTD.

Table of contents

1. GENERAL 7

1.1 General Description 7

1.2 Principal Dimensions 8

1.3 General Arrangement Plan 9

1.4 Tank Capacity Plan 10

1.5 Freeboard Mark and Deadweight Scale 11

2. MASTER'S INSTRUCTION 12

2.1 General Precautions Against Capsizing 13

2.2 Loading Criteria and Instruction 14

2.3 Displacement Calculation from Draught Reading 17

2.4 Trim and Stability Calculation 18

2.5 Stability Criteria and Evaluation 24

3. STABILITY DATA 31

3.1 Stability Cross Curves Data 31

3.2 Stability Limiting Curve 37

4. PART CONDITIONS 38

Part condition No 1: 100% bunkers etc. 39

Part condition No 2: 50% bunkers etc. 41

Part condition No 3: 10% bunkers etc. 43

5. WATER BALLAST GROUPS 44

Load group No1: WB No1 100% 45

Load group No2: WB No2 100%……46

Load group No3: WB No3 100%……47

Load group No4: WB No4 100%……48

Load group No5: WB No5 100%……49

Load group No6: WB No6 100%……50

Load group No7: WB No7 100%……51

Load group No8: WB No8 100%……52

Load group No9: WB No9 100%……53

6. LOADING CONDITIONS……54

Loading Conditions summary from……55

Condition No1: Light ship (Not seagoing condition)……56

Condition No2: Normal ballast-Departure bunkers……61

Condition No3: Normal ballast-50% bunkers……69

Condition No4: Normal ballast-Arrival bunkers……77

Condition No5: Heavy ballast-Departure bunkers……84

Condition No6: Heavy ballast-50% bunkers……92

Condition No7: Heavy ballast-Arrival bunkers……100

Condition No8: Homogeneous cargo (0.792t/m^3) at Design draught-

Departure bunkers……108

Condition No9: Homogeneous cargo (0.792t/m^3)

(Design draught)-50% bunkers……115

Condition No10: Homogeneous cargo (0.792t/m^3)

(Design draught)-Arrival bunkers……122

Condition No11: Homogeneous cargo (0.900t/m^3) at Scantling draught-Departure bunkers·····129

Condition No12: Homogeneous cargo (0.900t/m^3) (Scantling draught)-50% bunkers·····136

Condition No13: Homogeneous cargo (0.900t/m^3) (Scantling draught)-Arrival bunkers·····143

Condition No14: Fully loaded with coal (42 cft/longton)-Departure bunkers···150

Condition No15: Fully loaded with coal (42 cft/longton)-50% bunkers·····157

Condition No16: Fully loaded with coal (42 cft/longton)-Arrival bunkers·····164

Condition No17: Fully loaded with ore (12 cft/longton) in alternative holds at Design draught-Departure bunkers·····171

Condition No18: Fully loaded with ore (12 cft/longton) in alternative holds (Design draught)-50% bunkers·····178

Condition No19: Fully loaded with ore (12 cft/longton) in alternative holds (Design draught)-Attival bunkers·····185

Condition No20: Fully loaded with ore (12 cft/longton) in alternative holds at Scantling draught-Departure bunkers·····191

Condition No21: Fully loaded with ore (12 cft/longton) in alternative holds (Scantling draught)-50% bunkers·····198

Condition No22: Fully loaded with ore (12 cft/longton) in alternative holds (Scantling draught)-Attival bunkers·····205

Condition No23: Homogeneous ore cargo (12 cft/longton) at Scantling draught-Departure bunkers·····212

Condition No24: Homogeneous ore cargo (12 cft/longton)
(Scantling draught) 50% bunkers·····219
Condition No25: Homogeneous ore cargo (12 cft/longton)
(Scantling draught)-Arrival bunkers·····226
Condition No26: Multiple port loading-First port-(42 cft/longton)-
Departure bunkers (Second port see condition No 14)·····233
Condition No27: Multiple port loading-First port-(42 cft/longton)-
50% bunkers (Second port see condition No 15)·····240
Condition No28: Multiple port loading-First port-(42 cft/longton)-
Arrival bunkers (Second port see condition No 16)·····247
Condition No29: Harbour condition for Dunkirk East-d>12.95m-
Arrival bunkers (Not seagoing condition)·····254
Condition No30: Docking condition-Arrival bunkers (Not seagoing condition)···262
Condition No31: Stern tube seal renewal condition- Arrival bunkers
(Not seagoing condition)·····268
7. Ballast exchange sequence based on loading condition No3·····274
8. Ballast exchange sequence based on loading condition No6·····354
9. Loading/unloading sequence: From cond. No4 to cond. No20·····452
10. Loading/unloading sequence: From cond. No22 to cond. No2·····518
11. Loading/unloading sequence: From cond. No4 to cond. No23·····585
12. Loading/unloading sequence: From cond. No25 to cond. No2·····641

APPENDIX

A1. Light ship weight distribution·····A1-1

A2. Addendum to Loading Manual Maximum Allowable and Minimum Required Mass of Cargo for Double Bottom·····A2-1

1. GENERAL

The vessel is designed as an ocean going single screw diesel driven double hull bulk carrier suitable for normal world wide service having a deadweight capacity of about 174,000 t

1.1 General Description

Name of ship.....NICOLE M

Type of ship ..... Bulk Carrier

Owner ..... TRANSMED SHIPPING LIMITED

Designer.....FARTYGSKONSTRUKTIONER AB

Builder ..... BOHAI SHIPBUILDING HEAVY INDUSTRY CO.,LTD

Builder's number BH410-2

Class ..... NK: NS* BC-A(Holds 2,4,6 & 8 empty) (ESP) /MNS*, PS-DA,PS-FA,M0,PSCM,IWS

Navigation area..... UNRESTRICTED SERVICE

Flag state..... Cyprus

Port of registry.....LIMASSOL

IMO number 9300374

Call sign.....C4GE2

Date of keel laid.....2004.11.17

Date of launch..... .2005.9.8

Date of delivery 2005.12.8

Class Number 054416

1.2 Principal Dimensions

Length over all abt. .....289,00 m

Length between perpendiculars.....278.20 m

Breadth(moulded) .....45.00 m

Depth (moulded) abt. 24.65m

Designed draft (moulded) .....16.50 m

Scantling draft (moulded) .....18.20 m

Freeboard type.....B-60 (International Load Line Convention 1966) + assigned freeboard corresponding to the scantling draught.

Frame space..... .Fore peak & aft peak and engine room 800 mm
Cargo hold area 910 mm

Cargo hold capacity .....186160 m^3

Gross tonnage .....89659 t

Net tonnage ..... 55787 t

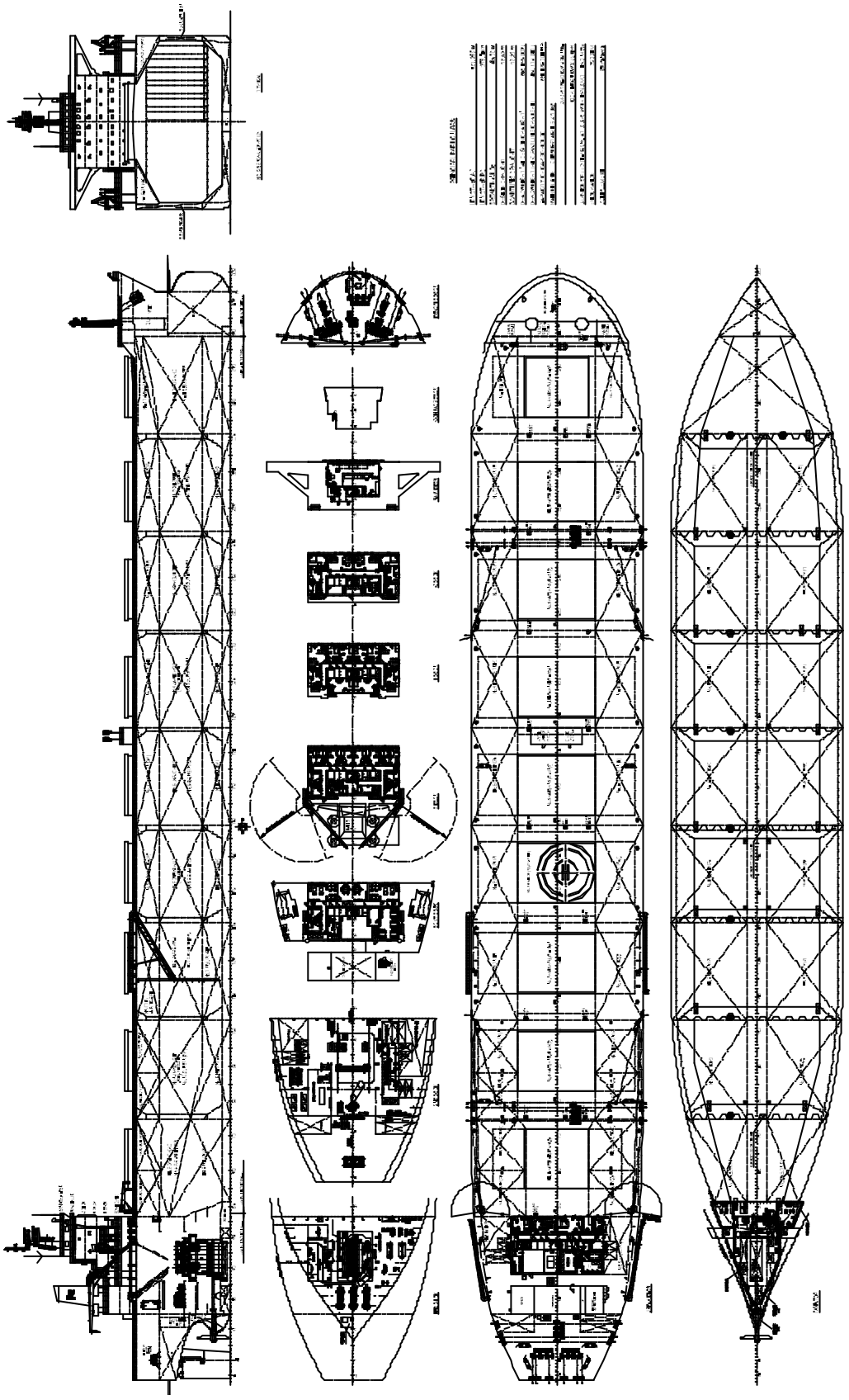
Final Light ship weight …… 24819.29 t

L.C.G. ……132.898 m from AP

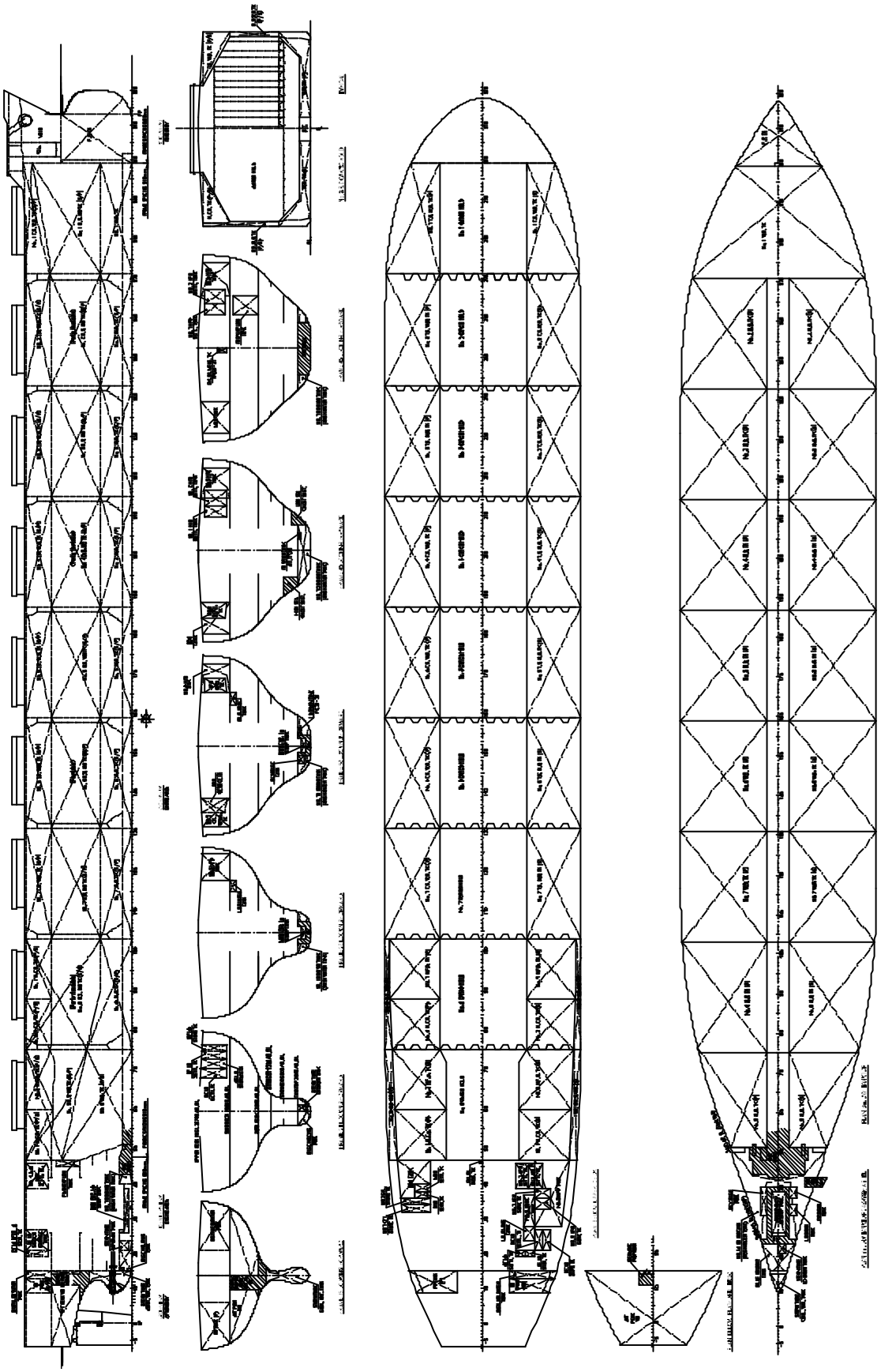
T.C.G. …… 0.00 m from CL

V.C.G. ……13.667 m above BL

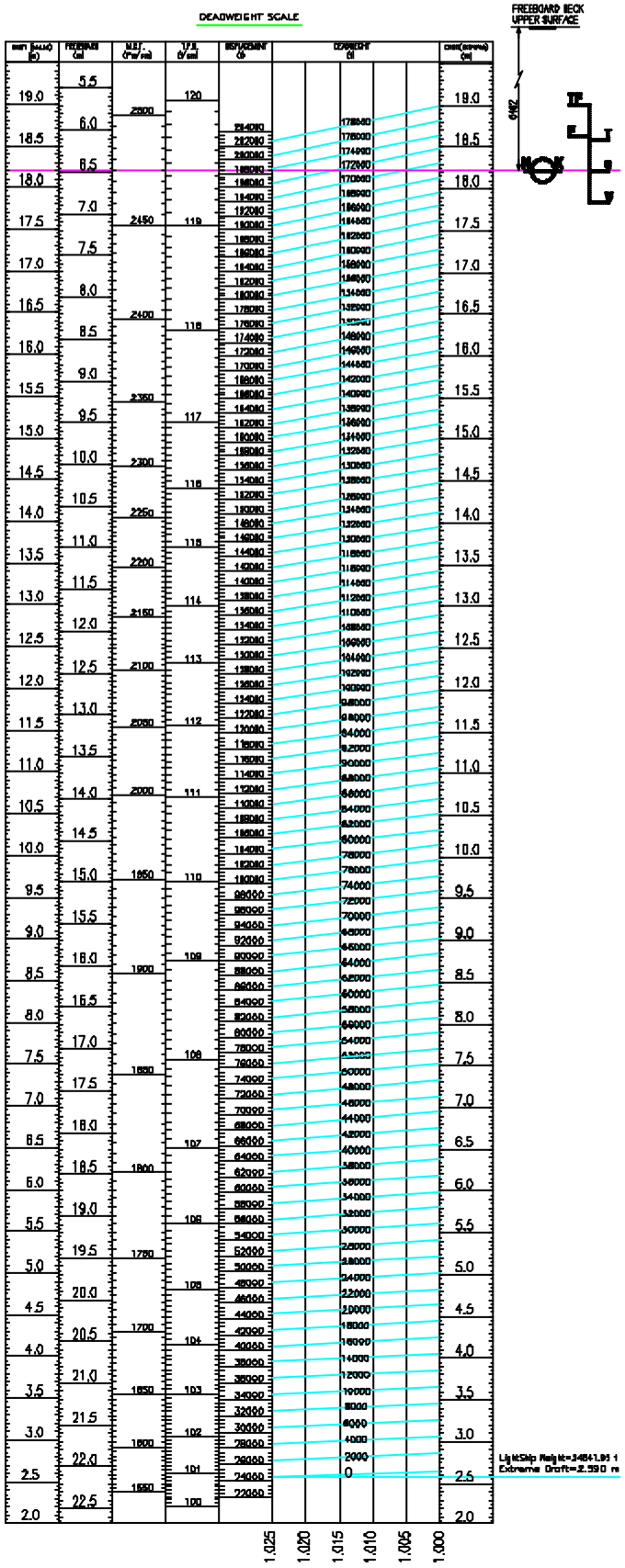
1.3 General Arrangement Plan



1.4 Tank Capacity Plan



1.5 Freeboard Mark and Deadweight Scale



ITEM LOADLINE	FREEBOARD (m)	DRAFT (MOULD) (m)	DISPLACEMENT (t)	DEADWEIGHT (t)
TROPICAL FRESH (TF)	5.663	19.999	202907.3	178265.4
FRESH WATER (F)	6.042	19.62.0	198472.0	173530.1
TROPICAL (T)	6.013	19.579	202942.7	178300.8
SUMMER (S)	6.462	19.200	198408.0	173794.1
WINTER (W)	6.881	17.821	193679.9	168938.0

NOTE: (A) LIGHT SHIP WEIGHT: 24841.93 t

(S) +------ SPECIFIC GRAVITY 1.000

OTHER--- SPECIFIC GRAVITY 1.025

2. MASTER'S INSTRUCTION

(1). This booklet is made for the guidance to the master so as to have the safe operation and proper loading of the vessel.

(2). This booklet presents the necessary data and instructions to arrange cargoes, fuel oil, ballast water etc., in such a way that the vessel can be free from creation of unacceptable stress in hull structure and also can maintain sufficient stability and loading condition in service.

(3). The master of the vessel should read the whole of this booklet and be well familiar with the characteristics of the vessel before placing it into service. Any ignorance of the instructions in this booklet may jeopardize the safety of crew and vessel.

(4). It should be noted that, however, this booklet cannot cover unusual condition or contingencies.

(5). Calculation in the booklet is obtained from NAPA program of NAPA Oy, Helsinki, Finland. Napa Release 2004.2.

(6). Coordinate system in the booklet is right-handed. Origin of the coordinate system is located at the intersection of frame0 (AP), centerline of ship (CL) and the base line (BL).

	Positive value (+)	Negative value (-)	Remark
X	Forward	aftward	All parameter in x direction,such as LCG,LCB…
Y	Port side	Starboard side	All parameter in y direction,such as TCG…
Z	Upward	—	All parameter in z direction,such as VCG,KM…
Trim	By stern	By bow	
Heel	Port side	Starboard side	Such as heeling angle

(7). SI units to be adopted in the booklet.

2.1 General Precautions Against Capsizing

Compliance with the stability criteria indicated below does not ensure immunity against capsizing regardless of the circumstances or absolve the master from his responsibilities. Master should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts and the navigational zone and should take the appropriate action as to speed and course warranted by the prevailing circumstances as well as IMO regulation for subdivision and damage stability.

(1). Care should be taken to ensure that the cargo allocated to the ship is capable of being stowed so that compliance with the criteria can be achieved. If necessary, the amount should be limited to the extend that ballast weight may be required.

(2). Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed so as to minimize the possibility of both longitudinal and lateral shifting while at sea, under the effect of acceleration caused by rolling and pitching.

(3). In all loading conditions, the initial metacentric height should be corrected for the free surface effect of each kind of tank which having larger free surface, and selecting the greatest free surface moment of these tanks. The correction of stability curves to be adopted the method by increasing the gravity center of ship, the increasing value is the corrective value for sum of free surface of tanks.

(4). Weathertight or watertight closing devices such as doors, hatch, etc. shall be kept closed during navigation. No modification or removal can be allowed for doorsills, hatch coamings, air piping, ventilators and other piping systems without authorization by the classification society.

(5). Master should note that stability can be adversely affected by such influences as beam wind on ships with large windage area, water trapped on deck and in deck cargo, rolling characteristics and following seas.

(8). Master should be aware of the possibility that the carriage of certain bulk cargoes may adversely affect the stability of the ship. In this respect, attention is drawn to the recommendations made in the IMO’s code of safe practice for solid bulk cargoes.

2.2 Loading Criteria and Instruction

The standard loading conditions described in this booklet are fully warranted for all of trim, stability and longitudinal strength. In case to make the loading conditions deviated from these standards conditions, the master should comply with the criteria of stability and longitudinal strength in any conditions. It is recommended to comply with the requirement as follows:

(1). Allowable local loading

Cargo hold	Allowable cargo weight	Uniform load for inner bottom
#	[t]	[t/m^2]
1	27793	39.9
2	23880	26.6
3	36972	39.9
4	24670	26.6
5	36972	39.9
6	24123	26.6
7	36972	39.9
8	24257	26.6
9	30153	39.9

(2). Stability

It is necessary to comply with the requirement of stability criteria of IMO document about intact stability shown in 2.5 Stability Criteria and Evaluation. To keep necessary righting lever and metacentric height, care should be taken on the following points.

a). Cargo weight distribution in vertical direction.

b). Filling up proper ballast water in tanks.

c). Minimizing free surface effect of ballast water and oil.

In any loading conditions, the GM value corrected for effect of free surface of liquids in tanks shall be above a GM limits curve obtained from the intact stability criteria. Besides intact stability, to keep the vessel at a satisfactory safety level in view of damage stability, the GM value shall also be above a GM limits curve obtained from the damage stability criteria.

(3). Damage stability for SOLAS74 Chapter XII

Additional safety measures for bulk carriers" of "Amendments to the annex to the International Convention for the Safety of Life at Sea, 1974".

The following criteria in condition of equilibrium after flooding have been considered:

1) The final waterline after flooding, taking into account sinkage, heel and trim, is below the lower edge of any opening through which progressive downflooding may take place.

2) The angle of heel due to unsymmetrical flooding does not exceed 15^0 . If no part of the deck is immersed, an angle of heel of up to 17^0 may be accepted.

3) The metacentric height in the flooded condition is positive.

4) When any part of the deck outside the compartment assumed flooded in a particular case of damage is immersed, or in any case where the margin of stability in the flooded condition may be considered doubtful, the residual stability is to be investigated :

1. The rigting lever curve should have a minimum range of 20^0 beyond the position of equilibrium

with a maximum righting lever of at least 0.1m within this range.

2. The area under the righting lever curve within this range shall be not less than 0.0175m-rad.

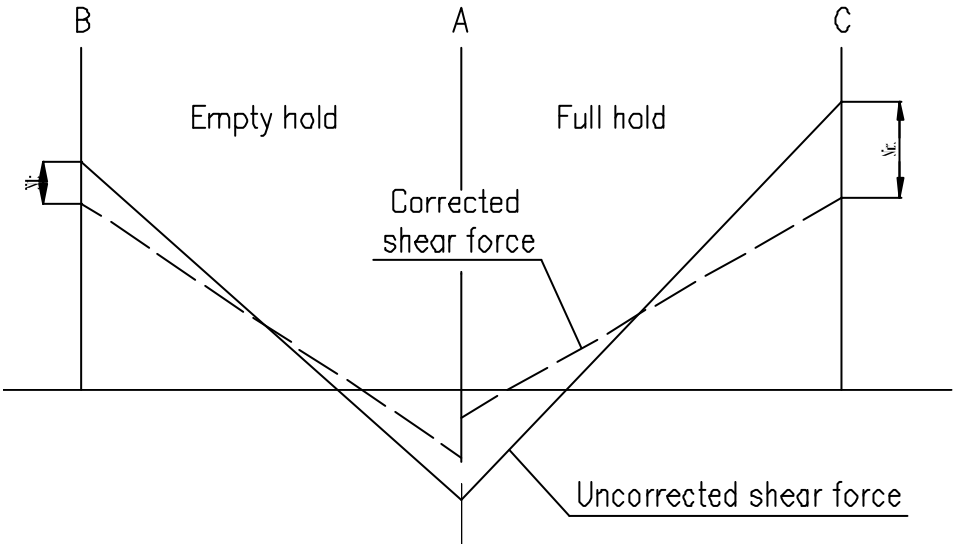
(4). Longitudinal strength

In any loading conditions and flooded conditions, the BM and SF value shall not more than the maximum permissible values.

From A.P.[m]	0	37.6	62.97	88.45	113.93	139.41	164.89	190.37	215.85	241.33	266.81	278.2
Frame No.	0	47	75	103	131	159	187	215	243	271	299	~313
Allowable Bending Moment (Sea-Going)												
Ms (+)[kNm]	0	1960000	3290000	4630000	4630000	4630000	4630000	4630000	3280000	1940000	600000	0
Ms (-)[kNm]	0	-1820000	-3060000	-4300000	-4300000	-4300000	-4300000	-4300000	-3050000	-1800000	-557000	0
Allowable Bending Moment (Harbour)												
Ms (+)[kNm]	0	3060000	5120000	7200000	7200000	7200000	7200000	7200000	5110000	3020000	933000	0
Ms (-)[kNm]	0	-3010000	-5050000	-7100000	-7100000	-7100000	-7100000	-7100000	-5040000	-2980000	-920000	0
Allowable Shearing Force (Sea-Going)												
Fs(+)[kN]		110000	106000	110000	110000	110000	110000	102000	99000	106000	110000	
Fs(-)[kN]		-110000	-103000-	-110000	-110000	-110000	-110000	-104000	-102000	-108000	-110000	
Allowable Shearing Force (Harbour)												
Fs(+)[kN]		128000	132000	135000	131000	131000	131000	131000	129000	133000	118000	
Fs(-)[kN]		-129000	-131000	-135000	-131000	-131000	-131000	-131000	-131000	-133000	-117000	

(5). Shear Force Correction

Shear force corrections (see following figure) should be taken into accounted for alternate loading or heavy inhomogeneous loading.



Formulas of shear force corrections Nk and Nm as following:

$N_k = F \mid F_{sA} - F_{sB} \mid \quad t$

$N_m = F \mid F_{sA} - F_{sC} \mid \quad t$

Where:

F_{sA}、F_{sB}、F_{sC}--------shear forces for section A、B and C respectively, obtained by calculation with hull girder loads, t;

F--------still water shear force correction, formulas as following:

$$F = \frac{B_0}{2.2(B + l_H)}$$

Bo--------the average breadth for the flat part of top tank in the calculated hold m

B-------- the mould breadth =32.26 m

l_H--------the length of the calculated hold m

(6). Local strength

Base on the design loading criteria for local strength, hold mass curve, viz. maximum allowable and minimum required mass of cargo and double bottom content of each hold as a function of the draught at mid-hold position are shown. In any loading conditions, mass of cargo in every hold should be in the allowable area of the hold mass curve.

(7). Permissible deck load

Not allowed load on the deck and hatch cover.

(8). Minimum draft forward

The scantling is approved for operation at draft forward not less than 8.00 meters for ice class B. If, in the opinion of the master, sea conditions are likely to cause regular slamming, then other appropriate measures such as a change in speed, heading or an increase in the draught forward may also need to be taken.

(9). Bridge visibility

The view of the sea surface from the conning position in wheel house should not be obscured more than two ship lengths, forward of the bow to 10o on either side irrespective of the ship's draft, trim .

(10). Propeller size, location and immersion

Propeller diameter D = m;

Height of the shaft line from the base line = m;

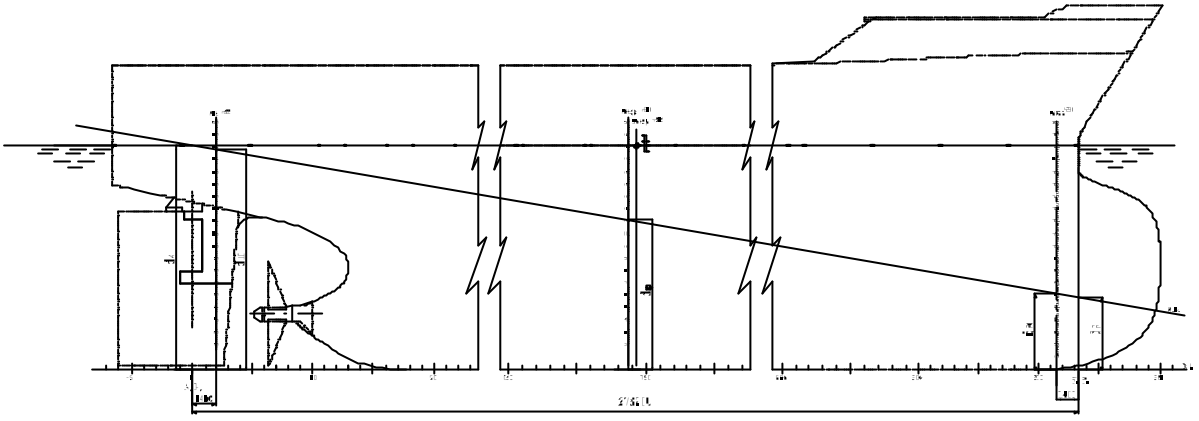
X-coordinate of the propeller from AP = m.

To keep efficiency of propeller, draft aft should be deep enough to take sufficient propeller immersion. For the heavy ballast condition the propeller immersion I/D is to be at least 60%

where: I : the distance from propeller centerline to waterline

D : propeller diameter.

2.3 Displacement Calculation from Draught Reading



(1). Read the draught at aft, midship and fore draught marks, then minus the thickness of keel plate 0.020m, transfer to the draught moulded. (dA_0, $d\otimes$, dF_0)

(2). Calculate the draught at "AP" (dA), midship ($d\otimes$.) and "FP" (dF).

Fore draft mark position:	1.40m after F.P.
Amidship draft mark position:	even at Amidships
Aft draft mark position:	1.40m before A.P.
Length between Fore and Aft draft marks:	$L_{f\text{-}a}=L_{bp}-1.4-1.4=275.40m$
Draft at F.P.:	$d_F = d_{F0}-1.4(d_{A0}-d_{F0})/L_{f\text{-}a}$
Draft at Amid:	$d\otimes = d\otimes$
Draft at A.P.:	$d_A=d_{A0}+1.4(d_{A0}-d_{F0})/L_{f\text{-}a}$

Where:

D_{F0}, $d\otimes$ and d_{A0} are the mean drafts at Fore, Amid and Aft Mark respectively.

(3). Trim (tr), mean draught (dM) and deflection (δ d) are obtained as follows:

tr = dF – dA in meter

dM = (dA + dF) / 2 in meter

δ d = $d\otimes$ - dM in meter

Where:

tr : (+) trim by the stern; (–) trim by the bow

δ d : (+) sagging condition; (–) hogging condition

(4). Displacement (D_1), tons per cm immersion (TCP) and center of floating (LCA) corresponding to "dM" are obtained from “Hydrostatic Data”.

(5). Displacement correction due to trim and deflection:

a). Correction due to trim (D2)

D2 = 100 × TCP × tr × LCF / Lpp in tons

b). Correction due to deflection (D_3)

D3 = 100 × TCP × (3/4) × δ d in tons

(6). Displacement correction due to specific gravity of seawater (D_4):

When the specific gravity (SG) of seawater differs from standard of 1.025, displacement should also be corrected by following formula:

D4 = (D1 + D2 + D3) × (SG –1.025) / 1.025 in tons

2.4 Trim and Stability Calculation

This calculation is made with the attached blank sheets of "Trim and Stability Calculation Sheet 1, 2 and 3".

At the first stage, the weight, the position of center of gravity and moment distribution are to be calculated using "Trim and Stability Calculation Sheet 1" and at the next stage, trim and stability calculations to be performed using "Trim and Stability Calculation Sheet 2 and 3".

(1). Put the weight of cargo, oil, water and ballast etc. into column (1), and also put the center of gravity of each loading weight of length from AP (L.C.G.), breadth from centerline (T.C.G.) and height above base line (V.C.G.) into column (2), (4) and (6) respectively.L.C.G., T.C.G. and V.C.G. corresponding to loading weight for each tank are obtained from "Tank Capacity Data".

(2). Multiply the value in column (1) and (2) (or (4), (6)), put the results into column (3) (or (5), (7)) as the longitudinal moment, transverse moment or vertical moment of each loading

(3). Put "Frs.Mom" (the product of specific gravity of liquid and moment of inertia of free surface of each tank) into column (8). Moment of inertia corresponding to loading weight of each tank is also to be obtained from " Tank Capacity Data".

(4). The total weight, longitudinal moment, transverse moment, vertical moment and free surface moment of inertia are to be put into each column in the 'Trim and Stability Calculation Sheet 2". Then sum up the columns (1)', (3)', (5)' and (7)' respectively and the results of grand total are to be written in the bottom line. These figures mean displacement, longitudinal moment from AP transverse moment from centerline, vertical moment above base line, and total of free surface effect of liquid in tanks respectively.

(5). The center of gravity corresponding to the calculated condition can be obtained as follows.

L.C.G. = Σ (3)' / Σ (1)' in meter

T.C.G. = Σ (5)' / Σ (1)' in meter

V.C.G. (KG) = Σ (7)' / Σ (1)' in meter

(6). Trim and draught corresponding to the displacement can be calculated as follows.

Trim : tr = Disp × (L.C.B.-L.C.G.) / (100 × MCT) in meter

Draught at AP : dA = do + (L/2+LCA)× tr./ L_{PP} in meter

Draught at FP : dF = do - (L/2-LCA)× tr / L_{PP} in meter

Draught mean : dm = (d_F + d_A) / 2 in meter

Where:

do = draught corresponding to the displacement obtained from " Hydrostatic Data" Notice: do, MTC, L.C.B. and L.C.A. correspond to the corrected displacement (D'= D×1.025 /S.G.) in "Hydrostatic Data" when the specific gravity (SG) of seawater differs from standard of 1.025.

(7). Ship's initial metacentric height GM can be calculated as follows.

GM = GMo + GMcorr in meter

GMo = KM - KG in meter

GMcorr = – Σ (Frs.mom) / Disp in meter

Where:

GM : initial metacentric height with correction due to free surface effect

GMo : initial metacentric height without correction due to free surface effect

KM : transverse metacenter height corresponding to displacement in "Hydrostatic Data"

KG : vertical position of center of gravity of the ship above base line (V.C.G.), obtained

from the result of "Trim and Stability Calculation Sheet 2"

GMcorr : free surface correction to the initial stability

Σ(Frs.mom): total of free surface moment obtained from the result of "Trim and Stability Calculation Sheet 2"

(8). Ship's heeling angle can be calculated as follows.

Heeling angle = arctg (Σ(T.mom.) / (Disp×GM)) = arctg (T.C.G./GM) in degree

(9). Ship's statically stability curve (righting lever vs. heeling angle) can be obtained by following method.

$GZ = GZo - KG' \times \sin\theta$

KG' = KG + KGcorr

KGcorr = Σ (Frs.mom) / Disp in meter

Where:

GZ : righting lever

GZo : righting lever when center of gravity of the ship is assumed on the base line (shown in " Stability Cross Curves Data")

KG' : vertical position of center of gravity of the ship above base line with correction due to free surface effect

KG : vertical position of center of gravity of the ship above base line (V.C.G.), obtained from the result of "Trim and Stability Calculation Sheet 2"

KGcorr : increase of height of center of gravity of the ship due to free surface

Σ(Frs.mom): total of free surface moment obtained from the result of "Trim and Stability Calculation Sheet 2"

Stability curve calculation to be performed using "Trim and Stability Calculation Sheet 3" Plot point (GZ vs. θ), GZ in ordinate and θ in abscissa, and connect these points by a fair curve. Thus the statically stability curve is obtained.

(10). About free surface correction

For all loading conditions, the initial metacentric height and the righting lever curve should be corrected for the effect of free surface of liquids in tanks.

In calculating the free surface effects in tanks containing consumable liquids, it should be assumed that for each type of liquids at least one transverse pair or a single center line tank has a free surface and the tank or combination of tanks taken into account should be those where the effects of free surface is the greatest.

Where water ballast tanks are to be filled or discharged during the course of a voyage, the free surface effects should be calculated to take account of the most onerous transitory stage relating to such operations. This of course assumes only one ballast tank or pair of tanks is allowed to

Trim and Stability Caculation Sheet 1
(Detail of deadweight distribution)

Loading conding:_____

		Weight (t) (1)	L.C.G. (m) (2)	L.Moment (t-m) (3)=(1)X(2)	T.C.G. (m) (4)	T.Moment (t-m) (5)=(1)x(4)	V.C.G. (m) (6)	V.Moment (t-m) (7)=(1)x(6)	Frs.Mom. (t-m) (8)
	CARGO								
R1.1	NO.1 CARGO HOLD								
R1.2	NO.2 CARGO HOLD								
R1.3	NO.3 CARGO HOLD								
R1.4	NO.4 CARGO HOLD								
R1.5	NO.5 CARGO HOLD								
R1.6	NO.6 CARGO HOLD								
R1.7	NO.7 CARGO HOLD								
R1.8	NO.8 CARGO HOLD								
R1.9	NO.9 CARGO HOLD								
	TOTAL								
	HEAVY FUEL OIL								
R3.1P	NO.1 H.F.O.TK(P)								
R3.1S	NO.1 H.F.O.TK(S)								
R3.2P	NO.2 H.F.O.TK(P)								
R3.2S	NO.2 H.F.O.TK(S)								
R3.3P	NO.3 H.F.O.TK(P)								
R3.3S	NO.3 H.F.O.TK(S)								
R3.4P	NO.4 H.F.O.TK(P)								
R3.4S	NO.4 H.F.O.TK(S)								
R.3.5	NO.5 H.F.O.TKP								
R.3.6	NO.1 H.F.O.SERV.TK								
R3.7	NO.2 H.F.O.SERV.TK								
R3.8	NO.1 H.F.O.SETT.TK								
R3.9	NO.2 H.F.O.SETT.TK								
	TOTAL								
	DIESEL OIL								
R4.1	M.D.O.TK								
R4.2	M.D.O.SERV.TK								
R4.3	M.D.O.SETT.TK								
	TOTAL								
	LUBRICATING OIL								
R6.01	G/E L.O.STOR.TK.								

		Weight (t) (1)	L.C.G. (m) (2)	L.Moment (t-m) (3)=(1)X(2)	T.C.G. (m) (4)	T.Moment (t-m) (5)=(1)x(4)	V.C.G. (m) (6)	V.Moment (t-m) (7)=(1)x(6)	Frs.Mom. (t-m) (8)
R6.02	G/E L.O.SETT.TK.								
R6.03	M/E L.O.STOR.TK.								
R6.04	M/E L.O.SETT.TK..								
R6.05	L.O.SLUDGE TK.								
R6.13	M/E L.O.SUMP TK.								
	TOTAL								
	FRESH WATER								
R5.1P	F.W.TK.(P)								
R5.1S	F.W.TK.(S)								
R6.17	DISTILLED WATER TK.								
	TOTAL								
	WATER BALLAST								
R2.01	NO.1 W.B TK.								
R2.02P	NO.2 W.B TK(P)								
R2.02S	NO.2 W.B TK(S)								
R2.03P	NO.3 W.B TK(P)								
R2.03S	NO.3 W.B TK(S)								
R2.04P	NO.4 W.B TK(P)								
R2.04S	NO.4 W.B TK(S)								
R2.05P	NO.5 W.B TK(P)								
R2.05S	NO.5 W.B TK(S)								
R2.06P	NO.6 W.B TK(P)								
R2.06S	NO.6 W.B TK(S)								
R2.07P	NO.7 W.B TK(P)								
R2.07S	NO.7 W.B TK(S)								
R2.08P	NO.8 W.B TK(P)								
R2.08S	NO.8 W.B TK(S)								
R2.09P	NO.9 W.B TK(P)								
R2.09S	NO.9 W.B TK(S)								
R2.11P	NO.1 D.S.W.B.TK(P)								
R2.11S	NO.1 D.S.W.B.TK(S)								
R2.12P	NO.2 D.S.W.B.TK(P)								
R2.12S	NO.2 D.S.W.B.TK(S)								
R2.13P	NO.3 D.S.W.B.TK(P)								
R2.13S	NO.3 D.S.W.B.TK(S)								
R2.14P	NO.4 D.S.W.B.TK(P)								
R2.14S	NO.4 D.S.W.B.TK(S)								

		Weight (t) (1)	L.C.G. (m) (2)	L.Moment (t-m) (3)=(1)X(2)	T.C.G. (m) (4)	T.Moment (t-m) (5)=(1)x(4)	V.C.G. (m) (6)	V.Moment (t-m) (7)=(1)x(6)	Frs.Mom. (t-m) (8)
R2.15P	NO.5 D.S.W.B.TK(P)								
R2.15S	NO.5 D.S.W.B.TK(S)								
R2.16P	NO.6 D.S.W.B.TK(P)								
R2.16S	NO.6 D.S.W.B.TK(S)								
R2.17P	NO.7 D.S.W.B.TK(P)								
R2.17S	NO.7 D.S.W.B.TK(S)								
R2.18P	NO.8 D.S.W.B.TK(P)								
R2.18S	NO.8 D.S.W.B.TK(S)								
R2.19P	NO.9 D.S.W.B.TK(P)								
R2.19S	NO.9 D.S.W.B.TK(S)								
R2.21P	NO.1 T.S.W.B.TK(P)								
R2.21S	NO.1 T.S.W.B.TK(S)								
R2.22P	NO.2 T.S.W.B.TK(P)								
R2.22S	NO.2 T.S.W.B.TK(S)								
R2.23P	NO.3 T.S.W.B.TK(P)								
R2.23S	NO.3 T.S.W.B.TK(S)								
R2.24P	NO.4 T.S.W.B.TK(P)								
R2.24S	NO.4 T.S.W.B.TK(S)								
R2.25P	NO.5 T.S.W.B.TK(P)								
R2.25S	NO.5 T.S.W.B.TK(S)								
R2.26P	NO.6 T.S.W.B.TK(P)								
R2.26S	NO.6 T.S.W.B.TK(S)								
R2.27P	NO.7 T.S.W.B.TK(P)								
R2.27S	NO.7 T.S.W.B.TK(S)								
R2.31	F.P.TK								
R2.32	A.P.TK.								
R2.33	NO.10 D.B.W.B.TK.								
	TOTAL								
	MISCELLANEOUS								
R6.12	STERN TUBE L.O.DRAN TK.								
R6.14	L.O.DRAIN TK.								
R6.15	F.O.DRAIN TK.								
R6.16	OILY BILGE TK.								
	TOTAL								
	COOLING W.TK.								
R6.18	STERN TUBE C.W.TK								
	TOTAL								

Trim and Stability Caculation Sheet 2
(Summary of deadweight distribution)

Loading conding:_____

	Weight (t) (1)'	L.C.G. (m) (2)'	L.Moment (t-m) (3)'=(1)'X(2)'	T.C.G. (m) (4)'	T.Moment (t-m) (5)'=(1)'x(4)'	V.C.G. (m) (6)'	V.Moment (t-m) (7)'=(1)'x(6)'	Frs.Mom. (t-m) (8)'
Cargo								
Heavy Fuel Oil								
Diesel Oil								
Lubrication Oil								
Fresh Water								
Miscellaneous								
Cooling Water								
Provision								
Crew & Effect								
Spare								
DEADWEIGHT								
LIGHTWEIGHT								
DISPLACEMENT								

Trim and Stability Caculation Sheet 3
(Stability Curve)

Loading conding:_____

θ (°)	sin θ (1)	Disp (t) (2)	KG (m) (3)	Σ (Frs.mom) (t-m) (4)	Kgcorr (m) (5)=(4)/(2)	KG' (m) (6)=(3)+(5)	GZ_0 (m) (7)	GZ (m) (8)=(7)-(6)x(1)
10	0.17365							
20	0.34202							
30	0.50000							
40	0.64279							
50	0.76604							
60	0.86603							
70	0.93969							
80	0.98481							

2.5 Stability Criteria and Evaluation

The following stability criteria (1) and (2) comply with the requirement of IMO Resolution A.749(18).

(1). General Stability Requirements (IMO Resolution A.749(18))

The stability curves are to comply with the following requirements in Fig.1

a). A1 ≥ 0.055m-rad

The area under the righting lever curve (GZ curve) up to θ =30° angle of heel should not be less than 0.055 metre-radians.

b). A1+A2 ≥ 0.09m-rad

The area under the righting lever curve (GZ curve) up to θ =40° or the angle of downflooding θ f if this angle is less than 40°, should not be less than 0.09 metre-radians.

c). A2 ≥ 0.03m-rad

The area under the righting lever curve (GZ curve) between the angles of heel 30° and 40° or between 30° and θ f if this angle is less than 40°, should not be less than 0.03 metre-radians.

d). GZ30 ≥ 0.2m

The righting lever GZ should be at least 0.20 m at an angle of heel equal or greater than 30°.

e). θ max ≥ 25°

The maximum righting arm should occur at an angle of heel preferably exceeding 30° but not less than 25°.

f). GM ≥ 0.15m

Initial metacentric height GM is to be not less than 0.15 m.

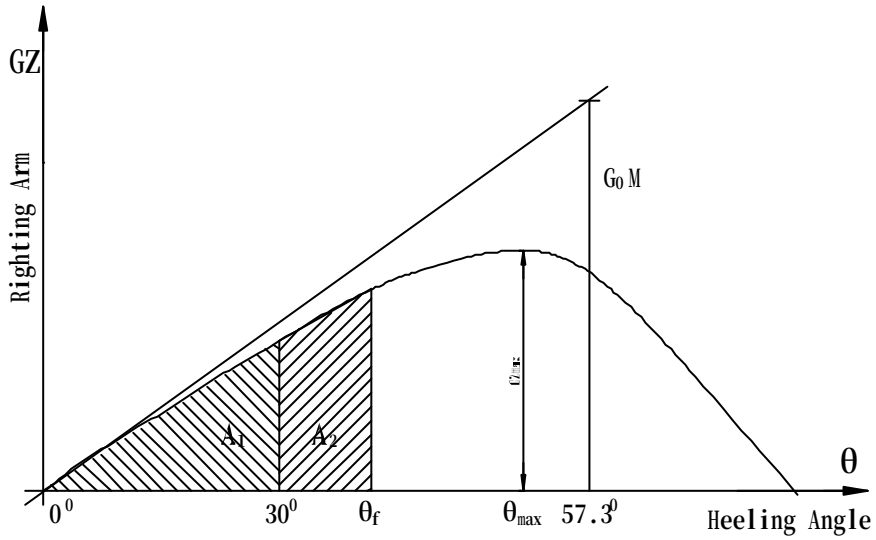


Fig.1 Stability Curve (General Stability Requirements)

Where:

A1 = Area under stability curve between 0°and 30° (m-rad)

A2 = Area under stability curve between 30°and θ f (m-rad)

θ f = Heeling angle (°) to be taken of whichever is less, flooding angle in relevant loading condition or 40°

Gzmax = Maximum righting lever (m)

θ max = Heeling angle at which righting arm reaches maximum (°)

GM = Initial metacentric height corrected by free surface effect (m)

(2). Stability Requirement in Wind and Waves (IMO Resolution A.749(18))

Stability Curves and wind-heeling moment lever curves of ships are to comply with the following requirement in Fig.2

a). Heeling angle caused by steady wind is to be less than 16° or an angle corresponding to 80% of immersing angle of deck edge, whichever is less. In general, deck edge means the point of intersection between the continuation of top of freeboard deck at ship's side of lowest point of freeboard deck and the outside of side shell.

b). Area "b" is to be not less than area "a".

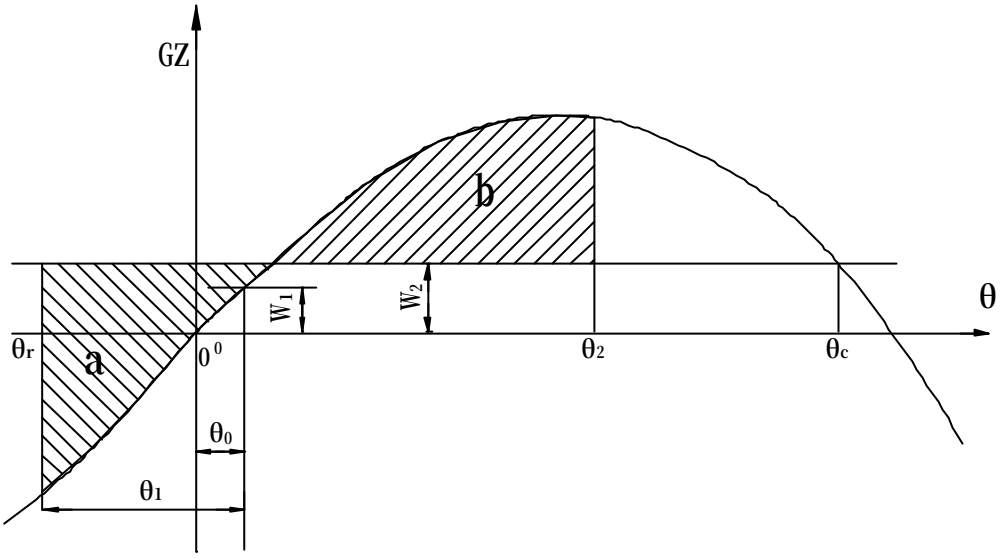


Fig.2 Stability and Wind-heeling Moment Lever Curve (Stability Requirement in Wind and Waves)

Where:

A_1 = Area under stability curve between 0°and 30° (m-rad)

lw_1 = Heeling moment lever caused by steady wind(m) given by the following formula:

$0.0514 \times A \times Z / D$

A = Projected lateral area of hull and cargoes on deck above waterline (m2)

Z = Vertical distance between the center of "A" and the center of underwater project lateral area of hull (m). In general, the center of underwater projected lateral area may be approximated to locate at half the mean draught.

D = Displacement (ext.) (t)

lw_2 = Heeling moment lever caused by gust (m) given by the following formula:

lw_2=1.5 lw1

a = Area encircled by stability curve, lw_2 and θ_r (m-rad)

b = Area encircled by stability curve, $_{lw2}$ and θ_2 (m-rad)

θ_r = Angle of rolling stop motion (°). In general, it may be given by the formula (θ_0 - θ_1)

θ_c = Heeling angle at the second intersection between heeling moment lever and stability curve(°)

θ_2 = Heeling angle (°) to be taken of whichever is least, downflooding angle, θ_c or 50°.

θ_0 = Angle of heel under action of steady wind (°)

θ_1 = Angle of roll to windward due to wave action (°) given by the following formula:

θ 1=109KX1X2 (rs)1/2

X_1 = Values obtained from Table1 according to the value of B/d'. In case the value of B/d' becomes intermediate, value are to be determined by interpolation.

B = Moulded breadth of the ship (m)

d' = Mean moulded draught of the ship (mld.) (m)

Table 1: values of factor X_1

B/d'	≤2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.4	≥3.5
X_1	1.0	0.98	0.96	0.95	0.93	0.91	0.90	0.88	0.86	0.84	0.82	0.80

X_2 = Values obtained from Table2 according to Cb.

In case Cb becomes intermediate, values are to be determined by interpolation.

C_b = Block coefficient given by the following formula: Cb = D / (1.025L'Bd')

L' = Length of the ship at waterline (m)

Table 2: values of X_2

C_b	≤0.45	0.50	0.55	0.60	0.65	≥0.70
X_2	0.75	0.82	0.89	0.95	0.97	1.00

K = Values determined as follows:

a. For round-bilged ships having neither bilge keels nor bar keels: 1.0

b. For ships with sharp bilge: 0.7

c. For ships with bilge keels and/or bar keels:

Values obtained from Table3 according to the value of 100Ak /(L'B).

In case 100Ak /(L'B) becomes intermediate, values are to be determined by interpolation.

Ak = Total area of bilge keels, projected lateral area of bar keels or sum of those areas (m2).

Table 3: values of factor K

$100A_k/(L'B)$	0	1.0	1.5	2.0	2.5	3.0	3.5	≥4.0
K	1.0	0.98	0.95	0.88	0.79	0.74	0.72	0.70

r = Values obtained from the following formula. However, the value of r need not be taken over than 1.0: r = 0.73 + 0.06OG/d'

OG = Distance between the center of gravity and the waterline (m), and is taken as positive when the center of gravity is above waterline.

s = Values obtained from Table4 according to the value of T.

In case T becomes intermediate, values are to be determined by interpolation.

Table 4: values of factors

T	≤6	7	8	12	14	16	18	≥20
S	0.100	0.098	0.093	0.065	0.053	0.044	0.038	0.035

T = Rolling period (sec.) obtained from the following formula.
[2B/(GoM)1/2](0.373+0.023B/d'-0.043L'/100)
GoM = Initial metacentric height corrected by free surface effect (m)

(3). Method of stability evaluation

a. Refer to "Trim and Stability Calculation (1)~(7)".
Calculate the displacement (Disp), draught corresponding to the displacement (d0), initial metacentric height with correction due to free surface effect (GM) and etc. in actual loading condition.

b. Refer to "Trim and Stability Calculation (9)".
Draw the statically stability curve in actual loading condition.

c. Refer to "General Stability Requirements (IMO Resolution A.749(18))".
Fill up "IMO A.749(18) Stability Criteria Calculation Sheet 1".
Check if the requirements of General Stability Requirements (A.749 (18)) are satisfied or not.

d. Refer to "Wind Profile Curves". Find out the projected lateral area above the water line (A) and vertical distance (Z) from the center of (A) to the center of the under water lateral area. In case D becomes intermediate, values are to be determined by interpolation. Calculate lw1 and lw2 using "IMO A.749(18) Stability Criteria Calculation Sheet 2" and draw them in the statically stability curve.

e. Refer to "Stability Requirement in Wind and Waves (IMO Resolution A.749(18))".
Find out θ_0, θ_W, θ_C (θ_W: Angle of heel under action of gust).
Find out θ_2 to be taken of whichever is least, down flooding angle, θ_C or 50°.

f. Refer to "Stability Requirement in Wind and Waves (IMO Resolution A.749(18))".
Calculate θ_1 using "IMO A.749(18) Stability Criteria Calculation Sheet 2" and draw it in the statically stability curve.

g. Fill up "IMO A.749(18) Stability Criteria Calculation Sheet 3".

h. Check if Stability Requirement in Wind and Waves (A.749 (18)) are satisfied or not.

IMO A.749(18) Stability Criteria Calculation Sheet 1

Loading condition: _____

Displacement = _____ t KM = _____ m Σ(Frs.mom) = _____ tm

KG = _____ m KGcorr = _____ m KG' = _____ m

GMo= _____ m GMcorr = _____ m GM = _____ m (≥0.15m)

	θ	0°	10°	20°	30°	40°
	sinθ	0.0000	0.1736	0.3420	0.5000	0.6428
(1)	KG'×Sinθ					
(2)	GZ_0					
(3)	GZ= (2) - (1)					
(4)	Coeff.					
(5)	(3)×(4)					
(6)	Coeff.					
(7)	(3)×(6)					

(8)	Σ (5) =
(9)	Σ (7) =

(10)	Σ (8) =
(11)	Σ (9) =

Area(0°~30°)=(10)=	≥0.055 m-rad
Area(0°~40°)=(11)-(12)=	≥0.09 m-rad
Area(30°~40°)=(11)-(10)-(12)=	≥0.03 m-rad

In case down flooding angle (θ_f) is 40° or over, (12) is taken 0.
In case down flooding angle (θ_f) is less than 40°, the correction (12) is taken as follows:

θ	40°	θ_f(°)
GZ	a	B
(12)	△Area=(a+b)×(40-Θu) ×π/360=	

The following criteria are to be found by using GZ curve.

$GZ_{30°}$=	≥0.2m
θ_{max}=	≥25°

IMO A.749(18) Stability Criteria Calculation Sheet 2

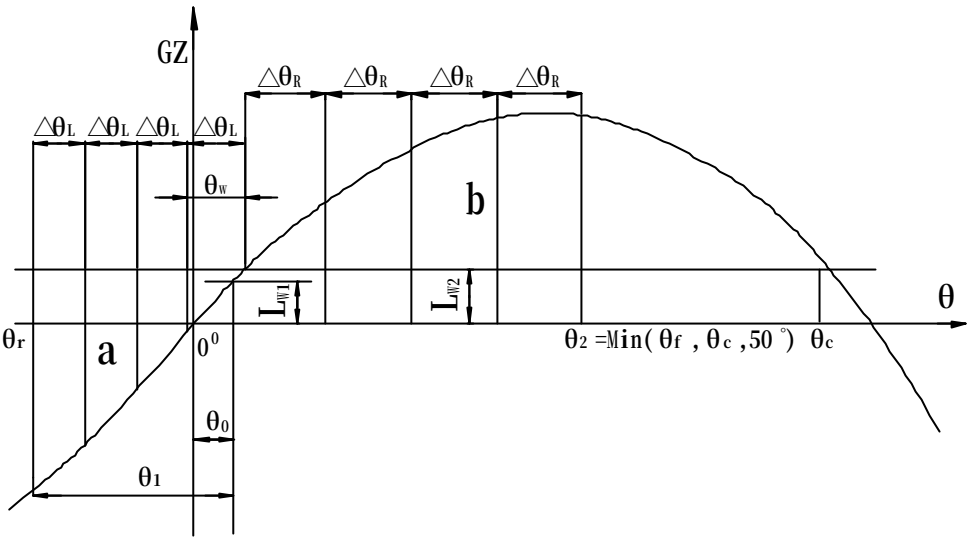
Loading condition: _____

Displacement (ext.) (Disp) = _____
Draught (ext.) (do) = _____
Draught (mld.) (d') = _____
Metacentric height (corrected by free surface effect) (GM) = _____
Center of gravity (KG) = _____
Center of gravity (corrected by free surface effect) (KG') = _____
Water line length (L') = _____
Breadth (mld.) (B) = _____
Depth (mld.) (D) = _____
Bilge keel area (A_k) = _____
Projected lateral area (A_f) = _____
Vertical distance (Z) = _____
0.0514AZ/D (lw_1) = _____
1.5lw_1 (lw_2) = _____
Flooding angle (θ_f) = _____
Min. (θ_f, θ_C, 50 °) (θ_2) = _____
B/d' = _____
X_1 (from table 1) = _____
C_b = Disp / (1.025Bd') = _____
X_2 (from table 2) = _____
100Ak /L'B = _____
K (from table 3) = _____
OG = KG – d' = _____
r = 0.73 + 0.6OG/d' = _____
Rolling period (T) = [2B/(GoM) 1/2]×(0.373+0.023B/d'–0.043L'/100)
= _____
s (from table 4) = _____
θ 1 = 109X1 X2 K (rs)1/2 = _____
80% of immersing angle of deck edge (0.8 θ d)
= [arctan (2(D'- d')/B')]×0.8 = _____
D' (from top of keel to top of freeboard deck at ship's side) = _____
B' (from outside of side shell to outside of another side shell.) = _____

IMO A.749(18) Stability Criteria Calculation Sheet 3

Loading condition: _____

θf=
θc=
θ1=
θ0=
θw=
θr=
θ2=
△θR=(θ2 - θw)/4=
△θL=(θw - θr)/4=



Area "b":

(1)	(deg.)	θw	θw+△θR	θw+2θR	θw+3△θR	θw+4△θR
(2)	GZ	-				
(3)	(GZ-lw2)	0				
(4)	Simpson's F	1	4	2	4	1
(5)	(3) ×(4)	0				

Σ(5)=_____

b =1/3×△θR×Σ(5) ×π/180=_____

Area "a":

(6)	(deg.)	θw	θw-△θL	θw-2θL	θw-3△θL	θw-4△θL
(7)	GZ	-				
(8)	lw2	-				
(9)	(lw2-GZ)	0				
(10)	Simpson's F	1	4	2	4	1
(11)	(4) ×(5)	0				

Σ(5)=_____

b =1/3×△θL×Σ(11) ×π/180=_____

Criteria:

b = _____ > a = _____

θ0= _____ < Min. (16 °, 0.8θd)

3. STABILITY DATA

3.1 Stability Cross Curves Data

Note:
1. The stability lever is calculated assuming that the center of gravity is on base line.
2. The calculation is based on Stabhull data (including deck house, forecastle, and hatch coaming).

Explanation of symbols:
T draught, moulded …….m
KN stability righting lever ……..m
TRIM (+):trim by stern, (-):trim by head…… m

Initial trim 6 m (aft)

		Heeling angles (deg.)								
		0.0	10.0	20.0	25.0	30.0	40.0	50.0	60.0	70.0	80.0
T	TK	KN	KN	KN	KN	KN	KN	KN	KN	KN	KN
m	m	m	m	m	m	m	m	m	m	m	m
2.000	2.020	0.000	9.106	12.825	13.808	14.487	15.236	15.419	15.307	14.872	14.049
2.500	2.520	0.000	8.434	12.420	13.505	14.273	15.176	15.519	15.531	15.112	14.216
3.000	3.020	0.000	7.759	11.993	13.181	14.039	15.108	15.616	15.739	15.329	14.361
3.500	3.520	0.000	7.119	11.553	12.842	13.794	15.035	15.705	15.915	15.504	14.473
4.000	4.020	0.000	6.542	11.120	12.504	13.546	14.953	15.788	16.063	15.637	14.557
4.500	4.520	0.000	6.036	10.697	12.174	13.302	14.872	15.850	16.172	15.729	14.609
5.000	5.020	0.000	5.604	10.290	11.854	13.066	14.794	15.891	16.241	15.784	14.643
5.500	5.520	0.000	5.241	9.904	11.544	12.836	14.716	15.917	16.275	15.809	14.661
6.000	6.020	0.000	4.935	9.539	11.247	12.612	14.643	15.920	16.281	15.808	14.666
6.500	6.520	0.000	4.676	9.194	10.966	12.399	14.572	15.899	16.261	15.790	14.662
7.000	7.020	0.000	4.457	8.871	10.699	12.196	14.496	15.858	16.218	15.771	14.661
7.500	7.520	0.000	4.269	8.571	10.446	12.003	14.412	15.801	16.159	15.727	14.628
8.000	8.020	0.000	4.109	8.294	10.205	11.819	14.324	15.725	16.088	15.669	14.602
8.500	8.520	0.000	3.971	8.042	9.977	11.644	14.219	15.629	16.005	15.598	14.571
9.000	9.020	0.000	3.852	7.816	9.761	11.477	14.119	15.520	15.911	15.529	14.539
9.500	9.520	0.000	3.749	7.616	9.565	11.329	13.998	15.398	15.813	15.456	14.507
10.000	10.020	0.000	3.661	7.442	9.379	11.182	13.869	15.271	15.693	15.379	14.469
10.500	10.520	0.000	3.585	7.289	9.209	11.038	13.732	15.129	15.577	15.300	14.430
11.000	11.020	0.000	3.519	7.157	9.054	10.893	13.586	14.981	15.449	15.213	14.390
11.500	11.520	0.000	3.463	7.043	8.915	10.753	13.432	14.826	15.321	15.123	14.349
12.000	12.020	0.000	3.416	6.945	8.792	10.614	13.272	14.667	15.188	15.031	14.307
12.500	12.520	0.000	3.376	6.861	8.683	10.476	13.105	14.503	15.054	14.939	14.263
13.000	13.020	0.000	3.343	6.788	8.587	10.338	12.933	14.336	14.916	14.844	14.215
13.500	13.520	0.000	3.316	6.729	8.501	10.200	12.755	14.163	14.775	14.746	14.166
14.000	14.020	0.000	3.294	6.682	8.421	10.062	12.571	13.989	14.631	14.647	14.116
14.500	14.520	0.000	3.277	6.642	8.344	9.926	12.383	13.812	14.484	14.544	14.063
15.000	15.020	0.000	3.265	6.610	8.269	9.791	12.191	13.632	14.337	14.439	14.009
15.500	15.520	0.000	3.256	6.581	8.194	9.656	11.995	13.450	14.188	14.331	13.955
16.000	16.020	0.000	3.251	6.553	8.118	9.522	11.795	13.266	14.037	14.222	13.897
16.500	16.520	0.000	3.249	6.523	8.039	9.387	11.595	13.078	13.883	14.111	13.840
17.000	17.020	0.000	3.250	6.491	7.955	9.252	11.395	12.889	13.727	14.001	13.780
17.500	17.520	0.000	3.253	6.453	7.867	9.116	11.197	12.697	13.569	13.890	13.719
18.000	18.020	0.000	3.258	6.409	7.773	8.979	11.001	12.503	13.408	13.777	13.661
18.500	18.520	0.000	3.265	6.357	7.673	8.839	10.808	12.306	13.246	13.662	13.600
19.000	19.020	0.000	3.273	6.295	7.567	8.698	10.620	12.108	13.082	13.547	13.540
19.500	19.520	0.000	3.278	6.222	7.455	8.554	10.438	11.910	12.915	13.431	13.479
20.000	20.020	0.000	3.276	6.137	7.333	8.409	10.254	11.717	12.747	13.312	13.418
20.500	20.520	0.000	3.265	6.042	7.206	8.260	10.074	11.525	12.581	13.192	13.354
21.000	21.020	0.000	3.242	5.934	7.072	8.107	9.897	11.341	12.415	13.072	13.295
21.500	21.520	0.000	3.203	5.814	6.929	7.947	9.720	11.160	12.256	12.953	13.229
22.000	22.020	0.000	3.143	5.681	6.771	7.777	9.544	10.989	12.099	12.838	13.168
22.500	22.520	0.000	3.066	5.525	6.603	7.601	9.367	10.821	11.955	12.727	13.108
23.000	23.020	0.000	2.970	5.373	6.440	7.432	9.205	10.678	11.826	12.633	13.057
23.500	23.520	0.000	2.875	5.231	6.292	7.284	9.064	10.549	11.726	12.557	13.016
24.000	24.020	0.000	2.769	5.086	6.144	7.140	8.936	10.445	11.639	12.491	12.977
24.500	24.520	0.000	2.683	4.969	6.022	7.019	8.827	10.355	11.569	12.440	12.948
24.650	24.670	0.000	2.657	4.937	5.989	6.987	8.797	10.330	11.550	12.427	12.939

Initial trim 4 m (aft)

		Heeling angles (deg.)								
		0.0	10.0	20.0	25.0	30.0	40.0	50.0	60.0	70.0	80.0
T	TK	KN	KN	KN	KN	KN	KN	KN	KN	KN	KN
m	m	m	m	m	m	m	m	m	m	m	m
2.000	2.020	0.000	9.885	13.478	14.381	14.975	15.522	15.468	15.090	14.642	13.953
2.500	2.520	0.000	8.992	12.931	13.959	14.664	15.416	15.571	15.411	15.028	14.223
3.000	3.020	0.000	8.153	12.387	13.534	14.349	15.309	15.664	15.694	15.330	14.420
3.500	3.520	0.000	7.400	11.866	13.126	14.043	15.197	15.753	15.936	15.569	14.554
4.000	4.020	0.000	6.743	11.370	12.738	13.752	15.091	15.833	16.124	15.734	14.642
4.500	4.520	0.000	6.183	10.906	12.366	13.476	14.991	15.906	16.261	15.838	14.697
5.000	5.020	0.000	5.718	10.472	12.015	13.211	14.893	15.968	16.352	15.895	14.728
5.500	5.520	0.000	5.332	10.065	11.684	12.959	14.804	16.004	16.399	15.917	14.744
6.000	6.020	0.000	5.009	9.683	11.374	12.721	14.718	16.017	16.407	15.918	14.746
6.500	6.520	0.000	4.738	9.324	11.082	12.497	14.635	16.008	16.384	15.911	14.738
7.000	7.020	0.000	4.508	8.986	10.806	12.285	14.559	15.977	16.340	15.874	14.721
7.500	7.520	0.000	4.313	8.670	10.545	12.084	14.478	15.929	16.284	15.813	14.699
8.000	8.020	0.000	4.147	8.379	10.297	11.891	14.389	15.835	16.200	15.754	14.671
8.500	8.520	0.000	4.004	8.115	10.062	11.715	14.308	15.735	16.113	15.688	14.643
9.000	9.020	0.000	3.881	7.881	9.848	11.557	14.203	15.620	16.012	15.615	14.602
9.500	9.520	0.000	3.775	7.673	9.635	11.398	14.083	15.500	15.899	15.538	14.563
10.000	10.020	0.000	3.684	7.491	9.441	11.239	13.963	15.366	15.785	15.460	14.525
10.500	10.520	0.000	3.605	7.332	9.263	11.096	13.828	15.222	15.662	15.369	14.483
11.000	11.020	0.000	3.537	7.195	9.100	10.957	13.684	15.071	15.534	15.278	14.441
11.500	11.520	0.000	3.479	7.076	8.955	10.821	13.531	14.915	15.398	15.185	14.394
12.000	12.020	0.000	3.431	6.973	8.827	10.685	13.371	14.755	15.263	15.090	14.347
12.500	12.520	0.000	3.389	6.886	8.716	10.549	13.204	14.590	15.125	14.995	14.298
13.000	13.020	0.000	3.355	6.813	8.622	10.413	13.031	14.422	14.986	14.897	14.251
13.500	13.520	0.000	3.327	6.751	8.537	10.276	12.852	14.251	14.842	14.798	14.202
14.000	14.020	0.000	3.304	6.700	8.460	10.138	12.668	14.077	14.697	14.696	14.151
14.500	14.520	0.000	3.286	6.658	8.388	10.001	12.479	13.898	14.553	14.591	14.098
15.000	15.020	0.000	3.273	6.625	8.318	9.865	12.285	13.718	14.405	14.487	14.041
15.500	15.520	0.000	3.264	6.600	8.248	9.729	12.087	13.535	14.256	14.379	13.983
16.000	16.020	0.000	3.258	6.577	8.175	9.593	11.886	13.349	14.104	14.272	13.925
16.500	16.520	0.000	3.255	6.554	8.098	9.457	11.683	13.161	13.951	14.163	13.867
17.000	17.020	0.000	3.254	6.529	8.016	9.321	11.477	12.969	13.796	14.052	13.808
17.500	17.520	0.000	3.257	6.498	7.928	9.184	11.274	12.776	13.638	13.941	13.753
18.000	18.020	0.000	3.261	6.458	7.835	9.045	11.073	12.579	13.478	13.829	13.693
18.500	18.520	0.000	3.267	6.410	7.735	8.905	10.876	12.381	13.315	13.716	13.634
19.000	19.020	0.000	3.274	6.351	7.630	8.763	10.684	12.181	13.150	13.601	13.576
19.500	19.520	0.000	3.284	6.281	7.517	8.618	10.497	11.980	12.983	13.483	13.515
20.000	20.020	0.000	3.291	6.199	7.397	8.471	10.314	11.780	12.815	13.366	13.455
20.500	20.520	0.000	3.292	6.106	7.271	8.322	10.136	11.585	12.644	13.247	13.394
21.000	21.020	0.000	3.281	6.000	7.137	8.169	9.957	11.397	12.473	13.126	13.330
21.500	21.520	0.000	3.256	5.883	6.996	8.013	9.781	11.213	12.307	13.004	13.269
22.000	22.020	0.000	3.213	5.754	6.847	7.850	9.607	11.041	12.147	12.883	13.209
22.500	22.520	0.000	3.146	5.612	6.687	7.678	9.431	10.871	11.994	12.765	13.143
23.000	23.020	0.000	3.050	5.450	6.510	7.497	9.256	10.713	11.851	12.656	13.082
23.500	23.520	0.000	2.930	5.269	6.322	7.309	9.078	10.561	11.726	12.556	13.030
24.000	24.020	0.000	2.797	5.094	6.144	7.134	8.923	10.426	11.626	12.484	12.987
24.500	24.520	0.000	2.669	4.936	5.988	6.983	8.792	10.326	11.549	12.430	12.955
24.650	24.670	0.000	2.639	4.896	5.946	6.943	8.757	10.299	11.528	12.417	12.947

Initial trim 2 m (aft)
Heeling angles (deg.)

T	TK	0.0	10.0	20.0	25.0	30.0	40.0	50.0	60.0	70.0	80.0
m	m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m
2.000	2.020	0.000	10.472	13.997	14.844	15.371	15.744	15.511	14.935	14.413	13.939
2.500	2.520	0.000	9.410	13.303	14.297	14.959	15.601	15.616	15.301	14.971	14.283
3.000	3.020	0.000	8.475	12.667	13.796	14.581	15.459	15.711	15.640	15.377	14.495
3.500	3.520	0.000	7.643	12.086	13.332	14.233	15.330	15.794	15.939	15.643	14.636
4.000	4.020	0.000	6.921	11.557	12.903	13.908	15.204	15.876	16.174	15.813	14.724
4.500	4.520	0.000	6.321	11.070	12.507	13.603	15.089	15.950	16.341	15.925	14.779
5.000	5.020	0.000	5.828	10.620	12.141	13.320	14.981	16.017	16.444	15.988	14.809
5.500	5.520	0.000	5.422	10.201	11.800	13.058	14.878	16.072	16.496	16.015	14.821
6.000	6.020	0.000	5.083	9.809	11.482	12.812	14.785	16.104	16.509	16.012	14.820
6.500	6.520	0.000	4.801	9.441	11.184	12.581	14.691	16.117	16.491	15.989	14.817
7.000	7.020	0.000	4.563	9.095	10.902	12.363	14.601	16.069	16.445	15.957	14.789
7.500	7.520	0.000	4.361	8.768	10.637	12.158	14.523	16.008	16.380	15.899	14.764
8.000	8.020	0.000	4.189	8.469	10.387	11.969	14.447	15.924	16.300	15.838	14.734
8.500	8.520	0.000	4.041	8.195	10.151	11.790	14.364	15.830	16.201	15.770	14.699
9.000	9.020	0.000	3.914	7.949	9.926	11.620	14.278	15.712	16.099	15.694	14.665
9.500	9.520	0.000	3.805	7.734	9.713	11.459	14.168	15.584	15.984	15.619	14.624
10.000	10.020	0.000	3.711	7.546	9.512	11.305	14.048	15.445	15.858	15.528	14.581
10.500	10.520	0.000	3.630	7.382	9.326	11.159	13.915	15.301	15.735	15.430	14.535
11.000	11.020	0.000	3.560	7.239	9.157	11.020	13.772	15.151	15.604	15.336	14.484
11.500	11.520	0.000	3.501	7.116	9.007	10.891	13.619	14.994	15.468	15.241	14.437
12.000	12.020	0.000	3.450	7.012	8.876	10.756	13.459	14.833	15.330	15.144	14.387
12.500	12.520	0.000	3.408	6.921	8.759	10.622	13.292	14.667	15.189	15.047	14.336
13.000	13.020	0.000	3.372	6.844	8.659	10.485	13.117	14.499	15.046	14.943	14.284
13.500	13.520	0.000	3.343	6.780	8.573	10.348	12.938	14.327	14.903	14.841	14.231
14.000	14.020	0.000	3.319	6.726	8.499	10.209	12.752	14.152	14.757	14.737	14.178
14.500	14.520	0.000	3.300	6.682	8.431	10.070	12.561	13.974	14.611	14.632	14.122
15.000	15.020	0.000	3.286	6.647	8.365	9.931	12.365	13.792	14.462	14.527	14.066
15.500	15.520	0.000	3.275	6.619	8.297	9.792	12.165	13.607	14.313	14.421	14.009
16.000	16.020	0.000	3.268	6.599	8.225	9.653	11.961	13.419	14.161	14.314	13.950
16.500	16.520	0.000	3.263	6.581	8.147	9.515	11.754	13.228	14.007	14.206	13.892
17.000	17.020	0.000	3.262	6.561	8.065	9.376	11.546	13.034	13.852	14.097	13.837
17.500	17.520	0.000	3.262	6.533	7.976	9.237	11.338	12.838	13.694	13.985	13.780
18.000	18.020	0.000	3.265	6.497	7.882	9.096	11.132	12.639	13.533	13.873	13.722
18.500	18.520	0.000	3.270	6.449	7.781	8.954	10.930	12.438	13.372	13.759	13.666
19.000	19.020	0.000	3.276	6.390	7.674	8.810	10.733	12.235	13.205	13.645	13.606
19.500	19.520	0.000	3.284	6.320	7.560	8.664	10.542	12.032	13.038	13.529	13.547
20.000	20.020	0.000	3.294	6.239	7.440	8.516	10.356	11.831	12.867	13.411	13.490
20.500	20.520	0.000	3.304	6.147	7.314	8.364	10.174	11.631	12.695	13.291	13.428
21.000	21.020	0.000	3.305	6.043	7.179	8.209	9.996	11.438	12.523	13.170	13.367
21.500	21.520	0.000	3.290	5.927	7.038	8.052	9.823	11.252	12.350	13.048	13.306
22.000	22.020	0.000	3.254	5.799	6.889	7.890	9.646	11.078	12.180	12.924	13.241
22.500	22.520	0.000	3.194	5.659	6.732	7.724	9.472	10.905	12.020	12.800	13.178
23.000	23.020	0.000	3.110	5.505	6.565	7.550	9.299	10.746	11.878	12.682	13.118
23.500	23.520	0.000	3.000	5.334	6.383	7.363	9.123	10.592	11.745	12.572	13.051
24.000	24.020	0.000	2.855	5.138	6.182	7.166	8.945	10.442	11.629	12.481	12.996
24.500	24.520	0.000	2.682	4.925	5.972	6.964	8.773	10.309	11.532	12.419	12.960
24.650	24.670	0.000	2.627	4.868	5.915	6.910	8.721	10.272	11.513	12.409	12.955

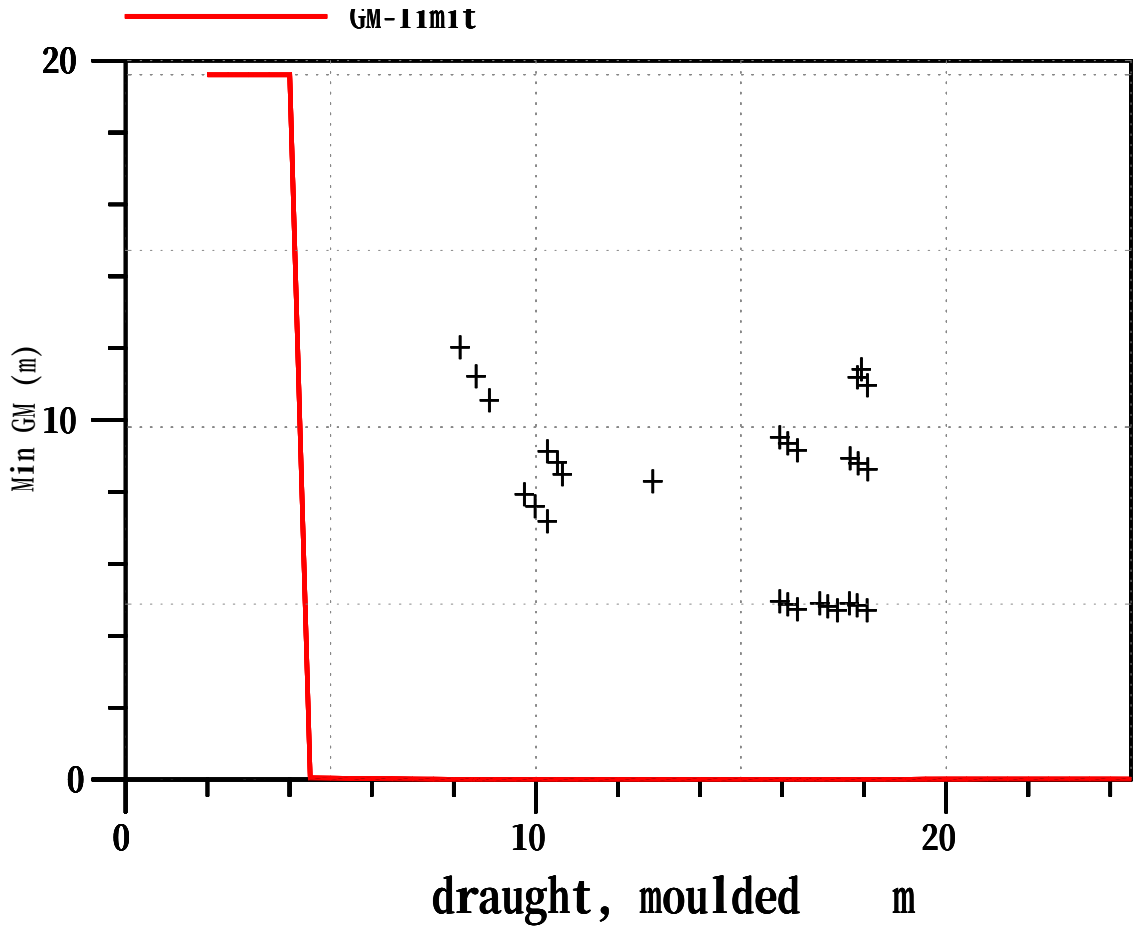
Initial trim 0 m (level)
Heeling angles (deg.)

T	TK	0.0	10.0	20.0	25.0	30.0	40.0	50.0	60.0	70.0	80.0
m	m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m	KN m
2.000	2.020	0.000	10.784	14.245	15.064	15.554	15.862	15.554	14.894	14.287	14.004
2.500	2.520	0.000	9.668	13.482	14.468	15.110	15.699	15.662	15.281	14.966	14.360
3.000	3.020	0.000	8.698	12.812	13.928	14.709	15.549	15.753	15.626	15.421	14.578
3.500	3.520	0.000	7.837	12.215	13.442	14.338	15.409	15.842	15.941	15.708	14.716
4.000	4.020	0.000	7.075	11.677	13.002	13.996	15.283	15.917	16.218	15.887	14.798
4.500	4.520	0.000	6.447	11.185	12.601	13.685	15.161	15.988	16.405	15.999	14.852
5.000	5.020	0.000	5.935	10.731	12.232	13.398	15.048	16.059	16.516	16.059	14.881
5.500	5.520	0.000	5.511	10.309	11.890	13.131	14.936	16.125	16.572	16.089	14.891
6.000	6.020	0.000	5.159	9.914	11.570	12.882	14.830	16.169	16.601	16.089	14.886
6.500	6.520	0.000	4.866	9.543	11.269	12.647	14.733	16.171	16.571	16.077	14.873
7.000	7.020	0.000	4.619	9.193	10.988	12.433	14.646	16.136	16.529	16.025	14.860
7.500	7.520	0.000	4.411	8.862	10.724	12.233	14.567	16.074	16.461	15.973	14.829
8.000	8.020	0.000	4.233	8.554	10.474	12.043	14.494	15.991	16.372	15.914	14.796
8.500	8.520	0.000	4.081	8.275	10.236	11.862	14.421	15.899	16.281	15.845	14.760
9.000	9.020	0.000	3.950	8.025	10.009	11.691	14.341	15.781	16.162	15.758	14.720
9.500	9.520	0.000	3.838	7.805	9.792	11.528	14.237	15.653	16.044	15.677	14.677
10.000	10.020	0.000	3.741	7.610	9.586	11.373	14.120	15.515	15.927	15.585	14.626
10.500	10.520	0.000	3.659	7.440	9.396	11.231	13.990	15.369	15.797	15.490	14.581
11.000	11.020	0.000	3.588	7.295	9.223	11.089	13.848	15.218	15.663	15.389	14.528
11.500	11.520	0.000	3.527	7.168	9.069	10.953	13.697	15.062	15.525	15.290	14.475
12.000	12.020	0.000	3.475	7.058	8.931	10.822	13.536	14.901	15.385	15.191	14.421
12.500	12.520	0.000	3.431	6.964	8.810	10.690	13.368	14.735	15.243	15.087	14.366
13.000	13.020	0.000	3.394	6.884	8.705	10.553	13.192	14.565	15.099	14.982	14.311
13.500	13.520	0.000	3.364	6.816	8.615	10.414	13.010	14.391	14.954	14.877	14.256
14.000	14.020	0.000	3.338	6.759	8.539	10.273	12.823	14.215	14.807	14.771	14.199
14.500	14.520	0.000	3.318	6.713	8.474	10.130	12.629	14.035	14.659	14.666	14.142
15.000	15.020	0.000	3.302	6.675	8.409	9.987	12.431	13.851	14.509	14.561	14.086
15.500	15.520	0.000	3.289	6.644	8.340	9.845	12.228	13.664	14.358	14.455	14.030
16.000	16.020	0.000	3.280	6.621	8.266	9.703	12.020	13.473	14.205	14.348	13.975
16.500	16.520	0.000	3.274	6.605	8.187	9.561	11.809	13.280	14.050	14.240	13.920
17.000	17.020	0.000	3.271	6.587	8.102	9.419	11.596	13.083	13.894	14.131	13.862
17.500	17.520	0.000	3.270	6.560	8.010	9.276	11.384	12.885	13.734	14.021	13.805
18.000	18.020	0.000	3.271	6.522	7.913	9.132	11.175	12.684	13.574	13.909	13.748
18.500	18.520	0.000	3.275	6.473	7.810	8.987	10.969	12.480	13.410	13.795	13.692
19.000	19.020	0.000	3.280	6.413	7.700	8.840	10.768	12.275	13.244	13.680	13.634
19.500	19.520	0.000	3.287	6.341	7.585	8.691	10.573	12.068	13.078	13.563	13.578
20.000	20.020	0.000	3.296	6.259	7.462	8.540	10.384	11.863	12.906	13.445	13.518
20.500	20.520	0.000	3.306	6.165	7.333	8.386	10.199	11.662	12.733	13.327	13.456
21.000	21.020	0.000	3.314	6.060	7.198	8.229	10.017	11.467	12.558	13.204	13.398
21.500	21.520	0.000	3.303	5.944	7.055	8.070	9.839	11.281	12.382	13.080	13.334
22.000	22.020	0.000	3.269	5.816	6.905	7.907	9.665	11.098	12.210	12.955	13.272
22.500	22.520	0.000	3.212	5.675	6.748	7.739	9.492	10.926	12.043	12.833	13.208
23.000	23.020	0.000	3.129	5.522	6.580	7.565	9.316	10.764	11.895	12.705	13.142
23.500	23.520	0.000	3.019	5.352	6.402	7.385	9.142	10.607	11.758	12.583	13.075
24.000	24.020	0.000	2.882	5.165	6.210	7.192	8.964	10.457	11.637	12.487	13.015
24.500	24.520	0.000	2.712	4.955	5.995	6.982	8.784	10.314	11.534	12.413	12.958
24.650	24.670	0.000	2.650	4.886	5.926	6.917	8.729	10.272	11.503	12.396	12.944

Initial trim -2 m (forward

		Heeling angles (deg.)								
		0.0	10.0	20.0	25.0	30.0	40.0	50.0	60.0	70.0	80.0
T	TK	KN	KN	KN	KN	KN	KN	KN	KN	KN	KN
m	m	m	m	m	m	m	m	m	m	m	m
2.000	2.020	0.000	10.769	14.117	14.949	15.477	15.853	15.605	15.007	14.440	14.098
2.500	2.520	0.000	9.726	13.439	14.411	15.071	15.705	15.712	15.363	15.024	14.450
3.000	3.020	0.000	8.792	12.817	13.913	14.692	15.571	15.803	15.690	15.461	14.666
3.500	3.520	0.000	7.951	12.248	13.457	14.344	15.438	15.884	15.992	15.756	14.792
4.000	4.020	0.000	7.199	11.727	13.035	14.021	15.317	15.962	16.253	15.944	14.874
4.500	4.520	0.000	6.558	11.247	12.646	13.720	15.194	16.035	16.444	16.057	14.920
5.000	5.020	0.000	6.030	10.802	12.286	13.440	15.078	16.101	16.563	16.120	14.941
5.500	5.520	0.000	5.594	10.385	11.949	13.172	14.965	16.161	16.625	16.143	14.951
6.000	6.020	0.000	5.233	9.994	11.635	12.934	14.866	16.201	16.656	16.143	14.946
6.500	6.520	0.000	4.931	9.625	11.341	12.710	14.774	16.206	16.625	16.134	14.929
7.000	7.020	0.000	4.678	9.278	11.065	12.500	14.688	16.177	16.576	16.095	14.916
7.500	7.520	0.000	4.463	8.950	10.803	12.300	14.610	16.118	16.522	16.040	14.885
8.000	8.020	0.000	4.281	8.642	10.553	12.111	14.538	16.047	16.425	15.980	14.841
8.500	8.520	0.000	4.125	8.359	10.314	11.931	14.469	15.939	16.326	15.890	14.808
9.000	9.020	0.000	3.991	8.105	10.087	11.760	14.394	15.833	16.223	15.807	14.767
9.500	9.520	0.000	3.876	7.879	9.870	11.597	14.295	15.707	16.097	15.725	14.723
10.000	10.020	0.000	3.778	7.680	9.664	11.448	14.179	15.571	15.971	15.634	14.671
10.500	10.520	0.000	3.693	7.508	9.474	11.298	14.051	15.426	15.844	15.531	14.618
11.000	11.020	0.000	3.621	7.356	9.296	11.154	13.910	15.274	15.711	15.436	14.563
11.500	11.520	0.000	3.558	7.225	9.136	11.017	13.759	15.117	15.571	15.333	14.509
12.000	12.020	0.000	3.505	7.112	8.993	10.886	13.599	14.955	15.431	15.226	14.452
12.500	12.520	0.000	3.459	7.014	8.868	10.753	13.429	14.789	15.288	15.120	14.394
13.000	13.020	0.000	3.421	6.931	8.759	10.614	13.252	14.618	15.143	15.013	14.334
13.500	13.520	0.000	3.389	6.860	8.665	10.472	13.068	14.443	14.996	14.907	14.276
14.000	14.020	0.000	3.362	6.800	8.585	10.327	12.877	14.263	14.847	14.801	14.217
14.500	14.520	0.000	3.339	6.750	8.516	10.180	12.681	14.080	14.696	14.695	14.161
15.000	15.020	0.000	3.321	6.708	8.447	10.033	12.479	13.893	14.544	14.588	14.104
15.500	15.520	0.000	3.307	6.675	8.375	9.886	12.273	13.704	14.391	14.481	14.050
16.000	16.020	0.000	3.296	6.649	8.297	9.740	12.062	13.511	14.236	14.374	13.995
16.500	16.520	0.000	3.288	6.629	8.214	9.594	11.847	13.315	14.080	14.266	13.940
17.000	17.020	0.000	3.283	6.607	8.125	9.447	11.630	13.117	13.922	14.157	13.886
17.500	17.520	0.000	3.280	6.575	8.030	9.300	11.414	12.915	13.762	14.046	13.831
18.000	18.020	0.000	3.280	6.534	7.929	9.152	11.201	12.712	13.599	13.935	13.773
18.500	18.520	0.000	3.282	6.481	7.822	9.002	10.992	12.506	13.434	13.820	13.715
19.000	19.020	0.000	3.286	6.418	7.709	8.851	10.788	12.298	13.268	13.707	13.657
19.500	19.520	0.000	3.292	6.345	7.589	8.699	10.591	12.089	13.100	13.588	13.599
20.000	20.020	0.000	3.300	6.258	7.465	8.544	10.398	11.881	12.929	13.469	13.540
20.500	20.520	0.000	3.308	6.162	7.332	8.388	10.209	11.678	12.756	13.348	13.482
21.000	21.020	0.000	3.309	6.054	7.195	8.230	10.024	11.481	12.581	13.225	13.420
21.500	21.520	0.000	3.294	5.935	7.049	8.068	9.842	11.291	12.405	13.103	13.357
22.000	22.020	0.000	3.257	5.804	6.897	7.902	9.665	11.109	12.231	12.978	13.294
22.500	22.520	0.000	3.196	5.661	6.737	7.730	9.488	10.936	12.063	12.851	13.230
23.000	23.020	0.000	3.109	5.505	6.566	7.553	9.311	10.769	11.908	12.725	13.165
23.500	23.520	0.000	2.996	5.332	6.382	7.366	9.133	10.608	11.768	12.606	13.100
24.000	24.020	0.000	2.853	5.137	6.182	7.168	8.953	10.455	11.644	12.506	13.035
24.500	24.520	0.000	2.681	4.928	5.973	6.966	8.777	10.314	11.542	12.436	12.988
24.650	24.670	0.000	2.621	4.869	5.915	6.908	8.728	10.278	11.518	12.419	12.982

3.2 Stability Cross Curves Data



4. PART CONDITIONS

LOADING CONDITION - PCON1 : 100% BUNKERS ETC.

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6185.2		50.47	-0.76	22.82	
Total weight		30827.1		115.65	-0.15	15.51	

F L O A T I N G P O S I T I O N

Draught moulded	3.407	m	KM	49.97 m
Trim	7.052	m	KG	15.51 m
Heel, PS=+	-0.3	deg		
TA	6.933	m	GM0	34.46 m
TF	-0.118	m	GMCORR	0.00 m
Trimming moment	******	tonm	GM	34.46 m

LOADING CONDITION - PCON2 : 50% BUNKERS ETC.

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		3237.0		38.77	-2.48	22.46	
Total weight		27878.8		121.18	-0.29	14.69	

F L O A T I N G P O S I T I O N

Draught moulded	3.078	m	KM	54.52 m
Trim	5.443	m	KG	14.69 m
Heel, PS=+	-0.4	deg		
TA	5.800	m	GM0	39.83 m
TF	0.356	m	GMCORR	-0.11 m
Trimming moment	-835487	tonm	GM	39.72 m

LOADING CONDITION - PCON3 : 10% BUNKERS ETC.

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		868.1		30.69	-7.22	22.97	
Total weight		25509.9		128.56	-0.25	13.99	

F L O A T I N G P O S I T I O N

Draught moulded	2.797 m	KM	58.96 m	
Trim	3.809 m	KG	13.99 m	
Heel, PS=+	-0.3 deg			
TA	4.701 m	GM0	44.97 m	
TF	0.892 m	GMCORR	-0.03 m	
Trimming moment	-590369 tonm	GM	44.94 m	

5. WATER BALLAST GROUPS

LOADING CONDITION - LG1 : WB NO.1 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
SUBTOTAL		7259.7		253.70	-0.00	12.08	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		7259.7		253.70	-0.00	12.08	
Total weight		31901.6		159.70	-0.00	13.31	

F L O A T I N G P O S I T I O N

Draught moulded	3.206 m	KM	48.56 m	
Trim	-1.553 m	KG	13.31 m	
Heel, PS=+	0.0 deg			
TA	2.430 m	GM0	35.25 m	
TF	3.983 m	GMCORR	0.00 m	
Trimming moment	253735 tonm	GM	35.25 m	

LOADING CONDITION - LG2 : WB NO.2 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
SUBTOTAL		6736.0		228.54	-0.00	11.11	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6736.0		228.54	-0.00	11.11	
Total weight		31377.8		152.73	-0.00	13.12	

F L O A T I N G P O S I T I O N

Draught moulded	3.219	m	KM	49.38 m
Trim	-0.179	m	KG	13.12 m
Heel, PS=+	0.0	deg		
TA	3.129	m	GM0	36.26 m
TF	3.308	m	GMCORR	0.00 m
Trimming moment	29249	tonm	GM	36.26 m

LOADING CONDITION - LG3 : WB NO.3 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
SUBTOTAL		6572.1		203.10	0.00	10.87	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6572.1		203.10	0.00	10.87	
Total weight		31214.0		146.98	0.00	13.08	

F L O A T I N G P O S I T I O N

Draught moulded	3.251	m	KM	49.67 m
Trim	0.931	m	KG	13.08 m
Heel, PS=+	0.0	deg		
TA	3.716	m	GM0	36.59 m
TF	2.785	m	GMCORR	0.00 m
Trimming moment	-151965	tonm	GM	36.59 m

LOADING CONDITION - LG4 : WB NO.4 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
SUBTOTAL		6578.2		177.63	0.00	10.86	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6578.2		177.63	0.00	10.86	
Total weight		31220.0		141.62	0.00	13.08	

F L O A T I N G P O S I T I O N

Draught moulded	3.293 m	KM	49.72 m	
Trim	1.966 m	KG	13.08 m	
Heel, PS=+	0.0 deg			
TA	4.275 m	GM0	36.64 m	
TF	2.310 m	GMCORR	0.00 m	
Trimming moment	-319757 tonm	GM	36.64 m	

LOADING CONDITION - LG5 : WB NO.5 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
SUBTOTAL		6578.2		152.15	-0.00	10.86	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6578.2		152.15	-0.00	10.86	
Total weight		31220.0		136.25	-0.00	13.08	

F L O A T I N G P O S I T I O N

Draught moulded	3.331 m	KM	49.74 m	
Trim	3.007 m	KG	13.08 m	
Heel, PS=+	0.0 deg			
TA	4.835 m	GM0	36.67 m	
TF	1.828 m	GMCORR	0.00 m	
Trimming moment	-486270 tonm	GM	36.67 m	

LOADING CONDITION - LG6 : WB NO.6 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
SUBTOTAL		6578.2		126.67	0.00	10.86	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6578.2		126.67	0.00	10.86	
Total weight		31220.0		130.89	0.00	13.08	

F L O A T I N G P O S I T I O N

Draught moulded	3.367	m	KM	49.74 m
Trim	4.056	m	KG	13.08 m
Heel, PS=+	0.0	deg		
TA	5.395	m	GM0	36.66 m
TF	1.339	m	GMCORR	0.00 m
Trimming moment	-650105	tonm	GM	36.66 m

LOADING CONDITION - LG7 : WB NO.7 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
SUBTOTAL		6548.8		101.23	0.00	10.90	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		6548.8		101.23	0.00	10.90	
Total weight		31190.7		125.55	0.00	13.09	

F L O A T I N G P O S I T I O N

Draught moulded	3.396	m	KM	49.73 m
Trim	5.106	m	KG	13.09 m
Heel, PS=+	0.0	deg		
TA	5.949	m	GM0	36.64 m
TF	0.843	m	GMCORR	0.00 m
Trimming moment	-808056	tonm	GM	36.64 m

LOADING CONDITION - LG8 : WB NO.8 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
SUBTOTAL		4275.7		75.52	-0.00	6.08	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		4275.7		75.52	-0.00	6.08	
Total weight		28917.6		123.66	-0.00	12.55	

F L O A T I N G P O S I T I O N

Draught moulded	3.173	m	KM	52.93 m
Trim	5.141	m	KG	12.55 m
Heel, PS=+	0.0	deg		
TA	5.744	m	GM0	40.38 m
TF	0.603	m	GMCORR	0.00 m
Trimming moment	-800102	tonm	GM	40.38 m

LOADING CONDITION - LG9 : WB NO.9 100%

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
SUBTOTAL		4348.3		51.27	0.00	8.63	0.00
Lightweight		24641.9		132.01	0.00	13.67	
Deadweight		4348.3		51.27	0.00	8.63	
Total weight		28990.1		119.90	0.00	12.92	

F L O A T I N G P O S I T I O N

Draught moulded	3.199	m	KM	52.73 m
Trim	5.863	m	KG	12.92 m
Heel, PS=+	0.0	deg		
TA	6.131	m	GM0	39.81 m
TF	0.267	m	GMCORR	0.00 m
Trimming moment	-900551	tonm	GM	39.81 m

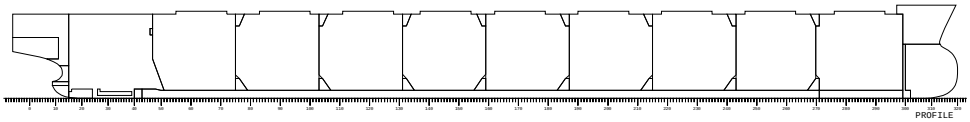
6. LOADING CONDITIONS

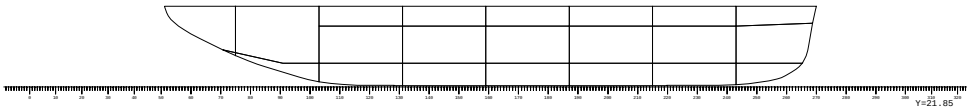
Loading conditions:174 000 DWT DOUBLE SKIN BULK CARRIER (W=24641.9ton,LCG=132.014m&vcg=13.667m)

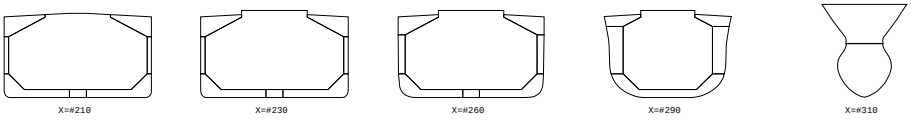
No.	Condition	Bunkers	Ballast	DWT	Displacement	d_{aft}	$d_{L/2}$	d_{fwd}	Trim	Max BM*	Max SF*	VCG	G'M
			[t]	[t]	[t]	[m]	[m]	[m]	[m]	[t·m]	[t]	[m]	[m]
1	Light ship (Not seagoing condition)		0.0	0.0	24641.9	4.26	2.69	1.12	3.14	238595.2	3702.0	13.67	47.12
2	Normal ballast-Departure bunkers	Departure	60974.2	67159.0	91801.3	9.98	8.97	7.96	2.02	463872.7	5830.2	12.20	10.58
3	Normal ballast-50% bunkers	50%	60974.2	64211.0	88853.10	9.00	8.65	8.31	0.69	42.6760.5	6029.0	11.84	11.26
4	Normal ballast-Arrival bunkers	Arrival	59192.8	60060.9	84702.7	8.52	8.27	8.02	0.50	318651.0	4669.6	11.53	12.07
5	Heavy ballast-Departure bunkers	Departure	59352.0	86700.8	111343.0	12.29	10.76	9.22	3.07	178874.7	9058.0	12.42	8.52
6	Heavy ballast-50% bunkers	50%	60974.2	85374.7	110017.0	12.13	10.63	9.14	2.99	188387.8	8452.8	12.19	8.85
7	Heavy ballast-Arrival bunkers	Arrival	60974.2	83005.9	107648.0	11.27	10.39	9.51	1.76	165079.0	8710.2	11.97	9.15
8	Homogeneous cargo (0.792t/m3)-at Design draught-Departure bunkers	Departure	0.0	153624.0	178266.0	17.45	16.49	15.52	1.93	-140634.0	-4255.7	14.04	4.75
9	Homogeneous cargo (0.792t/m3)-at Design draught-50% bunkers	50%	0.0	150676.0	175318.0	16.75	16.24	15.73	1.02	-153518.0	-3883.7	13.89	4.88
10	Homogeneous cargo (0.792t/m3)-at Design draught-Arrival bunkers	Arrival	0.0	148307.0	172949.0	16.07	16.05	16.02	0.05	-189926.0	-3768.6	13.78	4.97
11	Homogeneous cargo (0.900t/m3) -at Scantling draught-Departure bunkers	Departure	0.0	173730.0	198372.0	18.57	18.18	17.79	0.78	-238720.0	-5912.4	14.01	4.72
12	Homogeneous cargo (0.900t/m3) -at Scantling draught-50% bunkers	50%	1622.2	172404.0	197046.0	18.45	18.07	17.68	0.77	-196279.0	-4602.6	13.89	4.82
13	Homogeneous cargo (0.900t/m3) -at Scantling draught-Arrival bunkers	Arrival	3300.8	171713.0	196355.0	18.08	18.02	17.95	0.13	-250779.0	-6435.9	13.63	5.08
14	Fully loaded with coal (42 cft/longton)-Departure bunkers	Departure	0.0	165166.0	189808.0	18.10	17.46	16.82	1.28	-195624.0	-5202.4	14.03	4.71
15	Fully loaded with coal (42 cft/longton)-50% bunkers	50%	0.0	162218.0	186860.0	17.40	17.22	17.03	0.37	-206052.0	-4815.8	13.88	4.83
16	Fully loaded with coal (42 cft/longton)-Arrival bunkers	Arrival	1622.2	161471.0	186113.0	17.31	17.15	16.99	0.32	-190349.0	-3749.5	13.80	4.91
17	Fully loaded with ore (12 cft/longton) in alternative holds-at Design draught-Departure bunkers	Departure	0.0	153608.0	178250.0	17.61	16.48	15.35	2.26	209316.9	9926.0	9.65	9.16
18	Fully loaded with ore (12 cft/longton) in alternative holds-at Design draught-50% bunkers	50%	0.0	150660.0	175307.0	16.92	16.24	15.56	1.36	194709.50	10376.5	9.4	9.36
19	Fully loaded with ore (12 cft/longton) in alternative holds-at Design draught-Arrival bunkers	Arrival	0.0	148291.0	172933.0	16.25	16.05	15.84	0.41	168080.0	9359.6	9.25	9.51
20	Fully loaded with ore (12 cft/longton) in alternative holds-at Scantling draught-Departure bunkers	Departure	0.0	173959.0	198601.0	18.77	18.20	17.62	1.15	162727.8	10257.6	10.11	8.63
21	Fully loaded with ore (12 cft/longton) in alternative holds-at Scantling draught-50% bunkers	50%	0.0	171011.0	195653.0	18.09	17.96	17.82	0.27	148927.5	10374.8	9.91	8.80
22	Fully loaded with ore (12 cft/longton) in alternative holds-at Scantling draught-Arrival bunkers	Arrival	1622.2	170264.0	194906.0	18.01	17.90	17.78	0.23	157612.7	9833.5	9.81	8.89
23	Homogeneous ore cargo (12 cft/longton)-at Scantling draught-Departure bunkers	Departure	0.0	173786.0	198427.0	18.54	18.19	17.83	0.71	-260536.0	-6133.3	7.75	10.98
24	Homogeneous ore cargo (12 cft/longton)-at Scantling draught-50% bunkers	50%	1622.2	172460.0	197101.0	18.42	18.07	17.72	0.70	-214431.0	-4815.4	7.59	11.12
25	Homogeneous ore cargo (12 cft/longton)-at Scantling draught-Arrival bunkers	Arrival	3614.8	172083.0	196725.0	18.14	18.04	17.94	0.20	-265977.0	-6539.1	7.29	11.42
26	Multiple port loading-First port-(42 cft/longton)-Departure bunkers (Second port see condition No 14)	Departure	3562.9	82624.2	107266.0	11.79	10.39	8.99	2.80	150049.0	6601.0	14.03	7.20
27	Multiple port loading-First port-(42 cft/longton)-50% bunkers (Second port see condition No 15)	50%	3562.9	79675.9	104318.0	10.87	10.09	9.30	1.57	116913.9	5574.0	13.77	7.61
28	Multiple port loading-First port-(42 cft/longton)-Arrival bunkers (Second port see condition No 16)	Arrival	3562.9	77307.0	101949.0	9.97	9.84	9.70	0.27	63912.6	4987.8	13.58	7.95
29	Harbour condition for Dunkirk East-d>12.95m-Arrival bunkers (Not seagoing condition)	Arrival	60974.2	112303.0	136944.0	12.95	12.95	12.95	0.00	-176675.0	-7872.2	11.06	8.32
30	Docking condition-Arrival bunkers (Not seagoing condition)	Arrival	5189.2	6057.2	30699.1	3.16	3.16	3.16	0.00	407317.8	-5550.4	12.74	35.26
31	Stern tube seal renewal condition- Arrival bunkers(Not seagoing condition)	Arrival	7702.6	8570.6	33212.5	2.59	3.33	4.13	-1.60	465985.2	-7097.3	12.37	34.54

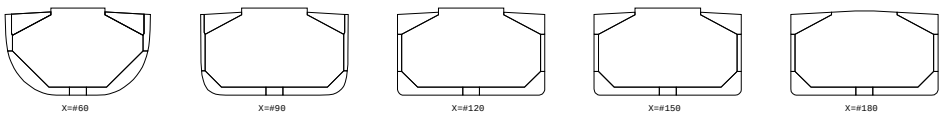
*Positive numbers corresponds to Hogging and negative numbers corresponds to Sagging

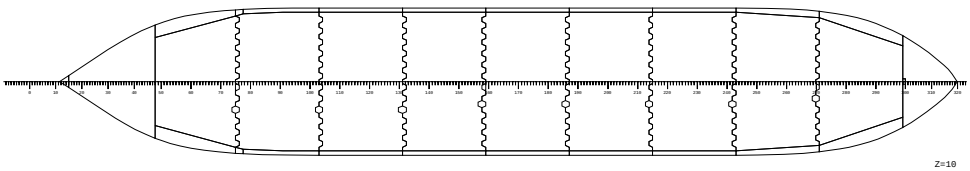
LOADING CONDITION - COND01 : LIGHTSHIP

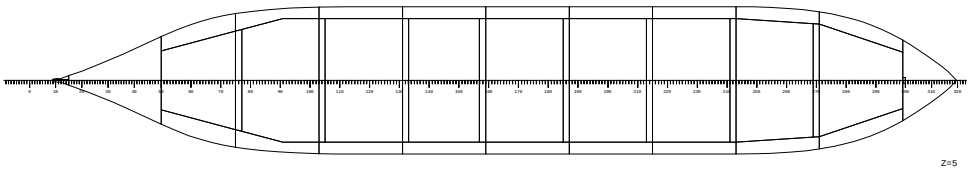


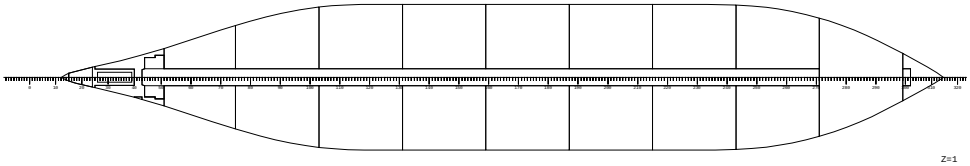










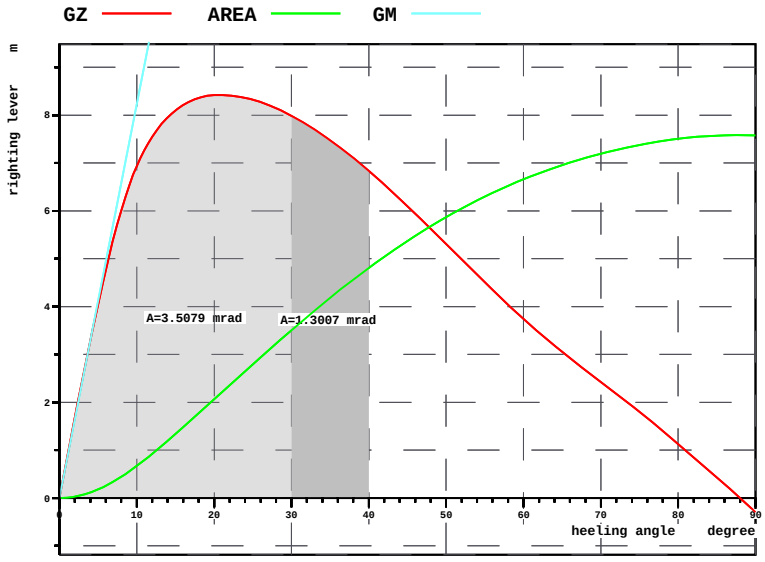


Lightweight	24641.9	132.01	0.00	13.67
Deadweight	0.0	0.00	0.00	0.00
Total weight	24641.9	132.01	0.00	13.67

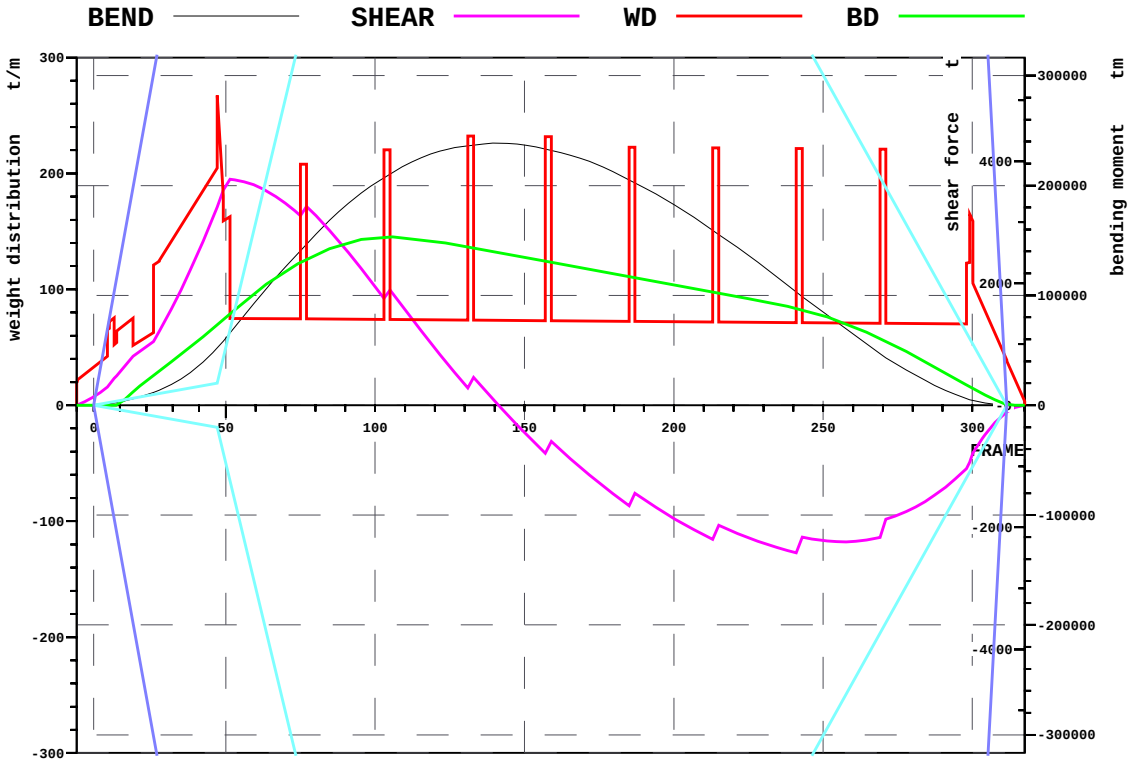
F L O A T I N G P O S I T I O N

Draught moulded	2.689	m	KM	60.79 m
Trim	3.146	m	KG	13.67 m
Heel, PS=+	0.0	deg		
TA	4.262	m	GM0	47.12 m
TF	1.116	m	GMCORR	0.00 m
Trimming moment	-488799	tonm	GM	47.12 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	0.00	0.000	0.0	0.000
5.0	4.05	0.180	0.0	0.000
10.0	6.94	0.673	0.0	0.000
20.0	8.42	2.063	0.0	0.000
30.0	7.99	3.508	0.0	0.000
40.0	6.83	4.809	0.0	0.000
45.0	6.10	5.373	0.0	0.000
50.0	5.30	5.871	0.0	0.000
55.0	4.51	6.299	0.0	0.000
60.0	3.74	6.659	0.0	0.000
65.0	3.05	6.955	0.0	0.000
70.0	2.42	7.193	0.0	0.000
75.0	1.80	7.377	0.0	0.000
80.0	1.13	7.505	0.0	0.000
85.0	0.43	7.574	0.0	0.000
90.0	-0.29	7.580	0.0	0.000



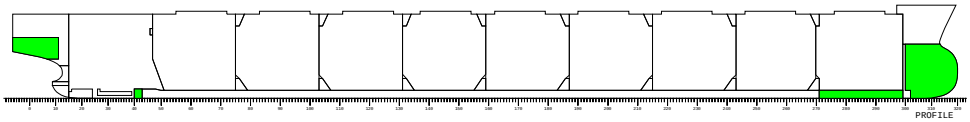
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	3.508	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	4.809	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.301	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.987	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	20.768	deg	NOT MET
GM0.15	GM > 0.15 m	0.150	47.116	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	1.059		OK

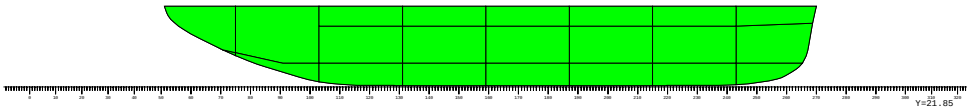


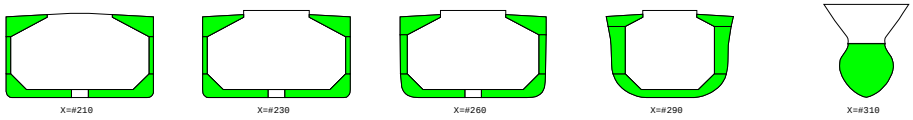
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	421	142	32.90	0.00
8.80	11.00	3538	600	68.09	3.69
12.00	15.00	5800	805	75.15	11.76
37.60	47.00	52147	3250	267.19	69.36
41.13	51.00	64537	3672	161.98	77.91
62.97	75.00	140922	3105	74.47	123.13
88.45	103.00	207182	1751	220.31	144.67
113.93	131.00	236107	282	73.37	136.61
139.41	159.00	231797	-591	231.66	123.37
164.89	187.00	202284	-1443	222.51	110.05
190.37	215.00	155208	-1969	71.72	96.70
215.85	243.00	98912	-2163	221.41	82.03
241.33	271.00	43220	-1866	220.86	54.83
266.81	299.00	5126	-937	123.00	15.98
278.01	313.00	192	-118	40.17	0.89

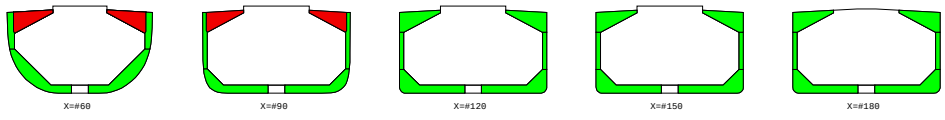
				X	FRAME
SHEAR FORCE (MIN)	-2415.2 t	(23.2%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	3702.0 t	(33.2%)		41.5 m	51
SAGGING MOMENT	-1.6 tm			283.5 m	320
HOGGING MOMENT	238595.2 tm	(50.5%)		123.2 m	141

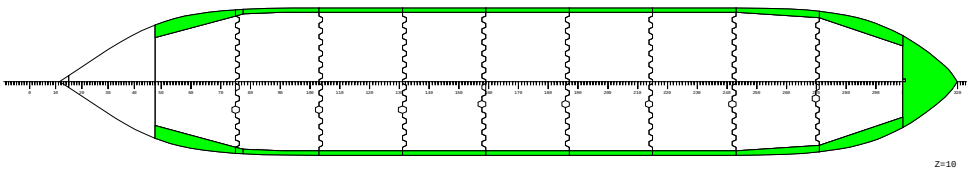
LOADING CONDITION - COND02 : NORMAL BALLAST-DEPARTURE BUNKERS

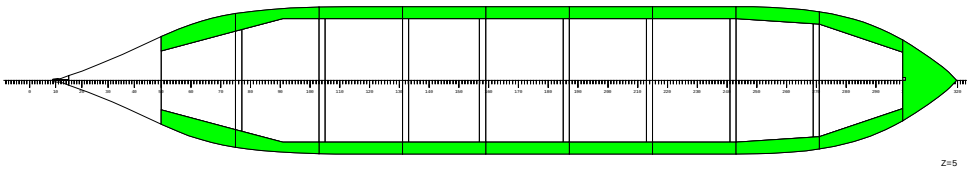


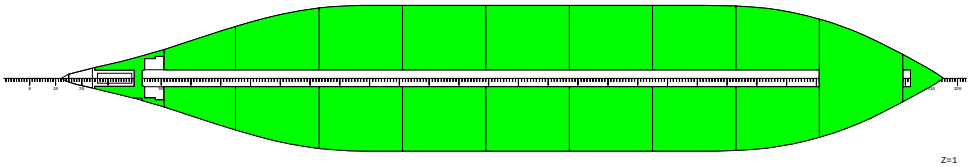












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

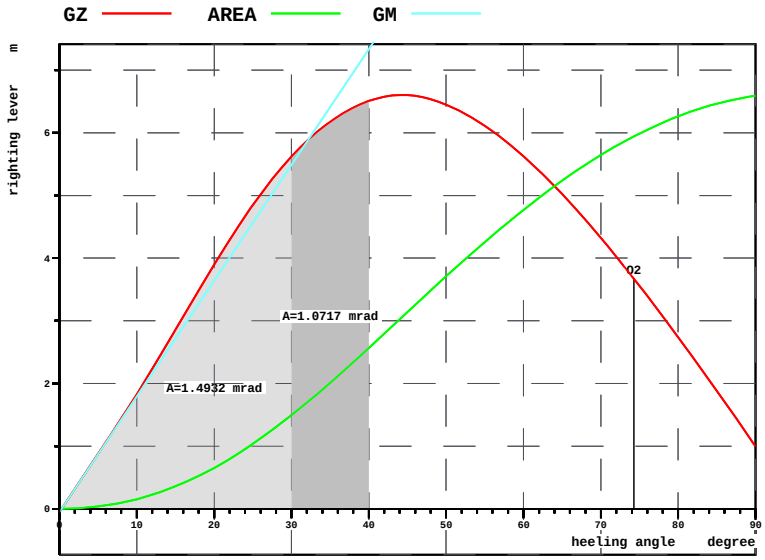
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	67159.5	152.30	-0.07 11.67
Total weight	91801.3	146.85	-0.05 12.20

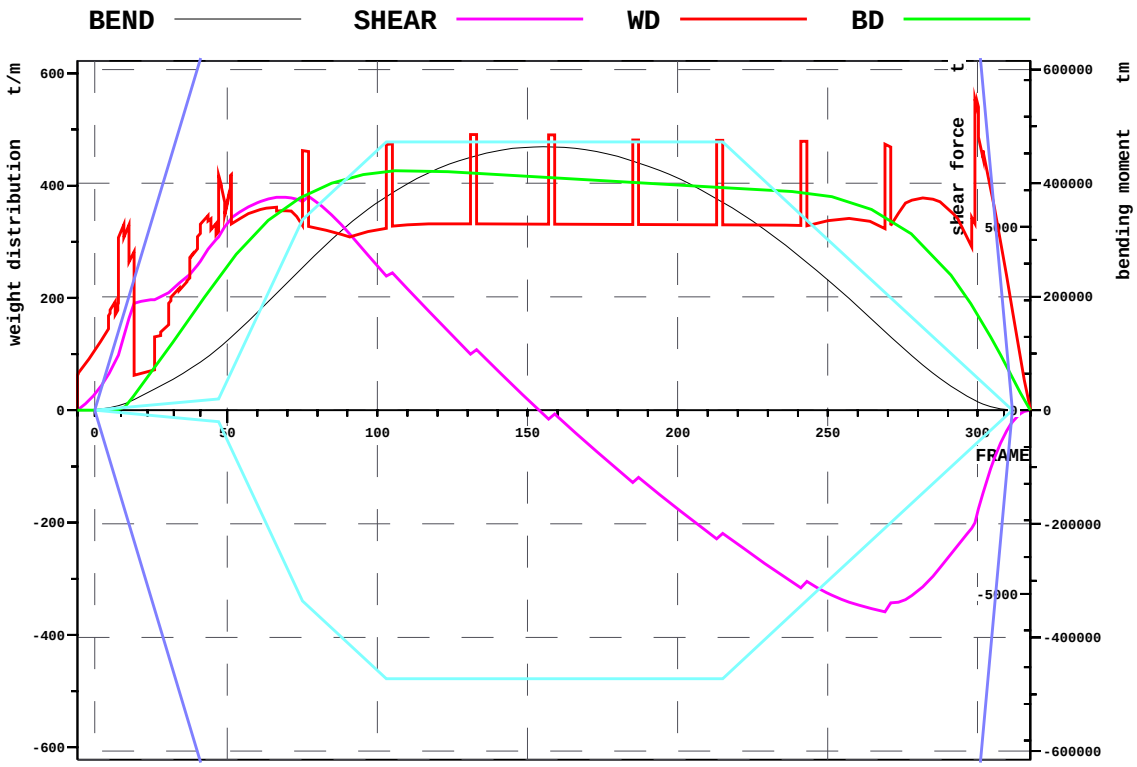
F L O A T I N G P O S I T I O N

Draught moulded	8.971	m	KM	22.79 m
Trim	2.027	m	KG	12.20 m
Heel, PS=+	-0.3	deg		
TA	9.984	m	GM0	10.58 m
TF	7.957	m	GMCORR	0.00 m
Trimming moment	-394576	tonm	GM	10.58 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.88	0.036	0.0	0.000
10.0	1.83	0.153	0.0	0.000
20.0	3.90	0.652	0.0	0.000
30.0	5.62	1.493	0.0	0.000
40.0	6.51	2.565	0.0	0.000
45.0	6.60	3.139	0.0	0.000
50.0	6.45	3.710	0.0	0.000
55.0	6.11	4.259	0.0	0.000
60.0	5.63	4.772	0.0	0.000
65.0	5.02	5.238	0.0	0.000
70.0	4.33	5.646	0.0	0.000
75.0	3.56	5.991	0.0	0.000
80.0	2.74	6.266	0.0	0.000
85.0	1.88	6.467	0.0	0.000
90.0	0.99	6.593	0.0	0.000



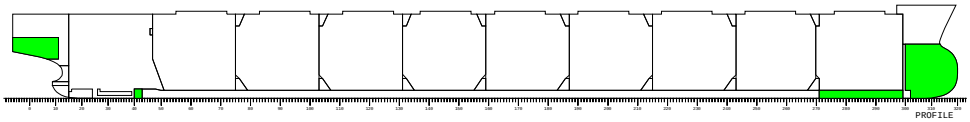
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.493	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.565	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.072	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.604	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.363	deg	OK
GM0.15	GM > 0.15 m	0.150	10.582	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.008		OK

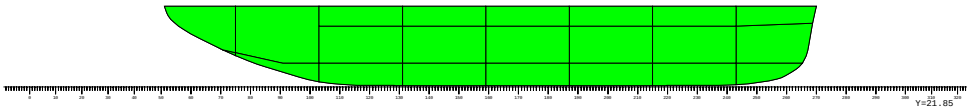


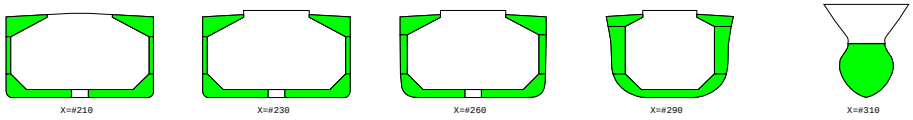
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1157	443	106.41	0.00
8.80	11.00	10658	2015	331.01	5.28
12.00	15.00	18650	2914	62.53	27.06
37.60	47.00	109775	4723	418.74	235.58
41.13	51.00	127388	5195	417.90	263.71
62.97	75.00	250990	5686	396.05	380.13
88.45	103.00	376047	3661	473.44	424.62
113.93	131.00	445224	1530	490.87	422.41
139.41	159.00	463331	-98	490.32	413.91
164.89	187.00	435441	-1826	481.16	405.34
190.37	215.00	366115	-3350	405.42	396.74
215.85	243.00	260570	-4667	328.47	386.09
241.33	271.00	129115	-5250	331.37	336.83
266.81	299.00	17118	-3070	332.73	177.94
278.01	313.00	589	-335	182.47	60.50

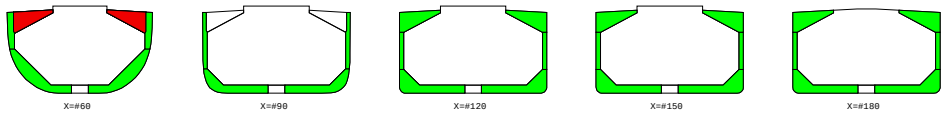
				X	FRAME
SHEAR FORCE (MIN)	-5488.6 t	(50.0%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	5830.2 t	(53.8%)		64.8 m	77
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	463872.7 tm	(98.2%)		135.9 m	155

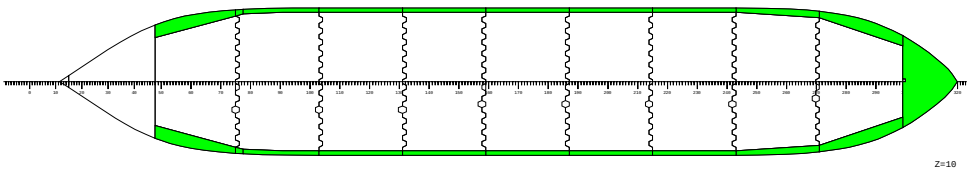
LOADING CONDITION - COND03 : NORMAL BALLAST-50% BUNKERS

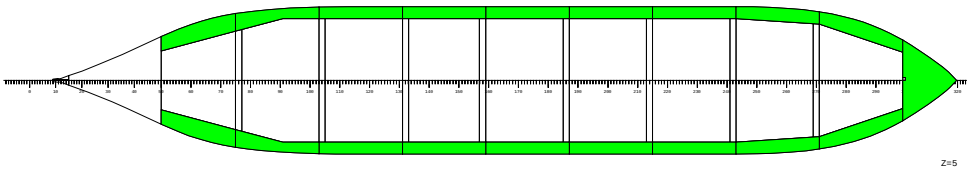


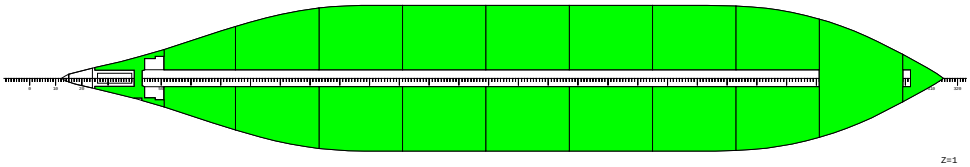












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

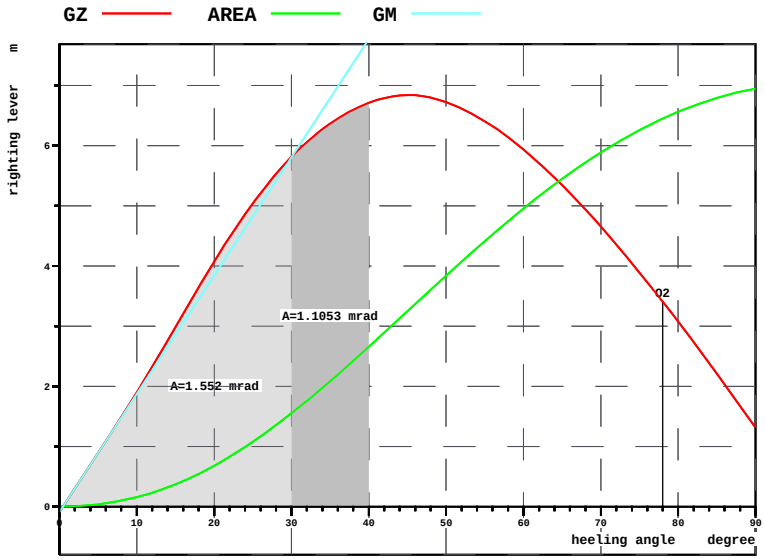
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00	13.67
Deadweight	64211.2	156.38	-0.13	11.14
Total weight	88853.1	149.62	-0.09	11.84

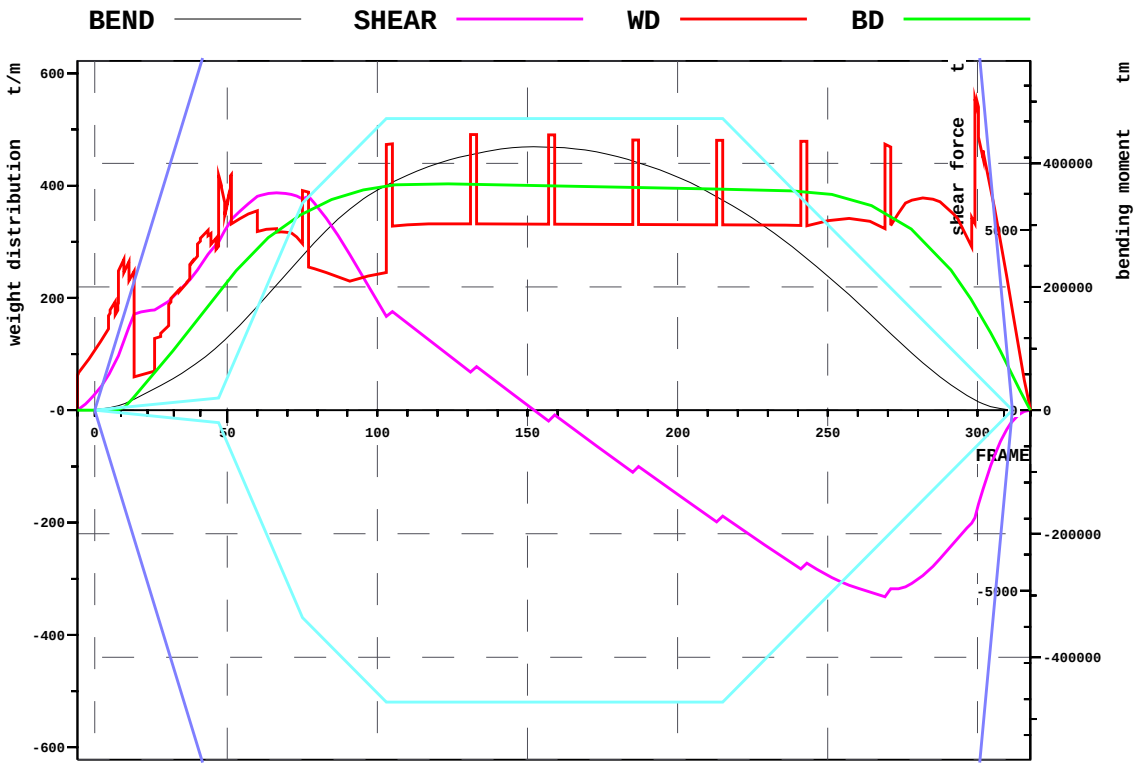
F L O A T I N G P O S I T I O N

Draught moulded	8.654	m	KM	23.10 m
Trim	0.698	m	KG	11.84 m
Heel, PS=+	-0.5	deg		
TA	9.003	m	GM0	11.26 m
TF	8.306	m	GMCORR	0.00 m
Trimming moment	-133550	tonm	GM	11.26 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.09	0.000	0.0	0.000
5.0	0.89	0.035	273.1	0.003
10.0	1.90	0.156	572.7	0.006
20.0	4.07	0.678	1197.1	0.013
30.0	5.82	1.552	1623.0	0.018
40.0	6.71	2.657	1920.4	0.022
45.0	6.84	3.250	2017.0	0.023
50.0	6.72	3.844	2076.3	0.023
55.0	6.41	4.418	2095.0	0.024
60.0	5.94	4.958	2072.0	0.023
65.0	5.34	5.451	2014.9	0.023
70.0	4.66	5.888	1930.4	0.022
75.0	3.89	6.261	1823.1	0.021
80.0	3.08	6.565	1696.6	0.019
85.0	2.21	6.796	1553.9	0.017
90.0	1.31	6.950	1399.3	0.016



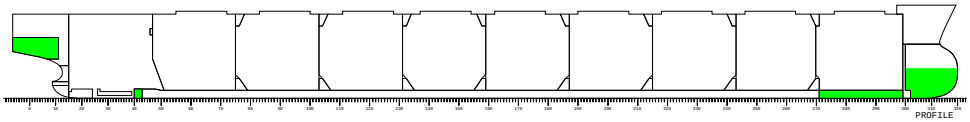
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.552	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.657	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.105	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.844	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.089	deg	OK
GM0.15	GM > 0.15 m	0.150	11.259	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.913		OK

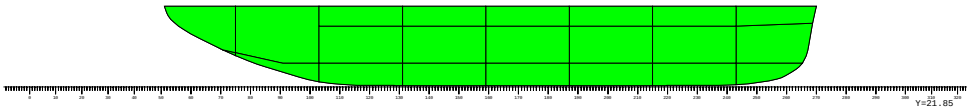


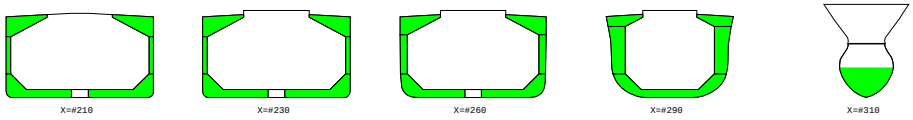
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1116	444	106.41	0.00
8.80	11.00	10470	1918	268.47	5.28
12.00	15.00	17841	2664	59.79	24.23
37.60	47.00	104895	4657	418.43	209.16
41.13	51.00	122416	5225	417.60	235.44
62.97	75.00	250069	5861	344.16	349.78
88.45	103.00	364325	2601	473.45	398.88
113.93	131.00	414034	1054	490.88	402.27
139.41	159.00	425749	-131	490.33	399.39
164.89	187.00	400902	-1561	481.17	396.43
190.37	215.00	340403	-2930	405.43	393.45
215.85	243.00	245826	-4234	328.48	388.41
241.33	271.00	123850	-4944	331.37	344.28
266.81	299.00	16700	-2984	332.74	186.14
278.01	313.00	605	-322	182.48	64.81

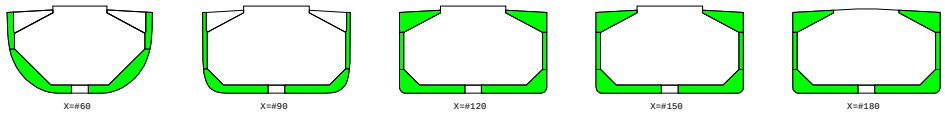
				X	FRAME
SHEAR FORCE (MIN)	-5170.2 t	(47.1%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	6029.0 t	(55.1%)		55.1 m	66
SAGGING MOMENT	-				
HOGGING MOMENT	426760.5 tm	(90.3%)		133.0 m	152

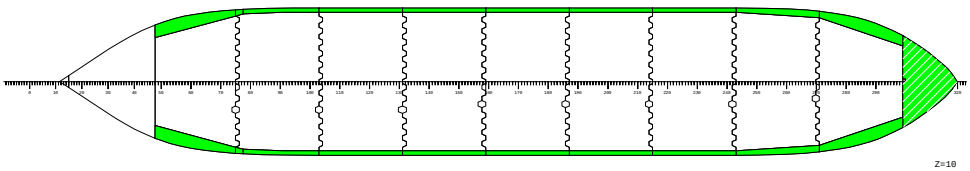
LOADING CONDITION - COND04 : NORMAL BALLAST-ARRIVAL BUNKERS

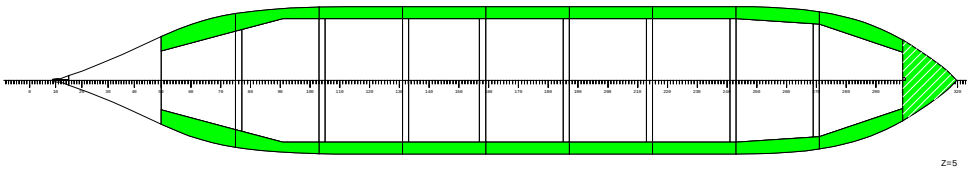


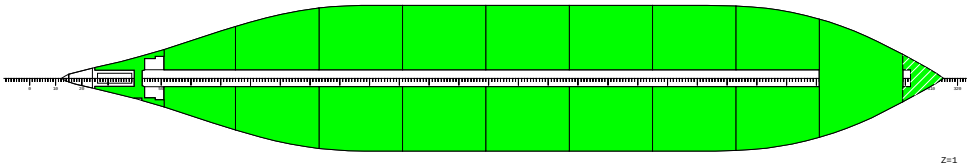












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00

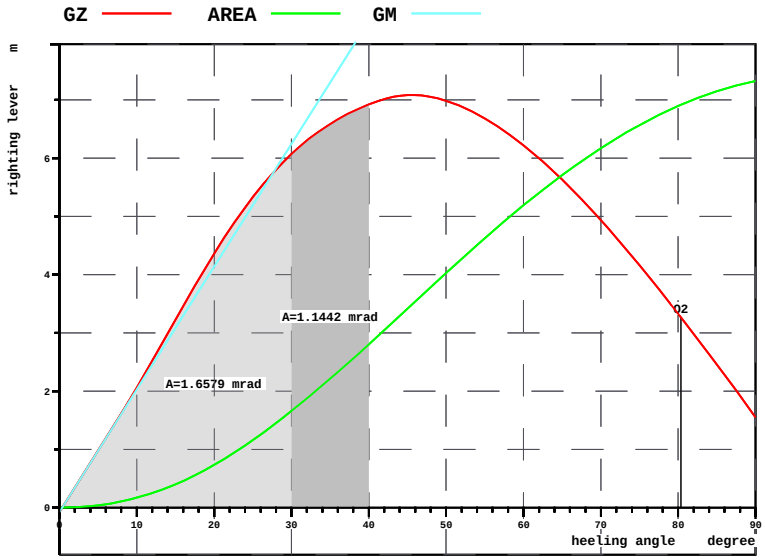
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	1781.5	50.0	272.58	-0.00	5.53	9845.88
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		59192.8		159.33	-0.00	10.47	9845.88

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	60060.9	157.47	-0.11 10.65
Total weight	84702.7	150.06	-0.08 11.53

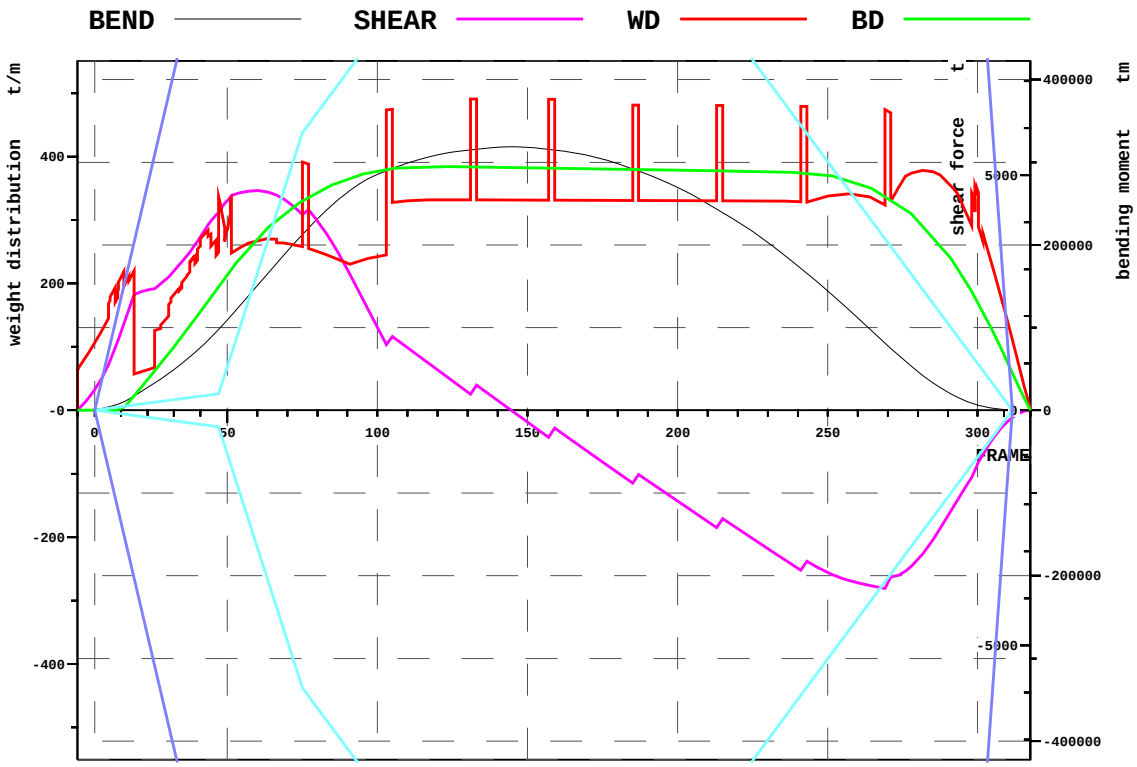
F L O A T I N G P O S I T I O N

Draught moulded	8.267	m	KM	23.71 m
Trim	0.504	m	KG	11.53 m
Heel, PS=+	-0.4	deg		
TA	8.519	m	GM0	12.18 m
TF	8.015	m	GMCORR	-0.12 m
Trimming moment	-95450	tonm	GM	12.07 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.08	0.000	0.0	0.000
5.0	0.98	0.040	921.4	0.011
10.0	2.06	0.172	1830.8	0.022
20.0	4.37	0.734	3540.9	0.042
30.0	6.07	1.658	5101.6	0.060
40.0	6.92	2.802	6414.3	0.076
45.0	7.08	3.415	6919.2	0.082
50.0	6.99	4.030	7287.0	0.086
55.0	6.68	4.628	7536.4	0.089
60.0	6.22	5.192	7686.6	0.091
65.0	5.63	5.710	7751.5	0.092
70.0	4.94	6.171	7731.2	0.091
75.0	4.16	6.569	7618.3	0.090
80.0	3.33	6.896	7436.0	0.088
85.0	2.46	7.149	7197.6	0.085
90.0	1.55	7.324	6913.4	0.082



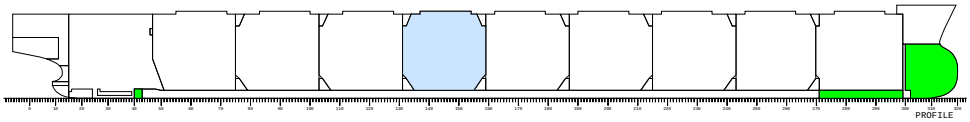
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.658	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.802	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.144	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.084	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.610	deg	OK
GM0.15	GM > 0.15 m	0.150	12.066	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.686		OK

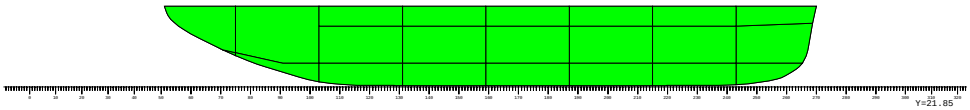


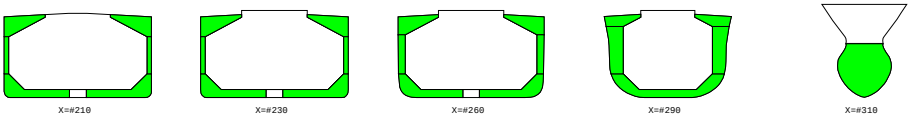
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1107	444	106.41	0.00
8.80	11.00	10384	1840	218.55	5.28
12.00	15.00	17272	2462	57.06	23.04
37.60	47.00	97686	4198	336.99	194.56
41.13	51.00	113161	4528	334.21	219.45
62.97	75.00	212486	4158	257.92	330.18
88.45	103.00	289706	1400	473.46	379.35
113.93	131.00	315030	341	490.89	383.54
139.41	159.00	314524	-378	490.34	381.49
164.89	187.00	289083	-1362	481.18	379.35
190.37	215.00	239085	-2305	405.44	377.19
215.85	243.00	165571	-3206	328.48	372.99
241.33	271.00	74645	-3539	331.38	330.48
266.81	299.00	7327	-1281	332.75	177.74
278.01	313.00	315	-159	114.76	61.18

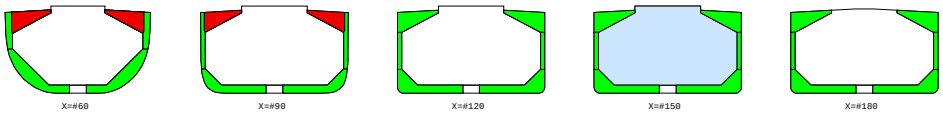
				X	FRAME
SHEAR FORCE (MIN)	-3790.0 t	(34.5%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	4669.6 t	(42.3%)		49.5 m	60
SAGGING MOMENT	-				
HOGGING MOMENT	318651.0 tm	(67.4%)		126.0 m	144

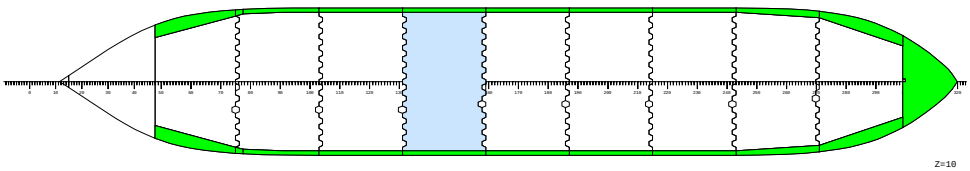
LOADING CONDITION - COND05 : HEAVY BALLAST-DEPARTURE BUNKERS

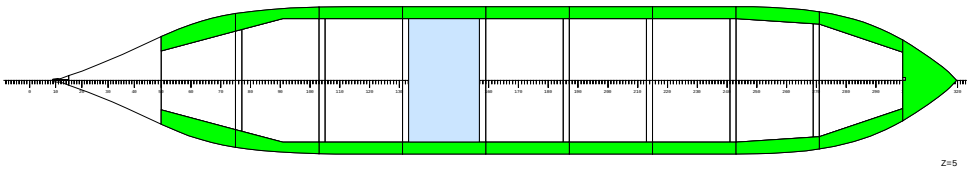


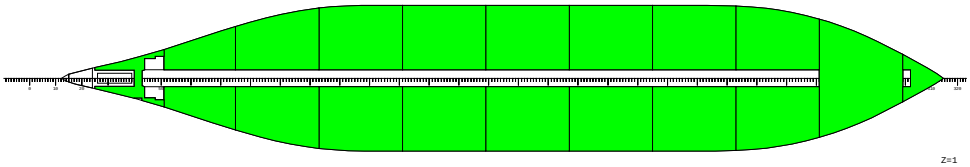












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00

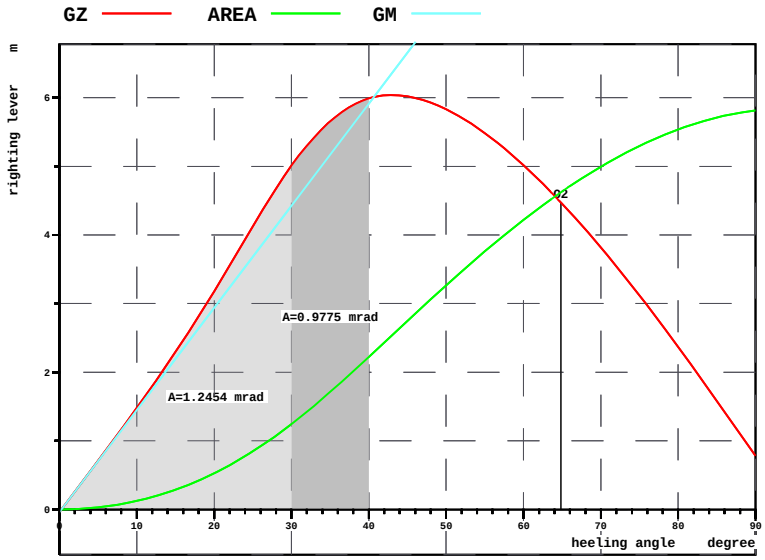
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		59352.0		166.94	0.00	10.38	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	86700.8	148.79	-0.05 12.06
Total weight	111342.6	145.08	-0.04 12.42

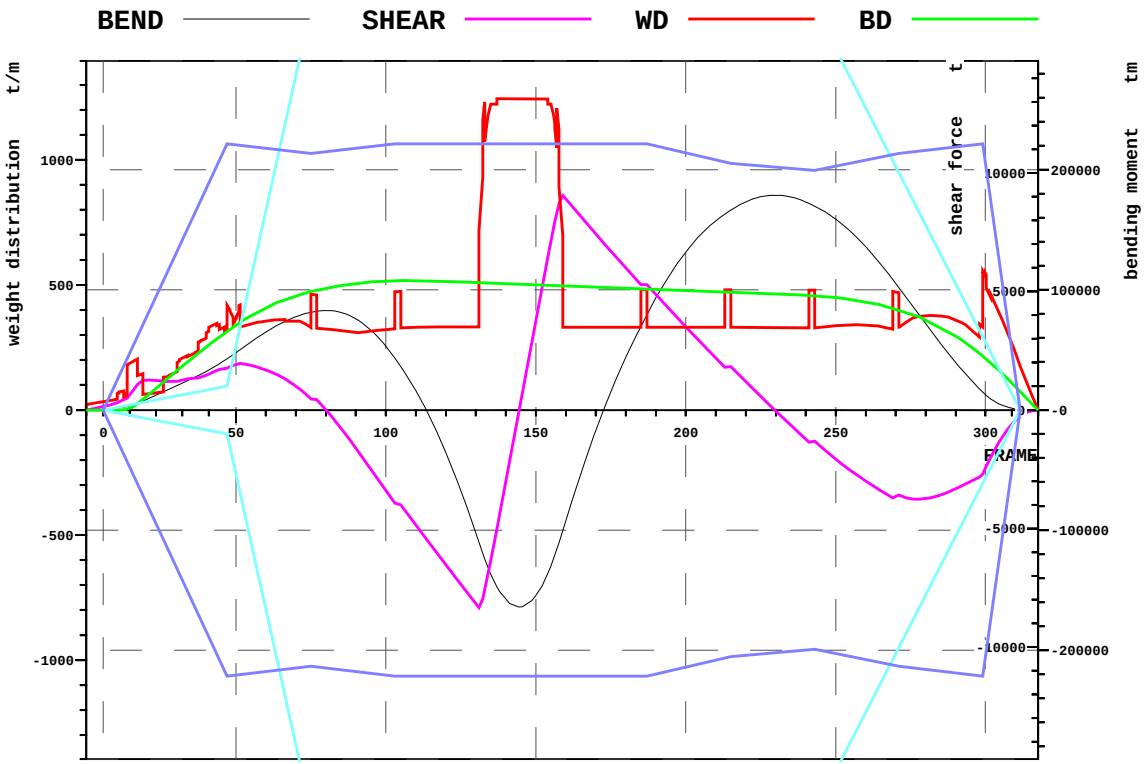
F L O A T I N G P O S I T I O N

Draught moulded	10.756	m	KM	20.94 m
Trim	3.068	m	KG	12.42 m
Heel, PS=+	-0.2	deg		
TA	12.290	m	GM0	8.52 m
TF	9.221	m	GMCORR	0.00 m
Trimming moment	-643201	tonm	GM	8.52 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.71	0.029	0.0	0.000
10.0	1.48	0.124	0.0	0.000
20.0	3.18	0.527	0.0	0.000
30.0	5.01	1.245	0.0	0.000
40.0	5.99	2.223	0.0	0.000
45.0	6.02	2.749	0.0	0.000
50.0	5.83	3.267	0.0	0.000
55.0	5.49	3.762	0.0	0.000
60.0	5.02	4.221	0.0	0.000
65.0	4.45	4.635	0.0	0.000
70.0	3.82	4.996	0.0	0.000
75.0	3.12	5.299	0.0	0.000
80.0	2.37	5.539	0.0	0.000
85.0	1.58	5.712	0.0	0.000
90.0	0.78	5.815	0.0	0.000



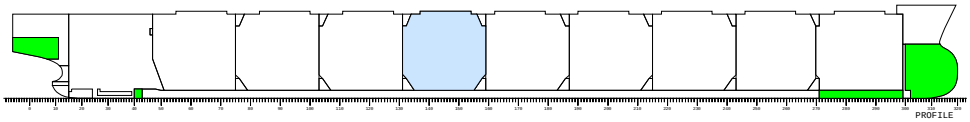
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.245	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.223	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.978	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.039	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	42.928	deg	OK
GM0.15	GM > 0.15 m	0.150	8.523	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.295		OK

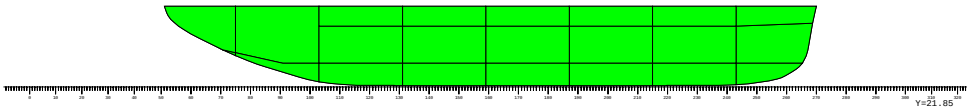


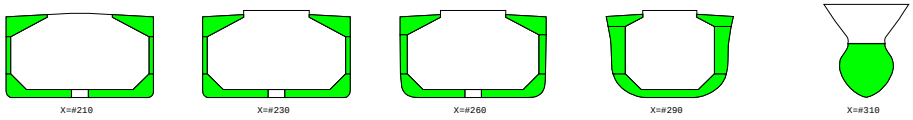
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	480	142	32.92	0.00
8.80	11.00	3816	784	193.06	11.08
12.00	15.00	7235	1246	62.52	42.30
37.60	47.00	42845	1756	418.67	313.64
41.13	51.00	49531	1943	417.84	346.56
62.97	75.00	81297	467	395.98	473.69
88.45	103.00	43375	-3928	473.37	516.02
113.93	131.00	-109704	-8333	715.78	509.41
139.41	159.00	-99971	9058	697.00	496.48
164.89	187.00	79403	5281	481.09	483.48
190.37	215.00	166583	1821	405.36	470.44
215.85	243.00	169667	-1318	328.42	455.27
241.33	271.00	102023	-3583	331.31	398.17
266.81	299.00	14780	-2718	332.68	214.97
278.01	313.00	421	-286	182.44	76.47

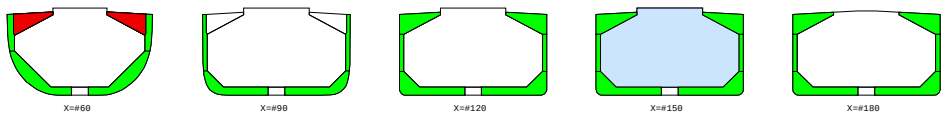
				X	FRAME
SHEAR FORCE (MIN)	-8333.2 t	(74.2%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	9058.0 t	(80.7%)		139.4 m	159
SAGGING MOMENT	-164176 tm	(37.4%)		126.1 m	144
HOGGING MOMENT	178874.7 tm	(44.3%)		203.1 m	229

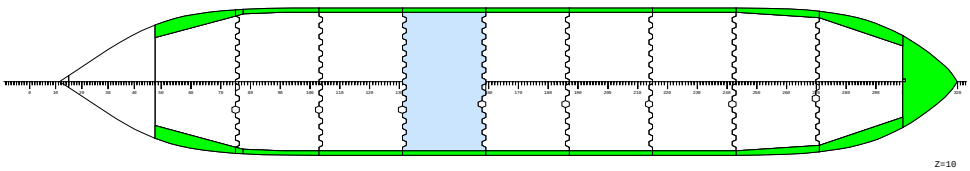
LOADING CONDITION - COND06 : HEAVY BALLAST-50% BUNKERS

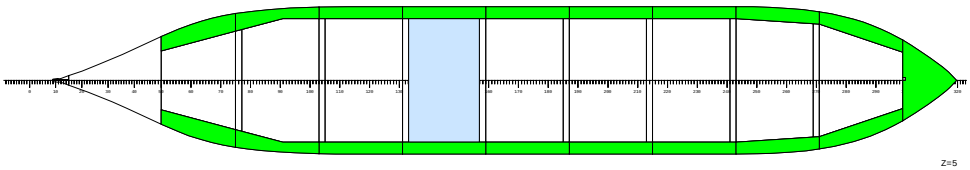


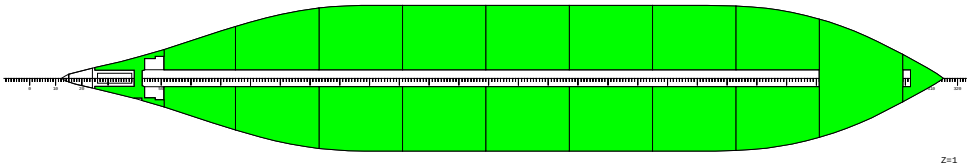












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

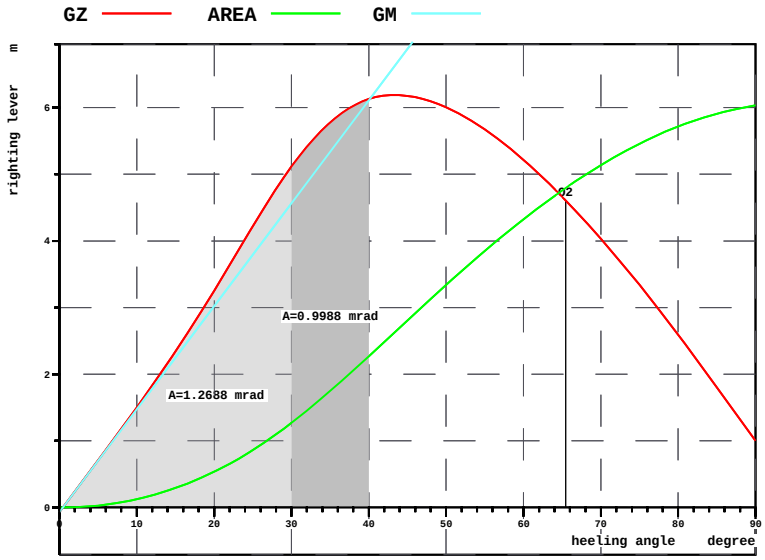
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	85374.7	149.01	-0.09 11.76
Total weight	110016.6	145.20	-0.07 12.19

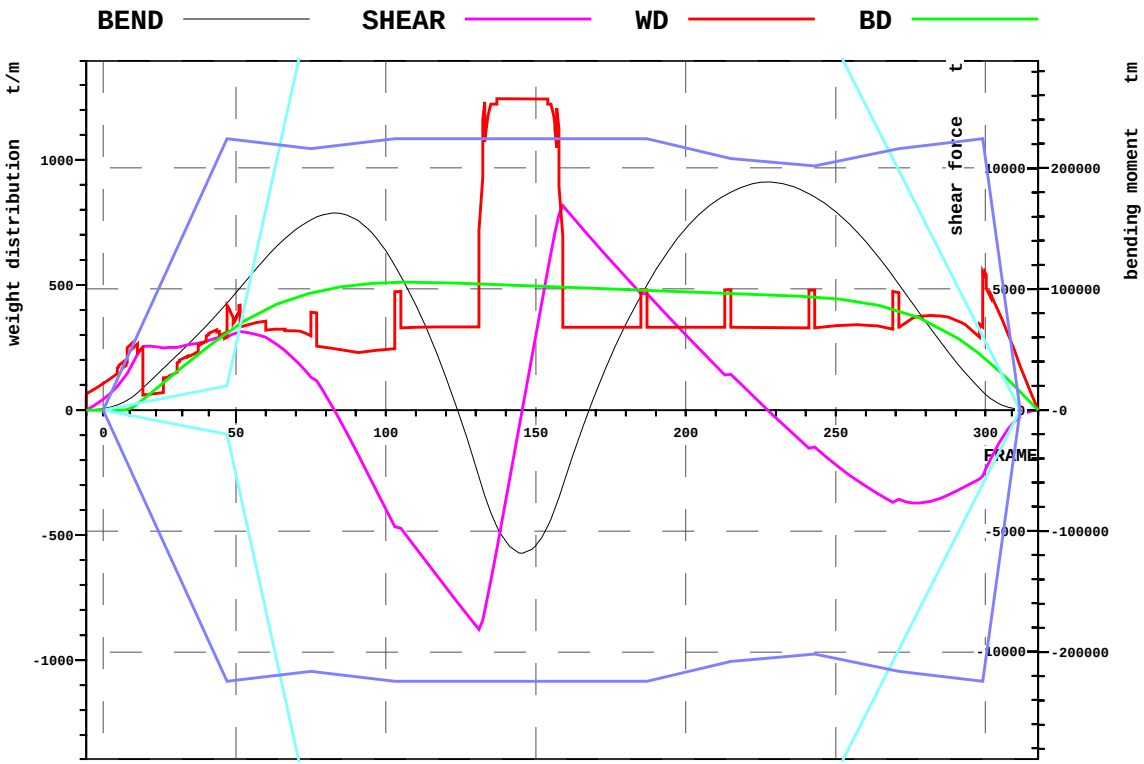
F L O A T I N G P O S I T I O N

Draught moulded	10.636	m	KM	21.04	m
Trim	2.994	m	KG	12.19	m
Heel, PS=+	-0.4	deg			
TA	12.133	m	GM0	8.85	m
TF	9.139	m	GMCORR	0.00	m
Trimming moment	-623243	tonm	GM	8.85	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.70	0.028	273.1	0.002
10.0	1.50	0.123	572.7	0.005
20.0	3.25	0.534	1197.1	0.011
30.0	5.12	1.269	1623.0	0.015
40.0	6.13	2.267	1920.4	0.017
45.0	6.18	2.806	2017.0	0.018
50.0	6.00	3.339	2076.3	0.019
55.0	5.67	3.849	2095.0	0.019
60.0	5.21	4.325	2072.0	0.019
65.0	4.66	4.757	2014.9	0.018
70.0	4.03	5.136	1930.4	0.018
75.0	3.34	5.458	1823.1	0.017
80.0	2.59	5.717	1696.6	0.015
85.0	1.80	5.909	1553.9	0.014
90.0	1.00	6.032	1399.3	0.013



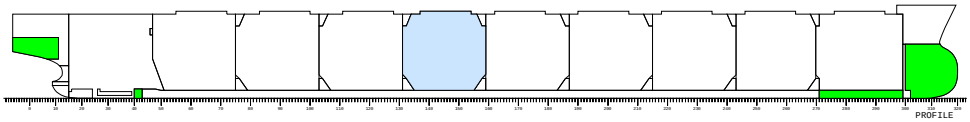
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.269	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.268	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.999	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.188	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.279	deg	OK
GM0.15	GM > 0.15 m	0.150	8.848	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.258		OK

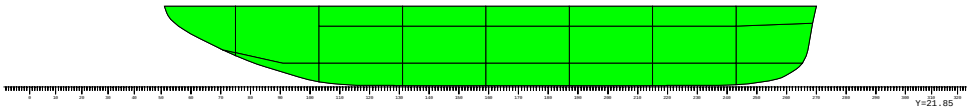


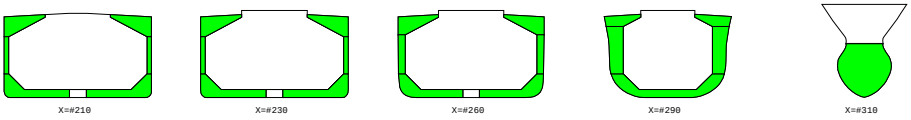
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1201	443	106.39	0.00
8.80	11.00	10688	1911	268.49	10.06
12.00	15.00	18046	2624	59.78	40.56
37.60	47.00	88533	3010	418.35	308.09
41.13	51.00	99666	3217	417.52	340.76
62.97	75.00	157071	1361	343.61	467.35
88.45	103.00	119383	-4829	473.36	509.87
113.93	131.00	-54815	-9082	715.77	503.58
139.41	159.00	-62421	8453	696.99	490.96
164.89	187.00	103189	4812	481.08	478.27
190.37	215.00	179979	1481	405.35	465.55
215.85	243.00	175843	-1538	328.41	450.70
241.33	271.00	103932	-3692	331.31	394.13
266.81	299.00	14986	-2741	332.67	212.54
278.01	313.00	450	-290	182.44	75.42

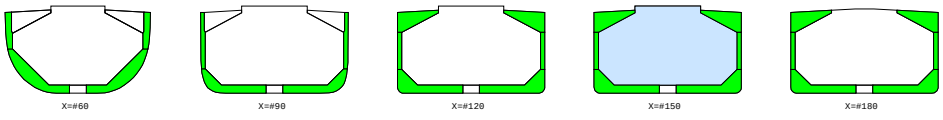
				X	FRAME
SHEAR FORCE (MIN)	-9082.5 t	(80.9%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	8452.8 t	(75.3%)		139.4 m	159
SAGGING MOMENT	-118390 tm	(27.0%)		127.1 m	145
HOGGING MOMENT	188387.8 tm	(46.2%)		202.3 m	228

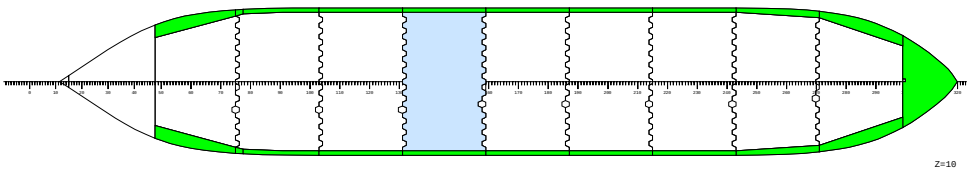
LOADING CONDITION - COND07 : HEAVY BALLAST-ARRIVAL BUNKERS

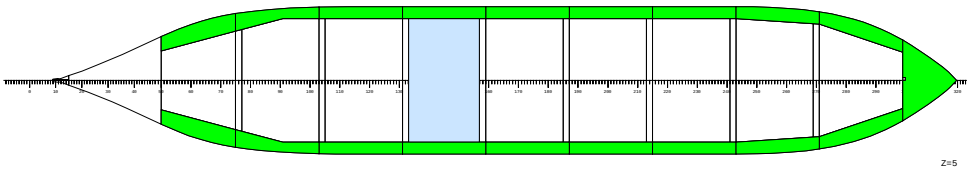


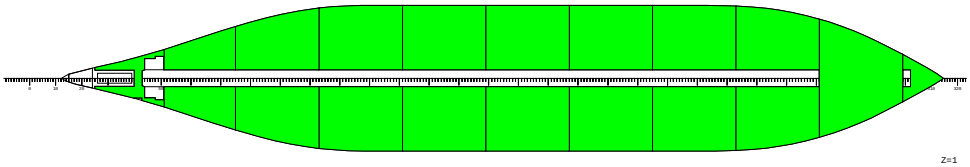












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

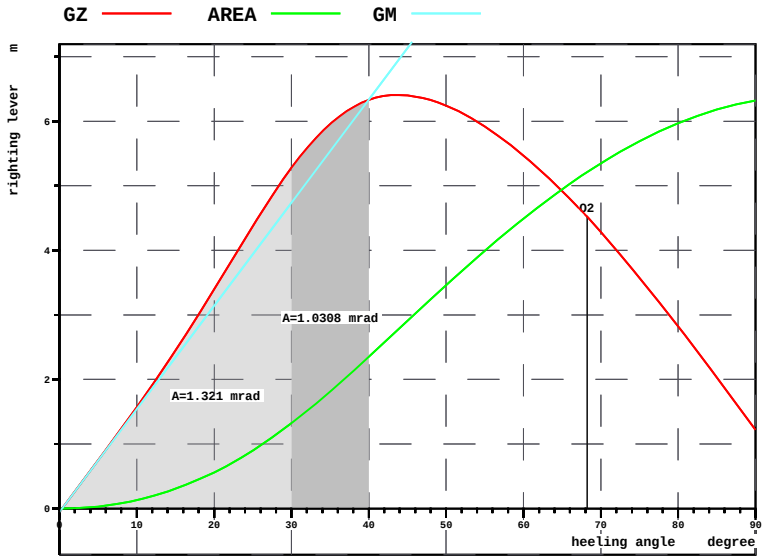
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	83005.9	152.07	-0.07 11.46
Total weight	107647.7	147.48	-0.06 11.97

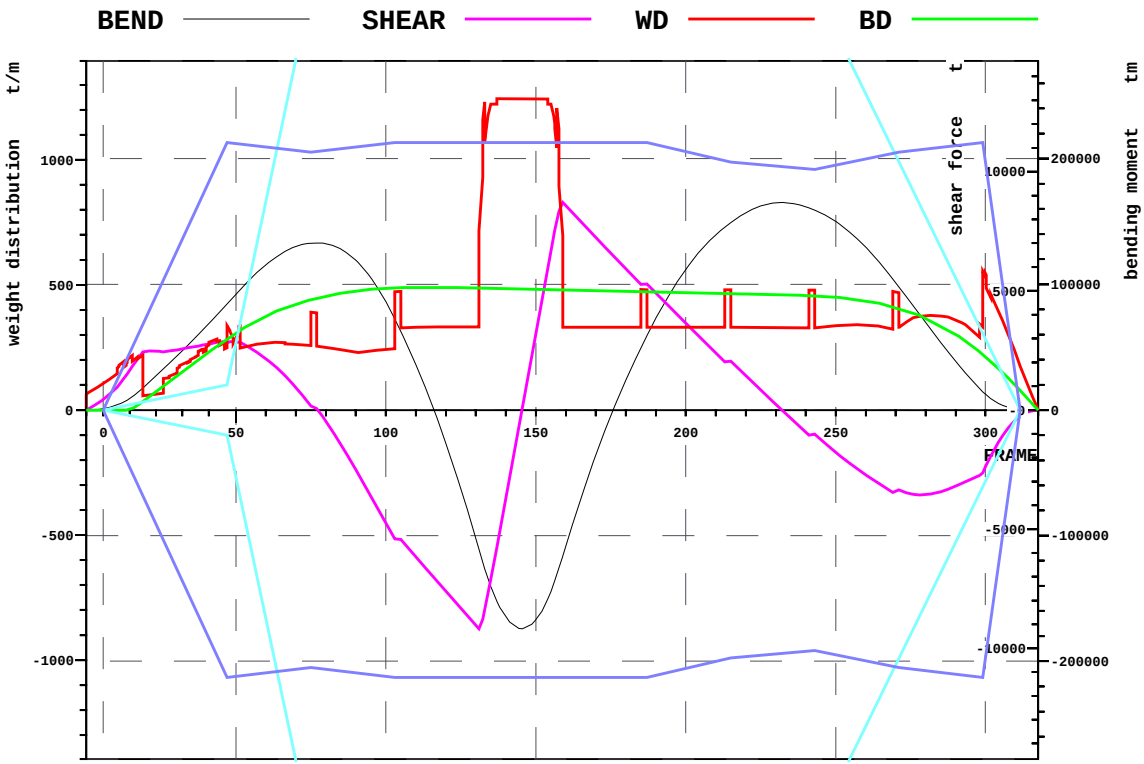
F L O A T I N G P O S I T I O N

Draught moulded	10.391	m	KM	21.12 m
Trim	1.769	m	KG	11.97 m
Heel, PS=+	-0.3	deg		
TA	11.276	m	GM0	9.15 m
TF	9.507	m	GMCORR	0.00 m
Trimming moment	-358248	tonm	GM	9.15 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.75	0.030	64.4	0.001
10.0	1.57	0.131	128.6	0.001
20.0	3.39	0.560	221.7	0.002
30.0	5.28	1.321	296.1	0.003
40.0	6.33	2.352	401.9	0.004
45.0	6.40	2.909	456.9	0.004
50.0	6.24	3.462	503.3	0.005
55.0	5.92	3.994	540.1	0.005
60.0	5.47	4.491	568.2	0.005
65.0	4.92	4.945	588.6	0.005
70.0	4.28	5.347	603.4	0.006
75.0	3.58	5.690	614.6	0.006
80.0	2.82	5.970	620.8	0.006
85.0	2.03	6.182	620.0	0.006
90.0	1.22	6.324	613.4	0.006



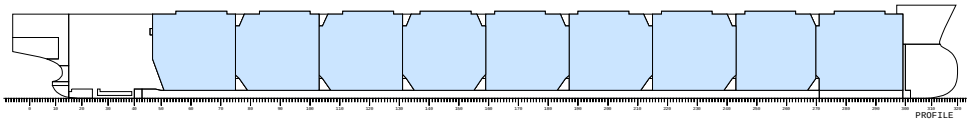
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.321	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.352	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.031	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.406	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.684	deg	OK
GM0.15	GM > 0.15 m	0.150	9.151	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.146		OK

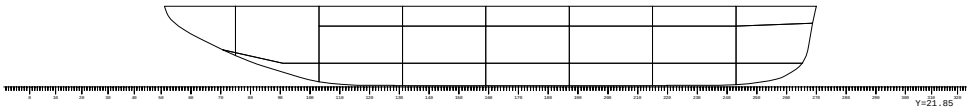


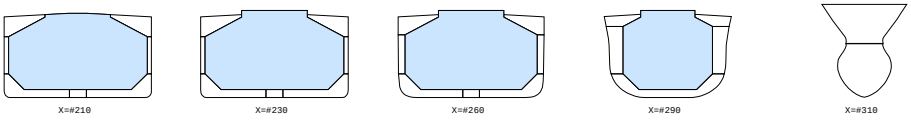
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1153	443	106.39	0.00
8.80	11.00	10504	1839	218.54	6.64
12.00	15.00	17394	2443	57.05	33.52
37.60	47.00	84582	2829	336.92	282.41
41.13	51.00	94692	2836	334.14	314.14
62.97	75.00	132710	164	257.87	441.02
88.45	103.00	72374	-5412	473.36	488.21
113.93	131.00	-110431	-9179	715.77	487.11
139.41	159.00	-115945	8710	697.00	479.71
164.89	187.00	59071	5290	481.08	472.22
190.37	215.00	149192	2047	405.35	464.71
215.85	243.00	158931	-1017	328.41	455.06
241.33	271.00	98078	-3346	331.31	403.16
266.81	299.00	14539	-2645	332.68	221.63
278.01	313.00	472	-275	182.44	80.13

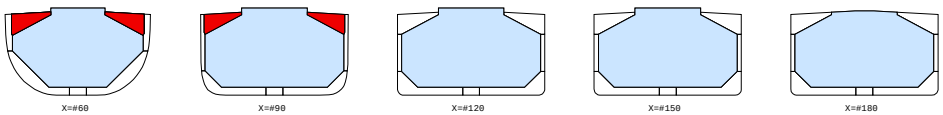
				X	FRAME
SHEAR FORCE (MIN)	-9179.1 t	(81.8%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	8710.2 t	(77.6%)		139.4 m	159
SAGGING MOMENT	-174047 tm	(39.7%)		127.1 m	145
HOGGING MOMENT	165079.0 tm	(42.6%)		206.0 m	232

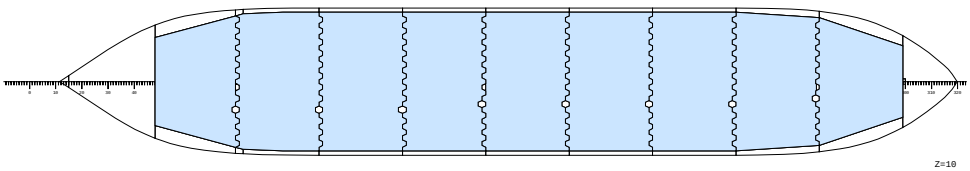
LOADING CONDITION - COND08 : HOMOGENEOUS (0.792 t/m3 DESIGN DRAUGHT)-DEP BUNK

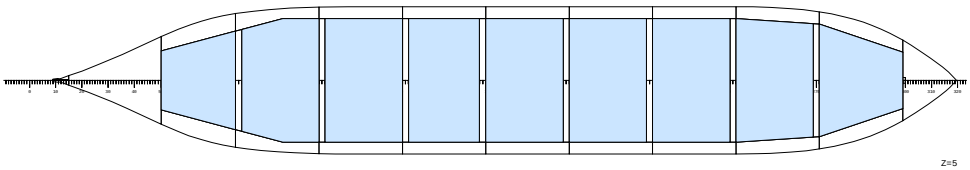


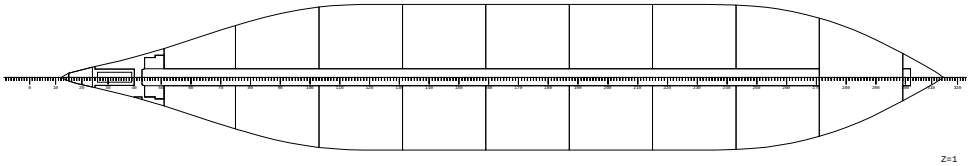












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.792)							
R1.1	NO.1 CARGO HOLD	13361.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	16801.8	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	17357.3	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	17357.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	17356.6	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	16352.7	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	17372.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	17072.9	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	14406.4	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		147439.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

CONTENTS= (RHO=1)

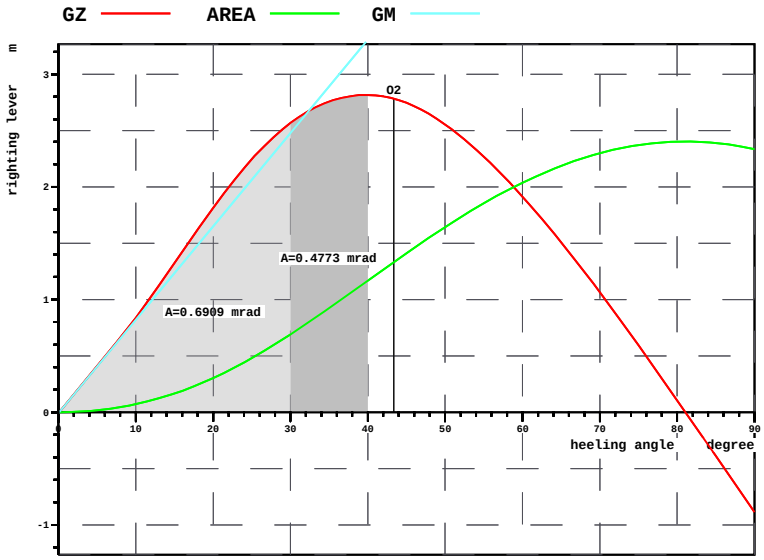
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	153624.4	147.19	-0.01 14.10
Total weight	178266.2	145.09	-0.01 14.04

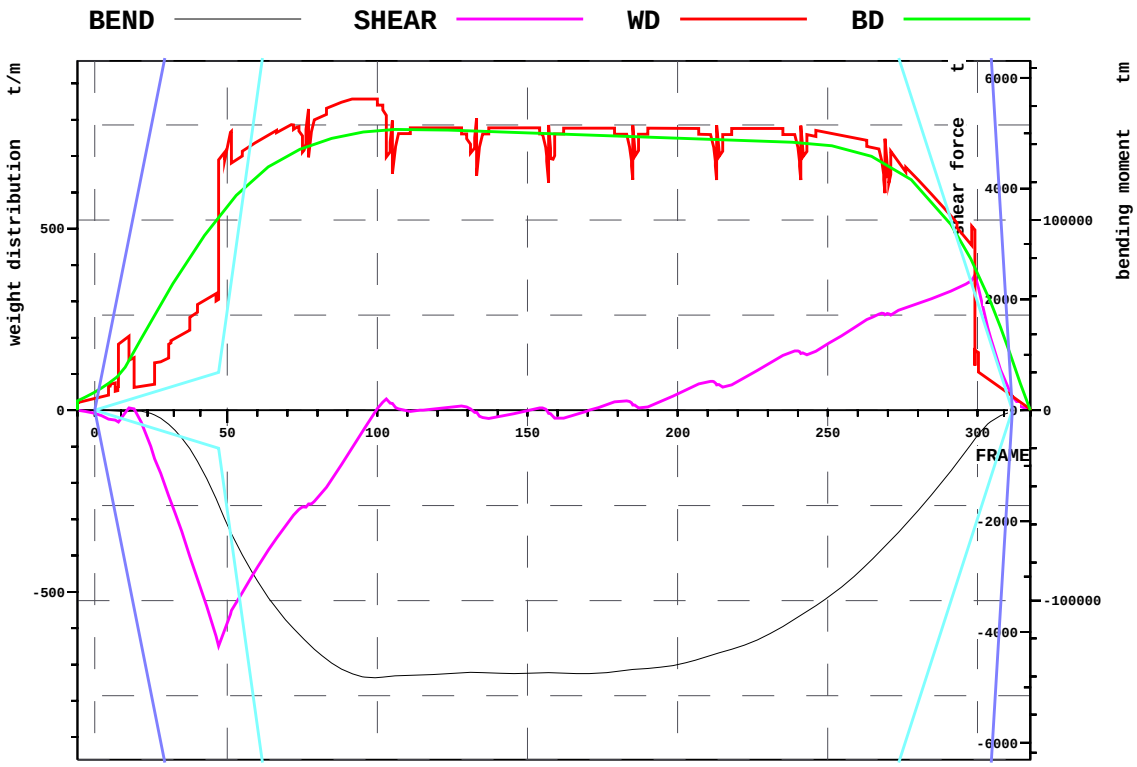
F L O A T I N G P O S I T I O N

Draught moulded	16.486	m	KM	18.79 m
Trim	1.933	m	KG	14.04 m
Heel, PS=+	-0.1	deg		
TA	17.453	m	GM0	4.75 m
TF	15.520	m	GMCORR	0.00 m
Trimming moment	-473921	tonm	GM	4.75 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.41	0.017	0.0	0.000
10.0	0.84	0.071	0.0	0.000
20.0	1.81	0.302	0.0	0.000
30.0	2.57	0.691	0.0	0.000
40.0	2.82	1.168	0.0	0.000
45.0	2.75	1.412	0.0	0.000
50.0	2.55	1.644	0.0	0.000
55.0	2.27	1.855	0.0	0.000
60.0	1.91	2.038	0.0	0.000
65.0	1.51	2.187	0.0	0.000
70.0	1.06	2.300	0.0	0.000
75.0	0.59	2.372	0.0	0.000
80.0	0.10	2.403	0.0	0.000
85.0	-0.39	2.390	0.0	0.000
90.0	-0.89	2.334	0.0	0.000



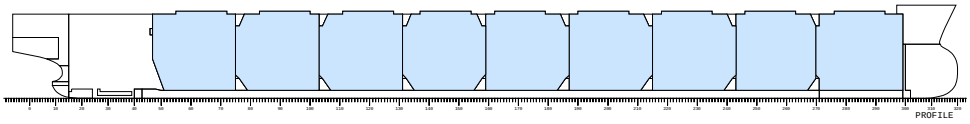
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.691	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.168	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.477	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.816	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	39.806	deg	OK
GM0.15	GM > 0.15 m	0.150	4.747	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.414		OK

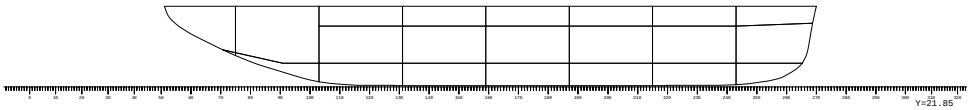


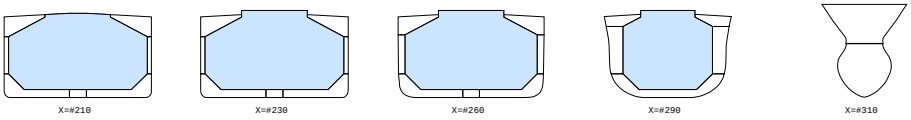
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	138	-56	32.90	50.04
8.80	11.00	-585	-78	192.91	113.39
12.00	15.00	-392	27	62.47	163.11
37.60	47.00	-49725	-4256	689.88	531.11
41.13	51.00	-63565	-3683	765.61	571.56
62.97	75.00	-119563	-1742	733.53	721.47
88.45	103.00	-140110	203	697.51	771.37
113.93	131.00	-137835	27	732.99	769.60
139.41	159.00	-137961	-142	701.09	761.49
164.89	187.00	-136059	42	760.22	753.29
190.37	215.00	-126996	420	759.67	745.07
215.85	243.00	-105780	1002	712.76	734.40
241.33	271.00	-68308	1720	649.49	668.27
266.81	299.00	-15100	2438	496.78	392.77
278.01	313.00	-837	272	40.17	141.79

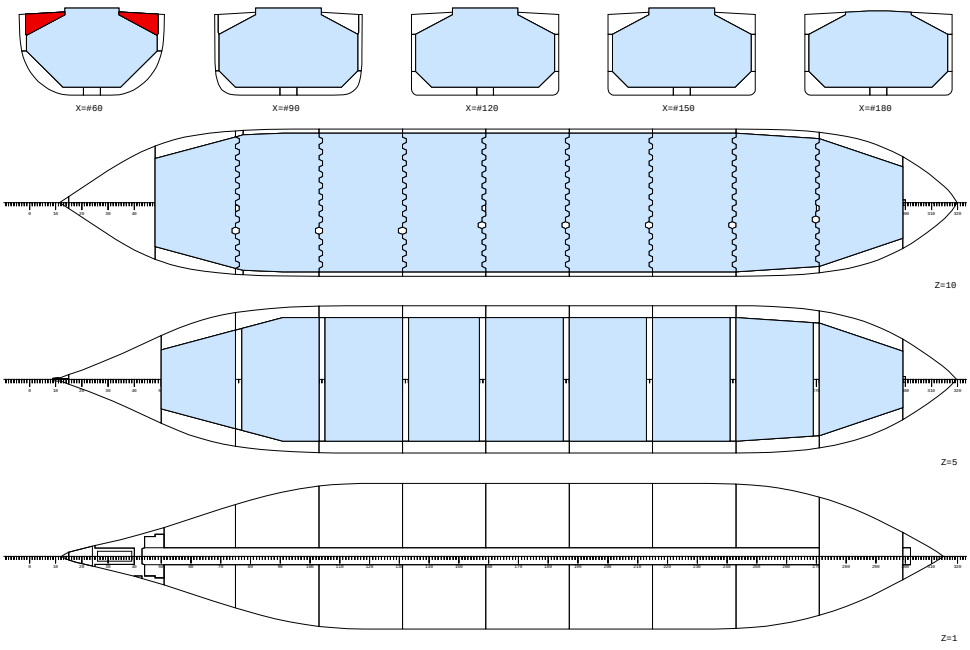
				X	FRAME
SHEAR FORCE (MIN)	-4255.7 t	(37.9%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2438.4 t	(21.7%)		266.8 m	299
SAGGING MOMENT	-140634 tm	(33.5%)		84.7 m	99
HOGGING MOMENT	146.0 tm			-0.4 m	-0

LOADING CONDITION - COND09 : HOMOGENEOUS (0.792 t/m3 DESIGN DRAUGHT)-50% BUNK









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.792)							
R1.1	NO.1 CARGO HOLD	13361.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	16801.8	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	17357.3	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	17357.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	17356.6	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	16352.7	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	17372.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	17072.9	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	14406.4	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		147439.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

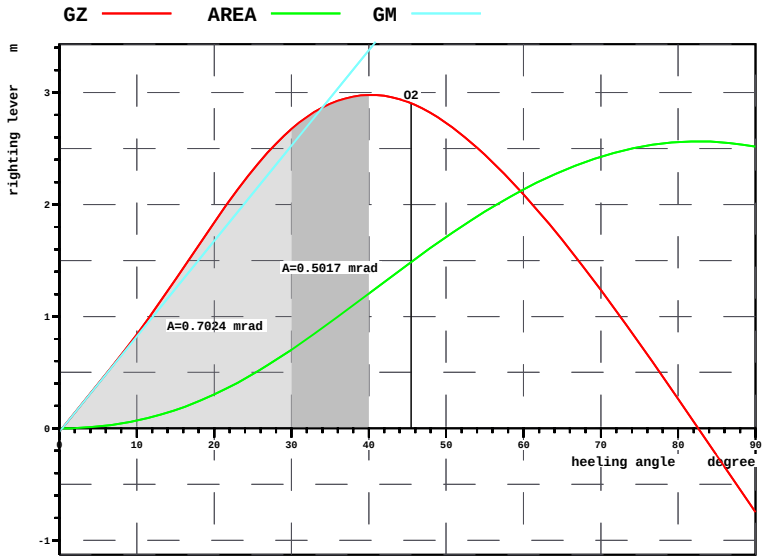
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	150676.1	148.83	-0.03 13.93
Total weight	175317.9	146.46	-0.03 13.89

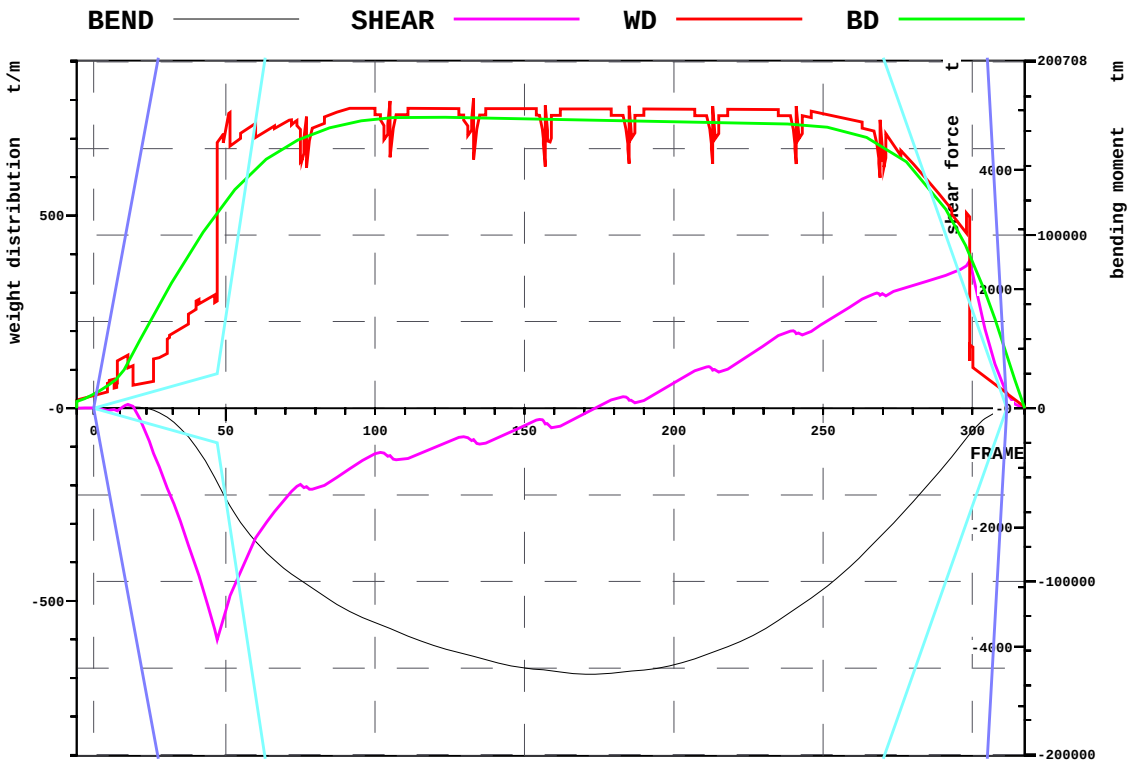
F L O A T I N G P O S I T I O N

Draught moulded	16.242	m	KM	18.77 m
Trim	1.020	m	KG	13.89 m
Heel, PS=+	-0.3	deg		
TA	16.752	m	GM0	4.88 m
TF	15.732	m	GMCORR	0.00 m
Trimming moment	-247248	tonm	GM	4.88 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.40	0.016	273.1	0.002
10.0	0.84	0.070	572.7	0.003
20.0	1.84	0.303	1197.1	0.007
30.0	2.68	0.702	1623.0	0.009
40.0	2.98	1.204	1920.4	0.011
45.0	2.92	1.462	2017.0	0.012
50.0	2.73	1.709	2076.3	0.012
55.0	2.44	1.936	2095.0	0.012
60.0	2.09	2.134	2072.0	0.012
65.0	1.68	2.299	2014.9	0.011
70.0	1.24	2.427	1930.4	0.011
75.0	0.76	2.514	1823.1	0.010
80.0	0.26	2.559	1696.6	0.010
85.0	-0.24	2.560	1553.9	0.009
90.0	-0.75	2.517	1399.3	0.008



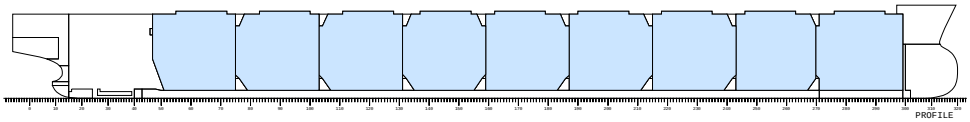
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.702	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.204	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.502	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.978	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	40.326	deg	OK
GM0.15	GM > 0.15 m	0.150	4.880	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.903		OK

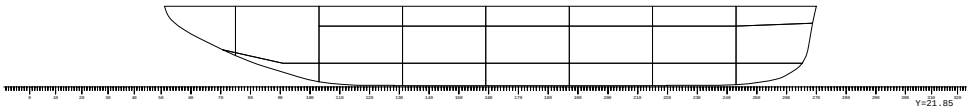


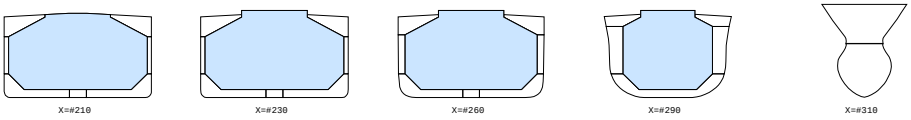
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	229	2	32.90	37.04
8.80	11.00	414	22	130.43	95.05
12.00	15.00	699	31	59.74	143.26
37.60	47.00	-42872	-3884	689.56	506.43
41.13	51.00	-55285	-3225	765.29	546.86
62.97	75.00	-99302	-1276	723.47	698.77
88.45	103.00	-125742	-753	697.50	752.39
113.93	131.00	-142368	-495	732.98	754.50
139.41	159.00	-151598	-329	701.09	750.27
164.89	187.00	-151530	92	760.22	745.95
190.37	215.00	-139527	607	759.67	741.60
215.85	243.00	-113122	1228	712.75	734.80
241.33	271.00	-70724	1887	649.49	672.32
266.81	299.00	-15106	2478	496.78	397.66
278.01	313.00	-776	273	40.17	142.84

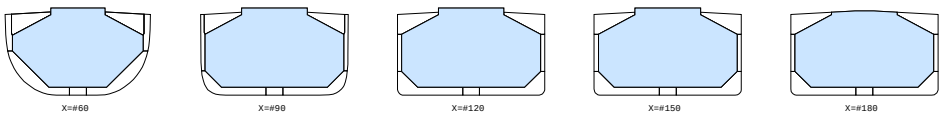
				X	FRAME
SHEAR FORCE (MIN)	-3883.7 t	(34.6%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2477.7 t	(22.1%)		266.8 m	299
SAGGING MOMENT	-153518 tm	(35.0%)		152.8 m	174
HOGGING MOMENT	699.0 tm	(11.0%)		12.0 m	15

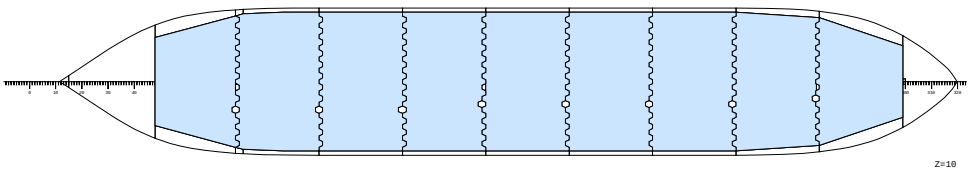
LOADING CONDITION - COND10 : HOMOGENEOUS (0.792 t/m3 DESIGN DRAUGHT)-ARRIVAL

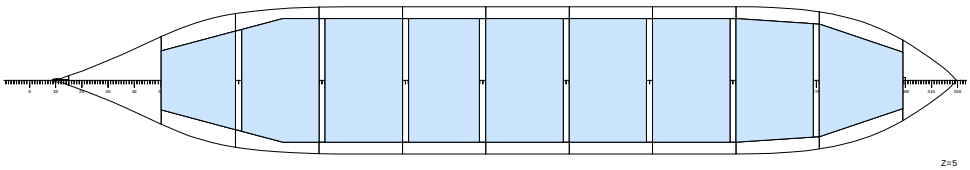


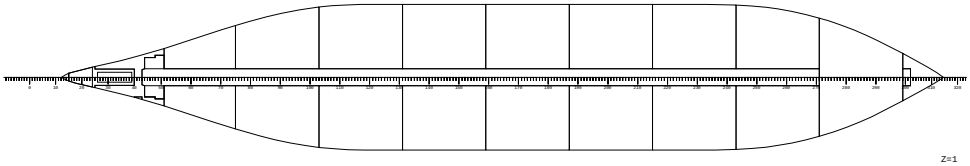












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.792)							
R1.1	NO.1 CARGO HOLD	13361.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	16801.8	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	17357.3	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	17357.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	17356.6	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	16352.7	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	17372.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	17072.9	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	14406.4	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		147439.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61

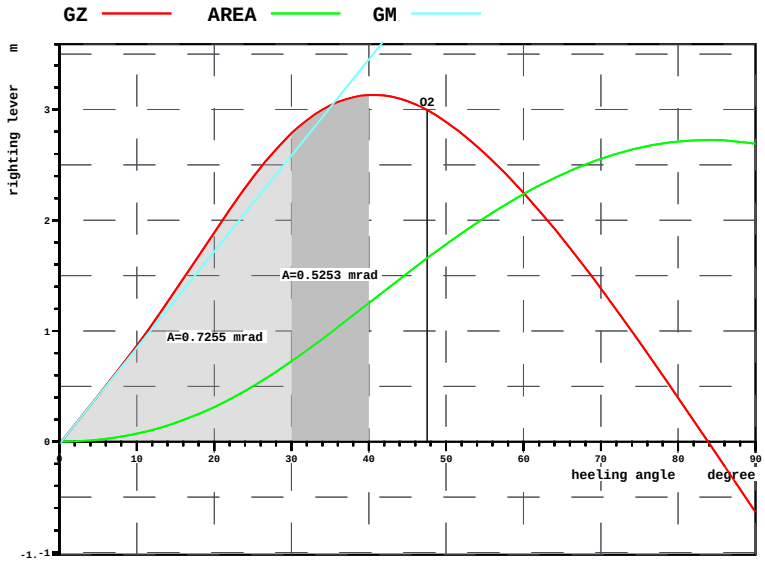
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	148307.2	150.54	-0.02 13.79
Total weight	172949.1	147.90	-0.02 13.78

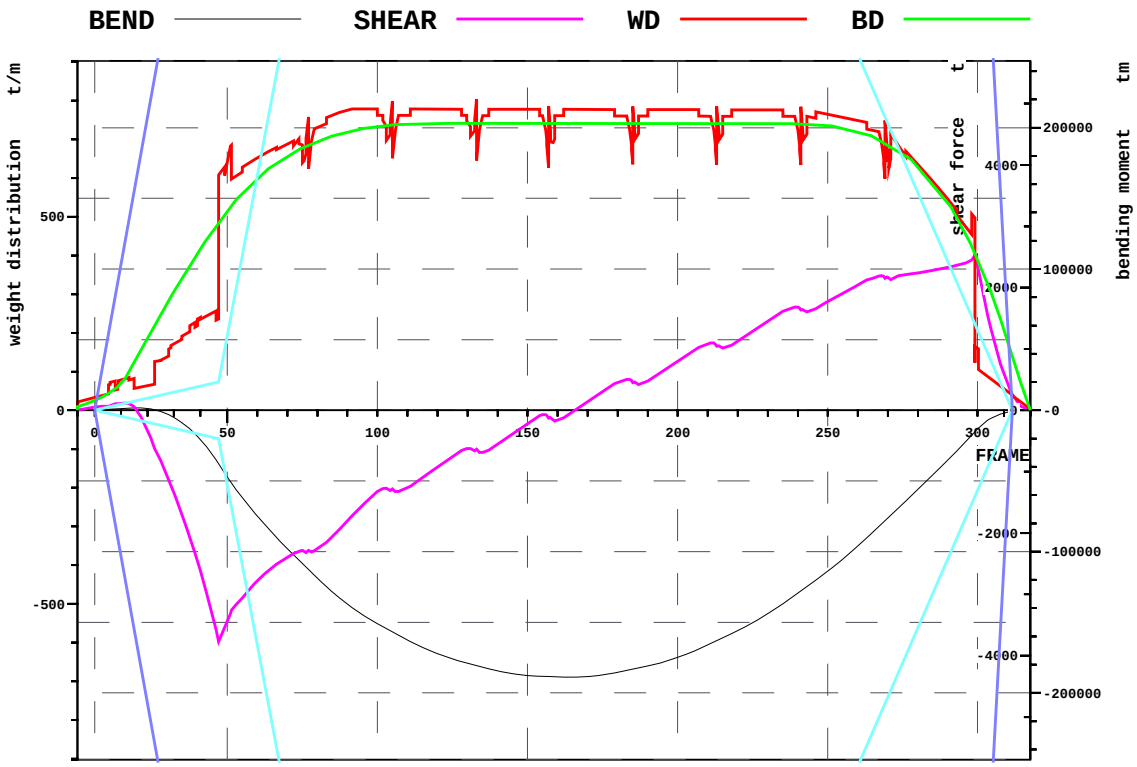
F L O A T I N G P O S I T I O N

Draught moulded	16.045	m	KM	18.74 m
Trim	0.055	m	KG	13.78 m
Heel, PS=+	-0.2	deg		
TA	16.072	m	GM0	4.97 m
TF	16.017	m	GMCORR	0.00 m
Trimming moment	-13087	tonm	GM	4.97 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.42	0.017	64.4	0.000
10.0	0.87	0.073	128.6	0.001
20.0	1.89	0.312	221.7	0.001
30.0	2.79	0.725	296.1	0.002
40.0	3.13	1.251	401.9	0.002
45.0	3.08	1.523	456.9	0.003
50.0	2.89	1.784	503.3	0.003
55.0	2.60	2.024	540.1	0.003
60.0	2.25	2.236	568.2	0.003
65.0	1.84	2.415	588.6	0.003
70.0	1.38	2.556	603.4	0.003
75.0	0.90	2.655	614.6	0.004
80.0	0.40	2.712	620.8	0.004
85.0	-0.12	2.724	620.0	0.004
90.0	-0.64	2.692	613.4	0.004



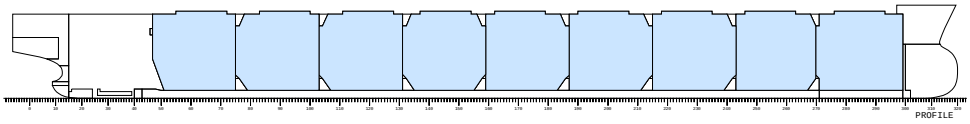
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.726	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.251	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.525	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.131	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	40.633	deg	OK
GM0.15	GM > 0.15 m	0.150	4.969	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.331		OK

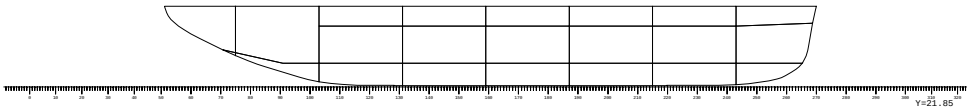


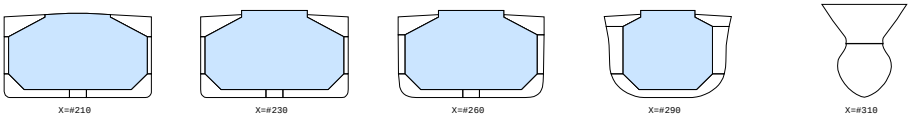
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	300	51	32.90	25.68
8.80	11.00	1277	119	80.54	78.27
12.00	15.00	1696	58	57.01	124.92
37.60	47.00	-37987	-3769	608.19	483.13
41.13	51.00	-50397	-3318	681.97	523.55
62.97	75.00	-108542	-2285	684.66	677.59
88.45	103.00	-154503	-1272	697.50	735.15
113.93	131.00	-179460	-626	732.98	741.34
139.41	159.00	-188479	-177	701.08	741.19
164.89	187.00	-182309	423	760.21	740.96
190.37	215.00	-160967	1013	759.67	740.69
215.85	243.00	-124636	1605	712.75	737.98
241.33	271.00	-74363	2133	649.48	679.21
266.81	299.00	-15186	2531	496.77	404.48
278.01	313.00	-720	275	40.17	144.13

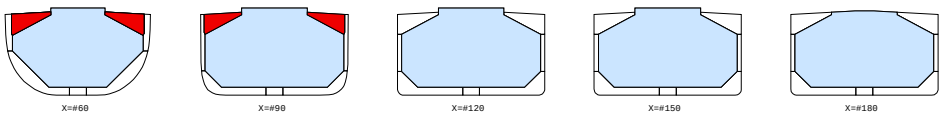
				X	FRAME
SHEAR FORCE (MIN)	-3768.6 t	(33.6%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2530.8 t	(22.5%)		266.8 m	299
SAGGING MOMENT	-188926 tm	(43.1%)		145.0 m	165
HOGGING MOMENT	1703.8 tm	(22.8%)		14.1 m	18

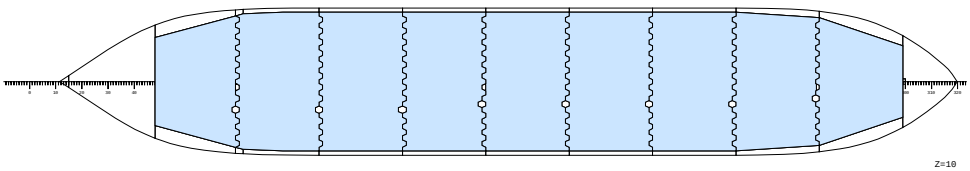
LOADING CONDITION - COND11 : HOMOGENEOUS (0.9 t/m3 SCANTLING DRAUGHT)-DEP BUN

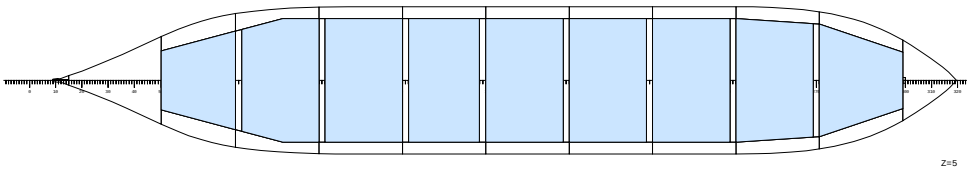


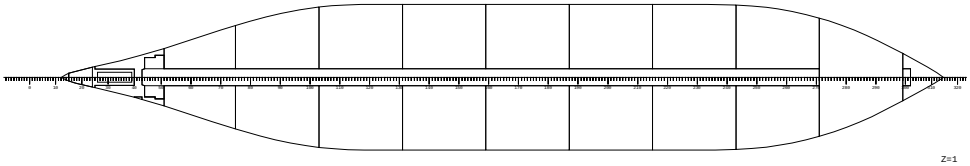












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.9)							
R1.1	NO.1 CARGO HOLD	15183.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	19092.9	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	19724.2	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	19724.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	19723.4	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	18582.6	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	19741.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	19401.1	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	16370.9	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		167544.5		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

CONTENTS= (RHO=1)

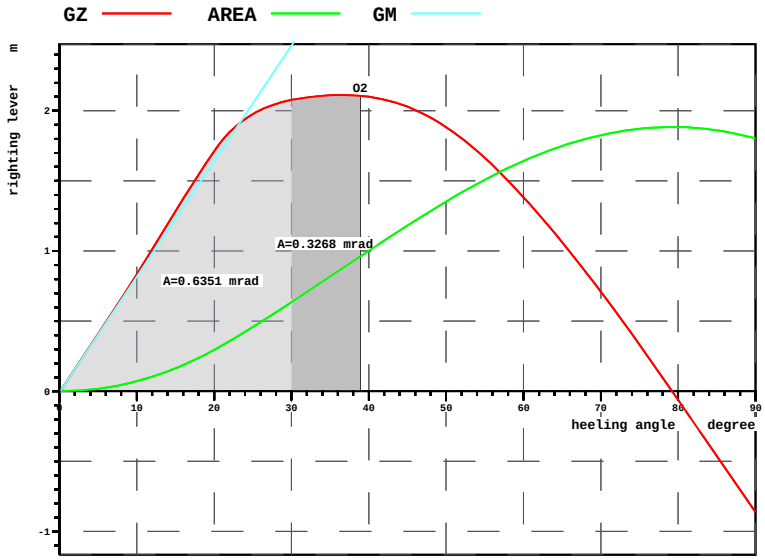
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	173729.7	147.66	-0.01 14.06
Total weight	198371.5	145.71	-0.01 14.01

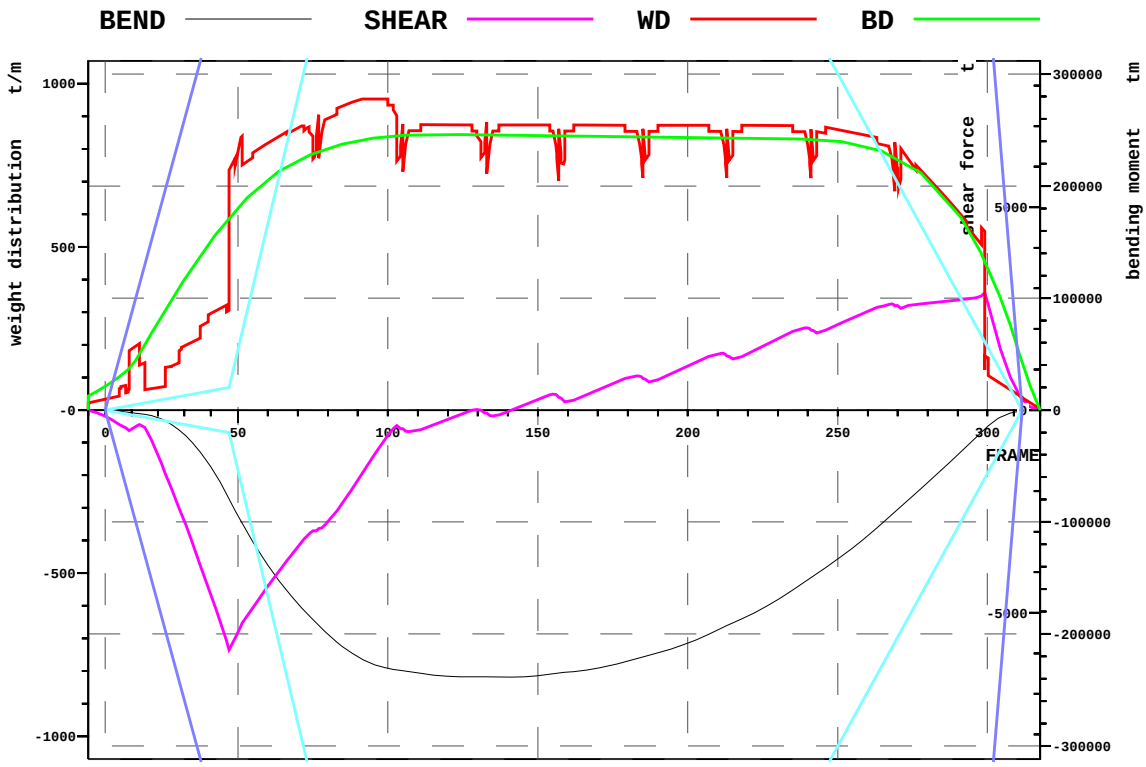
F L O A T I N G P O S I T I O N

Draught moulded	18.178	m	KM	18.73 m
Trim	0.778	m	KG	14.01 m
Heel, PS=+	-0.1	deg		
TA	18.567	m	GM0	4.72 m
TF	17.789	m	GMCORR	0.00 m
Trimming moment	-193684	tonm	GM	4.72 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.41	0.017	0.0	0.000
10.0	0.83	0.071	0.0	0.000
20.0	1.71	0.295	0.0	0.000
30.0	2.08	0.635	0.0	0.000
40.0	2.10	1.002	0.0	0.000
45.0	2.03	1.183	0.0	0.000
50.0	1.88	1.354	0.0	0.000
55.0	1.66	1.509	0.0	0.000
60.0	1.39	1.643	0.0	0.000
65.0	1.06	1.750	0.0	0.000
70.0	0.71	1.828	0.0	0.000
75.0	0.33	1.873	0.0	0.000
80.0	-0.06	1.885	0.0	0.000
85.0	-0.46	1.862	0.0	0.000
90.0	-0.86	1.804	0.0	0.000



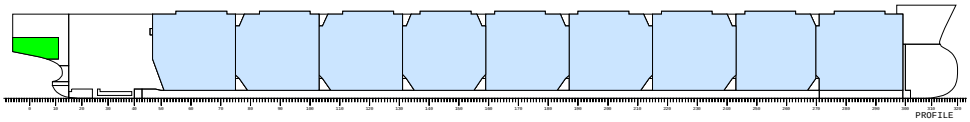
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.635	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	0.962	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.327	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.113	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	36.406	deg	OK
GM0.15	GM > 0.15 m	0.150	4.718	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.194		OK

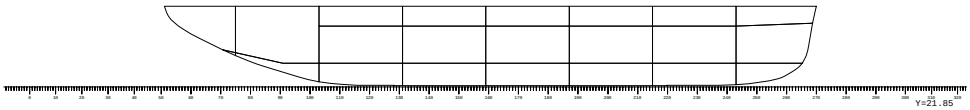


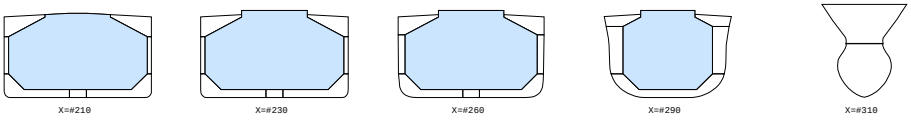
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-165	-158	32.90	72.69
8.80	11.00	-2859	-425	192.91	146.49
12.00	15.00	-3978	-432	62.47	199.60
37.60	47.00	-79603	-5912	735.57	585.96
41.13	51.00	-99318	-5328	835.70	628.04
62.97	75.00	-187269	-2976	804.58	784.70
88.45	103.00	-232005	-387	762.57	839.85
113.93	131.00	-238279	0	822.94	842.98
139.41	159.00	-234349	199	765.10	839.76
164.89	187.00	-219028	693	854.03	836.47
190.37	215.00	-190219	1256	853.49	833.14
215.85	243.00	-146120	1900	779.75	827.27
241.33	271.00	-86247	2508	707.94	761.54
266.81	299.00	-17055	2893	547.75	455.16
278.01	313.00	-803	284	40.17	150.56

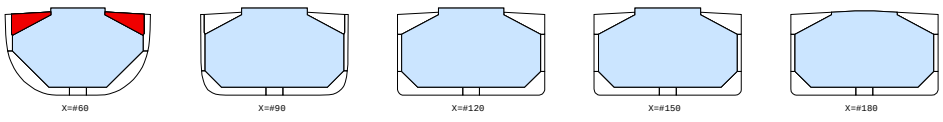
				X	FRAME
SHEAR FORCE (MIN)	-5912.4 t	(52.7%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2892.6 t	(25.8%)		266.8 m	299
SAGGING MOMENT	-238720 tm	(54.4%)		122.3 m	140
HOGGING MOMENT	31.6 tm			-2.8 m	-3

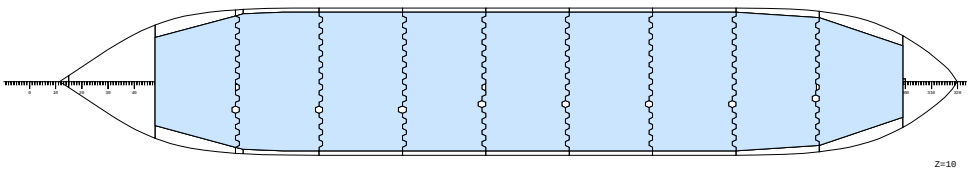
LOADING CONDITION - COND12 : HOMOGENEOUS (0.9 t/m3 SCANTLING DRAUGHT)-50% BUN

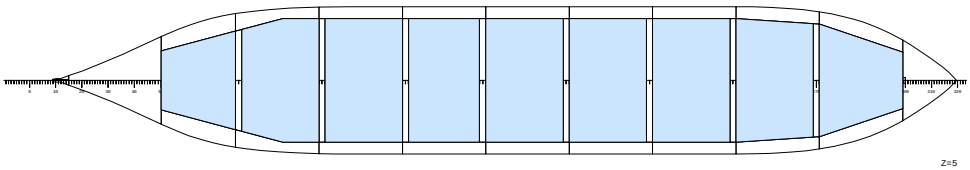


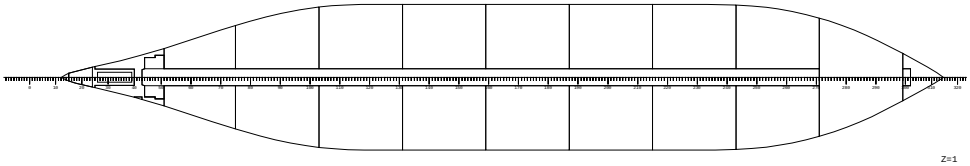












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.9)							
R1.1	NO.1 CARGO HOLD	15183.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	19092.9	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	19724.2	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	19724.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	19723.4	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	18582.6	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	19741.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	19401.1	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	16370.9	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		167544.5		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

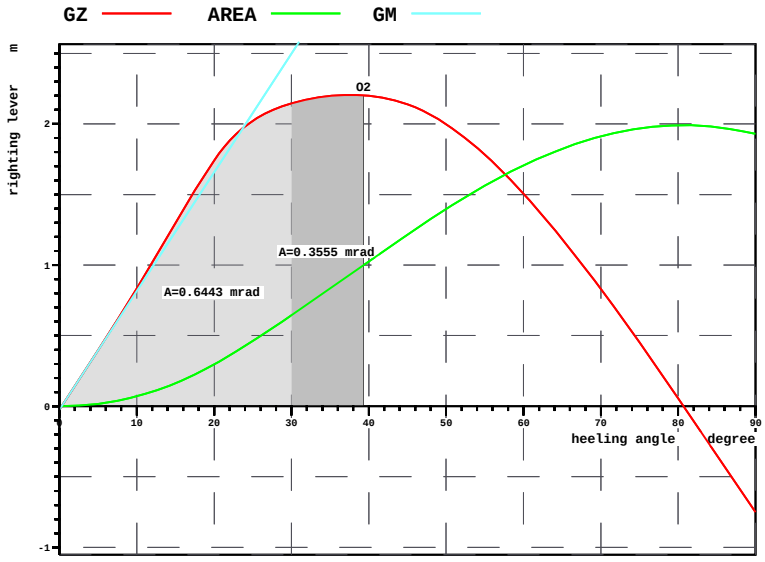
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	172403.7	147.76	-0.03 13.93
Total weight	197045.5	145.79	-0.02 13.89

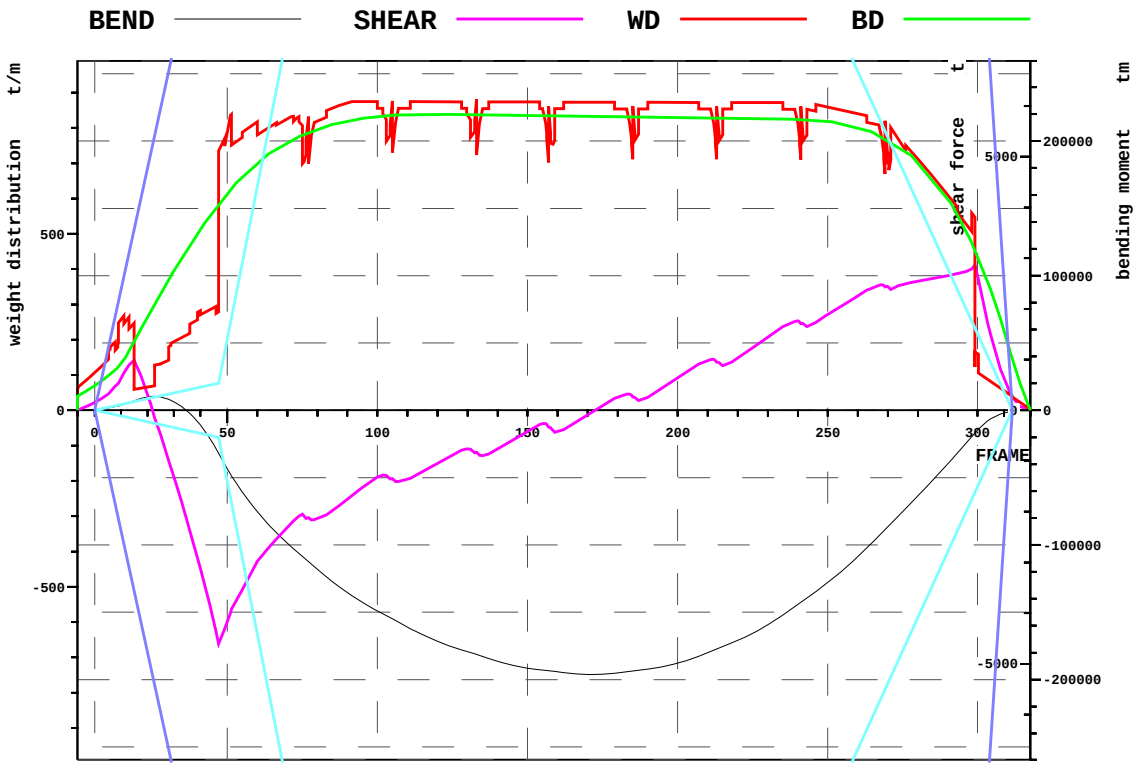
F L O A T I N G P O S I T I O N

Draught moulded	18.067	m	KM	18.73 m
Trim	0.766	m	KG	13.89 m
Heel, PS=+	-0.3	deg		
TA	18.450	m	GM0	4.83 m
TF	17.684	m	GMCORR	-0.02 m
Trimming moment	-190265	tonm	GM	4.82 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.40	0.016	273.1	0.001
10.0	0.83	0.070	572.7	0.003
20.0	1.74	0.296	1197.1	0.006
30.0	2.15	0.644	1623.0	0.008
40.0	2.20	1.026	1920.4	0.010
45.0	2.14	1.216	2017.0	0.010
50.0	2.00	1.397	2076.3	0.011
55.0	1.78	1.563	2095.0	0.011
60.0	1.51	1.706	2072.0	0.011
65.0	1.18	1.824	2014.9	0.010
70.0	0.83	1.912	1930.4	0.010
75.0	0.45	1.968	1823.1	0.009
80.0	0.06	1.990	1696.6	0.009
85.0	-0.35	1.978	1553.9	0.008
90.0	-0.75	1.930	1399.3	0.007



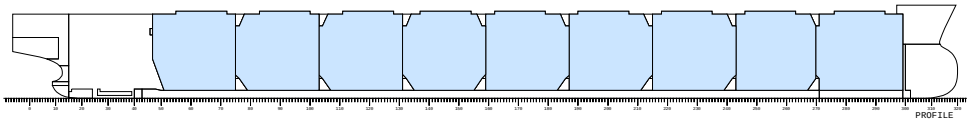
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.644	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.000	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.355	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.205	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	37.655	deg	OK
GM0.15	GM > 0.15 m	0.150	4.818	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.302		OK

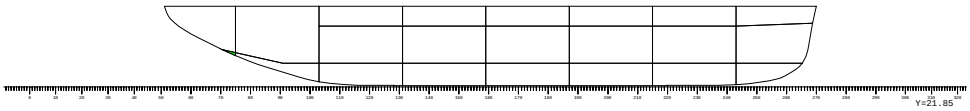


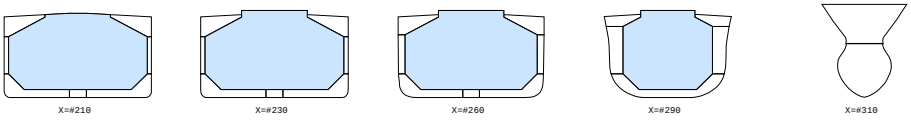
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	602	155	106.31	70.22
8.80	11.00	4275	736	268.22	143.05
12.00	15.00	7221	987	59.74	195.87
37.60	47.00	-32101	-4603	735.24	580.96
41.13	51.00	-47171	-4002	835.38	622.96
62.97	75.00	-108532	-2050	806.78	779.45
88.45	103.00	-152496	-1282	762.56	834.63
113.93	131.00	-179942	-767	822.92	837.81
139.41	159.00	-193846	-433	765.09	834.65
164.89	187.00	-193038	190	854.02	831.41
190.37	215.00	-175465	881	853.47	828.13
215.85	243.00	-139360	1651	779.74	822.32
241.33	271.00	-84283	2383	707.92	756.86
266.81	299.00	-16922	2871	547.74	452.18
278.01	313.00	-791	283	40.17	150.20

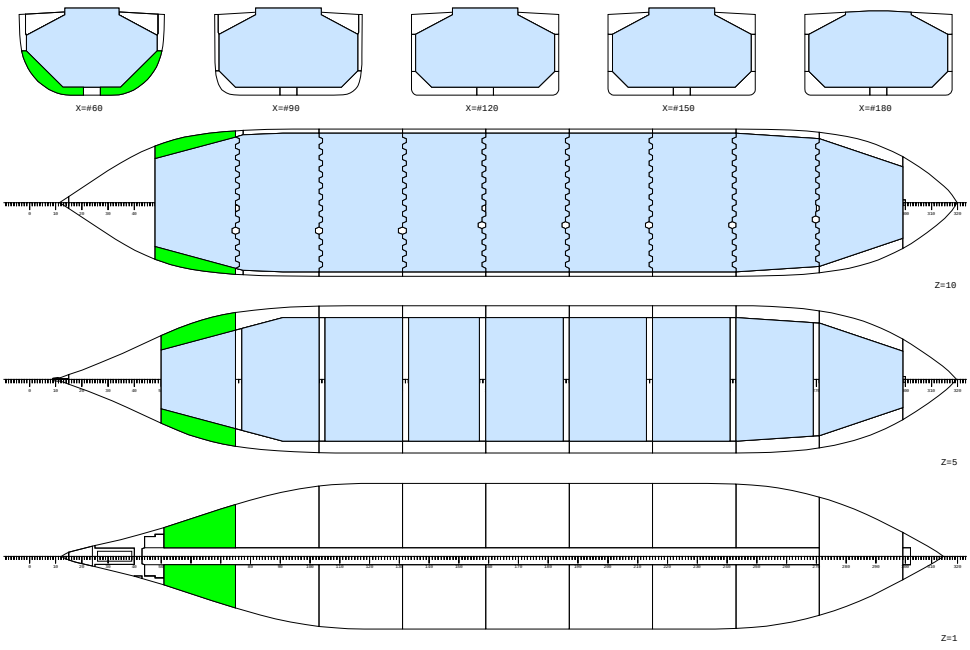
				X	FRAME
SHEAR FORCE (MIN)	-4602.6 t	(41.0%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2871.2 t	(25.6%)		266.8 m	299
SAGGING MOMENT	-196279 tm	(44.7%)		149.9 m	171
HOGGING MOMENT	10323.2 tm	(106.7%)		18.2 m	23

LOADING CONDITION - COND13 : HOMOGENEOUS (0.9 t/m3 SCANTLING DRAUGHT)-ARRIVAL









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.9)							
R1.1	NO.1 CARGO HOLD	15183.2	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	19092.9	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	19724.2	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	19724.8	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	19723.4	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	18582.6	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	19741.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	19401.1	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	16370.9	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		167544.5		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61

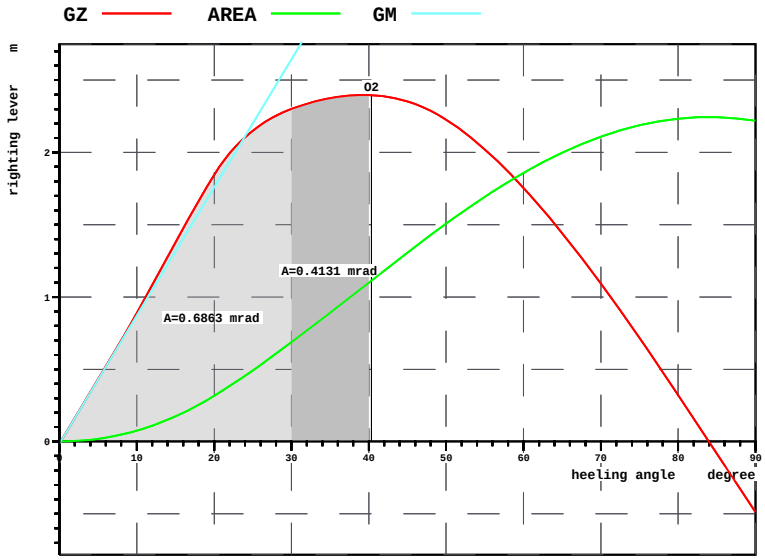
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
SUBTOTAL		3300.8		51.30	0.00	5.34	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	171713.3	148.71	-0.02 13.62
Total weight	196355.2	146.62	-0.02 13.63

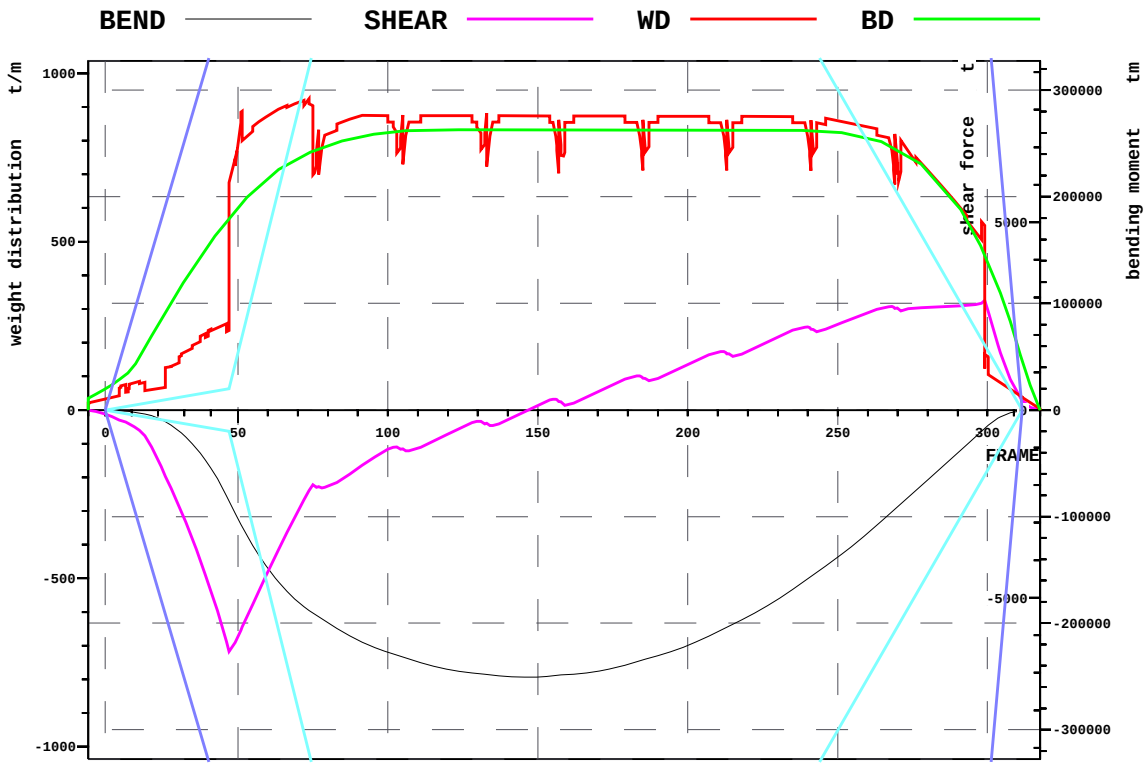
F L O A T I N G P O S I T I O N

Draught moulded	18.014	m	KM	18.71 m
Trim	0.127	m	KG	13.63 m
Heel, PS=+	-0.2	deg		
TA	18.077	m	GM0	5.08 m
TF	17.950	m	GMCORR	-0.00 m
Trimming moment	-31494	tonm	GM	5.08 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.43	0.018	64.4	0.000
10.0	0.89	0.075	128.6	0.001
20.0	1.85	0.315	221.7	0.001
30.0	2.30	0.686	296.1	0.002
40.0	2.40	1.099	401.9	0.002
45.0	2.35	1.307	456.9	0.002
50.0	2.23	1.508	503.3	0.003
55.0	2.02	1.693	540.1	0.003
60.0	1.75	1.859	568.2	0.003
65.0	1.44	1.998	588.6	0.003
70.0	1.09	2.109	603.4	0.003
75.0	0.72	2.188	614.6	0.003
80.0	0.32	2.233	620.8	0.003
85.0	-0.08	2.244	620.0	0.003
90.0	-0.49	2.219	613.4	0.003



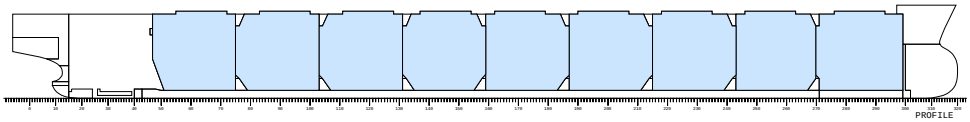
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.686	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.099	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.413	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.397	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	39.260	deg	OK
GM0.15	GM > 0.15 m	0.150	5.079	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.394		OK

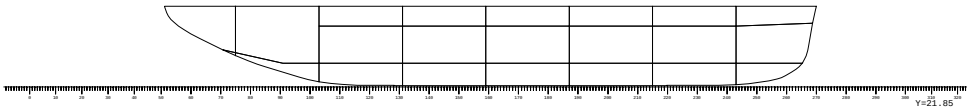


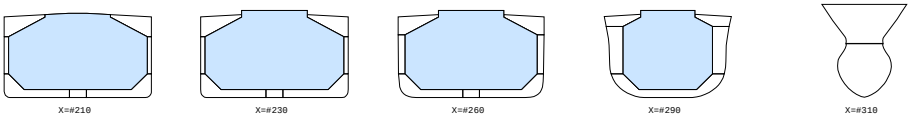
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-72	-111	32.90	62.44
8.80	11.00	-2098	-446	80.54	132.71
12.00	15.00	-3751	-690	57.01	184.85
37.60	47.00	-85570	-6436	675.16	568.51
41.13	51.00	-107252	-5873	852.93	610.65
62.97	75.00	-190426	-1991	903.78	768.92
88.45	103.00	-229820	-987	762.60	826.76
113.93	131.00	-247566	-305	822.96	832.67
139.41	159.00	-248407	126	765.12	832.23
164.89	187.00	-232938	777	854.06	831.71
190.37	215.00	-200875	1427	853.51	831.15
215.85	243.00	-152216	2086	779.78	828.06
241.33	271.00	-88272	2640	707.96	764.92
266.81	299.00	-17106	2923	547.76	459.05
278.01	313.00	-774	284	40.17	151.16

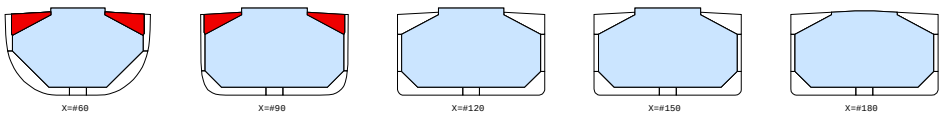
				X	FRAME
SHEAR FORCE (MIN)	-6435.9 t	(57.3%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2922.7 t	(26.0%)		266.8 m	299
SAGGING MOMENT	-250779 tm	(57.2%)		127.1 m	145
HOGGING MOMENT	42.4 tm			-2.8 m	-3

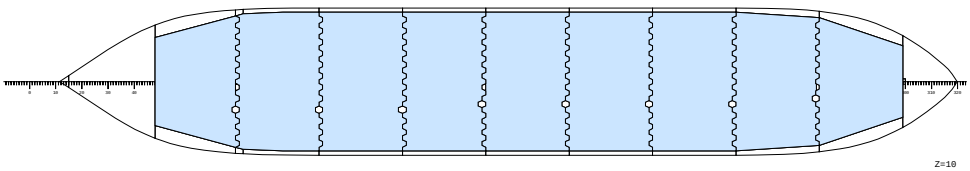
LOADING CONDITION - COND14 : FULLY LOADED COAL (42 cft/longton)-DEP BUNKERS

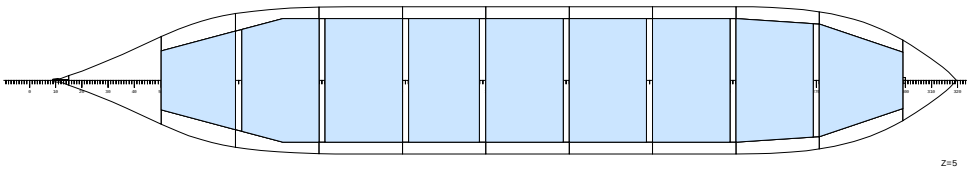


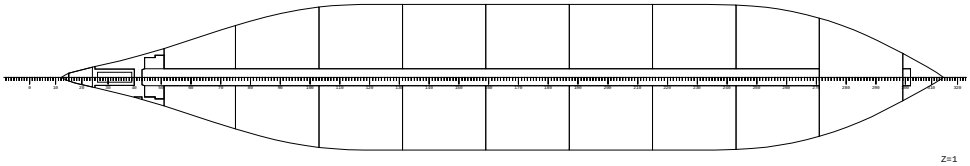












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.1	NO.1 CARGO HOLD	14407.1	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	18716.1	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	18715.3	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	18732.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	15534.1	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		158981.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

CONTENTS= (RHO=1)

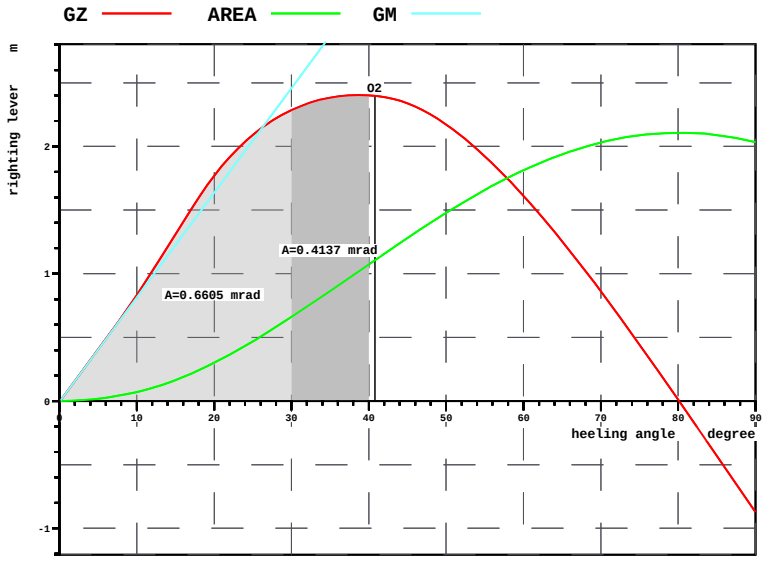
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	165166.3	147.47	-0.01 14.08
Total weight	189808.2	145.46	-0.01 14.03

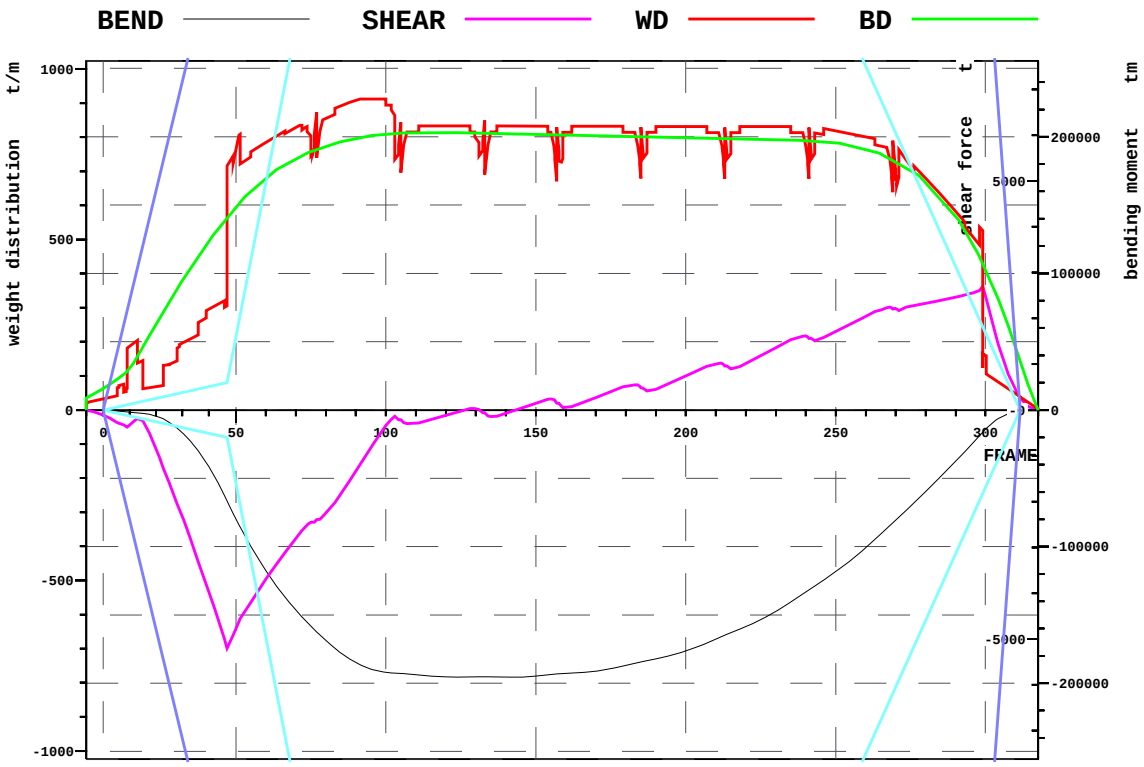
F L O A T I N G P O S I T I O N

Draught moulded	17.458	m	KM	18.73 m
Trim	1.273	m	KG	14.03 m
Heel, PS=+	-0.1	deg		
TA	18.095	m	GM0	4.71 m
TF	16.822	m	GMCORR	0.00 m
Trimming moment	-314693	tonm	GM	4.71 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.40	0.017	0.0	0.000
10.0	0.83	0.071	0.0	0.000
20.0	1.77	0.299	0.0	0.000
30.0	2.29	0.660	0.0	0.000
40.0	2.40	1.074	0.0	0.000
45.0	2.34	1.282	0.0	0.000
50.0	2.17	1.479	0.0	0.000
55.0	1.92	1.658	0.0	0.000
60.0	1.61	1.813	0.0	0.000
65.0	1.25	1.939	0.0	0.000
70.0	0.86	2.031	0.0	0.000
75.0	0.44	2.088	0.0	0.000
80.0	0.01	2.108	0.0	0.000
85.0	-0.43	2.090	0.0	0.000
90.0	-0.87	2.033	0.0	0.000



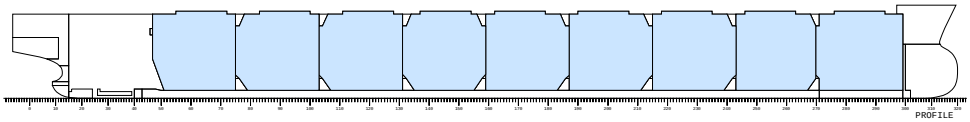
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.660	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.074	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.414	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.404	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	38.658	deg	OK
GM0.15	GM > 0.15 m	0.150	4.708	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.689		OK

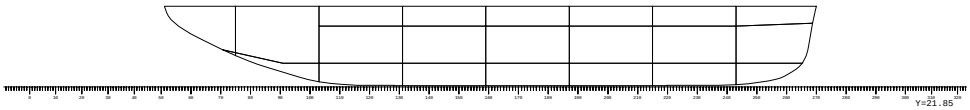


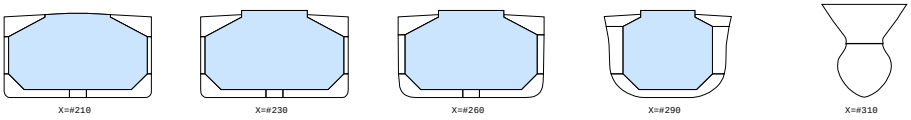
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-32	-113	32.90	62.85
8.80	11.00	-1867	-274	192.91	132.24
12.00	15.00	-2417	-233	62.47	183.92
37.60	47.00	-66731	-5202	716.11	562.62
41.13	51.00	-83927	-4623	805.85	604.02
62.97	75.00	-158183	-2448	774.32	757.85
88.45	103.00	-192560	-135	734.86	810.76
113.93	131.00	-195182	9	784.63	811.78
139.41	159.00	-193013	52	737.84	806.46
164.89	187.00	-183466	413	814.08	801.07
190.37	215.00	-163138	897	813.53	795.64
215.85	243.00	-128860	1514	751.22	787.70
241.33	271.00	-78596	2169	683.04	721.76
266.81	299.00	-16240	2700	526.04	428.40
278.01	313.00	-820	279	40.17	147.32

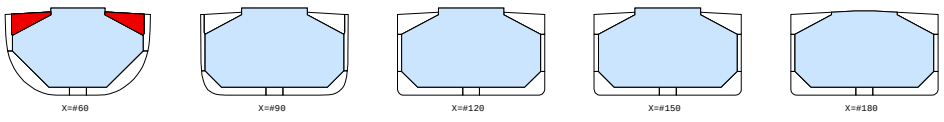
				X	FRAME
SHEAR FORCE (MIN)	-5202.4 t	(46.3%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2699.6 t	(24.1%)		266.8 m	299
SAGGING MOMENT	-195624 tm	(44.6%)		123.3 m	141
HOGGING MOMENT	63.1 tm			-2.8 m	-3

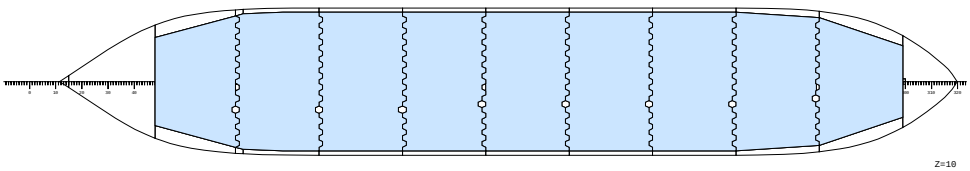
LOADING CONDITION - COND15 : FULLY LOADED COAL (42 cft/longton)-50% BUNKERS

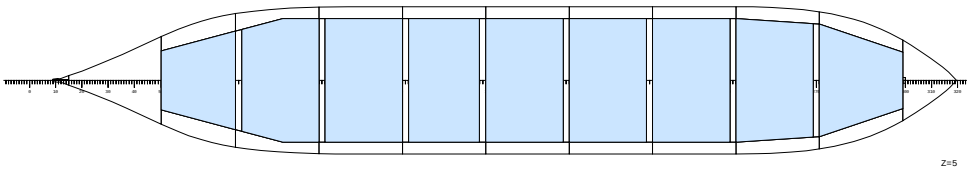


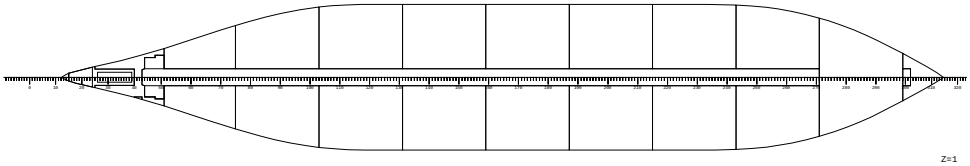












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.1	NO.1 CARGO HOLD	14407.1	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	18716.1	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	18715.3	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	18732.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	15534.1	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		158981.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

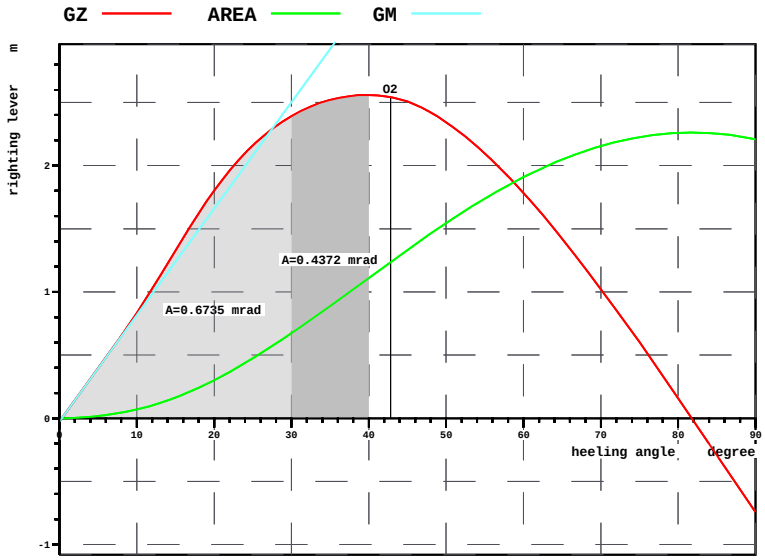
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)						
(OIL&WAT.	150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	162218.0	149.00	-0.03 13.91
Total weight	186859.9	146.76	-0.03 13.88

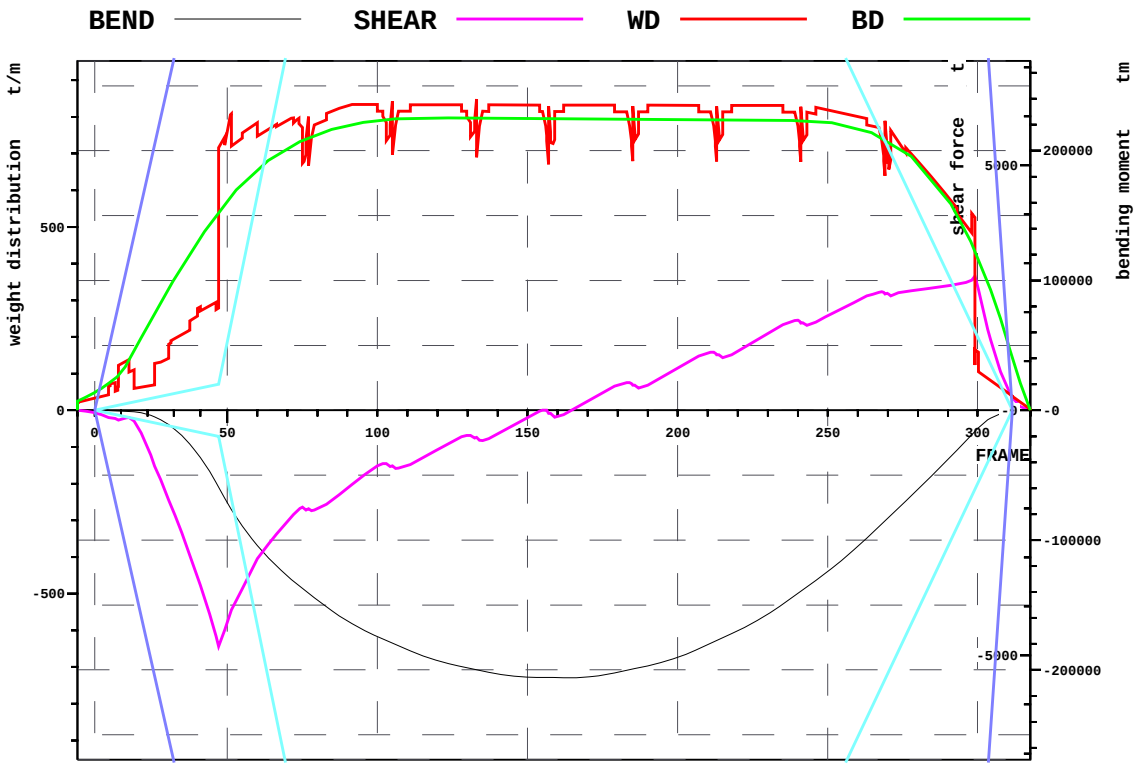
F L O A T I N G P O S I T I O N

Draught moulded	17.217	m	KM	18.71	m
Trim	0.374	m	KG	13.88	m
Heel, PS=+	-0.3	deg			
TA	17.404	m	GM0	4.83	m
TF	17.029	m	GMCORR	0.00	m
Trimming moment	-91720	tonm	GM	4.83	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.40	0.016	273.1	0.001
10.0	0.83	0.069	572.7	0.003
20.0	1.80	0.300	1197.1	0.006
30.0	2.39	0.673	1623.0	0.009
40.0	2.56	1.111	1920.4	0.010
45.0	2.51	1.332	2017.0	0.011
50.0	2.34	1.545	2076.3	0.011
55.0	2.10	1.739	2095.0	0.011
60.0	1.78	1.909	2072.0	0.011
65.0	1.42	2.049	2014.9	0.011
70.0	1.02	2.156	1930.4	0.010
75.0	0.60	2.227	1823.1	0.010
80.0	0.16	2.260	1696.6	0.009
85.0	-0.29	2.254	1553.9	0.008
90.0	-0.74	2.209	1399.3	0.007



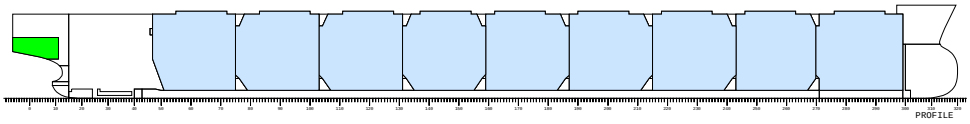
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.673	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.111	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.437	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.559	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	39.762	deg	OK
GM0.15	GM > 0.15 m	0.150	4.828	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.091		OK

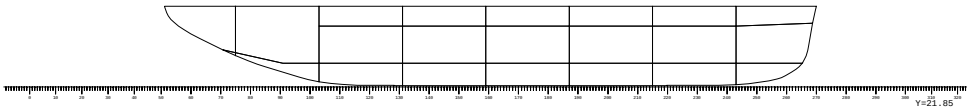


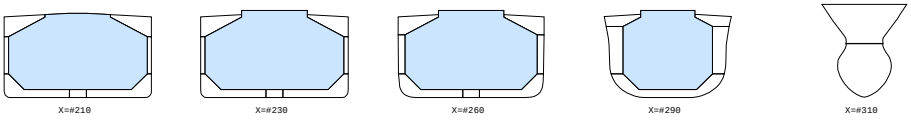
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	73	-51	32.90	49.08
8.80	11.00	-788	-165	130.42	113.37
12.00	15.00	-1211	-218	59.74	163.64
37.60	47.00	-59391	-4816	715.79	538.06
41.13	51.00	-75109	-4151	805.53	579.50
62.97	75.00	-137133	-1973	771.51	735.41
88.45	103.00	-177259	-1089	734.86	792.01
113.93	131.00	-198789	-516	784.62	796.85
139.41	159.00	-205837	-141	737.84	795.35
164.89	187.00	-198295	454	814.07	793.77
190.37	215.00	-175235	1075	813.52	792.15
215.85	243.00	-135968	1733	751.21	788.03
241.33	271.00	-80927	2332	683.04	725.69
266.81	299.00	-16236	2737	526.04	433.21
278.01	313.00	-761	280	40.17	148.03

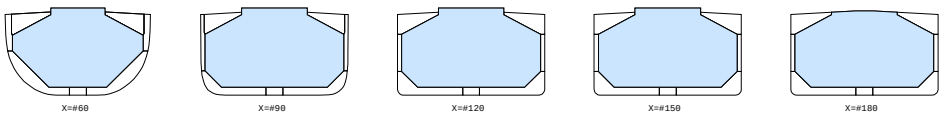
				X	FRAME
SHEAR FORCE (MIN)	-4815.8 t	(42.9%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2737.0 t	(24.4%)		266.8 m	299
SAGGING MOMENT	-206052 tm	(47.0%)		142.1 m	162
HOGGING MOMENT	84.5 tm			-0.4 m	-0

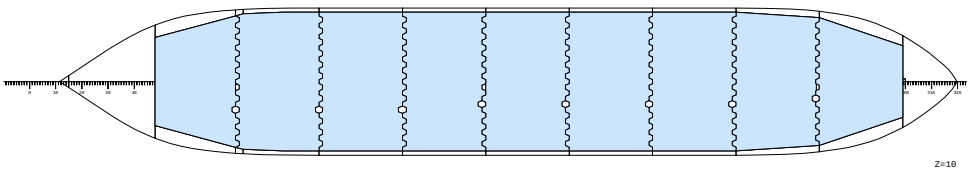
LOADING CONDITION - COND16 : FULLY LOADED COAL (42 cft/longton)-ARRIVAL BUNKE

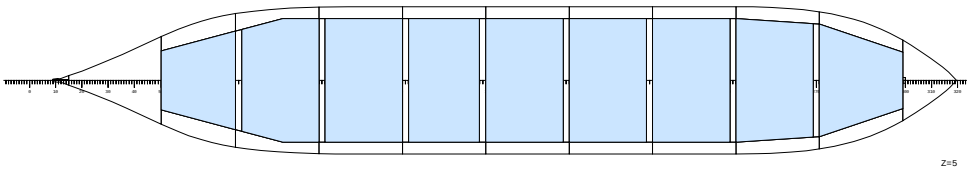


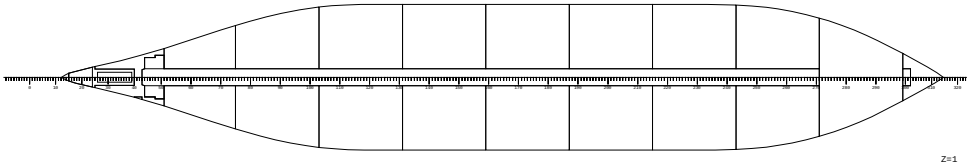












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.1	NO.1 CARGO HOLD	14407.1	100.0	252.76	0.00	13.97	0.00
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.3	NO.3 CARGO HOLD	18716.1	100.0	202.39	0.02	13.60	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.5	NO.5 CARGO HOLD	18715.3	100.0	151.43	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.7	NO.7 CARGO HOLD	18732.4	100.0	101.86	0.02	13.60	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
R1.9	NO.9 CARGO HOLD	15534.1	100.0	51.19	0.03	14.44	0.00
SUBTOTAL		158981.1		151.24	0.02	13.74	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61

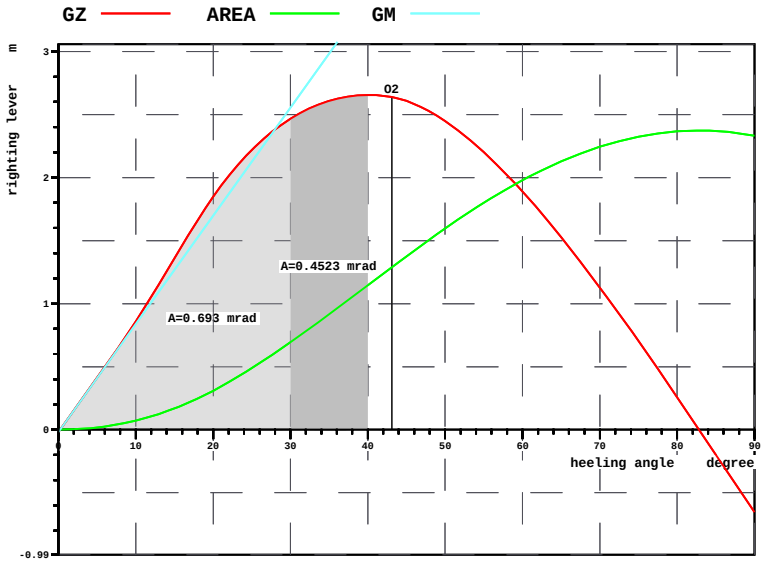
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	161471.4	149.13	-0.02 13.81
Total weight	186113.3	146.86	-0.02 13.80

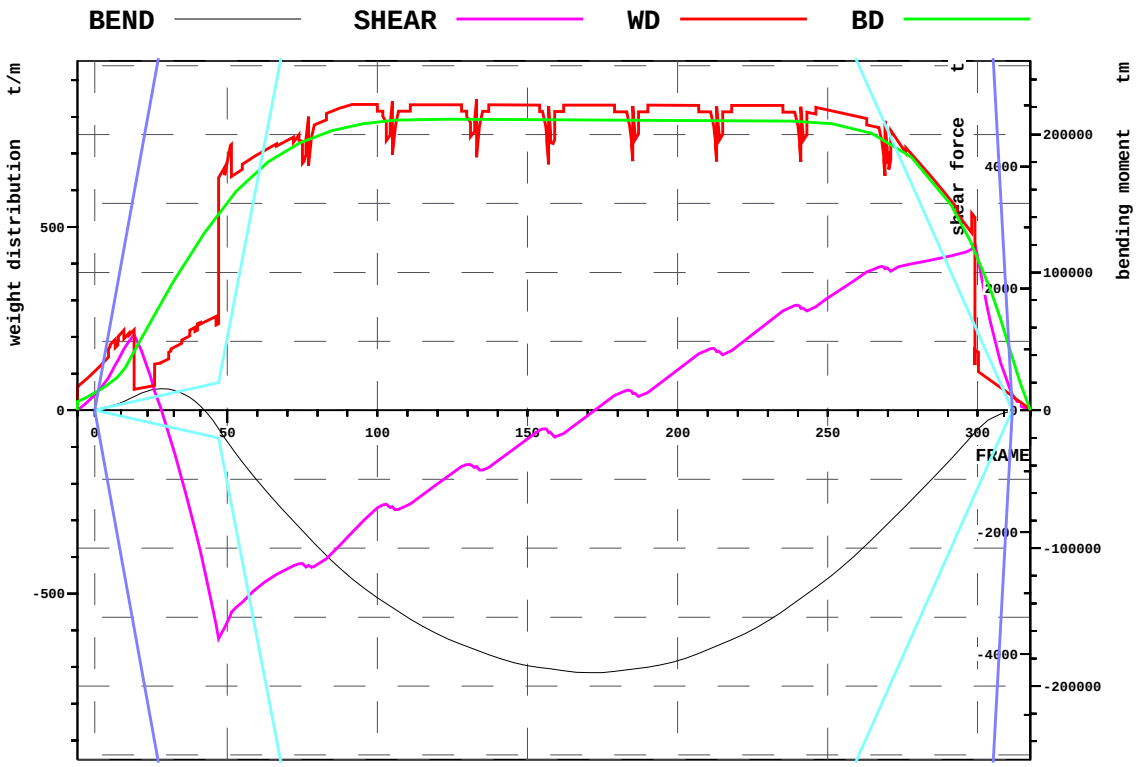
F L O A T I N G P O S I T I O N

Draught moulded	17.154	m	KM	18.71 m
Trim	0.325	m	KG	13.80 m
Heel, PS=+	-0.2	deg		
TA	17.317	m	GM0	4.91 m
TF	16.992	m	GMCORR	-0.00 m
Trimming moment	-79549	tonm	GM	4.91 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.41	0.017	64.4	0.000
10.0	0.86	0.072	128.6	0.001
20.0	1.85	0.309	221.7	0.001
30.0	2.47	0.693	296.1	0.002
40.0	2.65	1.145	401.9	0.002
45.0	2.61	1.376	456.9	0.002
50.0	2.45	1.597	503.3	0.003
55.0	2.20	1.800	540.1	0.003
60.0	1.89	1.979	568.2	0.003
65.0	1.53	2.129	588.6	0.003
70.0	1.13	2.245	603.4	0.003
75.0	0.70	2.325	614.6	0.003
80.0	0.26	2.367	620.8	0.003
85.0	-0.20	2.369	620.0	0.003
90.0	-0.65	2.332	613.4	0.003



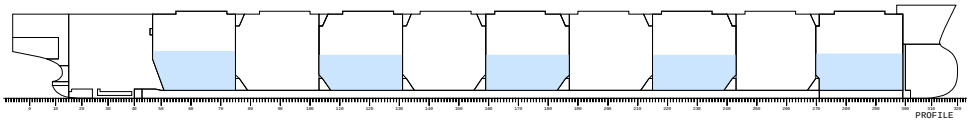
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	0.693	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.145	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.452	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.655	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	40.161	deg	OK
GM0.15	GM > 0.15 m	0.150	4.909	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.116		OK

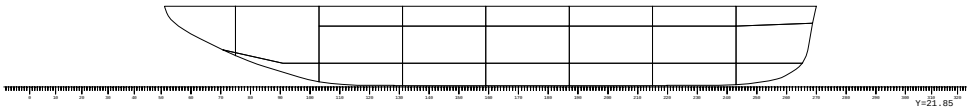


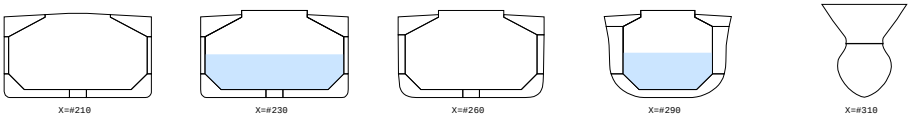
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	828	258	106.31	47.41
8.80	11.00	6281	1004	218.34	111.00
12.00	15.00	10001	1237	57.01	161.05
37.60	47.00	-13412	-3750	634.41	534.61
41.13	51.00	-25864	-3363	722.20	576.00
62.97	75.00	-87184	-2514	732.42	731.92
88.45	103.00	-140055	-1544	734.84	788.71
113.93	131.00	-172196	-890	784.61	793.76
139.41	159.00	-187855	-440	737.82	792.47
164.89	187.00	-187061	227	814.06	791.09
190.37	215.00	-169023	913	813.51	789.69
215.85	243.00	-133184	1630	751.20	785.78
241.33	271.00	-80119	2283	683.03	723.74
266.81	299.00	-16165	2729	526.03	432.09
278.01	313.00	-749	280	40.17	147.90

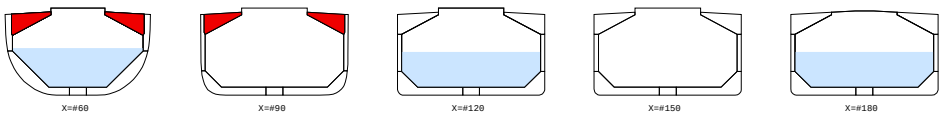
				X	FRAME
SHEAR FORCE (MIN)	-3749.5 t	(33.4%)	POSITION:	37.6 m	47
SHEAR FORCE (MAX)	2729.1 t	(24.3%)		266.8 m	299
SAGGING MOMENT	-190349 tm	(43.4%)		149.9 m	171
HOGGING MOMENT	15637.7 tm	(147.0%)		20.0 m	25

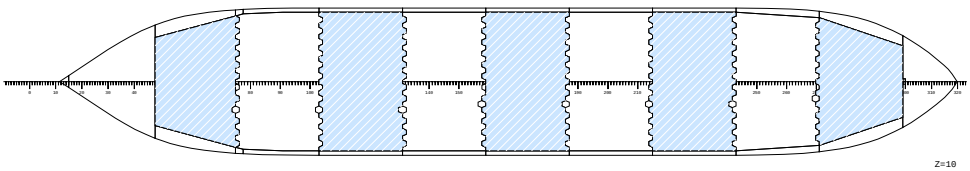
LOADING CONDITION - COND17 : FULLY LOADED ORE (DESIGN DRAUGHT)-DEP BUNKERS

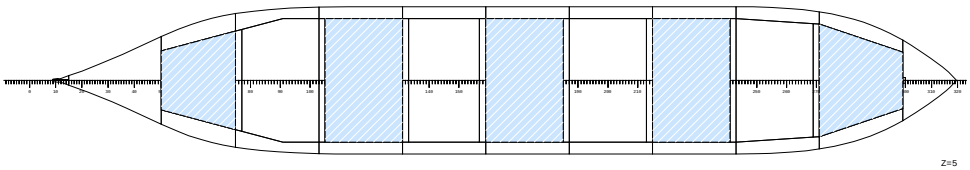


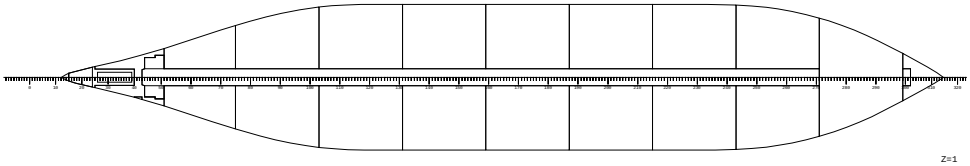












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.984)							
R1.1	NO.1 CARGO HOLD	24666.9	49.0	252.56	0.00	8.41	0.00
R1.3	NO.3 CARGO HOLD	32044.4	49.0	202.37	0.02	8.18	0.00
R1.5	NO.5 CARGO HOLD	32043.1	49.0	151.41	0.02	8.18	0.00
R1.7	NO.7 CARGO HOLD	32072.3	49.0	101.89	0.02	8.18	0.00
R1.9	NO.9 CARGO HOLD	26596.5	49.0	51.77	0.03	9.33	0.00
SUBTOTAL		147423.2		150.66	0.02	8.43	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

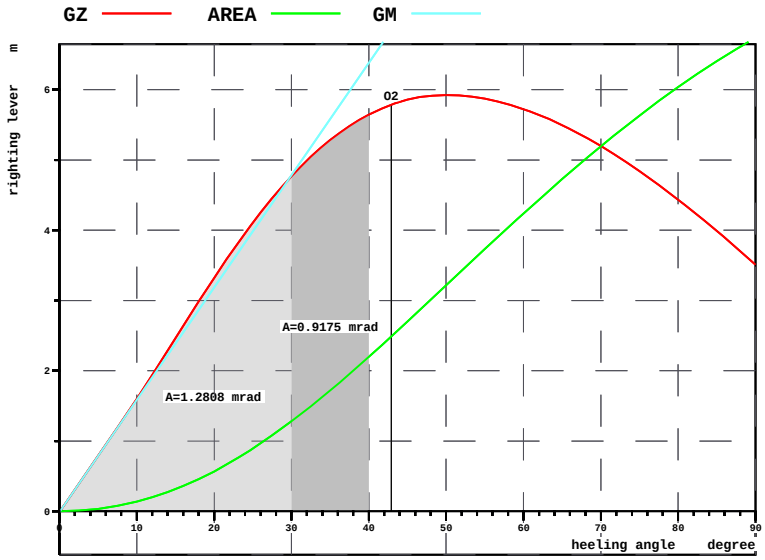
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)						
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	153608.4	146.63	-0.01 9.00
Total weight	178250.3	144.61	-0.01 9.65

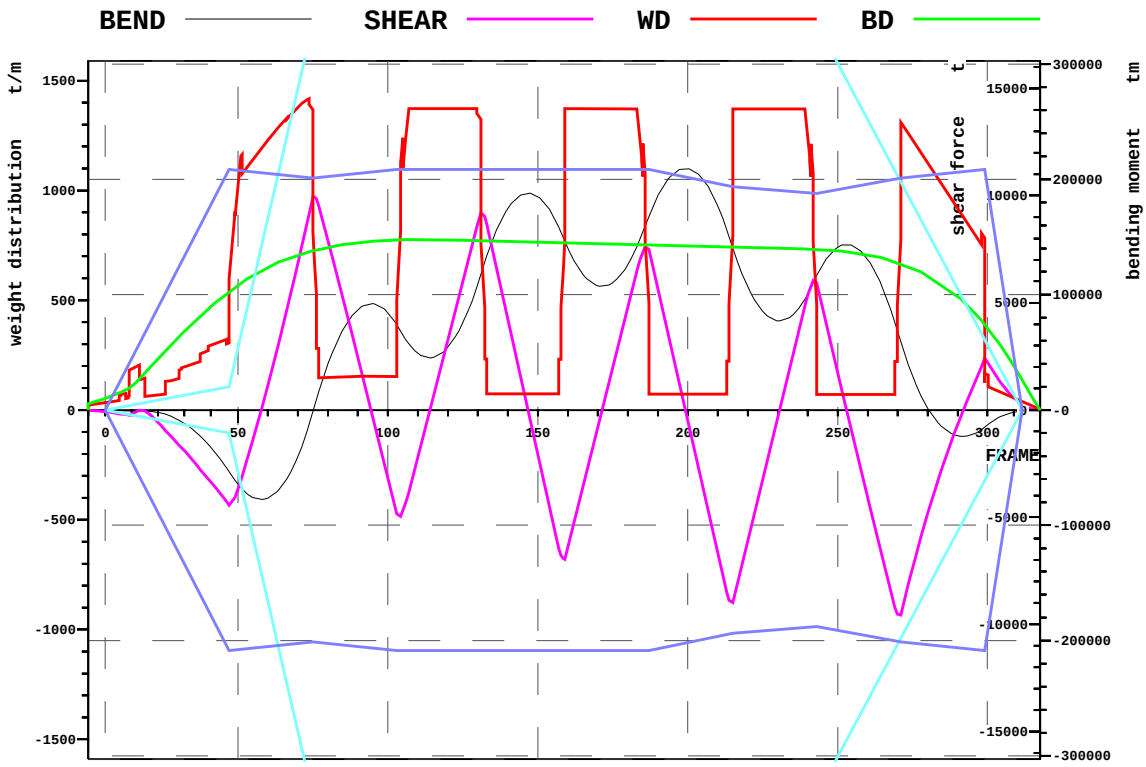
F L O A T I N G P O S I T I O N

Draught moulded	16.483	m	KM	18.81	m
Trim	2.262	m	KG	9.65	m
Heel, PS=+	-0.1	deg			
TA	17.614	m	GM0	9.16	m
TF	15.352	m	GMCORR	0.00	m
Trimming moment	-556055	tonm	GM	9.16	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.79	0.034	0.0	0.000
10.0	1.60	0.138	0.0	0.000
20.0	3.32	0.568	0.0	0.000
30.0	4.77	1.281	0.0	0.000
40.0	5.65	2.198	0.0	0.000
45.0	5.86	2.701	0.0	0.000
50.0	5.93	3.217	0.0	0.000
55.0	5.87	3.732	0.0	0.000
60.0	5.72	4.239	0.0	0.000
65.0	5.50	4.729	0.0	0.000
70.0	5.20	5.196	0.0	0.000
75.0	4.84	5.634	0.0	0.000
80.0	4.43	6.039	0.0	0.000
85.0	3.99	6.407	0.0	0.000
90.0	3.51	6.735	0.0	0.000



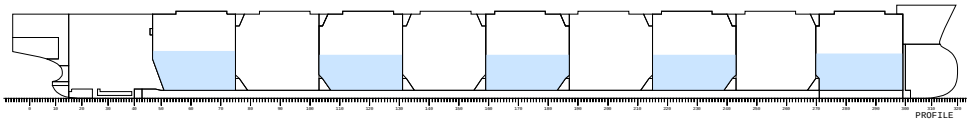
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.281	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.198	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.918	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.925	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.103	deg	OK
GM0.15	GM > 0.15 m	0.150	9.157	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.118		OK

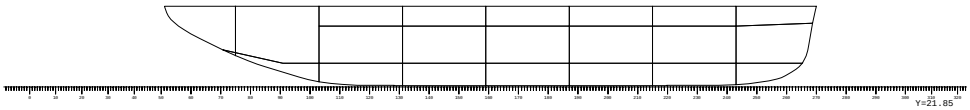


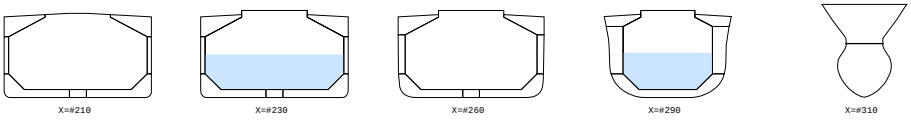
X	FR	BEND	SHEAR	WD	BD
m	#	tm	t	t/m	t/m
0.00	0.00	56	-71	32.90	53.19
8.80	11.00	-1005	-125	192.94	117.63
12.00	15.00	-1012	-34	62.48	167.63
37.60	47.00	-53725	-4443	595.75	536.14
41.13	51.00	-67621	-32931156.84		576.50
62.97	75.00	-2257	99181090.68		725.47
88.45	103.00	73780	-4840	502.15	774.00
113.93	131.00	117116	92051324.60		770.83
139.41	159.00	146321	-6961	771.62	761.32
164.89	187.00	167975	7480	486.78	751.73
190.37	215.00	140597	-89821370.92		742.11
215.85	243.00	116644	5926	481.48	730.05
241.33	271.00	59293	-95651309.43		662.80
266.81	299.00	-14794	2405	783.82	388.46
278.01	313.00	-787	271	40.18	140.91

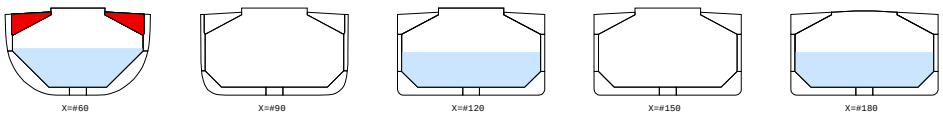
				X	FRAME
SHEAR FORCE (MIN)	-9564.6 t	(86.8%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	9926.0 t	(91.7%)		63.5 m	76
SAGGING MOMENT	-77657.2 tm	(32.9%)		47.7 m	58
HOGGING MOMENT	209316.9 tm	(44.3%)		177.1 m	200

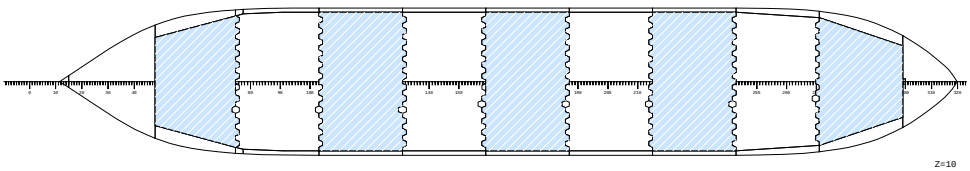
LOADING CONDITION - COND18 : FULLY LOADED ORE (DESIGN DRAUGHT)-50% BUNKERS

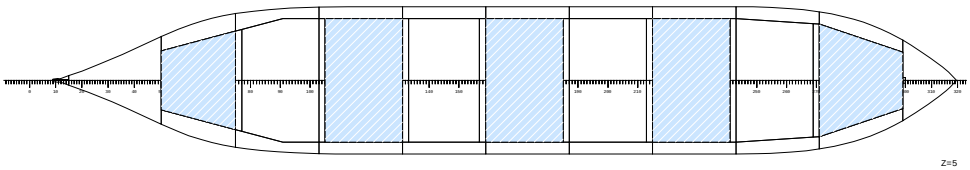


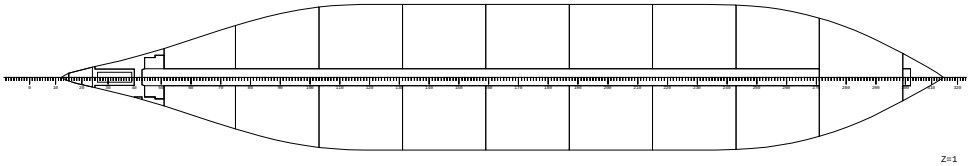












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.984)							
R1.1	NO.1 CARGO HOLD	24666.9	49.0	252.56	0.00	8.41	0.00
R1.3	NO.3 CARGO HOLD	32044.4	49.0	202.37	0.02	8.18	0.00
R1.5	NO.5 CARGO HOLD	32043.1	49.0	151.41	0.02	8.18	0.00
R1.7	NO.7 CARGO HOLD	32072.3	49.0	101.89	0.02	8.18	0.00
R1.9	NO.9 CARGO HOLD	26596.5	49.0	51.77	0.03	9.33	0.00
SUBTOTAL		147423.2		150.66	0.02	8.43	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

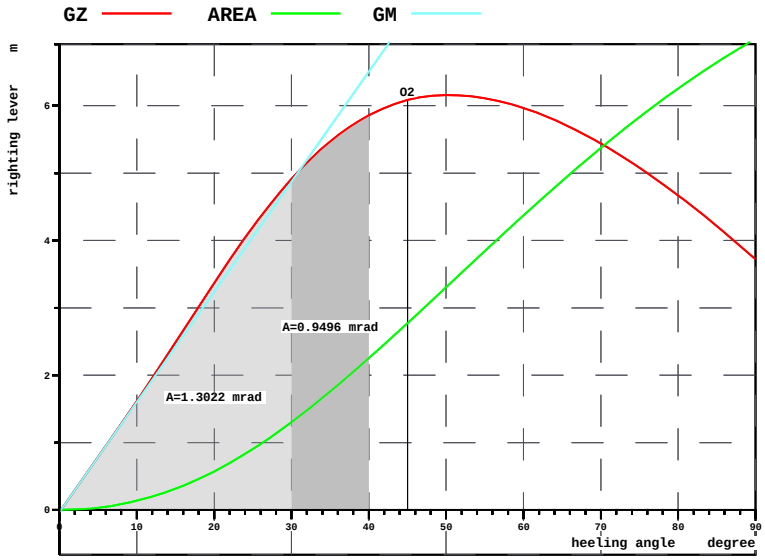
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	150660.1	148.26	-0.04 8.73
Total weight	175302.0	145.97	-0.03 9.42

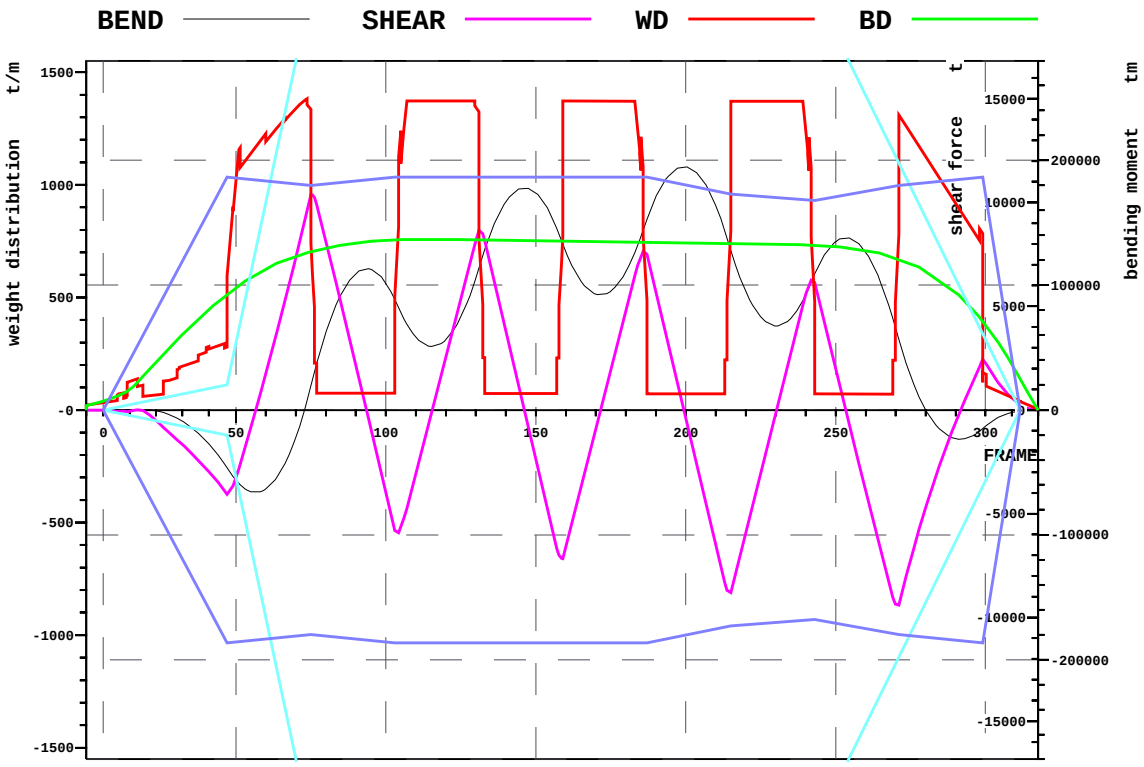
F L O A T I N G P O S I T I O N

Draught moulded	16.239	m	KM	18.79 m
Trim	1.364	m	KG	9.42 m
Heel, PS=+	-0.2	deg		
TA	16.921	m	GM0	9.36 m
TF	15.558	m	GMCORR	0.00 m
Trimming moment	-331640	tonm	GM	9.36 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.79	0.033	273.1	0.002
10.0	1.62	0.138	572.7	0.003
20.0	3.37	0.572	1197.1	0.007
30.0	4.92	1.302	1623.0	0.009
40.0	5.86	2.252	1920.4	0.011
45.0	6.09	2.774	2017.0	0.012
50.0	6.16	3.309	2076.3	0.012
55.0	6.11	3.846	2095.0	0.012
60.0	5.97	4.373	2072.0	0.012
65.0	5.74	4.885	2014.9	0.011
70.0	5.44	5.373	1930.4	0.011
75.0	5.08	5.832	1823.1	0.010
80.0	4.67	6.258	1696.6	0.010
85.0	4.21	6.646	1553.9	0.009
90.0	3.72	6.993	1399.3	0.008



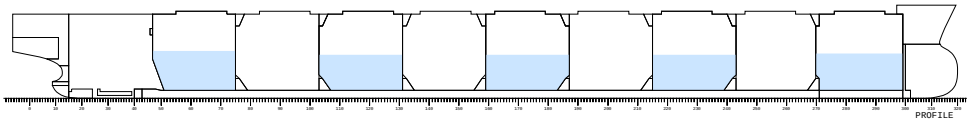
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.302	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.252	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.950	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.160	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.320	deg	OK
GM0.15	GM > 0.15 m	0.150	9.364	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.629		OK

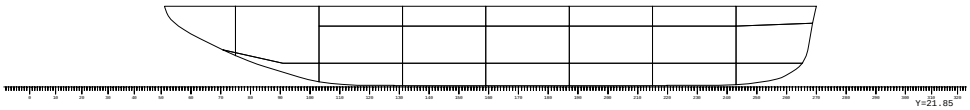


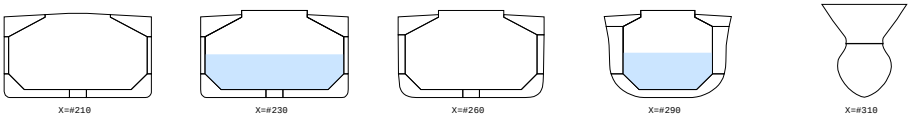
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	196	-12	32.90	40.07
8.80	11.00	131	-24	130.46	99.29
12.00	15.00	249	-29	59.75	147.82
37.60	47.00	-46478	-4074	595.44	511.65
41.13	51.00	-58926	-28381156.52		552.01
62.97	75.00	18503	103771335.03		702.96
88.45	103.00	88614	-5809	502.15	755.16
113.93	131.00	112923	86681324.59		755.81
139.41	159.00	132846	-7164	771.61	750.12
164.89	187.00	152485	7514	486.78	744.35
190.37	215.00	127883	-88101370.92		738.54
215.85	243.00	109019	6141	481.48	730.30
241.33	271.00	56597	-94041309.43		666.65
266.81	299.00	-14946	2443	783.82	393.19
278.01	313.00	-776	272	40.18	141.99

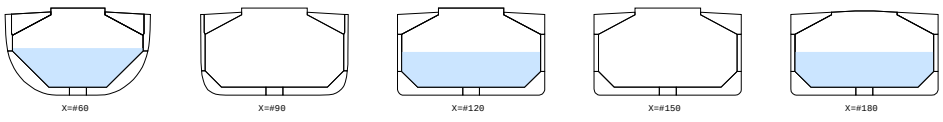
				X	FRAME
SHEAR FORCE (MIN)	-9403.7 t	(85.3%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	10376.5 t	(95.9%)		63.0 m	75
SAGGING MOMENT	-65496.4 tm	(29.6%)		44.8 m	55
HOGGING MOMENT	194709.5 tm	(41.2%)		177.1 m	200

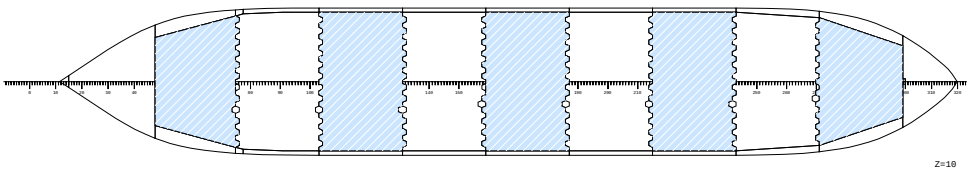
LOADING CONDITION - COND19 : FULLY LOADED ORE (DESIGN DRAUGHT)-ARRIVAL BUNKER

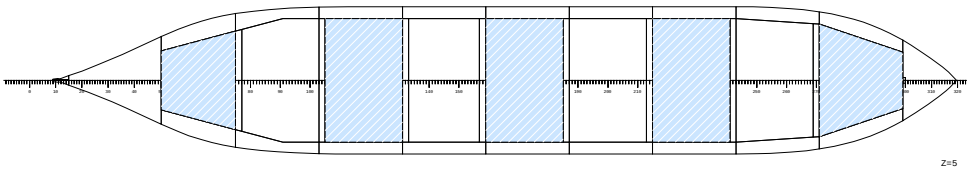


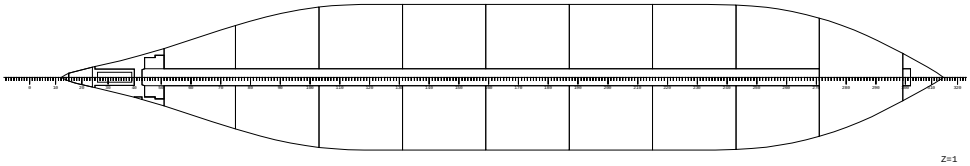












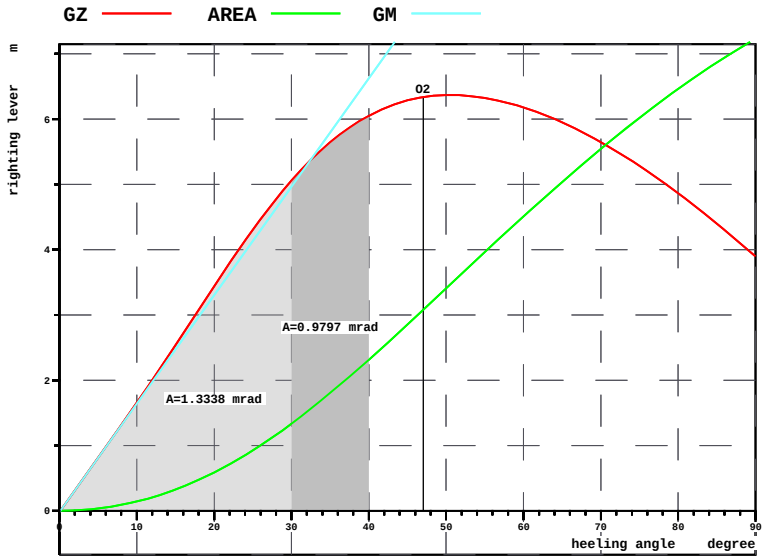
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.984)							
R1.1	NO.1 CARGO HOLD	24666.9	49.0	252.56	0.00	8.41	0.00
R1.3	NO.3 CARGO HOLD	32044.4	49.0	202.37	0.02	8.18	0.00
R1.5	NO.5 CARGO HOLD	32043.1	49.0	151.41	0.02	8.18	0.00
R1.7	NO.7 CARGO HOLD	32072.3	49.0	101.89	0.02	8.18	0.00
R1.9	NO.9 CARGO HOLD	26596.5	49.0	51.77	0.03	9.33	0.00
SUBTOTAL		147423.2		150.66	0.02	8.43	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	148291.2	149.96	-0.02 8.51
Total weight	172933.1	147.40	-0.02 9.25

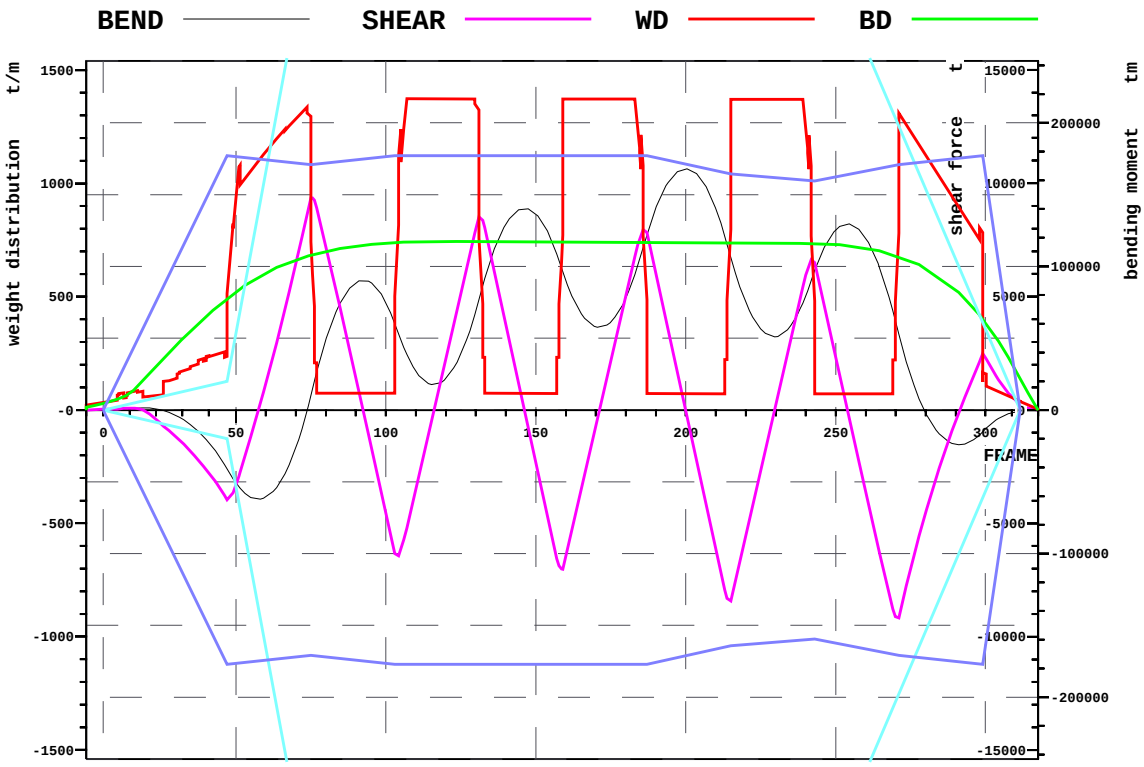
F L O A T I N G P O S I T I O N

Draught moulded	16.043	m	KM	18.76	m
Trim	0.414	m	KG	9.25	m
Heel, PS=+	-0.1	deg			
TA	16.250	m	GM0	9.51	m
TF	15.835	m	GMCORR	0.00	m
Trimming moment	-99490	tonm	GM	9.51	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.81	0.034	64.4	0.000
10.0	1.66	0.142	128.6	0.001
20.0	3.44	0.586	221.7	0.001
30.0	5.06	1.334	296.1	0.002
40.0	6.05	2.313	401.9	0.002
45.0	6.29	2.853	456.9	0.003
50.0	6.37	3.406	503.3	0.003
55.0	6.32	3.961	540.1	0.003
60.0	6.18	4.507	568.2	0.003
65.0	5.95	5.037	588.6	0.003
70.0	5.65	5.543	603.4	0.003
75.0	5.28	6.020	614.6	0.004
80.0	4.86	6.463	620.8	0.004
85.0	4.40	6.868	620.0	0.004
90.0	3.90	7.230	613.4	0.004



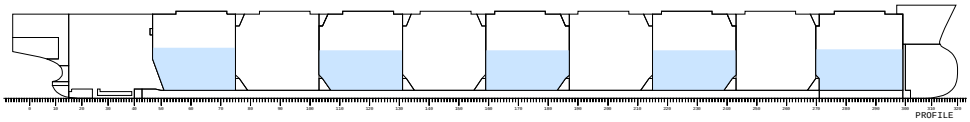
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.334	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.313	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.980	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.369	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.474	deg	OK
GM0.15	GM > 0.15 m	0.150	9.514	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.106		OK

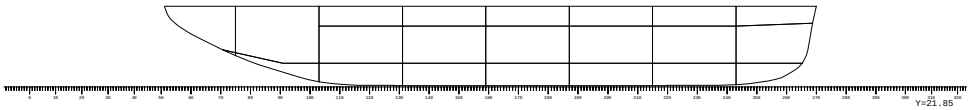


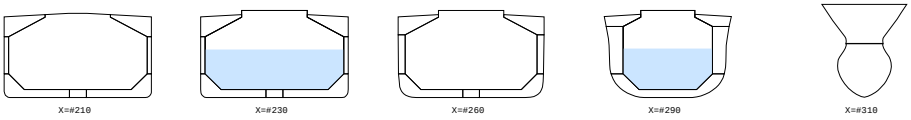
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	319	39	32.90	28.51
8.80	11.00	1144	76	80.56	82.47
12.00	15.00	1433	1	57.01	129.50
37.60	47.00	-41146	-3960	514.05	488.58
41.13	51.00	-53566	-29341073.19		528.94
62.97	75.00	9846	93601296.36		682.00
88.45	103.00	60409	-6340	502.15	738.08
113.93	131.00	76254	85211324.59		742.75
139.41	159.00	96195	-7030	771.61	741.07
164.89	187.00	121726	7827	486.77	739.32
190.37	215.00	106277	-84191370.91		737.53
215.85	243.00	97223	6506	481.48	733.30
241.33	271.00	52668	-91641309.42		673.31
266.81	299.00	-15178	2495	783.82	399.83
278.01	313.00	-772	274	40.18	143.32

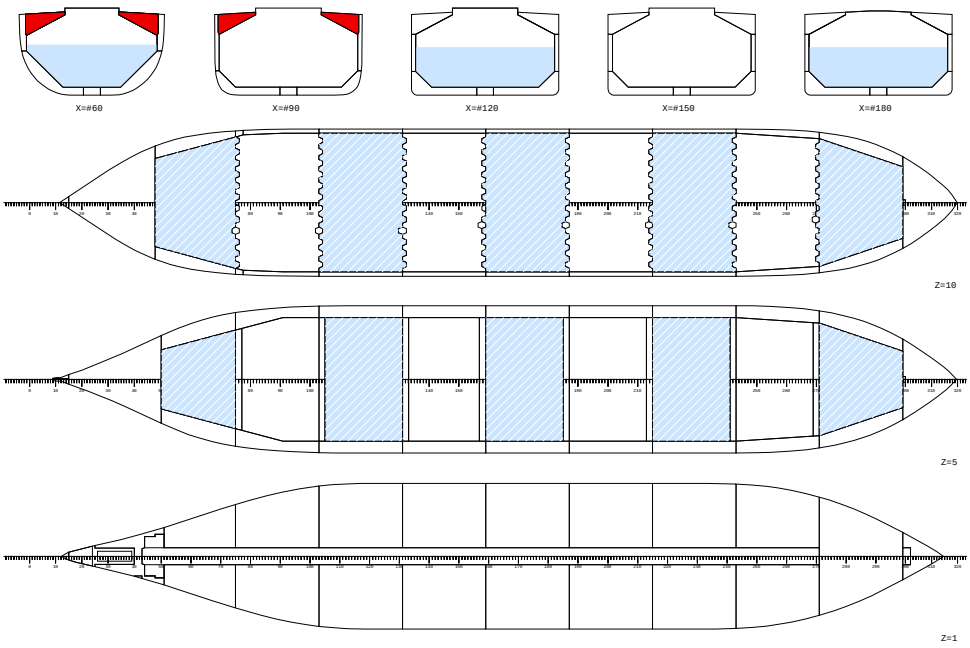
				X	FRAME
SHEAR FORCE (MIN)	-9164.5 t	(83.2%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	9359.6 t	(86.5%)		63.0 m	75
SAGGING MOMENT	-62073.9 tm	(26.3%)		47.7 m	58
HOGGING MOMENT	168080.0 tm	(35.6%)		177.1 m	200

LOADING CONDITION - COND20 : FULLY LOADED ORE (SCANTLING DRAUGHT)-DEP BUNKERS









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	29913.2	55.0	51.66	0.03	9.94	0.00
SUBTOTAL		167774.1		150.66	0.02	9.12	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

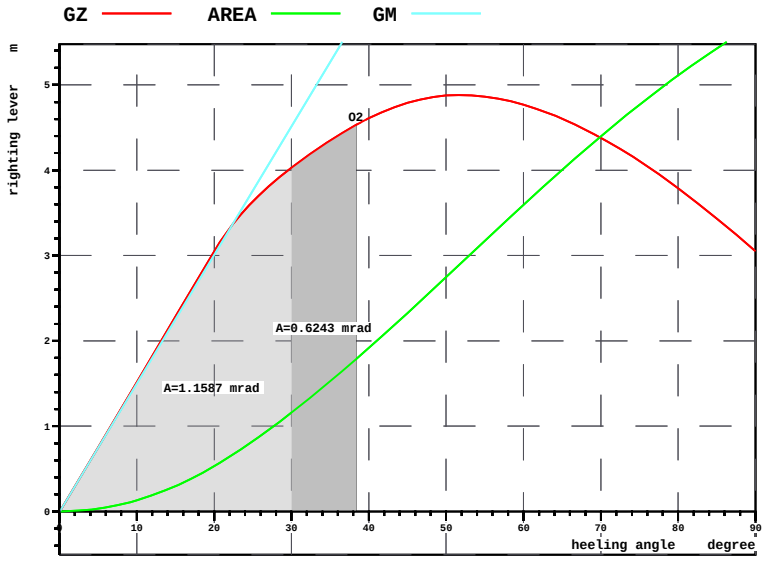
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)						
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	173959.3	147.10	-0.01 9.60
Total weight	198601.2	145.22	-0.01 10.11

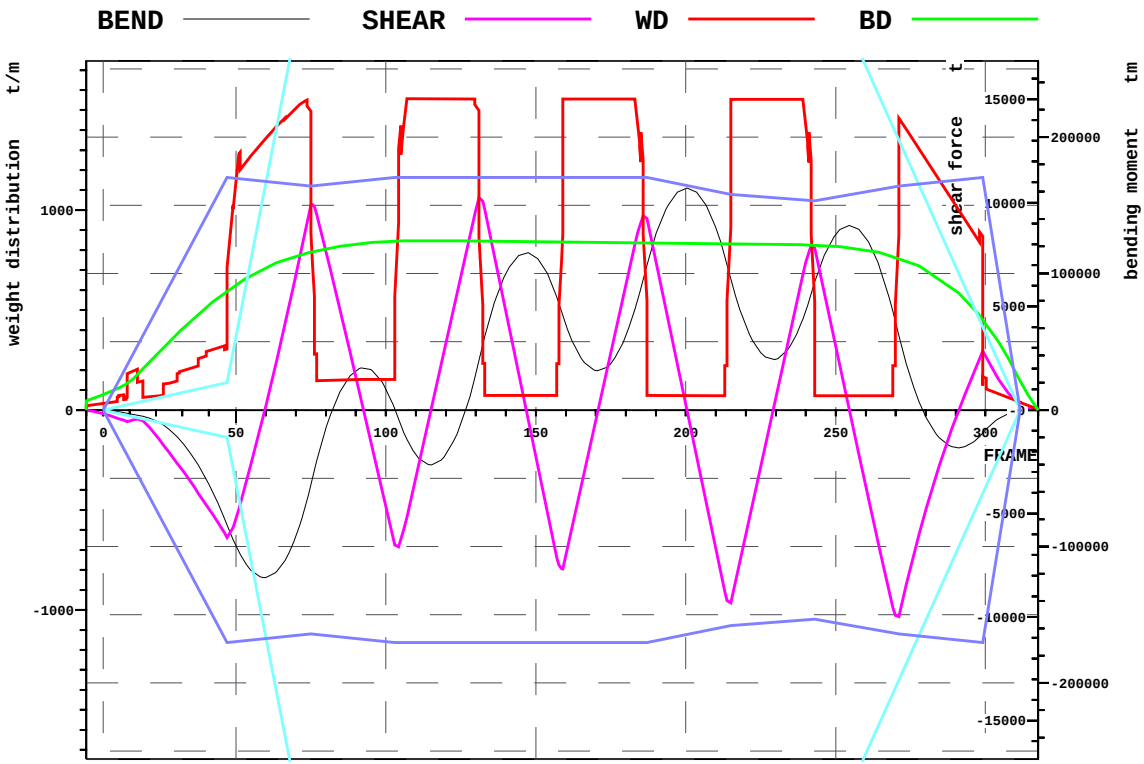
F L O A T I N G P O S I T I O N

Draught moulded	18.194	m	KM	18.74	m
Trim	1.153	m	KG	10.11	m
Heel, PS=+	-0.0	deg			
TA	18.771	m	GM0	8.63	m
TF	17.617	m	GMCORR	0.00	m
Trimming moment	-287380	tonm	GM	8.63	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.75	0.032	0.0	0.000
10.0	1.51	0.131	0.0	0.000
20.0	3.05	0.530	0.0	0.000
30.0	4.03	1.159	0.0	0.000
40.0	4.61	1.916	0.0	0.000
45.0	4.79	2.326	0.0	0.000
50.0	4.88	2.749	0.0	0.000
55.0	4.86	3.174	0.0	0.000
60.0	4.77	3.595	0.0	0.000
65.0	4.60	4.005	0.0	0.000
70.0	4.38	4.397	0.0	0.000
75.0	4.11	4.767	0.0	0.000
80.0	3.79	5.112	0.0	0.000
85.0	3.43	5.428	0.0	0.000
90.0	3.05	5.710	0.0	0.000



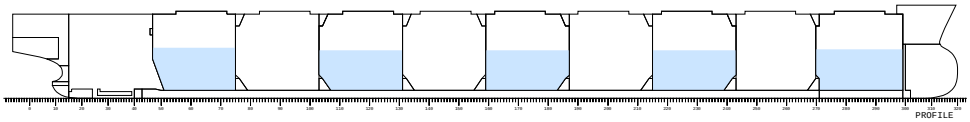
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.159	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.783	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.624	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.881	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.759	deg	OK
GM0.15	GM > 0.15 m	0.150	8.633	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.050		OK

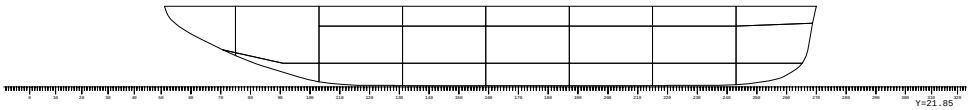


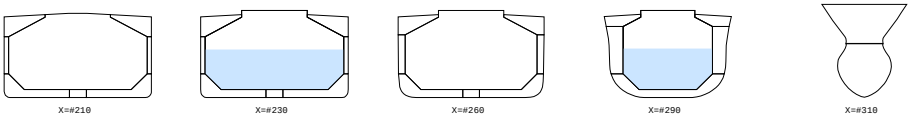
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-204	-178	32.90	77.04
8.80	11.00	-3237	-489	192.94	152.18
12.00	15.00	-4584	-515	62.48	205.62
37.60	47.00	-84388	-6163	712.38	592.58
41.13	51.00	-103957	-47901280.75		634.57
62.97	75.00	-55032	99031190.89		790.16
88.45	103.00	656	-6509	568.90	843.75
113.93	131.00	35367	102581498.11		845.30
139.41	159.00	69258	-7670	880.14	840.49
164.89	187.00	106111	9247	548.41	835.62
190.37	215.00	96129	-93161554.20		830.70
215.85	243.00	93909	7829	543.10	823.25
241.33	271.00	54612	-9985	874.75	756.20
266.81	299.00	-16944	2860	867.96	450.76
278.01	313.00	-816	283	40.18	149.97

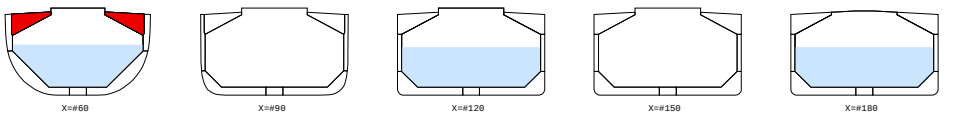
				X	FRAME
SHEAR FORCE (MIN)	-9985.1 t	(90.6%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	10257.6 t	(91.4%)		113.9 m	131
SAGGING MOMENT	-122830 tm	(51.0%)		48.6 m	59
HOGGING MOMENT	162727.8 tm	(34.4%)		177.1 m	200

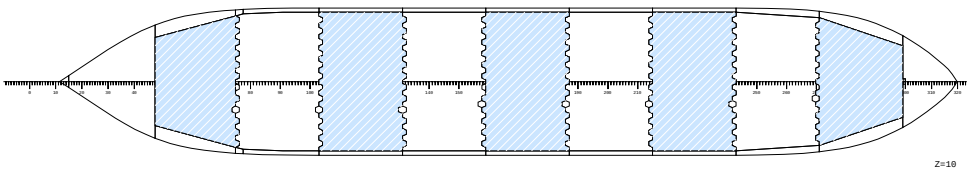
LOADING CONDITION - COND21 : FULLY LOADED ORE (SCANTLING DRAUGHT)-50% BUNKERS

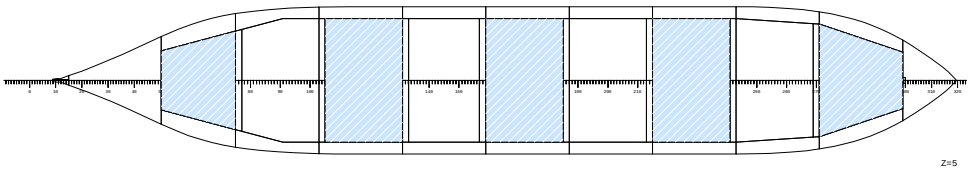


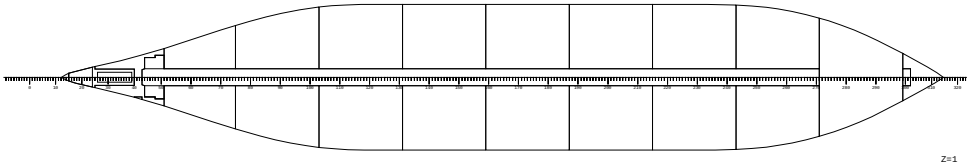












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	29913.2	55.0	51.66	0.03	9.94	0.00
SUBTOTAL		167774.1		150.66	0.02	9.12	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

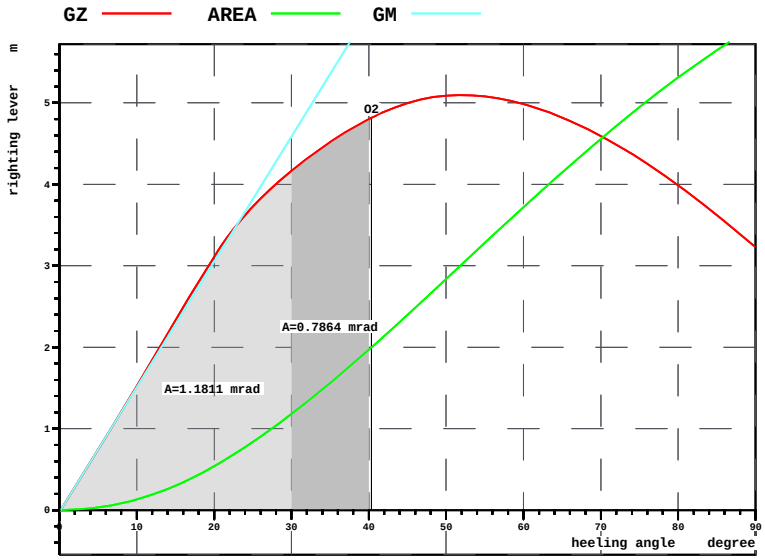
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	171011.0	148.54	-0.03 9.37
Total weight	195652.9	146.46	-0.02 9.91

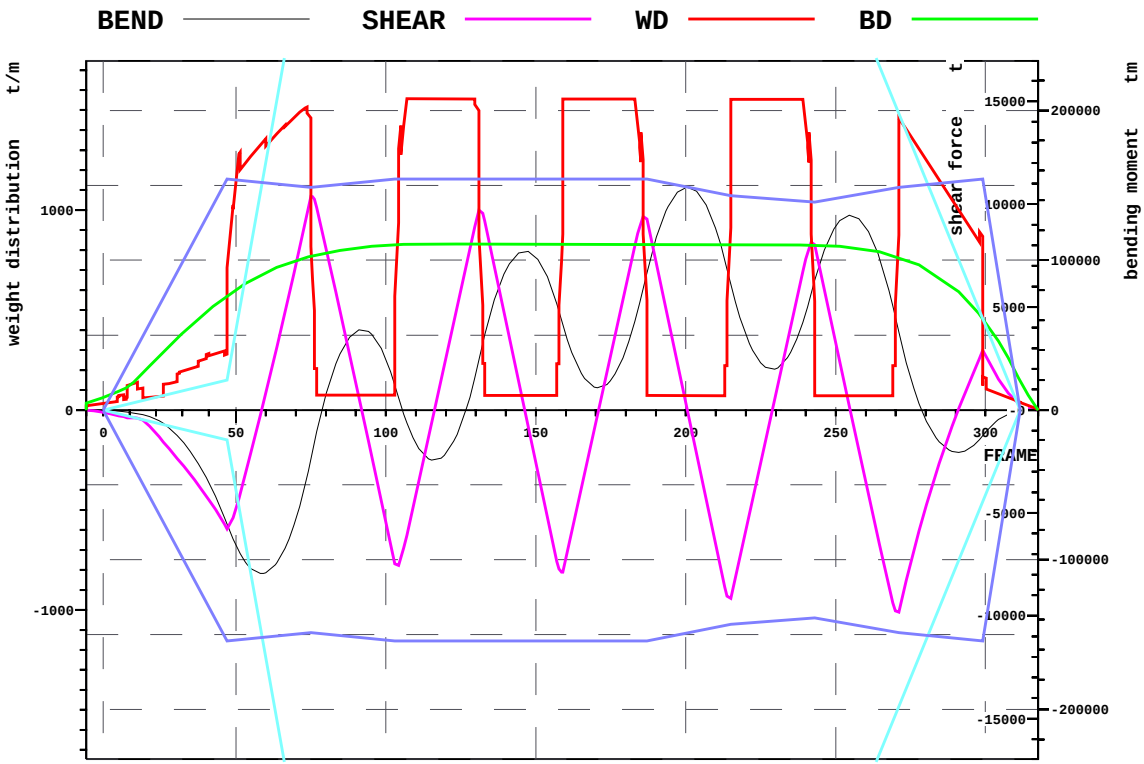
F L O A T I N G P O S I T I O N

Draught moulded	17.954	m	KM	18.71	m
Trim	0.275	m	KG	9.91	m
Heel, PS=+	-0.2	deg			
TA	18.091	m	GM0	8.80	m
TF	17.817	m	GMCORR	0.00	m
Trimming moment	-68081	tonm	GM	8.80	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.74	0.031	273.1	0.001
10.0	1.52	0.130	572.7	0.003
20.0	3.11	0.536	1197.1	0.006
30.0	4.17	1.181	1623.0	0.008
40.0	4.80	1.967	1920.4	0.010
45.0	5.00	2.396	2017.0	0.010
50.0	5.09	2.837	2076.3	0.011
55.0	5.08	3.281	2095.0	0.011
60.0	4.99	3.721	2072.0	0.011
65.0	4.82	4.149	2014.9	0.010
70.0	4.59	4.560	1930.4	0.010
75.0	4.31	4.949	1823.1	0.009
80.0	3.99	5.312	1696.6	0.009
85.0	3.62	5.644	1553.9	0.008
90.0	3.23	5.943	1399.3	0.007



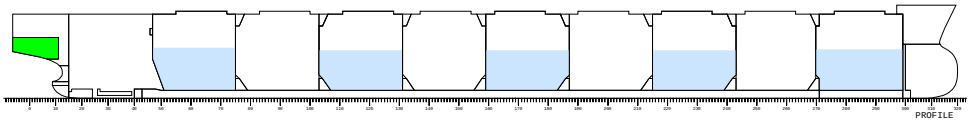
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.181	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.968	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.786	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.097	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.941	deg	OK
GM0.15	GM > 0.15 m	0.150	8.804	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.424		OK

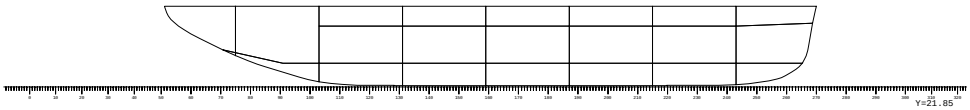


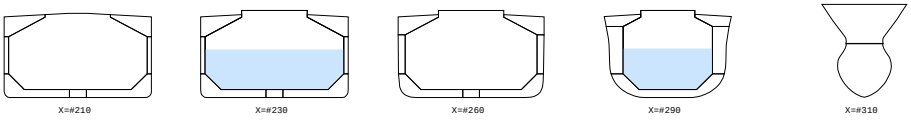
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-42	-112	32.90	62.76
8.80	11.00	-1976	-373	130.43	133.03
12.00	15.00	-3144	-492	59.75	185.14
37.60	47.00	-76388	-5771	712.07	568.30
41.13	51.00	-94429	-4314	1280.43	610.36
62.97	75.00	-33060	10375	816.53	768.07
88.45	103.00	16921	-7474	568.89	825.27
113.93	131.00	32594	9716	1498.11	830.54
139.41	159.00	57024	-7884	880.13	829.46
164.89	187.00	91582	9268	548.41	828.30
190.37	215.00	84052	-9156	1554.19	827.11
215.85	243.00	86620	8033	543.10	823.37
241.33	271.00	52011	-9831	874.75	759.84
266.81	299.00	-17089	2896	867.95	455.38
278.01	313.00	-809	284	40.18	150.66

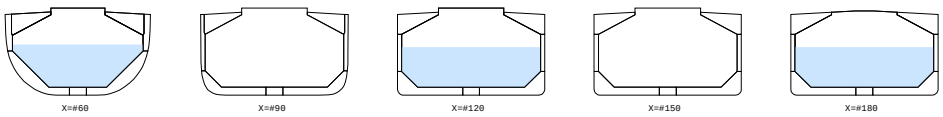
				X	FRAME
SHEAR FORCE (MIN)	-9831.0 t	(89.2%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	10374.8 t	(95.9%)		63.0 m	75
SAGGING MOMENT	-109179 tm	(46.3%)		47.7 m	58
HOGGING MOMENT	148927.5 tm	(31.5%)		177.1 m	200

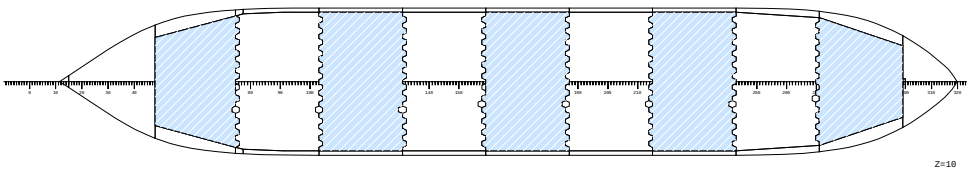
LOADING CONDITION - COND22 : FULLY LOADED ORE (SCANTLING DRAUGHT)-ARRIVAL BUN

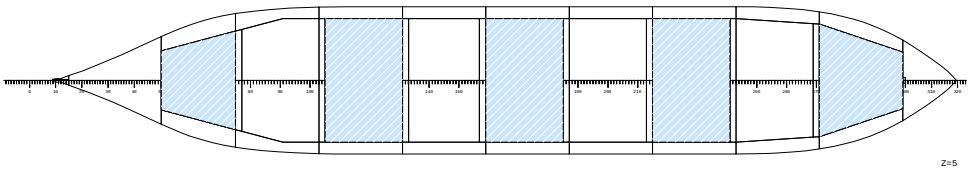


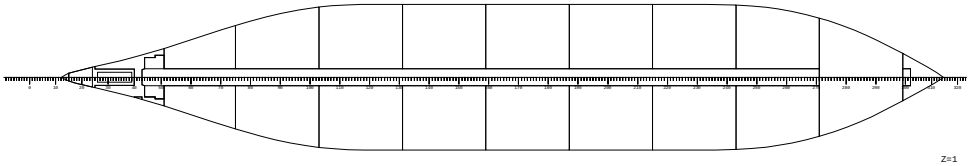












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	29913.2	55.0	51.66	0.03	9.94	0.00
SUBTOTAL		167774.1		150.66	0.02	9.12	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

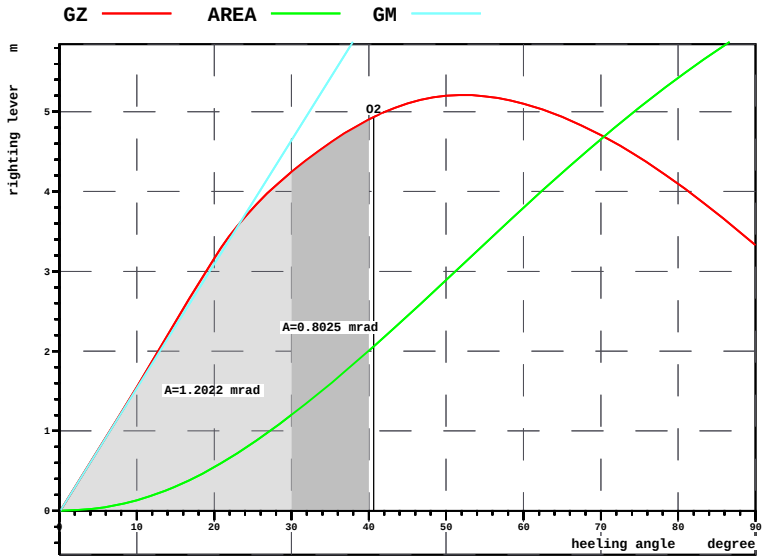
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	170264.4	148.66	-0.02 9.25
Total weight	194906.3	146.55	-0.02 9.81

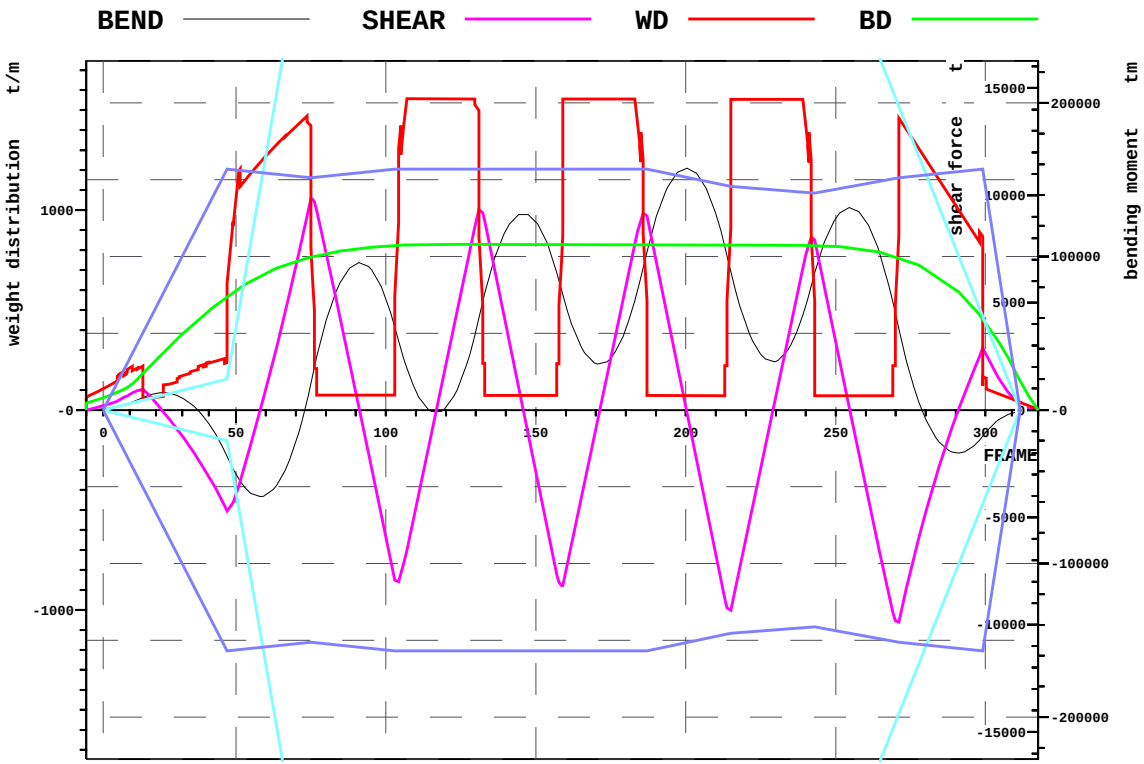
F L O A T I N G P O S I T I O N

Draught moulded	17.892	m	KM	18.71	m
Trim	0.230	m	KG	9.81	m
Heel, PS=+	-0.1	deg			
TA	18.007	m	GM0	8.90	m
TF	17.777	m	GMCORR	-0.00	m
Trimming moment	-56815	tonm	GM	8.89	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.76	0.033	64.4	0.000
10.0	1.55	0.133	128.6	0.001
20.0	3.16	0.546	221.7	0.001
30.0	4.25	1.202	296.1	0.002
40.0	4.90	2.005	401.9	0.002
45.0	5.11	2.442	456.9	0.002
50.0	5.20	2.893	503.3	0.003
55.0	5.19	3.347	540.1	0.003
60.0	5.10	3.797	568.2	0.003
65.0	4.94	4.235	588.6	0.003
70.0	4.71	4.656	603.4	0.003
75.0	4.43	5.055	614.6	0.003
80.0	4.10	5.427	620.8	0.003
85.0	3.73	5.769	620.0	0.003
90.0	3.33	6.077	613.4	0.003



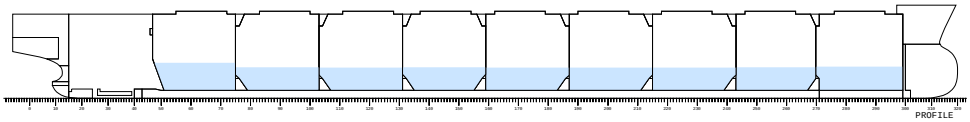
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.202	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.005	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.803	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.210	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	52.062	deg	OK
GM0.15	GM > 0.15 m	0.150	8.895	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.481		OK

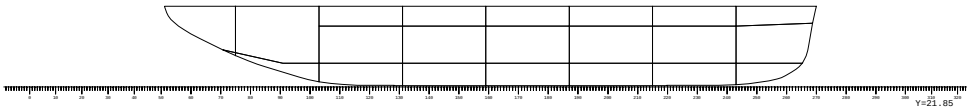


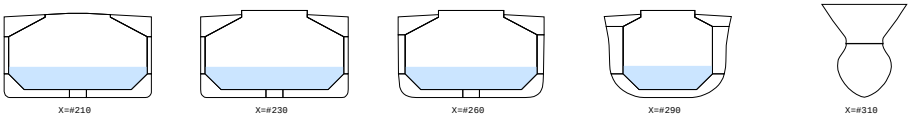
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	717	197	106.32	61.01
8.80	11.00	5110	797	218.36	130.59
12.00	15.00	8089	964	57.01	182.51
37.60	47.00	-30336	-4703	630.68	564.87
41.13	51.00	-45104	-35251197.09		606.89
62.97	75.00	17023	98341422.56		764.62
88.45	103.00	54269	-7930	568.89	822.00
113.93	131.00	59320	93401498.09		827.47
139.41	159.00	75114	-8184	880.12	826.59
164.89	187.00	102891	9038	548.40	825.63
190.37	215.00	90308	-93211554.17		824.63
215.85	243.00	89419	7929	543.09	821.10
241.33	271.00	52815	-9881	874.74	757.86
266.81	299.00	-17024	2888	867.94	454.22
278.01	313.00	-800	283	40.18	150.53

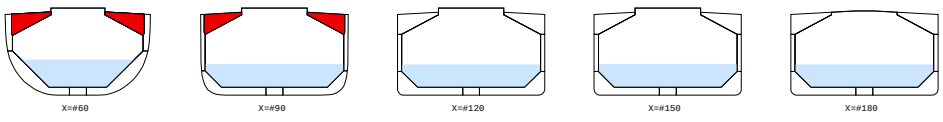
				X	FRAME
SHEAR FORCE (MIN)	-9880.9 t	(89.7%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	9833.5 t	(90.9%)		63.0 m	75
SAGGING MOMENT	-56438.2 tm	(23.9%)		47.7 m	58
HOGGING MOMENT	157612.7 tm	(33.4%)		177.1 m	200

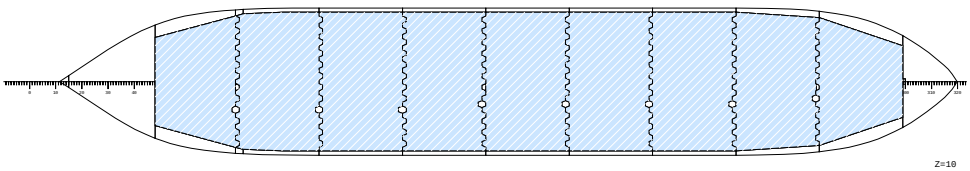
LOADING CONDITION - COND23 : HOMOGENEOUS ORE (SCANTLING DRAUGHT)-DEPAR. BUN.

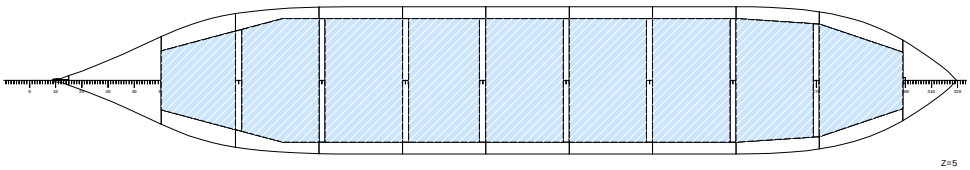


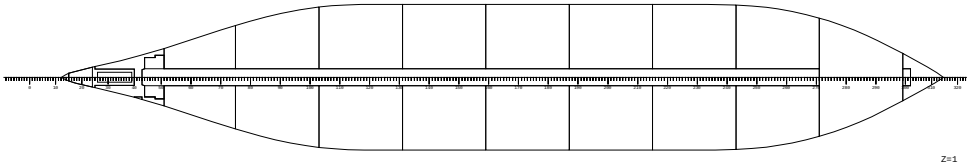












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.2	NO.2 CARGO HOLD	19099.3	30.0	227.55	0.01	6.22	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.8	NO.8 CARGO HOLD	19407.5	30.0	76.84	0.02	6.27	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		167600.3		151.32	0.01	6.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

CONTENTS= (RHO=1)

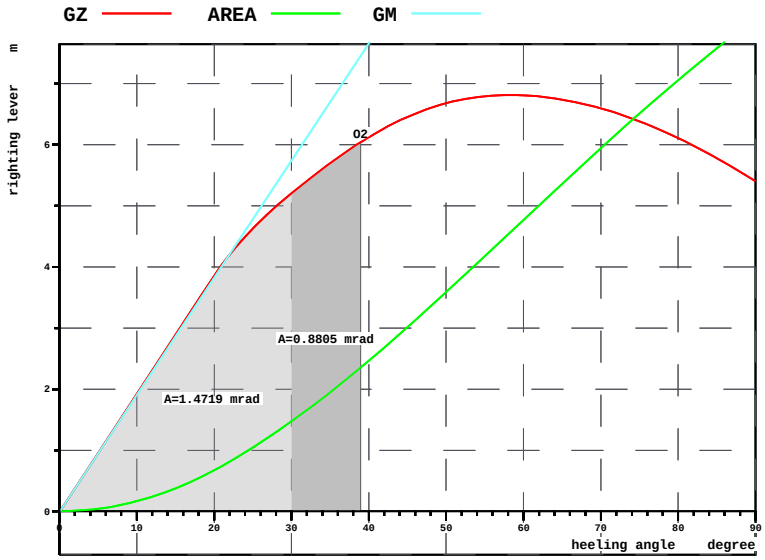
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	173785.5	147.73	-0.01 6.91
Total weight	198427.4	145.78	-0.01 7.75

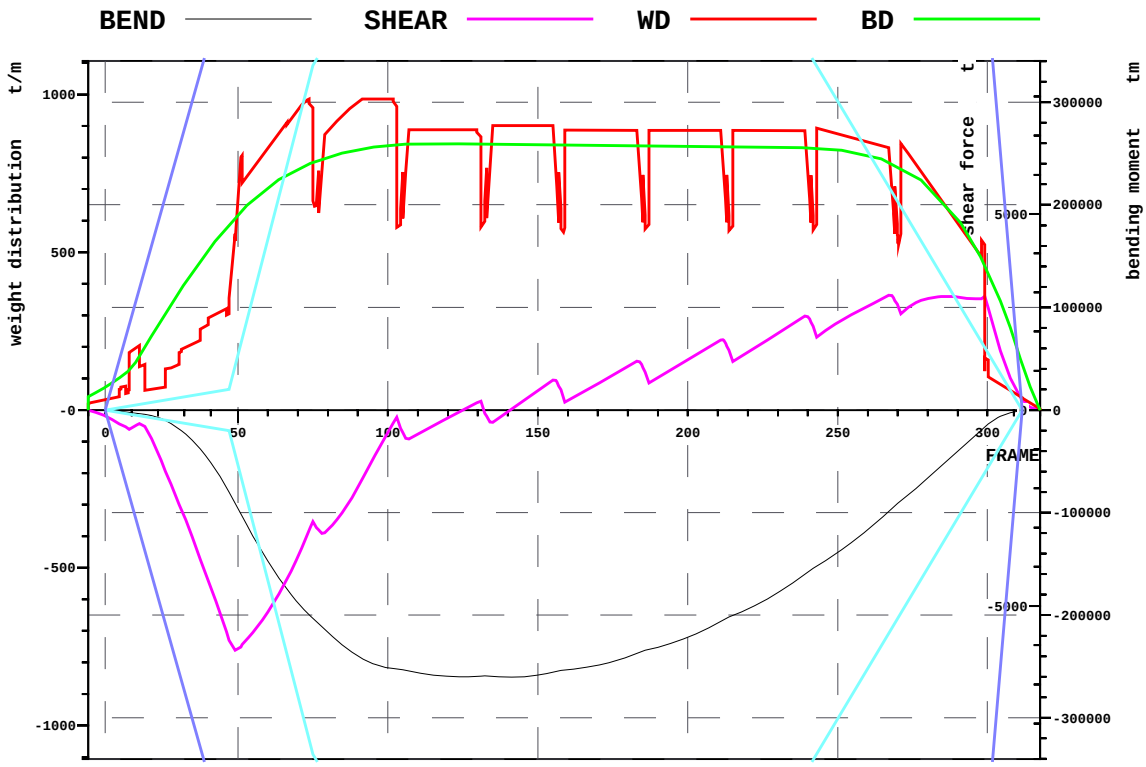
F L O A T I N G P O S I T I O N

Draught moulded	18.183	m	KM	18.73	m
Trim	0.710	m	KG	7.75	m
Heel, PS=+	-0.1	deg			
TA	18.538	m	GM0	10.98	m
TF	17.828	m	GMCORR	0.00	m
Trimming moment	-176768	tonm	GM	10.98	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.95	0.041	0.0	0.000
10.0	1.92	0.166	0.0	0.000
20.0	3.85	0.670	0.0	0.000
30.0	5.20	1.472	0.0	0.000
40.0	6.12	2.464	0.0	0.000
45.0	6.45	3.013	0.0	0.000
50.0	6.68	3.587	0.0	0.000
55.0	6.79	4.175	0.0	0.000
60.0	6.81	4.769	0.0	0.000
65.0	6.74	5.361	0.0	0.000
70.0	6.59	5.943	0.0	0.000
75.0	6.38	6.510	0.0	0.000
80.0	6.11	7.055	0.0	0.000
85.0	5.78	7.574	0.0	0.000
90.0	5.40	8.062	0.0	0.000



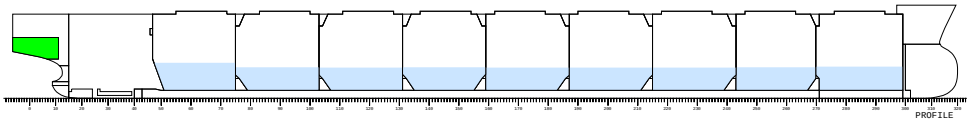
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.472	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.352	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.880	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.812	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	58.416	deg	OK
GM0.15	GM > 0.15 m	0.150	10.983	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.507		OK

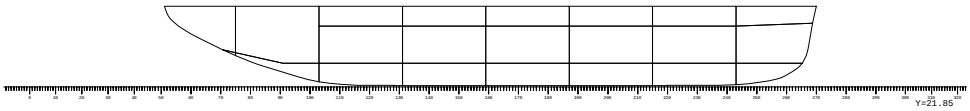


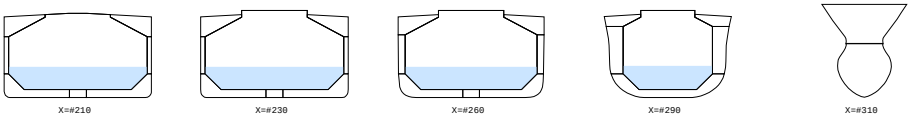
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-299	-155	32.89	72.08
8.80	11.00	-3182	-416	192.87	145.70
12.00	15.00	-4357	-421	62.46	198.76
37.60	47.00	-80111	-5880	354.76	585.11
41.13	51.00	-101404	-6045	800.92	627.22
62.97	75.00	-202928	-2850	810.94	784.08
88.45	103.00	-252326	-188	578.50	839.52
113.93	131.00	-259342	221	865.47	842.94
139.41	159.00	-253278	196	577.14	840.01
164.89	187.00	-233378	689	588.00	837.00
190.37	215.00	-200188	1235	885.74	833.96
215.85	243.00	-152213	1860	587.73	828.37
241.33	271.00	-88667	2456	558.24	762.87
266.81	299.00	-16633	2900	524.16	456.20
278.01	313.00	-651	284	40.16	150.70

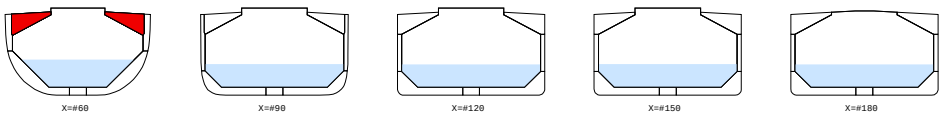
				X	FRAME
SHEAR FORCE (MIN)	-6133.3 t	(54.9%)	POSITION:	39.4 m	49
SHEAR FORCE (MAX)	2927.2 t	(27.3%)		237.7 m	267
SAGGING MOMENT	-260536 tm	(59.4%)		122.3 m	140
HOGGING MOMENT	2.7 tm			-5.0 m	-6

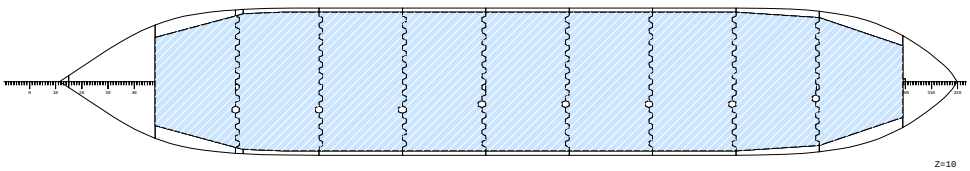
LOADING CONDITION - COND24 : HOMOGENEOUS ORE (SCANTLING DRAUGHT)-50% BUNKERS

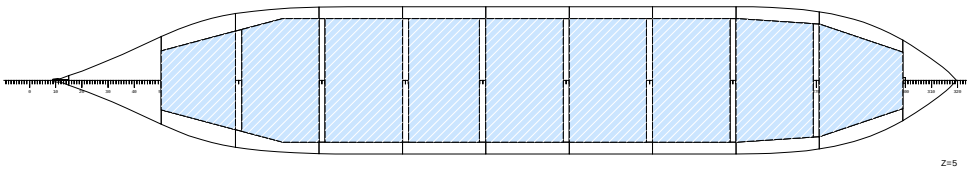


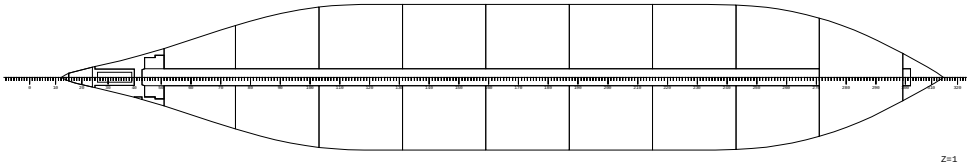












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.2	NO.2 CARGO HOLD	19099.3	30.0	227.55	0.01	6.22	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.8	NO.8 CARGO HOLD	19407.5	30.0	76.84	0.02	6.27	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		167600.3		151.32	0.01	6.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

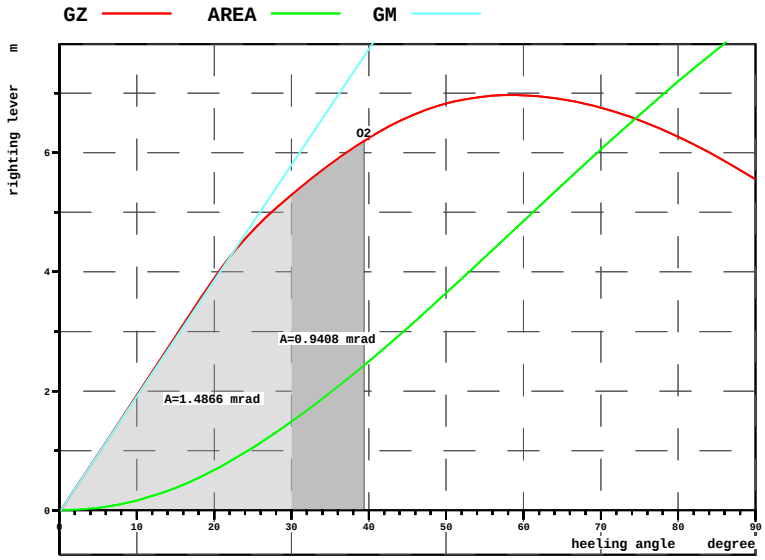
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	172459.5	147.83	-0.03 6.72
Total weight	197101.4	145.85	-0.03 7.59

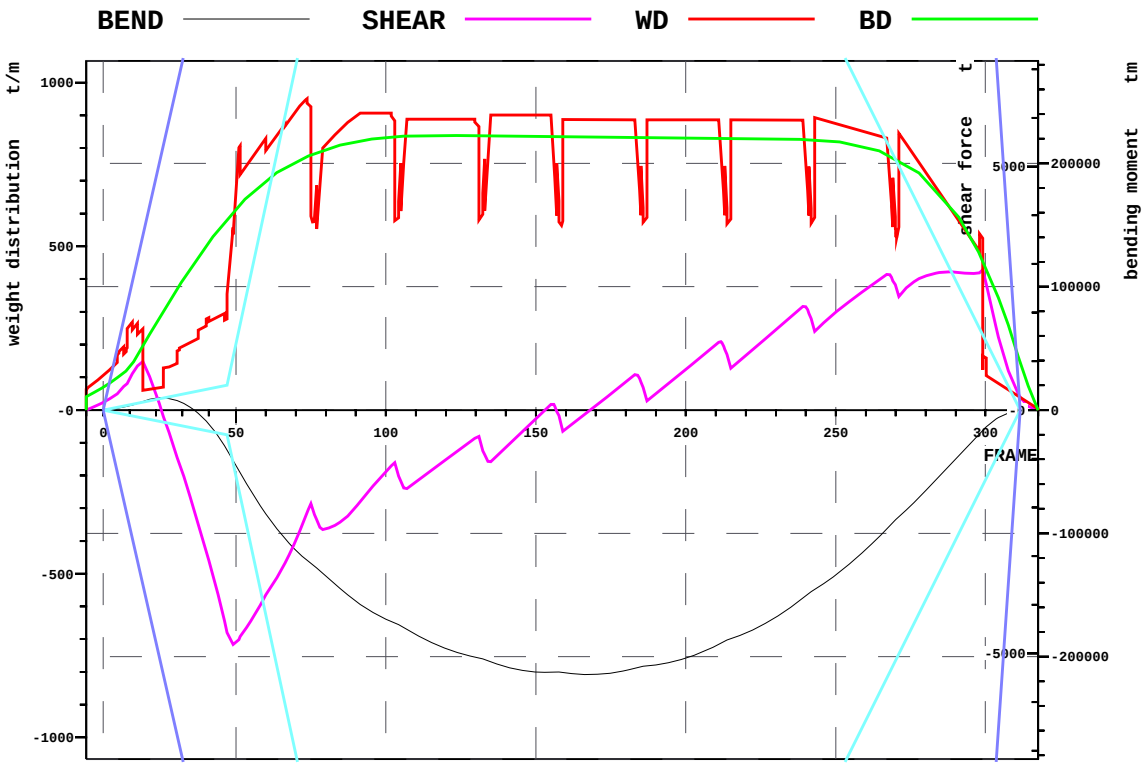
F L O A T I N G P O S I T I O N

Draught moulded	18.072	m	KM	18.73	m
Trim	0.698	m	KG	7.59	m
Heel, PS=+	-0.1	deg			
TA	18.421	m	GM0	11.14	m
TF	17.723	m	GMCORR	-0.02	m
Trimming moment	-173414	tonm	GM	11.12	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.94	0.040	273.1	0.001
10.0	1.92	0.165	572.7	0.003
20.0	3.89	0.673	1197.1	0.006
30.0	5.29	1.487	1623.0	0.008
40.0	6.25	2.498	1920.4	0.010
45.0	6.59	3.059	2017.0	0.010
50.0	6.82	3.645	2076.3	0.011
55.0	6.94	4.246	2095.0	0.011
60.0	6.96	4.854	2072.0	0.011
65.0	6.90	5.459	2014.9	0.010
70.0	6.75	6.055	1930.4	0.010
75.0	6.54	6.636	1823.1	0.009
80.0	6.27	7.195	1696.6	0.009
85.0	5.94	7.728	1553.9	0.008
90.0	5.56	8.230	1399.3	0.007



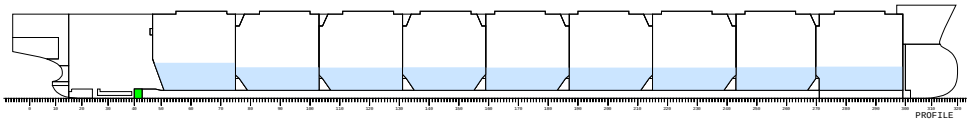
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.487	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.427	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.941	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.967	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	58.622	deg	OK
GM0.15	GM > 0.15 m	0.150	11.124	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.649		OK

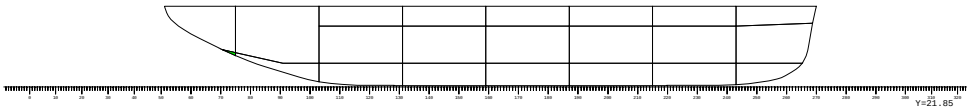


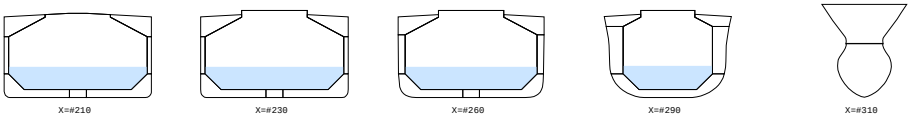
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	469	158	106.29	69.61
8.80	11.00	3951	745	268.16	142.27
12.00	15.00	6842	998	59.72	195.04
37.60	47.00	-32619	-4570	354.44	580.11
41.13	51.00	-49267	-4719	800.59	622.14
62.97	75.00	-124209	-1925	759.09	778.84
88.45	103.00	-172836	-1084	578.49	834.30
113.93	131.00	-201021	-542	865.45	837.77
139.41	159.00	-212781	-436	577.13	834.89
164.89	187.00	-207395	186	587.99	831.94
190.37	215.00	-185440	860	885.72	828.95
215.85	243.00	-145458	1611	587.72	823.42
241.33	271.00	-86707	2331	558.23	758.19
266.81	299.00	-16503	2879	524.15	453.21
278.01	313.00	-639	283	40.16	150.33

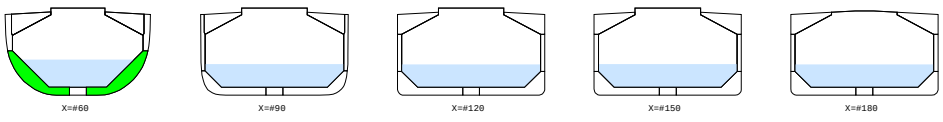
				X	FRAME
SHEAR FORCE (MIN)	-4815.4 t	(43.1%)	POSITION:	39.4 m	49
SHEAR FORCE (MAX)	2879.0 t	(25.6%)		266.8 m	299
SAGGING MOMENT	-214431 tm	(48.9%)		148.9 m	169
HOGGING MOMENT	9860.0 tm	(101.9%)		18.2 m	23

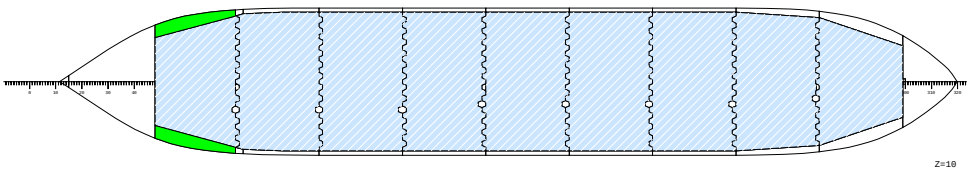
LOADING CONDITION - COND25 : HOMOGENEOUS ORE (SCANTLING DRAUGHT)-ARRIVAL BUNK

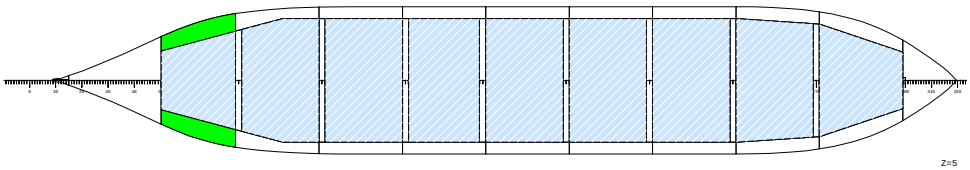


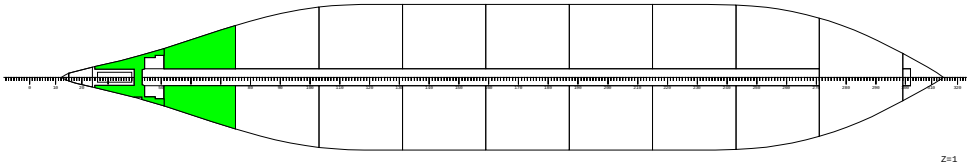












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.2	NO.2 CARGO HOLD	19099.3	30.0	227.55	0.01	6.22	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.8	NO.8 CARGO HOLD	19407.5	30.0	76.84	0.02	6.27	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		167600.3		151.32	0.01	6.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61

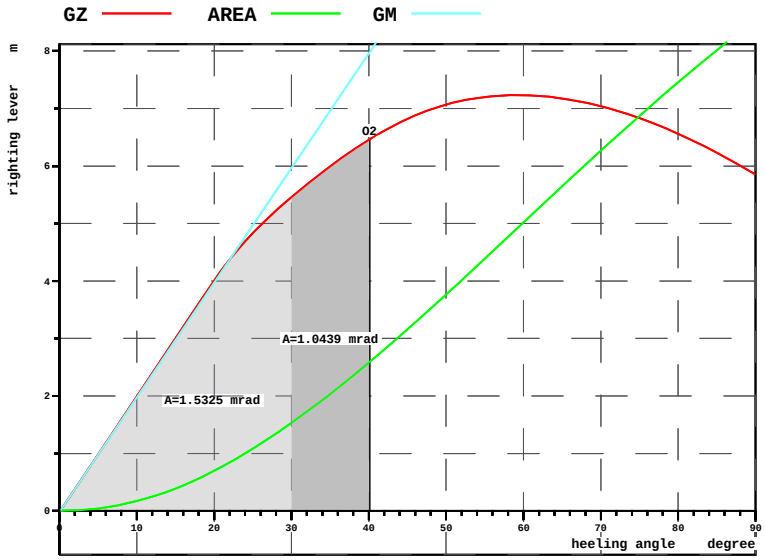
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		3614.8		49.58	0.01	5.02	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	172083.1	148.57	-0.02 6.38
Total weight	196725.0	146.50	-0.02 7.29

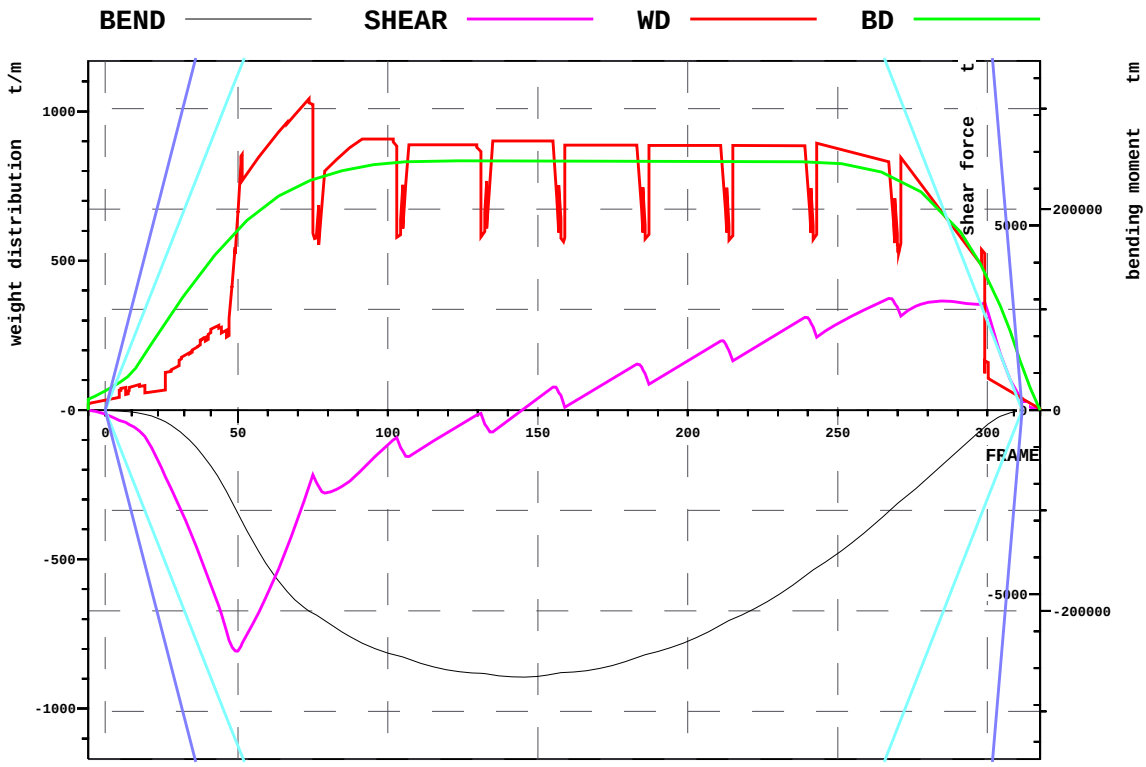
F L O A T I N G P O S I T I O N

Draught moulded	18.044	m	KM	18.72 m
Trim	0.201	m	KG	7.29 m
Heel, PS=+	-0.1	deg		
TA	18.145	m	GM0	11.42 m
TF	17.944	m	GMCORR	0.00 m
Trimming moment	-49870	tonm	GM	11.42 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.98	0.042	64.4	0.000
10.0	1.99	0.171	128.6	0.001
20.0	4.01	0.695	221.7	0.001
30.0	5.46	1.532	296.1	0.002
40.0	6.46	2.576	401.9	0.002
45.0	6.82	3.156	456.9	0.002
50.0	7.07	3.763	503.3	0.003
55.0	7.20	4.387	540.1	0.003
60.0	7.23	5.017	568.2	0.003
65.0	7.18	5.647	588.6	0.003
70.0	7.04	6.267	603.4	0.003
75.0	6.83	6.873	614.6	0.003
80.0	6.56	7.458	620.8	0.003
85.0	6.23	8.017	620.0	0.003
90.0	5.85	8.544	613.4	0.003



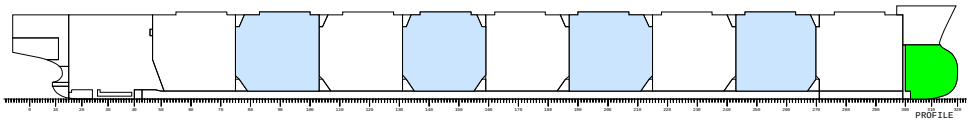
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.532	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.576	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.044	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.235	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	59.234	deg	OK
GM0.15	GM > 0.15 m	0.150	11.424	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.843		OK

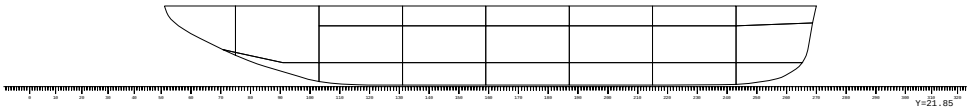


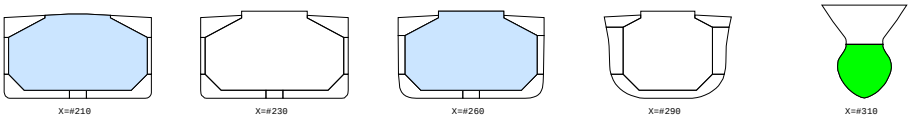
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-182	-117	32.89	63.85
8.80	11.00	-2483	-467	80.53	134.64
12.00	15.00	-4271	-718	57.00	186.92
37.60	47.00	-86045	-6248	307.22	571.04
41.13	51.00	-108769	-6410	849.71	613.18
62.97	75.00	-202157	-1755	807.58	771.29
88.45	103.00	-244078	-747	578.52	828.82
113.93	131.00	-261976	-91	865.49	834.40
139.41	159.00	-261236	74	577.16	833.63
164.89	187.00	-242506	702	588.01	832.78
190.37	215.00	-207733	1329	885.76	831.90
215.85	243.00	-156826	1979	587.75	828.48
241.33	271.00	-90397	2546	558.26	765.02
266.81	299.00	-16802	2922	524.17	458.91
278.01	313.00	-674	284	40.16	151.13

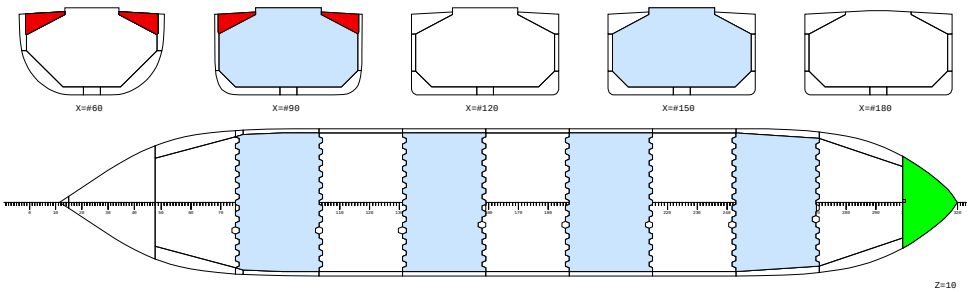
				X	FRAME
SHEAR FORCE (MIN)	-6539.1 t	(49.6%)	POSITION:	40.1 m	50
SHEAR FORCE (MAX)	3024.7 t	(22.3%)		237.7 m	267
SAGGING MOMENT	-265977 tm	(36.7%)		126.1 m	144
HOGGING MOMENT	3.6 tm			-5.0 m	-6

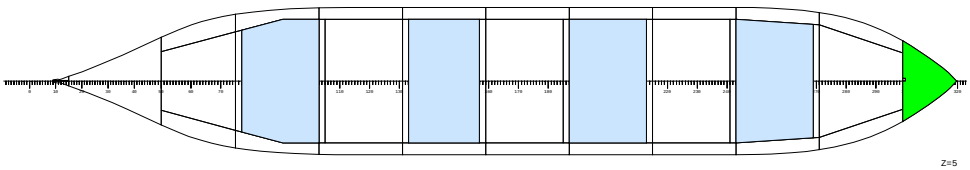
LOADING CONDITION - COND26 : MULTIPLE PORT-FIRST PORT-DEP (SECOND SEE COND14)

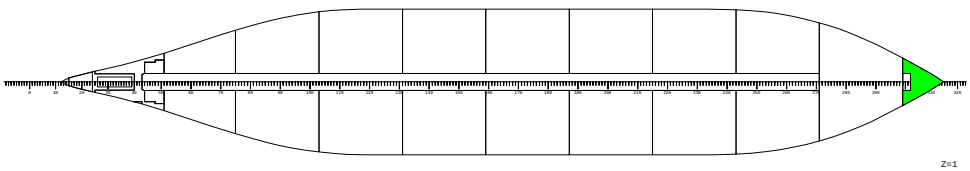












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
SUBTOTAL		72876.0		152.01	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00

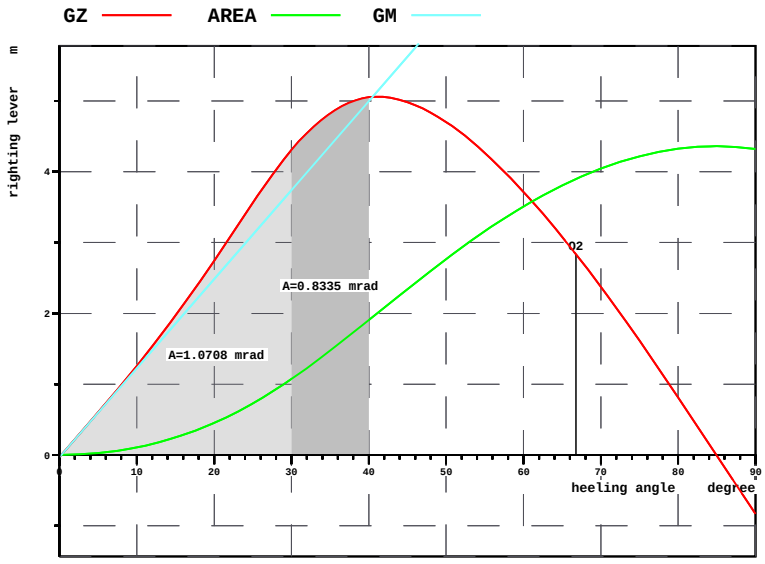
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

Lightweight	24641.9	132.01	0.00	13.67
Deadweight	82624.2	149.60	-0.04	14.14
Total weight	107266.0	145.56	-0.03	14.03

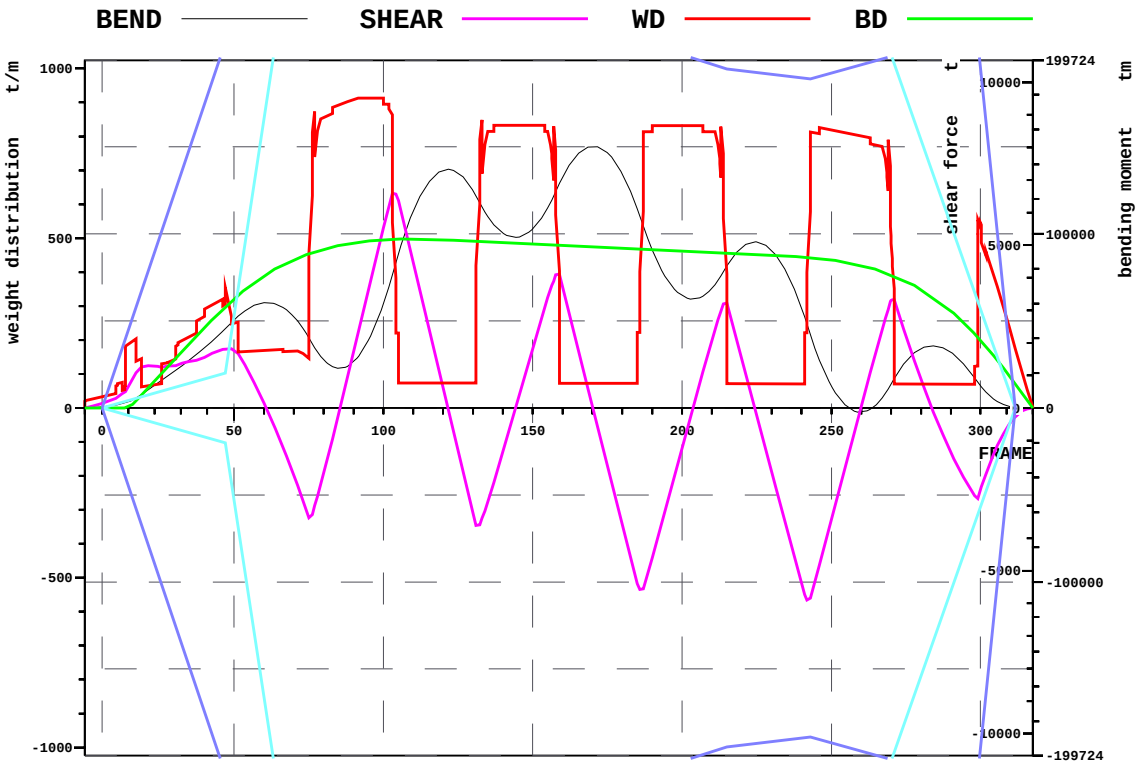
F L O A T I N G P O S I T I O N

Draught moulded	10.386	m	KM	21.24 m
Trim	2.799	m	KG	14.03 m
Heel, PS=+	-0.2	deg		
TA	11.786	m	GM0	7.20 m
TF	8.987	m	GMCORR	0.00 m
Trimming moment	-575311	tonm	GM	7.20 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.60	0.025	0.0	0.000
10.0	1.26	0.106	0.0	0.000
20.0	2.74	0.451	0.0	0.000
30.0	4.31	1.071	0.0	0.000
40.0	5.05	1.904	0.0	0.000
45.0	4.98	2.344	0.0	0.000
50.0	4.70	2.768	0.0	0.000
55.0	4.27	3.160	0.0	0.000
60.0	3.72	3.510	0.0	0.000
65.0	3.08	3.807	0.0	0.000
70.0	2.38	4.045	0.0	0.000
75.0	1.62	4.219	0.0	0.000
80.0	0.81	4.326	0.0	0.000
85.0	-0.01	4.361	0.0	0.000
90.0	-0.83	4.324	0.0	0.000



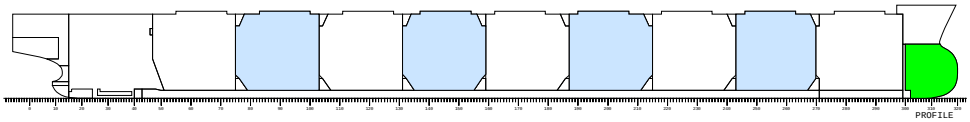
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.071	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.904	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.833	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.061	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	41.168	deg	OK
GM0.15	GM > 0.15 m	0.150	7.201	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.088		OK

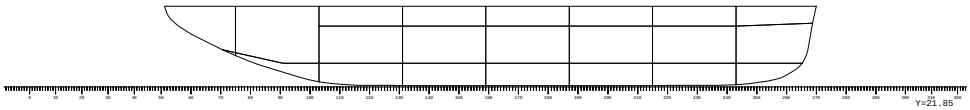


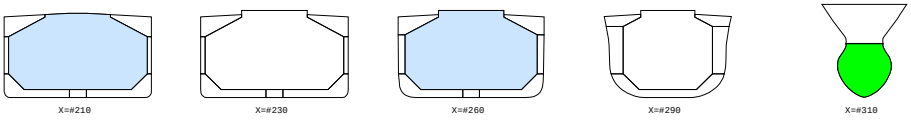
X	FR	BEND	SHEAR	WD	BD
m	#	tm	t	t/m	t/m
0.00	0.00	596	142	32.90	0.00
8.80	11.00	4135	789	192.91	8.25
12.00	15.00	7659	1263	62.47	37.19
37.60	47.00	45626	1807	354.83	296.07
41.13	51.00	52078	1698	251.56	328.15
62.97	75.00	40324	-3370	299.00	453.59
88.45	103.00	76581	6601	546.19	496.67
113.93	131.00	121955	-3611	73.37	491.20
139.41	159.00	129195	4098	403.81	479.42
164.89	187.00	104432	-5550	578.93	467.55
190.37	215.00	81590	3219	71.73	455.66
215.85	243.00	41847	-5843	577.83	441.65
241.33	271.00	16817	3320	352.08	386.33
266.81	299.00	14784	-2780	123.01	207.97
278.01	313.00	313	-295	182.30	73.47

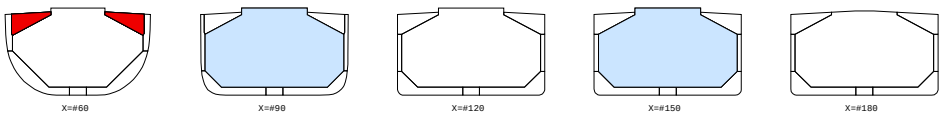
				X	FRAME
SHEAR FORCE (MIN)	-5893.9 t	(56.6%)	POSITION:	214.8 m	242
SHEAR FORCE (MAX)	6601.0 t	(58.8%)		88.4 m	103
SAGGING MOMENT	-2327.9 tm	(1.0%)		230.2 m	259
HOGGING MOMENT	150049.0 tm	(31.8%)		151.0 m	172

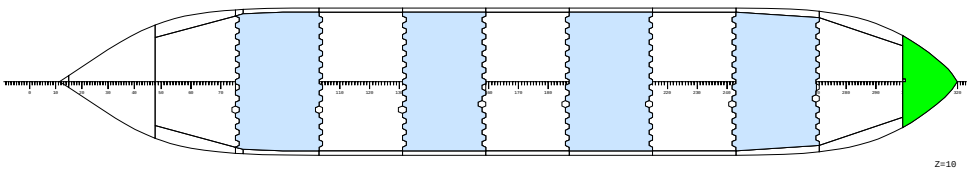
LOADING CONDITION - COND27 : MULTIPLE PORT-FIRST PORT-50% (SECOND SEE COND15)

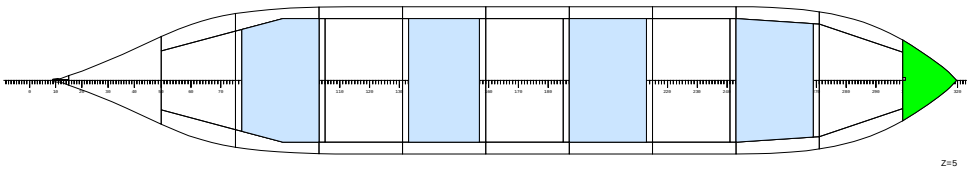


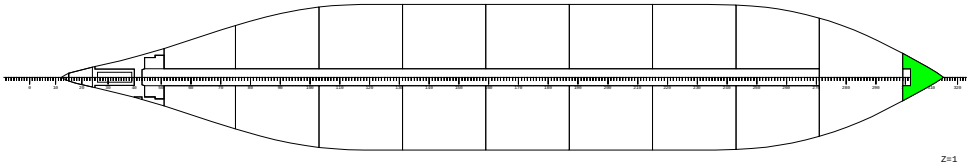












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
SUBTOTAL		72876.0		152.01	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

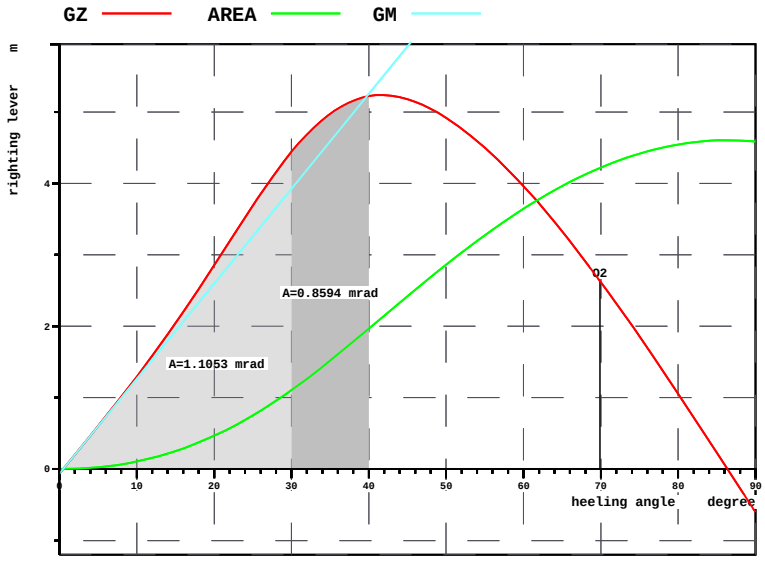
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	79675.9	152.80	-0.08 13.81
Total weight	104317.7	147.89	-0.06 13.77

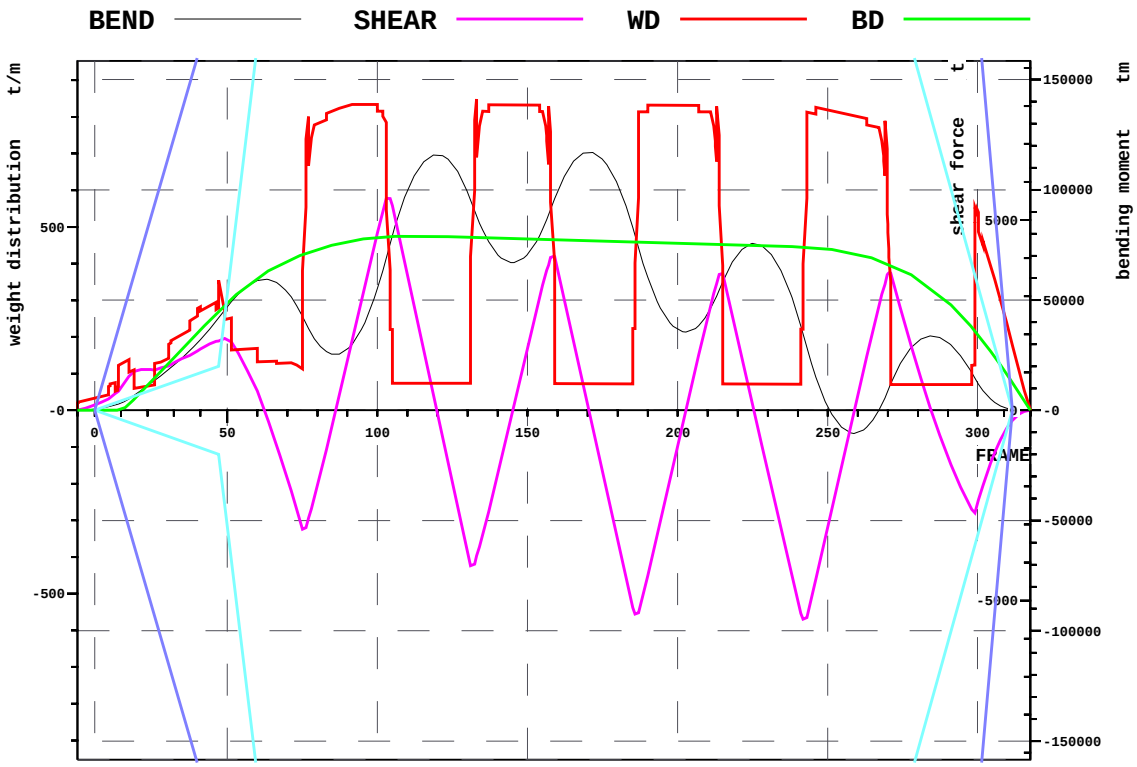
F L O A T I N G P O S I T I O N

Draught moulded	10.086	m	KM	21.39 m
Trim	1.562	m	KG	13.77 m
Heel, PS=+	-0.5	deg		
TA	10.867	m	GM0	7.61 m
TF	9.305	m	GMCORR	0.00 m
Trimming moment	-312510	tonm	GM	7.61 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.60	0.023	273.1	0.003
10.0	1.29	0.106	572.7	0.005
20.0	2.85	0.464	1197.1	0.011
30.0	4.44	1.105	1623.0	0.016
40.0	5.22	1.964	1920.4	0.018
45.0	5.18	2.420	2017.0	0.019
50.0	4.92	2.862	2076.3	0.020
55.0	4.50	3.274	2095.0	0.020
60.0	3.96	3.644	2072.0	0.020
65.0	3.33	3.962	2014.9	0.019
70.0	2.62	4.222	1930.4	0.019
75.0	1.86	4.418	1823.1	0.017
80.0	1.06	4.545	1696.6	0.016
85.0	0.23	4.601	1553.9	0.015
90.0	-0.60	4.585	1399.3	0.013



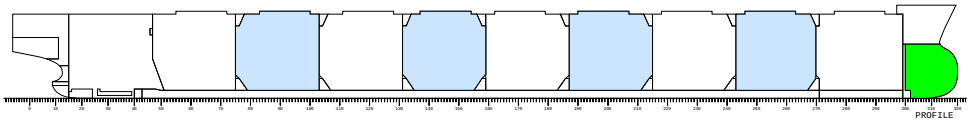
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.105	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.965	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.859	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.236	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	41.637	deg	OK
GM0.15	GM > 0.15 m	0.150	7.614	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.915		OK

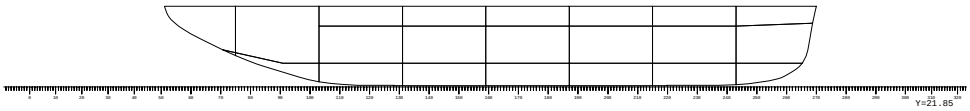


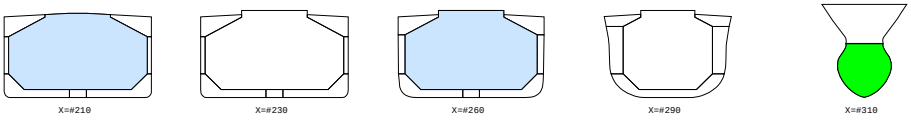
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	535	142	32.90	0.00
8.80	11.00	3894	694	130.45	5.88
12.00	15.00	6808	1026	59.74	31.07
37.60	47.00	41970	1826	354.51	268.64
41.13	51.00	48624	1814	251.24	299.54
62.97	75.00	42597	-3125	113.05	424.66
88.45	103.00	69263	5574	546.19	472.34
113.93	131.00	95491	-4085	73.37	472.12
139.41	159.00	95967	4043	403.80	465.60
164.89	187.00	73407	-5320	578.92	458.99
190.37	215.00	58324	3600	71.72	452.36
215.85	243.00	28479	-5445	577.82	443.60
241.33	271.00	12089	3603	352.07	393.13
266.81	299.00	14465	-2699	123.01	215.63
278.01	313.00	354	-282	182.30	77.55

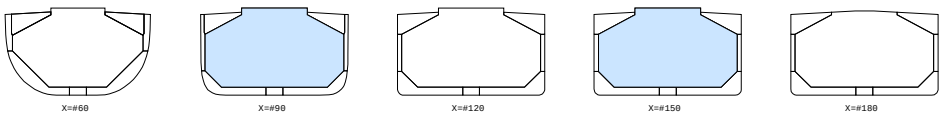
				X	FRAME
SHEAR FORCE (MIN)	-5494.2 t	(52.7%)	POSITION:	214.8 m	242
SHEAR FORCE (MAX)	5574.0 t	(49.7%)		88.4 m	103
SAGGING MOMENT	-10467.9 tm	(4.4%)		230.2 m	259
HOGGING MOMENT	116913.9 tm	(24.7%)		151.0 m	172

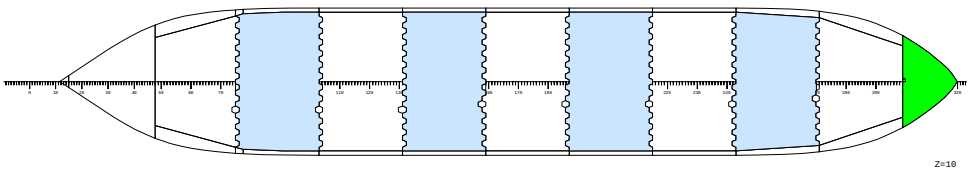
LOADING CONDITION - COND28 : MULTIPLEPORT-FIRSTPORT-ARRIVAL(SECOND SEE COND16

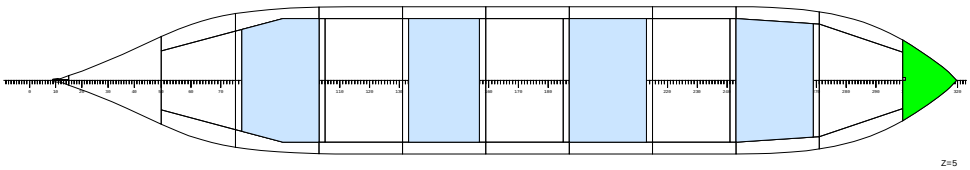


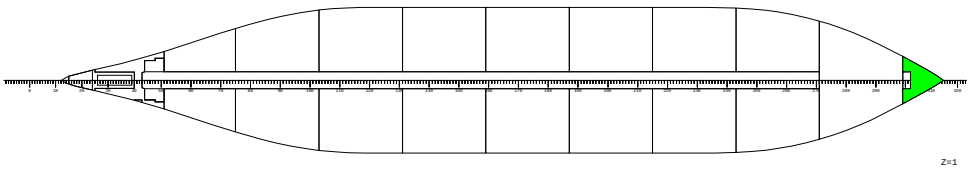












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=0.854)							
R1.2	NO.2 CARGO HOLD	18117.1	100.0	227.71	0.01	13.68	0.00
R1.4	NO.4 CARGO HOLD	18716.7	100.0	176.91	0.02	13.60	0.00
R1.6	NO.6 CARGO HOLD	17632.8	100.0	126.64	0.02	13.65	0.00
R1.8	NO.8 CARGO HOLD	18409.4	100.0	76.50	0.02	13.69	0.00
SUBTOTAL		72876.0		152.01	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

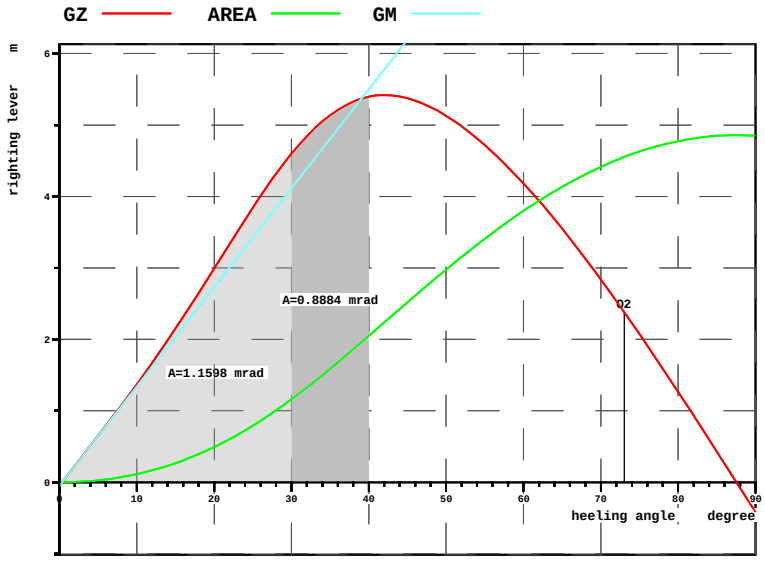
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	77307.0	156.20	-0.06 13.55
Total weight	101948.9	150.35	-0.05 13.58

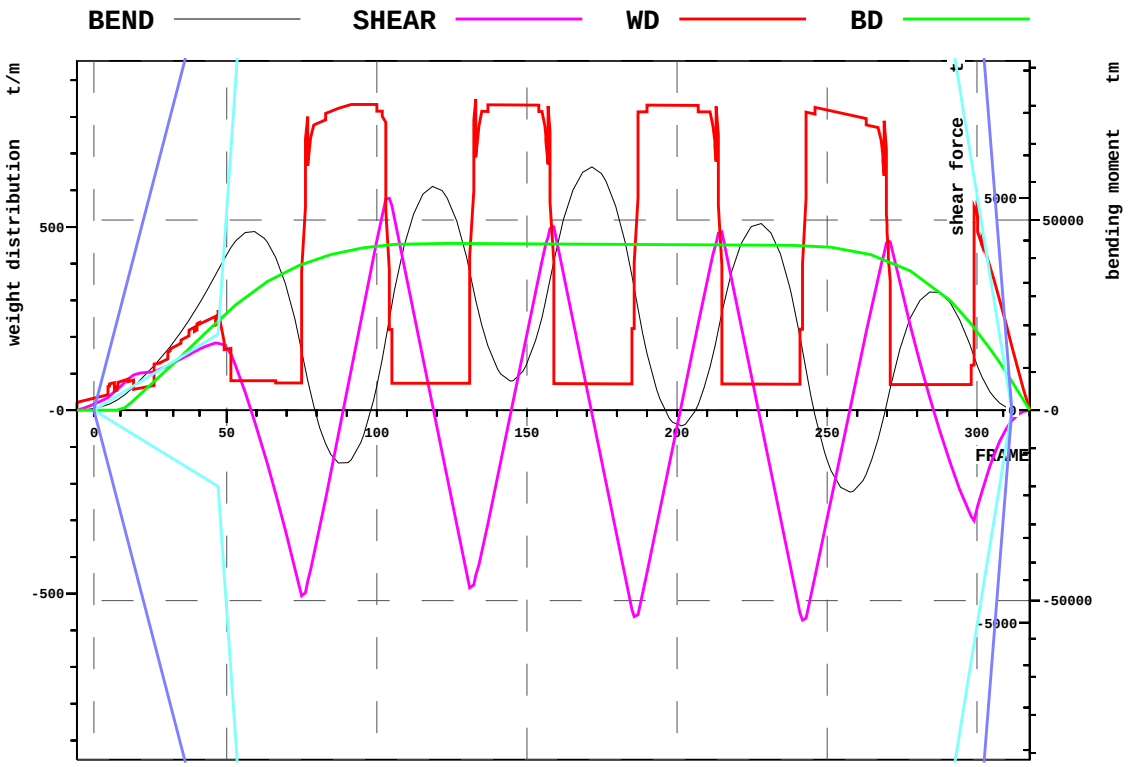
F L O A T I N G P O S I T I O N

Draught moulded	9.831	m	KM	21.52	m
Trim	0.272	m	KG	13.58	m
Heel, PS=+	-0.3	deg			
TA	9.967	m	GM0	7.95	m
TF	9.695	m	GMCORR	0.00	m
Trimming moment	-53333	tonm	GM	7.95	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.65	0.026	64.4	0.001
10.0	1.37	0.114	128.6	0.001
20.0	2.99	0.492	221.7	0.002
30.0	4.60	1.160	296.1	0.003
40.0	5.40	2.048	401.9	0.004
45.0	5.38	2.520	456.9	0.004
50.0	5.13	2.980	503.3	0.005
55.0	4.72	3.410	540.1	0.005
60.0	4.18	3.800	568.2	0.006
65.0	3.55	4.138	588.6	0.006
70.0	2.84	4.417	603.4	0.006
75.0	2.07	4.631	614.6	0.006
80.0	1.26	4.777	620.8	0.006
85.0	0.43	4.851	620.0	0.006
90.0	-0.42	4.851	613.4	0.006



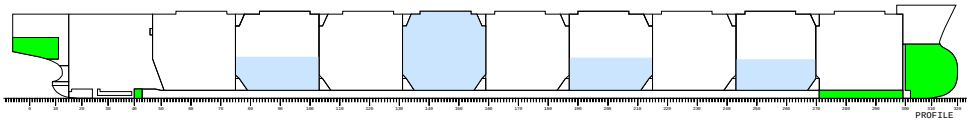
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.160	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.048	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.888	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.420	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	42.031	deg	OK
GM0.15	GM > 0.15 m	0.150	7.946	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.735		OK

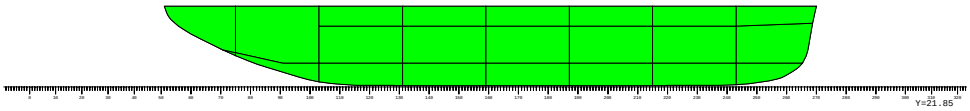


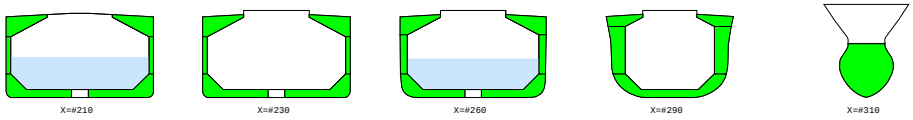
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	494	142	32.90	0.00
8.80	11.00	3723	618	80.54	5.29
12.00	15.00	6143	830	57.00	27.28
37.60	47.00	36825	1578	273.14	243.24
41.13	51.00	42211	1368	167.93	272.78
62.97	75.00	15594	-4369	74.47	397.24
88.45	103.00	18798	4972	546.18	449.66
113.93	131.00	36255	-4179	73.37	454.89
139.41	159.00	39143	4318	403.80	453.83
164.89	187.00	26631	-4814	578.92	452.69
190.37	215.00	25686	4197	71.72	451.51
215.85	243.00	10523	-4897	577.82	448.21
241.33	271.00	5837	3968	352.07	402.63
266.81	299.00	13961	-2598	123.01	225.21
278.01	313.00	366	-267	182.30	82.51

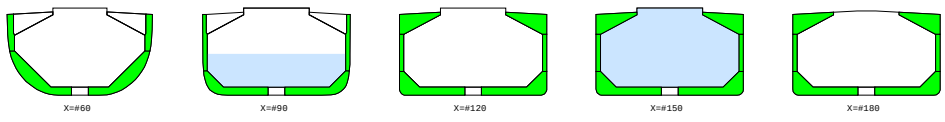
				X	FRAME
SHEAR FORCE (MIN)	-4940.8 t	(47.4%)	POSITION:	214.8 m	242
SHEAR FORCE (MAX)	4987.8 t	(44.4%)		89.6 m	104
SAGGING MOMENT	-21513.0 tm	(8.8%)		229.2 m	258
HOGGING MOMENT	63912.6 tm	(13.5%)		151.0 m	172

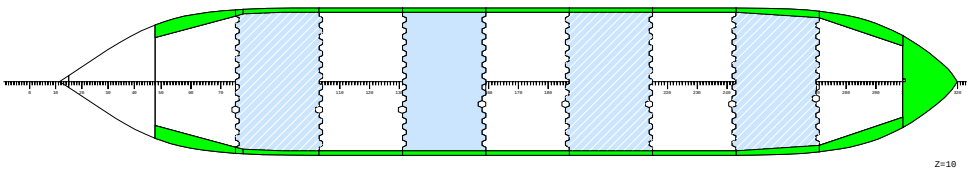
LOADING CONDITION - COND29 : HARBOUR COND FOR DUNKIRK EAST-d>12.95m-ARRIVAL

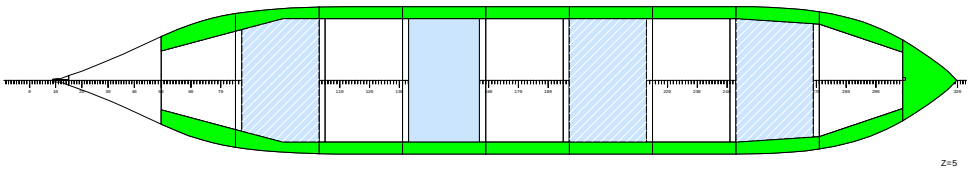


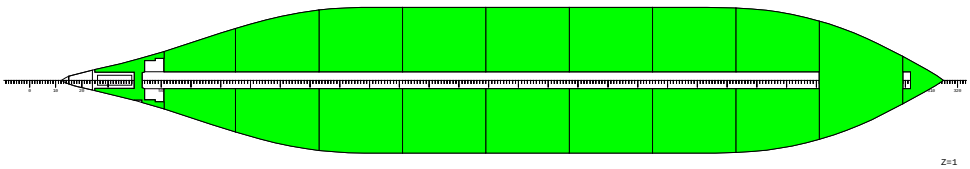












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.2	NO.2 CARGO HOLD	9132.8	42.0	227.62	0.01	7.50	0.00
R1.4	NO.4 CARGO HOLD	10109.0	45.0	176.87	0.02	7.76	0.00
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
R1.8	NO.8 CARGO HOLD	10055.0	45.5	76.65	0.02	7.93	0.00
SUBTOTAL		50460.3		145.02	0.02	10.22	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

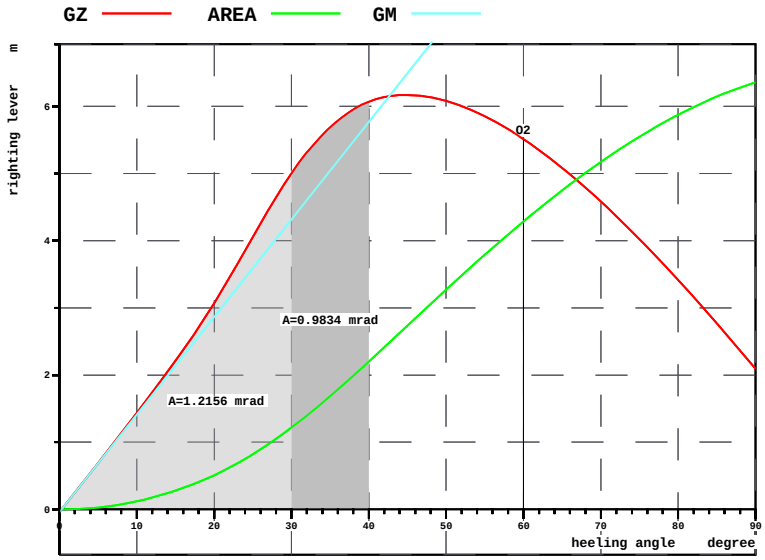
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	112302.6	153.70	-0.05 10.49
Total weight	136944.5	149.79	-0.04 11.06

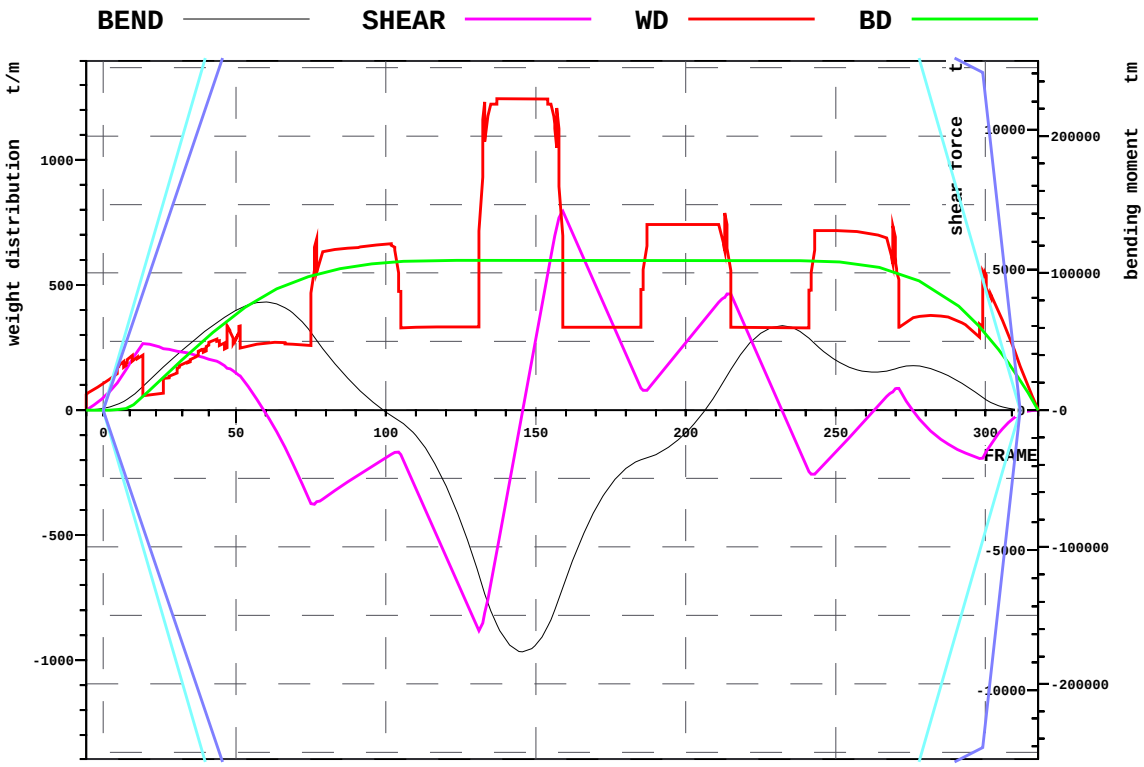
F L O A T I N G P O S I T I O N

Draught moulded	12.945	m	KM	19.38	m
Trim	0.000	m	KG	11.06	m
Heel, PS=+	-0.3	deg			
TA	12.945	m	GM0	8.32	m
TF	12.945	m	GMCORR	-0.01	m
Trimming moment	-66	tonm	GM	8.32	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.69	0.028	64.4	0.000
10.0	1.44	0.121	128.6	0.001
20.0	3.07	0.509	221.7	0.002
30.0	5.00	1.216	296.1	0.002
40.0	6.07	2.199	401.9	0.003
45.0	6.17	2.734	456.9	0.003
50.0	6.08	3.270	503.3	0.004
55.0	5.85	3.791	540.1	0.004
60.0	5.51	4.288	568.2	0.004
65.0	5.08	4.751	588.6	0.004
70.0	4.58	5.173	603.4	0.004
75.0	4.02	5.548	614.6	0.004
80.0	3.41	5.873	620.8	0.005
85.0	2.77	6.143	620.0	0.005
90.0	2.09	6.355	613.4	0.004



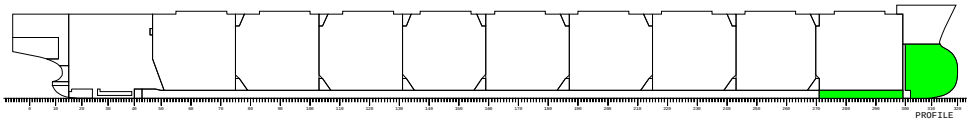
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.216	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.199	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.983	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.167	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.786	deg	OK
GM0.15	GM > 0.15 m	0.150	8.317	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.782		OK

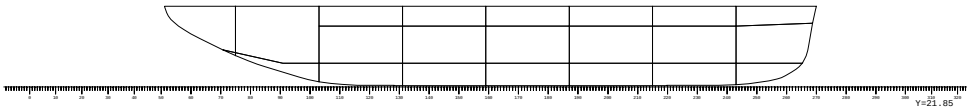


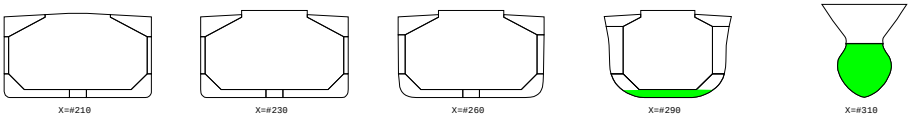
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1123	443	106.37	0.01
8.80	11.00	10369	1810	218.46	18.00
12.00	15.00	17077	2365	57.04	53.31
37.60	47.00	68941	1492	336.86	355.16
41.13	51.00	73850	1235	334.09	391.56
62.97	75.00	56651	-3338	257.83	535.09
88.45	103.00	-5884	-1510	639.63	591.42
113.93	131.00	-120461	-7872	715.66	597.84
139.41	159.00	-129633	7096	696.88	597.92
164.89	187.00	-35021	707	656.15	597.92
190.37	215.00	29451	4140	443.97	597.89
215.85	243.00	48387	-2287	678.02	595.56
241.33	271.00	30850	770	331.26	543.76
266.81	299.00	9086	-1723	332.62	318.02
278.01	313.00	257	-151	182.41	121.78

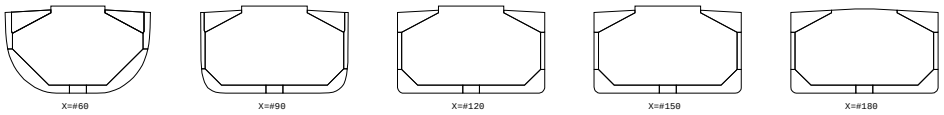
				X	FRAME
SHEAR FORCE (MIN)	-7872.2 t	(58.9%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	7095.9 t	(53.1%)		139.4 m	159
SAGGING MOMENT	-176675 tm	(24.4%)		127.1 m	145
HOGGING MOMENT	78808.6 tm	(19.2%)		49.5 m	60

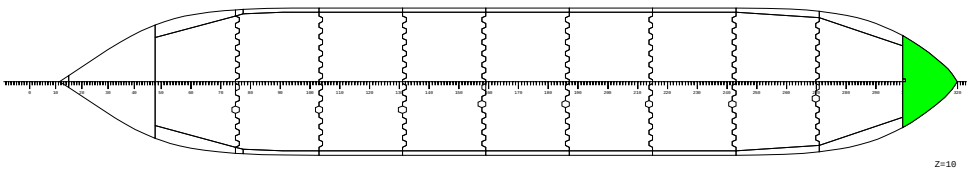
LOADING CONDITION - COND30 : DOCKING CONDITION-ARRIVAL BUNKERS

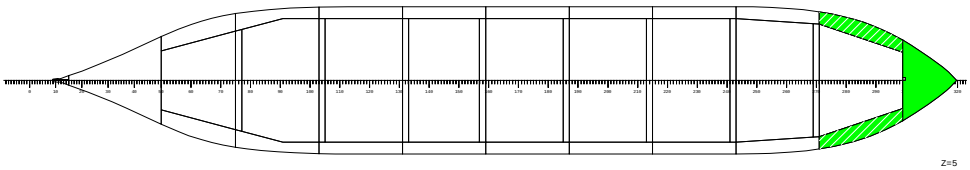


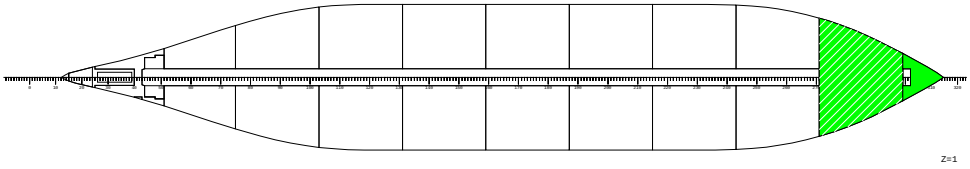












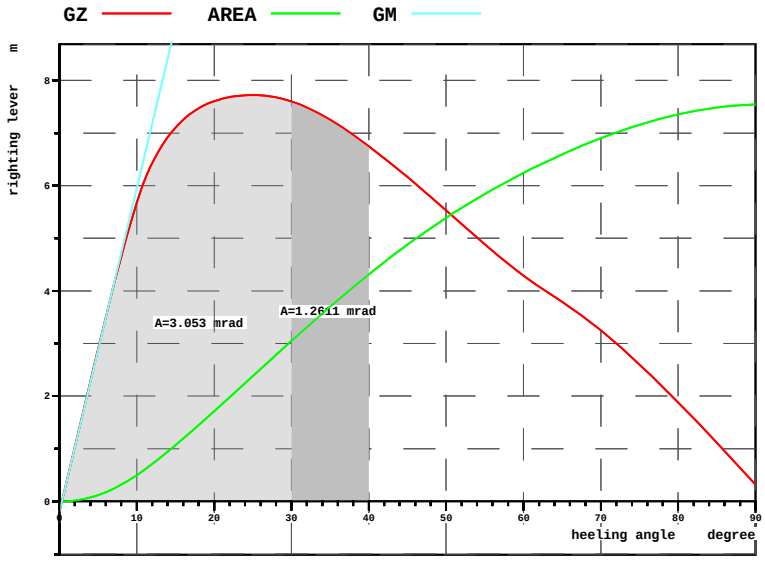
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	1626.3	57.4	252.36	0.00	1.27	70368.66
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		5189.2		266.15	-0.00	6.62	70368.66

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	6057.2	232.41	-1.04 8.96
Total weight	30699.1	151.82	-0.20 12.74

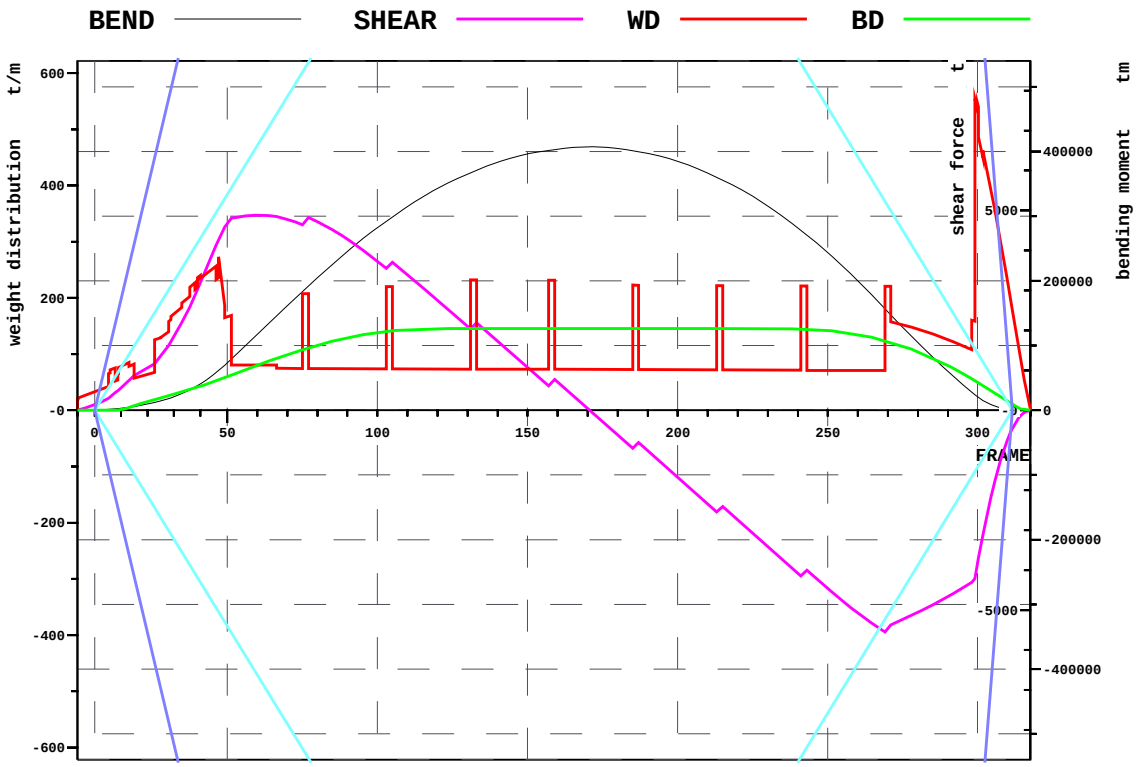
F L O A T I N G P O S I T I O N

Draught moulded	3.161	m	KM	50.31	m
Trim	0.000	m	KG	12.74	m
Heel, PS=+	-0.3	deg			
TA	3.161	m	GM0	37.57	m
TF	3.161	m	GMCORR	-2.32	m
Trimming moment	-3	tonm	GM	35.26	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.20	0.000	0.0	0.000
5.0	2.90	0.118	5132.0	0.167
10.0	5.68	0.497	9693.6	0.316
20.0	7.61	1.712	14055.7	0.458
30.0	7.60	3.052	13520.9	0.440
40.0	6.75	4.313	12513.5	0.408
45.0	6.16	4.877	11866.1	0.387
50.0	5.53	5.388	11121.3	0.362
55.0	4.89	5.842	10284.8	0.335
60.0	4.29	6.242	9364.4	0.305
65.0	3.79	6.594	8378.3	0.273
70.0	3.25	6.902	7311.9	0.238
75.0	2.60	7.158	6193.7	0.202
80.0	1.88	7.354	5029.2	0.164
85.0	1.11	7.485	3824.7	0.125
90.0	0.32	7.547	2590.2	0.084



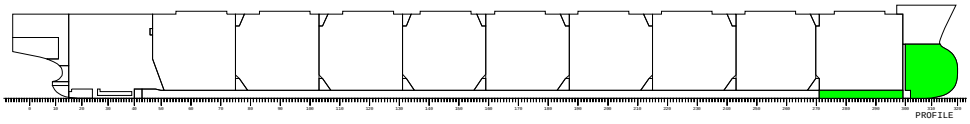
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	3.053	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	4.314	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.261	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.601	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	24.964	deg	NOT MET
GM0.15	GM > 0.15 m	0.150	35.257	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	1.281		OK

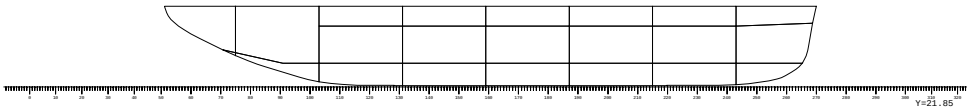


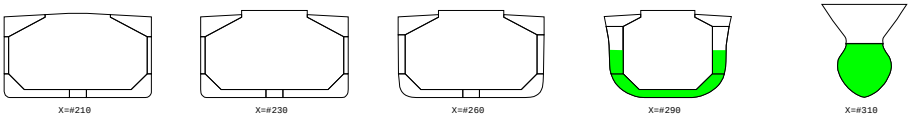
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	389	142	32.90	0.00
8.80	11.00	3443	621	80.55	2.01
12.00	15.00	5849	866	57.02	8.34
37.60	47.00	61579	4270	273.19	54.58
41.13	51.00	77689	4767	167.96	62.03
62.97	75.00	183074	4646	74.48	107.80
88.45	103.00	292381	3557	220.36	140.12
113.93	131.00	367203	2035	73.39	145.29
139.41	159.00	403166	773	231.71	145.37
164.89	187.00	399655	-809	222.56	145.37
190.37	215.00	355668	-2404	71.74	145.33
215.85	243.00	270902	-4005	221.47	143.63
241.33	271.00	147167	-5369	220.92	119.80
266.81	299.00	23882	-4206	158.47	52.72
278.01	313.00	902	-479	182.33	11.13

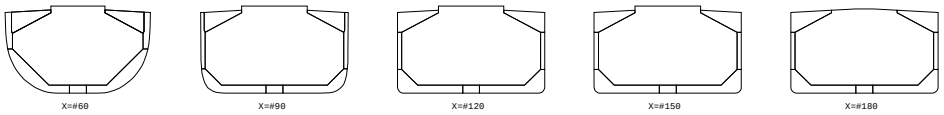
				X	FRAME
SHEAR FORCE (MIN)	-5550.4 t	(40.9%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	4878.9 t	(36.9%)		48.7 m	59
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	407317.8 tm	(55.4%)		151.0 m	172

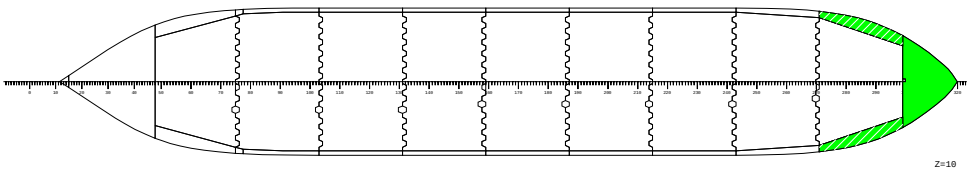
LOADING CONDITION - COND31 : STERN TUBE SEAL RENEWAL COND-ARRIVAL

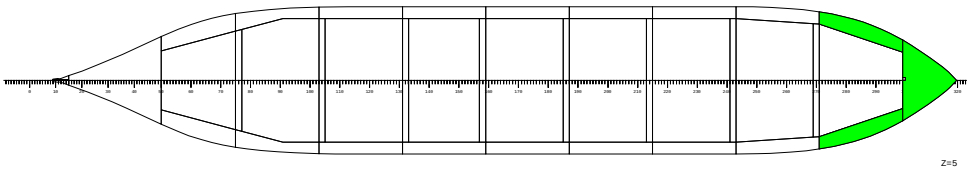


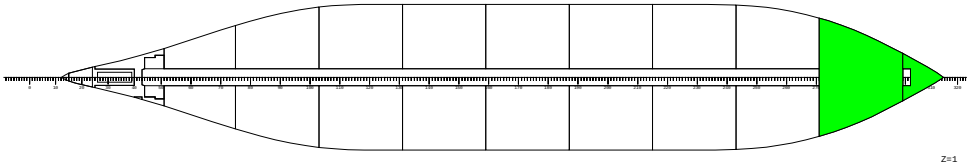












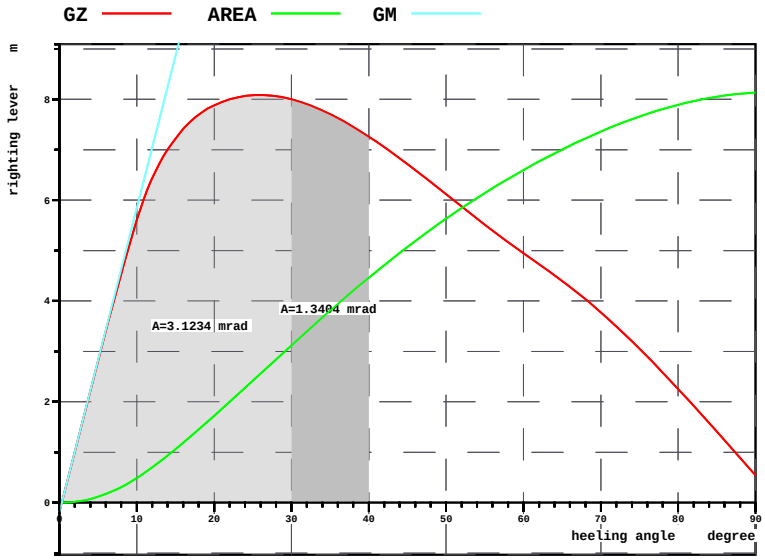
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	653.2	50.0	254.85	16.73	10.98	549.56
R2.11S	NO.1 D.S.W.B.TK(S)	653.2	50.0	254.85	-16.73	10.98	549.56
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		7702.6		262.33	-0.00	7.04	1099.11

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	8570.6	238.86	-0.73 8.65
Total weight	33212.5	159.58	-0.19 12.37

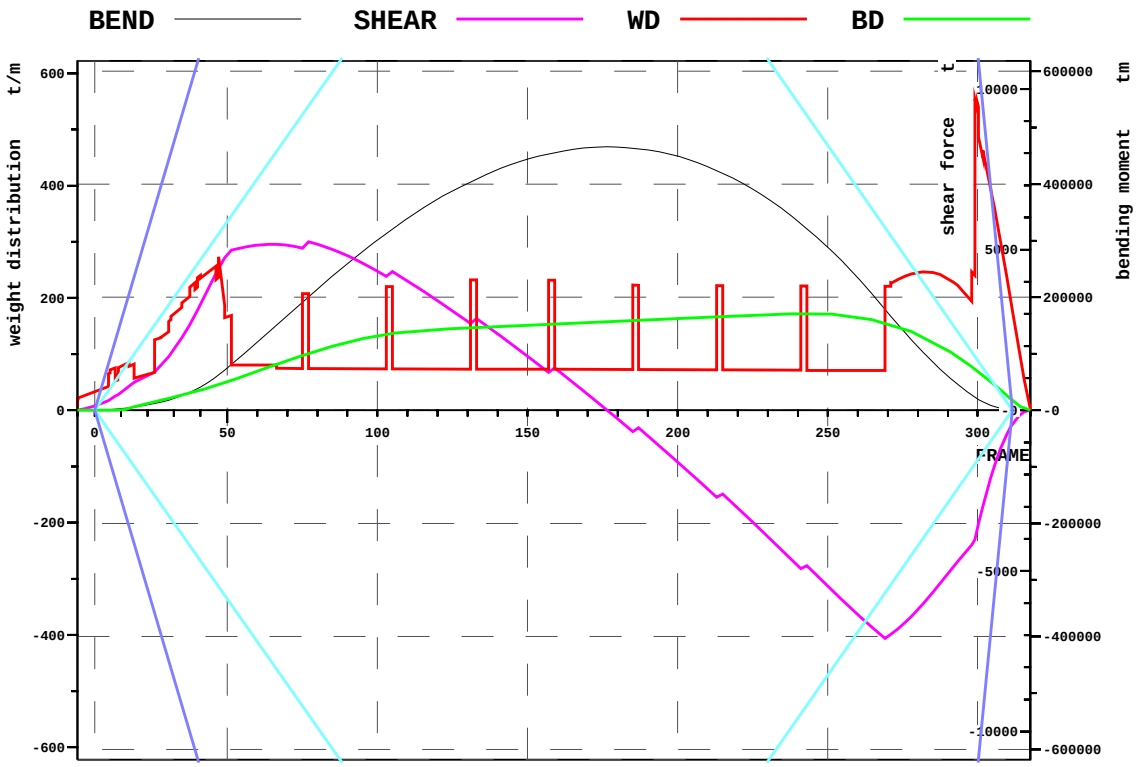
F L O A T I N G P O S I T I O N

Draught moulded	3.332	m	KM	46.95 m
Trim	-1.587	m	KG	12.37 m
Heel, PS=+	-0.3	deg		
TA	2.539	m	GM0	34.57 m
TF	4.126	m	GMCORR	-0.03 m
Trimming moment	260798	tonm	GM	34.54 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.19	0.000	0.0	0.000
5.0	2.82	0.115	160.5	0.005
10.0	5.62	0.486	322.4	0.010
20.0	7.88	1.721	622.7	0.019
30.0	8.00	3.123	938.6	0.028
40.0	7.26	4.463	1359.3	0.041
45.0	6.71	5.073	1627.2	0.049
50.0	6.12	5.633	1946.7	0.059
55.0	5.51	6.141	2324.6	0.070
60.0	4.95	6.597	2788.2	0.084
65.0	4.40	7.004	3291.0	0.099
70.0	3.77	7.362	3774.4	0.114
75.0	3.05	7.660	4198.5	0.126
80.0	2.25	7.892	4584.9	0.138
85.0	1.41	8.052	4886.2	0.147
90.0	0.54	8.137	5249.0	0.158



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	3.123	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	4.464	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.340	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	8.001	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	25.892	deg	OK
GM0.15	GM > 0.15 m	0.150	34.540	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	1.337		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	316	142	32.93	0.00
8.80	11.00	3251	622	80.57	1.12
12.00	15.00	5623	872	57.06	6.29
37.60	47.00	62609	4419	273.42	46.07
41.13	51.00	79254	4948	168.10	52.79
62.97	75.00	190696	5051	74.55	97.60
88.45	103.00	312880	4169	220.55	134.78
113.93	131.00	403998	2702	73.45	146.60
139.41	159.00	455455	1322	231.91	153.40
164.89	187.00	462275	-547	222.75	160.12
190.37	215.00	419089	-2603	71.80	166.80
215.85	243.00	321230	-4834	221.65	171.75
241.33	271.00	166329	-6973	221.10	151.16
266.81	299.00	23087	-4027	240.48	73.17
278.01	313.00	941	-458	182.49	18.82

				X	FRAME
SHEAR FORCE (MIN)	-7097.3 t	(52.4%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	5249.2 t	(38.9%)		64.8 m	77
SAGGING MOMENT	-1.2 tm			-5.0 m	-6
HOGGING MOMENT	465985.2 tm	(63.4%)		155.3 m	176

7. Ballast exchange sequence based on loading condition No3

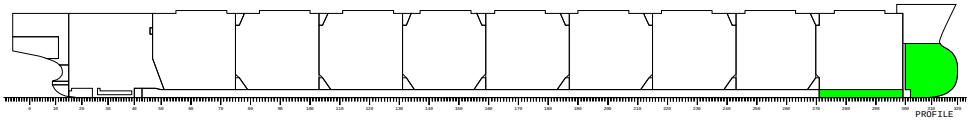
Ballast sequence:174000 DWT DOUBLE SKIN BULK CARRIER (W=24641.93ton,LCG=132.014m&vcg=13.667m)

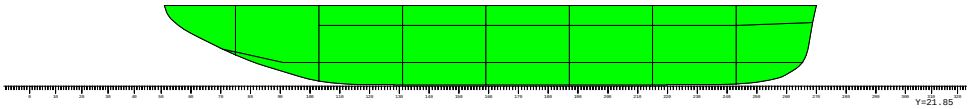
Loading condition No 3: Normal Ballast – 50% Bunkers

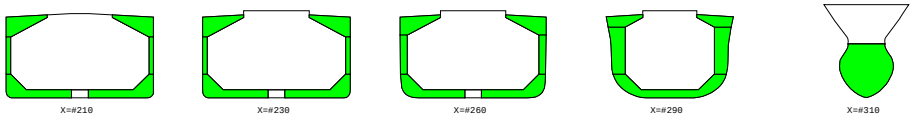
							Line of sight
No.	Condition	Bunkers	Stability	Long. Strength	Draught Aft	>8.5 m	<500 m
			[OK/Not OK]	[OK/Not OK]	[m]	[Yes/No]	[Yes/No]
1	Ballast sequence – WB10+A.P. empty	50%	OK	OK	8.05	No[1)]	Yes
2	Ballast sequence –WB1+WB10+A.P. empty	50%	OK	OK	9.55	Yes	No[3)]
3	Ballast sequence –WB2+WB10+A.P. empty	50%	OK	OK	8.96	Yes	Yes
4	Ballast sequence –WB3+WB10+A.P. empty	50%	OK	OK	8.46	No[2)]	Yes
5	Ballast sequence –F.P. +WB10+A.P. empty	50%	OK	OK	8.98	Yes	Yes
6	Ballast sequence – WB4+F.P.+WB10+A.P. empty	50%	OK	OK	8.91	Yes	No[3)]
7	Ballast sequence –WB5+F.P.+WB10+A.P. empty	50%	OK	OK	8.42	No[2)]	Yes
8	Ballast sequence –WB6+F.P.+WB10+A.P. empty	50%	OK	OK	7.94	No[2)]	Yes
9	Ballast sequence –F.P. empty	50%	OK	OK	9.92	Yes	No[3)]
10	Ballast sequence –WB7+F.P. empty	50%	OK	OK	8.43	No[2)]	Yes
11	Ballast sequence –WB8+F.P. empty	50%	OK	OK	8.64	Yes	Yes
12	Ballast sequence –WB9+F.P. empty	50%	OK	OK	8.31	No[2)]	Yes
13	Ballast sequence –Back to LC3 empty	50%	OK	OK	9.00	Yes	Yes

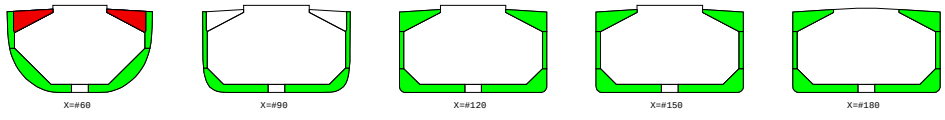
1) The propeller can stay immersed by combining sequence step No.1 and 2.
2) The propeller can stay immersed for condition 4,7,8,10 and 12 by splitting the condition into three operations:exchange bottom tanks, double side Tanks and top side tanks after each other.
3) The exchange is to be carried out in an open sea confirming that no other vessel is likely in the visibility. Condition 2 and 6 above can be splitted into three operations:exchange bottom tanks, double side tanks and top side tanks after each other. Line of sight will than be less than 500 m.

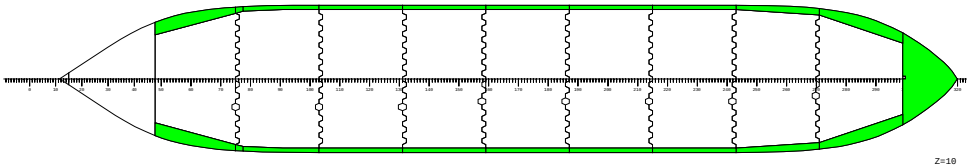
LOADING CONDITION - BALCON3-01 : BALLAST SEQUENCE LC3-WB10+AFT EMPTY-50% BUNKERS

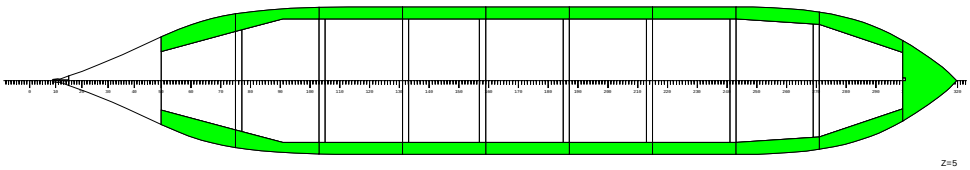


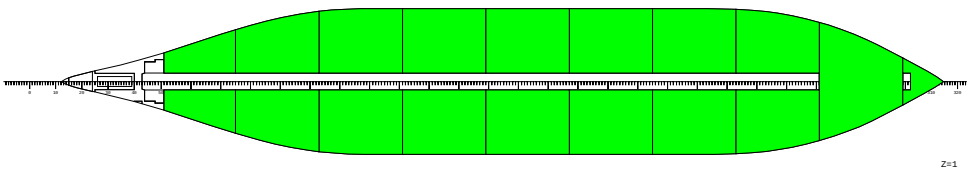












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

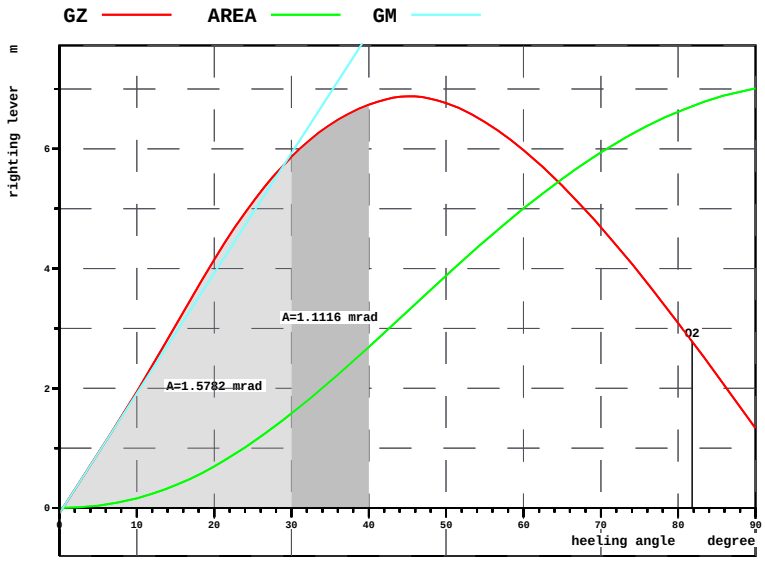
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		59038.0		167.66	-0.00	10.42	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	62275.0	160.96	-0.13 11.05
Total weight	86916.9	152.75	-0.09 11.79

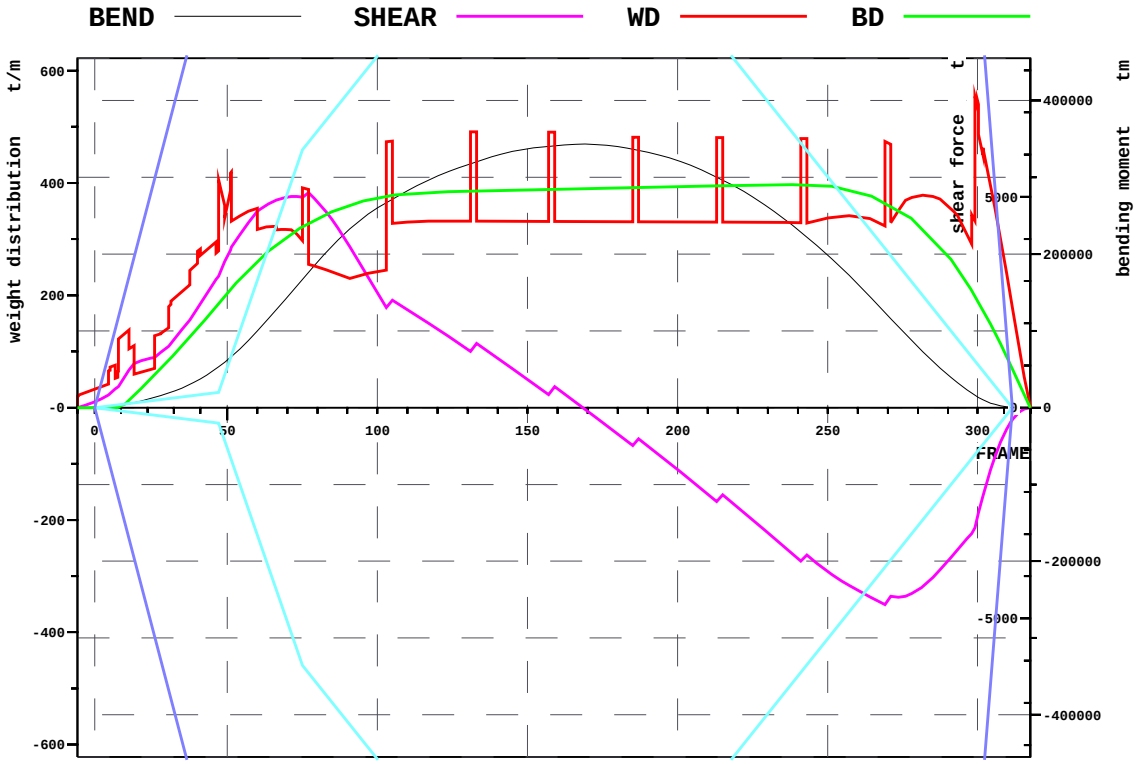
F L O A T I N G P O S I T I O N

Draught moulded	8.422	m	KM	23.30 m
Trim	-0.749	m	KG	11.79 m
Heel, PS=+	-0.5	deg		
TA	8.047	m	GM0	11.51 m
TF	8.797	m	GMCORR	0.00 m
Trimming moment	141550	tonm	GM	11.51 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.09	0.000	0.0	0.000
5.0	0.91	0.036	273.1	0.003
10.0	1.94	0.160	572.7	0.007
20.0	4.15	0.692	1197.1	0.014
30.0	5.87	1.578	1623.0	0.019
40.0	6.74	2.689	1920.4	0.022
45.0	6.88	3.285	2017.0	0.023
50.0	6.77	3.882	2076.3	0.024
55.0	6.45	4.460	2095.0	0.024
60.0	5.98	5.004	2072.0	0.024
65.0	5.39	5.501	2014.9	0.023
70.0	4.69	5.941	1930.4	0.022
75.0	3.92	6.317	1823.1	0.021
80.0	3.09	6.623	1696.6	0.020
85.0	2.22	6.856	1553.9	0.018
90.0	1.33	7.011	1399.3	0.016



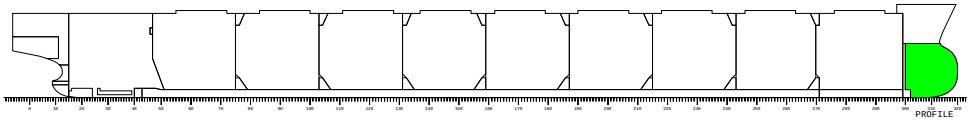
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.578	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.690	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.112	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.880	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.277	deg	OK
GM0.15	GM > 0.15 m	0.150	11.507	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.812		OK

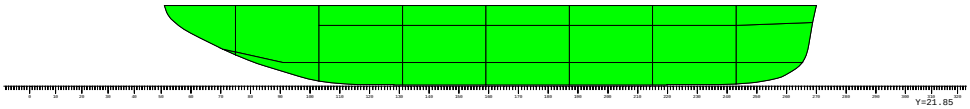


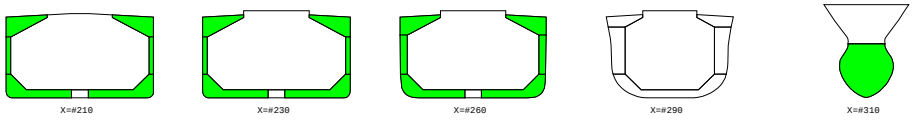
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	350	142	32.93	0.00
8.80	11.00	3399	696	130.51	5.28
12.00	15.00	6218	1041	59.80	22.08
37.60	47.00	53014	3116	405.57	185.25
41.13	51.00	65152	3735	386.25	209.65
62.97	75.00	166704	4979	297.52	321.85
88.45	103.00	266958	2372	473.48	375.97
113.93	131.00	317497	1332	490.91	385.47
139.41	159.00	340960	497	490.35	388.72
164.89	187.00	334794	-739	481.19	391.90
190.37	215.00	295936	-2069	405.45	395.04
215.85	243.00	221978	-3492	328.49	396.12
241.33	271.00	115654	-4471	331.39	357.32
266.81	299.00	15989	-2857	332.76	198.33
278.01	313.00	594	-302	182.49	70.99

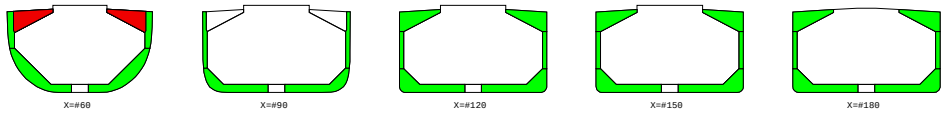
				X	FRAME
SHEAR FORCE (MIN)	-4673.4 t	(42.6%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	5098.4 t	(47.0%)		64.8 m	77
SAGGING MOMENT	-				
HOGGING MOMENT	343064.5 tm	(72.6%)		148.7 m	169

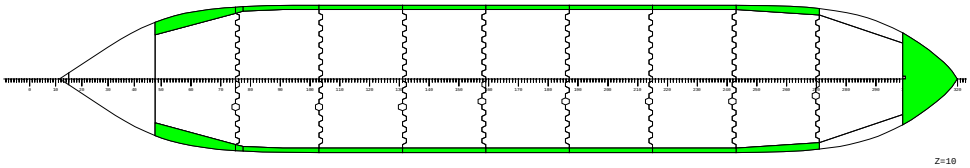
LOADING CONDITION - BALCON3-02 : BALLAST SEQUENCE LC3-WB1+WB10+AFT EMPTY-50% BUN

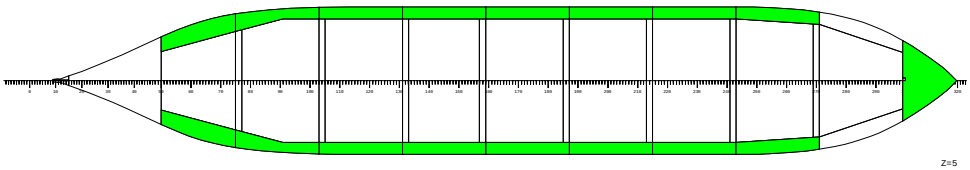


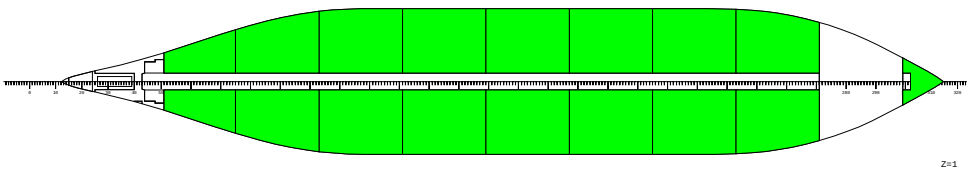












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00

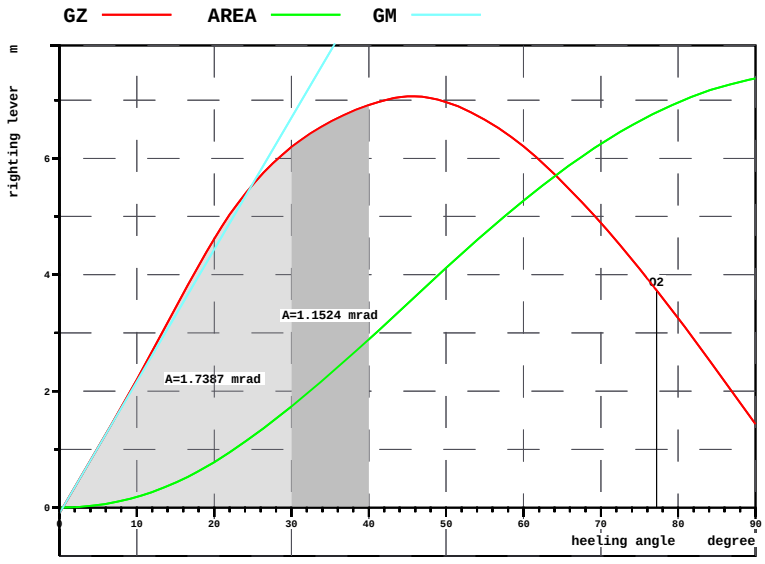
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		51778.3		155.59	-0.00	10.19	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	55015.3	148.72	-0.15 10.91
Total weight	79657.1	143.55	-0.10 11.77

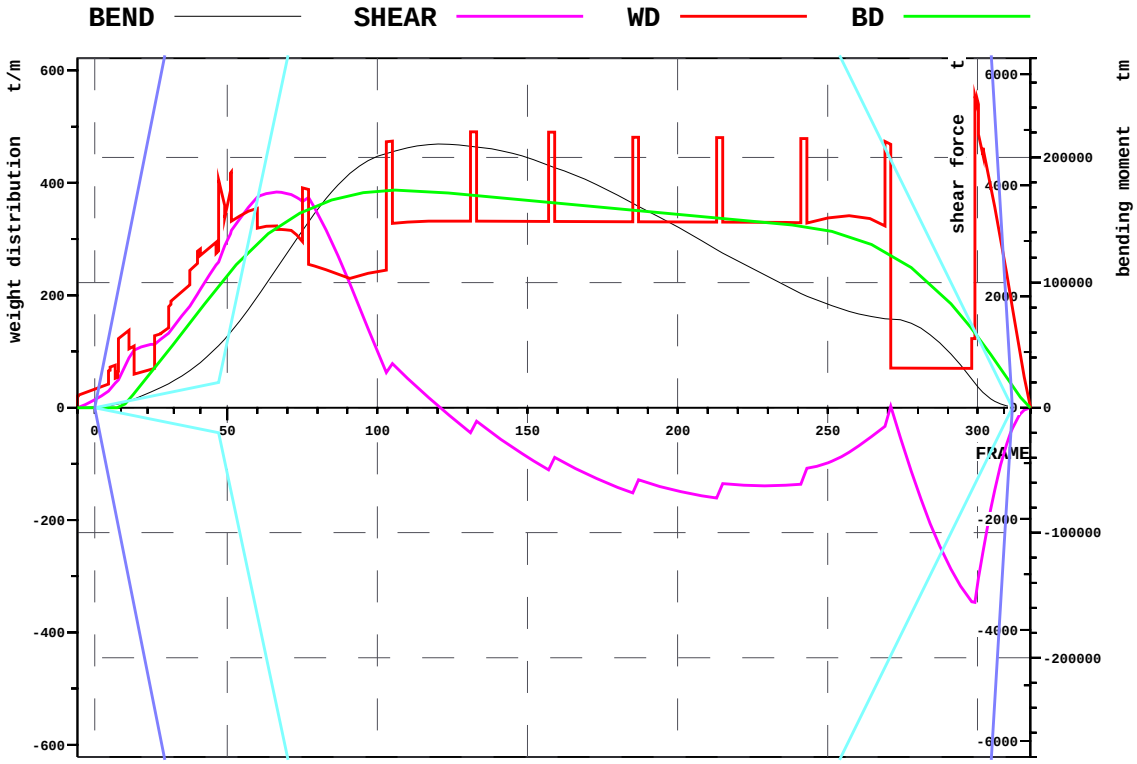
F L O A T I N G P O S I T I O N

Draught moulded	7.906	m	KM	24.76 m
Trim	3.288	m	KG	11.77 m
Heel, PS=+	-0.4	deg		
TA	9.550	m	GM0	12.99 m
TF	6.261	m	GMCORR	0.00 m
Trimming moment	-625656	tonm	GM	12.99 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.03	0.041	273.1	0.003
10.0	2.20	0.181	572.7	0.007
20.0	4.62	0.778	1197.1	0.015
30.0	6.19	1.738	1623.0	0.020
40.0	6.92	2.891	1920.4	0.024
45.0	7.06	3.502	2017.0	0.025
50.0	6.97	4.116	2076.3	0.026
55.0	6.66	4.712	2095.0	0.026
60.0	6.21	5.275	2072.0	0.026
65.0	5.60	5.791	2014.9	0.025
70.0	4.89	6.249	1930.4	0.024
75.0	4.10	6.642	1823.1	0.023
80.0	3.25	6.963	1696.6	0.021
85.0	2.36	7.208	1553.9	0.020
90.0	1.44	7.373	1399.3	0.018



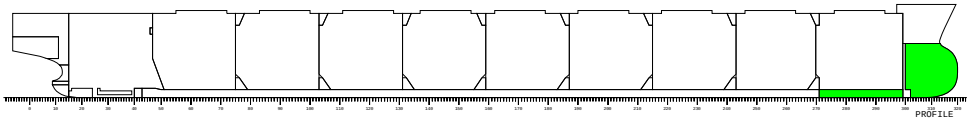
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.739	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.891	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.152	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.066	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.653	deg	OK
GM0.15	GM > 0.15 m	0.150	12.993	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.316		OK

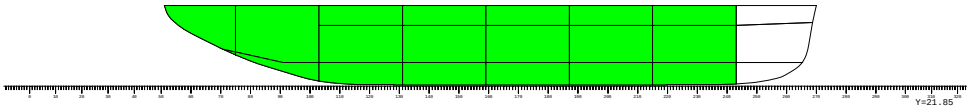


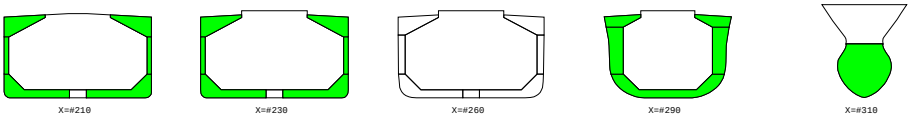
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	523	142	32.91	0.00
8.80	11.00	3865	696	130.56	5.27
12.00	15.00	6786	1036	59.77	25.45
37.60	47.00	49515	2618	405.36	215.41
41.13	51.00	59821	3127	386.05	241.19
62.97	75.00	141422	3709	295.66	347.98
88.45	103.00	203307	632	473.22	386.02
113.93	131.00	208895	-451	490.64	378.43
139.41	159.00	191862	-894	490.09	364.58
164.89	187.00	160758	-1301	330.62	350.65
190.37	215.00	123618	-1368	405.23	336.70
215.85	243.00	89135	-1091	328.32	320.78
241.33	271.00	70884	25	70.66	270.33
266.81	299.00	19414	-3505	123.07	132.69
278.01	313.00	595	-396	182.39	40.40

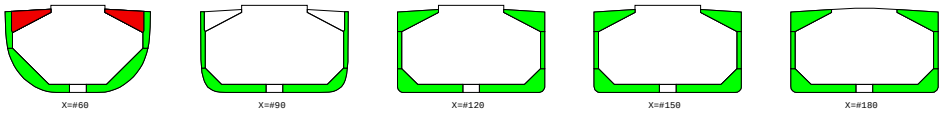
				X	FRAME
SHEAR FORCE (MIN)	-3505.2 t	(31.2%)	POSITION:	266.8 m	299
SHEAR FORCE (MAX)	3880.4 t	(35.5%)		55.1 m	66
SAGGING MOMENT	-19.8 tm			281.8 m	318
HOGGING MOMENT	210669.8 tm	(44.6%)		104.1 m	120

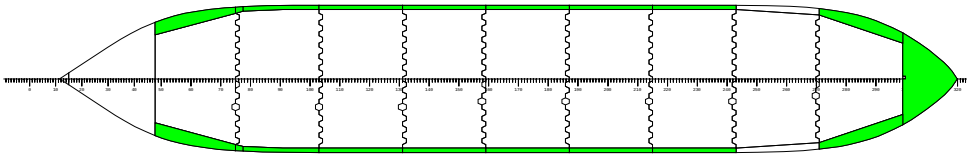
LOADING CONDITION - BALCON3-03 : BALLAST SEQUENCE LC3-WB2+WB10+AFT EMPTY-50% BUN

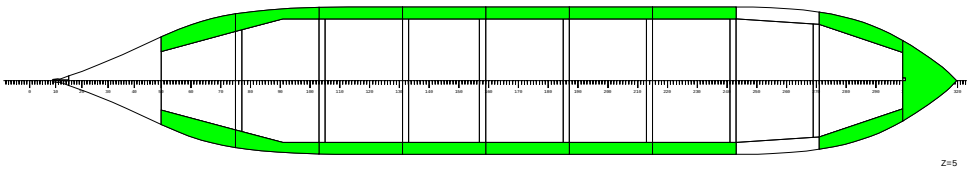


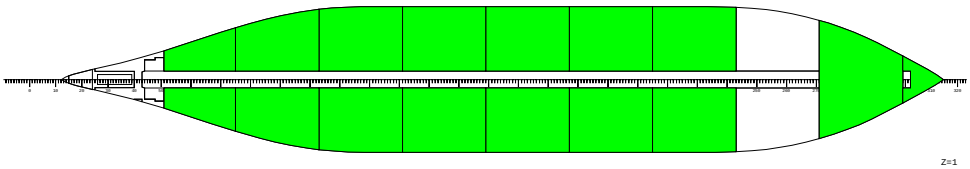












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00

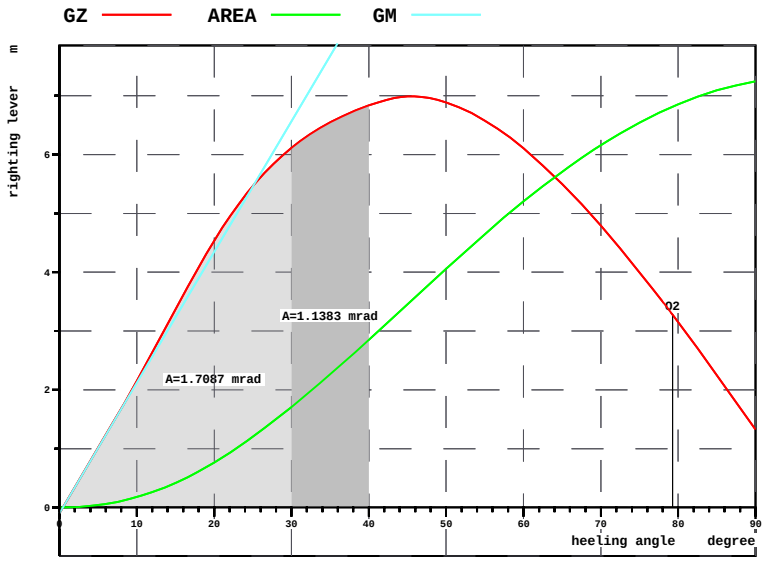
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		52302.1		159.82	-0.00	10.34	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	55539.0	152.76	-0.14 11.04
Total weight	80180.9	146.38	-0.10 11.85

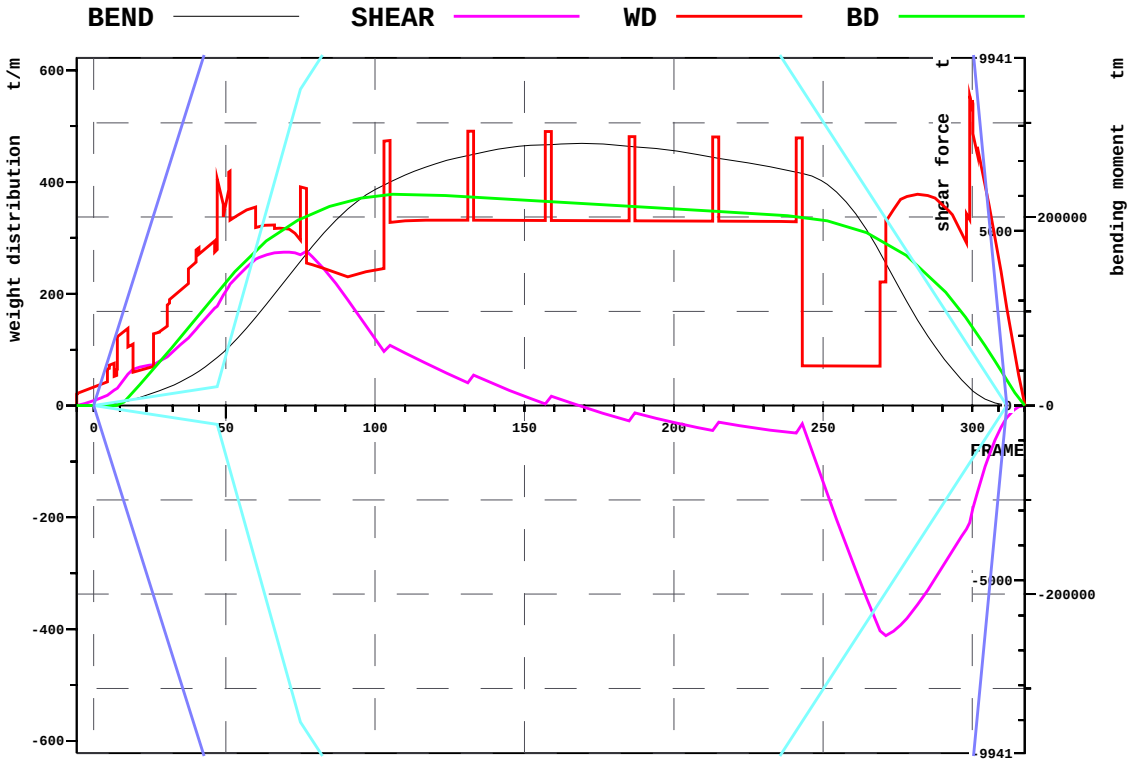
F L O A T I N G P O S I T I O N

Draught moulded	7.911	m	KM	24.59 m
Trim	2.093	m	KG	11.85 m
Heel, PS=+	-0.4	deg		
TA	8.958	m	GM0	12.74 m
TF	6.865	m	GMCORR	0.00 m
Trimming moment	-396127	tonm	GM	12.74 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.01	0.040	273.1	0.003
10.0	2.15	0.177	572.7	0.007
20.0	4.54	0.763	1197.1	0.015
30.0	6.11	1.708	1623.0	0.020
40.0	6.83	2.847	1920.4	0.024
45.0	6.99	3.451	2017.0	0.025
50.0	6.89	4.058	2076.3	0.026
55.0	6.58	4.647	2095.0	0.026
60.0	6.10	5.201	2072.0	0.026
65.0	5.50	5.708	2014.9	0.025
70.0	4.79	6.158	1930.4	0.024
75.0	3.99	6.541	1823.1	0.023
80.0	3.15	6.854	1696.6	0.021
85.0	2.25	7.090	1553.9	0.019
90.0	1.33	7.246	1399.3	0.017



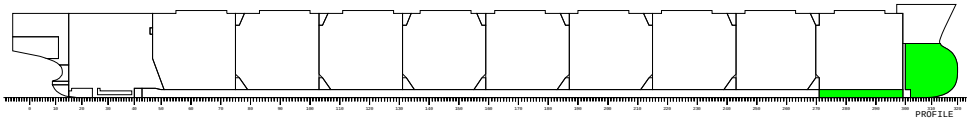
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.709	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.847	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.138	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.989	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.608	deg	OK
GM0.15	GM > 0.15 m	0.150	12.737	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.335		OK

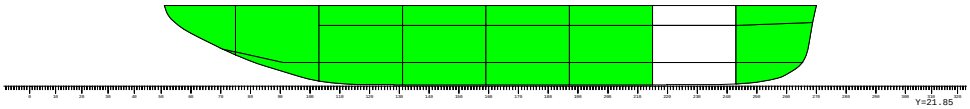


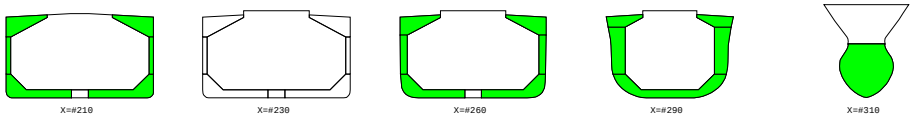
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	430	142	32.93	0.00
8.80	11.00	3616	696	130.57	5.28
12.00	15.00	6482	1038	59.79	23.95
37.60	47.00	50946	2842	405.51	201.53
41.13	51.00	62068	3402	386.19	226.48
62.97	75.00	152898	4315	343.83	333.68
88.45	103.00	233979	1549	473.40	376.26
113.93	131.00	265148	654	490.83	373.72
139.41	159.00	276832	270	490.28	364.94
164.89	187.00	274297	-208	481.12	356.07
190.37	215.00	262294	-473	405.39	347.17
215.85	243.00	246230	-524	478.92	336.29
241.33	271.00	150469	-6578	221.07	289.59
266.81	299.00	18824	-3360	332.70	147.67
278.01	313.00	654	-375	182.46	47.28

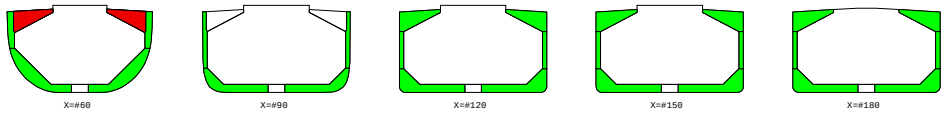
				X	FRAME
SHEAR FORCE (MIN)	-6577.9 t	(59.7%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	4413.1 t	(40.7%)		64.8 m	77
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	278062.8 tm	(58.9%)		148.7 m	169

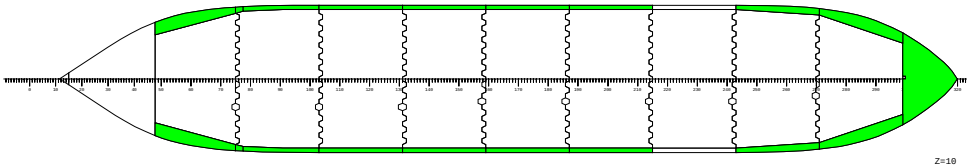
LOADING CONDITION - BALCON3-04 : BALLAST SEQUENCE LC3-WB3+WB10+AFT EMPTY-50% BUN

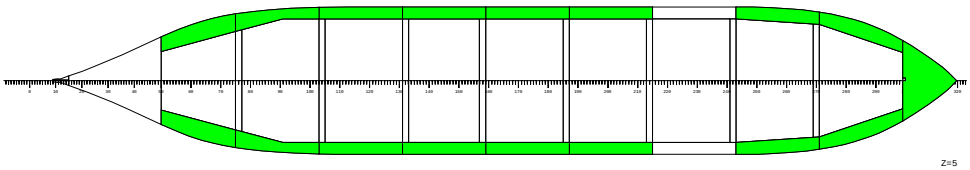


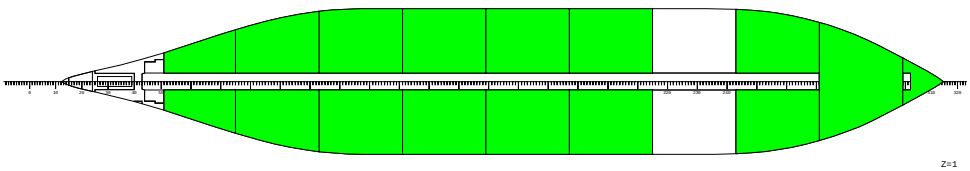












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00

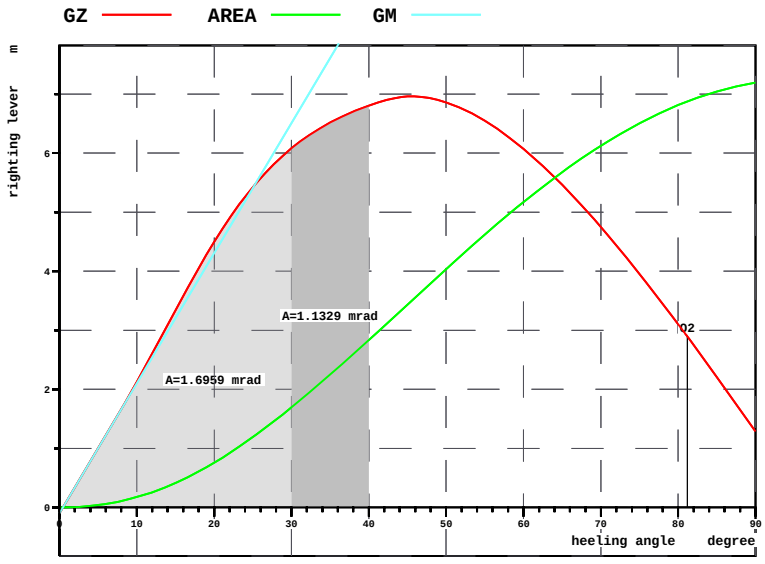
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		52465.9		163.22	-0.00	10.37	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	55702.9	155.99	-0.14 11.07
Total weight	80344.7	148.63	-0.10 11.87

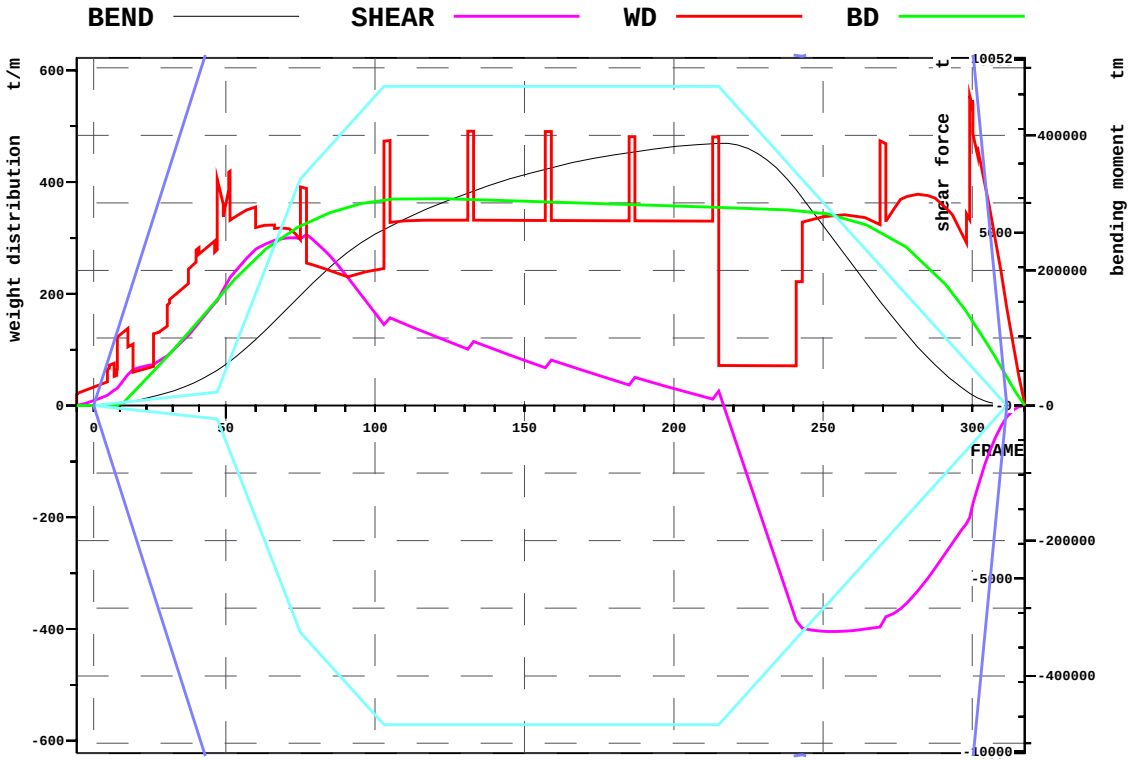
F L O A T I N G P O S I T I O N

Draught moulded	7.890	m	KM	24.50 m
Trim	1.128	m	KG	11.87 m
Heel, PS=+	-0.4	deg		
TA	8.455	m	GM0	12.63 m
TF	7.326	m	GMCORR	0.00 m
Trimming moment	-212527	tonm	GM	12.63 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.00	0.039	273.1	0.003
10.0	2.13	0.175	572.7	0.007
20.0	4.50	0.757	1197.1	0.015
30.0	6.08	1.695	1623.0	0.020
40.0	6.80	2.828	1920.4	0.024
45.0	6.96	3.430	2017.0	0.025
50.0	6.86	4.035	2076.3	0.026
55.0	6.55	4.621	2095.0	0.026
60.0	6.07	5.173	2072.0	0.026
65.0	5.46	5.677	2014.9	0.025
70.0	4.75	6.123	1930.4	0.024
75.0	3.95	6.503	1823.1	0.023
80.0	3.10	6.811	1696.6	0.021
85.0	2.21	7.043	1553.9	0.019
90.0	1.29	7.196	1399.3	0.017



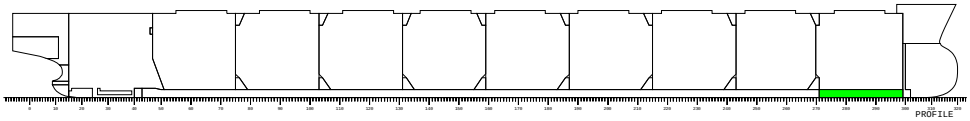
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.696	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.829	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.133	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.960	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.607	deg	OK
GM0.15	GM > 0.15 m	0.150	12.627	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.349		OK

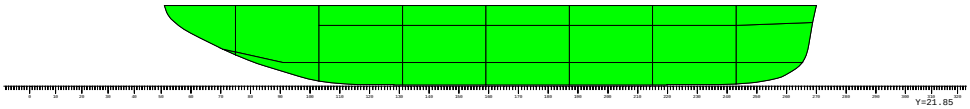


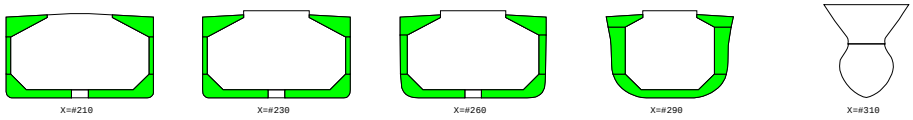
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	384	142	32.93	0.00
8.80	11.00	3491	696	130.58	5.28
12.00	15.00	6331	1040	59.80	22.82
37.60	47.00	52336	3025	405.61	189.86
41.13	51.00	64147	3627	386.28	214.00
62.97	75.00	162828	4827	296.75	321.05
88.45	103.00	260500	2342	473.51	367.20
113.93	131.00	314167	1628	490.95	368.73
139.41	159.00	351642	1321	490.39	364.03
164.89	187.00	375539	817	481.23	359.25
190.37	215.00	387953	420	276.25	354.44
215.85	243.00	307783	-6446	328.52	347.63
241.33	271.00	142853	-6109	331.42	304.08
266.81	299.00	18267	-3245	332.78	159.26
278.01	313.00	671	-359	182.50	52.75

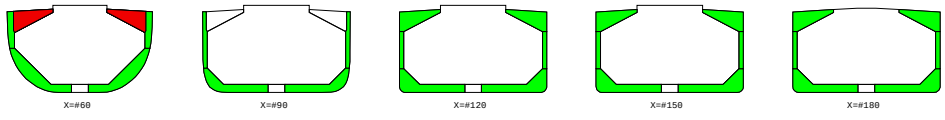
				X	FRAME
SHEAR FORCE (MIN)	-6537.5 t	(61.3%)	POSITION:	226.3 m	255
SHEAR FORCE (MAX)	4948.3 t	(45.6%)		64.8 m	77
SAGGING MOMENT	-				
HOGGING MOMENT	388002.3 tm	(84.9%)		193.3 m	218

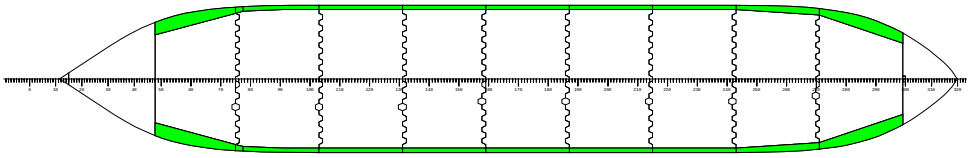
LOADING CONDITION - BALCON3-05 : BALLAST SEQUENCE LC3-FORE+WB10+AFT EMPTY-50%BUN

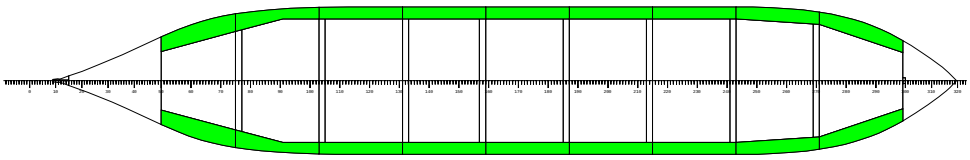


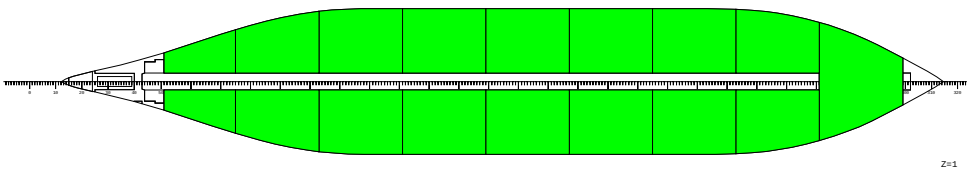












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

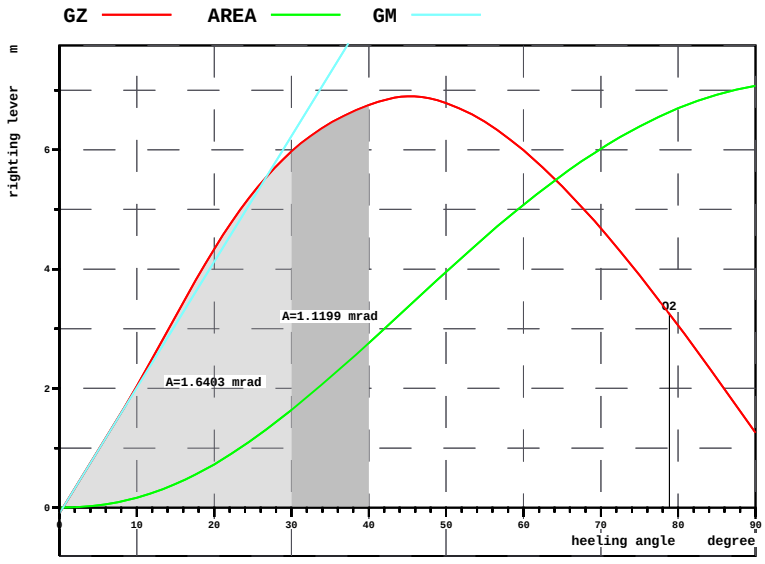
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
SUBTOTAL		55475.1		160.93	-0.00	10.51	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	58712.1	154.19	-0.14 11.17
Total weight	83354.0	147.64	-0.10 11.91

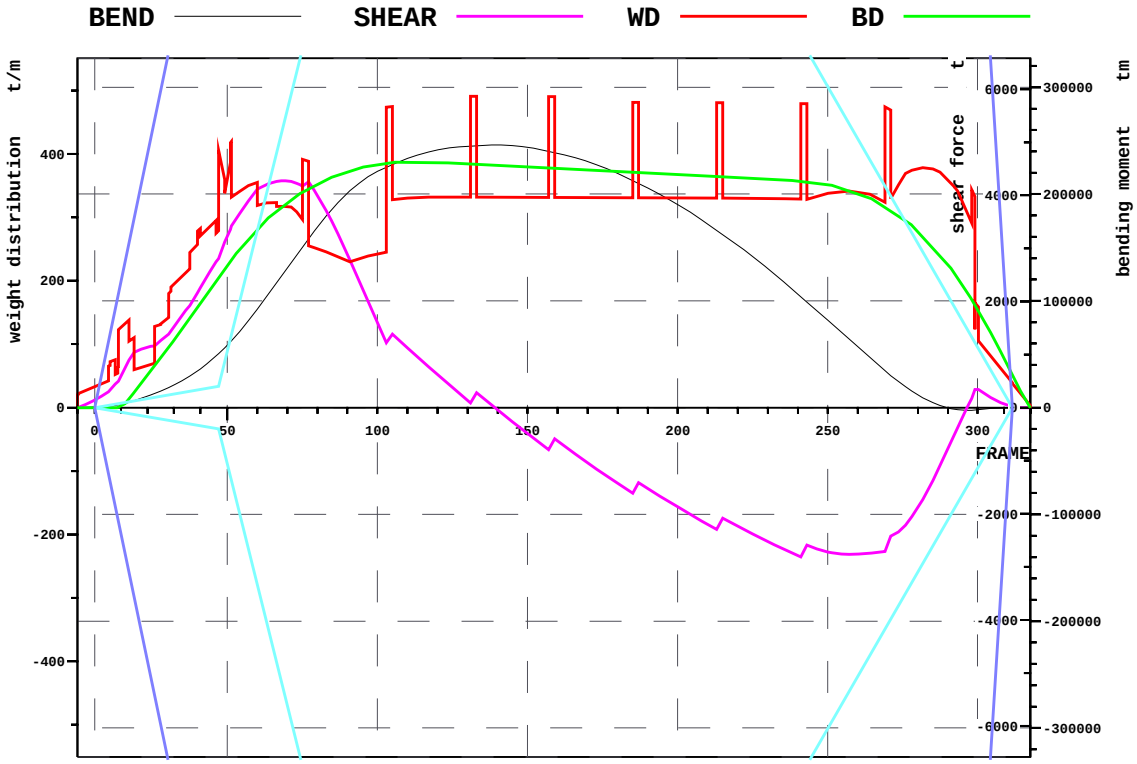
F L O A T I N G P O S I T I O N

Draught moulded	8.184	m	KM	24.00 m
Trim	1.586	m	KG	11.91 m
Heel, PS=+	-0.5	deg		
TA	8.977	m	GM0	12.09 m
TF	7.390	m	GMCORR	0.00 m
Trimming moment	-301699	tonm	GM	12.09 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	0.96	0.038	273.1	0.003
10.0	2.04	0.168	572.7	0.007
20.0	4.34	0.726	1197.1	0.014
30.0	5.97	1.640	1623.0	0.019
40.0	6.75	2.760	1920.4	0.023
45.0	6.90	3.357	2017.0	0.024
50.0	6.78	3.956	2076.3	0.025
55.0	6.47	4.535	2095.0	0.025
60.0	5.99	5.080	2072.0	0.025
65.0	5.39	5.577	2014.9	0.024
70.0	4.68	6.017	1930.4	0.023
75.0	3.90	6.392	1823.1	0.022
80.0	3.05	6.696	1696.6	0.020
85.0	2.17	6.924	1553.9	0.019
90.0	1.26	7.073	1399.3	0.017



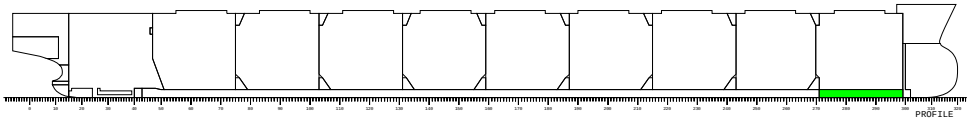
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.640	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.760	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.120	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.897	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.378	deg	OK
GM0.15	GM > 0.15 m	0.150	12.090	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.527		OK

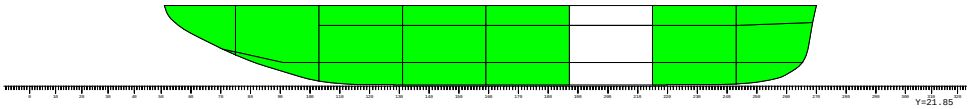


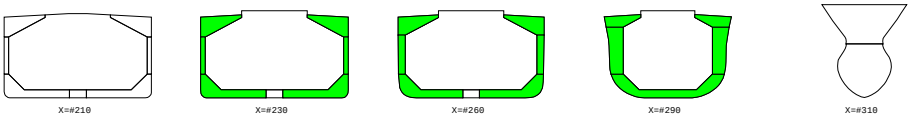
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	402	142	32.93	0.00
8.80	11.00	3539	696	130.58	5.28
12.00	15.00	6388	1038	59.80	24.06
37.60	47.00	50443	2809	405.58	204.38
41.13	51.00	61413	3359	386.26	229.82
62.97	75.00	150166	4170	296.54	339.64
88.45	103.00	225225	1222	473.48	384.60
113.93	131.00	245000	89	490.91	384.21
139.41	159.00	238516	-589	490.36	377.57
164.89	187.00	209647	-1414	330.80	370.85
190.37	215.00	161749	-2082	405.45	364.10
215.85	243.00	98837	-2591	328.49	355.34
241.33	271.00	29936	-2424	331.39	309.53
266.81	299.00	-2128	338	332.76	161.57
278.01	313.00	-79	25	40.21	53.53

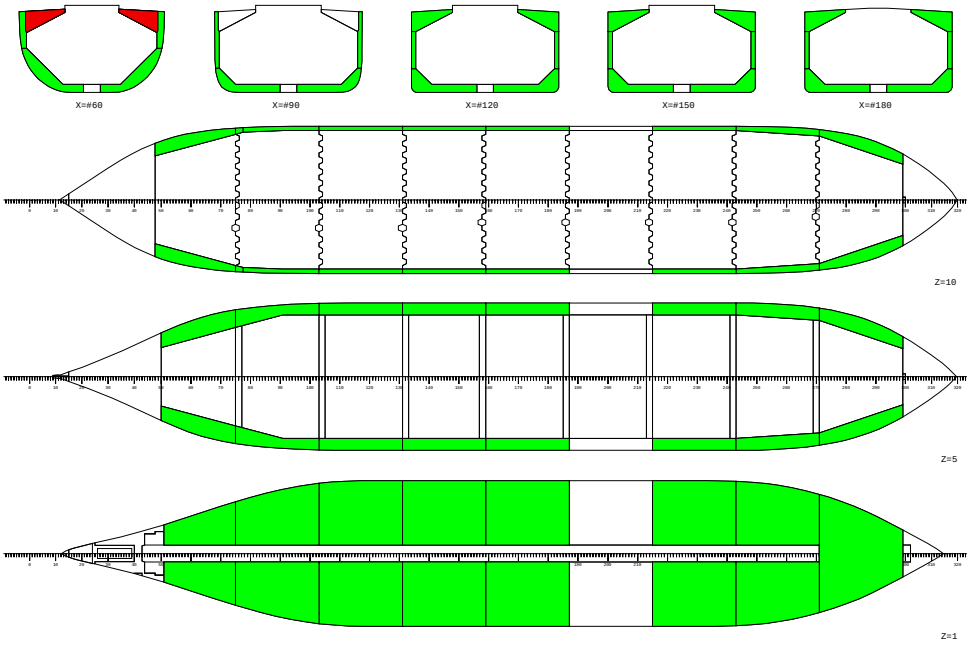
				X	FRAME
SHEAR FORCE (MIN)	-2814.9 t	(27.0%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	4272.9 t	(39.2%)		57.5 m	69
SAGGING MOMENT	-2517.2 tm	(3.7%)		264.4 m	296
HOGGING MOMENT	246147.0 tm	(52.1%)		120.3 m	138

LOADING CONDITION - BALCON3-06 : BAL. SEQU. LC3-WB4+FORE+WB10+AFT EMPTY-50% BUNK









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

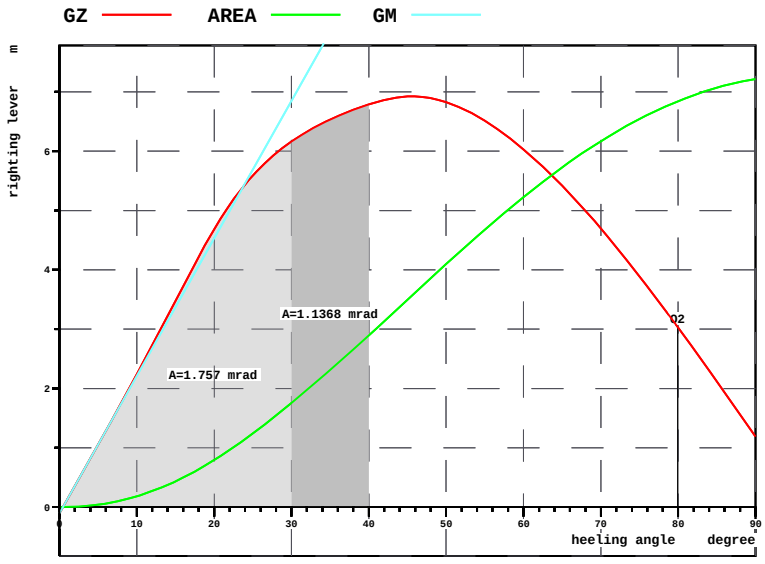
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
SUBTOTAL		48896.9		158.68	-0.00	10.46	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	52133.9	151.24	-0.15 11.21
Total weight	76775.8	145.07	-0.10 12.00

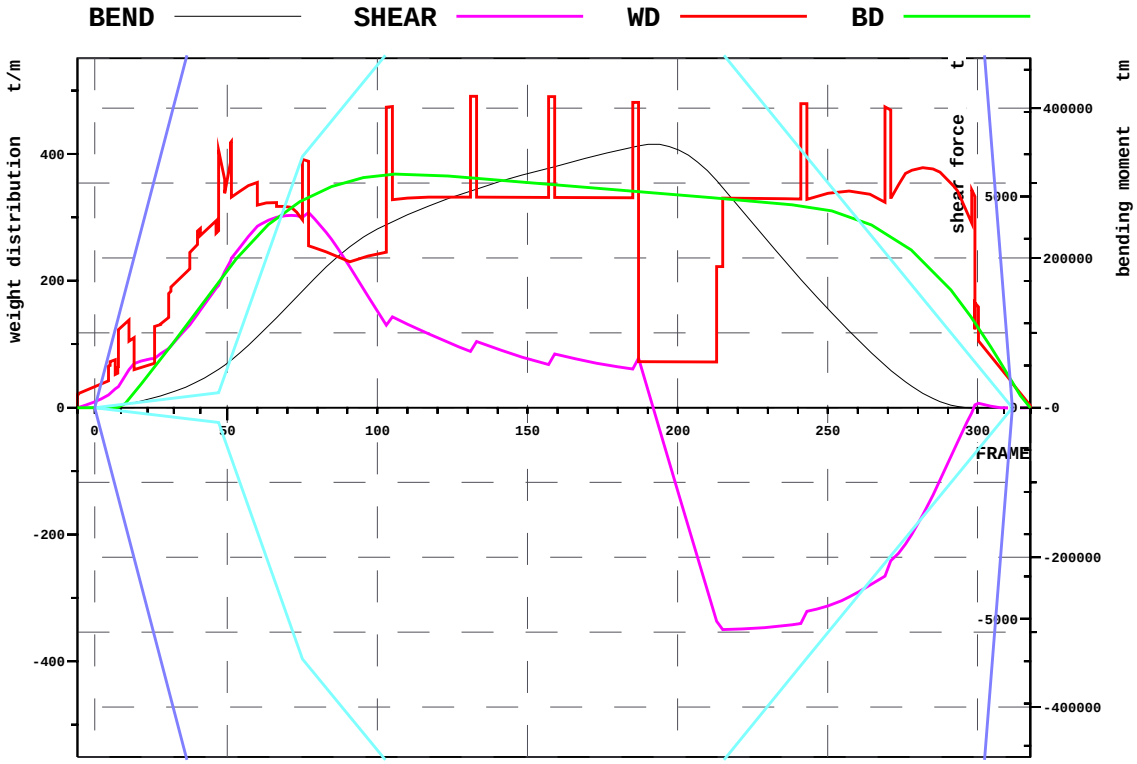
F L O A T I N G P O S I T I O N

Draught moulded	7.616	m	KM	25.28 m
Trim	2.585	m	KG	12.00 m
Heel, PS=+	-0.4	deg		
TA	8.909	m	GM0	13.28 m
TF	6.324	m	GMCORR	0.00 m
Trimming moment	-486173	tonm	GM	13.28 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.05	0.041	273.1	0.004
10.0	2.24	0.185	572.7	0.007
20.0	4.68	0.791	1197.1	0.016
30.0	6.16	1.757	1623.0	0.021
40.0	6.79	2.893	1920.4	0.025
45.0	6.92	3.493	2017.0	0.026
50.0	6.82	4.094	2076.3	0.027
55.0	6.51	4.677	2095.0	0.027
60.0	6.03	5.226	2072.0	0.027
65.0	5.42	5.726	2014.9	0.026
70.0	4.70	6.168	1930.4	0.025
75.0	3.89	6.543	1823.1	0.024
80.0	3.02	6.845	1696.6	0.022
85.0	2.12	7.069	1553.9	0.020
90.0	1.19	7.214	1399.3	0.018



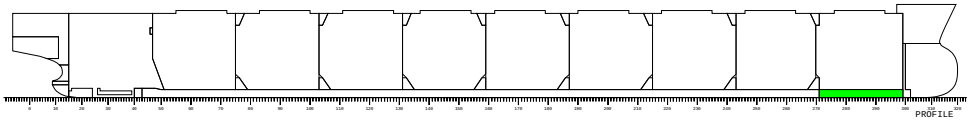
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.757	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.894	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.137	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.923	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.561	deg	OK
GM0.15	GM > 0.15 m	0.150	13.280	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.108		OK

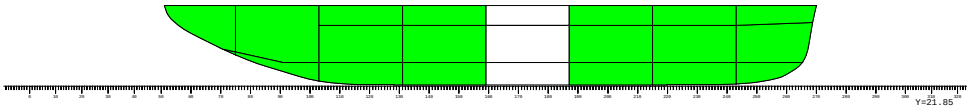


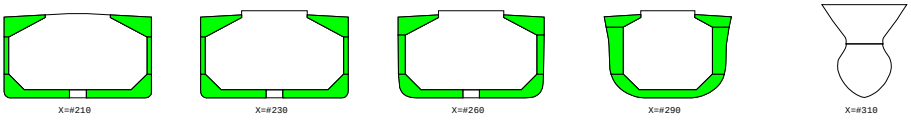
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	409	142	32.93	0.00
8.80	11.00	3559	696	130.62	5.28
12.00	15.00	6414	1039	59.80	23.77
37.60	47.00	51193	2890	405.61	197.81
41.13	51.00	62494	3464	386.29	222.18
62.97	75.00	155897	4506	296.14	326.53
88.45	103.00	244356	1955	473.52	366.75
113.93	131.00	289113	1331	490.96	362.12
139.41	159.00	321972	1272	490.40	351.26
164.89	187.00	349552	1171	72.35	340.31
190.37	215.00	294049	-5250	276.25	329.33
215.85	243.00	162224	-4818	328.52	316.40
241.33	271.00	49615	-3623	331.42	268.97
266.81	299.00	-489	67	332.79	133.55
278.01	313.00	-9	-13	40.21	41.07

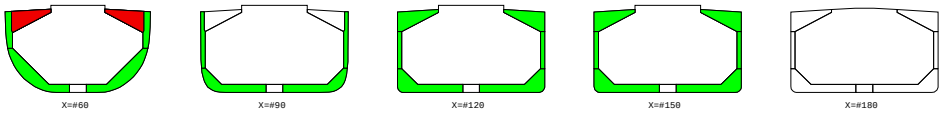
				X	FRAME
SHEAR FORCE (MIN)	-5249.6 t	(49.5%)	POSITION:	190.4 m	215
SHEAR FORCE (MAX)	4617.5 t	(42.6%)		64.8 m	77
SAGGING MOMENT	-488.6 tm	(0.9%)		266.8 m	299
HOGGING MOMENT	351919.2 tm	(74.5%)		170.7 m	193

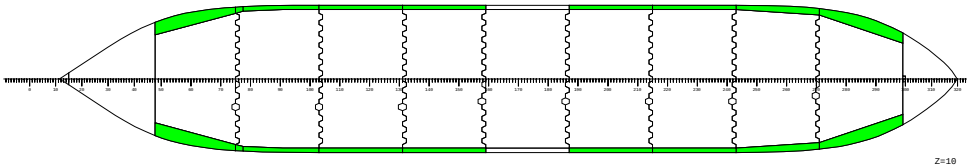
LOADING CONDITION - BALCON3-07 : BAL. SEQU. LC3-WB5+FORE+WB10+AFT EMPTY-50% BUNK

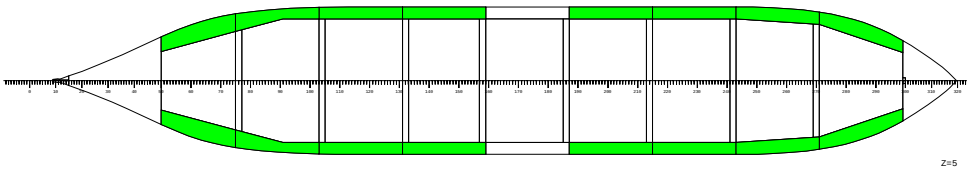


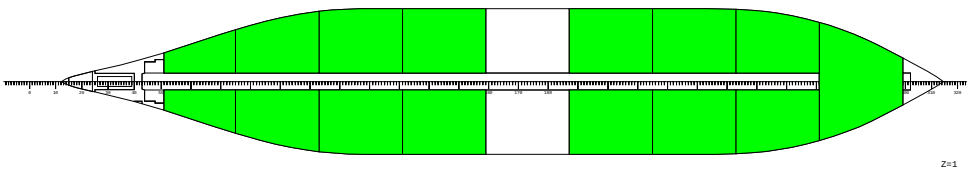












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

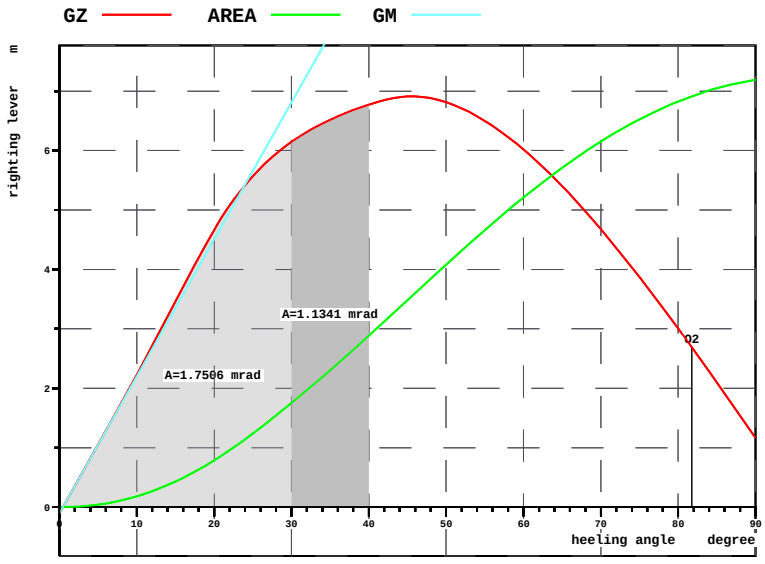
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
SUBTOTAL		48897.0		162.11	-0.00	10.46	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	52133.9	154.45	-0.15 11.21
Total weight	76775.8	147.25	-0.10 12.00

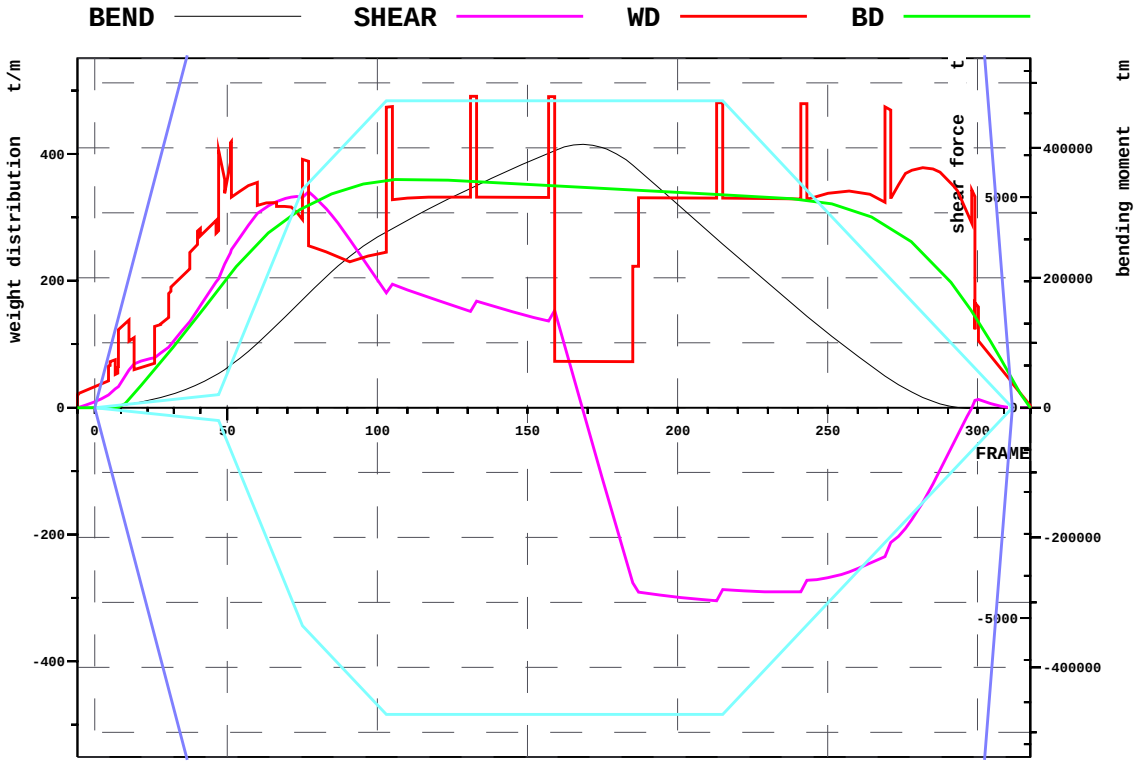
F L O A T I N G P O S I T I O N

Draught moulded	7.583	m	KM	25.22 m
Trim	1.681	m	KG	12.00 m
Heel, PS=+	-0.4	deg		
TA	8.424	m	GM0	13.22 m
TF	6.742	m	GMCORR	0.00 m
Trimming moment	-314956	tonm	GM	13.22 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.05	0.041	273.1	0.004
10.0	2.23	0.184	572.7	0.007
20.0	4.67	0.788	1197.1	0.016
30.0	6.15	1.750	1623.0	0.021
40.0	6.77	2.884	1920.4	0.025
45.0	6.91	3.483	2017.0	0.026
50.0	6.81	4.083	2076.3	0.027
55.0	6.50	4.665	2095.0	0.027
60.0	6.02	5.213	2072.0	0.027
65.0	5.40	5.712	2014.9	0.026
70.0	4.68	6.153	1930.4	0.025
75.0	3.87	6.526	1823.1	0.024
80.0	3.00	6.826	1696.6	0.022
85.0	2.10	7.049	1553.9	0.020
90.0	1.17	7.192	1399.3	0.018



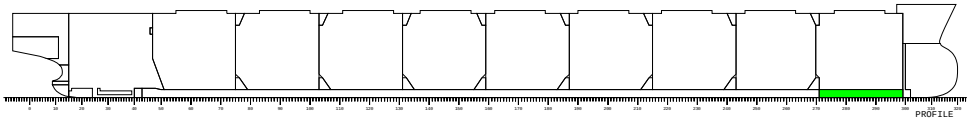
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.751	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.885	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.134	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.913	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.609	deg	OK
GM0.15	GM > 0.15 m	0.150	13.216	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.111		OK

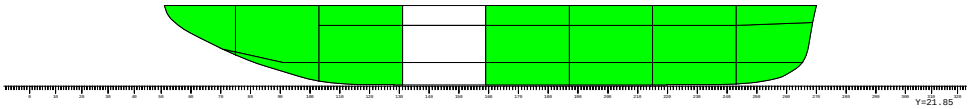


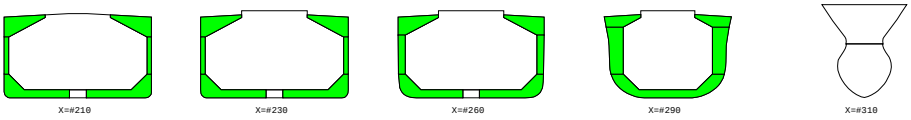
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	393	142	32.93	0.00
8.80	11.00	3516	696	130.60	5.28
12.00	15.00	6362	1040	59.80	22.69
37.60	47.00	52745	3064	405.61	186.53
41.13	51.00	64723	3680	386.29	210.10
62.97	75.00	165784	5000	296.52	314.13
88.45	103.00	270444	2726	473.52	357.63
113.93	131.00	337299	2286	490.96	356.81
139.41	159.00	395705	2313	490.40	349.77
164.89	187.00	365893	-4384	222.77	342.65
190.37	215.00	251480	-4328	405.49	335.49
215.85	243.00	140646	-4102	328.52	326.37
241.33	271.00	42589	-3203	331.42	281.88
266.81	299.00	-1082	169	332.79	143.85
278.01	313.00	-38	0	40.21	45.84

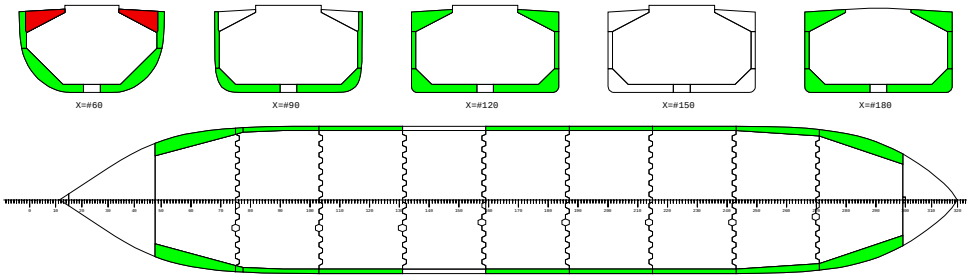
				X	FRAME
SHEAR FORCE (MIN)	-4591.7 t	(43.1%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	5133.4 t	(47.3%)		64.8 m	77
SAGGING MOMENT	-1163.3 tm	(1.9%)		265.9 m	298
HOGGING MOMENT	405437.9 tm	(85.8%)		148.1 m	169

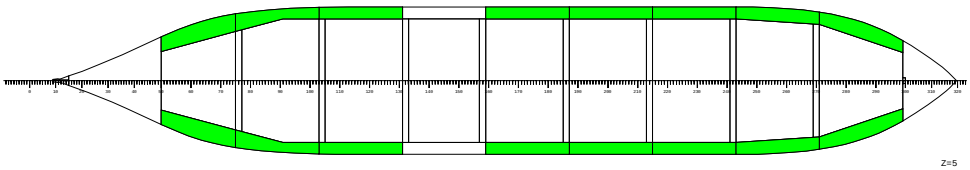
LOADING CONDITION - BALCON3-08 : BAL. SEQU. LC3-WB6+FORE+WB10+AFT EMPTY-50% BUNK

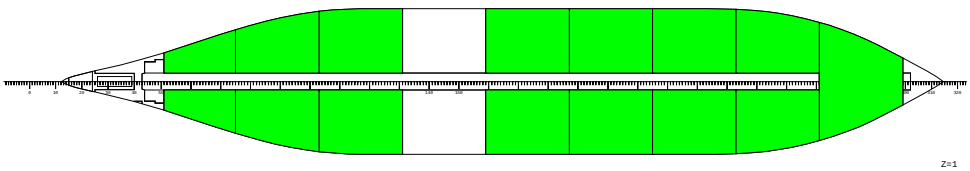












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

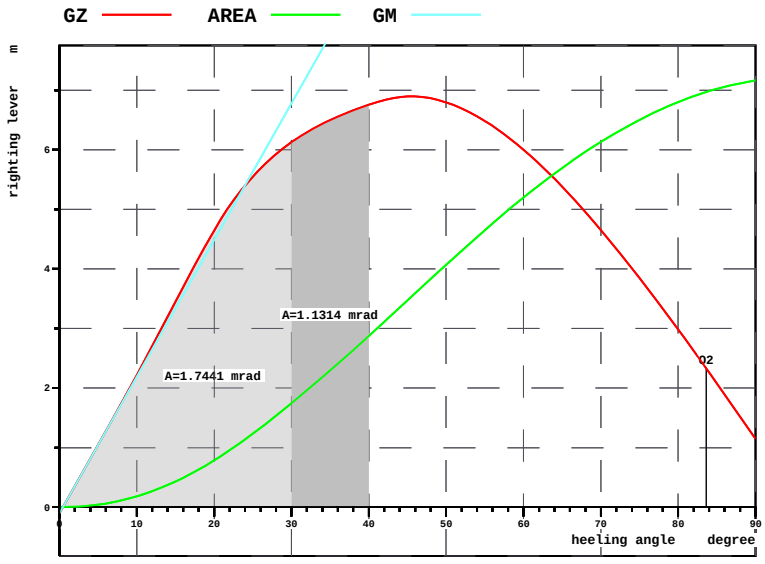
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
SUBTOTAL		48897.0		165.54	-0.00	10.46	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	52133.9	157.67	-0.15 11.21
Total weight	76775.8	149.43	-0.10 12.00

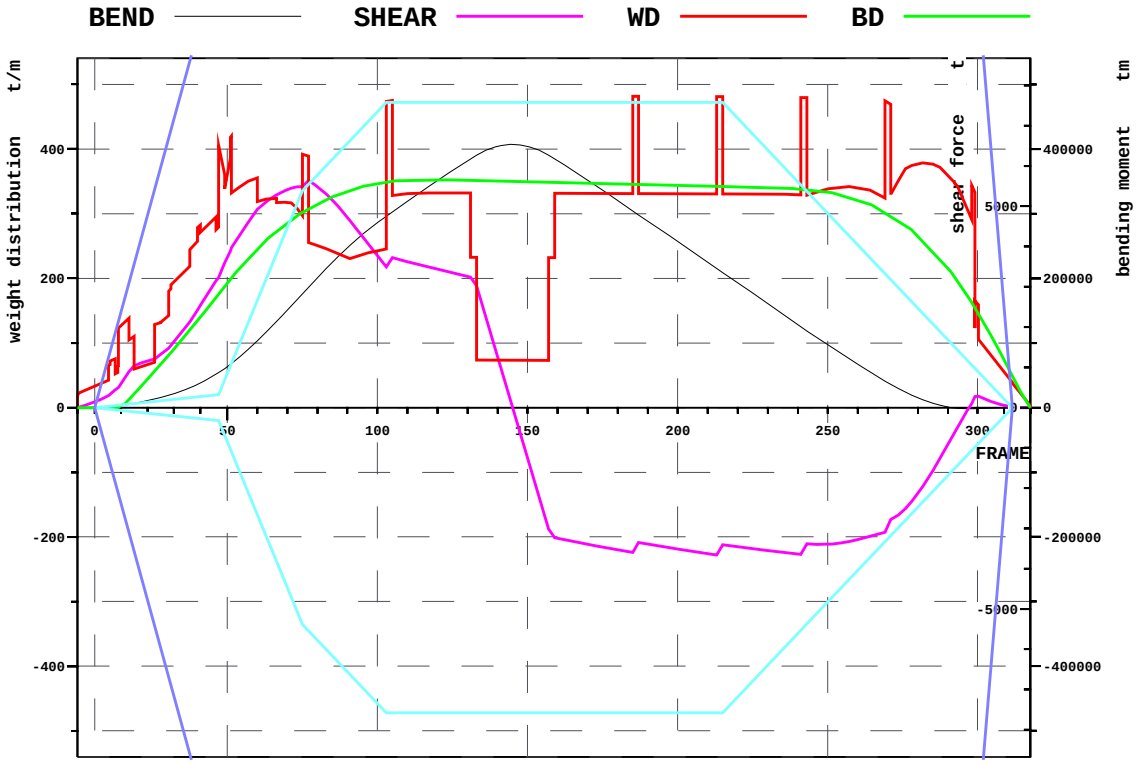
F L O A T I N G P O S I T I O N

Draught moulded	7.548	m	KM	25.16 m
Trim	0.775	m	KG	12.00 m
Heel, PS=+	-0.4	deg		
TA	7.935	m	GM0	13.16 m
TF	7.161	m	GMCORR	0.00 m
Trimming moment	-144486	tonm	GM	13.16 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.04	0.041	273.1	0.004
10.0	2.22	0.183	572.7	0.007
20.0	4.65	0.784	1197.1	0.016
30.0	6.13	1.744	1623.0	0.021
40.0	6.76	2.875	1920.4	0.025
45.0	6.90	3.472	2017.0	0.026
50.0	6.80	4.072	2076.3	0.027
55.0	6.49	4.653	2095.0	0.027
60.0	6.00	5.199	2072.0	0.027
65.0	5.38	5.696	2014.9	0.026
70.0	4.65	6.135	1930.4	0.025
75.0	3.85	6.507	1823.1	0.024
80.0	2.98	6.805	1696.6	0.022
85.0	2.08	7.026	1553.9	0.020
90.0	1.15	7.167	1399.3	0.018



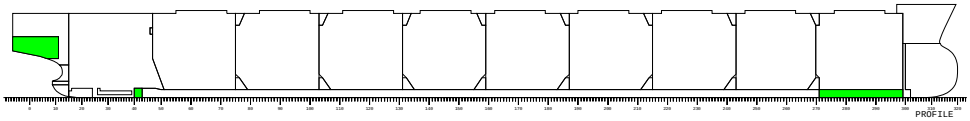
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.744	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.875	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.131	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.900	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.609	deg	OK
GM0.15	GM > 0.15 m	0.150	13.157	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.116		OK

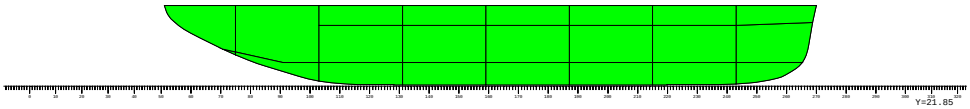


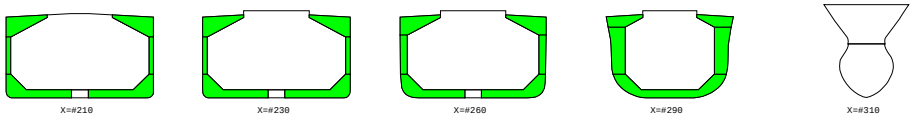
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	376	142	32.93	0.00
8.80	11.00	3471	696	130.57	5.28
12.00	15.00	6307	1041	59.80	21.68
37.60	47.00	54228	3234	405.61	175.42
41.13	51.00	66864	3890	386.29	198.13
62.97	75.00	175449	5487	296.90	301.64
88.45	103.00	296189	3493	473.52	348.41
113.93	131.00	385089	3239	232.48	351.42
139.41	159.00	385136	-3231	231.93	348.22
164.89	187.00	297973	-3352	481.24	344.93
190.37	215.00	208627	-3402	405.49	341.62
215.85	243.00	118884	-3381	328.52	336.32
241.33	271.00	35484	-2780	331.42	294.82
266.81	299.00	-1670	273	332.79	154.28
278.01	313.00	-40	17	40.21	50.76

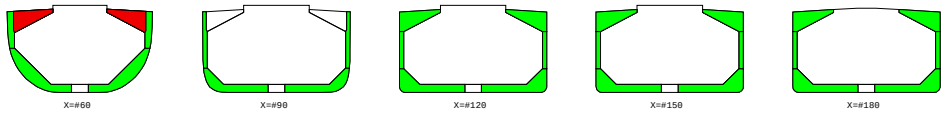
				X	FRAME
SHEAR FORCE (MIN)	-3655.4 t	(34.3%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	5643.7 t	(52.0%)		64.8 m	77
SAGGING MOMENT	-1877.3 tm	(3.0%)		265.6 m	298
HOGGING MOMENT	407248.8 tm	(86.2%)		126.0 m	144

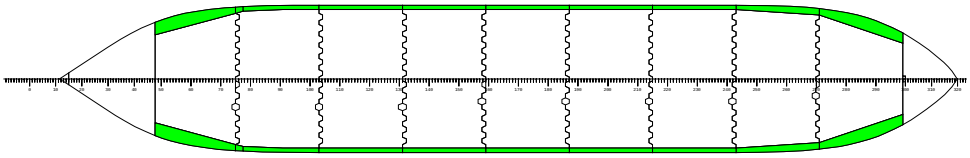
LOADING CONDITION - BALCON3-09 : BALLAST SEQUENCE LC3-FORE PEAK EMPTY-50% BUNKER

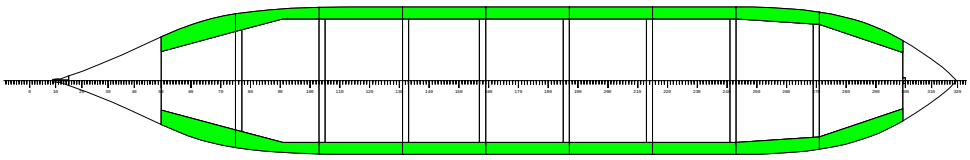


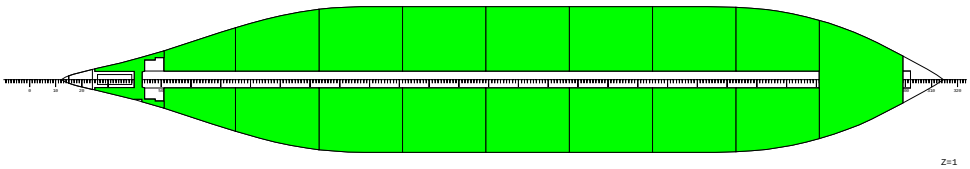












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

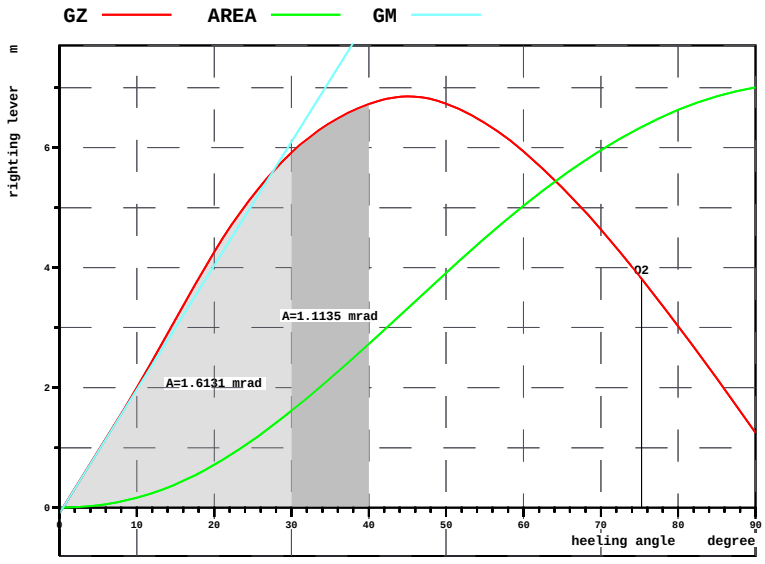
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		57411.3		155.81	-0.00	10.63	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	60648.3	149.57	-0.13 11.26
Total weight	85290.2	144.49	-0.10 11.95

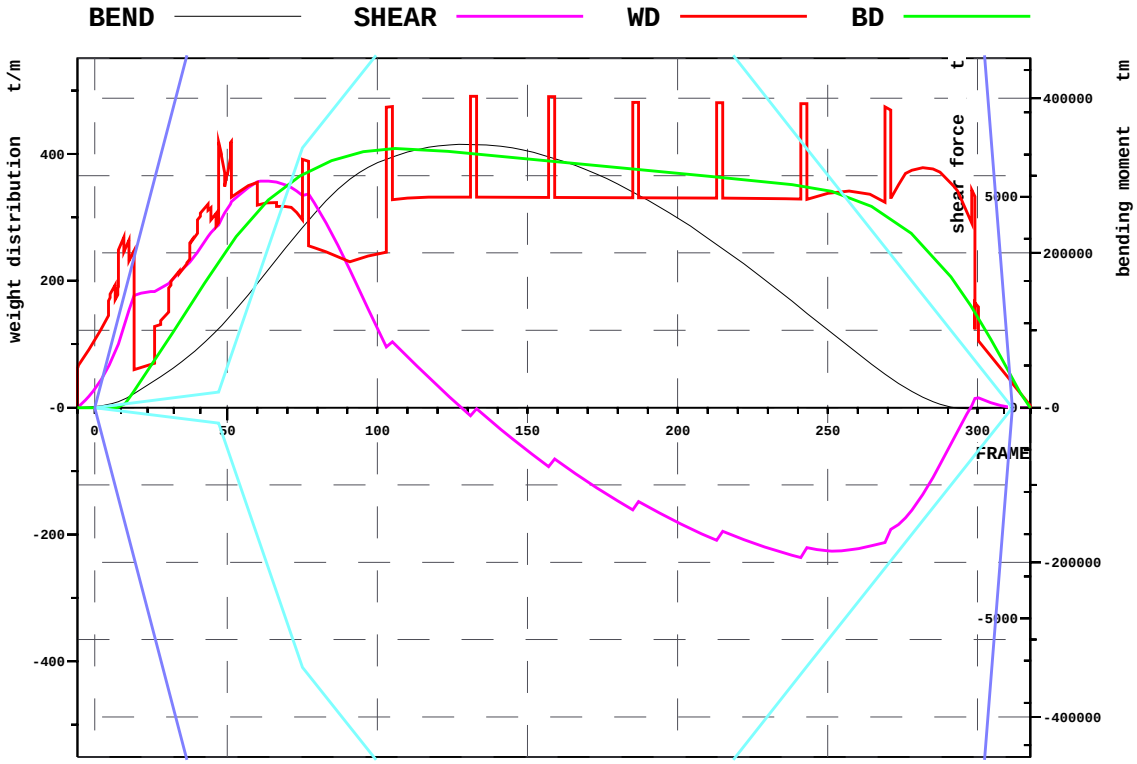
F L O A T I N G P O S I T I O N

Draught moulded	8.411	m	KM	23.78 m
Trim	3.015	m	KG	11.95 m
Heel, PS=+	-0.5	deg		
TA	9.919	m	GM0	11.83 m
TF	6.903	m	GMCORR	0.00 m
Trimming moment	-581535	tonm	GM	11.83 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	0.94	0.037	273.1	0.003
10.0	2.00	0.164	572.7	0.007
20.0	4.26	0.712	1197.1	0.014
30.0	5.92	1.613	1623.0	0.019
40.0	6.72	2.726	1920.4	0.023
45.0	6.85	3.320	2017.0	0.024
50.0	6.73	3.915	2076.3	0.024
55.0	6.41	4.489	2095.0	0.025
60.0	5.94	5.029	2072.0	0.024
65.0	5.34	5.522	2014.9	0.024
70.0	4.63	5.958	1930.4	0.023
75.0	3.86	6.329	1823.1	0.021
80.0	3.02	6.629	1696.6	0.020
85.0	2.15	6.855	1553.9	0.018
90.0	1.25	7.004	1399.3	0.016



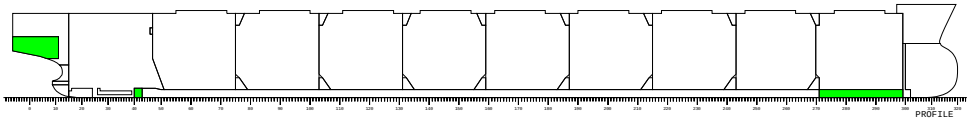
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.613	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.727	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.114	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.854	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.123	deg	OK
GM0.15	GM > 0.15 m	0.150	11.830	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.633		OK

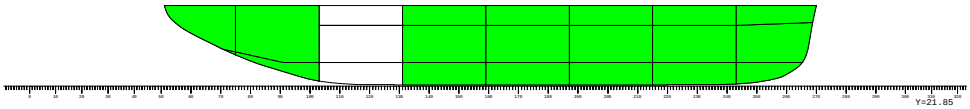


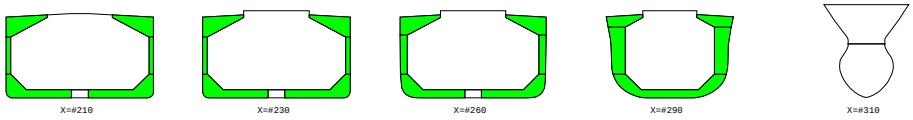
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1174	443	106.41	0.00
8.80	11.00	10625	1918	268.54	5.27
12.00	15.00	18029	2660	59.80	26.69
37.60	47.00	102081	4328	418.44	228.90
41.13	51.00	118354	4824	417.60	256.02
62.97	75.00	232684	5029	295.92	367.26
88.45	103.00	321297	1437	473.46	407.12
113.93	131.00	340037	-194	490.89	400.69
139.41	159.00	321810	-1214	490.33	387.99
164.89	187.00	274422	-2229	481.17	375.21
190.37	215.00	205162	-2931	405.43	362.41
215.85	243.00	121977	-3320	328.48	347.60
241.33	271.00	37788	-2885	331.38	296.60
266.81	299.00	-1484	218	332.74	149.79
278.01	313.00	-82	3	40.21	47.75

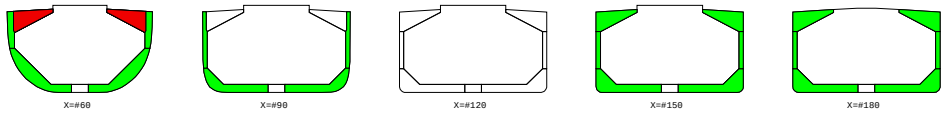
				X	FRAME
SHEAR FORCE (MIN)	-3557.5 t	(34.1%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	5370.7 t	(48.9%)		52.1 m	63
SAGGING MOMENT	-1632.0 tm	(2.6%)		265.6 m	298
HOGGING MOMENT	340240.7 tm	(72.0%)		110.4 m	127

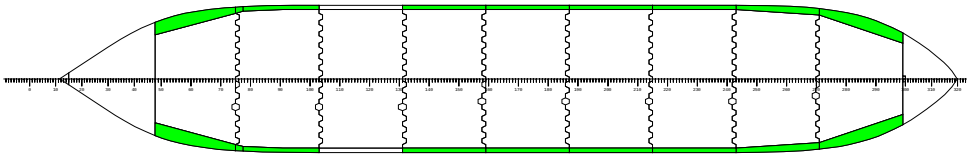
LOADING CONDITION - BALCON3-10 : BALLAST SEQUENCE LC3-WB7+FORE EMPTY-50% BUNKERS

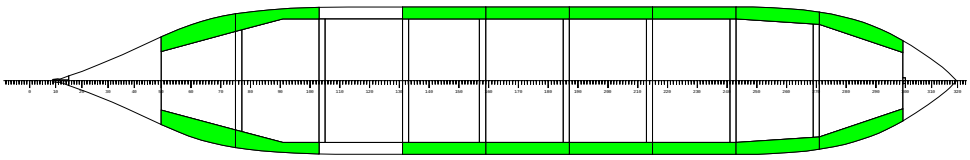


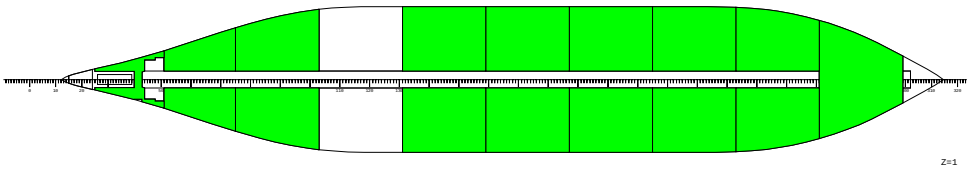












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

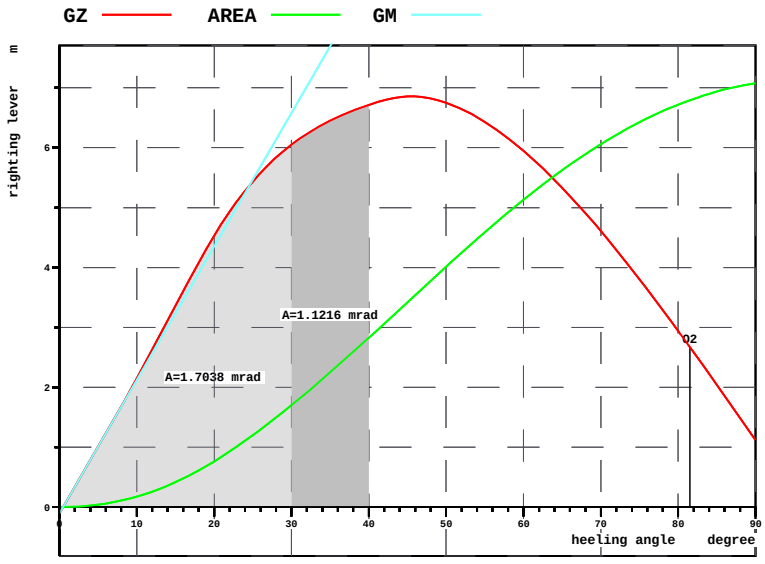
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		50862.5		162.84	-0.00	10.59	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	54099.4	155.42	-0.15 11.30
Total weight	78741.3	148.09	-0.10 12.04

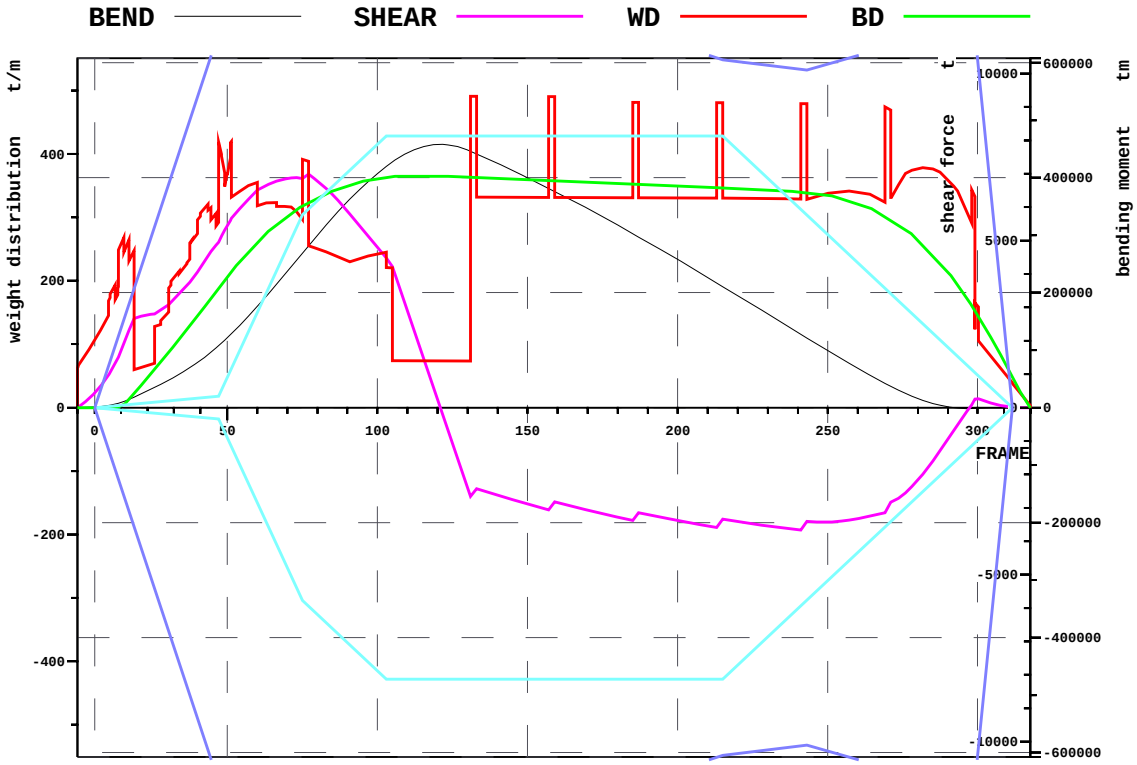
F L O A T I N G P O S I T I O N

Draught moulded	7.751	m	KM	24.81 m
Trim	1.349	m	KG	12.04 m
Heel, PS=+	-0.5	deg		
TA	8.426	m	GM0	12.77 m
TF	7.077	m	GMCORR	0.00 m
Trimming moment	-253464	tonm	GM	12.77 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	1.01	0.040	273.1	0.003
10.0	2.15	0.177	572.7	0.007
20.0	4.53	0.763	1197.1	0.015
30.0	6.05	1.703	1623.0	0.021
40.0	6.71	2.825	1920.4	0.024
45.0	6.86	3.418	2017.0	0.026
50.0	6.75	4.014	2076.3	0.026
55.0	6.43	4.590	2095.0	0.027
60.0	5.95	5.131	2072.0	0.026
65.0	5.33	5.624	2014.9	0.026
70.0	4.61	6.059	1930.4	0.025
75.0	3.80	6.426	1823.1	0.023
80.0	2.94	6.721	1696.6	0.022
85.0	2.04	6.939	1553.9	0.020
90.0	1.12	7.077	1399.3	0.018



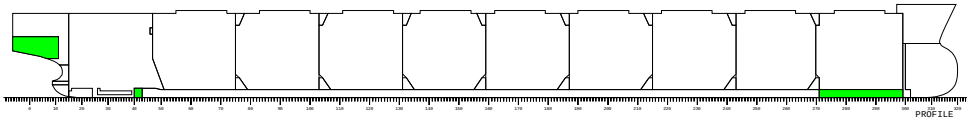
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.704	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.825	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.122	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.857	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.498	deg	OK
GM0.15	GM > 0.15 m	0.150	12.765	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.223		OK

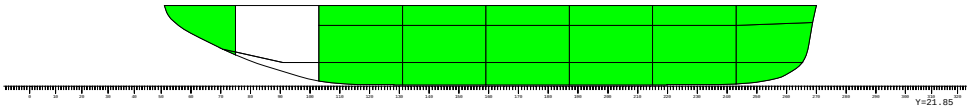


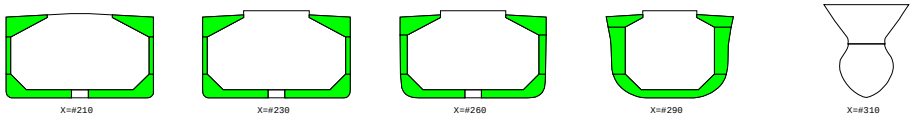
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1123	444	106.42	0.00
8.80	11.00	10489	1918	268.51	5.28
12.00	15.00	17866	2666	59.80	22.73
37.60	47.00	107652	4951	418.46	188.01
41.13	51.00	126351	5597	417.62	211.87
62.97	75.00	268675	6859	296.64	317.55
88.45	103.00	419367	4477	220.55	362.62
113.93	131.00	445767	-2662	490.91	363.22
139.41	159.00	375799	-2817	490.36	357.58
164.89	187.00	296425	-3146	481.20	351.87
190.37	215.00	210347	-3343	405.45	346.12
215.85	243.00	120928	-3407	328.50	338.39
241.33	271.00	36488	-2828	331.39	294.51
266.81	299.00	-1567	254	332.76	152.64
278.01	313.00	-42	13	40.21	49.77

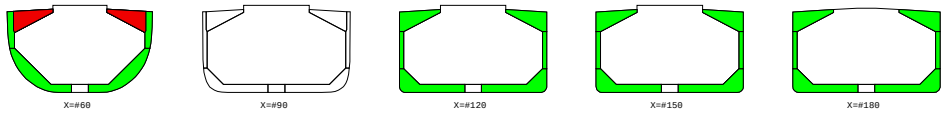
				X	FRAME
SHEAR FORCE (MIN)	-3661.8 t	(35.1%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	6986.7 t	(64.4%)		64.8 m	77
SAGGING MOMENT	-1751.4 tm	(2.8%)		265.6 m	298
HOGGING MOMENT	457853.4 tm	(96.9%)		105.5 m	122

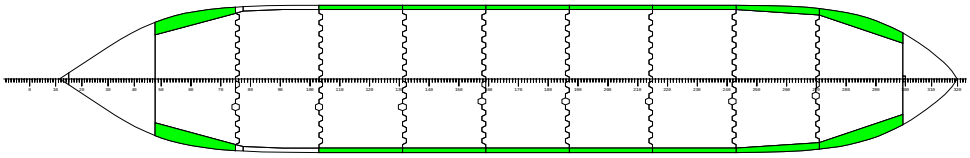
LOADING CONDITION - BALCON3-11 : BALLAST SEQUENCE LC3-WB8+FORE EMPTY-50% BUNKERS

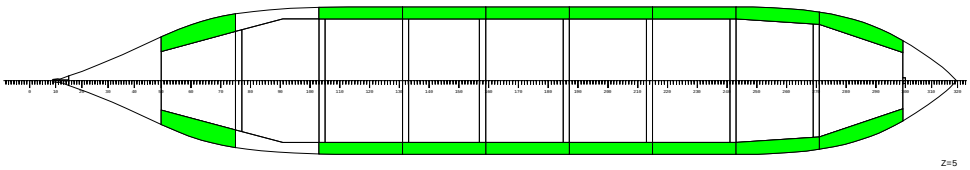


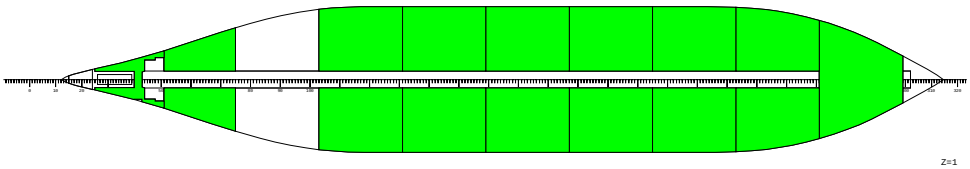












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

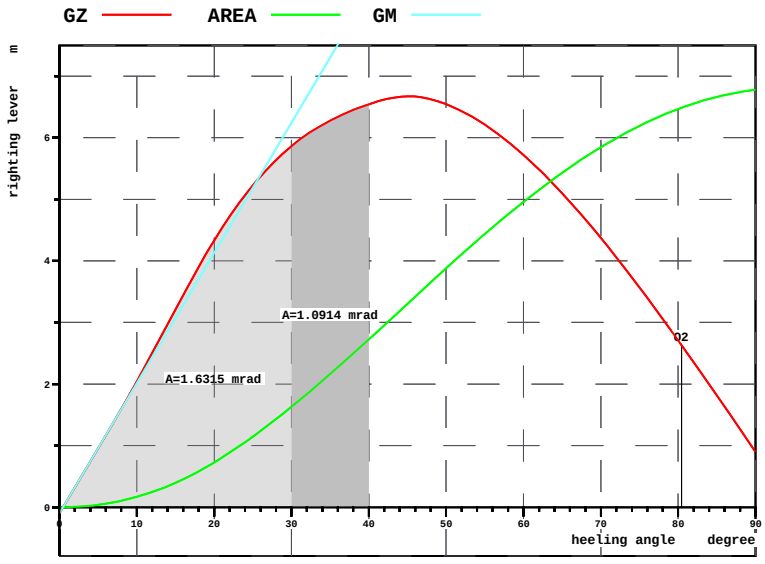
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		53135.6		162.27	-0.00	10.99	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	56372.6	155.18	-0.14 11.65
Total weight	81014.4	148.13	-0.10 12.26

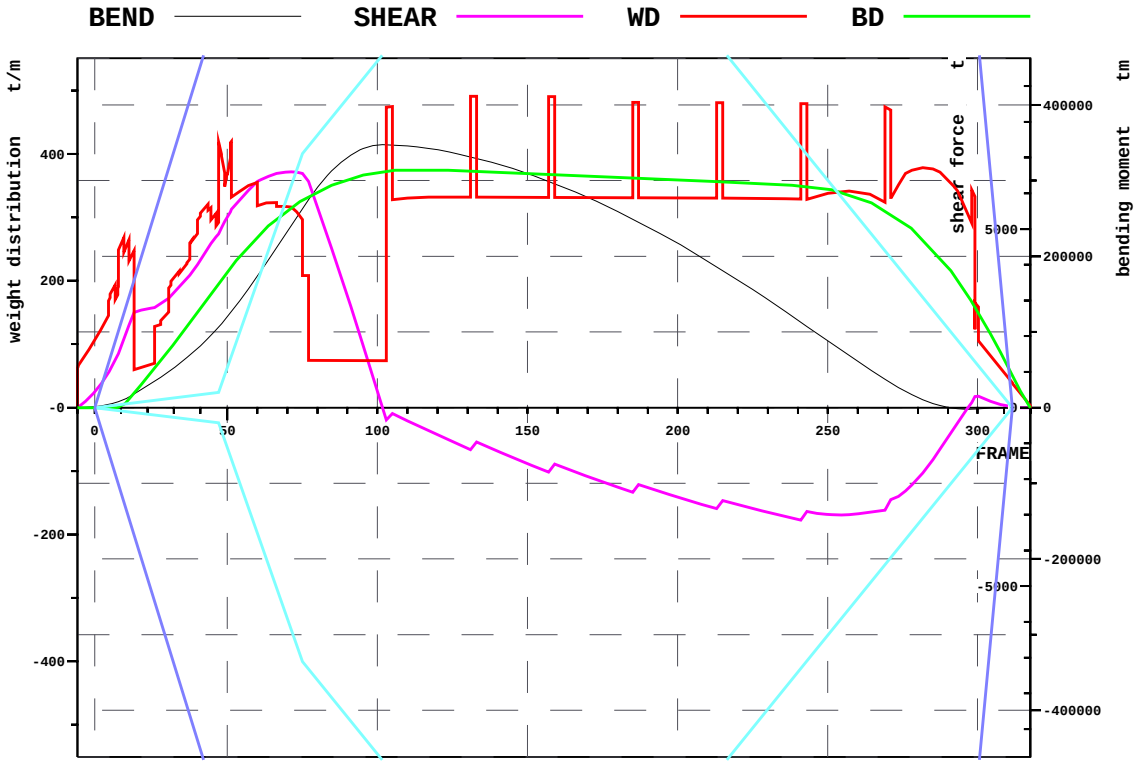
F L O A T I N G P O S I T I O N

Draught moulded	7.960	m	KM	24.39 m
Trim	1.350	m	KG	12.26 m
Heel, PS=+	-0.5	deg		
TA	8.635	m	GM0	12.12 m
TF	7.285	m	GMCORR	0.00 m
Trimming moment	-254946	tonm	GM	12.12 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	0.96	0.037	273.1	0.003
10.0	2.04	0.168	572.7	0.007
20.0	4.33	0.727	1197.1	0.015
30.0	5.86	1.631	1623.0	0.020
40.0	6.54	2.723	1920.4	0.024
45.0	6.67	3.301	2017.0	0.025
50.0	6.55	3.879	2076.3	0.026
55.0	6.21	4.437	2095.0	0.026
60.0	5.72	4.959	2072.0	0.026
65.0	5.10	5.432	2014.9	0.025
70.0	4.37	5.845	1930.4	0.024
75.0	3.57	6.193	1823.1	0.023
80.0	2.71	6.467	1696.6	0.021
85.0	1.81	6.665	1553.9	0.019
90.0	0.90	6.783	1399.3	0.017



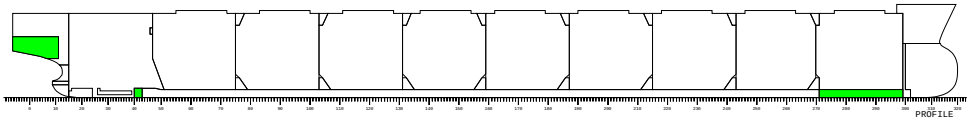
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.632	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.723	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.091	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.673	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.171	deg	OK
GM0.15	GM > 0.15 m	0.150	12.124	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.329		OK

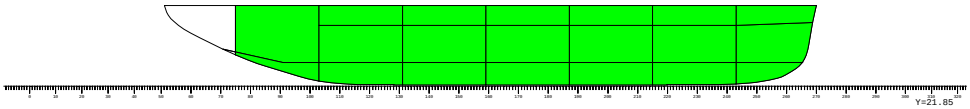


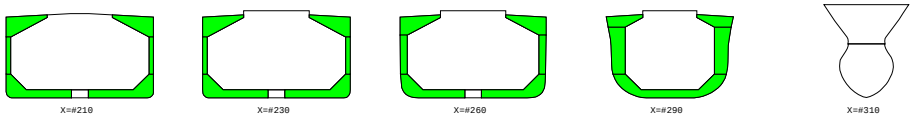
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1126	444	106.41	0.00
8.80	11.00	10497	1918	268.49	5.28
12.00	15.00	17875	2665	59.80	23.23
37.60	47.00	106807	4858	418.44	194.55
41.13	51.00	125136	5480	417.60	219.10
62.97	75.00	262951	6556	296.62	326.83
88.45	103.00	347541	-346	473.46	372.28
113.93	131.00	331154	-1179	490.89	372.89
139.41	159.00	295852	-1581	490.34	367.25
164.89	187.00	244855	-2157	481.18	361.53
190.37	215.00	180868	-2600	405.44	355.79
215.85	243.00	107256	-2910	328.48	348.04
241.33	271.00	32370	-2574	331.38	303.59
266.81	299.00	-1905	309	332.75	158.39
278.01	313.00	-56	21	40.21	52.26

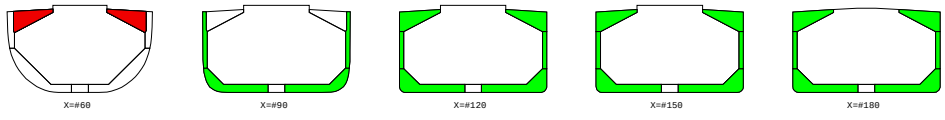
				X	FRAME
SHEAR FORCE (MIN)	-3147.8 t	(30.2%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	6602.0 t	(60.7%)		59.6 m	71
SAGGING MOMENT	-2206.5 tm	(3.2%)		264.4 m	296
HOGGING MOMENT	347735.1 tm	(74.6%)		87.2 m	102

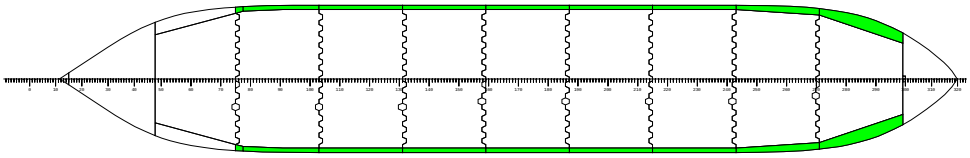
LOADING CONDITION - BALCON3-12 : BALLAST SEQUENCE LC3-WB9+FORE EMPTY-50% BUNKERS

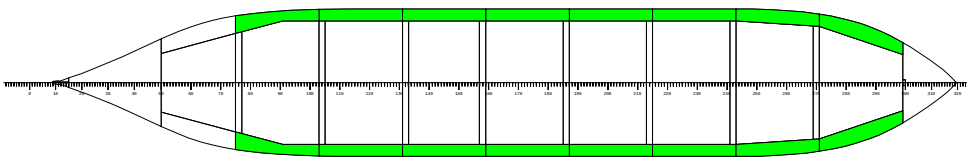


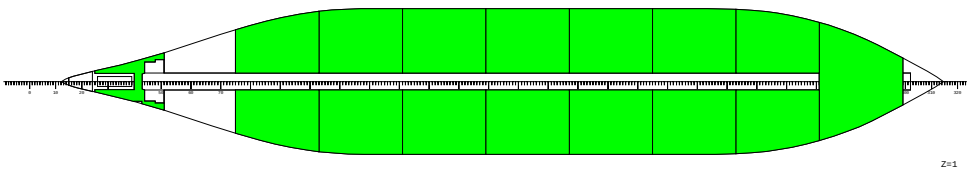












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

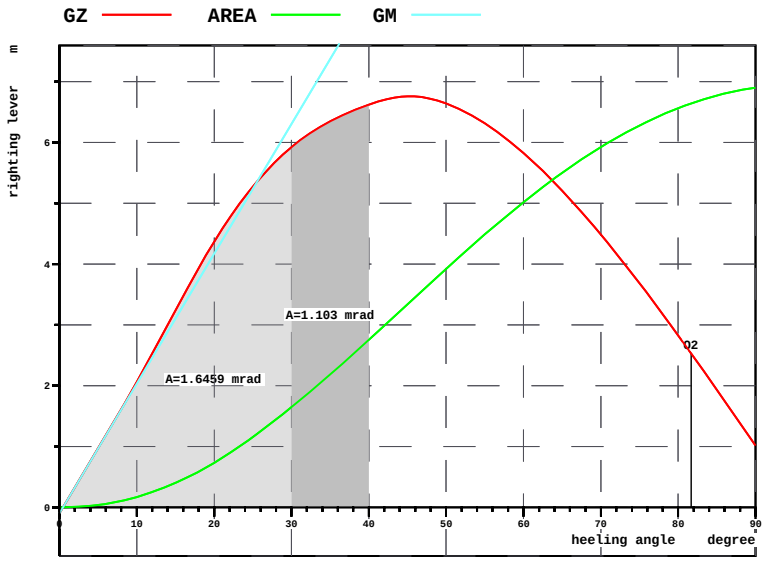
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		53063.1		164.38	-0.00	10.79	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	56300.0	157.16	-0.14 11.46
Total weight	80941.9	149.50	-0.10 12.13

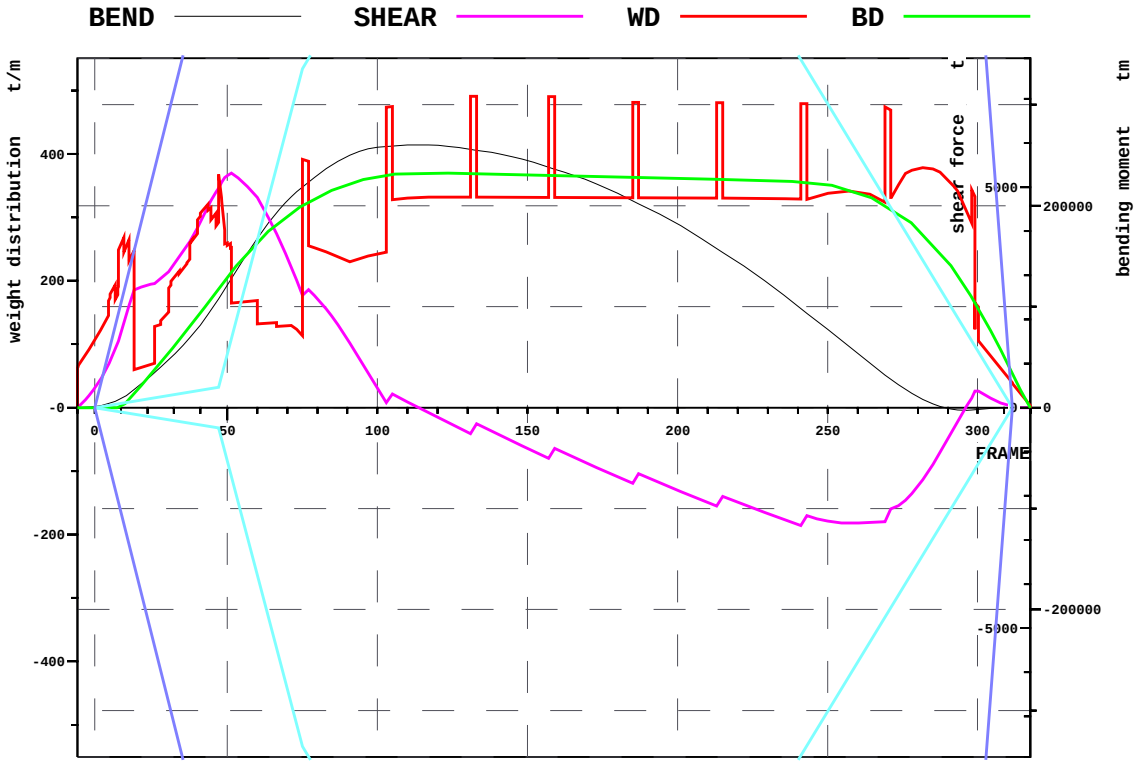
F L O A T I N G P O S I T I O N

Draught moulded	7.931	m	KM	24.36 m
Trim	0.754	m	KG	12.13 m
Heel, PS=+	-0.5	deg		
TA	8.308	m	GM0	12.23 m
TF	7.554	m	GMCORR	0.00 m
Trimming moment	-141991	tonm	GM	12.23 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.10	0.000	0.0	0.000
5.0	0.97	0.038	273.1	0.003
10.0	2.06	0.169	572.7	0.007
20.0	4.37	0.733	1197.1	0.015
30.0	5.92	1.646	1623.0	0.020
40.0	6.62	2.749	1920.4	0.024
45.0	6.76	3.334	2017.0	0.025
50.0	6.64	3.920	2076.3	0.026
55.0	6.31	4.486	2095.0	0.026
60.0	5.82	5.017	2072.0	0.026
65.0	5.21	5.499	2014.9	0.025
70.0	4.48	5.922	1930.4	0.024
75.0	3.68	6.279	1823.1	0.023
80.0	2.83	6.564	1696.6	0.021
85.0	1.93	6.771	1553.9	0.019
90.0	1.02	6.900	1399.3	0.017



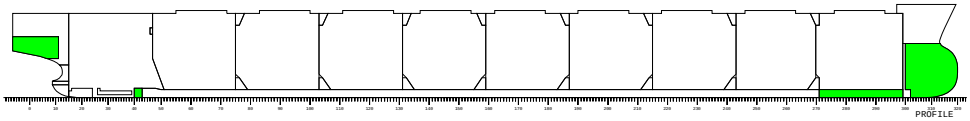
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.646	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.749	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.103	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.757	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.307	deg	OK
GM0.15	GM > 0.15 m	0.150	12.231	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.350		OK

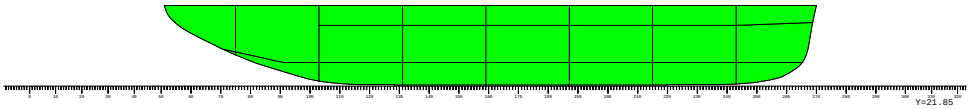


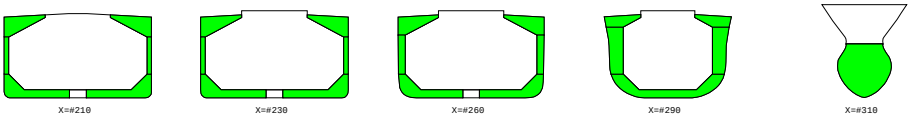
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1096	443	106.40	0.00
8.80	11.00	10414	1917	268.45	5.28
12.00	15.00	17774	2666	59.79	22.52
37.60	47.00	107668	4974	367.70	186.91
41.13	51.00	125952	5303	251.46	210.89
62.97	75.00	218523	2549	113.48	318.32
88.45	103.00	258712	114	473.41	365.93
113.93	131.00	255669	-590	490.84	369.05
139.41	159.00	236178	-927	490.29	365.94
164.89	187.00	201824	-1503	481.13	362.74
190.37	215.00	153674	-2010	405.39	359.51
215.85	243.00	93458	-2448	328.45	354.28
241.33	271.00	27914	-2300	331.34	311.82
266.81	299.00	-2223	377	332.71	165.14
278.01	313.00	-36	31	40.21	55.50

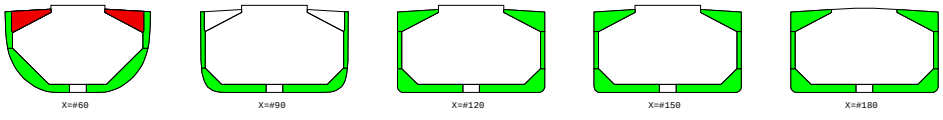
				X	FRAME
SHEAR FORCE (MIN)	-2674.1 t	(25.7%)	POSITION:	214.0 m	241
SHEAR FORCE (MAX)	5317.5 t	(47.6%)		41.5 m	51
SAGGING MOMENT	-2693.3 tm	(3.9%)		264.4 m	296
HOGGING MOMENT	260304.4 tm	(55.1%)		97.7 m	113

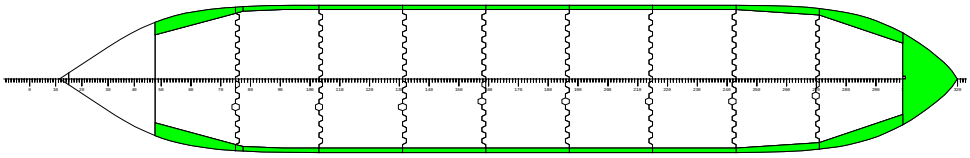
LOADING CONDITION - BALCON3-13 : BALLAST SEQUENCE LC3-NORMAL BAL. -50% BUNKERS

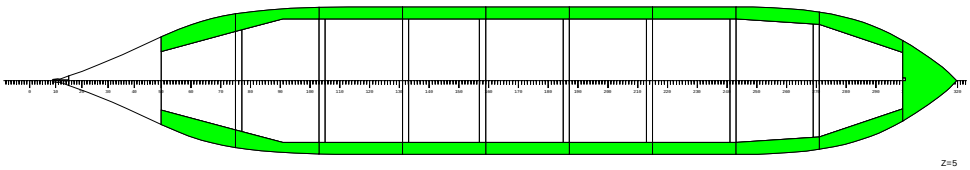


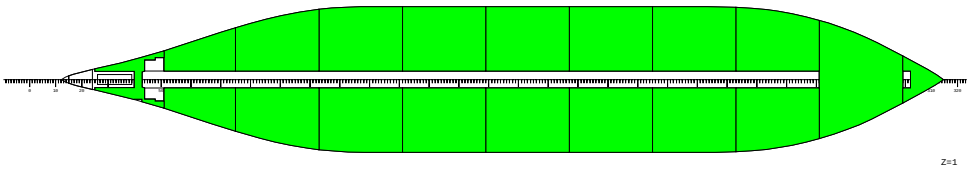












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00

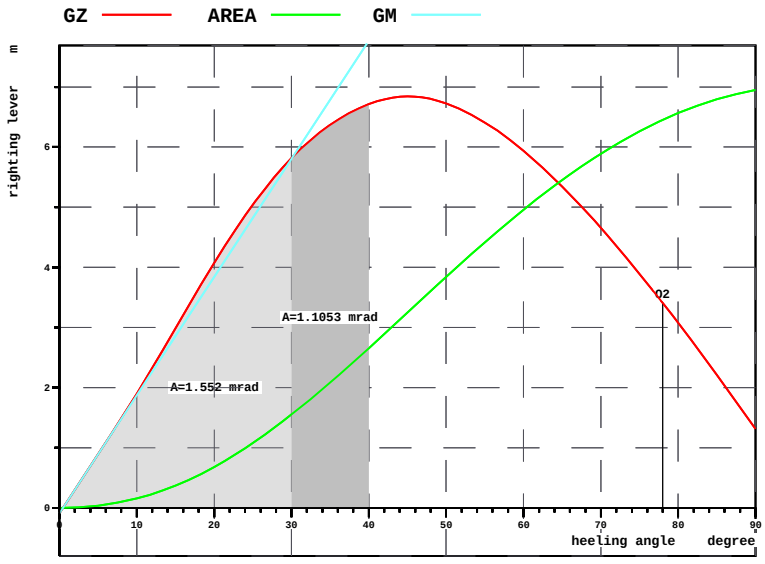
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	64211.2	156.38	-0.13 11.14
Total weight	88853.1	149.62	-0.09 11.84

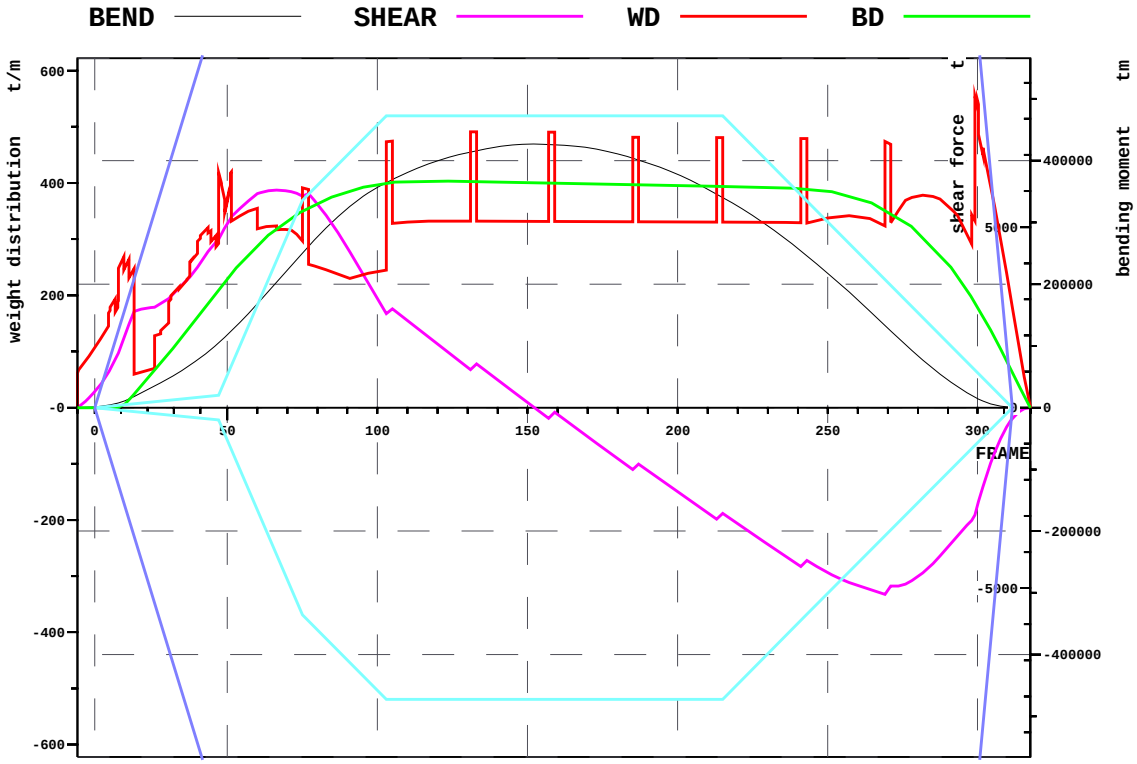
F L O A T I N G P O S I T I O N

Draught moulded	8.654	m	KM	23.10 m
Trim	0.698	m	KG	11.84 m
Heel, PS=+	-0.5	deg		
TA	9.003	m	GM0	11.26 m
TF	8.306	m	GMCORR	0.00 m
Trimming moment	-133550	tonm	GM	11.26 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.09	0.000	0.0	0.000
5.0	0.89	0.035	273.1	0.003
10.0	1.90	0.156	572.7	0.006
20.0	4.07	0.678	1197.1	0.013
30.0	5.82	1.552	1623.0	0.018
40.0	6.71	2.657	1920.4	0.022
45.0	6.84	3.250	2017.0	0.023
50.0	6.72	3.844	2076.3	0.023
55.0	6.41	4.418	2095.0	0.024
60.0	5.94	4.958	2072.0	0.023
65.0	5.34	5.451	2014.9	0.023
70.0	4.66	5.888	1930.4	0.022
75.0	3.89	6.261	1823.1	0.021
80.0	3.08	6.565	1696.6	0.019
85.0	2.21	6.796	1553.9	0.017
90.0	1.31	6.950	1399.3	0.016



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.552	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.657	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.105	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.844	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.089	deg	OK
GM0.15	GM > 0.15 m	0.150	11.259	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.913		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1116	444	106.41	0.00
8.80	11.00	10470	1918	268.47	5.28
12.00	15.00	17841	2664	59.79	24.23
37.60	47.00	104895	4657	418.43	209.16
41.13	51.00	122416	5225	417.60	235.44
62.97	75.00	250069	5861	344.16	349.78
88.45	103.00	364325	2601	473.45	398.88
113.93	131.00	414034	1054	490.88	402.27
139.41	159.00	425749	-131	490.33	399.39
164.89	187.00	400902	-1561	481.17	396.43
190.37	215.00	340403	-2930	405.43	393.45
215.85	243.00	245826	-4234	328.48	388.41
241.33	271.00	123850	-4944	331.37	344.28
266.81	299.00	16700	-2984	332.74	186.14
278.01	313.00	605	-322	182.48	64.81

				X	FRAME
SHEAR FORCE (MIN)	-5170.2 t	(47.1%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	6029.0 t	(55.1%)		55.1 m	66
SAGGING MOMENT	-				
HOGGING MOMENT	426760.5 tm	(90.3%)		133.0 m	152

8. Ballast exchange sequence based on loading condition No6

Ballast sequence:174000 DWT DOUBLE SKIN BULK CARRIER (W=24641.93ton,LCG=132.014m&vcg=13.667m)

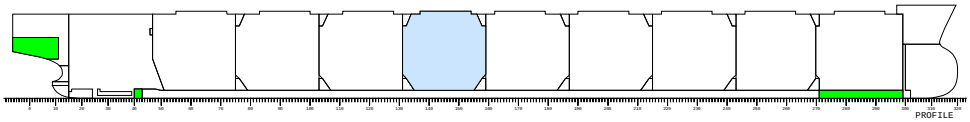
Loading condition No.6 Heavy Ballast – 50% Bunkers

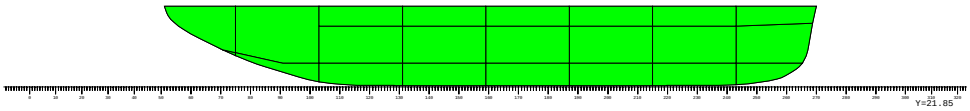
									Line of sight
No.	Condition	Bunkers	Stability	Long. Strength	Draught Aft	I/D 60%	Draught Fwd	>8.0 m	<500 m
			[OK/Not OK]	[OK/Not OK]	[m]	[Yes/No]	[m]	[Yes/No]	[Yes/No]
1	Ballast sequence –Fore peak empty	50%	OK	OK	12.95	Yes	7.79	No[1)]	No[2)]
2	Ballast sequence –WB1 empty	50%	OK	OK	13.45	Yes	6.70	No[1)]	No[2)]
3	Ballast sequence –WB2 empty	50%	OK	OK	12.94	Yes	7.26	No[1)]	No[2)]
4	Ballast sequence –WB3 empty	50%	OK	OK	12.51	Yes	7.69	No[1)]	No[2)]
5	Ballast sequence –WB4 empty	50%	OK	OK	12.08	Yes	8.08	Yes	No[2)]
6	Ballast sequence –WB5 empty	50%	OK	OK	11.64	Yes	8.47	Yes	No[2)]
7	Ballast sequence –WB6 empty	50%	OK	OK	11.20	Yes	8.86	Yes	Yes
8	Ballast sequence –WB7 TS/DS empty	50%	OK	OK	11.48	Yes	9.19	Yes	Yes
9	Ballast sequence –WB7 DB empty	50%	OK	OK	11.43	Yes	9.19	Yes	Yes
10	Ballast sequence –WB8empty	50%	OK	OK	10.95	Yes	9.47	Yes	Yes
11	Ballast sequence –WB9 empty	50%	OK	OK	10.65	Yes	9.73	Yes	Yes
12	Ballast sequence –WB10+A.P. empty	50%	OK	OK	11.27	Yes	9.59	Yes	Yes

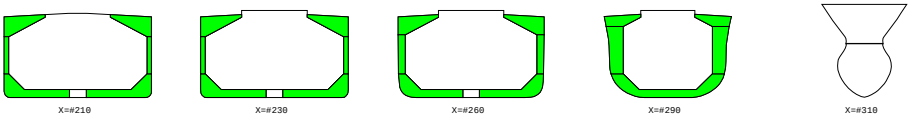
1) The exchange is not to be carried out in rough sea to avoid forward bottom slamming. Condition 2 to 4 above can be splitted into three operations: exchange bottom tanks, double side tanks and top side tanks after each other. Draught fwd will then be more than 8.0 m.

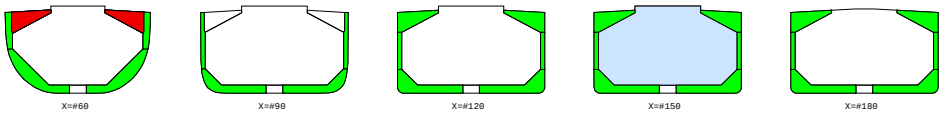
2) The exchange is not to be carried out in an open sea confirming that no other vessel is likely in the visibility. Condition 2 and 6 above can be splitted into three operations:exchange bottom tanks, double side tanks and top side tanks after each other. Line of sight will than be less than 500 m.

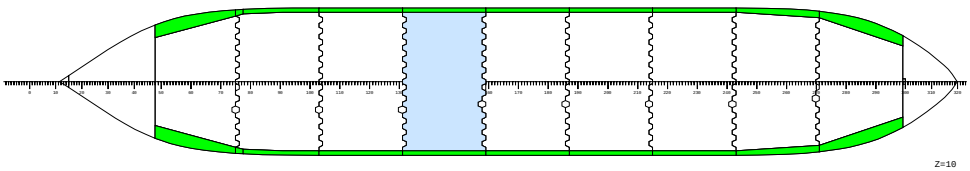
LOADING CONDITION - BALCON6-01 : BALLAST SEQUENCE LC6-FORE EMPTY-50% BUNKERS

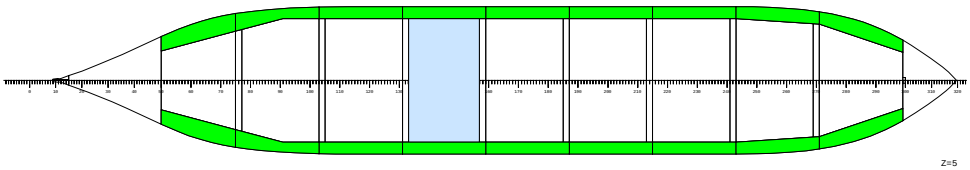


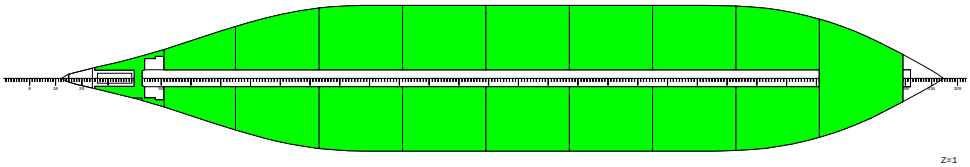












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

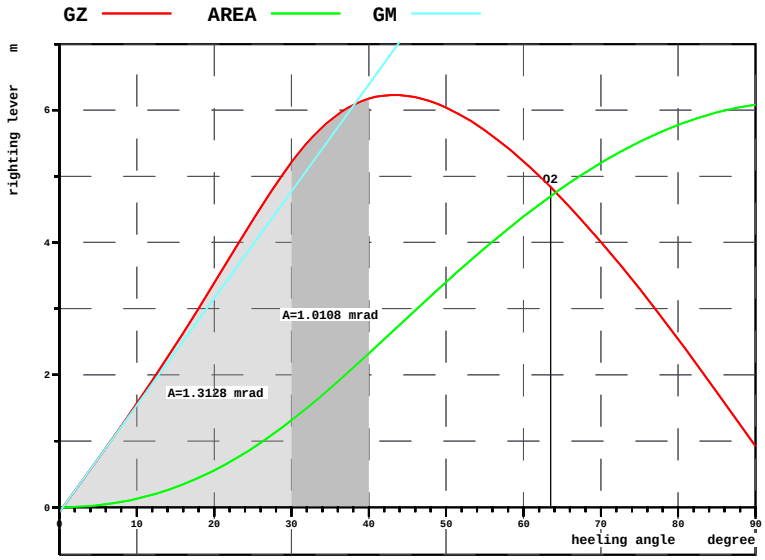
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		57411.3		155.81	-0.00	10.63	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	81811.8	143.64	-0.09 11.88
Total weight	106453.7	140.94	-0.07 12.29

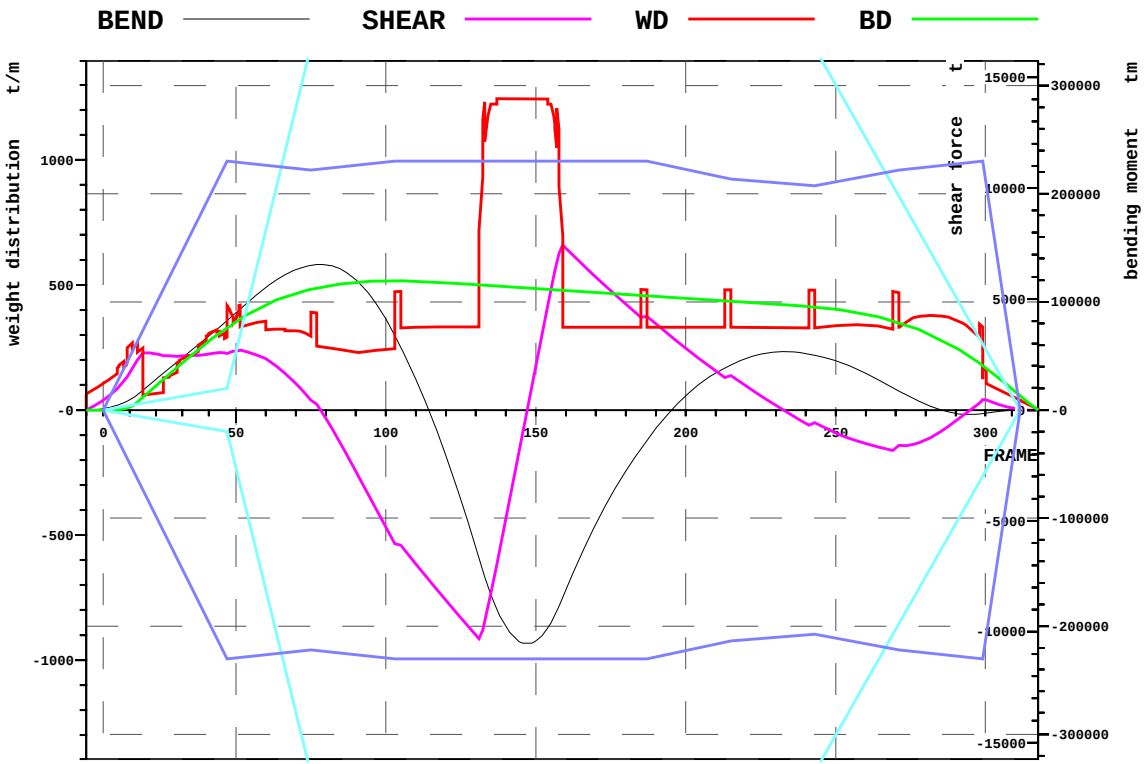
F L O A T I N G P O S I T I O N

Draught moulded	10.371	m	KM	21.54 m
Trim	5.166	m	KG	12.29 m
Heel, PS=+	-0.4	deg		
TA	12.954	m	GM0	9.25 m
TF	7.788	m	GMCORR	0.00 m
Trimming moment	******	tonm	GM	9.25 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.74	0.029	273.1	0.003
10.0	1.57	0.129	572.7	0.005
20.0	3.38	0.558	1197.1	0.011
30.0	5.21	1.313	1623.0	0.015
40.0	6.17	2.323	1920.4	0.018
45.0	6.21	2.865	2017.0	0.019
50.0	6.03	3.401	2076.3	0.020
55.0	5.69	3.914	2095.0	0.020
60.0	5.22	4.391	2072.0	0.019
65.0	4.66	4.823	2014.9	0.019
70.0	4.01	5.201	1930.4	0.018
75.0	3.30	5.521	1823.1	0.017
80.0	2.54	5.775	1696.6	0.016
85.0	1.74	5.962	1553.9	0.015
90.0	0.92	6.079	1399.3	0.013



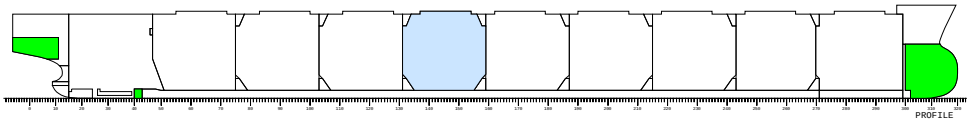
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.313	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.324	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.011	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.226	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.276	deg	OK
GM0.15	GM > 0.15 m	0.150	9.250	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.925		OK

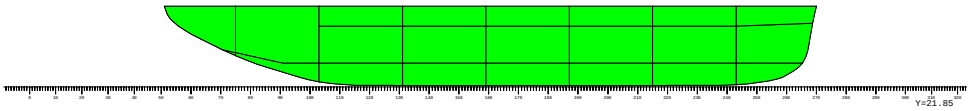


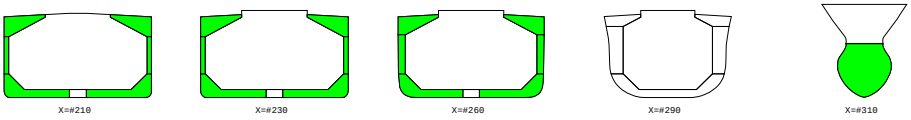
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1252	443	106.39	0.00
8.80	11.00	10789	1893	268.55	16.28
12.00	15.00	18083	2580	59.78	49.95
37.60	47.00	82884	2544	418.35	328.04
41.13	51.00	92280	2680	417.51	360.64
62.97	75.00	133756	435	343.15	482.22
88.45	103.00	68862	-6021	473.36	515.84
113.93	131.00	-136388	-10309	715.77	500.35
139.41	159.00	-172976	7423	696.99	478.55
164.89	187.00	-28312	4216	481.08	456.67
190.37	215.00	41563	1553	405.35	434.76
215.85	243.00	50508	-564	328.41	410.77
241.33	271.00	17511	-1603	331.31	347.68
266.81	299.00	-3184	464	332.67	176.85
278.01	313.00	-216	39	40.20	58.44

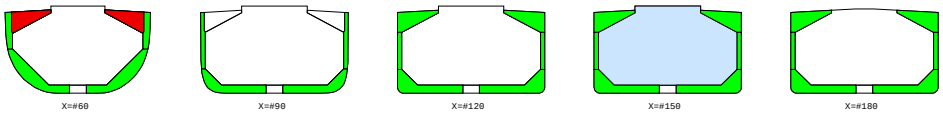
				X	FRAME
SHEAR FORCE (MIN)	-10309.0 t	(91.8%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	7423.3 t	(66.1%)		139.4 m	159
SAGGING MOMENT	-215677 tm	(49.2%)		130.0 m	149
HOGGING MOMENT	134592.2 tm	(37.9%)		66.6 m	79

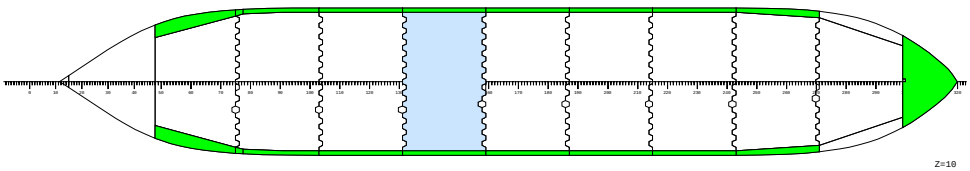
LOADING CONDITION - BALCON6-02 : BALLAST SEQUENCE LC6-WB1 EMPTY-50% BUNKERS

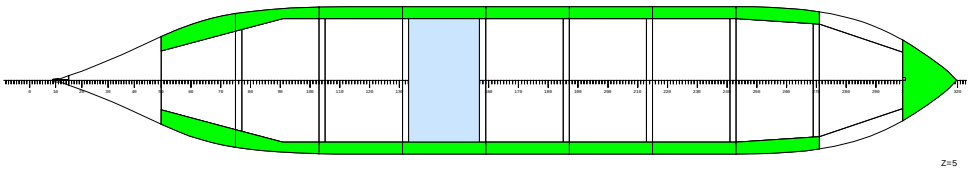


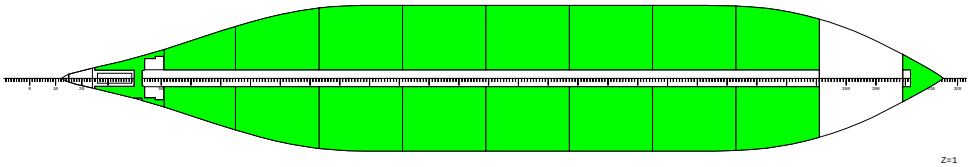












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

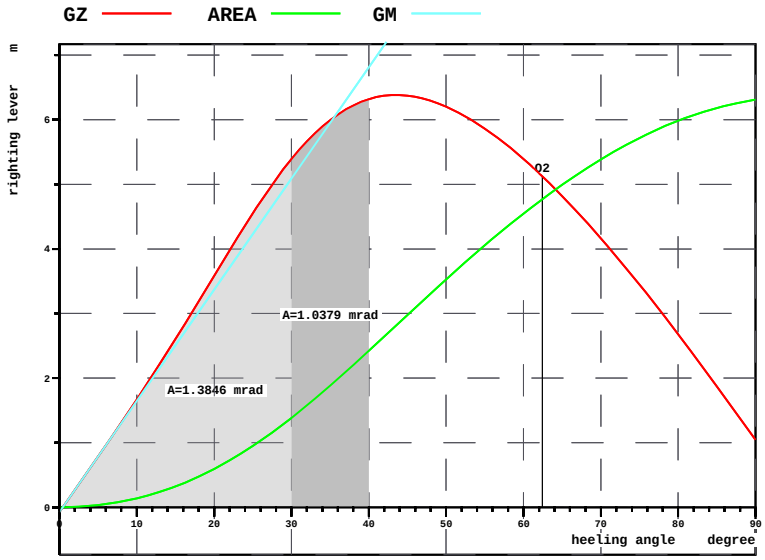
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		53714.5		150.32	-0.00	10.33	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78115.0	139.28	-0.10 11.73
Total weight	102756.8	137.54	-0.07 12.19

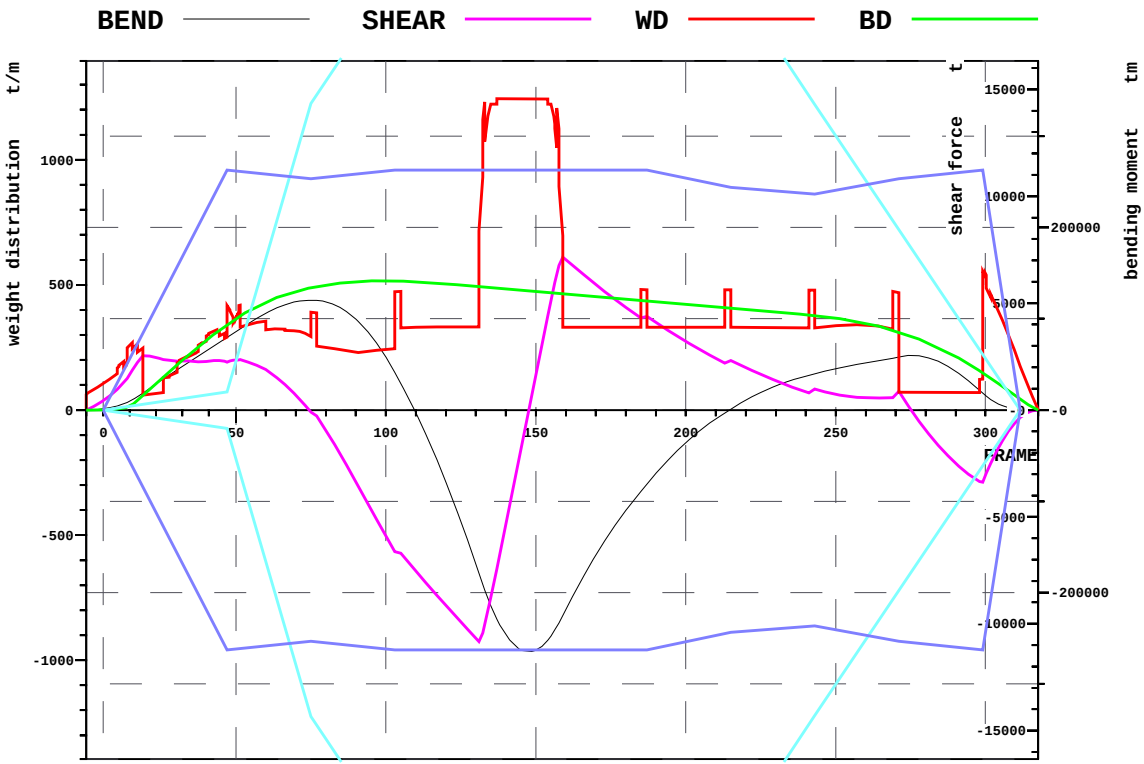
F L O A T I N G P O S I T I O N

Draught moulded	10.075	m	KM	22.06 m
Trim	6.753	m	KG	12.19 m
Heel, PS=+	-0.4	deg		
TA	13.451	m	GM0	9.86 m
TF	6.699	m	GMCORR	0.00 m
Trimming moment	******	tonm	GM	9.86 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.79	0.031	273.1	0.003
10.0	1.67	0.138	572.7	0.006
20.0	3.58	0.594	1197.1	0.012
30.0	5.39	1.384	1623.0	0.016
40.0	6.32	2.422	1920.4	0.019
45.0	6.37	2.978	2017.0	0.020
50.0	6.20	3.528	2076.3	0.020
55.0	5.86	4.055	2095.0	0.020
60.0	5.39	4.547	2072.0	0.020
65.0	4.82	4.994	2014.9	0.020
70.0	4.16	5.386	1930.4	0.019
75.0	3.45	5.719	1823.1	0.018
80.0	2.68	5.987	1696.6	0.017
85.0	1.87	6.186	1553.9	0.015
90.0	1.04	6.313	1399.3	0.014



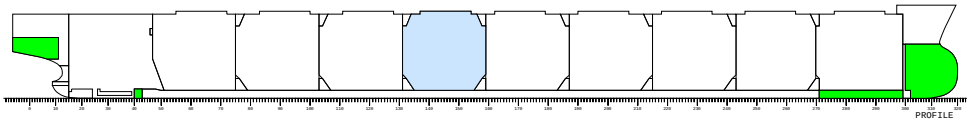
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.385	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.423	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.038	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.383	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.523	deg	OK
GM0.15	GM > 0.15 m	0.150	9.860	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.642		OK

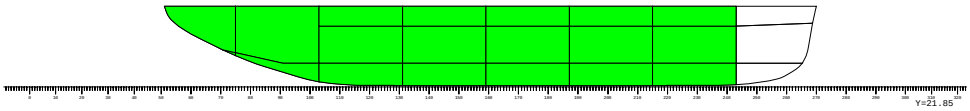


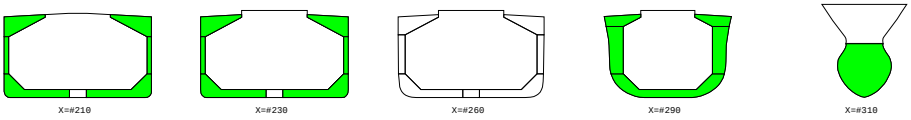
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1382	443	106.35	0.26
8.80	11.00	11061	1868	268.48	21.59
12.00	15.00	18324	2535	59.76	56.88
37.60	47.00	79542	2241	418.17	338.93
41.13	51.00	87890	2338	417.34	371.24
62.97	75.00	120214	-98	294.18	488.48
88.45	103.00	41028	-6633	473.16	515.50
113.93	131.00	-178356	-10829	715.46	493.30
139.41	159.00	-224732	7153	696.69	464.78
164.89	187.00	-81148	4379	480.87	436.18
190.37	215.00	861	2320	405.17	407.55
215.85	243.00	39509	978	328.27	376.89
241.33	271.00	58034	869	70.65	309.41
266.81	299.00	18301	-3375	123.05	148.49
278.01	313.00	438	-381	182.36	45.42

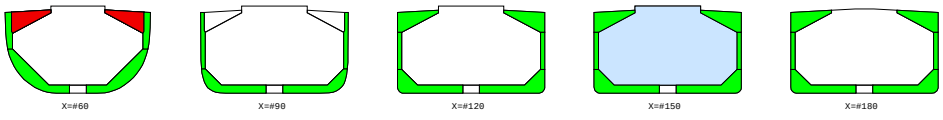
				X	FRAME
SHEAR FORCE (MIN)	-10829.2 t	(96.5%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	7153.3 t	(63.7%)		139.4 m	159
SAGGING MOMENT	-264585 tm	(60.3%)		130.0 m	149
HOGGING MOMENT	120214.1 tm	(35.8%)		63.0 m	75

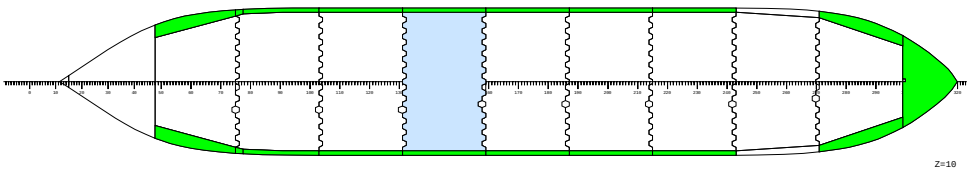
LOADING CONDITION - BALCON6-03 : BALLAST SEQUENCE LC6-WB2 EMPTY-50% BUNKERS

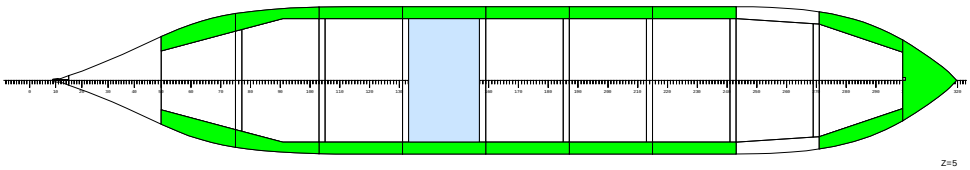


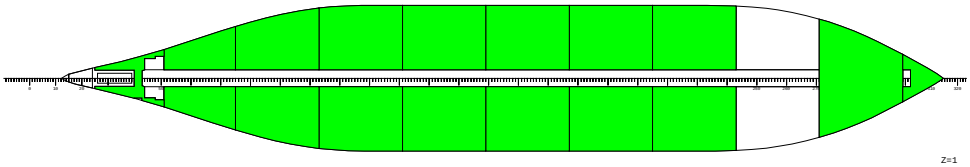












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00

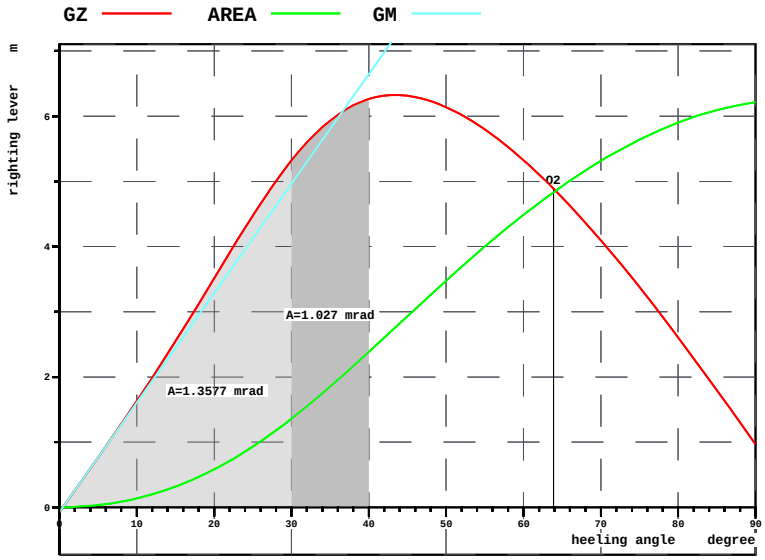
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		54238.3		154.44	-0.00	10.46	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78638.8	142.20	-0.10 11.82
Total weight	103280.6	139.77	-0.07 12.26

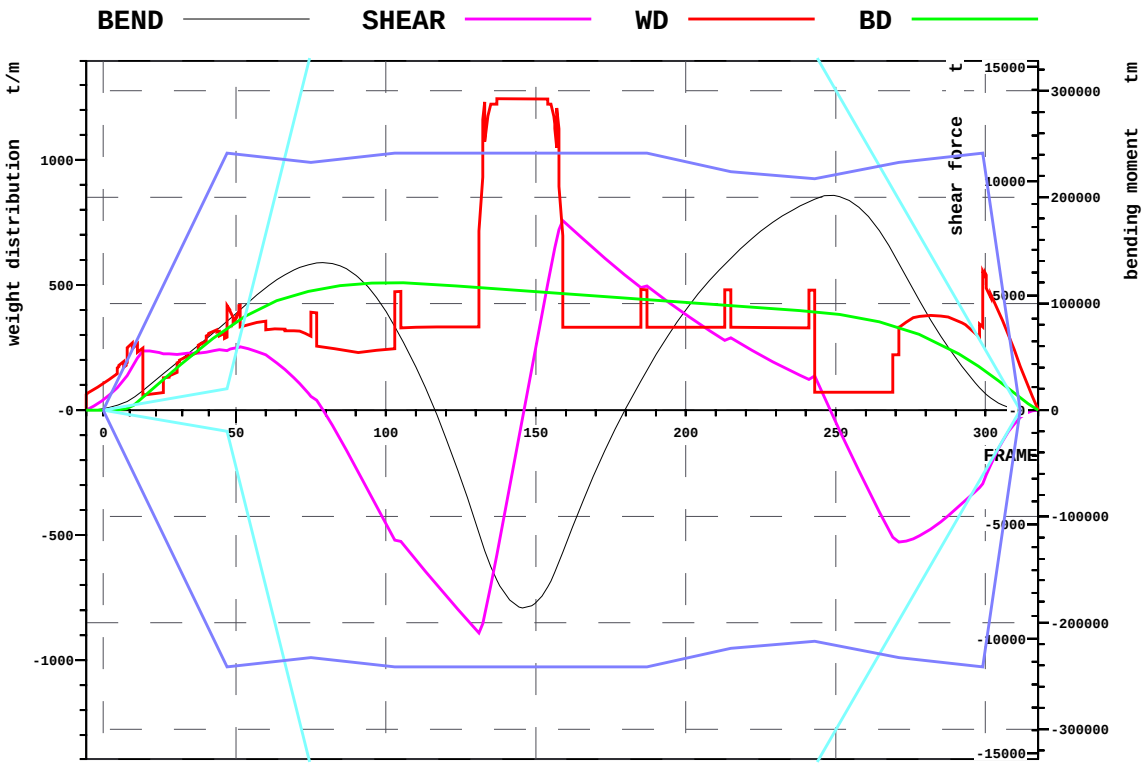
F L O A T I N G P O S I T I O N

Draught moulded	10.101	m	KM	21.88 m
Trim	5.685	m	KG	12.26 m
Heel, PS=+	-0.4	deg		
TA	12.944	m	GM0	9.62 m
TF	7.259	m	GMCORR	0.00 m
Trimming moment	******	tonm	GM	9.62 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.77	0.030	273.1	0.003
10.0	1.63	0.134	572.7	0.006
20.0	3.51	0.580	1197.1	0.012
30.0	5.32	1.357	1623.0	0.016
40.0	6.26	2.384	1920.4	0.019
45.0	6.31	2.935	2017.0	0.020
50.0	6.14	3.480	2076.3	0.020
55.0	5.80	4.002	2095.0	0.020
60.0	5.33	4.488	2072.0	0.020
65.0	4.75	4.928	2014.9	0.020
70.0	4.10	5.315	1930.4	0.019
75.0	3.38	5.642	1823.1	0.018
80.0	2.60	5.903	1696.6	0.016
85.0	1.80	6.095	1553.9	0.015
90.0	0.97	6.216	1399.3	0.014



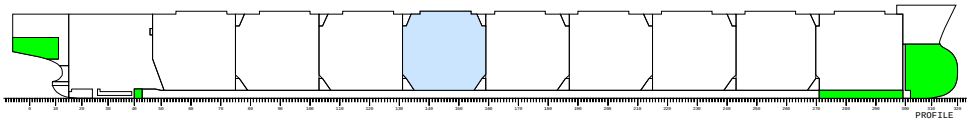
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.358	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.385	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.027	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.326	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.432	deg	OK
GM0.15	GM > 0.15 m	0.150	9.621	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.705		OK

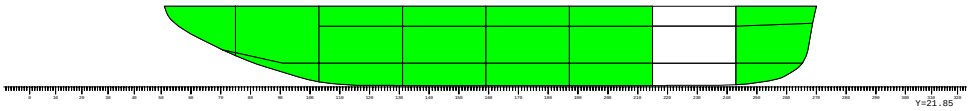


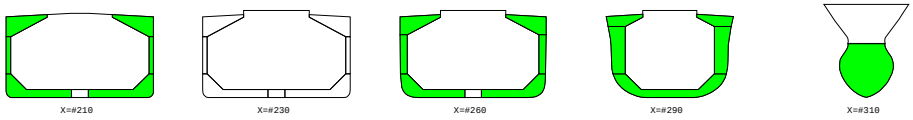
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1286	443	106.38	0.00
8.80	11.00	10879	1893	268.53	16.00
12.00	15.00	18197	2582	59.77	49.47
37.60	47.00	83612	2588	418.29	324.93
41.13	51.00	93209	2736	417.46	357.14
62.97	75.00	137074	592	294.71	476.43
88.45	103.00	78481	-5687	473.29	507.75
113.93	131.00	-115244	-9742	715.67	490.07
139.41	159.00	-133703	8276	696.89	466.07
164.89	187.00	37120	5414	481.01	441.99
190.37	215.00	142658	3151	405.29	417.88
215.85	243.00	198192	1491	478.81	391.72
241.33	271.00	138125	-5763	221.02	327.63
266.81	299.00	17761	-3236	332.63	162.70
278.01	313.00	503	-362	182.42	52.01

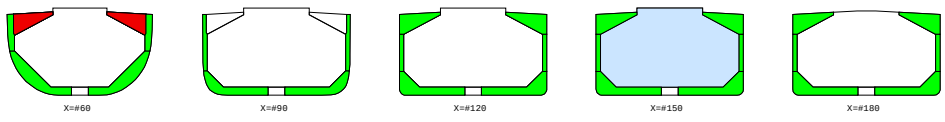
				X	FRAME
SHEAR FORCE (MIN)	-9742.0 t	(86.8%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	8275.7 t	(73.7%)		139.4 m	159
SAGGING MOMENT	-185978 tm	(42.4%)		127.1 m	145
HOGGING MOMENT	201712.3 tm	(66.4%)		221.6 m	249

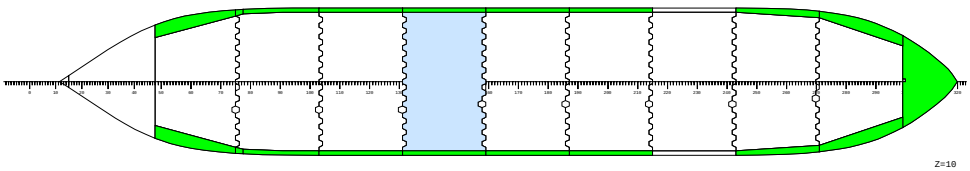
LOADING CONDITION - BALCON6-04 : BALLAST SEQUENCE LC6-WB3 EMPTY-50% BUNKERS

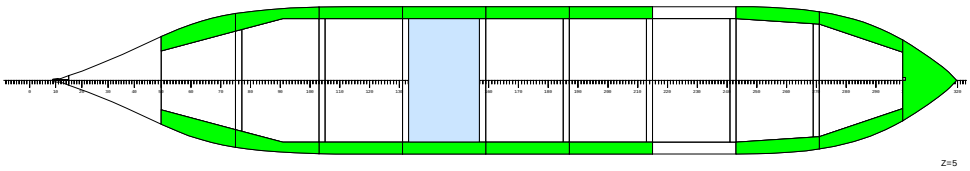


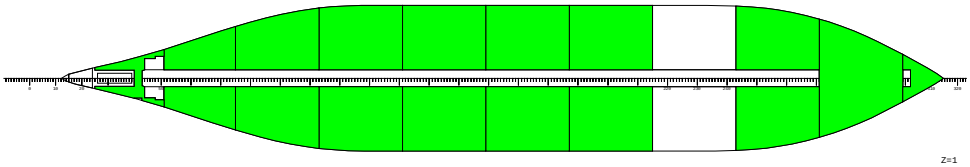












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00

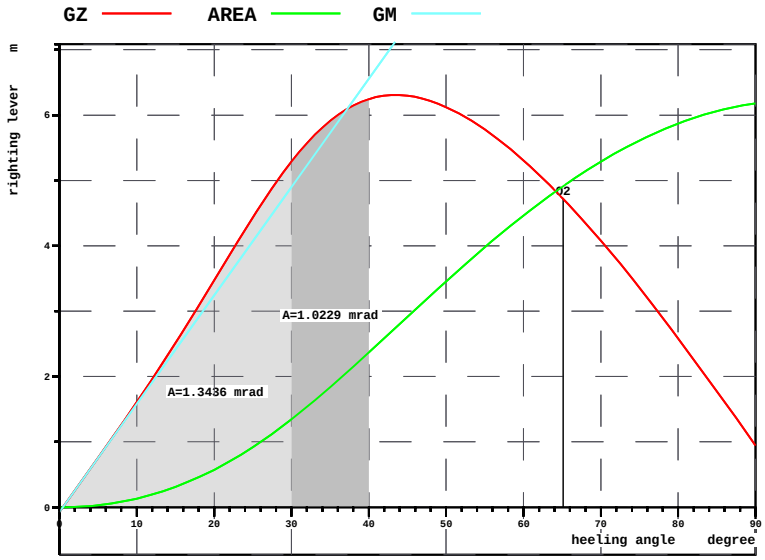
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		54402.1	157.74	-0.00	10.49		0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78802.6	144.50	-0.10 11.83
Total weight	103444.5	141.52	-0.07 12.27

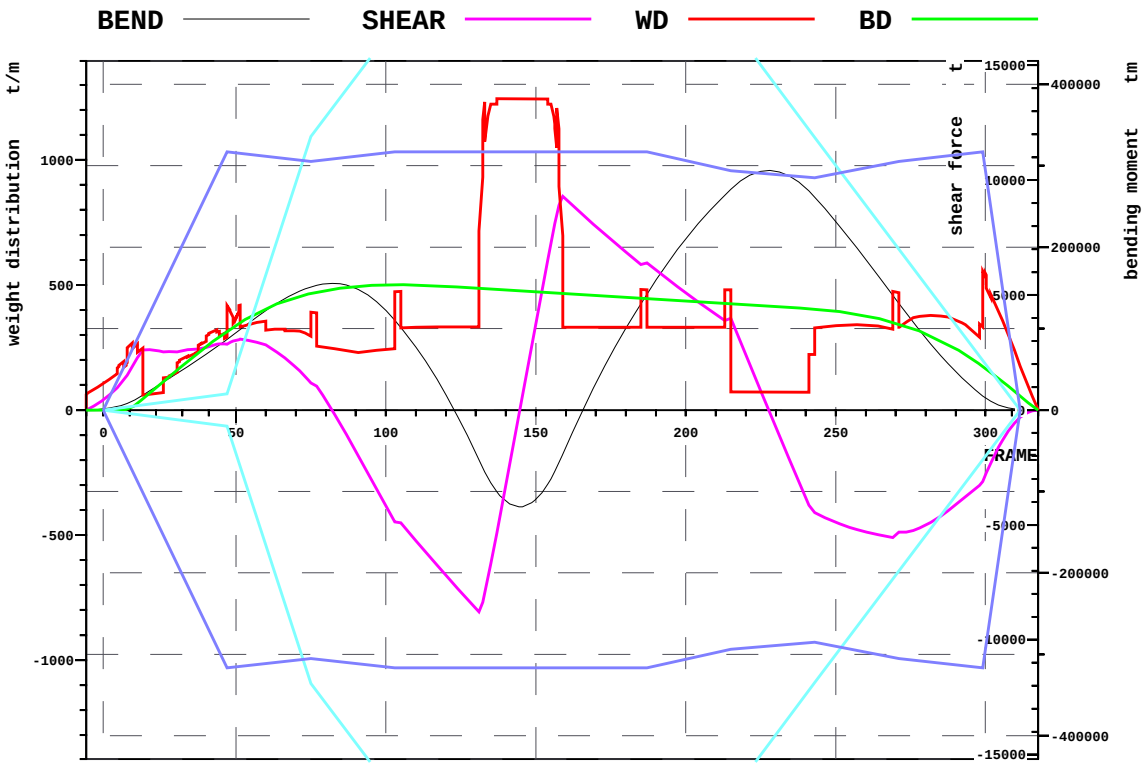
F L O A T I N G P O S I T I O N

Draught moulded	10.097	m	KM	21.77 m
Trim	4.818	m	KG	12.27 m
Heel, PS=+	-0.4	deg		
TA	12.506	m	GM0	9.50 m
TF	7.688	m	GMCORR	0.00 m
Trimming moment	******	tonm	GM	9.50 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.76	0.030	273.1	0.003
10.0	1.61	0.133	572.7	0.006
20.0	3.47	0.573	1197.1	0.012
30.0	5.29	1.343	1623.0	0.016
40.0	6.25	2.366	1920.4	0.019
45.0	6.30	2.915	2017.0	0.019
50.0	6.12	3.459	2076.3	0.020
55.0	5.78	3.979	2095.0	0.020
60.0	5.31	4.464	2072.0	0.020
65.0	4.73	4.902	2014.9	0.019
70.0	4.08	5.287	1930.4	0.019
75.0	3.36	5.612	1823.1	0.018
80.0	2.58	5.872	1696.6	0.016
85.0	1.78	6.062	1553.9	0.015
90.0	0.94	6.181	1399.3	0.014



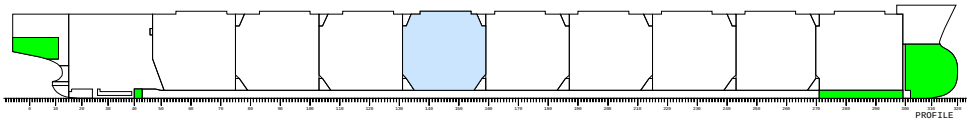
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.344	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.367	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.023	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.308	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.450	deg	OK
GM0.15	GM > 0.15 m	0.150	9.496	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.746		OK

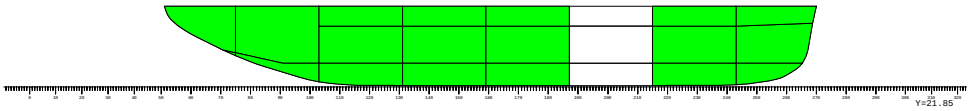


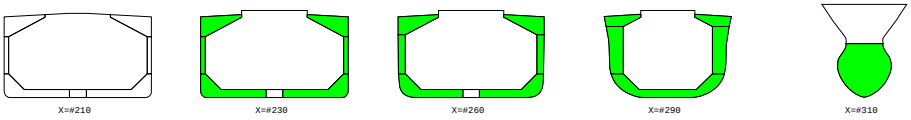
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1232	443	106.40	0.00
8.80	11.00	10764	1905	268.55	12.25
12.00	15.00	18110	2609	59.79	43.99
37.60	47.00	86791	2866	418.37	312.73
41.13	51.00	97409	3057	417.54	344.76
62.97	75.00	150967	1175	343.24	465.48
88.45	103.00	110156	-4868	473.38	500.27
113.93	131.00	-60908	-8777	715.80	486.25
139.41	159.00	-54127	9297	697.02	465.92
164.89	187.00	142131	6394	481.10	445.52
190.37	215.00	270851	3996	276.17	425.08
215.85	243.00	261109	-4469	328.43	402.58
241.33	271.00	130997	-5320	331.32	341.32
266.81	299.00	17263	-3127	332.69	173.65
278.01	313.00	529	-346	182.45	57.21

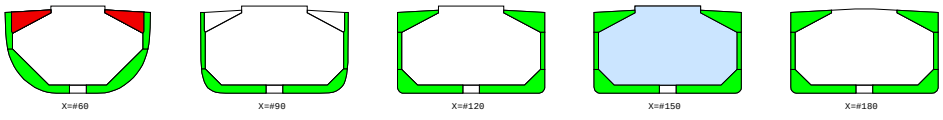
				X	FRAME
SHEAR FORCE (MIN)	-8777.4 t	(78.2%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	9297.4 t	(82.8%)		139.4 m	159
SAGGING MOMENT	-118964 tm	(27.1%)		126.1 m	144
HOGGING MOMENT	293926.8 tm	(72.0%)		202.3 m	228

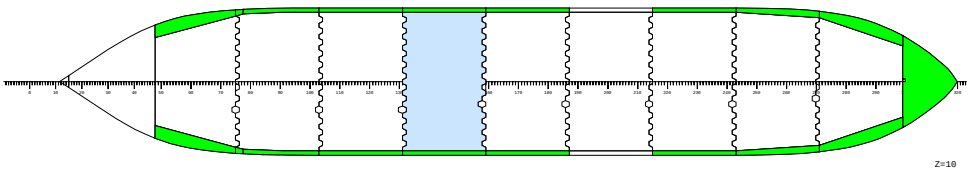
LOADING CONDITION - BALCON6-05 : BALLAST SEQUENCE LC6-WB4 EMPTY-50% BUNKERS

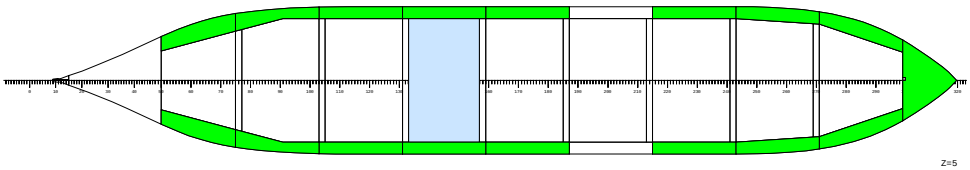


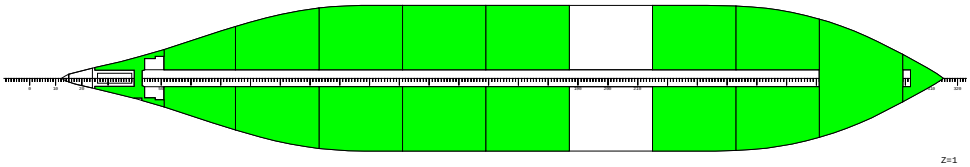












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00

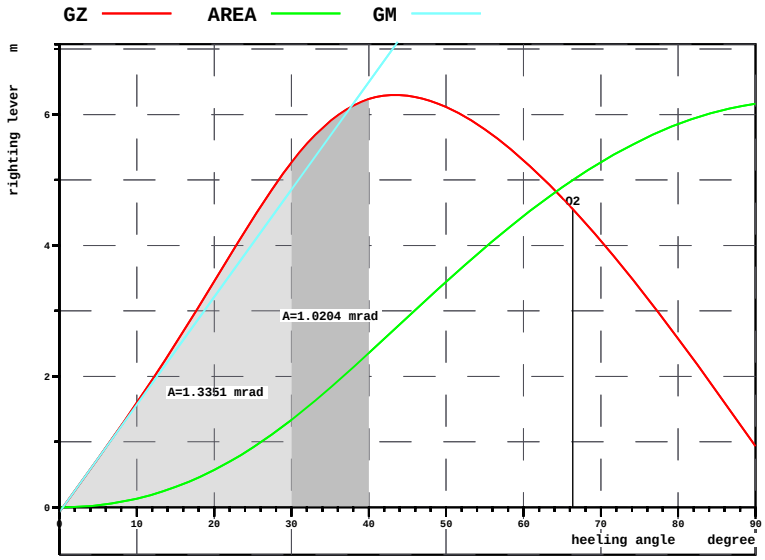
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		54396.0	160.81	-0.00	10.50		0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78796.5	146.62	-0.10 11.83
Total weight	103438.4	143.14	-0.07 12.27

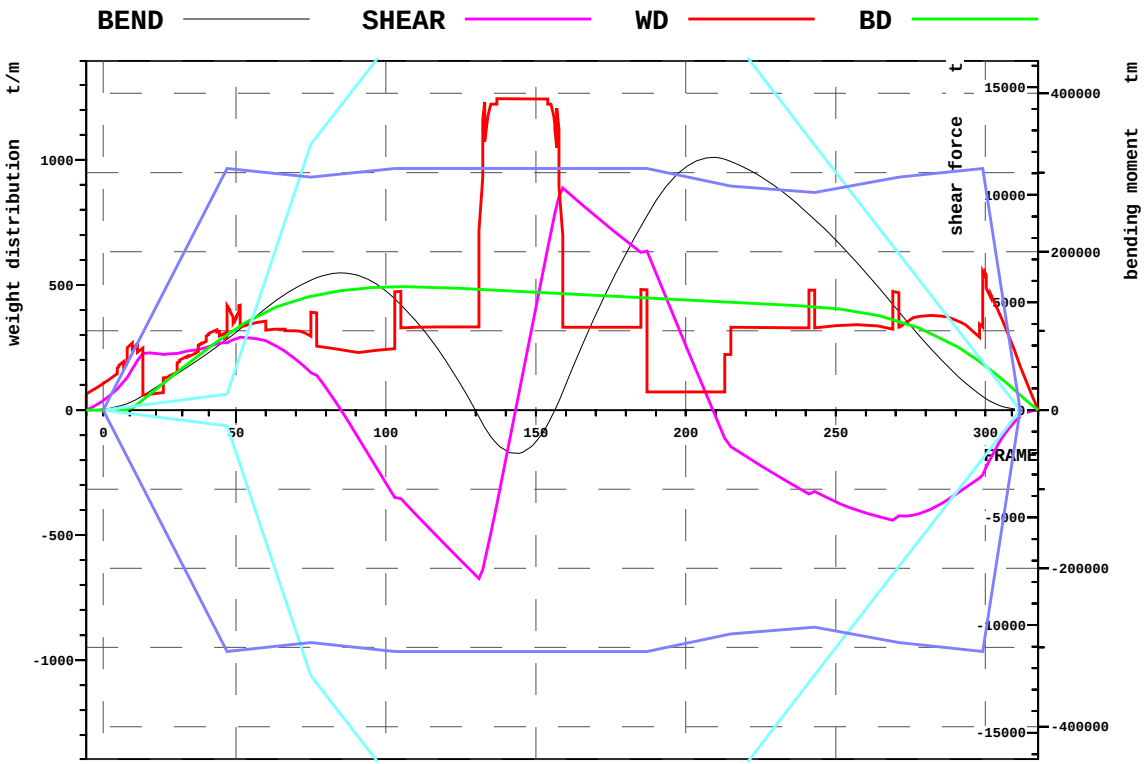
F L O A T I N G P O S I T I O N

Draught moulded	10.077	m	KM	21.68 m
Trim	4.003	m	KG	12.27 m
Heel, PS=+	-0.4	deg		
TA	12.078	m	GM0	9.41 m
TF	8.075	m	GMCORR	0.00 m
Trimming moment	-827254	tonm	GM	9.41 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.75	0.029	273.1	0.003
10.0	1.59	0.131	572.7	0.006
20.0	3.45	0.568	1197.1	0.012
30.0	5.27	1.335	1623.0	0.016
40.0	6.24	2.355	1920.4	0.019
45.0	6.29	2.904	2017.0	0.019
50.0	6.12	3.446	2076.3	0.020
55.0	5.77	3.966	2095.0	0.020
60.0	5.30	4.450	2072.0	0.020
65.0	4.73	4.888	2014.9	0.019
70.0	4.07	5.273	1930.4	0.019
75.0	3.35	5.597	1823.1	0.018
80.0	2.58	5.855	1696.6	0.016
85.0	1.76	6.045	1553.9	0.015
90.0	0.93	6.163	1399.3	0.014



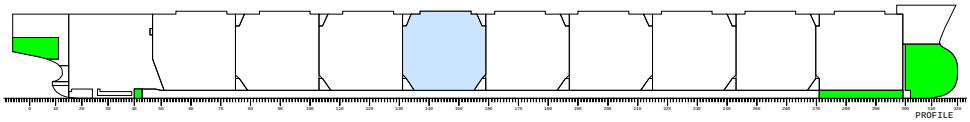
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.335	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.355	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.020	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.299	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.436	deg	OK
GM0.15	GM > 0.15 m	0.150	9.411	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.774		OK

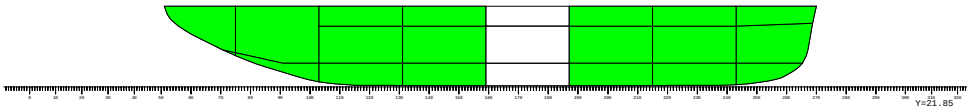


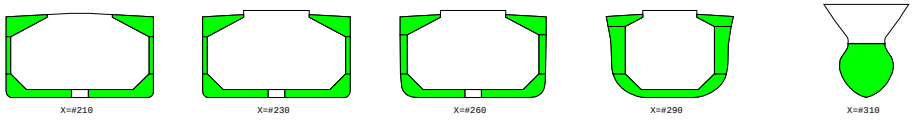
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1206	443	106.40	0.00
8.80	11.00	10703	1912	268.53	9.52
12.00	15.00	18071	2627	59.79	39.50
37.60	47.00	89663	3118	418.37	300.81
41.13	51.00	101227	3352	417.54	332.59
62.97	75.00	163969	1728	295.47	454.46
88.45	103.00	140363	-4074	473.38	492.47
113.93	131.00	-8452	-7829	715.81	481.91
139.41	159.00	23407	10313	697.03	465.03
164.89	187.00	245332	7388	72.33	448.08
190.37	215.00	314183	-1702	276.17	431.09
215.85	243.00	240619	-3788	328.43	412.03
241.33	271.00	124318	-4919	331.33	353.49
266.81	299.00	16729	-3027	332.69	183.57
278.01	313.00	528	-331	182.45	61.99

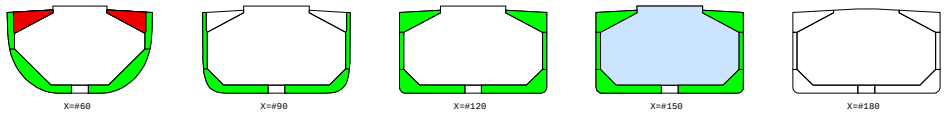
				X	FRAME
SHEAR FORCE (MIN)	-7829.1 t	(69.8%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	10313.3 t	(91.9%)		139.4 m	159
SAGGING MOMENT	-54702.8 tm	(12.5%)		126.1 m	144
HOGGING MOMENT	319124.6 tm	(67.5%)		185.7 m	210

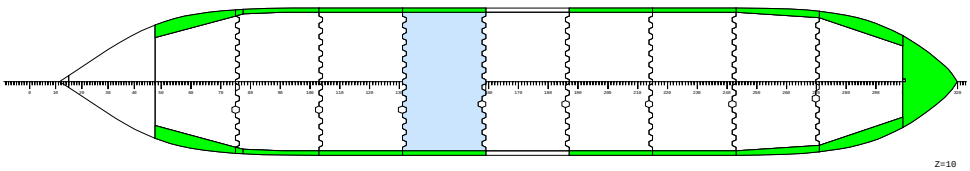
LOADING CONDITION - BALCON6-06 : BALLAST SEQUENCE LC6-WB5 EMPTY-50% BUNKERS

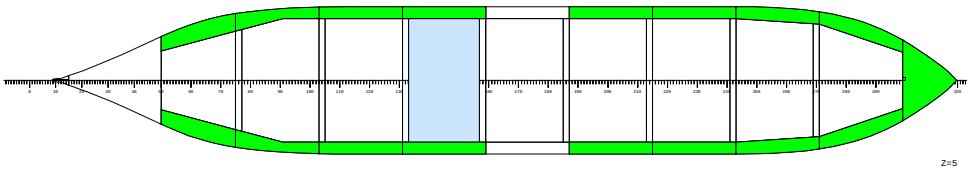


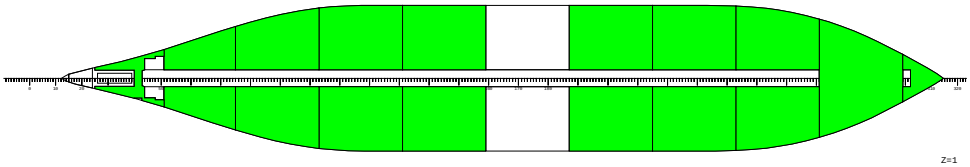












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00

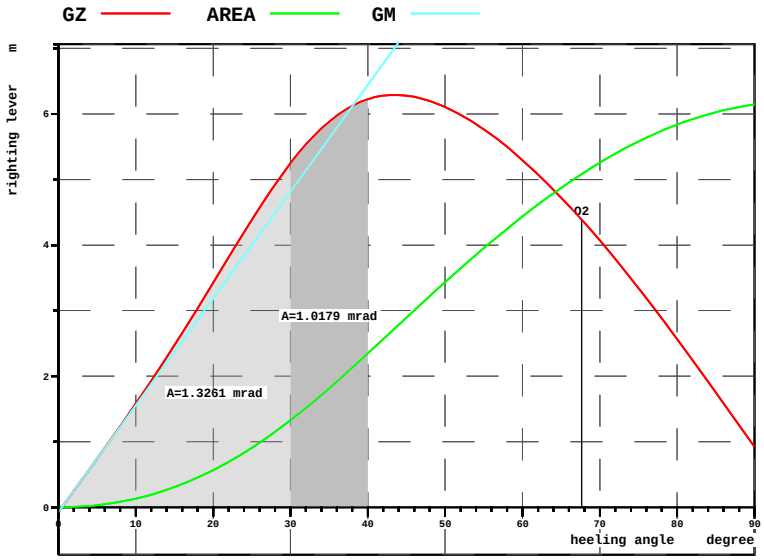
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		54396.1		163.89	-0.00	10.50	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78796.5	148.75	-0.10 11.83
Total weight	103438.4	144.76	-0.07 12.27

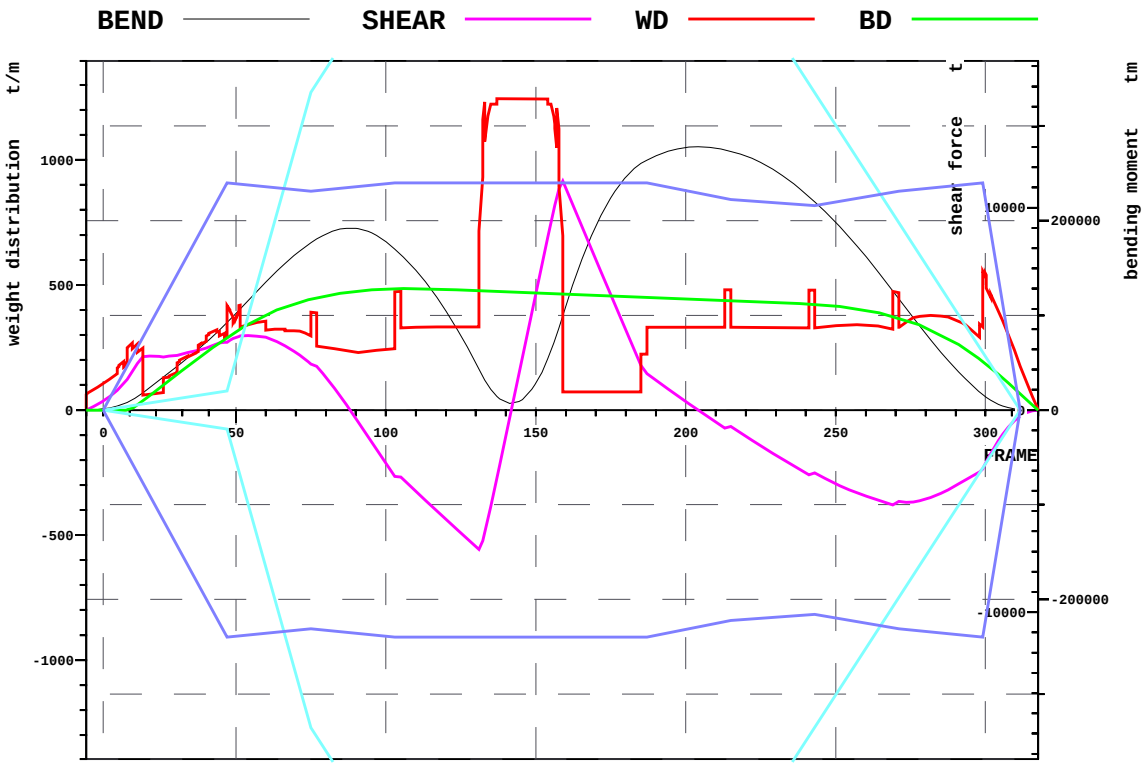
F L O A T I N G P O S I T I O N

Draught moulded	10.055	m	KM	21.60	m
Trim	3.176	m	KG	12.27	m
Heel, PS=+	-0.4	deg			
TA	11.643	m	GM0	9.33	m
TF	8.467	m	GMCORR	0.00	m
Trimming moment	-648089	tonm	GM	9.33	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.74	0.029	273.1	0.003
10.0	1.58	0.130	572.7	0.006
20.0	3.42	0.564	1197.1	0.012
30.0	5.25	1.326	1623.0	0.016
40.0	6.23	2.344	1920.4	0.019
45.0	6.28	2.891	2017.0	0.019
50.0	6.11	3.433	2076.3	0.020
55.0	5.77	3.952	2095.0	0.020
60.0	5.29	4.436	2072.0	0.020
65.0	4.72	4.873	2014.9	0.019
70.0	4.07	5.257	1930.4	0.019
75.0	3.34	5.581	1823.1	0.018
80.0	2.56	5.839	1696.6	0.016
85.0	1.75	6.028	1553.9	0.015
90.0	0.92	6.144	1399.3	0.014



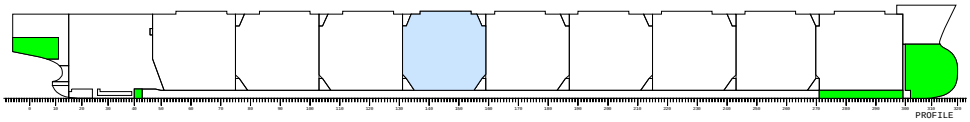
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.326	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.344	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.018	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.289	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.468	deg	OK
GM0.15	GM > 0.15 m	0.150	9.330	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.804		OK

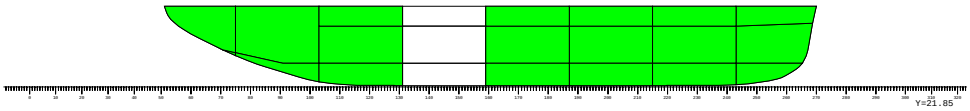


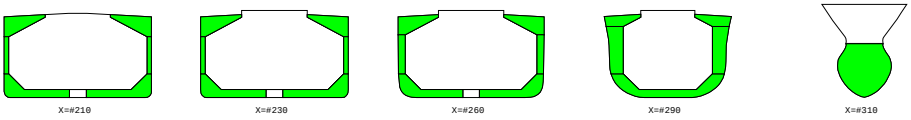
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1205	443	106.40	0.00
8.80	11.00	10705	1915	268.50	7.63
12.00	15.00	18096	2640	59.79	35.84
37.60	47.00	92408	3354	418.37	288.93
41.13	51.00	104880	3630	417.54	320.40
62.97	75.00	176587	2267	343.59	443.31
88.45	103.00	170007	-3291	473.38	484.55
113.93	131.00	43347	-6888	715.81	477.46
139.41	159.00	100257	11324	697.03	464.06
164.89	187.00	263994	1796	222.70	450.59
190.37	215.00	273020	-820	405.37	437.08
215.85	243.00	219612	-3103	328.43	421.48
241.33	271.00	117351	-4514	331.33	365.71
266.81	299.00	16102	-2926	332.69	193.59
278.01	313.00	498	-316	182.45	66.86

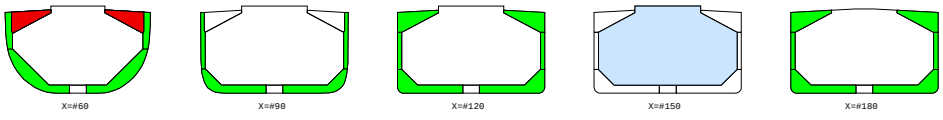
				X	FRAME
SHEAR FORCE (MIN)	-6887.9 t	(61.4%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	11324.1 t	(100.9%)		139.4 m	159
SAGGING MOMENT	-7.0 tm			281.8 m	318
HOGGING MOMENT	278068.8 tm	(58.9%)		180.5 m	204

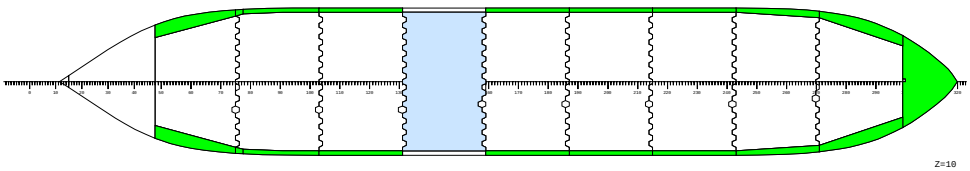
LOADING CONDITION - BALCON6-07 : BALLAST SEQUENCE LC6-WB6 EMPTY-50% BUNKERS

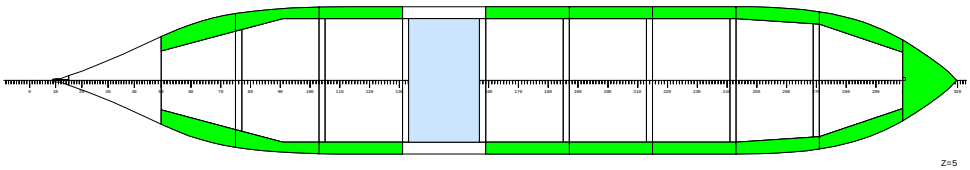


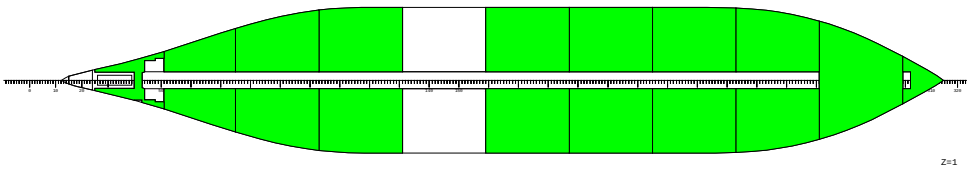












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00

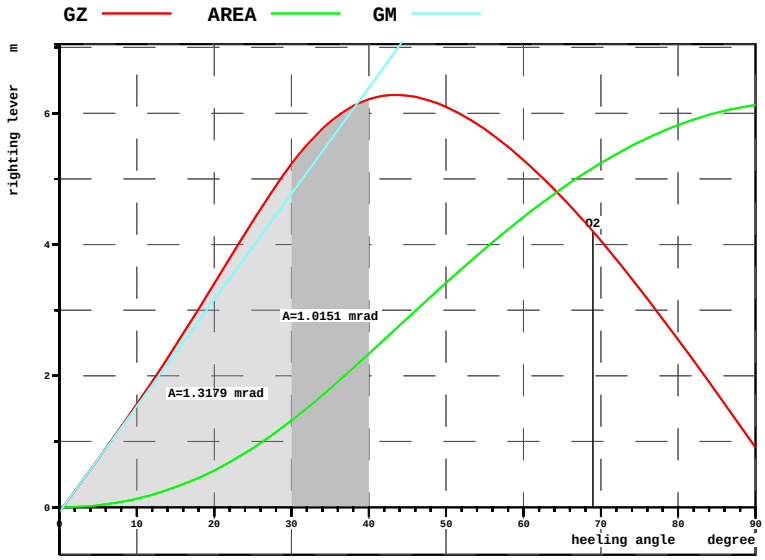
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL		54396.1	166.98	-0.00	10.50		0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	78796.5	150.88	-0.10 11.83
Total weight	103438.4	146.38	-0.07 12.27

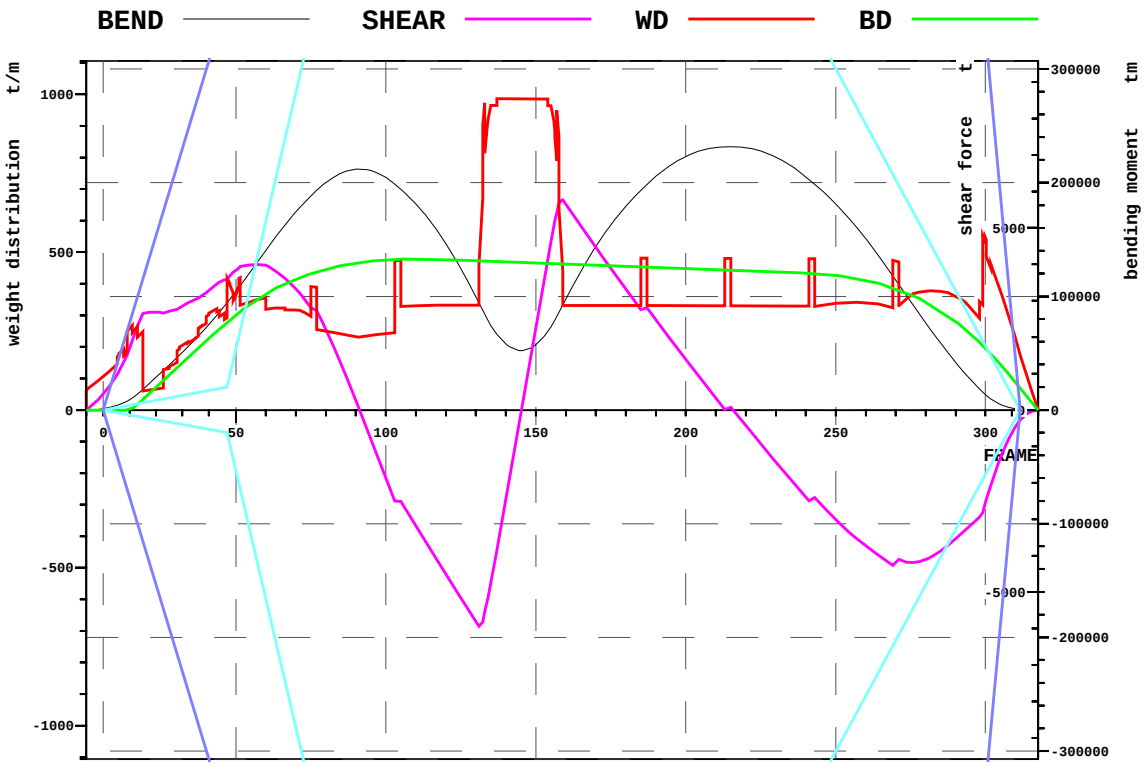
F L O A T I N G P O S I T I O N

Draught moulded	10.031	m	KM	21.53	m
Trim	2.341	m	KG	12.27	m
Heel, PS=+	-0.5	deg			
TA	11.201	m	GM0	9.26	m
TF	8.861	m	GMCORR	0.00	m
Trimming moment	-471934	tonm	GM	9.26	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.73	0.029	273.1	0.003
10.0	1.57	0.129	572.7	0.006
20.0	3.40	0.560	1197.1	0.012
30.0	5.23	1.318	1623.0	0.016
40.0	6.21	2.333	1920.4	0.019
45.0	6.27	2.879	2017.0	0.019
50.0	6.10	3.420	2076.3	0.020
55.0	5.76	3.938	2095.0	0.020
60.0	5.28	4.421	2072.0	0.020
65.0	4.72	4.858	2014.9	0.019
70.0	4.06	5.241	1930.4	0.019
75.0	3.33	5.563	1823.1	0.018
80.0	2.55	5.820	1696.6	0.016
85.0	1.74	6.008	1553.9	0.015
90.0	0.91	6.124	1399.3	0.014



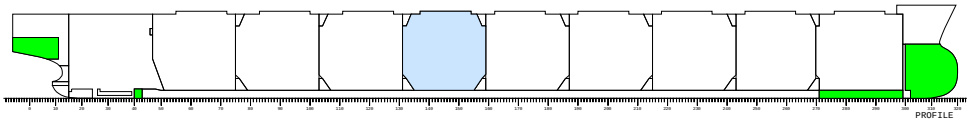
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.318	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.333	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.015	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.276	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.494	deg	OK
GM0.15	GM > 0.15 m	0.150	9.258	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.833		OK

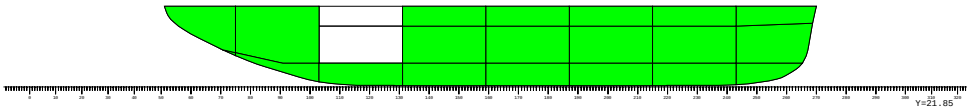


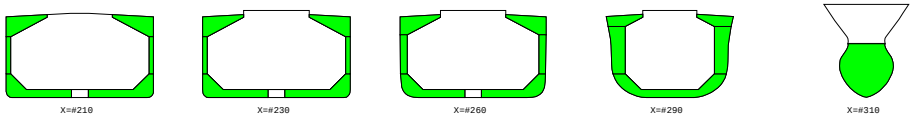
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1169	443	106.40	0.00
8.80	11.00	10611	1916	268.48	6.43
12.00	15.00	17993	2648	59.79	32.86
37.60	47.00	94627	3581	418.37	276.88
41.13	51.00	107951	3900	417.54	307.98
62.97	75.00	188357	2805	296.16	431.81
88.45	103.00	198691	-2501	473.38	476.36
113.93	131.00	94249	-5935	457.41	472.83
139.41	159.00	92515	5767	438.63	462.99
164.89	187.00	198304	2804	481.10	453.08
190.37	215.00	231636	79	405.37	443.13
215.85	243.00	198602	-2404	328.43	431.09
241.33	271.00	110501	-4100	331.33	378.15
266.81	299.00	15568	-2821	332.69	203.84
278.01	313.00	505	-300	182.45	71.87

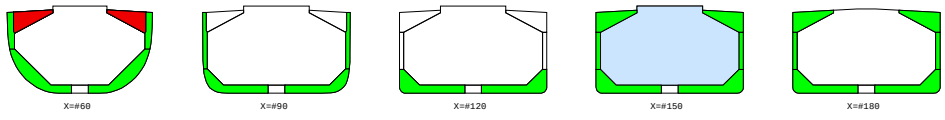
				X	FRAME
SHEAR FORCE (MIN)	-5934.9 t	(52.9%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	5768.1 t	(51.4%)		139.2 m	159
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	231636.0 tm	(49.0%)		190.4 m	215

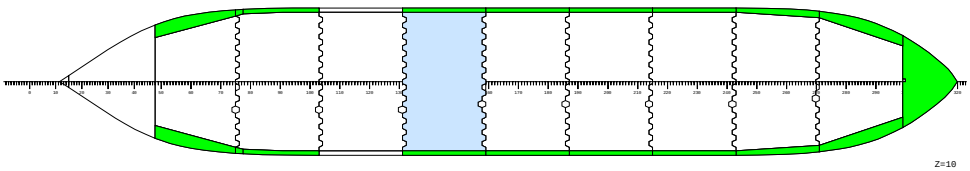
LOADING CONDITION - BALCON6-08 : BAL. SEQU.-WB7TS+WB7DS EMPTY-50% BUNKERS

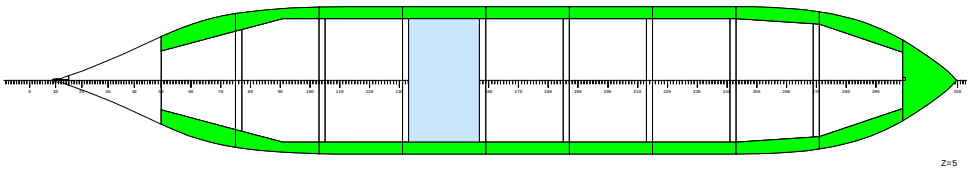


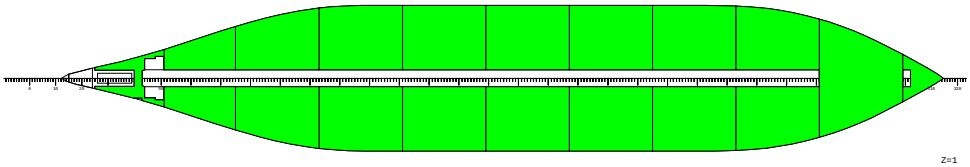












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

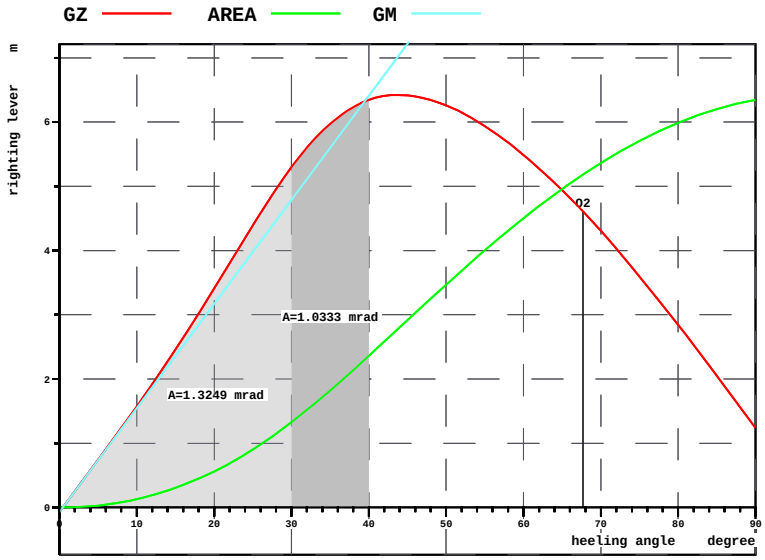
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		57815.1		165.98	-0.00	10.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	82215.6	150.85	-0.09 11.43
Total weight	106857.5	146.50	-0.07 11.95

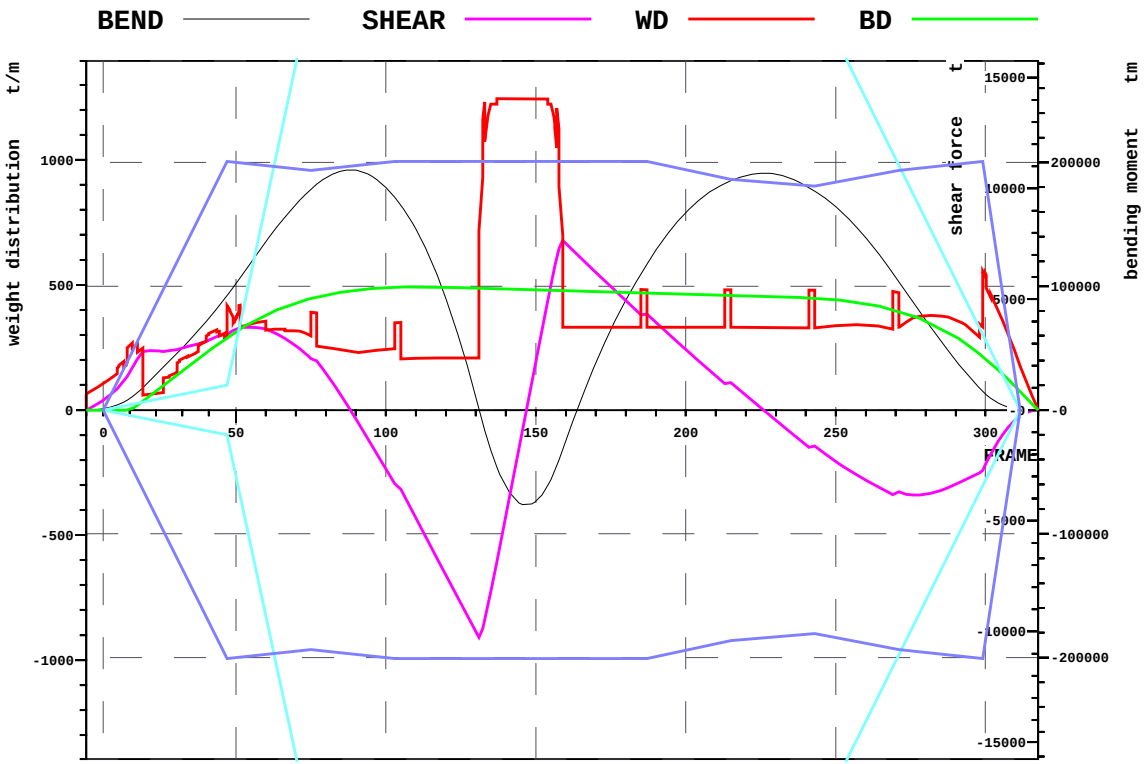
F L O A T I N G P O S I T I O N

Draught moulded	10.335	m	KM	21.22 m
Trim	2.288	m	KG	11.95 m
Heel, PS=+	-0.4	deg		
TA	11.479	m	GM0	9.28 m
TF	9.192	m	GMCORR	0.00 m
Trimming moment	-465781	tonm	GM	9.28 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.74	0.029	273.1	0.003
10.0	1.57	0.130	572.7	0.005
20.0	3.41	0.560	1197.1	0.011
30.0	5.30	1.325	1623.0	0.015
40.0	6.35	2.358	1920.4	0.018
45.0	6.41	2.917	2017.0	0.019
50.0	6.26	3.471	2076.3	0.019
55.0	5.94	4.004	2095.0	0.020
60.0	5.49	4.504	2072.0	0.019
65.0	4.94	4.959	2014.9	0.019
70.0	4.31	5.363	1930.4	0.018
75.0	3.60	5.708	1823.1	0.017
80.0	2.84	5.990	1696.6	0.016
85.0	2.05	6.203	1553.9	0.015
90.0	1.24	6.347	1399.3	0.013



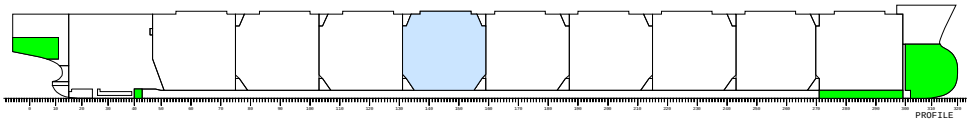
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.325	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.358	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.033	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.422	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.741	deg	OK
GM0.15	GM > 0.15 m	0.150	9.278	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.095		OK

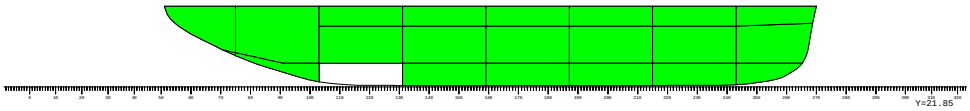


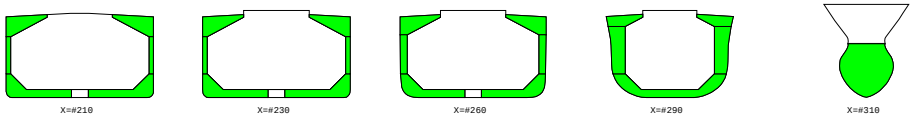
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1165	443	106.39	0.00
8.80	11.00	10599	1915	268.47	7.16
12.00	15.00	17970	2643	59.78	34.83
37.60	47.00	92733	3405	418.36	287.30
41.13	51.00	105364	3686	417.53	319.10
62.97	75.00	178197	2321	343.76	444.94
88.45	103.00	171827	-3328	349.28	490.01
113.93	131.00	1544	-10275	715.79	486.71
139.41	159.00	-31461	7653	697.01	477.10
164.89	187.00	117764	4327	330.72	467.41
190.37	215.00	185222	1235	405.36	457.69
215.85	243.00	176858	-1622	328.42	445.85
241.33	271.00	103868	-3692	331.32	392.35
266.81	299.00	15025	-2730	332.68	213.22
278.01	313.00	484	-288	182.44	76.09

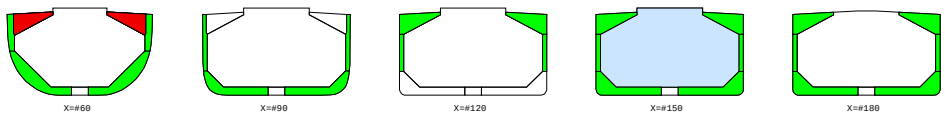
				X	FRAME
SHEAR FORCE (MIN)	-10275.1 t	(91.5%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	7652.8 t	(68.2%)		139.4 m	159
SAGGING MOMENT	-76431.7 tm	(17.4%)		127.1 m	145
HOGGING MOMENT	193935.2 tm	(49.3%)		73.8 m	87

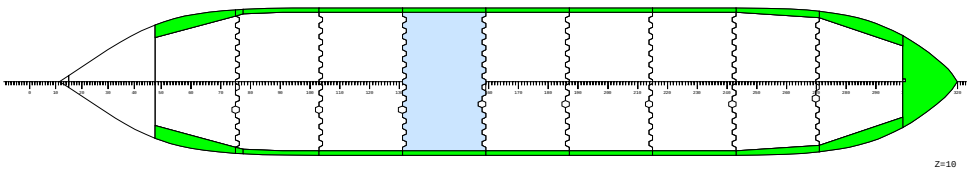
LOADING CONDITION - BALCON6-09 : BAL. SEQU.-WB7DB EMPTY-50% BUNKERS

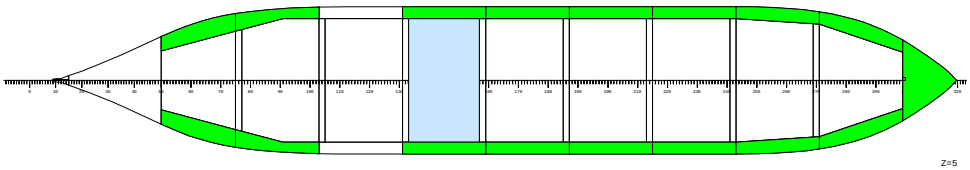


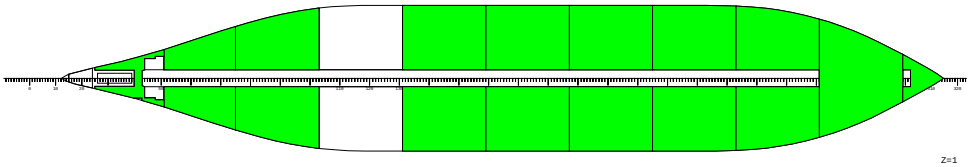












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00

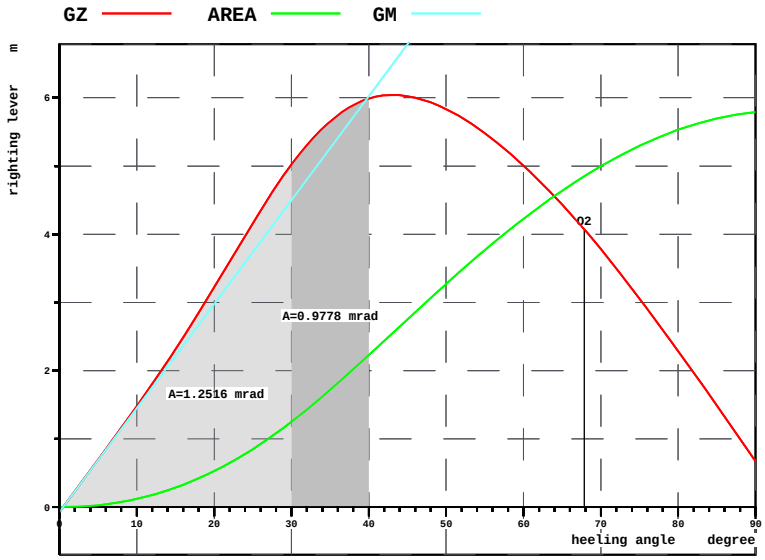
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		57584.5		166.24	-0.00	11.03	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	81985.0	150.99	-0.09 12.16
Total weight	106626.8	146.60	-0.07 12.51

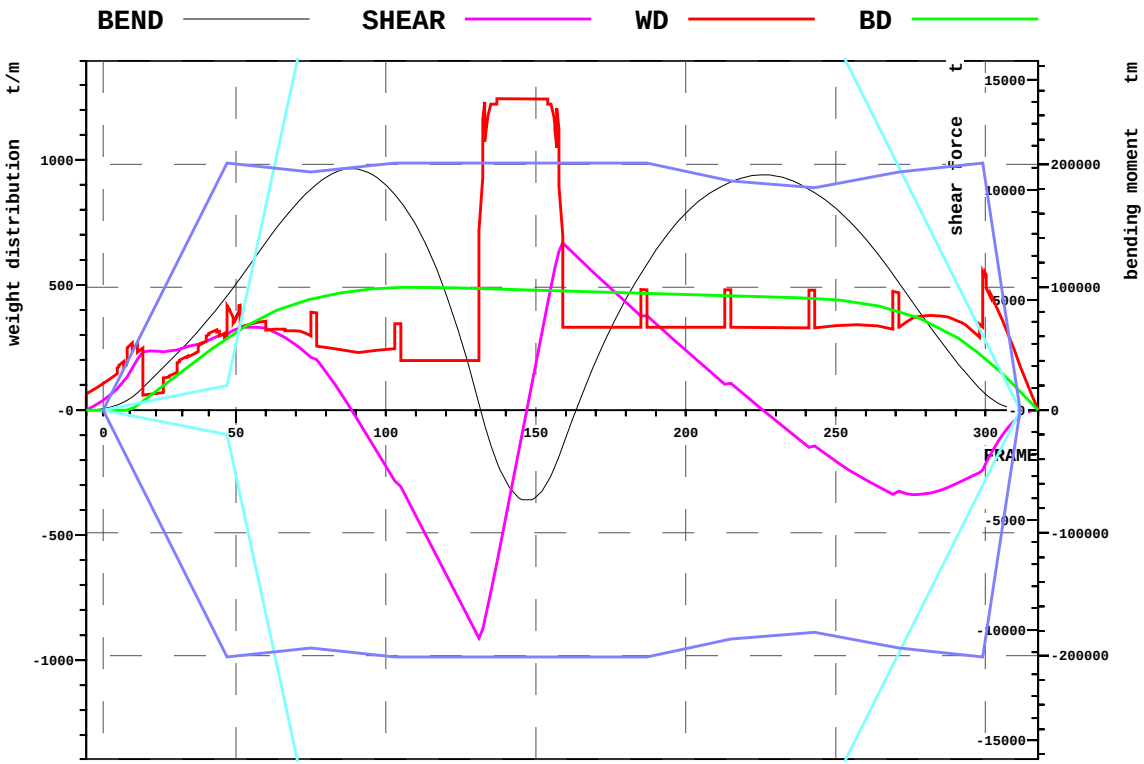
F L O A T I N G P O S I T I O N

Draught moulded	10.313	m	KM	21.24 m
Trim	2.239	m	KG	12.51 m
Heel, PS=+	-0.5	deg		
TA	11.433	m	GM0	8.73 m
TF	9.194	m	GMCORR	0.00 m
Trimming moment	-455184	tonm	GM	8.73 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.69	0.027	273.1	0.003
10.0	1.48	0.121	572.7	0.005
20.0	3.22	0.527	1197.1	0.011
30.0	5.02	1.251	1623.0	0.015
40.0	5.99	2.229	1920.4	0.018
45.0	6.02	2.755	2017.0	0.019
50.0	5.83	3.274	2076.3	0.019
55.0	5.48	3.768	2095.0	0.020
60.0	5.01	4.227	2072.0	0.019
65.0	4.43	4.639	2014.9	0.019
70.0	3.78	4.998	1930.4	0.018
75.0	3.06	5.297	1823.1	0.017
80.0	2.29	5.531	1696.6	0.016
85.0	1.49	5.696	1553.9	0.015
90.0	0.67	5.790	1399.3	0.013



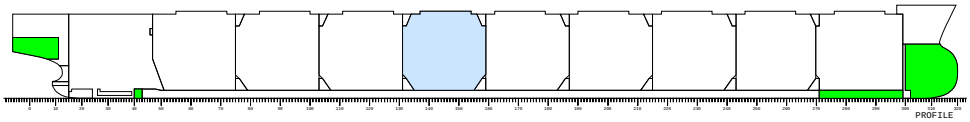
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.252	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.229	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.978	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.040	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.038	deg	OK
GM0.15	GM > 0.15 m	0.150	8.731	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.046		OK

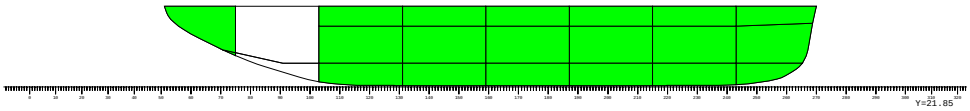


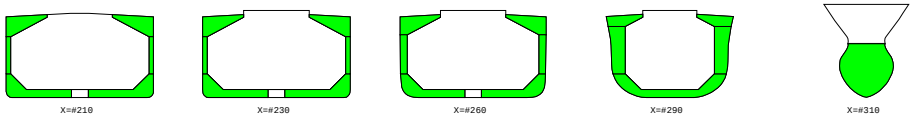
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1170	443	106.39	0.00
8.80	11.00	10612	1916	268.46	7.02
12.00	15.00	17988	2644	59.78	34.50
37.60	47.00	93058	3431	418.35	285.84
41.13	51.00	105795	3717	417.52	317.56
62.97	75.00	179714	2387	296.19	443.34
88.45	103.00	175562	-3223	344.58	488.58
113.93	131.00	5764	-10365	715.77	485.49
139.41	159.00	-29131	7591	696.99	476.08
164.89	187.00	118854	4289	481.08	466.60
190.37	215.00	185592	1214	405.35	457.09
215.85	243.00	176895	-1630	328.41	445.46
241.33	271.00	103821	-3693	331.31	392.18
266.81	299.00	15004	-2730	332.68	213.24
278.01	313.00	478	-287	182.44	76.13

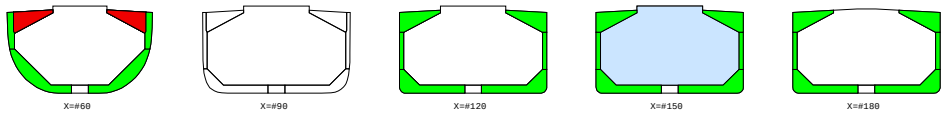
				X	FRAME
SHEAR FORCE (MIN)	-10364.6 t	(92.3%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	7591.2 t	(67.6%)		139.4 m	159
SAGGING MOMENT	-73278.9 tm	(16.7%)		127.1 m	145
HOGGING MOMENT	196363.2 tm	(48.0%)		76.7 m	90

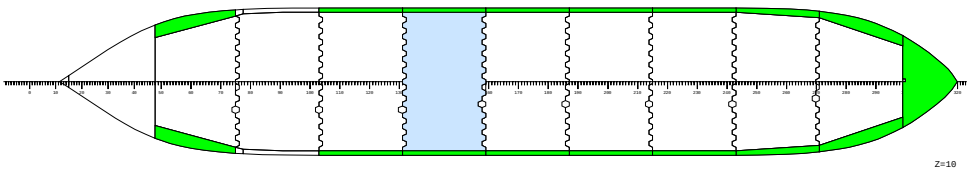
LOADING CONDITION - BALCON6-10 : BALLAST SEQUENCE LC6-WB8 EMPTY-50% BUNKERS

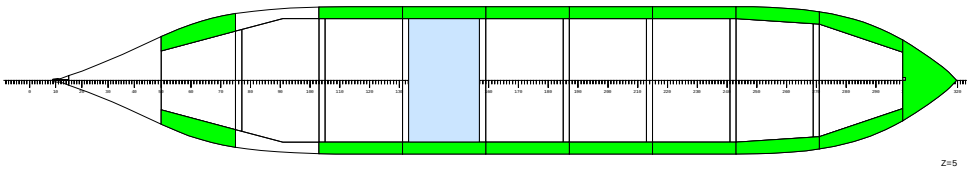


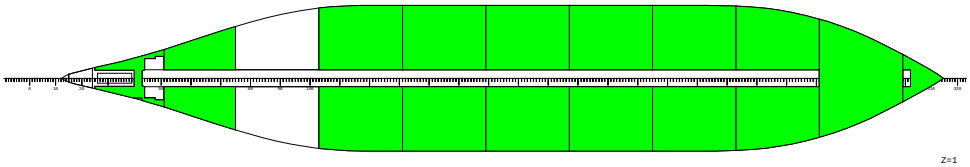












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

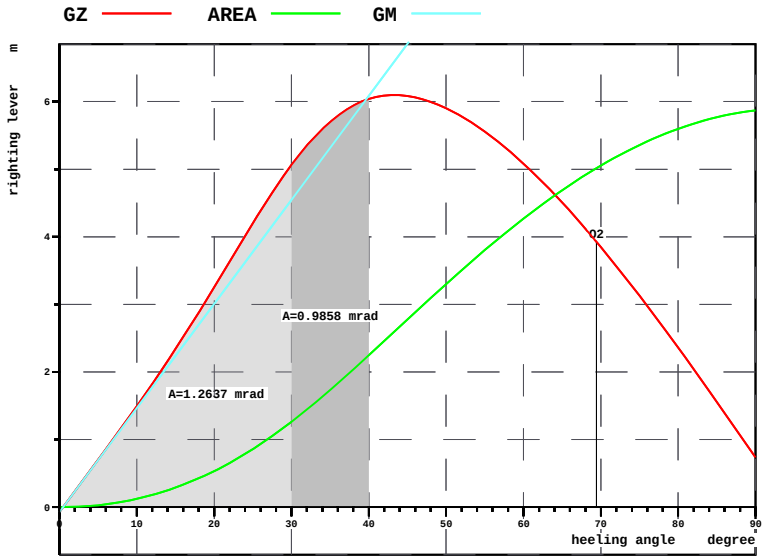
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		56698.5		169.20	-0.00	10.87	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	81099.0	152.89	-0.10 12.06
Total weight	105740.9	148.02	-0.07 12.43

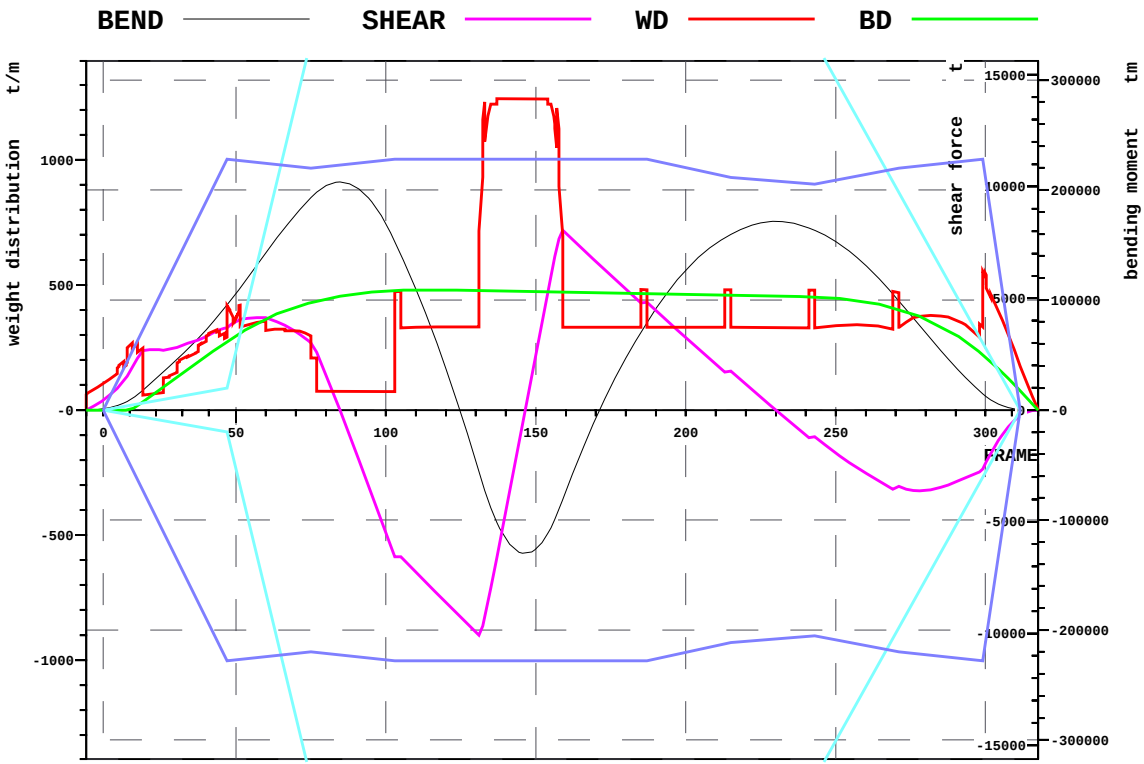
F L O A T I N G P O S I T I O N

Draught moulded	10.212	m	KM	21.26	m
Trim	1.485	m	KG	12.43	m
Heel, PS=+	-0.5	deg			
TA	10.954	m	GM0	8.82	m
TF	9.469	m	GMCORR	0.00	m
Trimming moment	-297914	tonm	GM	8.82	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.70	0.027	273.1	0.003
10.0	1.49	0.123	572.7	0.005
20.0	3.25	0.533	1197.1	0.011
30.0	5.06	1.263	1623.0	0.015
40.0	6.04	2.249	1920.4	0.018
45.0	6.08	2.780	2017.0	0.019
50.0	5.90	3.304	2076.3	0.020
55.0	5.55	3.805	2095.0	0.020
60.0	5.08	4.270	2072.0	0.020
65.0	4.51	4.689	2014.9	0.019
70.0	3.85	5.054	1930.4	0.018
75.0	3.13	5.359	1823.1	0.017
80.0	2.36	5.599	1696.6	0.016
85.0	1.56	5.770	1553.9	0.015
90.0	0.74	5.870	1399.3	0.013



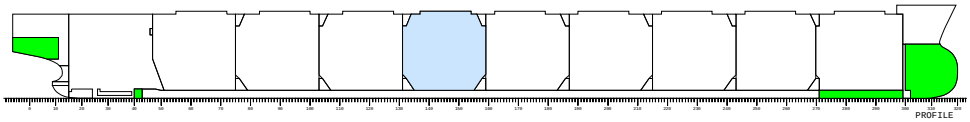
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.264	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.250	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.986	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.098	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.199	deg	OK
GM0.15	GM > 0.15 m	0.150	8.821	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.015		OK

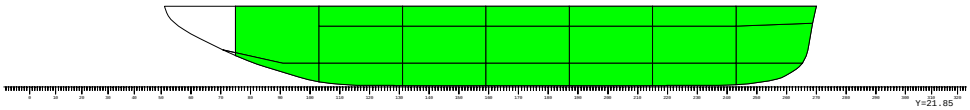


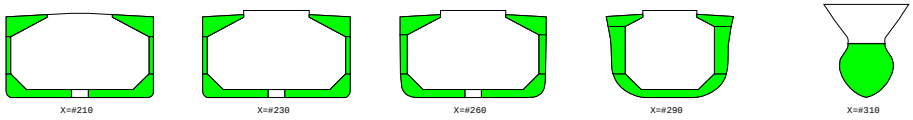
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1148	443	106.39	0.00
8.80	11.00	10553	1916	268.44	6.01
12.00	15.00	17927	2651	59.78	31.57
37.60	47.00	95543	3678	418.35	272.14
41.13	51.00	109224	4014	417.51	303.32
62.97	75.00	192971	2999	296.50	429.38
88.45	103.00	153245	-6566	473.36	477.50
113.93	131.00	-55661	-10076	715.77	477.61
139.41	159.00	-81109	8039	696.99	471.41
164.89	187.00	79356	4815	481.08	465.13
190.37	215.00	159525	1737	405.35	458.82
215.85	243.00	163128	-1193	328.41	450.39
241.33	271.00	99164	-3419	331.31	399.87
266.81	299.00	14623	-2657	332.67	220.23
278.01	313.00	480	-276	182.44	79.65

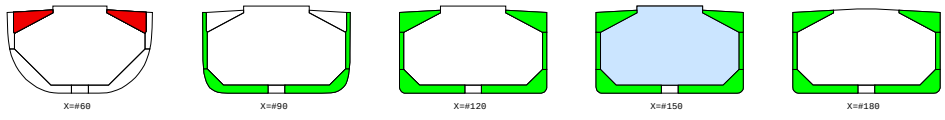
				X	FRAME
SHEAR FORCE (MIN)	-10076.4 t	(89.8%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	8039.5 t	(71.6%)		139.4 m	159
SAGGING MOMENT	-130319 tm	(29.7%)		127.1 m	145
HOGGING MOMENT	207274.3 tm	(54.1%)		71.8 m	85

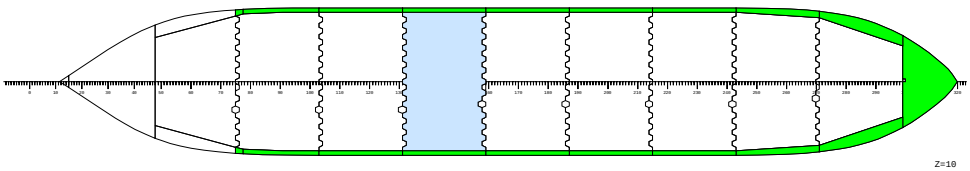
LOADING CONDITION - BALCON6-11 : BALLAST SEQUENCE LC6-WB9 EMPTY-50% BUNKERS

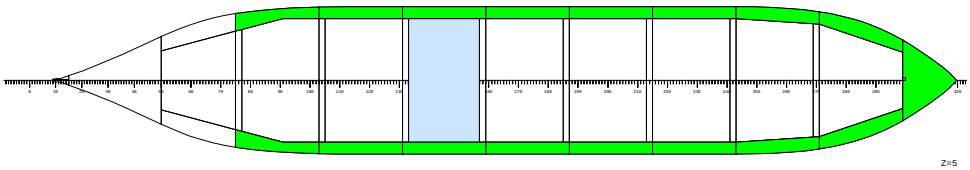


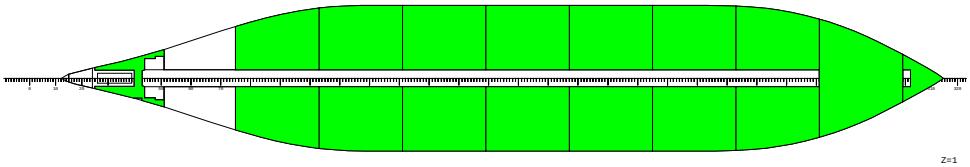












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00

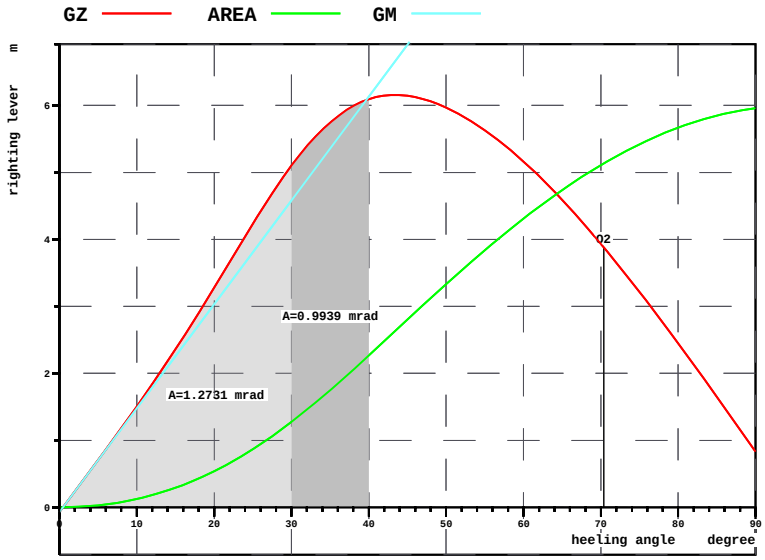
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		56626.0		171.18	-0.00	10.68	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	81026.5	154.26	-0.10 11.93
Total weight	105668.3	149.07	-0.07 12.33

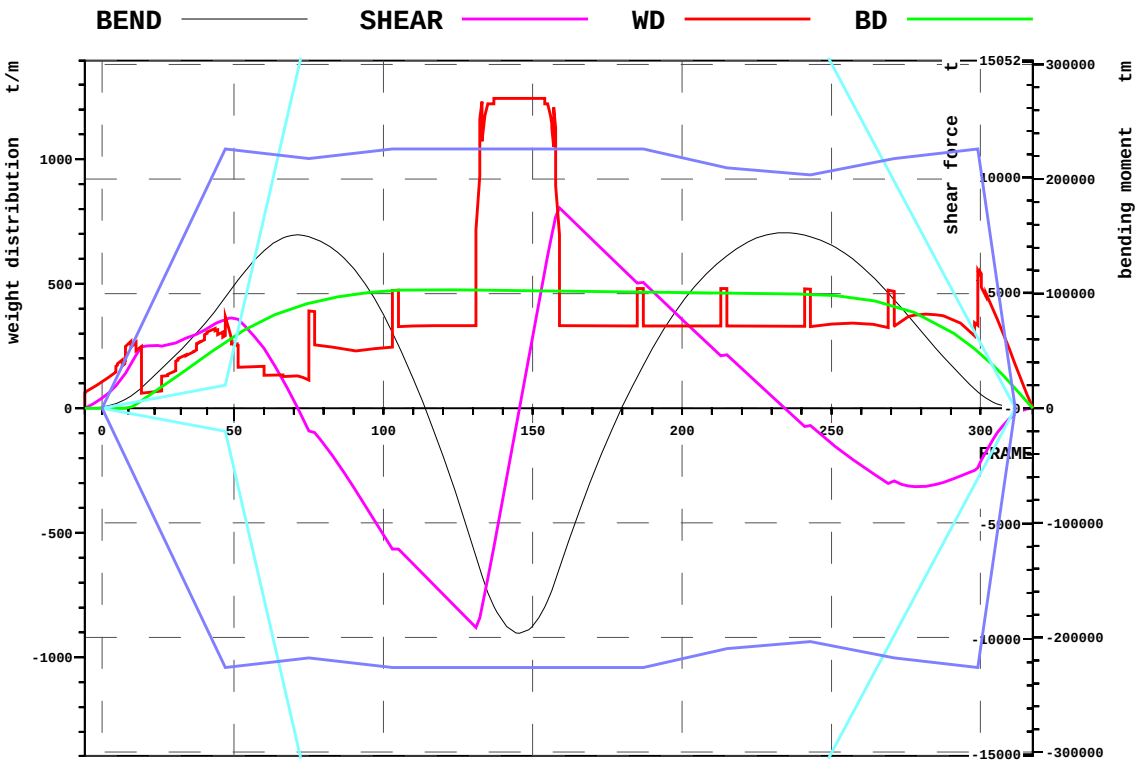
F L O A T I N G P O S I T I O N

Draught moulded	10.188	m	KM	21.22 m
Trim	0.925	m	KG	12.33 m
Heel, PS=+	-0.5	deg		
TA	10.651	m	GM0	8.89 m
TF	9.725	m	GMCORR	0.00 m
Trimming moment	-184352	tonm	GM	8.89 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.70	0.027	273.1	0.003
10.0	1.51	0.124	572.7	0.005
20.0	3.28	0.537	1197.1	0.011
30.0	5.10	1.273	1623.0	0.015
40.0	6.10	2.267	1920.4	0.018
45.0	6.14	2.803	2017.0	0.019
50.0	5.97	3.333	2076.3	0.020
55.0	5.63	3.840	2095.0	0.020
60.0	5.16	4.311	2072.0	0.020
65.0	4.59	4.738	2014.9	0.019
70.0	3.93	5.110	1930.4	0.018
75.0	3.22	5.423	1823.1	0.017
80.0	2.45	5.670	1696.6	0.016
85.0	1.65	5.849	1553.9	0.015
90.0	0.83	5.958	1399.3	0.013



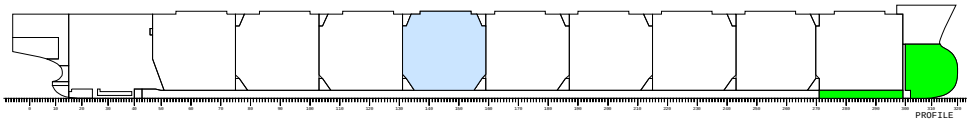
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.273	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.267	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.994	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.156	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.350	deg	OK
GM0.15	GM > 0.15 m	0.150	8.888	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.034		OK

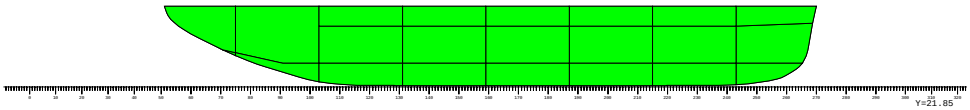


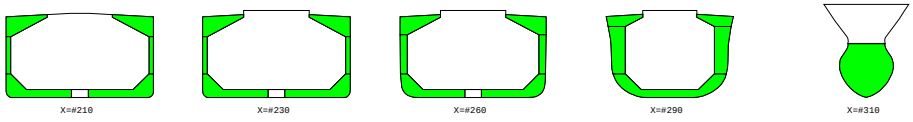
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1123	443	106.38	0.00
8.80	11.00	10485	1917	268.40	5.64
12.00	15.00	17847	2654	59.78	30.08
37.60	47.00	96831	3820	367.63	264.03
41.13	51.00	110565	3865	251.41	294.89
62.97	75.00	149666	-985	113.39	421.39
88.45	103.00	65957	-6099	473.32	471.69
113.93	131.00	-129536	-9491	715.71	474.17
139.41	159.00	-139373	8680	696.93	470.33
164.89	187.00	37363	5452	481.04	466.42
190.37	215.00	132949	2310	405.32	462.48
215.85	243.00	149576	-744	328.38	456.41
241.33	271.00	94727	-3151	331.28	407.80
266.81	299.00	14280	-2589	332.65	226.87
278.01	313.00	487	-266	182.43	82.92

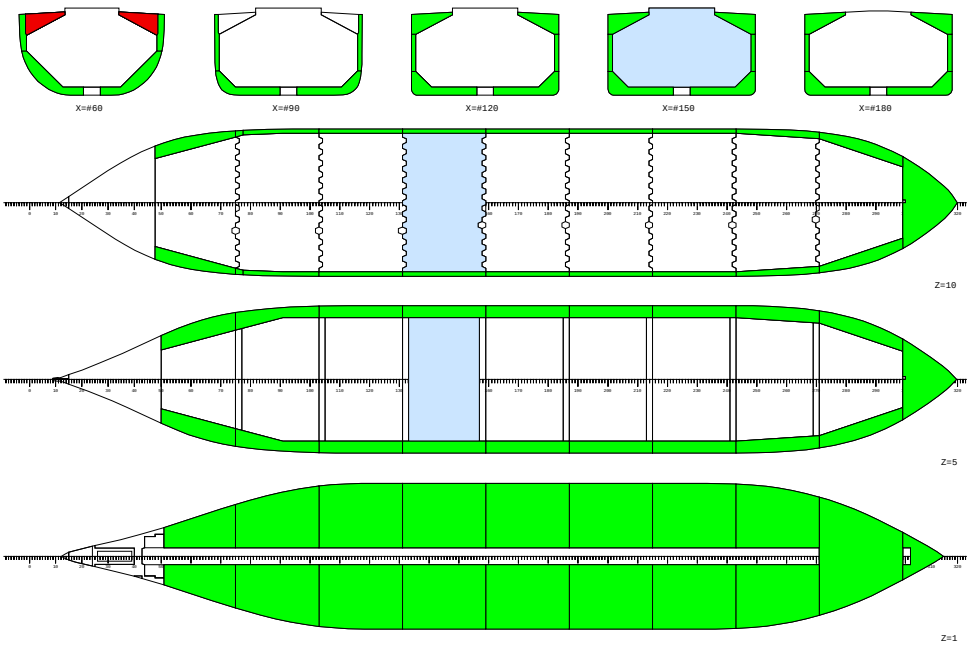
				X	FRAME
SHEAR FORCE (MIN)	-9491.5 t	(84.6%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	8679.7 t	(77.3%)		139.4 m	159
SAGGING MOMENT	-196285 tm	(44.7%)		127.1 m	145
HOGGING MOMENT	153255.2 tm	(41.2%)		208.9 m	235

LOADING CONDITION - BALCON6-12 : BAL. SEQU. LC6-WB10+AFT EMPTY-50% BUNKERS









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=1.025)							
R1.6	NO.6 CARGO HOLD	21163.5	100.0	126.64	0.02	13.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

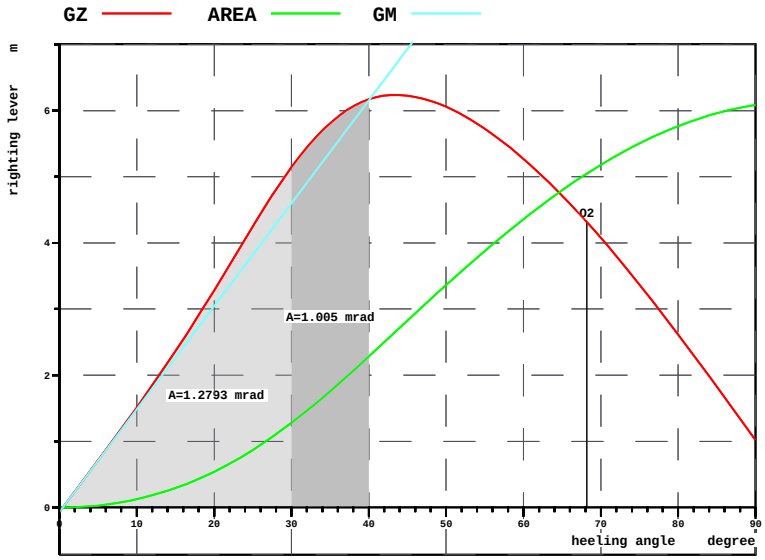
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
SUBTOTAL		59038.0		167.66	-0.00	10.42	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	83438.5	152.26	-0.09 11.71
Total weight	108080.4	147.64	-0.07 12.16

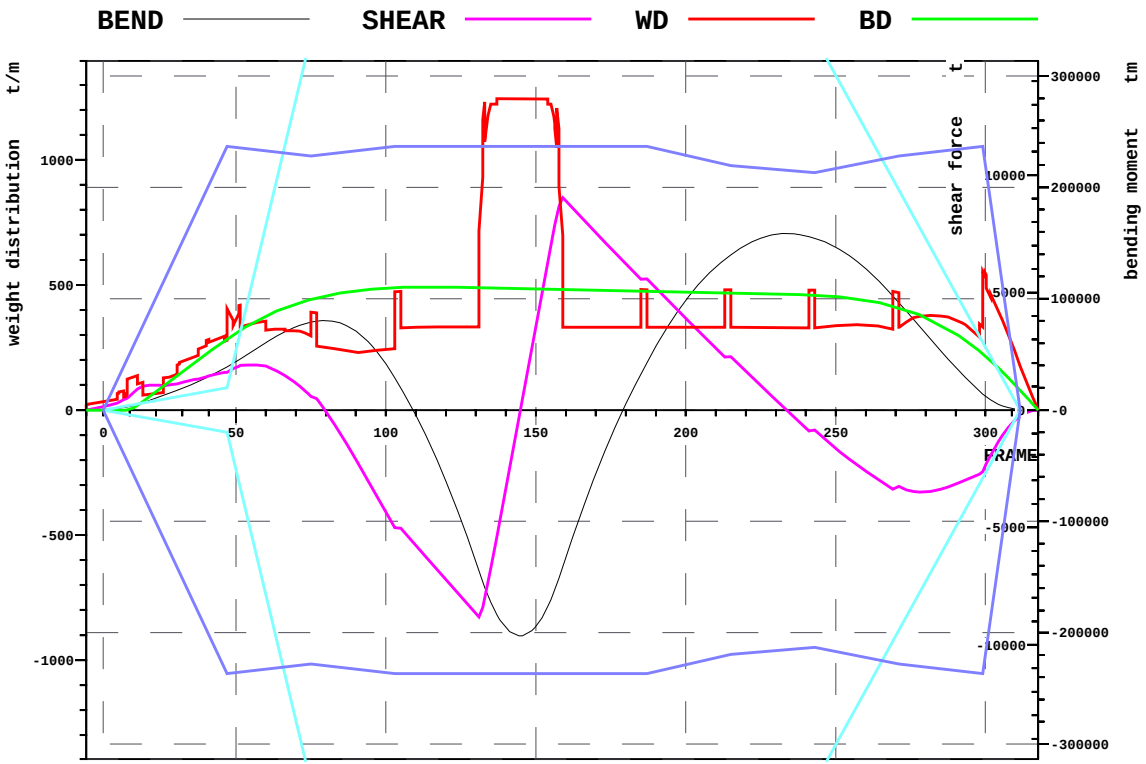
F L O A T I N G P O S I T I O N

Draught moulded	10.427	m	KM	21.07 m
Trim	1.683	m	KG	12.16 m
Heel, PS=+	-0.4	deg		
TA	11.269	m	GM0	8.92 m
TF	9.586	m	GMCORR	0.00 m
Trimming moment	-340934	tonm	GM	8.92 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.07	0.000	0.0	0.000
5.0	0.71	0.028	273.1	0.003
10.0	1.51	0.125	572.7	0.005
20.0	3.28	0.539	1197.1	0.011
30.0	5.15	1.279	1623.0	0.015
40.0	6.17	2.284	1920.4	0.018
45.0	6.23	2.827	2017.0	0.019
50.0	6.06	3.364	2076.3	0.019
55.0	5.73	3.879	2095.0	0.019
60.0	5.27	4.360	2072.0	0.019
65.0	4.72	4.796	2014.9	0.019
70.0	4.08	5.181	1930.4	0.018
75.0	3.37	5.506	1823.1	0.017
80.0	2.62	5.767	1696.6	0.016
85.0	1.83	5.961	1553.9	0.014
90.0	1.02	6.086	1399.3	0.013



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.279	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.284	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.005	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.236	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	43.429	deg	OK
GM0.15	GM > 0.15 m	0.150	8.918	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.184		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	420	142	32.92	0.00
8.80	11.00	3587	694	130.55	6.63
12.00	15.00	6424	1021	59.78	33.50
37.60	47.00	38650	1596	405.49	282.60
41.13	51.00	44845	1857	386.17	314.38
62.97	75.00	78767	583	296.43	441.60
88.45	103.00	28923	-5012	473.38	489.17
113.93	131.00	-144006	-8808	715.79	488.43
139.41	159.00	-140487	9044	697.01	481.39
164.89	187.00	42485	5576	481.09	474.27
190.37	215.00	139240	2276	405.36	467.11
215.85	243.00	154056	-853	328.42	457.83
241.33	271.00	96498	-3256	331.32	406.13
266.81	299.00	14381	-2624	332.69	223.79
278.01	313.00	457	-272	182.45	81.14

				X	FRAME
SHEAR FORCE (MIN)	-8808.3 t	(78.5%)	POSITION:	113.9 m	131
SHEAR FORCE (MAX)	9043.6 t	(80.6%)		139.4 m	159
SAGGING MOMENT	-202962 tm	(46.3%)		126.7 m	145
HOGGING MOMENT	158441.0 tm	(40.8%)		206.0 m	232

9. Loading/unloading sequence: From cond. No4 to cond. No20

LOADING/UNLOADING SEQUENCE SUMMARY FORM

Vessel name	Voyage No.	Condition	Yard	Id. Number

Port (specific or typical):	Condition at commencement of laoding/discharging:	
Total mass of cargo to be loaded/discharged:	Condition at end of loading/discharging:	
Dock water density (t/m^3):	Maximum loading/discharging rate:	Average Loading/discharging rate:
Number of loaders/dischargers:	Maximum Ballasting/Deballasting rate:	Average Ballasting/Deballasting rate:

Hold No.	9	8	7	6	5	4	3	2	1
volume of Hold, V(m^3)									
Height of hold, h (m)	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2

Note:During each pour it has to be controlled that allowable limits for hull girder shear force,bending moment and mass in holds are not exceeded.
Loading/discharging operations may have to be paused to allow for ballasting/deballasting in order to keep actual values within limits

Hold content at commencement of loading/discharging.

Cargo mass (t)									
Density, (t/m^3)									
Grade									

Ballast contents at commencement of loading/discharging.

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side											
Double side											
Double bottom											

Commencement of loading/discharging (sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)

CARGO OPERATIONS

BALLASTING OPERATIONS

Pour No./ Grade	Hold 9	Hold 8	Hold 7	Hold 6	Hold 5	Hold 4	Hold 3	Hold 2	Hold 1
1									
2									
3									
4									
5									
6									
7									
8									
9									

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side											
Double side											
Double bottom											

Values at end of pour (from harbour to sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)

Total cargo onboard (t): Remaining cargo to be loaded (t): Total amount of bunkers onboard (t):

Draft Survey n-1									
Draft Survey n									

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side											
Double side											
Double bottom											

Hold content at end of loading/discharging.

Ballast contents at end of loading/discharging.

Cargo mass (t)									

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side											
Double side											
Double bottom											

Values at end of loading/discharging (sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)

Total mass loaded/discharged (t)	

Net load on Double Bottom:										tons/m^2
Net load in two adjacent holds:										tons

Net load on double bottom = (Mh/V)×h-T[t/m^2]
where: Mh = Mass in hold + mass in DB (t)
V = Total volume of hold (m^3) T = draught (m)
h = height of hold from inner bottom to top of coaming (m)

LOADING/UNLOADING SEQUENCE SUMMARY FORM

Vessel name	Voyage No.	Condition	Yard	Id. Number

Port (specific or typical):	Condition at commencement of laoding/discharging:	
Total mass of cargo to be loaded/discharged:	Condition at end of loading/discharging:	
Dock water density (t/m^3):	Maximum loading/discharging rate:	Average Loading/discharging rate:
Number of loaders/dischargers:	Maximum Ballasting/Deballasting rate:	Average Ballasting/Deballasting rate:

Hold No.	9	8	7	6	5	4	3	2	1
volume of Hold, V(m^3)	18190	21557	21935	21647	21915	21917	21916	21214	16870
Height of hold, h (m)	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2

Note:During each pour it has to be controlled that allowable limits for hull girder shear force,bending moment and mass in holds are not exceeded.
Loading/discharging operations may have to be paused to allow for ballasting/deballasting in order to keep actual values within limits

Hold content at commencement of loading/discharging.

Cargo mass (t)	0	0	0	0	0	0	0	0	0
Density, (t/m^3)									
Grade									

Ballast contents at commencement of loading/discharging.

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0
Double side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0
Double bottom	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0

Commencement of loading/discharging (sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
8.52	0.50	8.02	43.30%	67.40%

CARGO OPERATIONS — BALLASTING OPERATIONS

Values at end of pour (from harbour to sea).

Pour No./ Grade	Hold 9	Hold 8	Hold 7	Hold 6	Hold 5	Hold 4	Hold 3	Hold 2	Hold 1	Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT	d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
1							25000t To38% of hold			Top side								Empty				7.92	-3.44	11.36	70.40%	39.50%
										Double side								Empty								
										Double bottom								Empty								
2			25000t To38% of hold							Top side				Empty								11.9	0.90	11.00	59.60%	29.00%
										Double side				Empty												
										Double bottom				Empty												
3									20000t To40% of hold	Top side									Empty	Empty		10.38	-3.01	13.39	60.00%	37.30%
										Double side									Empty	Empty						
										Double bottom									Empty	Empty						
4	20000t To40% of hold									Top side	Empty	Empty	Empty									14.24	2.40	11.84	77.60%	63.50%
										Double side	Empty	Empty	Empty													
										Double bottom	Empty	Empty	Empty													
5					36694.3t To56% of hold					Top side				Empty	Empty	Empty						15.33	1.69	13.64	81.10%	19.90%
										Double side				Empty	Empty	Empty										
										Double bottom				Empty	Empty	Empty										
6							11695.8t To56% of hold			Top side												14.74	-1.46	16.20	92.80%	29.00%
										Double side																
										Double bottom																
7			11727.8t To56% of hold							Top side												17.36	1.30	16.06	89.90%	38.70%
										Double side																
										Double bottom																
8									7743t To55% of hold	Top side												16.20	-2.35	18.55	75.70%	26.40%
										Double side																
										Double bottom																
9	9913.2t To55% of hold									Top side												18.77	1.15	17.62	76.70%	30.90%
										Double side																
										Double bottom																

Total cargo onboard (t): 167774.1 **Remaining cargo to be loaded (t): 0**

Total amount of bunkers onboard (t): 5975.2 (100%)

										Top side																
Draft Survey n-1										Double side																
										Double bottom																
										Top side																
Draft Survey n										Double side																
										Double bottom																

Hold content at end of loading/discharging.

Cargo mass (t)	29913	0	36728	0	36694	0	36696	0	27743

Total mass loaded/discharged (t):	1677743

Ballast contents at end of loading/discharging.

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	0	0	0	0	0	0	0	0	0	0	0
Double side	0	0	0	0	0	0	0	0	0	0	0
Double bottom	0	0	0	0	0	0	0	0	0	0	0

Values at end of loading/discharging (sea).

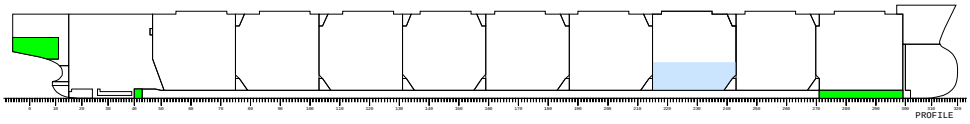
d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
18.77	1.15	17.62	91.40%	51.00%

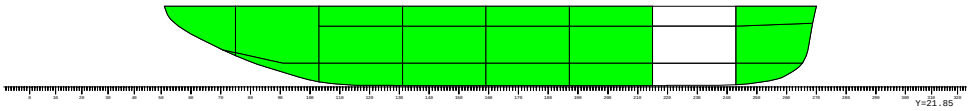
Net load on Double Bottom:	21.3	-18.2	22.3	-18.2	22.3	-18.2	22.5	-18.2	21.6	$tons/m^2$

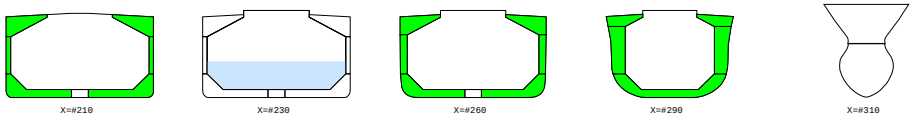
Net load in two adjacent holds:	29913.2	36728	36728	36694	36694	36696	36696	27743	tons

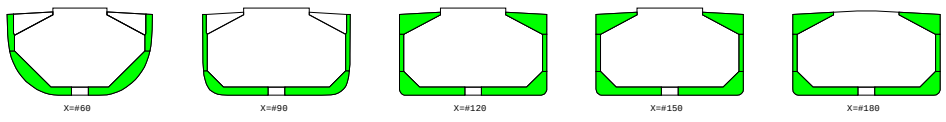
Net load on double bottom = (Mh/V)×h-T[t/m^2]
where: Mh = Mass in hold + mass in DB (t)
V = Total volume of hold (m^3) T = draught (m)
h = height of hold from inner bottom to top of coaming (m)

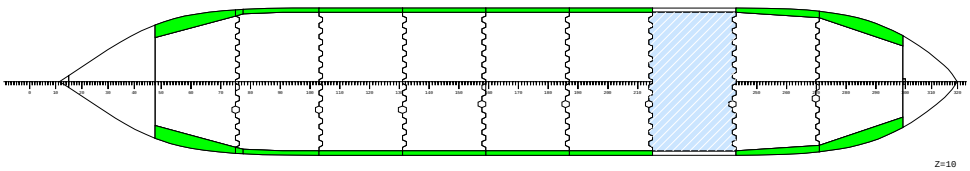
LOADING CONDITION - LSC4C20-01 : LOA. SEQU.LC4LC20 STEP1-GARGO3 25000t-ARR. BUN

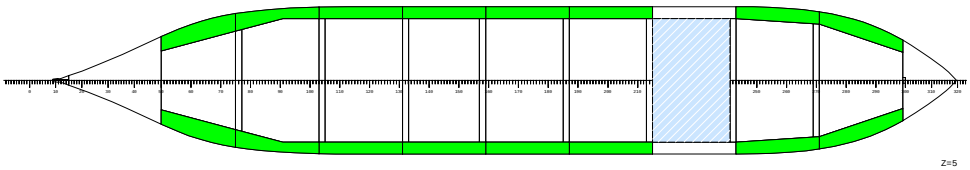


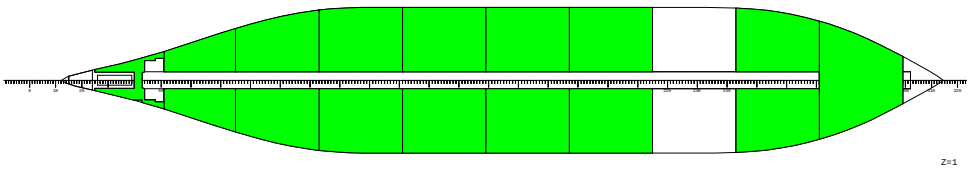












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.3	NO.3 CARGO HOLD	25000.0	38.2	202.32	0.01	7.05	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00

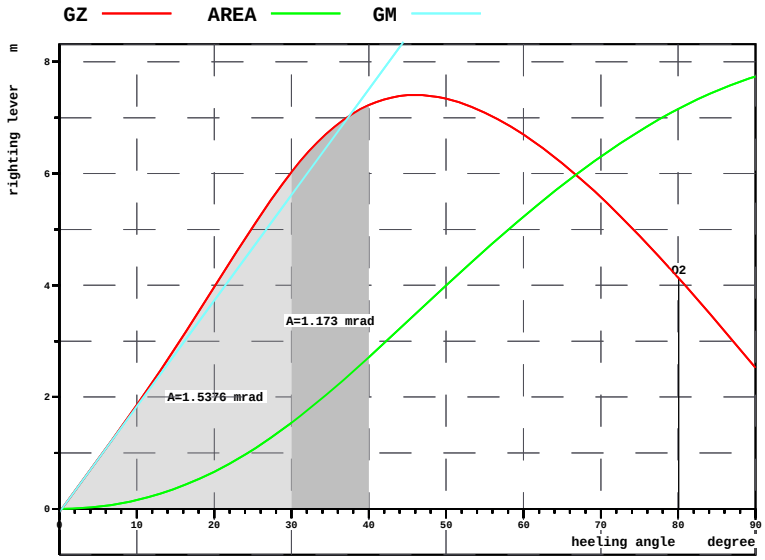
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		50839.2		149.70	-0.00	10.59	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	76707.3	165.50	-0.08 9.58
Total weight	101349.2	157.36	-0.06 10.57

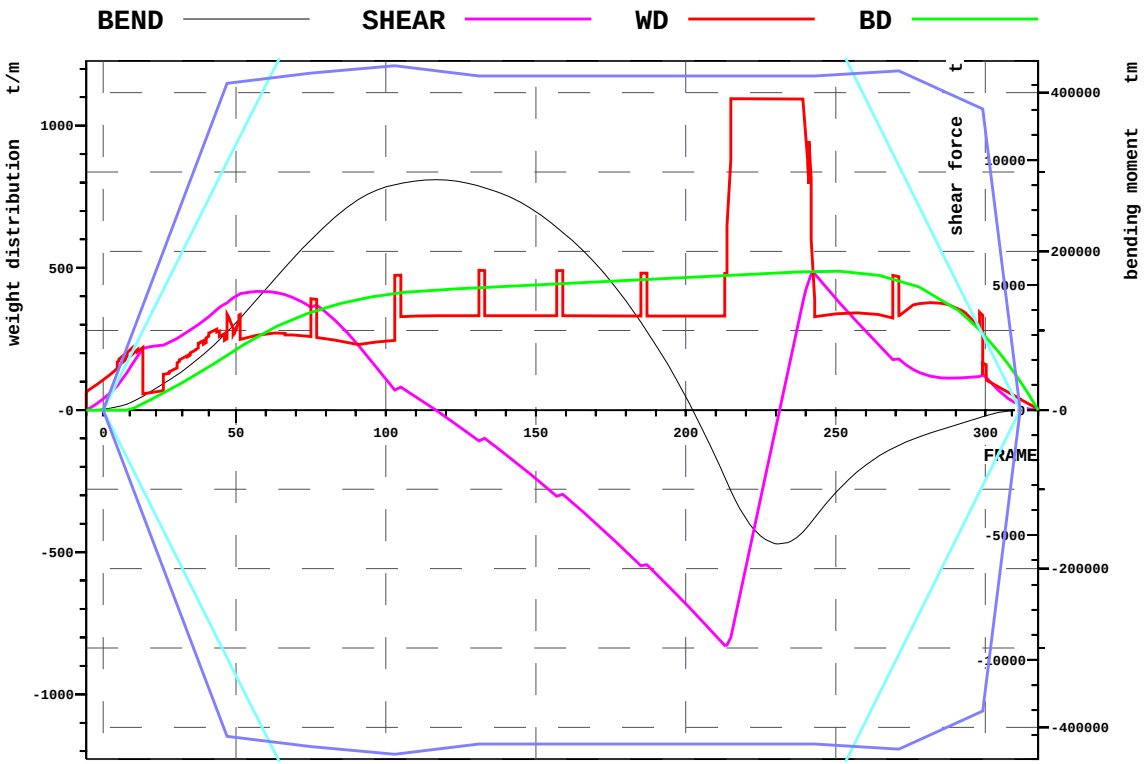
F L O A T I N G P O S I T I O N

Draught moulded	9.640	m	KM	21.41 m
Trim	-3.449	m	KG	10.57 m
Heel, PS=+	-0.3	deg		
TA	7.916	m	GM0	10.83 m
TF	11.364	m	GMCORR	0.00 m
Trimming moment	653715	tonm	GM	10.83 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.89	0.036	64.4	0.001
10.0	1.86	0.156	128.6	0.001
20.0	3.96	0.661	221.7	0.002
30.0	6.02	1.537	296.1	0.003
40.0	7.23	2.711	401.9	0.004
45.0	7.40	3.351	456.9	0.005
50.0	7.34	3.995	503.3	0.005
55.0	7.09	4.626	540.1	0.005
60.0	6.70	5.229	568.2	0.006
65.0	6.18	5.792	588.6	0.006
70.0	5.58	6.305	603.4	0.006
75.0	4.89	6.762	614.6	0.006
80.0	4.14	7.156	620.8	0.006
85.0	3.35	7.483	620.0	0.006
90.0	2.52	7.740	613.4	0.006



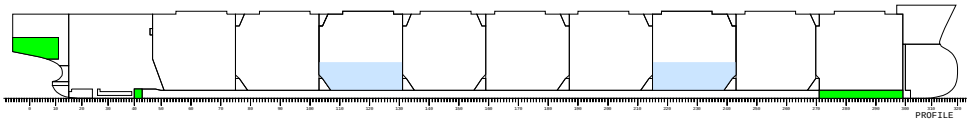
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.538	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.711	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.173	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.406	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	46.011	deg	OK
GM0.15	GM > 0.15 m	0.150	10.832	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.080		OK

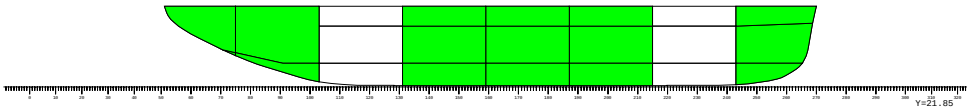


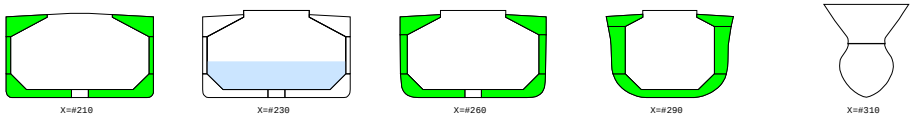
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1061	443	106.39	0.00
8.80	11.00	10257	1839	218.40	5.28
12.00	15.00	17115	2463	57.05	22.05
37.60	47.00	98265	4279	336.92	192.71
41.13	51.00	114004	4613	334.15	219.00
62.97	75.00	214110	4113	257.87	343.17
88.45	103.00	283836	802	473.37	409.57
113.93	131.00	282037	-1242	490.80	430.53
139.41	159.00	223864	-3373	490.24	445.21
164.89	187.00	99342	-6196	481.09	459.82
190.37	215.00	-101845	-9093	986.86	474.40
215.85	243.00	-135357	5446	328.42	486.80
241.33	271.00	-44557	2046	331.32	454.10
266.81	299.00	-8442	1401	332.68	268.64
278.01	313.00	-321	178	40.20	103.54

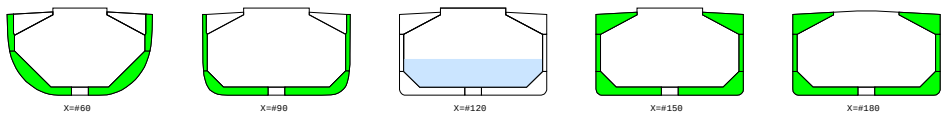
				X	FRAME
SHEAR FORCE (MIN)	-9416.5 t	(70.4%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	5446.4 t	(40.7%)		215.8 m	243
SAGGING MOMENT	-168912 tm	(27.9%)		204.7 m	231
HOGGING MOMENT	290169.9 tm	(39.5%)		100.6 m	116

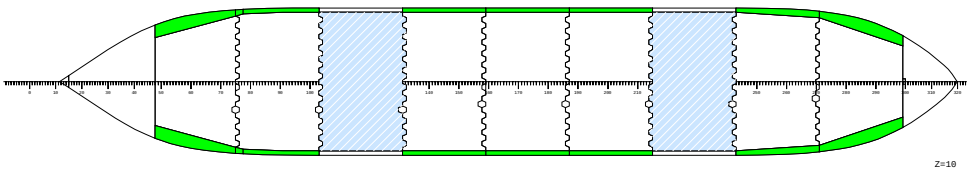
LOADING CONDITION - LSC4C20-02 : LOA. SEQU.LC4LC20 STEP2-GARGO7 25000t-ARR. BUN

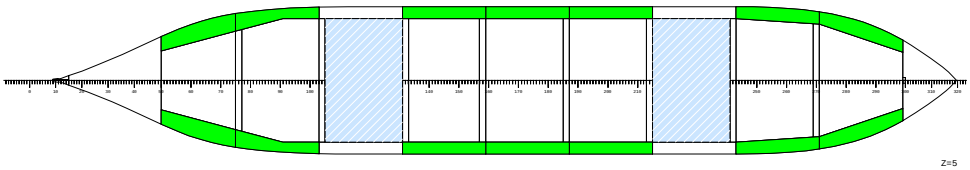


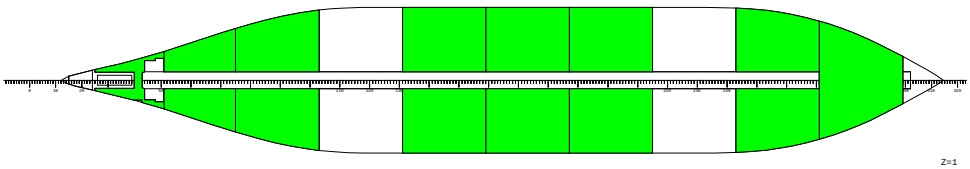












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.3	NO.3 CARGO HOLD	25000.0	38.2	202.32	0.01	7.05	0.00
R1.7	NO.7 CARGO HOLD	25000.0	38.1	101.94	0.02	7.04	0.00
SUBTOTAL		50000.0		152.13	0.02	7.04	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

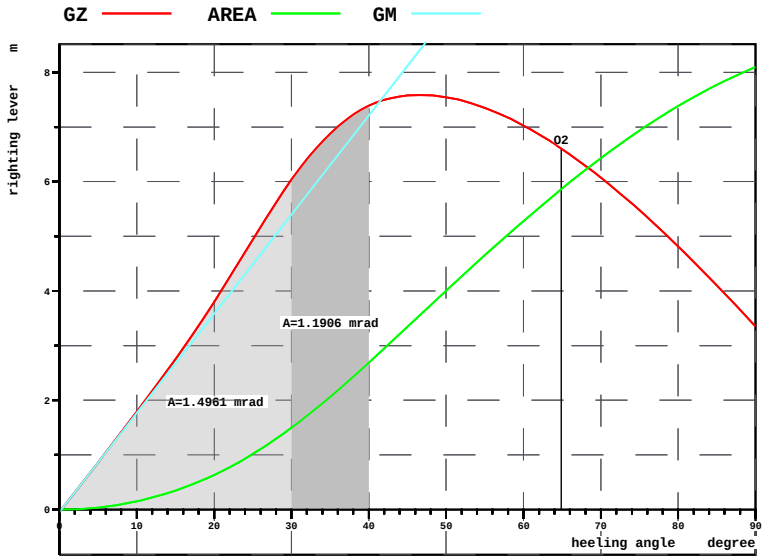
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		44290.4		156.87	-0.00	10.55	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	95158.4	153.23	-0.06 8.82
Total weight	119800.3	148.86	-0.05 9.82

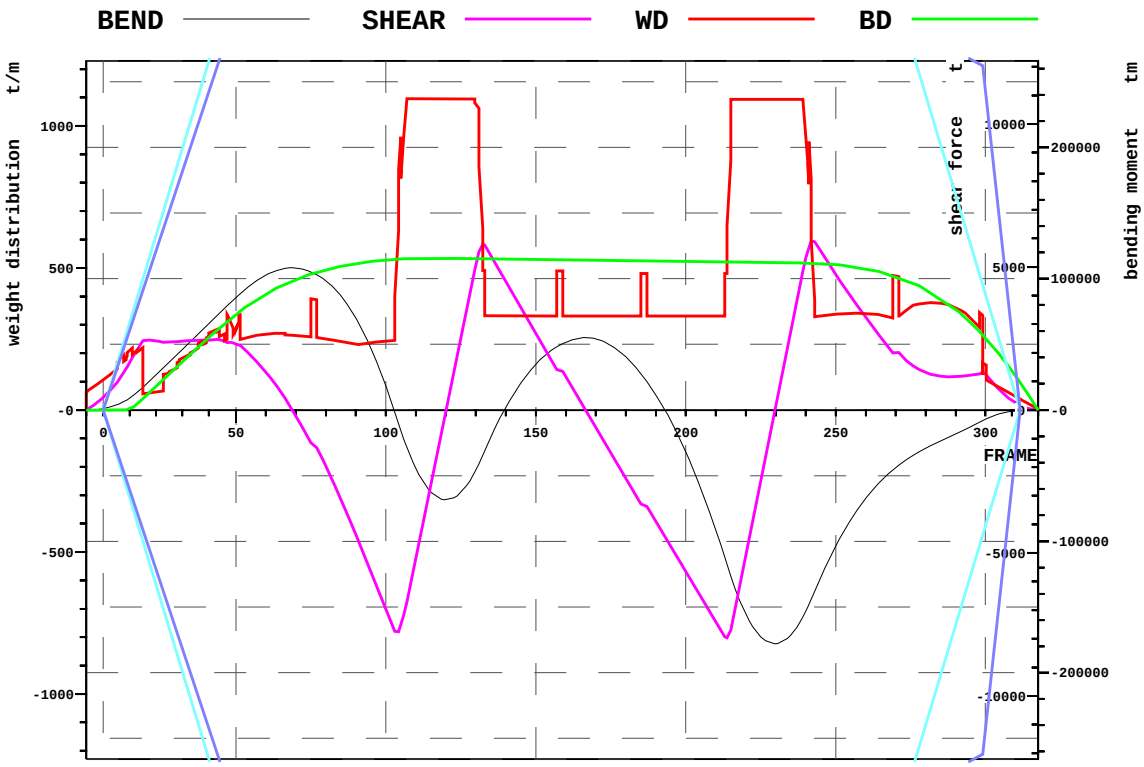
F L O A T I N G P O S I T I O N

Draught moulded	11.452	m	KM	20.21	m
Trim	0.903	m	KG	9.82	m
Heel, PS=+	-0.2	deg			
TA	11.903	m	GM0	10.39	m
TF	11.000	m	GMCORR	0.00	m
Trimming moment	-187353	tonm	GM	10.39	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.86	0.036	64.4	0.001
10.0	1.79	0.151	128.6	0.001
20.0	3.80	0.635	221.7	0.002
30.0	6.04	1.496	296.1	0.002
40.0	7.39	2.687	401.9	0.003
45.0	7.58	3.341	456.9	0.004
50.0	7.55	4.002	503.3	0.004
55.0	7.35	4.654	540.1	0.005
60.0	7.03	5.282	568.2	0.005
65.0	6.59	5.877	588.6	0.005
70.0	6.07	6.430	603.4	0.005
75.0	5.48	6.934	614.6	0.005
80.0	4.81	7.384	620.8	0.005
85.0	4.10	7.773	620.0	0.005
90.0	3.35	8.098	613.4	0.005



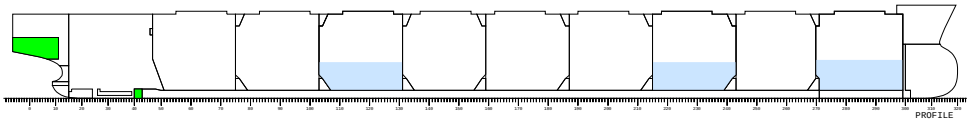
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.496	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.687	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.191	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.585	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	46.646	deg	OK
GM0.15	GM > 0.15 m	0.150	10.392	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.232		OK

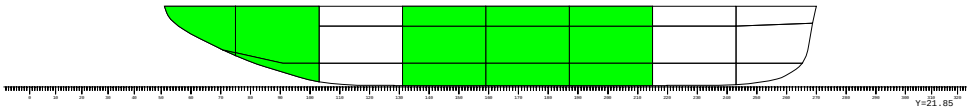


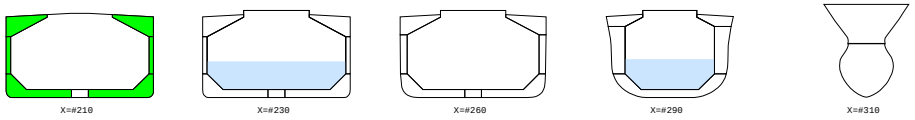
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1124	443	106.38	0.00
8.80	11.00	10421	1835	218.49	9.11
12.00	15.00	17264	2426	57.04	39.08
37.60	47.00	79320	2359	336.87	310.14
41.13	51.00	87577	2266	334.10	343.93
62.97	75.00	105157	-1152	257.83	478.37
88.45	103.00	-1525	-7741	400.01	529.98
113.93	131.00	-40894	5537	856.07	532.55
139.41	159.00	50530	1349	490.17	528.81
164.89	187.00	21482	-3371	481.01	524.98
190.37	215.00	-126575	-7695	986.71	521.13
215.85	243.00	-137359	5886	328.37	515.07
241.33	271.00	-42218	2003	331.26	463.60
266.81	299.00	-8123	1326	332.63	263.43
278.01	313.00	-359	164	40.20	99.11

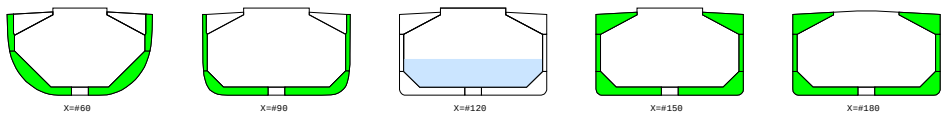
				X	FRAME
SHEAR FORCE (MIN)	-7961.3 t	(59.6%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	5909.8 t	(44.2%)		214.8 m	242
SAGGING MOMENT	-178130 tm	(29.0%)		203.7 m	230
HOGGING MOMENT	108288.7 tm	(22.7%)		57.5 m	69

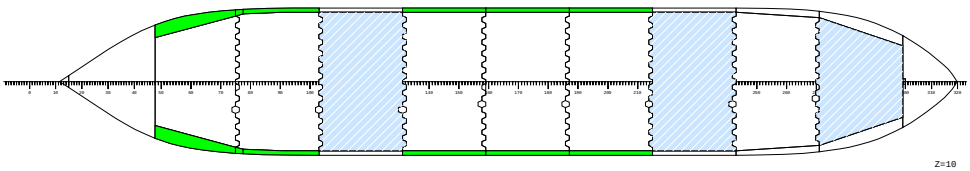
LOADING CONDITION - LSC4C20-03 : LOA. SEQU.LC4LC20 STEP3-GARGO1 20000t-ARR. BUN

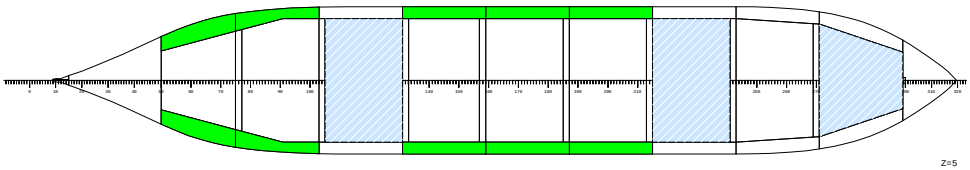


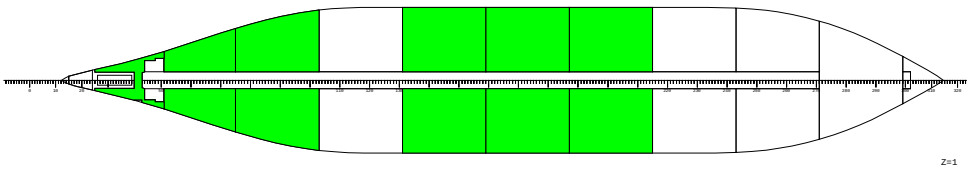












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20000.0	39.6	252.56	0.00	7.39	0.00
R1.3	NO.3 CARGO HOLD	25000.0	38.2	202.32	0.01	7.05	0.00
R1.7	NO.7 CARGO HOLD	25000.0	38.1	101.94	0.02	7.04	0.00
SUBTOTAL		70000.0		180.82	0.01	7.14	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

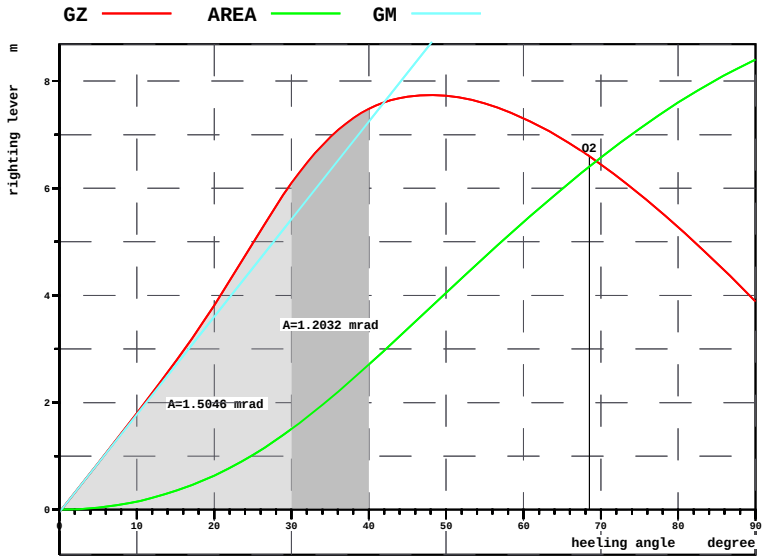
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		30294.7		117.72	-0.00	10.06	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	101162.8	160.64	-0.05 8.15
Total weight	125804.6	155.03	-0.04 9.23

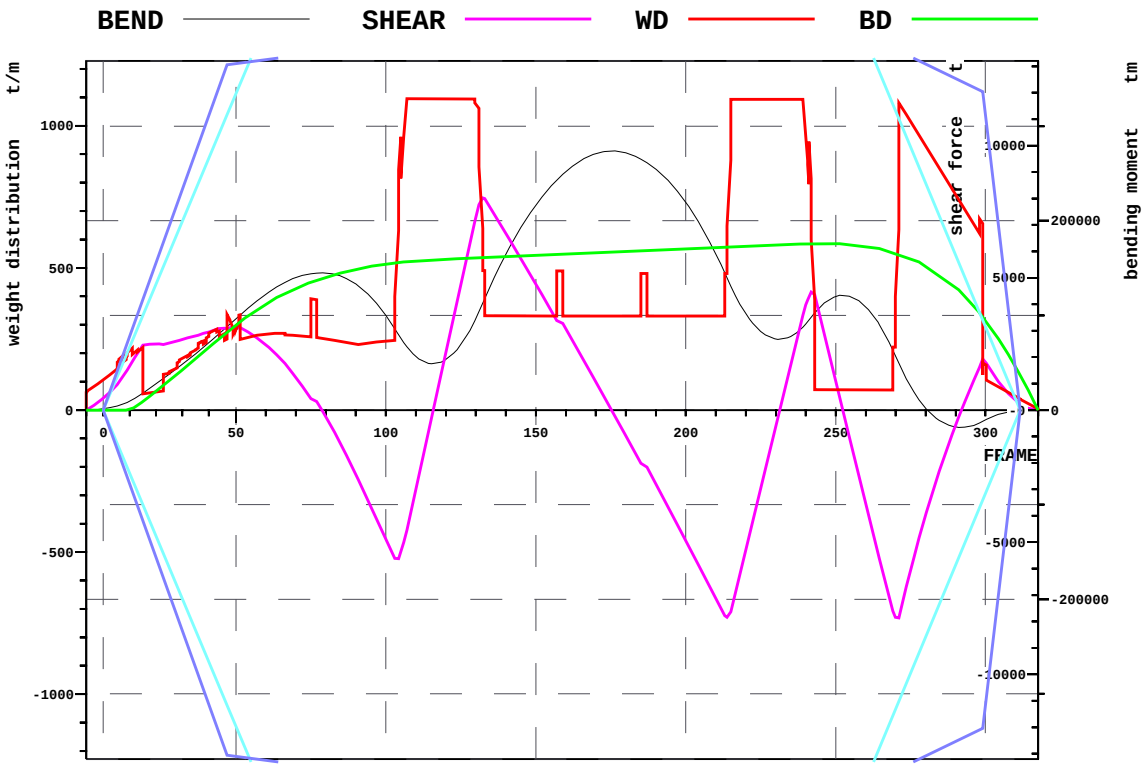
F L O A T I N G P O S I T I O N

Draught moulded	11.880	m	KM	19.67 m
Trim	-3.009	m	KG	9.23 m
Heel, PS=+	-0.2	deg		
TA	10.376	m	GM0	10.43 m
TF	13.385	m	GMCORR	0.00 m
Trimming moment	595177	tonm	GM	10.43 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.87	0.036	64.4	0.001
10.0	1.80	0.152	128.6	0.001
20.0	3.81	0.637	221.7	0.002
30.0	6.09	1.505	296.1	0.002
40.0	7.48	2.708	401.9	0.003
45.0	7.71	3.372	456.9	0.004
50.0	7.73	4.047	503.3	0.004
55.0	7.58	4.716	540.1	0.004
60.0	7.31	5.367	568.2	0.005
65.0	6.92	5.988	588.6	0.005
70.0	6.44	6.572	603.4	0.005
75.0	5.89	7.111	614.6	0.005
80.0	5.27	7.598	620.8	0.005
85.0	4.60	8.029	620.0	0.005
90.0	3.89	8.400	613.4	0.005



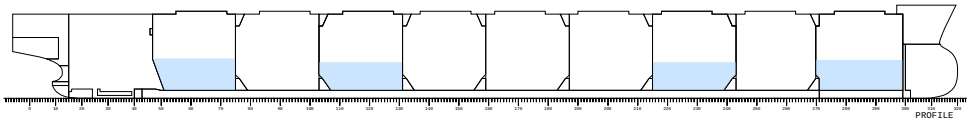
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.505	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.708	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.203	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.740	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.982	deg	OK
GM0.15	GM > 0.15 m	0.150	10.433	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.856		OK

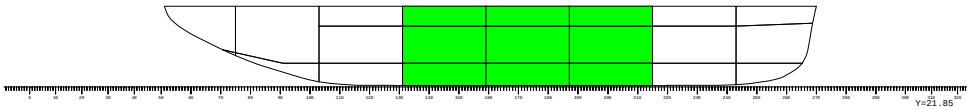


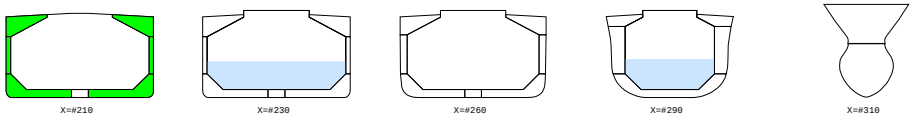
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1221	443	106.32	0.00
8.80	11.00	10683	1838	218.25	5.51
12.00	15.00	17623	2450	57.01	29.59
37.60	47.00	88052	3072	336.68	273.45
41.13	51.00	99119	3108	333.90	306.80
62.97	75.00	143934	432	257.68	449.22
88.45	103.00	85816	-5618	399.78	516.80
113.93	131.00	103303	7769	855.57	535.94
139.41	159.00	249096	3278	489.88	548.77
164.89	187.00	261332	-2166	330.48	561.52
190.37	215.00	130720	-7637	986.13	574.23
215.85	243.00	102624	4363	387.24	584.66
241.33	271.00	49416	-7871	634.97	545.42
266.81	299.00	-12085	1934	656.00	326.90
278.01	313.00	-629	242	40.17	126.15

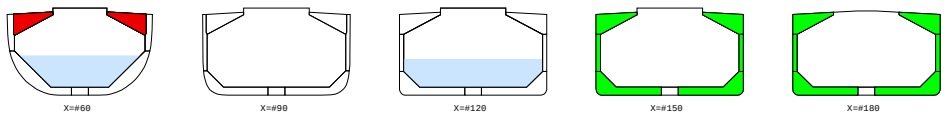
				X	FRAME
SHEAR FORCE (MIN)	-7870.9 t	(58.0%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	8022.3 t	(60.0%)		115.1 m	132
SAGGING MOMENT	-18358.3 tm	(11.9%)		259.5 m	291
HOGGING MOMENT	273759.7 tm	(37.3%)		155.0 m	176

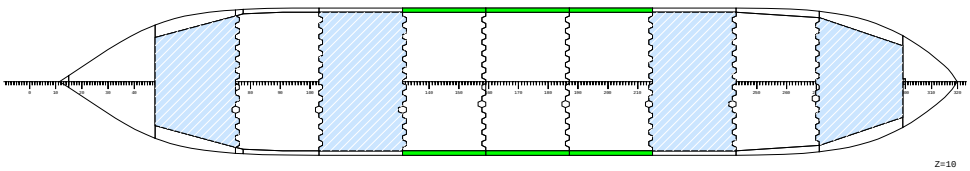
LOADING CONDITION - LSC4C20-04 : LOA. SEQU.LC4LC20 STEP4-GARGO9 20000t-50% BUN.

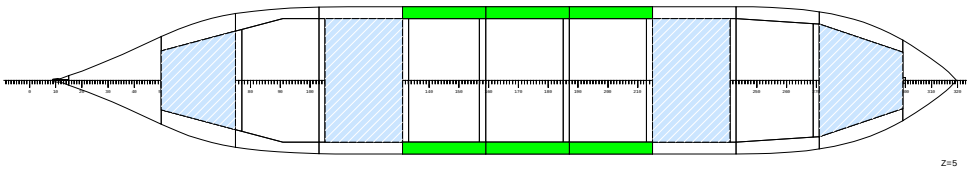


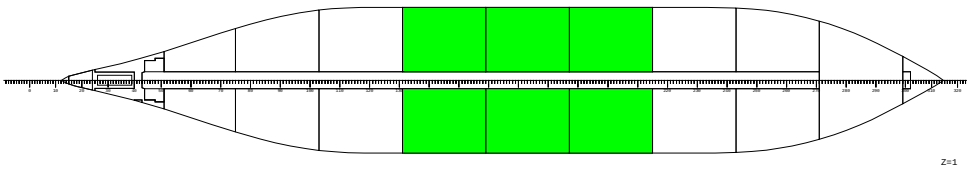












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20000.0	39.6	252.56	0.00	7.39	0.00
R1.3	NO.3 CARGO HOLD	25000.0	38.2	202.32	0.01	7.05	0.00
R1.7	NO.7 CARGO HOLD	25000.0	38.1	101.94	0.02	7.04	0.00
R1.9	NO.9 CARGO HOLD	20000.0	36.8	52.05	0.03	7.99	0.00
SUBTOTAL		90000.0		152.21	0.02	7.33	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

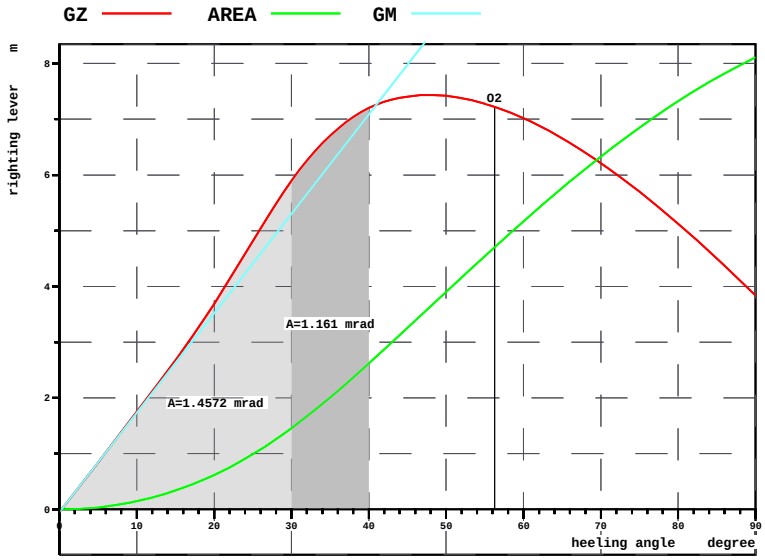
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
SUBTOTAL		19734.5		152.15	-0.00	10.86	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	112971.4	148.95	-0.06 8.38
Total weight	137613.3	145.91	-0.05 9.33

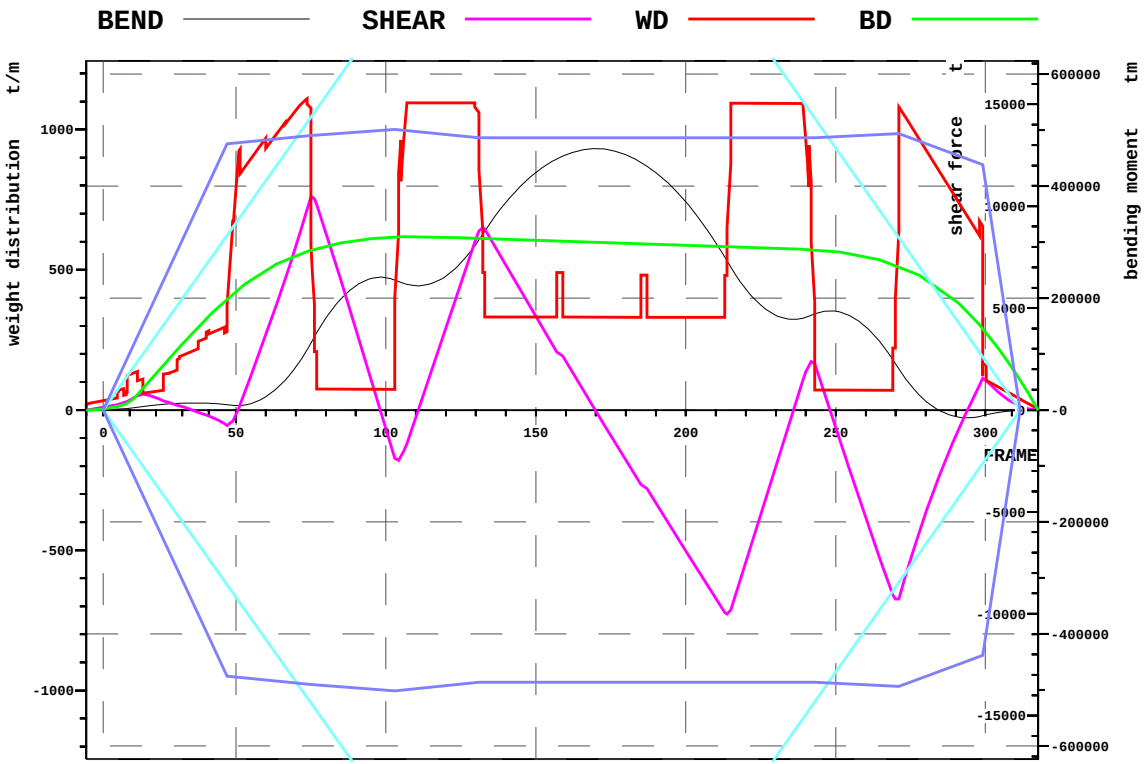
F L O A T I N G P O S I T I O N

Draught moulded	13.038	m	KM	19.54	m
Trim	2.395	m	KG	9.33	m
Heel, PS=+	-0.3	deg			
TA	14.235	m	GM0	10.21	m
TF	11.841	m	GMCORR	0.00	m
Trimming moment	-547482	tonm	GM	10.21	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.84	0.035	273.1	0.002
10.0	1.75	0.148	572.7	0.004
20.0	3.69	0.618	1197.1	0.009
30.0	5.90	1.457	1623.0	0.012
40.0	7.20	2.618	1920.4	0.014
45.0	7.41	3.257	2017.0	0.015
50.0	7.42	3.905	2076.3	0.015
55.0	7.28	4.548	2095.0	0.015
60.0	7.02	5.173	2072.0	0.015
65.0	6.66	5.770	2014.9	0.015
70.0	6.21	6.332	1930.4	0.014
75.0	5.70	6.852	1823.1	0.013
80.0	5.12	7.325	1696.6	0.012
85.0	4.50	7.745	1553.9	0.011
90.0	3.84	8.110	1399.3	0.010



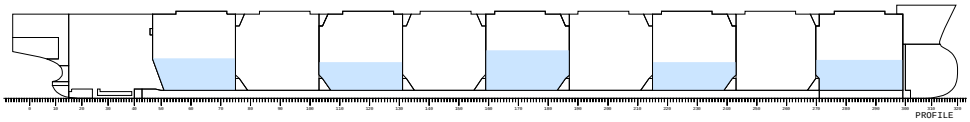
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.457	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.618	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.161	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.434	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.892	deg	OK
GM0.15	GM > 0.15 m	0.150	10.213	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.894		OK

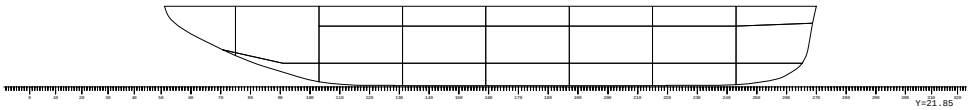


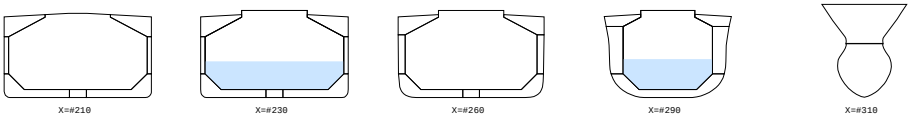
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	562	134	32.90	3.81
8.80	11.00	3536	561	130.50	36.21
12.00	15.00	5868	772	59.75	76.00
37.60	47.00	9314	-760	373.86	393.76
41.13	51.00	7902	96	923.67	430.40
62.97	75.00	119203	104571	076.81	569.23
88.45	103.00	232349	-2369	399.83	615.86
113.93	131.00	303449	8789	855.69	612.14
139.41	159.00	453467	2649	489.96	602.07
164.89	187.00	435356	-3860	480.80	591.93
190.37	215.00	254682	-9813	986.28	581.75
215.85	243.00	171710	2291	387.29	569.31
241.33	271.00	73584	-9272	635.06	509.21
266.81	299.00	-10155	1563	656.10	289.32
278.01	313.00	-631	194	40.18	109.39

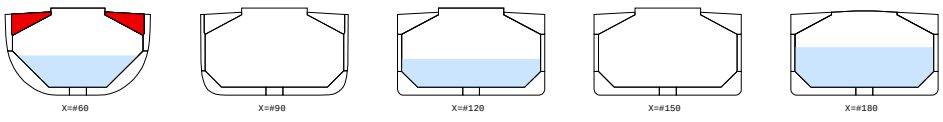
				X	FRAME
SHEAR FORCE (MIN)	-10012.7 t	(74.9%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	10457.2 t	(77.6%)		63.0 m	75
SAGGING MOMENT	-13834.4 tm	(10.0%)		261.4 m	293
HOGGING MOMENT	466817.5 tm	(63.5%)		148.7 m	169

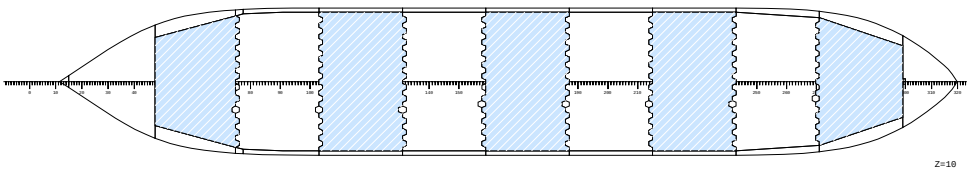
LOADING CONDITION - LSC4C20-05 : LOA. SEQU.LC4LC20 STEP5-GARGO5 LOAD 56%-50% BUN

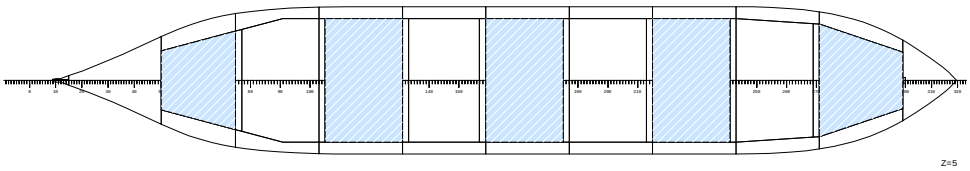


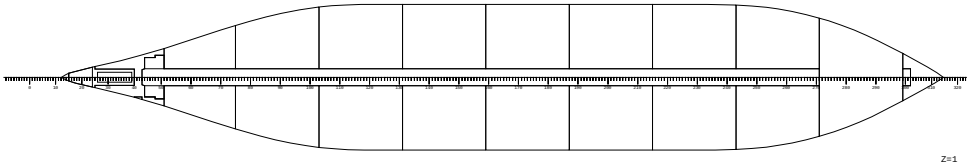












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20000.0	39.6	252.56	0.00	7.39	0.00
R1.3	NO.3 CARGO HOLD	25000.0	38.2	202.32	0.01	7.05	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	25000.0	38.1	101.94	0.02	7.04	0.00
R1.9	NO.9 CARGO HOLD	20000.0	36.8	52.05	0.03	7.99	0.00
SUBTOTAL		126694.3		151.98	0.02	7.79	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

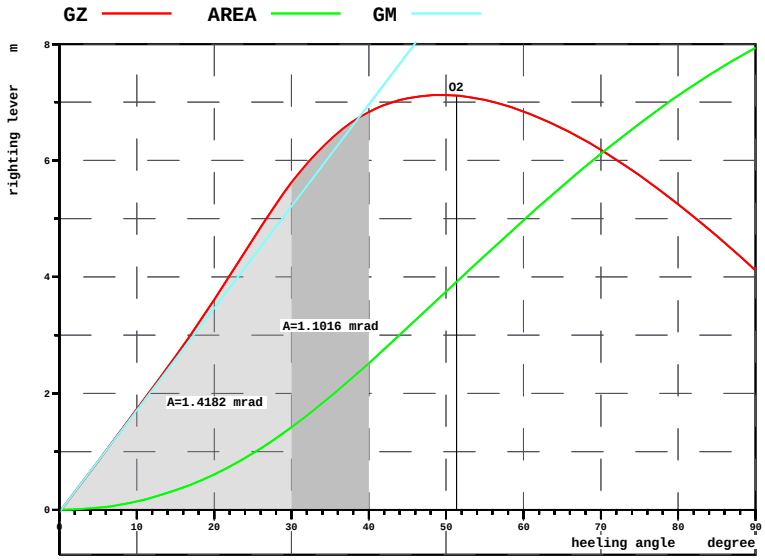
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	129931.2	149.16	-0.05 8.15
Total weight	154573.1	146.43	-0.04 9.03

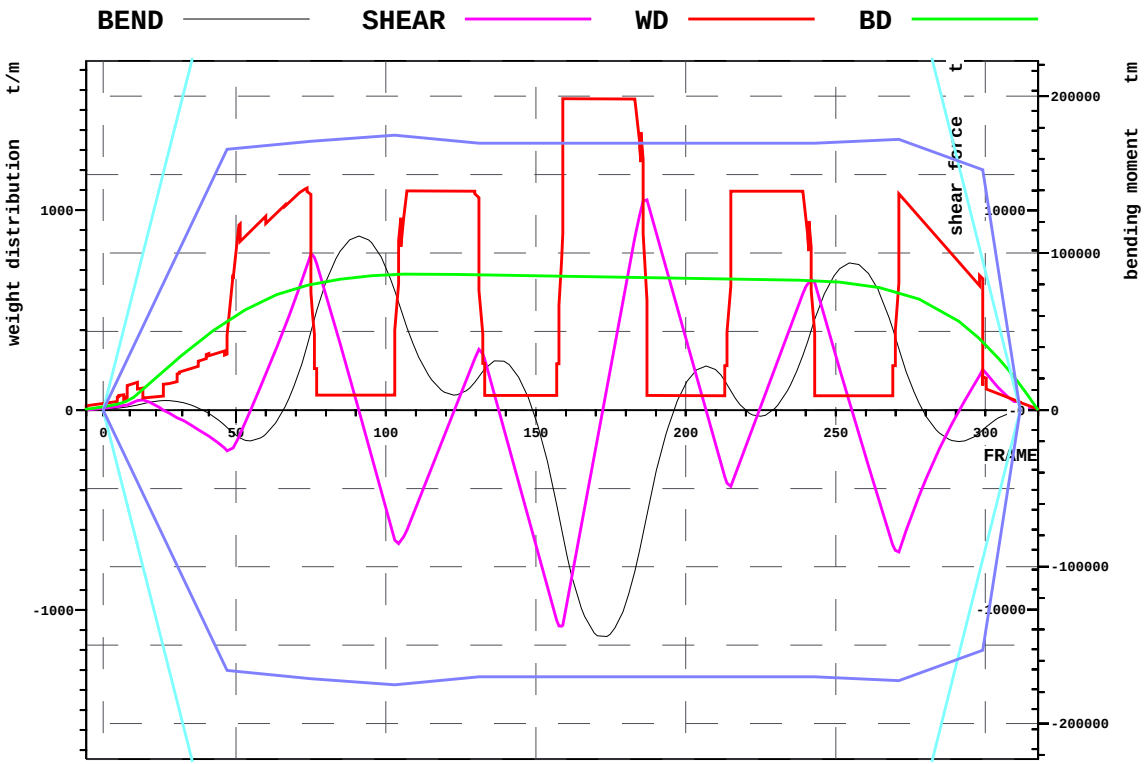
F L O A T I N G P O S I T I O N

Draught moulded	14.485	m	KM	19.06	m
Trim	1.686	m	KG	9.03	m
Heel, PS=+	-0.2	deg			
TA	15.328	m	GM0	10.02	m
TF	13.641	m	GMCORR	0.00	m
Trimming moment	-398623	tonm	GM	10.02	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.84	0.035	273.1	0.002
10.0	1.73	0.147	572.7	0.004
20.0	3.61	0.609	1197.1	0.008
30.0	5.62	1.418	1623.0	0.010
40.0	6.83	2.520	1920.4	0.012
45.0	7.07	3.128	2017.0	0.013
50.0	7.12	3.748	2076.3	0.013
55.0	7.04	4.366	2095.0	0.014
60.0	6.84	4.973	2072.0	0.013
65.0	6.55	5.558	2014.9	0.013
70.0	6.18	6.114	1930.4	0.012
75.0	5.74	6.634	1823.1	0.012
80.0	5.25	7.114	1696.6	0.011
85.0	4.70	7.549	1553.9	0.010
90.0	4.11	7.933	1399.3	0.009



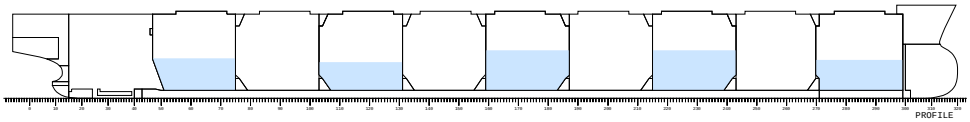
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.418	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.520	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.102	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.123	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	49.368	deg	OK
GM0.15	GM > 0.15 m	0.150	10.022	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.261		OK

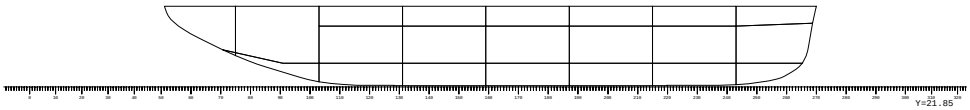


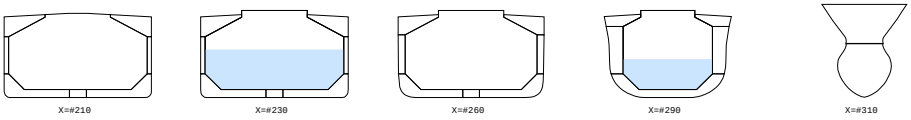
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	448	94	32.90	15.00
8.80	11.00	2454	370	130.47	59.21
12.00	15.00	4037	501	59.75	102.92
37.60	47.00	-11224	-2037	373.85	442.61
41.13	51.00	-17465	-1356	923.64	481.12
62.97	75.00	49264	7814	589.66	626.53
88.45	103.00	75919	-6522	399.82	676.76
113.93	131.00	21004	3046	597.44	676.03
139.41	159.00	-80013	-10788	880.16	668.97
164.89	187.00	-66845	10530	548.42	661.82
190.37	215.00	12194	-38221093.50		654.65
215.85	243.00	57774	6386	387.28	645.12
241.33	271.00	39103	-7121	635.04	584.20
266.81	299.00	-12757	2022	656.08	339.75
278.01	313.00	-720	248	40.18	128.62

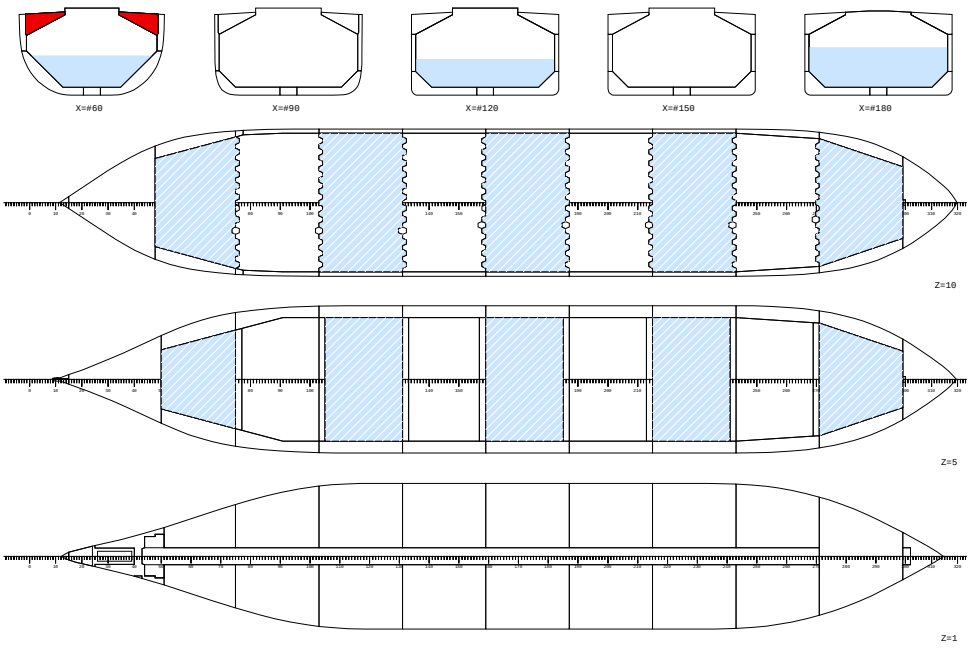
				X	FRAME
SHEAR FORCE (MIN)	-10845.3 t	(81.1%)	POSITION:	139.0 m	159
SHEAR FORCE (MAX)	10530.0 t	(78.8%)		164.9 m	187
SAGGING MOMENT	-144424 tm	(19.9%)		152.8 m	174
HOGGING MOMENT	110973.6 tm	(17.2%)		77.6 m	91

LOADING CONDITION - LSC4C20-06 : LOA. SEQU.LC4LC20 STEP6-GARGO3 LOAD 56%-50% BUN









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20000.0	39.6	252.56	0.00	7.39	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	25000.0	38.1	101.94	0.02	7.04	0.00
R1.9	NO.9 CARGO HOLD	20000.0	36.8	52.05	0.03	7.99	0.00
SUBTOTAL		138390.0		156.25	0.02	8.22	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

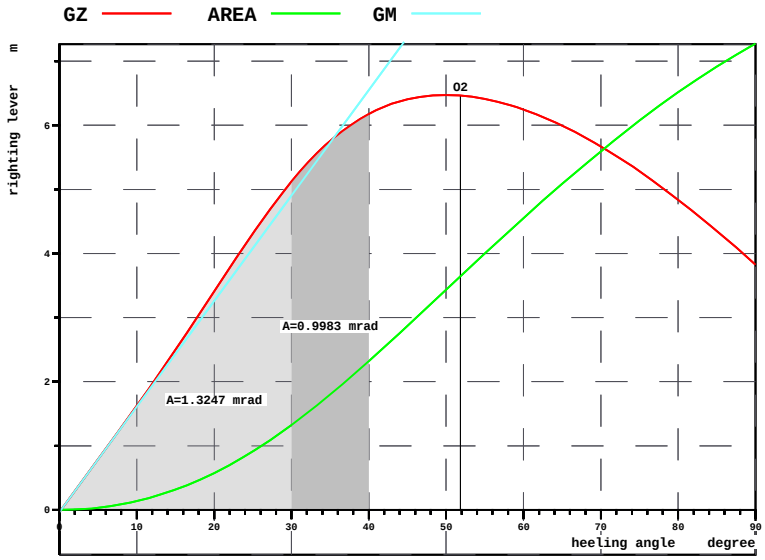
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00	13.67
Deadweight	141627.0	153.57	-0.04	8.54
Total weight	166268.8	150.37	-0.03	9.30

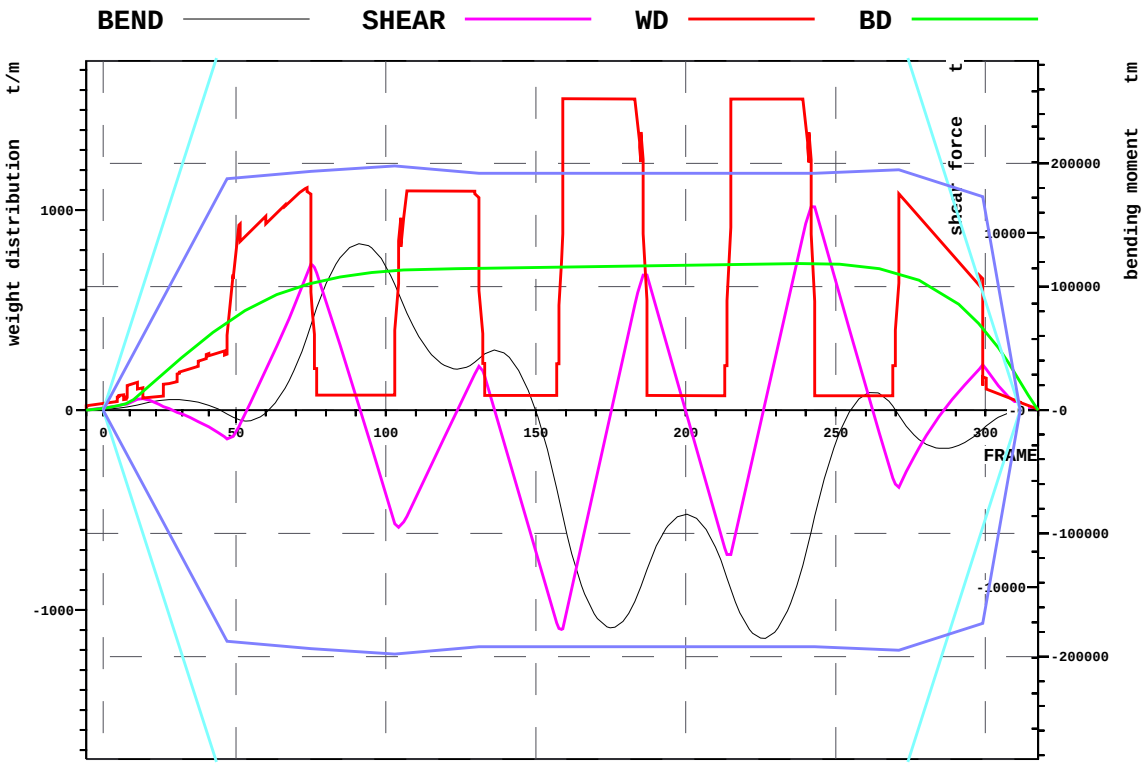
F L O A T I N G P O S I T I O N

Draught moulded	15.474	m	KM	18.73 m
Trim	-1.460	m	KG	9.30 m
Heel, PS=+	-0.2	deg		
TA	14.744	m	GM0	9.43 m
TF	16.203	m	GMCORR	0.00 m
Trimming moment	336687	tonm	GM	9.43 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.79	0.033	273.1	0.002
10.0	1.63	0.138	572.7	0.003
20.0	3.40	0.575	1197.1	0.007
30.0	5.13	1.325	1623.0	0.010
40.0	6.18	2.323	1920.4	0.012
45.0	6.41	2.873	2017.0	0.012
50.0	6.47	3.436	2076.3	0.012
55.0	6.41	3.999	2095.0	0.013
60.0	6.25	4.552	2072.0	0.012
65.0	6.00	5.087	2014.9	0.012
70.0	5.67	5.596	1930.4	0.012
75.0	5.28	6.075	1823.1	0.011
80.0	4.84	6.517	1696.6	0.010
85.0	4.35	6.918	1553.9	0.009
90.0	3.83	7.275	1399.3	0.008



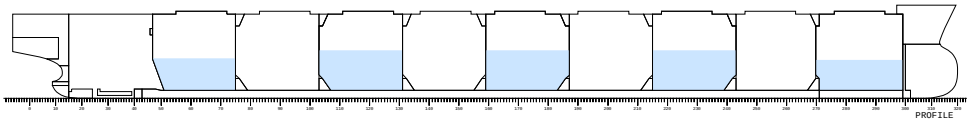
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.325	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.323	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.998	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.473	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	49.863	deg	OK
GM0.15	GM > 0.15 m	0.150	9.425	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.955		OK

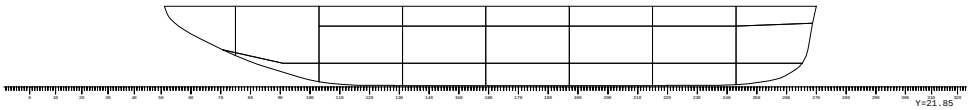


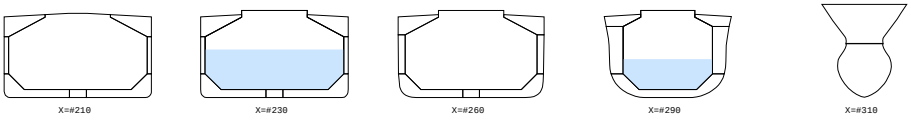
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	511	121	32.90	8.26
8.80	11.00	3061	474	130.38	48.63
12.00	15.00	5033	640	59.75	91.66
37.60	47.00	-3121	-1641	373.84	436.12
41.13	51.00	-7929	-940	923.63	476.12
62.97	75.00	68190	82231	078.39	632.43
88.45	103.00	101877	-6434	399.82	696.04
113.93	131.00	41479	2472	597.43	708.64
139.41	159.00	-86197	-12362	880.14	714.89
164.89	187.00	-129508	7615	548.41	721.07
190.37	215.00	-145290	-81701	554.21	727.22
215.85	243.00	-86120	11471	543.11	730.92
241.33	271.00	-4346	-4360	635.03	678.55
266.81	299.00	-15462	2560	656.07	407.91
278.01	313.00	-764	276	40.18	144.90

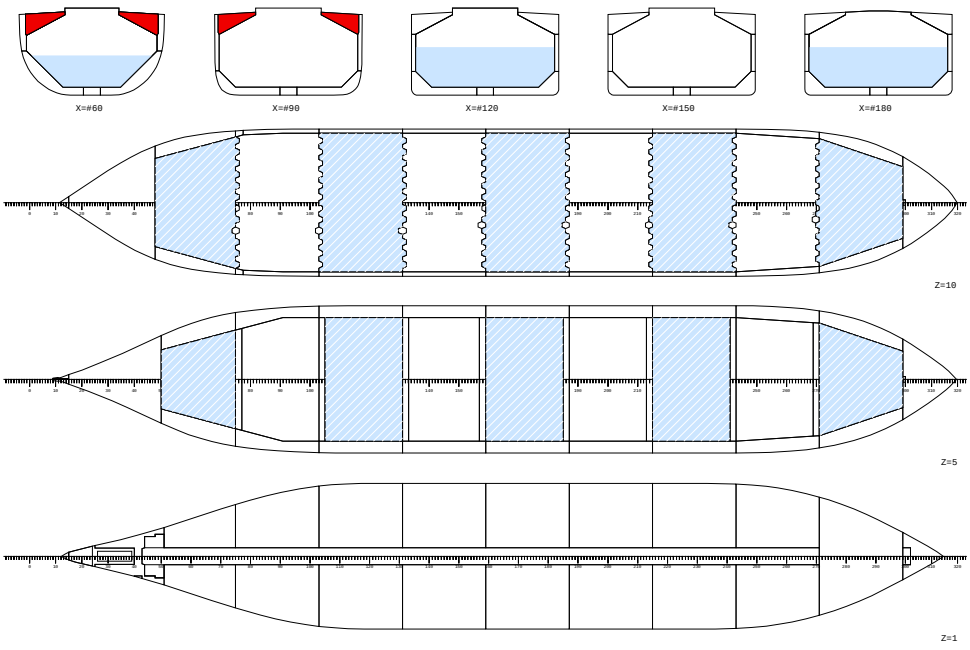
				X	FRAME
SHEAR FORCE (MIN)	-12400.8 t	(92.8%)	POSITION:	139.0 m	159
SHEAR FORCE (MAX)	11493.1 t	(86.0%)		214.8 m	242
SAGGING MOMENT	-185159 tm	(29.0%)		200.8 m	227
HOGGING MOMENT	134961.9 tm	(20.9%)		77.6 m	91

LOADING CONDITION - LSC4C20-07 : LOA. SEQU.LC4LC20 STEP7-GARGO7 LOAD 56%-DEP. BU









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20000.0	39.6	252.56	0.00	7.39	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	20000.0	36.8	52.05	0.03	7.99	0.00
SUBTOTAL		150117.8		151.99	0.02	8.58	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

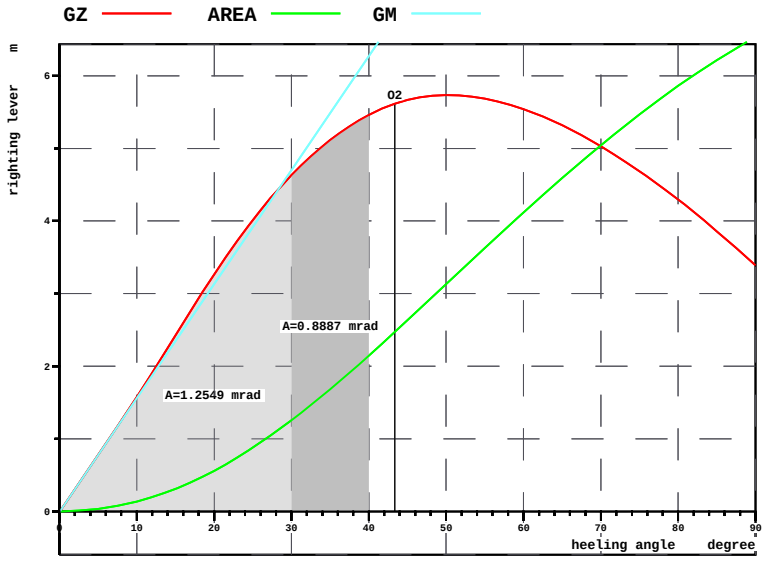
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)						
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	156303.1	147.98	-0.01 9.15
Total weight	180944.9	145.80	-0.01 9.76

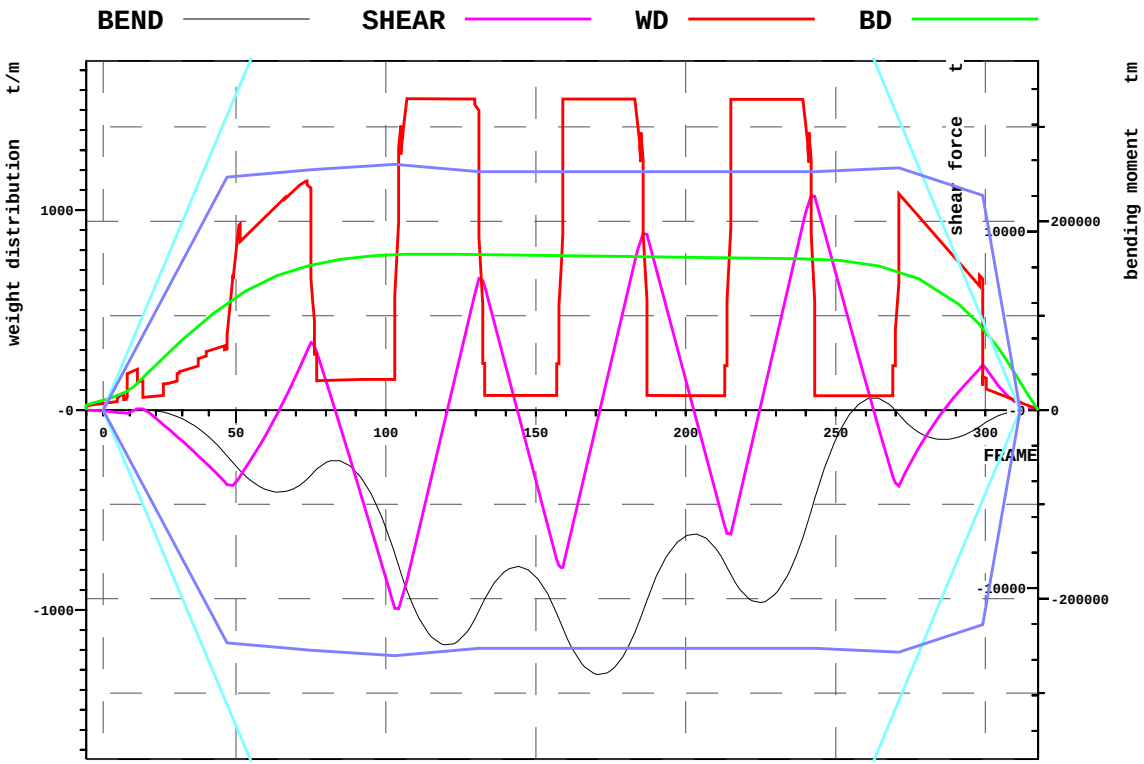
F L O A T I N G P O S I T I O N

Draught moulded	16.715	m	KM	18.75 m
Trim	1.301	m	KG	9.76 m
Heel, PS=+	-0.1	deg		
TA	17.365	m	GM0	8.99 m
TF	16.064	m	GMCORR	0.00 m
Trimming moment	-318443	tonm	GM	8.99 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.78	0.033	0.0	0.000
10.0	1.57	0.136	0.0	0.000
20.0	3.26	0.558	0.0	0.000
30.0	4.63	1.255	0.0	0.000
40.0	5.46	2.144	0.0	0.000
45.0	5.67	2.630	0.0	0.000
50.0	5.74	3.129	0.0	0.000
55.0	5.69	3.628	0.0	0.000
60.0	5.54	4.118	0.0	0.000
65.0	5.32	4.593	0.0	0.000
70.0	5.03	5.045	0.0	0.000
75.0	4.68	5.469	0.0	0.000
80.0	4.29	5.861	0.0	0.000
85.0	3.86	6.217	0.0	0.000
90.0	3.39	6.534	0.0	0.000



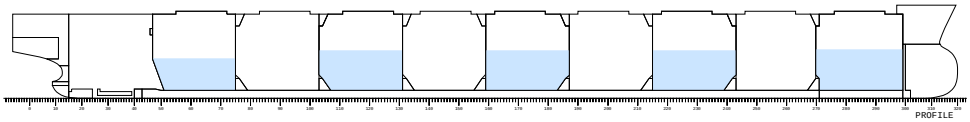
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.255	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.144	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.889	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.736	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.136	deg	OK
GM0.15	GM > 0.15 m	0.150	8.991	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.126		OK

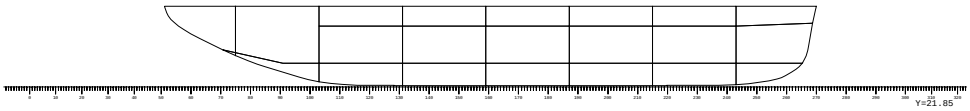


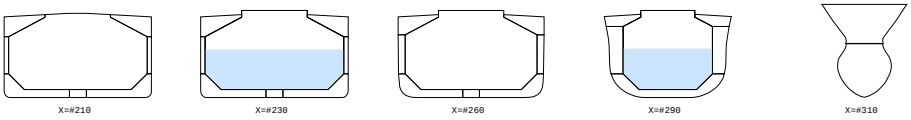
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	108	-48	32.90	48.35
8.80	11.00	-563	-54	192.93	111.51
12.00	15.00	-314	57	62.48	161.27
37.60	47.00	-48664	-4197	374.15	531.03
41.13	51.00	-63069	-3831	923.93	571.82
62.97	75.00	-69115	3782	885.41	724.02
88.45	103.00	-152595	-11118	568.90	776.61
113.93	131.00	-213575	73671498.11		777.52
139.41	159.00	-231378	-8826	880.13	772.09
164.89	187.00	-201807	9842	548.41	766.57
190.37	215.00	-174248	-69531554.20		761.02
215.85	243.00	-93690	11975	543.10	753.01
241.33	271.00	-4854	-4267	635.03	688.55
266.81	299.00	-15446	2546	656.06	407.23
278.01	313.00	-786	275	40.18	144.33

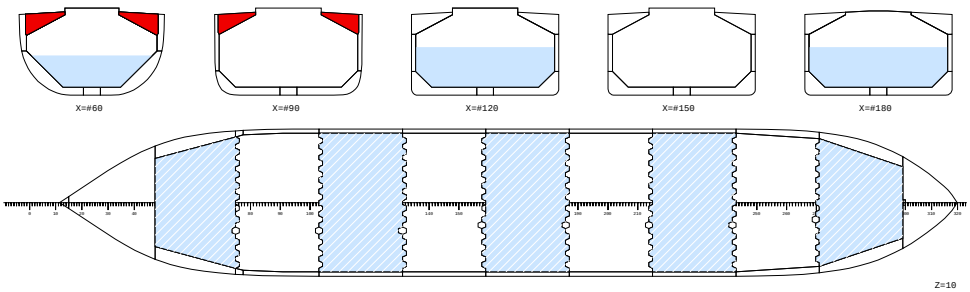
				X	FRAME
SHEAR FORCE (MIN)	-11145.8 t	(81.0%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	12021.4 t	(89.9%)		214.8 m	242
SAGGING MOMENT	-280384 tm	(38.7%)		149.9 m	171
HOGGING MOMENT	12646.7 tm	(3.5%)		235.0 m	264

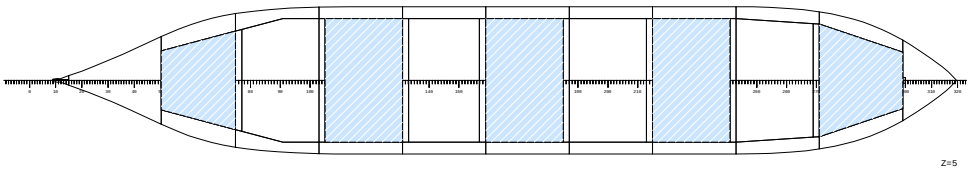
LOADING CONDITION - LSC4C20-08 : LOA. SEQU.LC4LC20 STEP8-GARGO1 LOAD 55%-DEP. BU

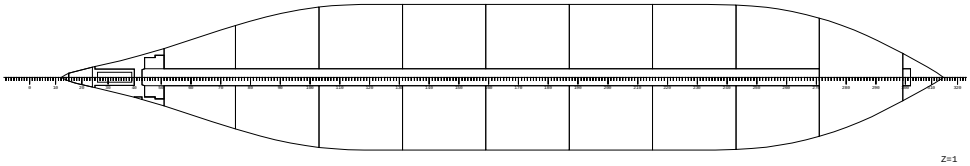












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	20000.0	36.8	52.05	0.03	7.99	0.00
SUBTOTAL		157860.9		156.92	0.02	8.82	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

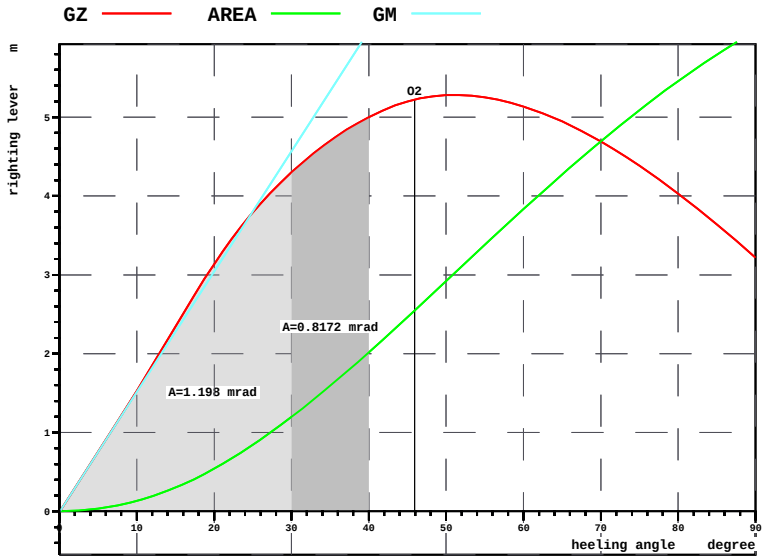
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)						
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00	13.67
Deadweight	164046.1	152.91	-0.01	9.34
Total weight	188688.0	150.18	-0.01	9.91

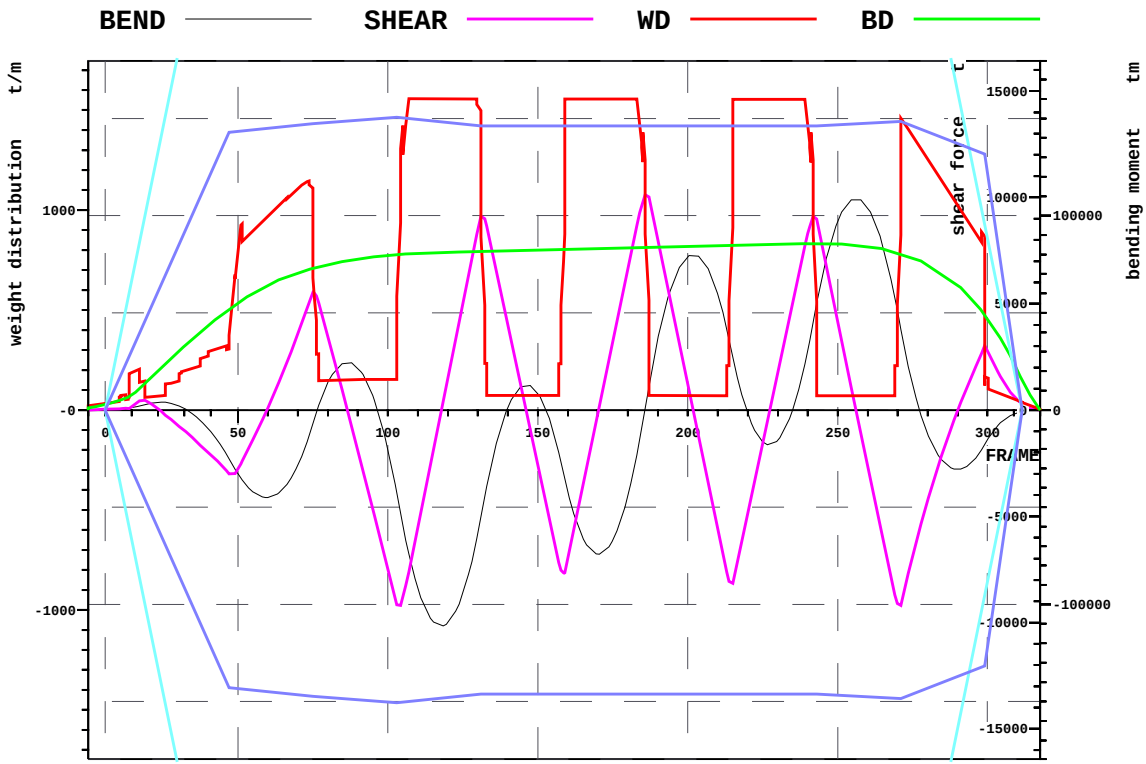
F L O A T I N G P O S I T I O N

Draught moulded	17.380	m	KM	18.64 m
Trim	-2.352	m	KG	9.91 m
Heel, PS=+	-0.1	deg		
TA	16.204	m	GM0	8.73 m
TF	18.557	m	GMCORR	0.00 m
Trimming moment	568201	tonm	GM	8.73 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.75	0.032	0.0	0.000
10.0	1.53	0.132	0.0	0.000
20.0	3.14	0.541	0.0	0.000
30.0	4.30	1.198	0.0	0.000
40.0	5.00	2.015	0.0	0.000
45.0	5.20	2.461	0.0	0.000
50.0	5.28	2.919	0.0	0.000
55.0	5.25	3.379	0.0	0.000
60.0	5.14	3.833	0.0	0.000
65.0	4.95	4.273	0.0	0.000
70.0	4.69	4.694	0.0	0.000
75.0	4.38	5.090	0.0	0.000
80.0	4.03	5.458	0.0	0.000
85.0	3.64	5.793	0.0	0.000
90.0	3.22	6.092	0.0	0.000



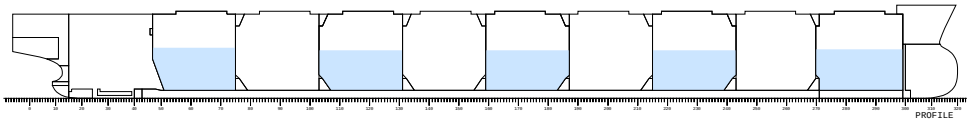
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.198	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.015	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.817	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.281	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.051	deg	OK
GM0.15	GM > 0.15 m	0.150	8.733	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.436		OK

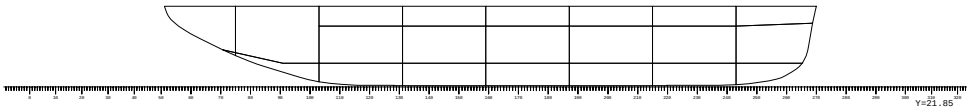


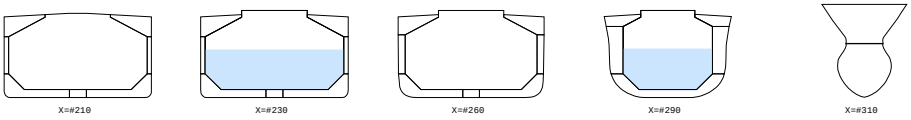
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	349	44	32.90	27.76
8.80	11.00	1394	255	192.93	83.51
12.00	15.00	2779	458	62.48	131.56
37.60	47.00	-25122	-3008	374.15	502.54
41.13	51.00	-35153	-2544	923.92	544.71
62.97	75.00	-7472	5539	885.40	708.64
88.45	103.00	-42837	-9165	568.89	776.63
113.93	131.00	-55717	91211498.10		793.04
139.41	159.00	-35502	-7664	880.13	803.10
164.89	187.00	11950	10016	548.40	813.09
190.37	215.00	27185	-81621554.18		823.04
215.85	243.00	55154	8988	543.10	830.44
241.33	271.00	41380	-9192	874.75	777.16
266.81	299.00	-17733	3036	867.95	473.48
278.01	313.00	-806	288	40.18	153.66

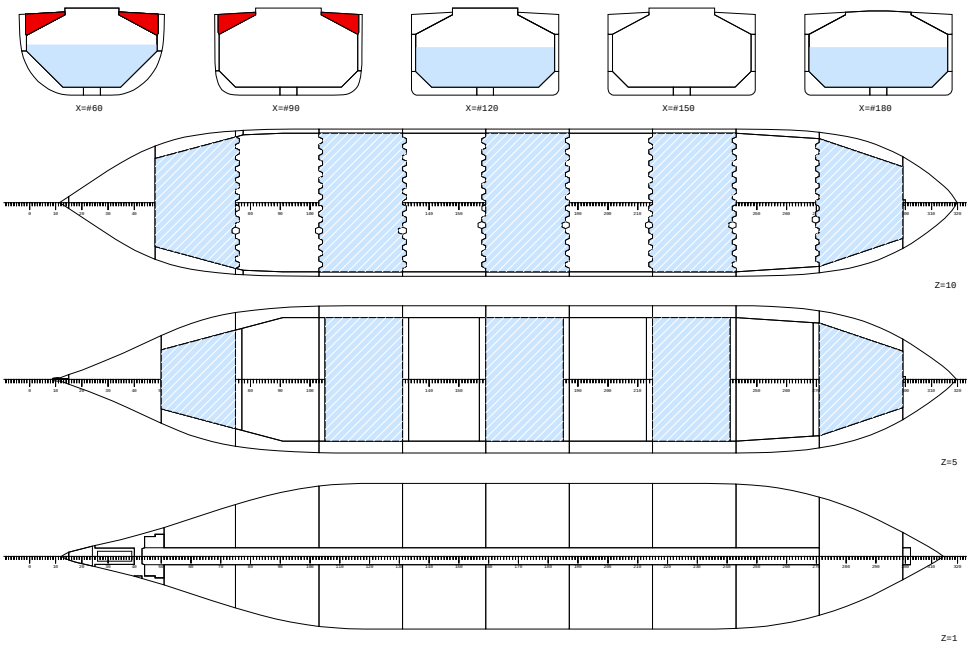
				X	FRAME
SHEAR FORCE (MIN)	-9192.8 t	(66.8%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	10124.4 t	(75.7%)		163.8 m	186
SAGGING MOMENT	-111096 tm	(15.3%)		102.5 m	118
HOGGING MOMENT	108140.9 tm	(26.4%)		229.2 m	258

LOADING CONDITION - LSC4C20-09 : LOA. SEQU.LC4LC20 STEP9-GARGO9 LOAD 55%-DEP. BU









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	29913.2	55.0	51.66	0.03	9.94	0.00
SUBTOTAL		167774.1		150.66	0.02	9.12	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00

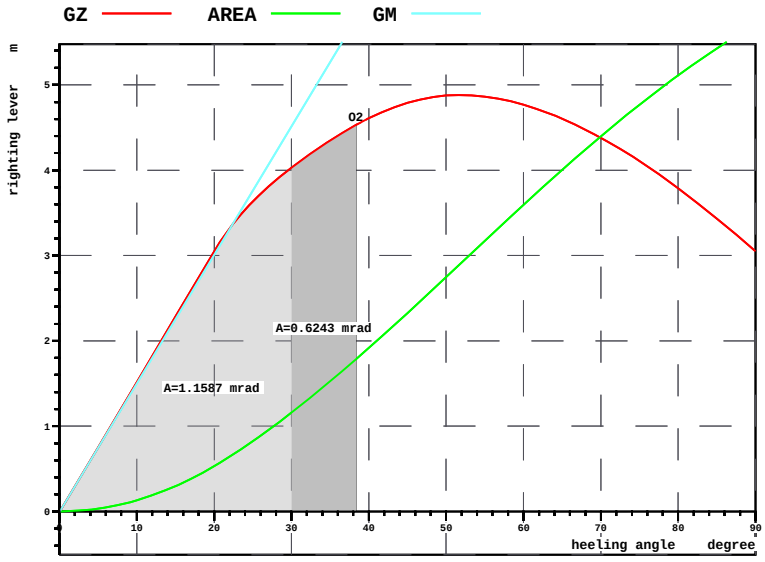
NAME DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
SUBTOTAL	4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)						
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	173959.3	147.10	-0.01 9.60
Total weight	198601.2	145.22	-0.01 10.11

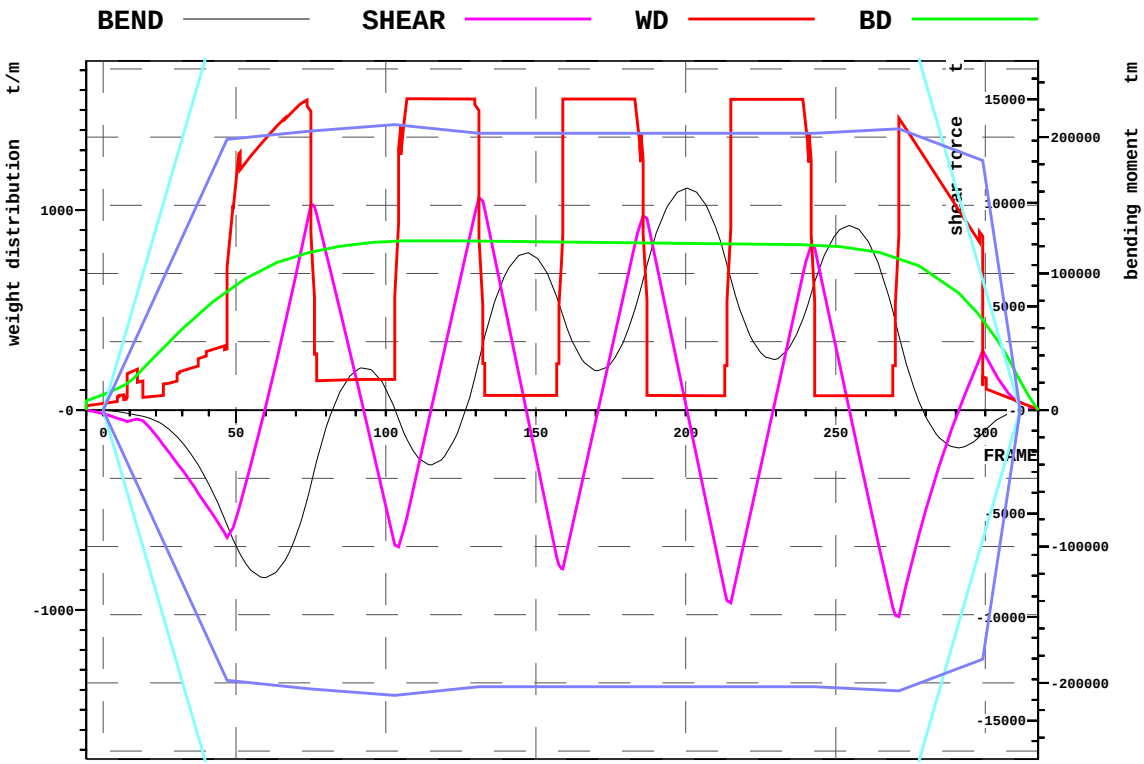
F L O A T I N G P O S I T I O N

Draught moulded	18.194	m	KM	18.74 m
Trim	1.153	m	KG	10.11 m
Heel, PS=+	-0.0	deg		
TA	18.771	m	GM0	8.63 m
TF	17.617	m	GMCORR	0.00 m
Trimming moment	-287390	tonm	GM	8.63 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.75	0.032	0.0	0.000
10.0	1.51	0.131	0.0	0.000
20.0	3.05	0.530	0.0	0.000
30.0	4.03	1.159	0.0	0.000
40.0	4.61	1.916	0.0	0.000
45.0	4.79	2.326	0.0	0.000
50.0	4.88	2.749	0.0	0.000
55.0	4.86	3.174	0.0	0.000
60.0	4.77	3.595	0.0	0.000
65.0	4.60	4.005	0.0	0.000
70.0	4.38	4.397	0.0	0.000
75.0	4.11	4.767	0.0	0.000
80.0	3.79	5.112	0.0	0.000
85.0	3.43	5.428	0.0	0.000
90.0	3.05	5.710	0.0	0.000



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.159	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	1.783	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.624	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.881	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.759	deg	OK
GM0.15	GM > 0.15 m	0.150	8.633	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.050		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-204	-178	32.90	77.04
8.80	11.00	-3237	-489	192.94	152.18
12.00	15.00	-4585	-515	62.48	205.62
37.60	47.00	-84389	-6163	712.38	592.58
41.13	51.00	-103958	-47901280.75		634.57
62.97	75.00	-55034	99031190.89		790.16
88.45	103.00	654	-6509	568.90	843.75
113.93	131.00	35365	102581498.12		845.30
139.41	159.00	69255	-7670	880.14	840.49
164.89	187.00	106108	9247	548.41	835.62
190.37	215.00	96126	-93161554.20		830.70
215.85	243.00	93906	7829	543.10	823.25
241.33	271.00	54610	-9985	874.75	756.20
266.81	299.00	-16945	2860	867.96	450.76
278.01	313.00	-816	283	40.18	149.97

				X	FRAME
SHEAR FORCE (MIN)	-9985.0 t	(73.6%)	POSITION:	241.3 m	271
SHEAR FORCE (MAX)	10257.7 t	(76.7%)		113.9 m	131
SAGGING MOMENT	-122831 tm	(30.9%)		48.6 m	59
HOGGING MOMENT	162725.2 tm	(22.1%)		177.1 m	200

10. Loading/unloading sequence: From cond. No22 to cond. No2

LOADING/UNLOADING SEQUENCE SUMMARY FORM

Vessel name	Voyage No.	Condition	Yard	Id. Number

Port (specific or typical):	Condition at commencement of laoding/discharging:	
Total mass of cargo to be loaded/discharged:	Condition at end of loading/discharging:	
Dock water density (t/m^3):	Maximum loading/discharging rate:	Average Loading/discharging rate:
Number of loaders/dischargers:	Maximum Ballasting/Deballasting rate:	Average Ballasting/Deballasting rate:

Hold No.	9	8	7	6	5	4	3	2	1
volume of Hold, V(m^3)	18190	21557	21935	21647	21915	21917	21916	21214	16870
Height of hold, h (m)	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2

Note:During each pour it has to be controlled that allowable limits for hull girder shear force,bending moment and mass in holds are not exceeded.
Loading/discharging operations may have to be paused to allow for ballasting/deballasting in order to keep actual values within limits

Hold content at commencement of loading/discharging.

Ballast contents at commencement of loading/discharging.

Commencement of loading/discharging (sea).

Cargo mass (t)	29913	0	36728	0	36694	0	36696	0	27743
Density, (t/m^3)	2.99		2.99		2.99		2.99		2.99
Grade									

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	0	0	0	0	0	0	0	0	0	0	0
Double side	0	0	0	0	0	0	0	0	0	0	0
Double bottom	0	0	0	0	0	0	0	0	0	0	0

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
18.01	0.23	17.78	90.90%	33.40%

CARGO OPERATIONS — BALLASTING OPERATIONS

Values at end of pour (from harbour to sea).

Pour No./ Grade	Hold 9	Hold 8	Hold 7	Hold 6	Hold 5	Hold 4	Hold 3	Hold 2	Hold 1	Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT	d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
1	75000t To41% of Hold									Top side												16.05	-2.45	18.50	72.80%	25.60%
										Double side																
										Double bottom																
2									75000t To40% of Hold	Top side												17.17	1.09	16.08	87.90%	34.40%
										Double side																
										Double bottom																
3			100000t To41% of Hold							Top side												15.57	-0.43	16.00	91.20%	25.70%
										Double side																
										Double bottom																
4							100000t To41% of Hold			Top side												16.73	3.19	13.54	82.20%	24.80%
										Double side																
										Double bottom																
5					3669430 To0% of Hold					Top side					100%	100%	100%					15.70	3.99	11.71	86.20%	72.70%
										Double side					100%	100%	100%									
										Double bottom					100%	100%	100%									
6	22413.2t To0% of Hold									Top side	100%	100%	100%									11.46	-1.98	13.44	66.80%	42.40%
										Double side	100%	100%	100%													
										Double bottom	100%	100%	100%													
7									20243t To0% of Hold	Top side									100%	100%		13.79	3.06	10.73	69.30%	28.10%
										Double side									100%	100%						
										Double bottom									100%	100%						
8			26727.8t To0% of Hold							Top side				100%								9.73	-1.26	10.99	82.90%	51.50%
										Double side				100%												
										Double bottom				100%												
9							26695.8t To0% of Hold			Top side								100%			100%	9.98	2.02	7.96	43.20%	63.10%
										Double side								100%			100%					
										Double bottom								100%			100%					

Total cargo onboard (t): 0 **Remaining cargo to be loaded (t): 0**

Total amount of bunkers onboard (t): 5975.2 (100%)

Draft Survey n-1										Top side
										Double side
										Double bottom
Draft Survey n										Top side
										Double side
										Double bottom

Hold content at end of loading/discharging.

Ballast contents at end of loading/discharging.

Values at end of loading/discharging (sea).

Cargo mass (t)	0	0	0	0	0	0	0	0	0

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Double side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Double bottom	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

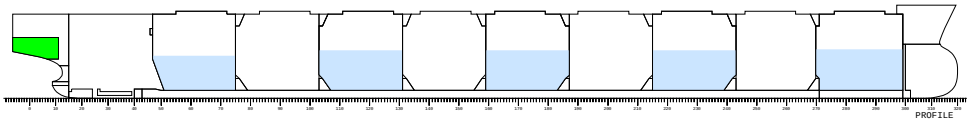
d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
9.98	2.02	7.96	53.80%	98.20%

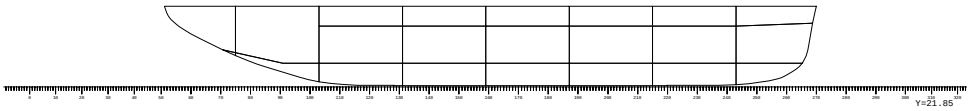
Total mass loaded/discharged (t): 167774.1

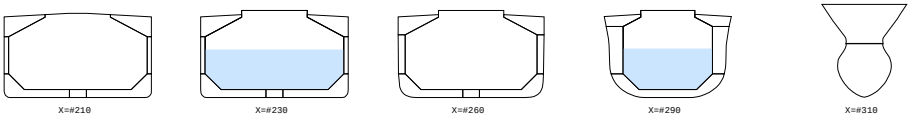
Net load on Double Bottom:	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	tons/m^2
Net load in two adjacent holds:	0	0	0	0	0	0	0	0	0	tons

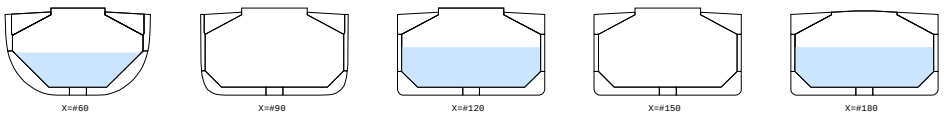
Net load on double bottom = (Mh/V)×h-T[t/m^2]
where: Mh = Mass in hold + mass in DB (t)
V = Total volume of hold (m^3) T = draught (m)
h = height of hold from inner bottom to top of coaming (m)

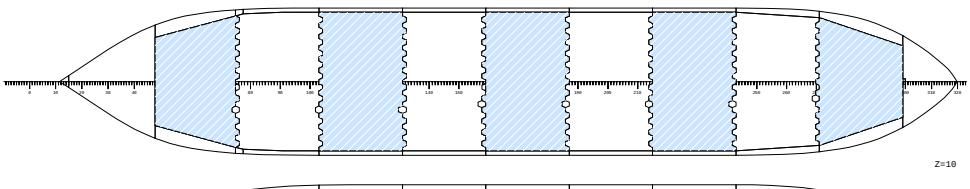
LOADING CONDITION - LSC22C2-01 : LOA. SEQU.LC22LC2 STEP1-GAR.9 UNLOAD 7500t-ARR.

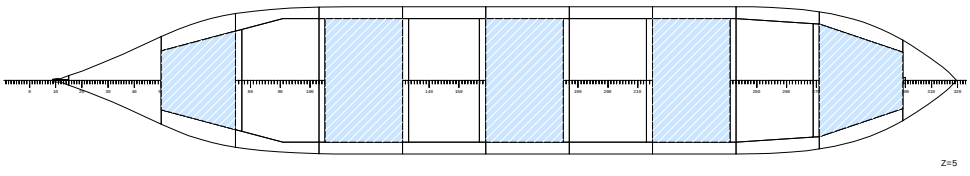


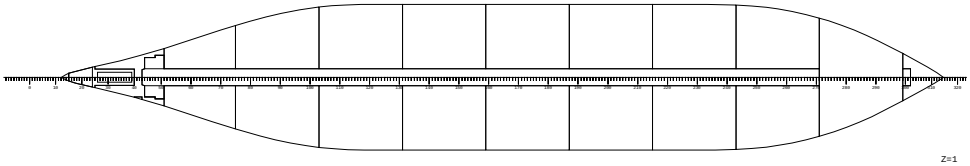












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	27743.0	55.0	252.56	0.00	9.06	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	22413.2	41.2	51.94	0.03	8.50	0.00
SUBTOTAL		160274.0		155.33	0.02	8.87	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

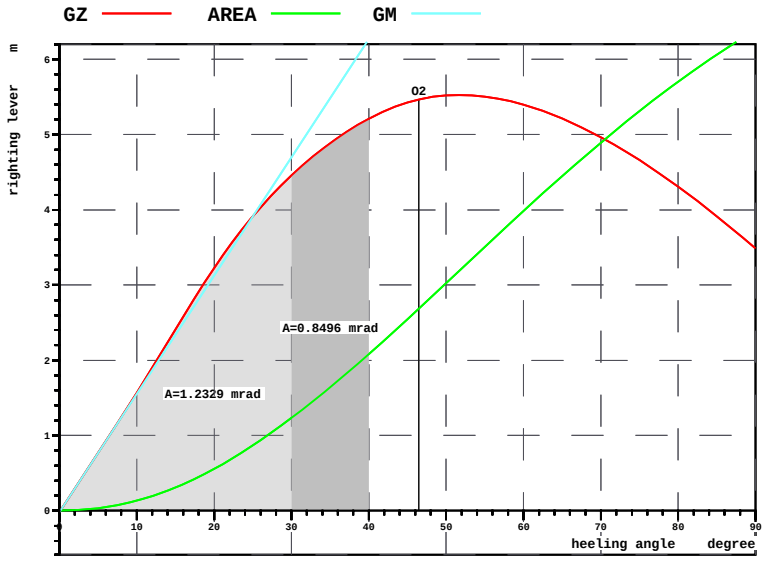
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00	13.67
Deadweight	162764.4	153.17	-0.02	9.02
Total weight	187406.2	150.39	-0.02	9.63

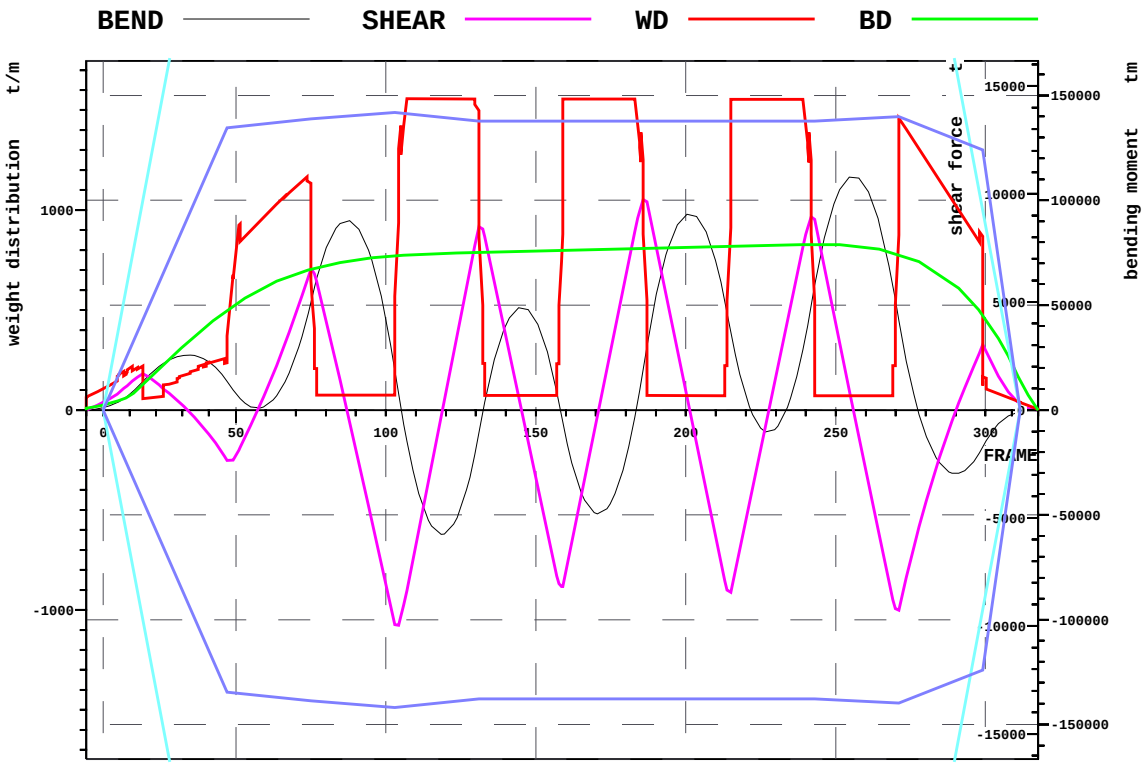
F L O A T I N G P O S I T I O N

Draught moulded	17.272	m	KM	18.64 m
Trim	-2.448	m	KG	9.63 m
Heel, PS=+	-0.1	deg		
TA	16.048	m	GM0	9.00 m
TF	18.496	m	GMCORR	-0.00 m
Trimming moment	589242	tonm	GM	9.00 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.77	0.033	64.4	0.000
10.0	1.57	0.134	128.6	0.001
20.0	3.23	0.554	221.7	0.001
30.0	4.46	1.233	296.1	0.002
40.0	5.21	2.082	401.9	0.002
45.0	5.43	2.547	456.9	0.002
50.0	5.52	3.026	503.3	0.003
55.0	5.50	3.508	540.1	0.003
60.0	5.40	3.984	568.2	0.003
65.0	5.22	4.448	588.6	0.003
70.0	4.97	4.892	603.4	0.003
75.0	4.66	5.313	614.6	0.003
80.0	4.31	5.704	620.8	0.003
85.0	3.91	6.063	620.0	0.003
90.0	3.49	6.386	613.4	0.003



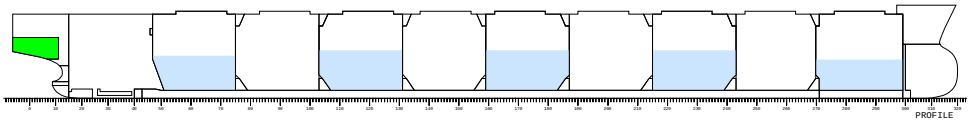
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.233	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.082	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.850	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.526	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.642	deg	OK
GM0.15	GM > 0.15 m	0.150	9.000	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.613		OK

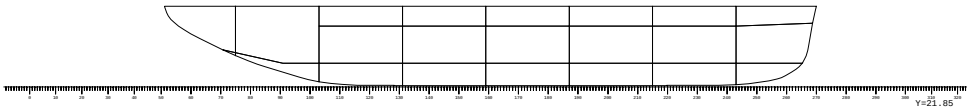


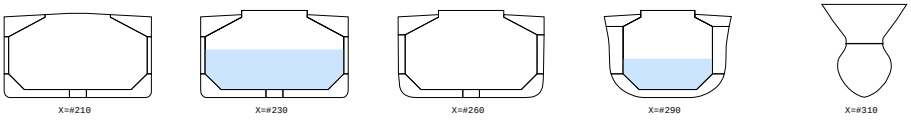
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1118	356	106.32	25.29
8.80	11.00	8484	1340	218.28	79.63
12.00	15.00	13469	1677	57.01	127.25
37.60	47.00	13654	-2343	371.09	496.46
41.13	51.00	5994	-1864	923.59	538.55
62.97	75.00	51165	65141	134.76	702.47
88.45	103.00	19324	-9944	568.88	770.83
113.93	131.00	-11573	84851	498.07	787.64
139.41	159.00	-5881	-8169	880.11	798.11
164.89	187.00	30278	9633	548.39	808.50
190.37	215.00	37189	-84331	554.15	818.86
215.85	243.00	59540	8818	543.09	826.67
241.33	271.00	42583	-9273	874.73	773.94
266.81	299.00	-17667	3023	867.93	471.63
278.01	313.00	-804	287	40.18	153.38

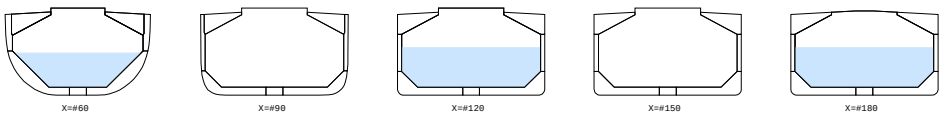
				X	FRAME
SHEAR FORCE (MIN)	-9965.1 t	(72.4%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	9736.2 t	(72.8%)		163.8 m	186
SAGGING MOMENT	-59305.9 tm	(8.2%)		102.5 m	118
HOGGING MOMENT	110931.2 tm	(25.6%)		226.3 m	255

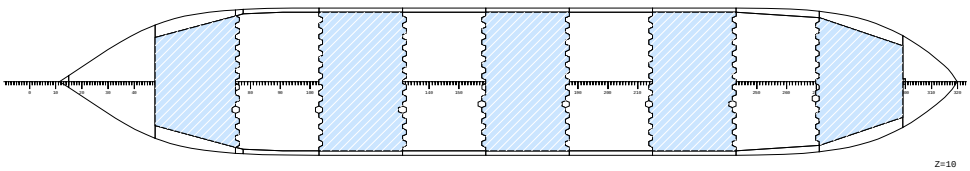
LOADING CONDITION - LSC22C2-02 : LOA. SEQU.LC22LC2 STEP1-GAR.9 UNLOAD 7500t-ARR.

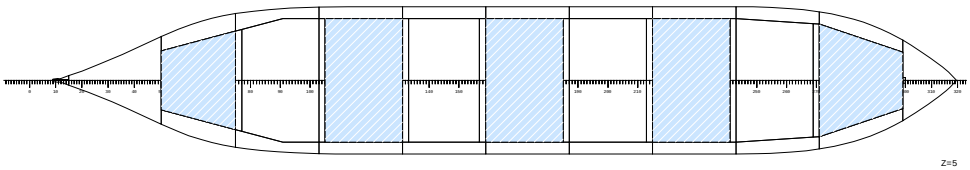


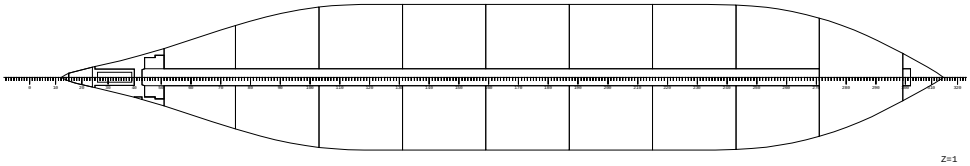












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20243.0	40.1	252.56	0.00	7.44	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	36727.8	56.0	101.87	0.02	8.91	0.00
R1.9	NO.9 CARGO HOLD	22413.2	41.2	51.94	0.03	8.50	0.00
SUBTOTAL		152774.0		150.56	0.02	8.65	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

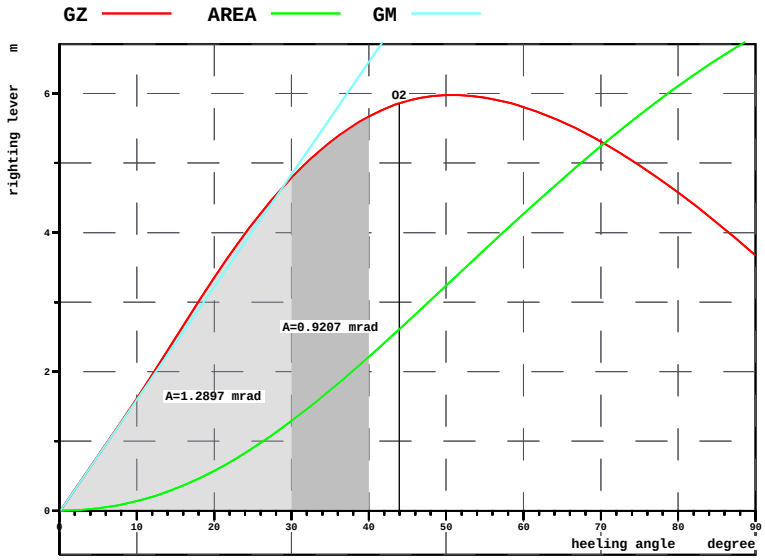
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	155264.4	148.37	-0.02 8.81
Total weight	179906.2	146.13	-0.02 9.48

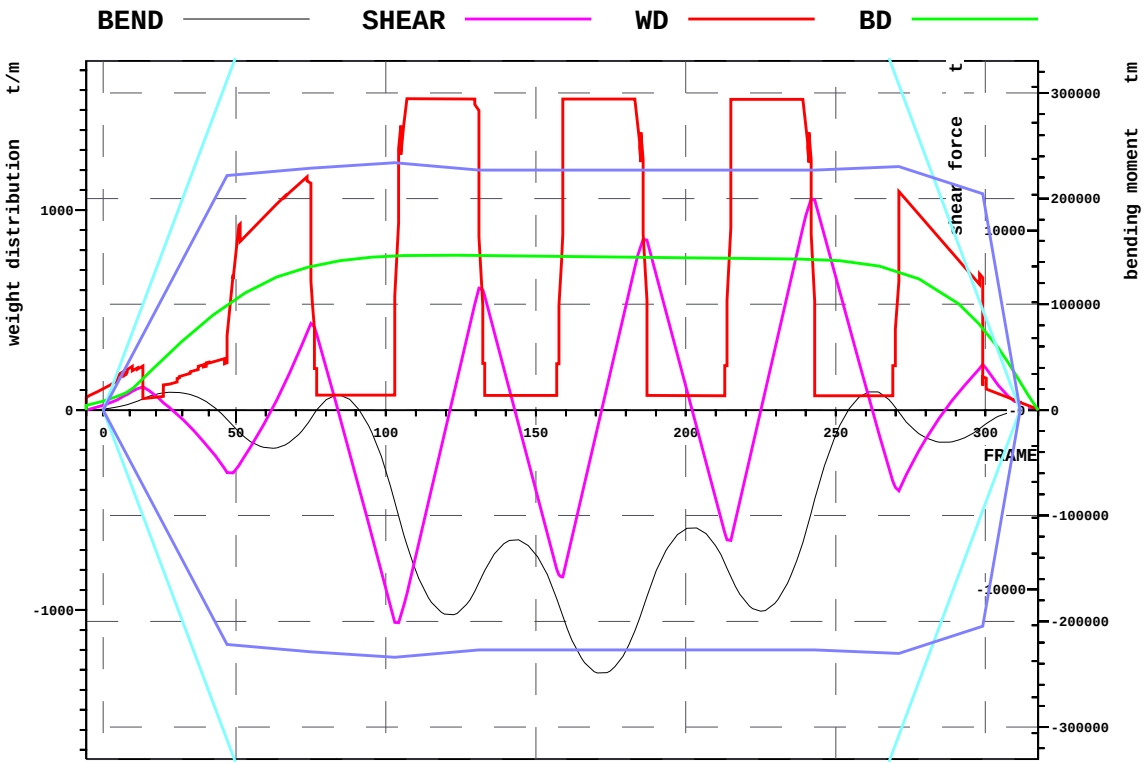
F L O A T I N G P O S I T I O N

Draught moulded	16.627	m	KM	18.75	m
Trim	1.095	m	KG	9.48	m
Heel, PS=+	-0.1	deg			
TA	17.174	m	GM0	9.27	m
TF	16.079	m	GMCORR	-0.00	m
Trimming moment	-267266	tonm	GM	9.27	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.79	0.034	64.4	0.000
10.0	1.61	0.138	128.6	0.001
20.0	3.35	0.572	221.7	0.001
30.0	4.79	1.290	296.1	0.002
40.0	5.67	2.210	401.9	0.002
45.0	5.90	2.716	456.9	0.003
50.0	5.98	3.235	503.3	0.003
55.0	5.94	3.756	540.1	0.003
60.0	5.81	4.269	568.2	0.003
65.0	5.59	4.767	588.6	0.003
70.0	5.31	5.243	603.4	0.003
75.0	4.97	5.691	614.6	0.003
80.0	4.58	6.108	620.8	0.003
85.0	4.14	6.489	620.0	0.003
90.0	3.67	6.830	613.4	0.003



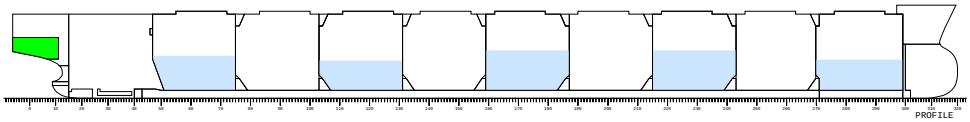
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.290	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.210	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.921	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.978	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.740	deg	OK
GM0.15	GM > 0.15 m	0.150	9.268	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.300		OK

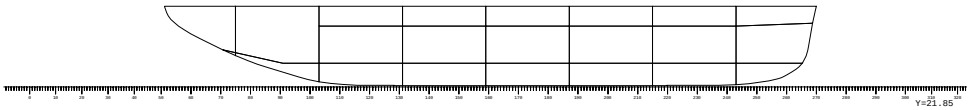


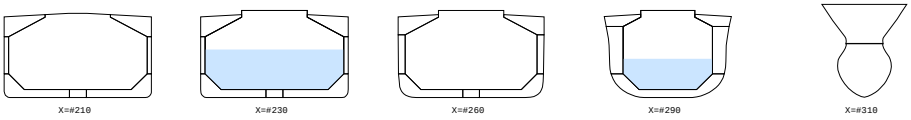
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	880	269	106.32	44.77
8.80	11.00	6601	1047	218.38	106.45
12.00	15.00	10498	1295	57.01	155.79
37.60	47.00	-8957	-3486	371.09	524.11
41.13	51.00	-20834	-3102	923.59	564.87
62.97	75.00	-8226	4819	1134.77	717.46
88.45	103.00	-86605	-11832	568.88	770.89
113.93	131.00	-164077	6787	1498.08	772.67
139.41	159.00	-195235	-9293	880.12	768.11
164.89	187.00	-176448	9465	548.40	763.47
190.37	215.00	-157669	-7263	1554.16	758.80
215.85	243.00	-84457	11710	543.09	751.66
241.33	271.00	-2098	-4502	1092.69	688.09
266.81	299.00	-15410	2549	662.70	407.47
278.01	313.00	-770	275	40.18	144.40

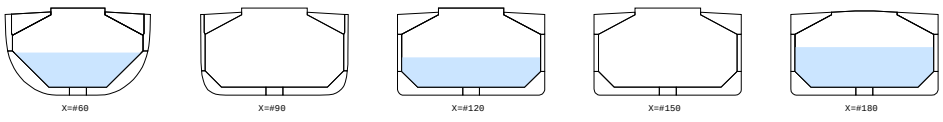
				X	FRAME
SHEAR FORCE (MIN)	-11852.8 t	(86.2%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	11755.1 t	(87.9%)		214.8 m	242
SAGGING MOMENT	-248947 tm	(34.4%)		149.9 m	171
HOGGING MOMENT	17276.4 tm	(4.5%)		232.1 m	261

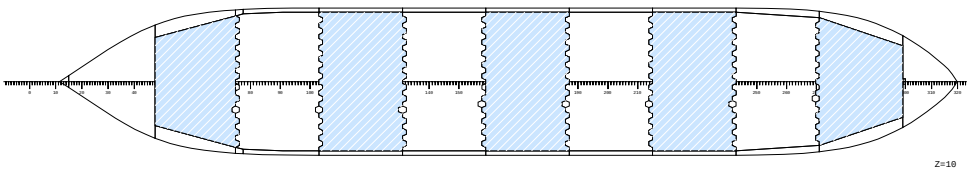
LOADING CONDITION - LSC22C2-03 : LOA. SEQU.LC22LC2 STEP3-GAR.7 UNLOAD 10000t-ARR

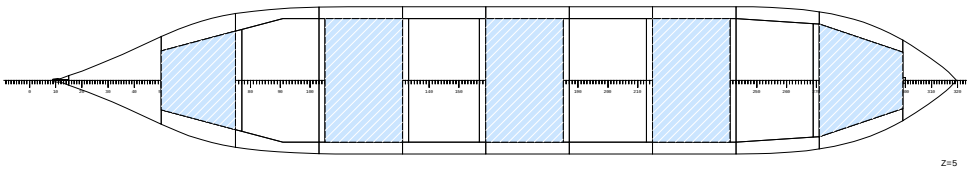


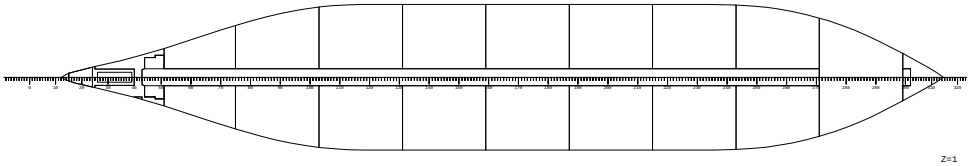












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20243.0	40.1	252.56	0.00	7.44	0.00
R1.3	NO.3 CARGO HOLD	36695.8	56.0	202.39	0.02	8.91	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	26727.8	40.8	101.93	0.02	7.32	0.00
R1.9	NO.9 CARGO HOLD	22413.2	41.2	51.94	0.03	8.50	0.00
SUBTOTAL		142774.0		153.98	0.02	8.34	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

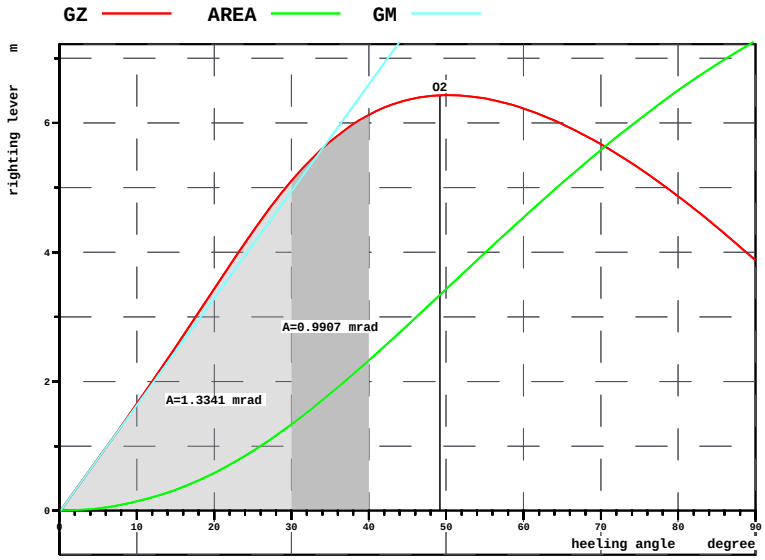
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	145264.3	151.58	-0.03 8.51
Total weight	169906.2	148.74	-0.02 9.26

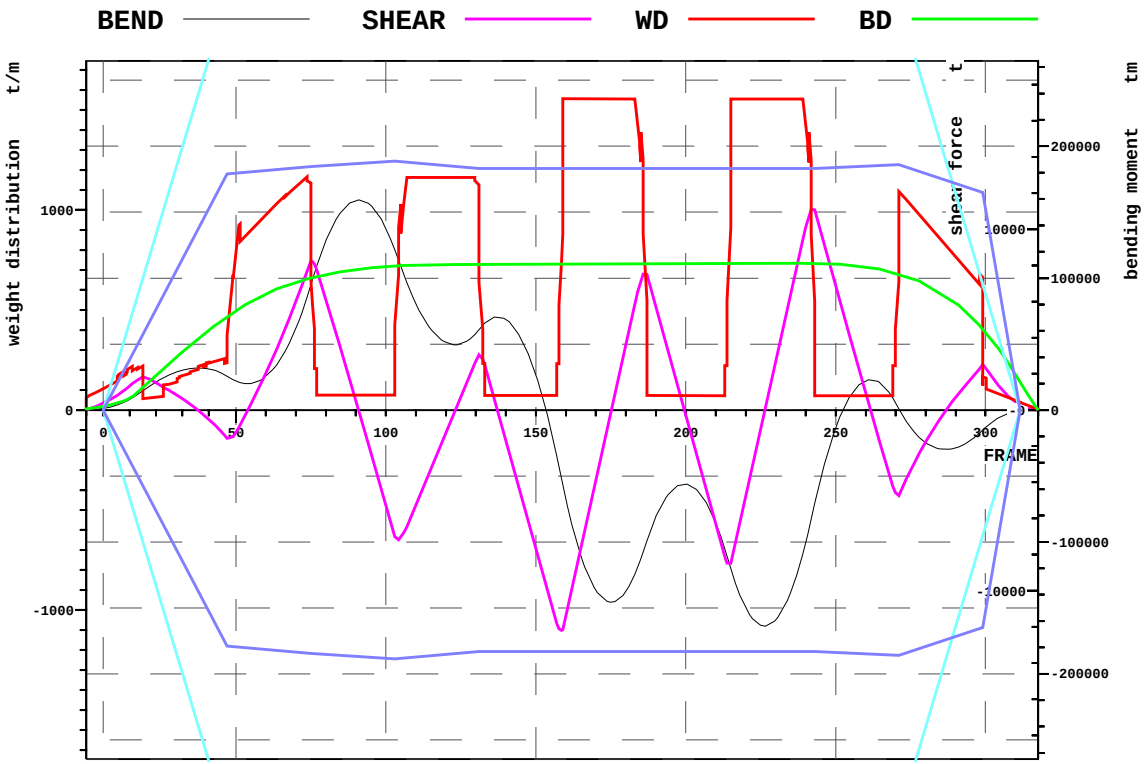
F L O A T I N G P O S I T I O N

Draught moulded	15.785	m	KM	18.74 m
Trim	-0.435	m	KG	9.26 m
Heel, PS=+	-0.1	deg		
TA	15.567	m	GM0	9.48 m
TF	16.002	m	GMCORR	-0.00 m
Trimming moment	102827	tonm	GM	9.48 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.81	0.034	64.4	0.000
10.0	1.65	0.141	128.6	0.001
20.0	3.43	0.583	221.7	0.001
30.0	5.10	1.334	296.1	0.002
40.0	6.13	2.325	401.9	0.002
45.0	6.36	2.871	456.9	0.003
50.0	6.43	3.430	503.3	0.003
55.0	6.38	3.990	540.1	0.003
60.0	6.23	4.540	568.2	0.003
65.0	5.99	5.074	588.6	0.003
70.0	5.67	5.583	603.4	0.004
75.0	5.30	6.062	614.6	0.004
80.0	4.87	6.506	620.8	0.004
85.0	4.39	6.910	620.0	0.004
90.0	3.88	7.271	613.4	0.004



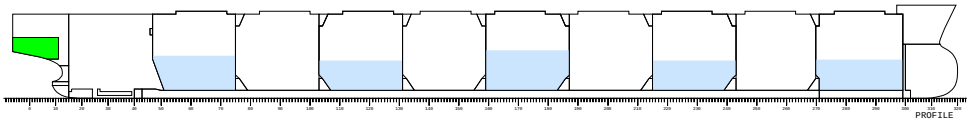
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.334	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.325	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.991	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.433	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.190	deg	OK
GM0.15	GM > 0.15 m	0.150	9.479	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.634		OK

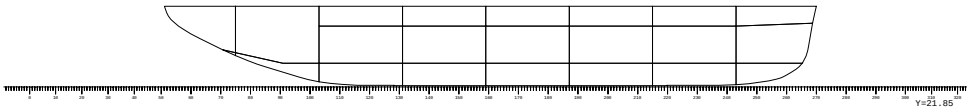


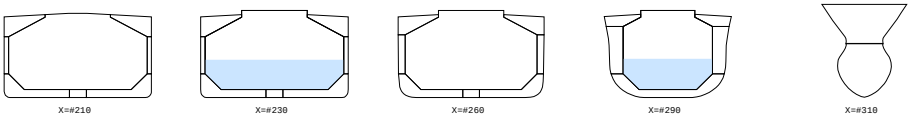
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1159	384	106.32	18.24
8.80	11.00	9074	1457	218.34	66.39
12.00	15.00	14487	1840	57.01	111.71
37.60	47.00	25710	-1565	371.10	464.75
41.13	51.00	20980	-970	923.60	504.99
62.97	75.00	94358	82471	134.78	659.62
88.45	103.00	121442	-7004	424.72	719.11
113.93	131.00	60633	30611	125.56	727.37
139.41	159.00	-56764	-12159	880.13	729.29
164.89	187.00	-99133	7506	548.40	731.13
190.37	215.00	-120510	-84801	554.18	732.94
215.85	243.00	-70658	11070	543.10	732.30
241.33	271.00	896	-47361	092.71	675.75
266.81	299.00	-15284	2525	662.71	403.53
278.01	313.00	-760	275	40.18	144.07

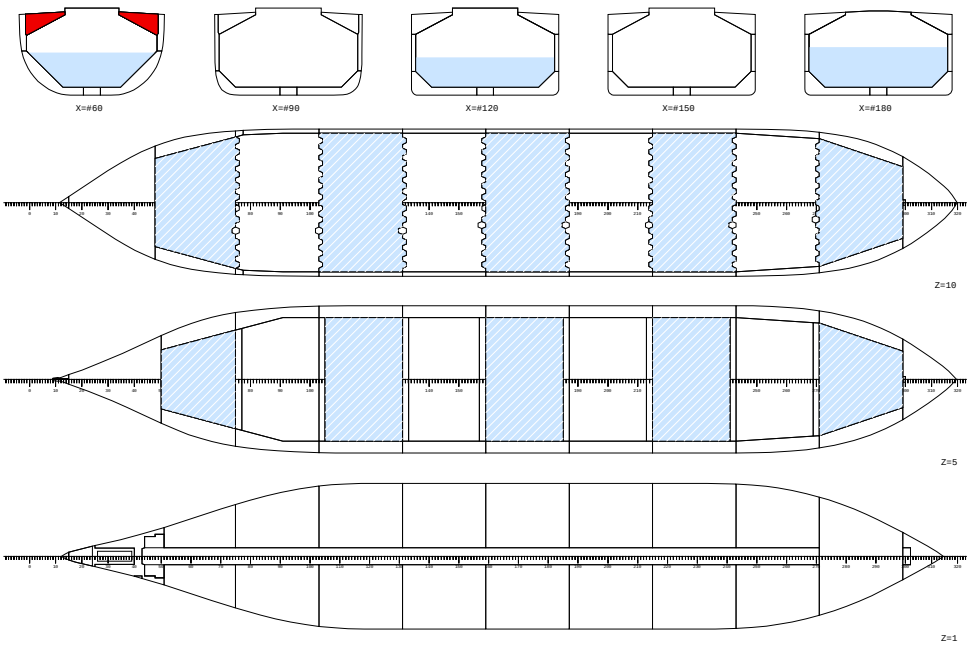
				X	FRAME
SHEAR FORCE (MIN)	-12192.1 t	(91.2%)	POSITION:	139.0 m	159
SHEAR FORCE (MAX)	11093.4 t	(83.0%)		214.8 m	242
SAGGING MOMENT	-163915 tm	(25.7%)		200.8 m	227
HOGGING MOMENT	159323.6 tm	(24.7%)		77.6 m	91

LOADING CONDITION - LSC22C2-04 : LOA. SEQU.LC22LC2 STEP4-GAR.3 UNLOAD 10000t-50%









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20243.0	40.1	252.56	0.00	7.44	0.00
R1.3	NO.3 CARGO HOLD	26695.8	40.7	202.33	0.01	7.32	0.00
R1.5	NO.5 CARGO HOLD	36694.3	56.0	151.43	0.02	8.91	0.00
R1.7	NO.7 CARGO HOLD	26727.8	40.8	101.93	0.02	7.32	0.00
R1.9	NO.9 CARGO HOLD	22413.2	41.2	51.94	0.03	8.50	0.00
SUBTOTAL		132774.1		150.32	0.02	7.98	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

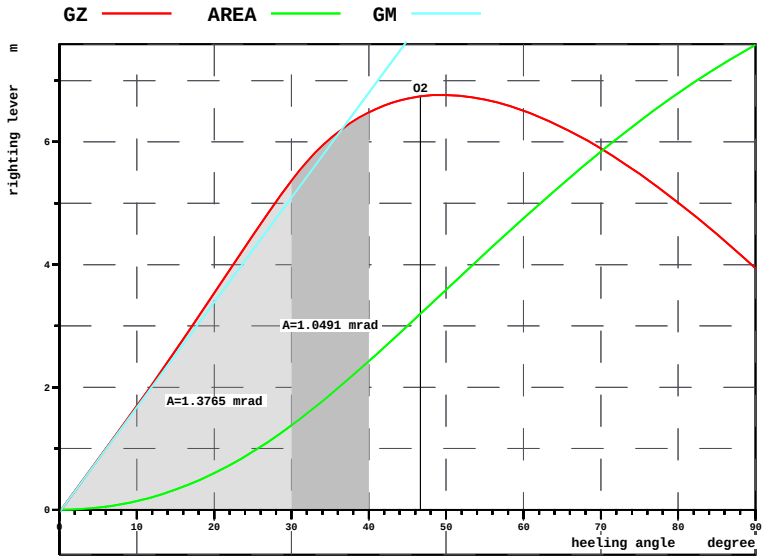
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	137633.3	145.99	-0.04 8.41
Total weight	162275.1	143.86	-0.04 9.21

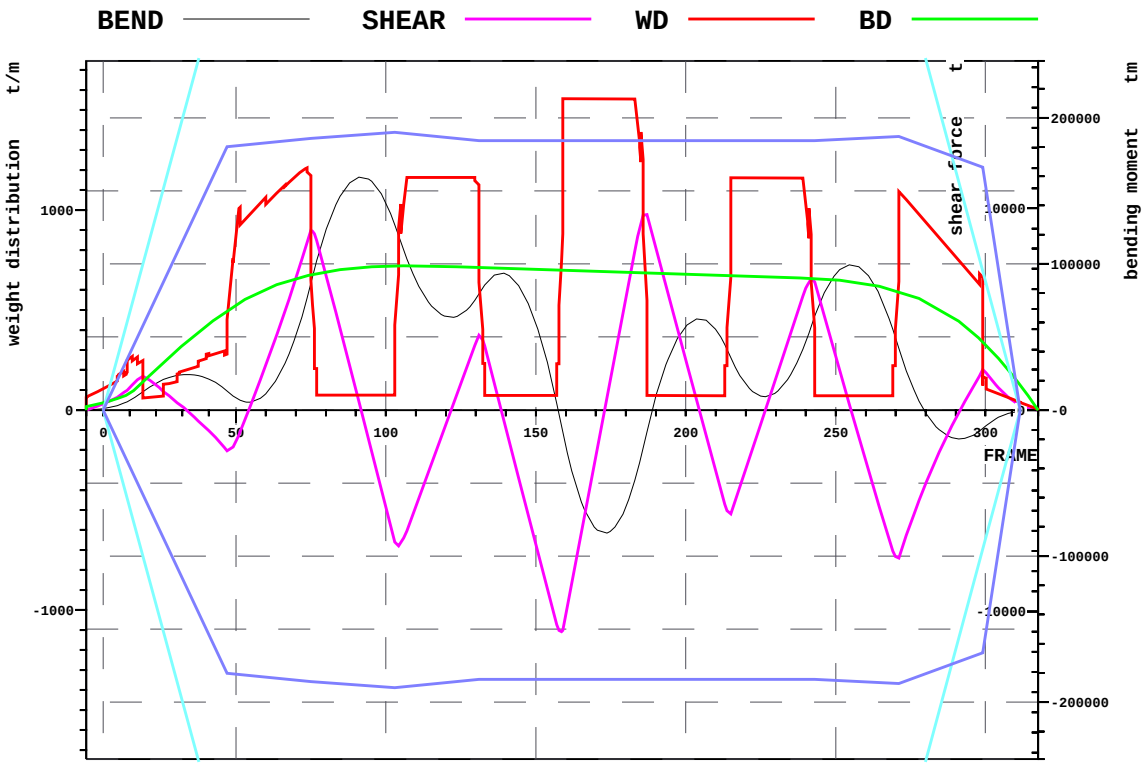
F L O A T I N G P O S I T I O N

Draught moulded	15.134	m	KM	19.02	m
Trim	3.188	m	KG	9.21	m
Heel, PS=+	-0.2	deg			
TA	16.728	m	GM0	9.81	m
TF	13.540	m	GMCORR	-0.02	m
Trimming moment	-778417	tonm	GM	9.79	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.82	0.034	273.1	0.002
10.0	1.69	0.144	572.7	0.004
20.0	3.53	0.597	1197.1	0.007
30.0	5.37	1.376	1623.0	0.010
40.0	6.48	2.426	1920.4	0.012
45.0	6.71	3.002	2017.0	0.012
50.0	6.76	3.591	2076.3	0.013
55.0	6.69	4.179	2095.0	0.013
60.0	6.51	4.756	2072.0	0.013
65.0	6.24	5.313	2014.9	0.012
70.0	5.89	5.843	1930.4	0.012
75.0	5.48	6.339	1823.1	0.011
80.0	5.01	6.797	1696.6	0.010
85.0	4.50	7.213	1553.9	0.010
90.0	3.94	7.581	1399.3	0.009



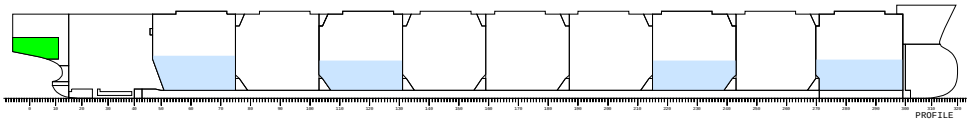
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.377	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.426	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.049	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.765	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	49.481	deg	OK
GM0.15	GM > 0.15 m	0.150	9.792	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.267		OK

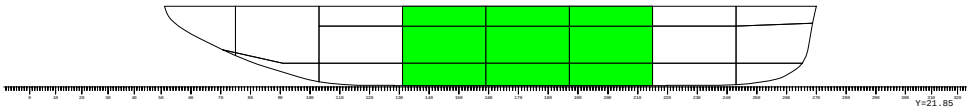


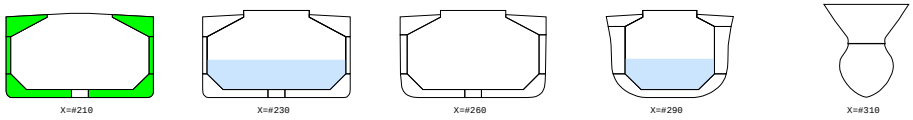
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	969	303	106.33	36.68
8.80	11.00	7440	1255	268.33	92.69
12.00	15.00	12308	1676	59.75	139.92
37.60	47.00	12966	-2028	452.49	493.04
41.13	51.00	6948	-12411006.94		532.00
62.97	75.00	86170	89101172.70		675.28
88.45	103.00	126781	-6546	424.73	719.55
113.93	131.00	79224	37031125.58		712.46
139.41	159.00	-15226	-10940	880.14	699.03
164.89	187.00	-15003	9691	548.41	685.52
190.37	215.00	35714	-51481160.28		671.98
215.85	243.00	64284	6359	409.87	656.08
241.33	271.00	42127	-73371092.73		588.62
266.81	299.00	-12675	2006	662.72	338.72
278.01	313.00	-724	246	40.18	127.72

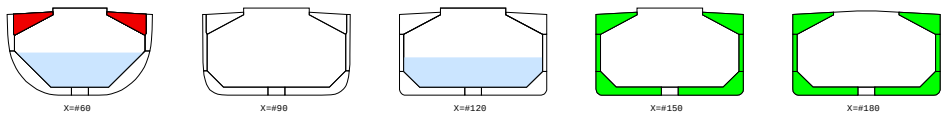
				X	FRAME
SHEAR FORCE (MIN)	-10985.8 t	(82.2%)	POSITION:	139.0 m	159
SHEAR FORCE (MAX)	9690.6 t	(72.5%)		164.9 m	187
SAGGING MOMENT	-84243.5 tm	(11.6%)		152.8 m	174
HOGGING MOMENT	159506.9 tm	(24.8%)		77.6 m	91

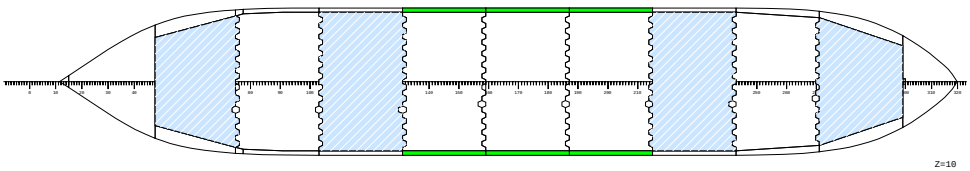
LOADING CONDITION - LSC22C2-05 : LOA. SEQU.LC22LC2 STEP5-GAR.5 EMPTY-50% BUNKERS

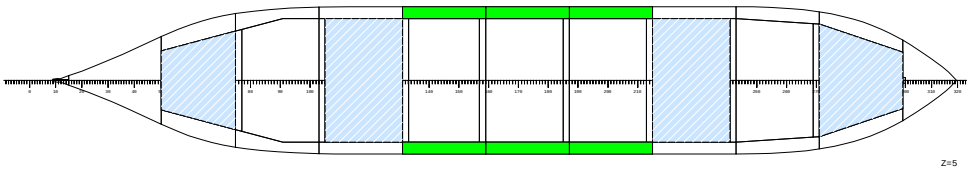


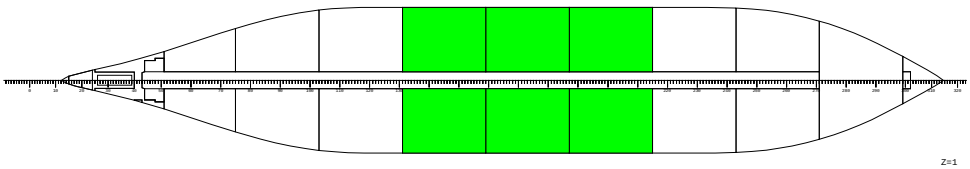












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20243.0	40.1	252.56	0.00	7.44	0.00
R1.3	NO.3 CARGO HOLD	26695.8	40.7	202.33	0.01	7.32	0.00
R1.7	NO.7 CARGO HOLD	26727.8	40.8	101.93	0.02	7.32	0.00
R1.9	NO.9 CARGO HOLD	22413.2	41.2	51.94	0.03	8.50	0.00
SUBTOTAL		96079.8		149.90	0.02	7.62	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

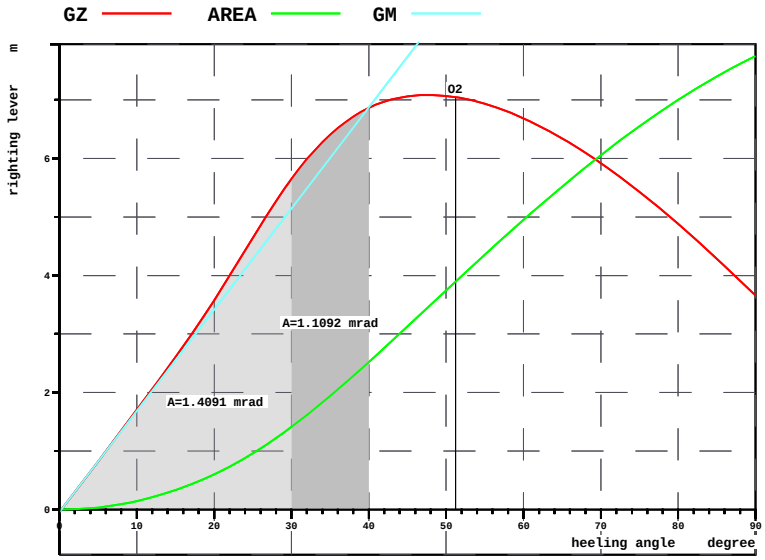
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
SUBTOTAL		21356.8		140.97	-0.01	11.27	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	120673.5	145.34	-0.05 8.66
Total weight	145315.3	143.08	-0.04 9.51

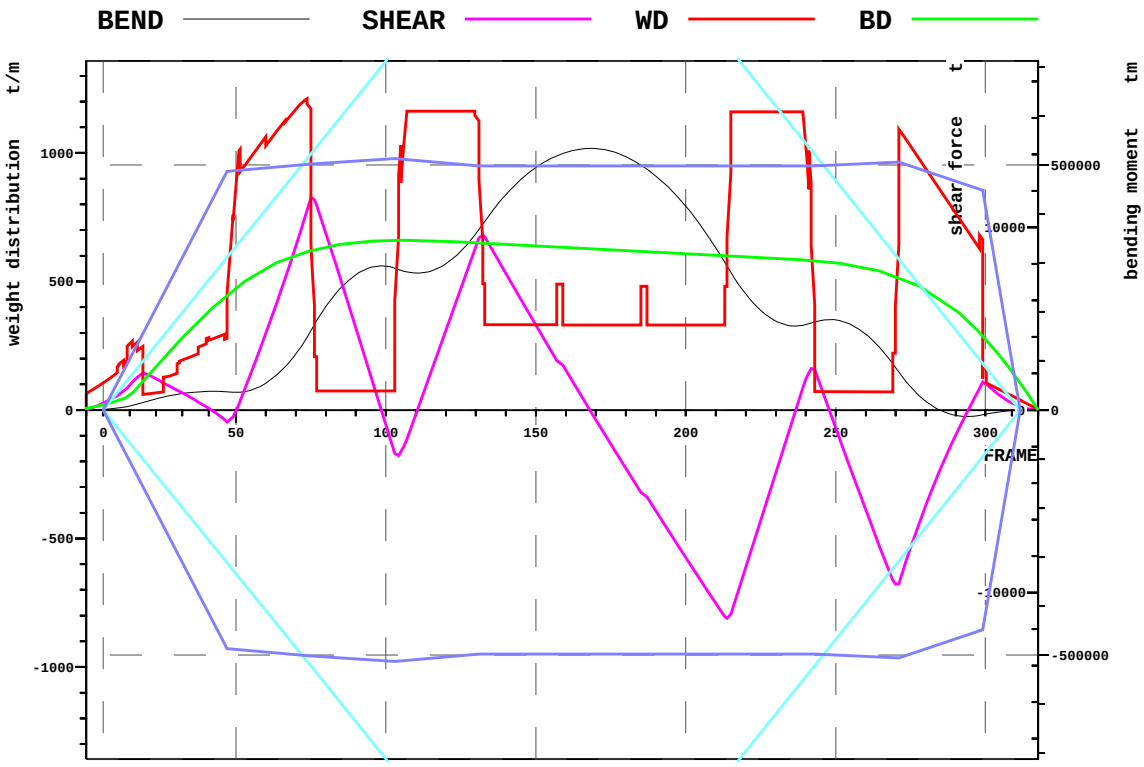
F L O A T I N G P O S I T I O N

Draught moulded	13.703	m	KM	19.44	m
Trim	3.989	m	KG	9.51	m
Heel, PS=+	-0.3	deg			
TA	15.697	m	GM0	9.93	m
TF	11.708	m	GMCORR	-0.02	m
Trimming moment	-958821	tonm	GM	9.91	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.82	0.034	273.1	0.002
10.0	1.70	0.144	572.7	0.004
20.0	3.58	0.600	1197.1	0.008
30.0	5.65	1.409	1623.0	0.011
40.0	6.87	2.518	1920.4	0.013
45.0	7.06	3.128	2017.0	0.014
50.0	7.07	3.745	2076.3	0.014
55.0	6.94	4.357	2095.0	0.014
60.0	6.69	4.953	2072.0	0.014
65.0	6.34	5.522	2014.9	0.014
70.0	5.92	6.057	1930.4	0.013
75.0	5.43	6.553	1823.1	0.013
80.0	4.88	7.003	1696.6	0.012
85.0	4.29	7.404	1553.9	0.011
90.0	3.66	7.751	1399.3	0.010



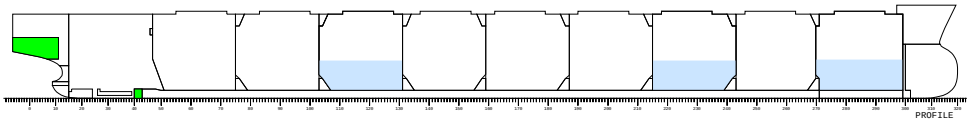
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.409	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.518	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.109	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.085	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.741	deg	OK
GM0.15	GM > 0.15 m	0.150	9.907	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.432		OK

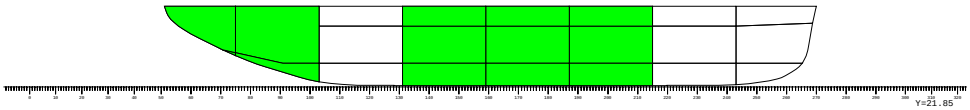


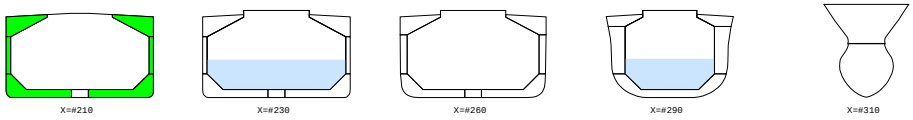
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1175	374	106.33	20.10
8.80	11.00	9083	1515	268.36	66.19
12.00	15.00	14945	2026	59.75	110.13
37.60	47.00	36831	-655	452.50	445.15
41.13	51.00	35983	3051	006.97	482.51
62.97	75.00	161480	116111	172.40	619.76
88.45	103.00	289535	-2378	424.74	660.17
113.93	131.00	367968	94271	010.37	649.68
139.41	159.00	523792	2454	489.95	632.85
164.89	187.00	491537	-4753	480.79	615.94
190.37	215.00	281156	-111961	040.82	599.00
215.85	243.00	179812	2214	409.88	579.79
241.33	271.00	77167	-95211	092.75	512.79
266.81	299.00	-10022	1537	662.73	287.48
278.01	313.00	-638	189	40.18	107.85

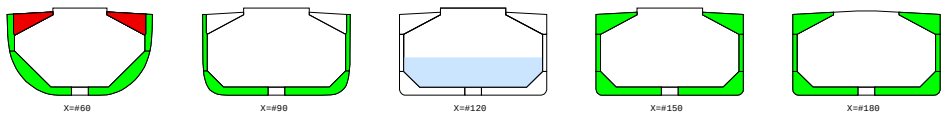
				X	FRAME
SHEAR FORCE (MIN)	-11411.8 t	(85.4%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	11611.3 t	(86.2%)		63.0 m	75
SAGGING MOMENT	-13444.7 tm	(9.7%)		261.4 m	293
HOGGING MOMENT	534083.1 tm	(72.7%)		148.7 m	169

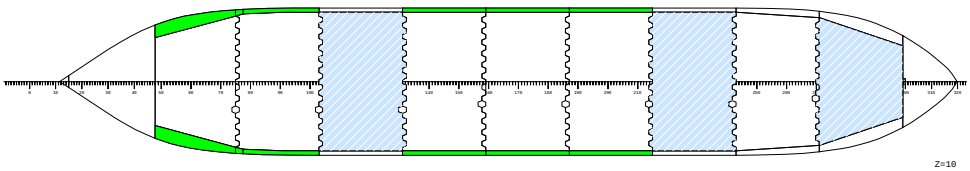
LOADING CONDITION - LSC22C2-06 : LOA. SEQU.LC22LC2 STEP6-GAR.9 EMPTY-50% BUNKERS

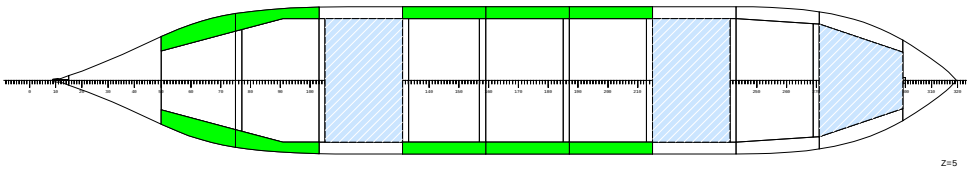


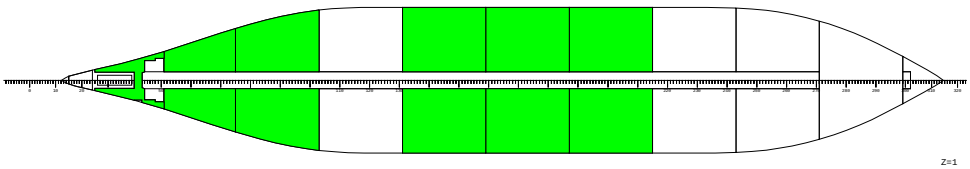












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.1	NO.1 CARGO HOLD	20243.0	40.1	252.56	0.00	7.44	0.00
R1.3	NO.3 CARGO HOLD	26695.8	40.7	202.33	0.01	7.32	0.00
R1.7	NO.7 CARGO HOLD	26727.8	40.8	101.93	0.02	7.32	0.00
SUBTOTAL		73666.6		179.71	0.01	7.35	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

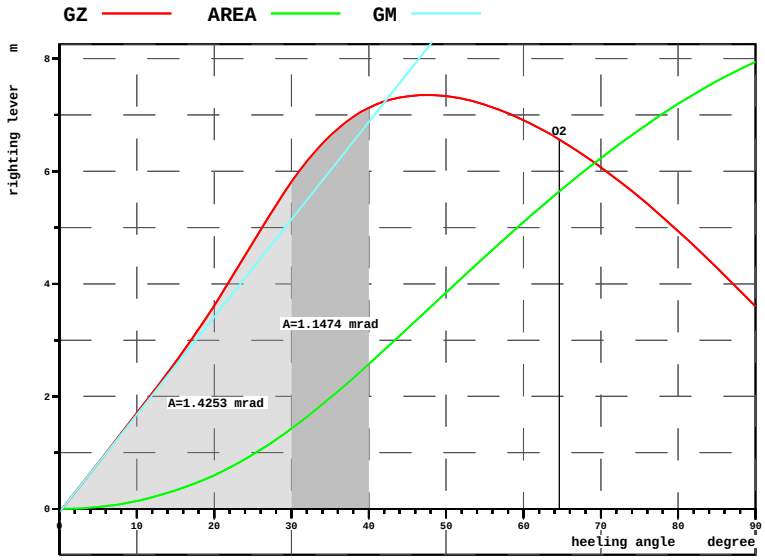
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		30294.7		117.72	-0.00	10.06	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	107198.2	157.93	-0.07 8.57
Total weight	131840.1	153.09	-0.05 9.53

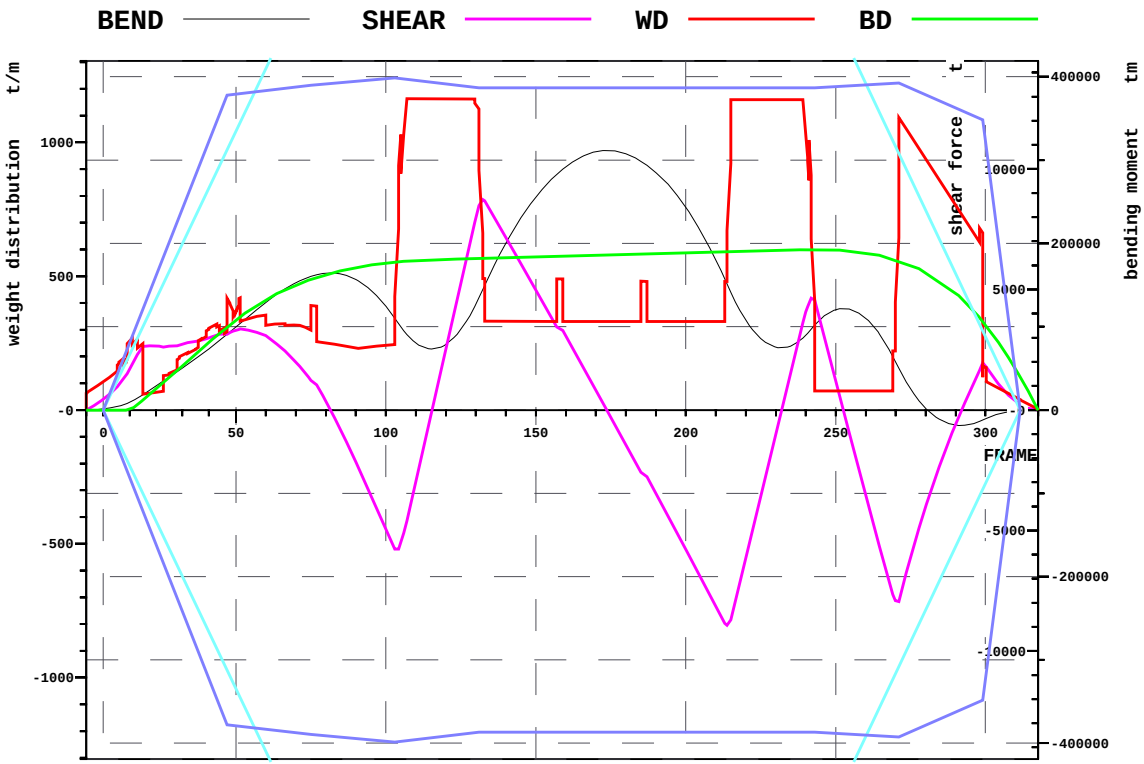
F L O A T I N G P O S I T I O N

Draught moulded	12.450	m	KM	19.45 m
Trim	-1.978	m	KG	9.53 m
Heel, PS=+	-0.3	deg		
TA	11.461	m	GM0	9.92 m
TF	13.439	m	GMCORR	0.00 m
Trimming moment	403342	tonm	GM	9.92 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.81	0.033	273.1	0.002
10.0	1.70	0.143	572.7	0.004
20.0	3.61	0.601	1197.1	0.009
30.0	5.81	1.425	1623.0	0.012
40.0	7.13	2.573	1920.4	0.015
45.0	7.33	3.205	2017.0	0.015
50.0	7.33	3.846	2076.3	0.016
55.0	7.18	4.481	2095.0	0.016
60.0	6.91	5.097	2072.0	0.016
65.0	6.53	5.684	2014.9	0.015
70.0	6.07	6.234	1930.4	0.015
75.0	5.54	6.741	1823.1	0.014
80.0	4.94	7.198	1696.6	0.013
85.0	4.29	7.601	1553.9	0.012
90.0	3.60	7.946	1399.3	0.011



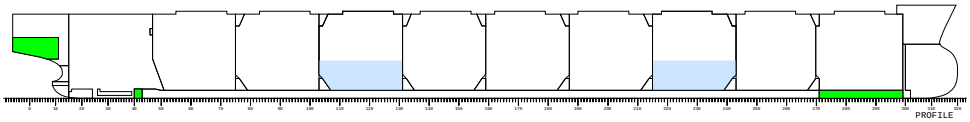
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.425	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.573	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.147	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.354	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.569	deg	OK
GM0.15	GM > 0.15 m	0.150	9.923	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	7.017		OK

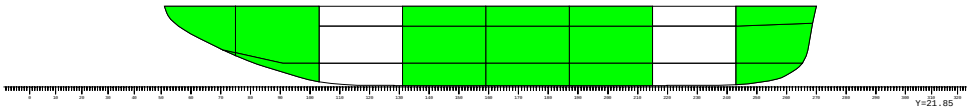


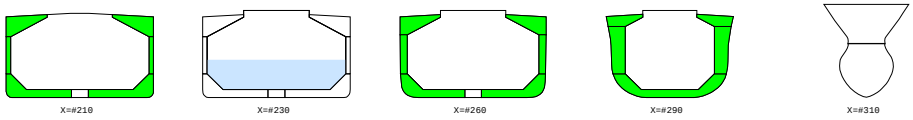
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1210	443	106.32	0.00
8.80	11.00	10713	1913	268.15	7.57
12.00	15.00	18101	2638	59.74	36.16
37.60	47.00	90699	3137	418.06	308.29
41.13	51.00	102277	3337	417.22	343.36
62.97	75.00	160236	1228	297.76	487.90
88.45	103.00	110126	-5765	424.69	551.87
113.93	131.00	134033	8475	1010.25	566.66
139.41	159.00	289173	3293	489.89	575.11
164.89	187.00	293673	-2766	480.74	583.49
190.37	215.00	141074	-8706	1040.69	591.84
215.85	243.00	102114	4562	409.83	597.88
241.33	271.00	50123	-7944	1092.62	553.88
266.81	299.00	-12143	1953	662.65	329.69
278.01	313.00	-619	243	40.17	126.69

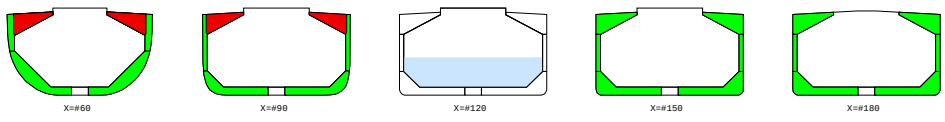
				X	FRAME
SHEAR FORCE (MIN)	-8930.5 t	(66.8%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	8728.1 t	(65.3%)		115.1 m	132
SAGGING MOMENT	-18439.7 tm	(12.0%)		259.5 m	291
HOGGING MOMENT	311468.1 tm	(42.4%)		152.1 m	173

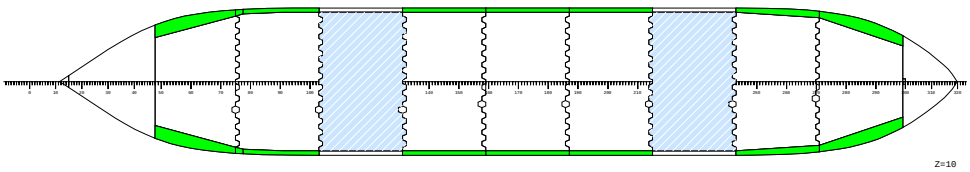
LOADING CONDITION - LSC22C2-07 : LOA. SEQU.LC22LC2 STEP7-GARGO1 EMPTY-DEP. BUN.

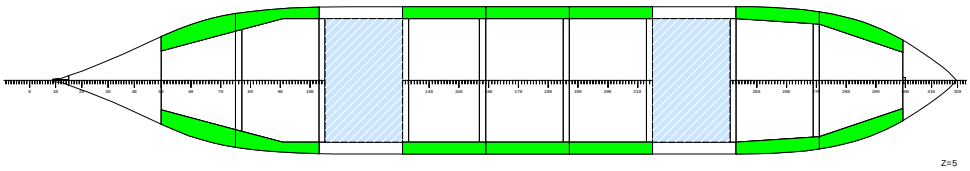


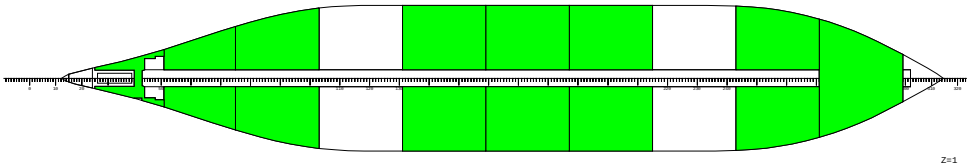












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.3	NO.3 CARGO HOLD	26695.8	40.7	202.33	0.01	7.32	0.00
R1.7	NO.7 CARGO HOLD	26727.8	40.8	101.93	0.02	7.32	0.00
SUBTOTAL		53423.6		152.10	0.02	7.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

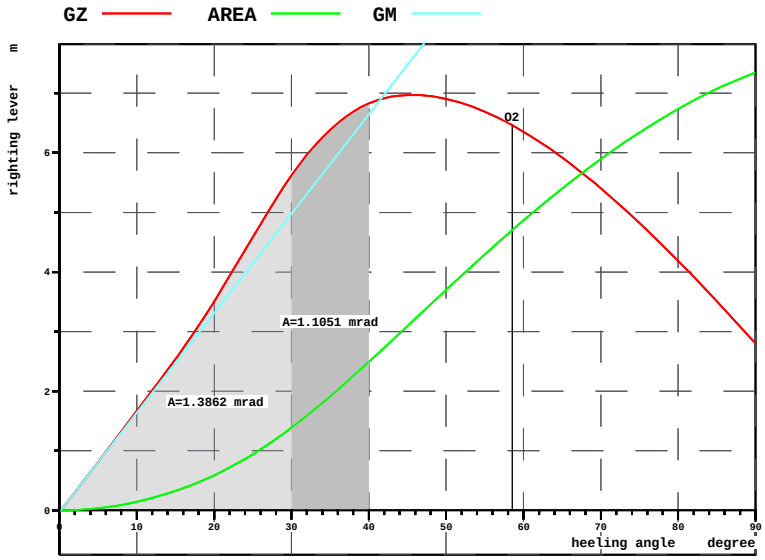
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		44290.4		156.87	-0.00	10.55	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	103899.2	148.08	-0.04 9.62
Total weight	128541.1	145.00	-0.03 10.40

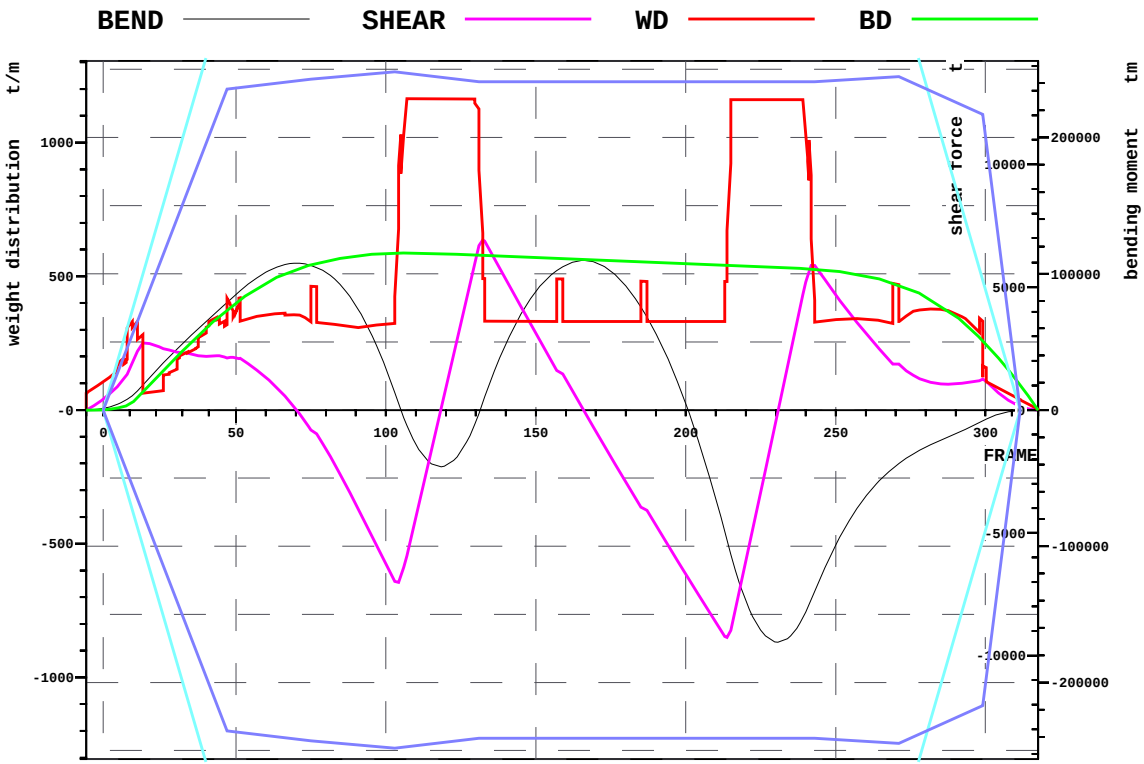
F L O A T I N G P O S I T I O N

Draught moulded	12.263	m	KM	19.95	m
Trim	3.059	m	KG	10.40	m
Heel, PS=+	-0.2	deg			
TA	13.792	m	GM0	9.55	m
TF	10.733	m	GMCORR	0.00	m
Trimming moment	-686008	tonm	GM	9.55	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.81	0.034	0.0	0.000
10.0	1.66	0.141	0.0	0.000
20.0	3.50	0.587	0.0	0.000
30.0	5.62	1.386	0.0	0.000
40.0	6.83	2.491	0.0	0.000
45.0	6.97	3.095	0.0	0.000
50.0	6.91	3.702	0.0	0.000
55.0	6.69	4.296	0.0	0.000
60.0	6.36	4.866	0.0	0.000
65.0	5.92	5.402	0.0	0.000
70.0	5.40	5.897	0.0	0.000
75.0	4.82	6.344	0.0	0.000
80.0	4.19	6.737	0.0	0.000
85.0	3.51	7.073	0.0	0.000
90.0	2.80	7.349	0.0	0.000



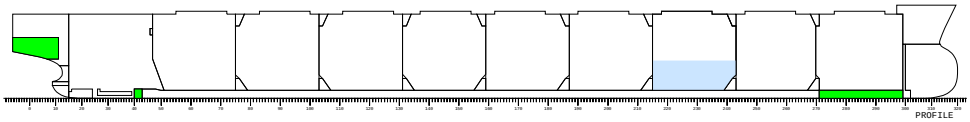
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.386	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.491	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.105	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.970	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.628	deg	OK
GM0.15	GM > 0.15 m	0.150	9.555	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.420		OK

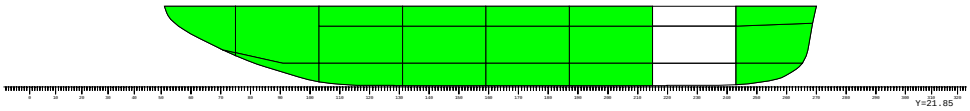


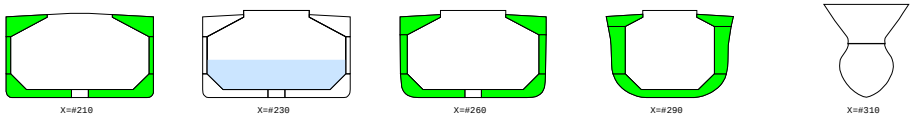
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1179	442	106.37	1.27
8.80	11.00	10482	1932	330.90	28.28
12.00	15.00	18077	2732	62.51	66.18
37.60	47.00	79082	2119	418.60	372.28
41.13	51.00	86652	2095	417.76	407.81
62.97	75.00	106139	-819	395.91	542.14
88.45	103.00	12831	-6979	424.92	585.62
113.93	131.00	-2776	66891	010.81	579.07
139.41	159.00	105068	1468	490.16	566.19
164.89	187.00	68198	-4087	330.66	553.23
190.37	215.00	-105997	-89791	041.27	540.23
215.85	243.00	-132786	5891	328.36	525.03
241.33	271.00	-39474	1870	331.26	464.45
266.81	299.00	-7899	1264	332.62	258.34
278.01	313.00	-398	153	40.19	95.78

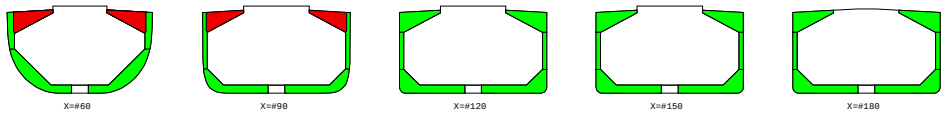
				X	FRAME
SHEAR FORCE (MIN)	-9260.2 t	(69.3%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	6928.0 t	(51.8%)		115.1 m	132
SAGGING MOMENT	-170479 tm	(28.1%)		204.7 m	231
HOGGING MOMENT	109774.7 tm	(14.9%)		145.8 m	166

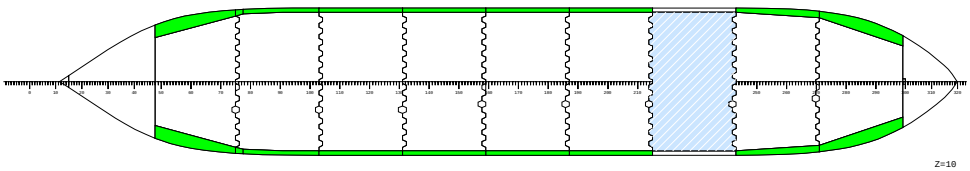
LOADING CONDITION - LSC22C2-08 : LOA. SEQU.LC22LC2 STEP8-GARGO7 EMPTY-DEP. BUN.

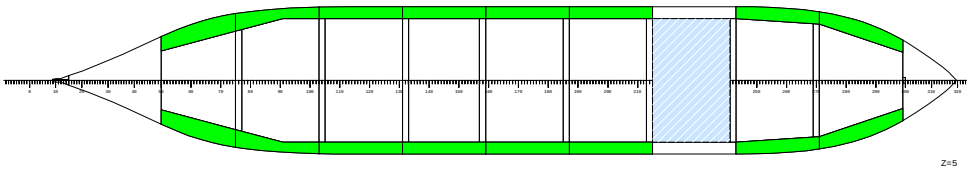


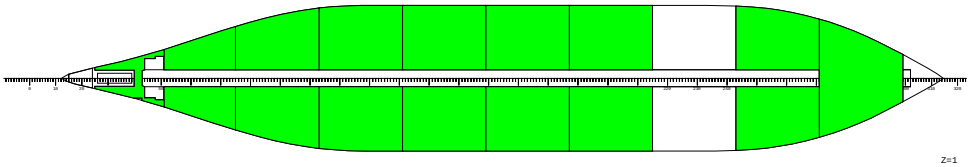












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=2.99)							
R1.3	NO.3 CARGO HOLD	26695.8	40.7	202.33	0.01	7.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00

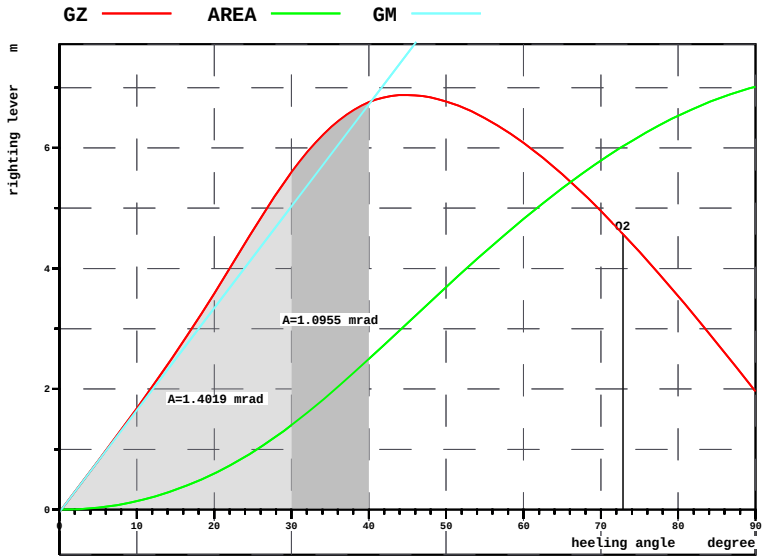
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		50839.2		149.70	-0.00	10.59	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	83720.3	159.15	-0.05 10.45
Total weight	108362.1	152.98	-0.04 11.19

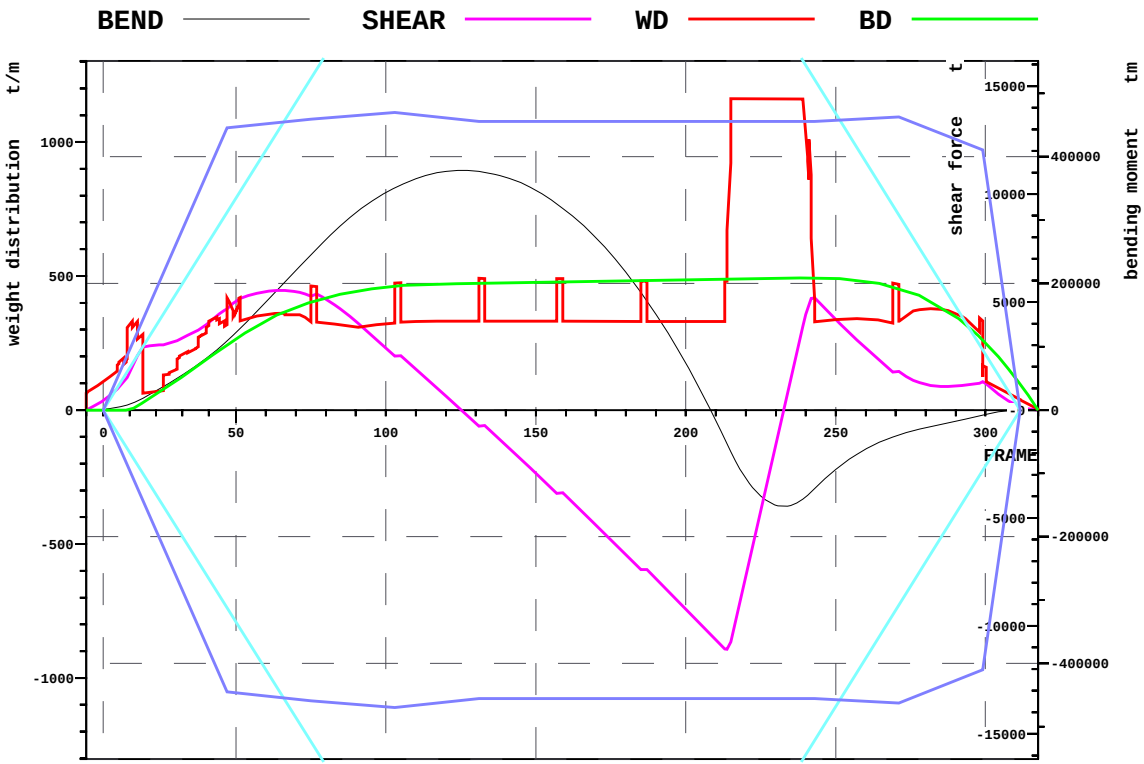
F L O A T I N G P O S I T I O N

Draught moulded	10.359	m	KM	20.86	m
Trim	-1.258	m	KG	11.19	m
Heel, PS=+	-0.2	deg			
TA	9.730	m	GM0	9.68	m
TF	10.988	m	GMCORR	0.00	m
Trimming moment	246135	tonm	GM	9.68	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.81	0.033	0.0	0.000
10.0	1.68	0.142	0.0	0.000
20.0	3.58	0.597	0.0	0.000
30.0	5.60	1.402	0.0	0.000
40.0	6.76	2.497	0.0	0.000
45.0	6.88	3.094	0.0	0.000
50.0	6.77	3.691	0.0	0.000
55.0	6.50	4.271	0.0	0.000
60.0	6.08	4.821	0.0	0.000
65.0	5.56	5.330	0.0	0.000
70.0	4.95	5.789	0.0	0.000
75.0	4.27	6.192	0.0	0.000
80.0	3.54	6.533	0.0	0.000
85.0	2.76	6.809	0.0	0.000
90.0	1.96	7.015	0.0	0.000



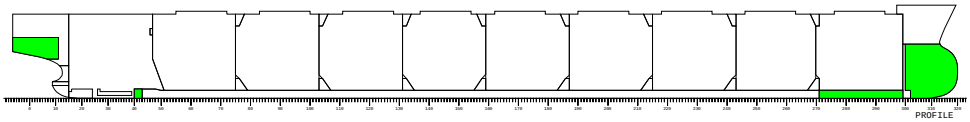
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.402	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.497	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.095	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.877	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.815	deg	OK
GM0.15	GM > 0.15 m	0.150	9.678	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.366		OK

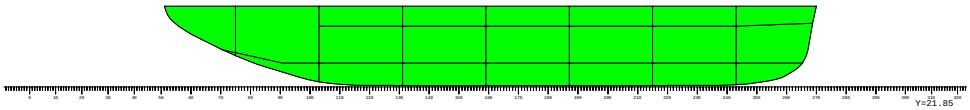


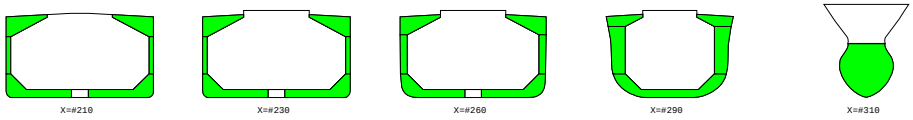
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1098	443	106.39	0.00
8.80	11.00	10497	2014	330.95	5.28
12.00	15.00	18452	2914	62.52	26.68
37.60	47.00	109084	4676	418.66	242.34
41.13	51.00	126445	5120	417.82	272.47
62.97	75.00	245012	5274	395.97	402.20
88.45	103.00	350468	2496	473.36	461.24
113.93	131.00	376253	-747	490.78	472.96
139.41	159.00	318107	-3842	490.23	478.37
164.89	187.00	172074	-7391	481.07	483.71
190.37	215.00	-66122	-107431	041.42	489.01
215.85	243.00	-123570	5203	328.41	492.13
241.33	271.00	-39519	1784	331.30	450.43
266.81	299.00	-7979	1310	332.67	260.49
278.01	313.00	-330	164	40.20	98.92

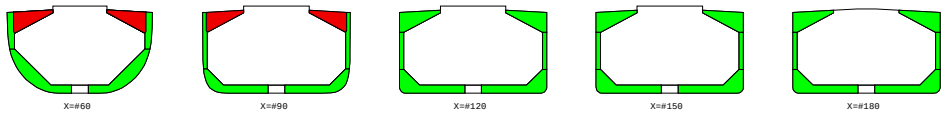
				X	FRAME
SHEAR FORCE (MIN)	-11081.5 t	(82.9%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	5543.7 t	(41.5%)		55.1 m	66
SAGGING MOMENT	-151687 tm	(26.0%)		207.6 m	234
HOGGING MOMENT	378160.5 tm	(51.5%)		107.6 m	124

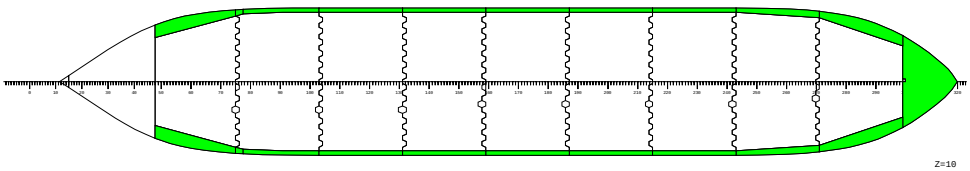
LOADING CONDITION - LSC22C2-09 : LOA. SEQU.LC22LC2 STEP9-GARGO3 EMPTY-DEP. BUN.

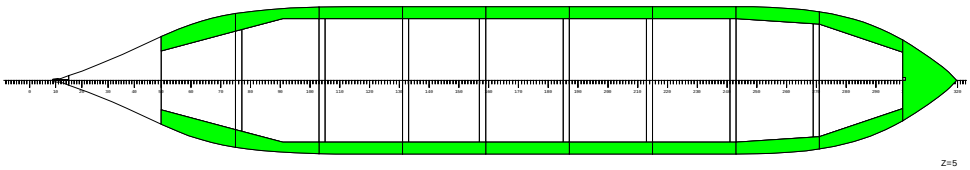


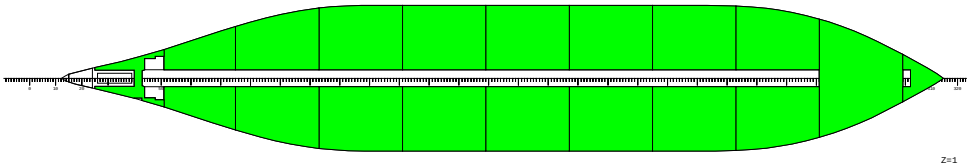












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00

CONTENTS=Water Ballast (RHO=1.025)

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00

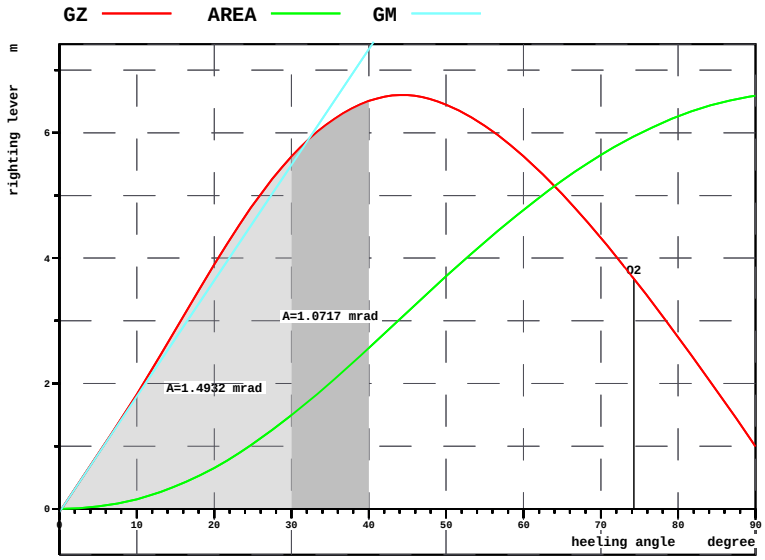
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	67159.5	152.30	-0.07 11.67
Total weight	91801.3	146.85	-0.05 12.20

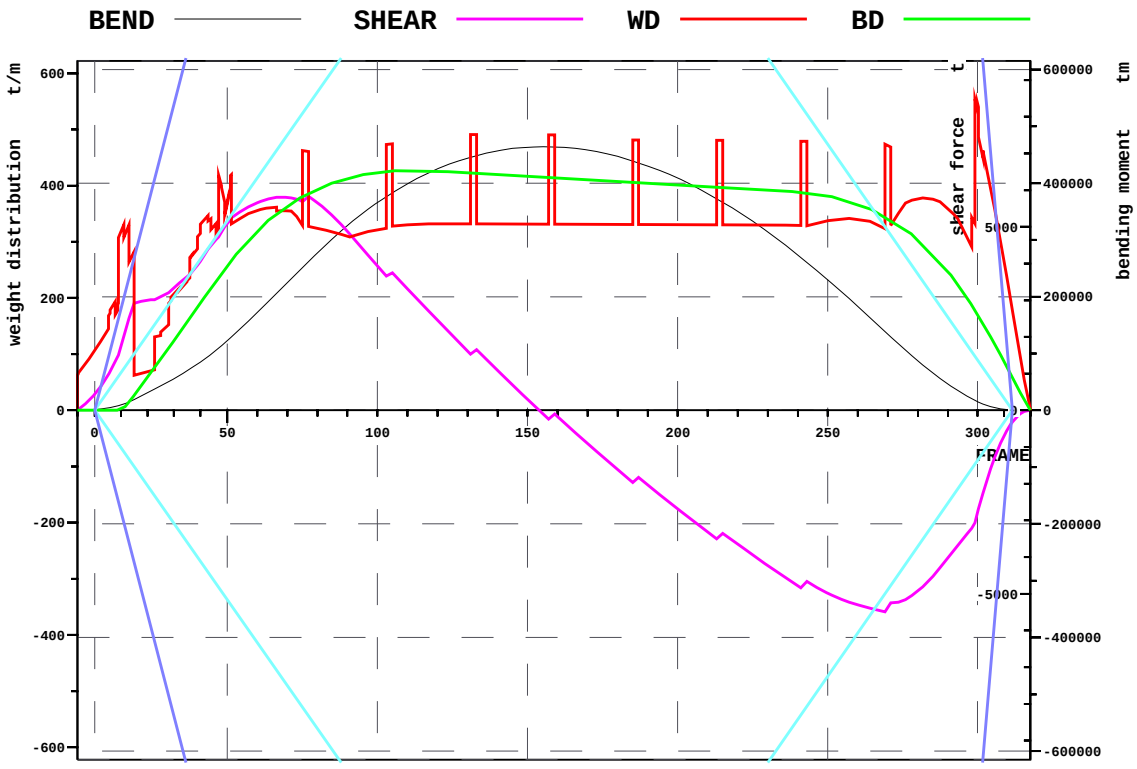
F L O A T I N G P O S I T I O N

Draught moulded	8.971	m	KM	22.79 m
Trim	2.027	m	KG	12.20 m
Heel, PS=+	-0.3	deg		
TA	9.984	m	GM0	10.58 m
TF	7.957	m	GMCORR	0.00 m
Trimming moment	-394576	tonm	GM	10.58 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.88	0.036	0.0	0.000
10.0	1.83	0.153	0.0	0.000
20.0	3.90	0.652	0.0	0.000
30.0	5.62	1.493	0.0	0.000
40.0	6.51	2.565	0.0	0.000
45.0	6.60	3.139	0.0	0.000
50.0	6.45	3.710	0.0	0.000
55.0	6.11	4.259	0.0	0.000
60.0	5.63	4.772	0.0	0.000
65.0	5.02	5.238	0.0	0.000
70.0	4.33	5.646	0.0	0.000
75.0	3.56	5.991	0.0	0.000
80.0	2.74	6.266	0.0	0.000
85.0	1.88	6.467	0.0	0.000
90.0	0.99	6.593	0.0	0.000



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.493	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.565	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.072	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.604	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.363	deg	OK
GM0.15	GM > 0.15 m	0.150	10.582	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.008		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1157	443	106.41	0.00
8.80	11.00	10658	2015	331.01	5.28
12.00	15.00	18650	2914	62.53	27.06
37.60	47.00	109775	4723	418.74	235.58
41.13	51.00	127388	5195	417.90	263.71
62.97	75.00	250990	5686	396.05	380.13
88.45	103.00	376047	3661	473.44	424.62
113.93	131.00	445224	1530	490.87	422.41
139.41	159.00	463331	-98	490.32	413.91
164.89	187.00	435441	-1826	481.16	405.34
190.37	215.00	366115	-3350	405.42	396.74
215.85	243.00	260570	-4667	328.47	386.09
241.33	271.00	129115	-5250	331.37	336.83
266.81	299.00	17118	-3070	332.73	177.94
278.01	313.00	589	-335	182.47	60.50

				X	FRAME
SHEAR FORCE (MIN)	-5488.6 t	(40.5%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	5830.2 t	(43.2%)		64.8 m	77
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	463872.7 tm	(63.1%)		135.9 m	155

11. Loading/unloading sequence: From cond. No4 to cond. No23

LOADING/UNLOADING SEQUENCE SUMMARY FORM

Vessel name	Voyage No.	Condition	Yard	Id. Number

Port (specific or typical):	Condition at commencement of laoding/discharging:	
Total mass of cargo to be loaded/discharged:	Condition at end of loading/discharging:	
Dock water density (t/m^3):	Maximum loading/discharging rate:	Average Loading/discharging rate:
Number of loaders/dischargers:	Maximum Ballasting/Deballasting rate:	Average Ballasting/Deballasting rate:

Hold No.	9	8	7	6	5	4	3	2	1
Volume of Hold, V(m^3)	18190	21557	21935	21647	21915	21917	21916	21214	16870
Height of hold, h (m)	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2

Note:During each pour it has to be controlled that allowable limits for hull girder shear force,bending moment and mass in holds are not exceeded.
Loading/discharging operations may have to be paused to allow for ballasting/deballasting in order to keep actual values within limits

Hold content at commencement of loading/discharging.

	9	8	7	6	5	4	3	2	1
Cargo mass (t)	0	0	0	0	0	0	0	0	0
Density, (t/m^3)									
Grade									

Ballast contents at commencement of loading/discharging.

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0
Double side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0
Double bottom	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0

Commencement of loading/discharging (sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
8.52	0.50	8.02	42.30%	67.40%

CARGO OPERATIONS — BALLASTING OPERATIONS

Pour No./ Grade	Hold 9	Hold 8	Hold 7	Hold 6	Hold 5	Hold 4	Hold 3	Hold 2	Hold 1	Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
1							19730t To30% of Hold			Top side								Empty			
										Double side								Empty			
										Double bottom								Empty			
2			19748t To30% of Hold							Top side				Empty							
										Double side				Empty							
										Double bottom				Empty							
3					19730t To30% of Hold					Top side						Empty					
										Double side						Empty					
										Double bottom						Empty					
4									15188t To30% of Hold	Top side										Empty	
										Double side										Empty	
										Double bottom										Empty	
5	16376t To30% of Hold									Top side	Empty	Empty									
										Double side	Empty	Empty									
										Double bottom	Empty	Empty									
6				18588t To30% of Hold						Top side					Empty						
										Double side					Empty						
										Double bottom					Empty						
7						19731t To30% of Hold				Top side							Empty				
										Double side							Empty				
										Double bottom							Empty				
8								19099t To30% of Hold		Top side									Empty		
										Double side									Empty		
										Double bottom									Empty		
9		19407t To30% of Hold								Top side			Empty								
										Double side			Empty								
										Double bottom			Empty								

Values at end of pour (from harbour to sea).

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
8.23	-1.97	10.20	55.90%	39.00%
11.11	1.18	9.93	48.00%	24.60%
12.14	0.91	11.23	62.40%	35.30%
11.55	-1.96	13.51	44.90%	27.80%
14.44	1.92	12.52	47.60%	17.80%
15.82	2.63	13.19	53.00%	29.90%
16.62	1.50	15.12	62.30%	53.30%
15.40	-3.05	18.45	47.40%	33.50%
18.54	0.71	17.83	46.50%	36.00%

Total cargo onboard (t): 167600.3 Remaining cargo to be loaded (t): 0

Total amount of bunkers onboard (t): 5975.2 (100%)

Draft Survey n-1	Top side											
	Double side											
	Double bottom											
Draft Survey n	Top side											
	Double side											
	Double bottom											

Hold content at end of loading/discharging.

Cargo mass (t)										Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
										Top side	0	0	0	0	0	0	0	0	0	0	0
										Double side	0	0	0	0	0	0	0	0	0	0	0
										Double bottom	0	0	0	0	0	0	0	0	0	0	0

Total mass loaded/discharged (t): 167600.3

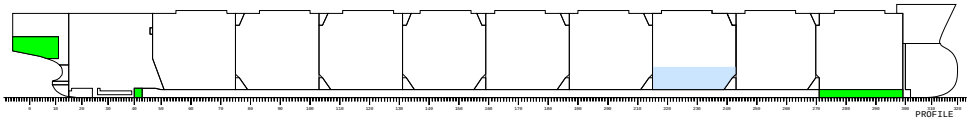
Values at end of loading/discharging (sea).

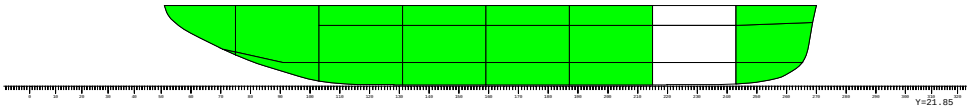
d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
18.54	0.71	17.83	54.80%	59.40%

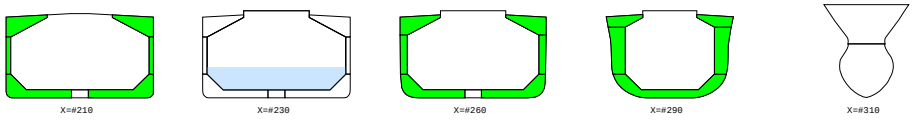
	9	8	7	6	5	4	3	2	1	
Net load on Double Bottom:	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	$tons/m^2$
Net load in two adjacent holds:	35783	39155	38336	38318	39461	39461	38829	34287		tons

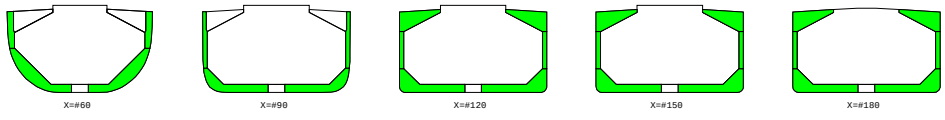
Net load on double bottom = (Mh/V)×h-T[t/m^2]
where: Mh = Mass in hold + mass in DB (t)
V = Total volume of hold (m^3) T = draught (m)
h = height of hold from inner bottom to top of coaming (m)

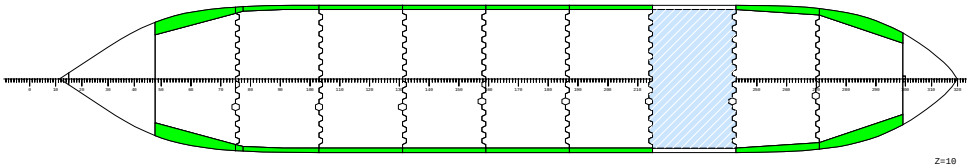
LOADING CONDITION - LSC4C23-01 : LOA. SEQU.LC4LC23 STEP1-GARGO 3-ARRIVAL BUNKERS

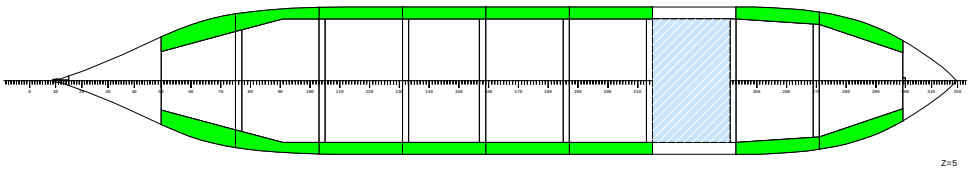


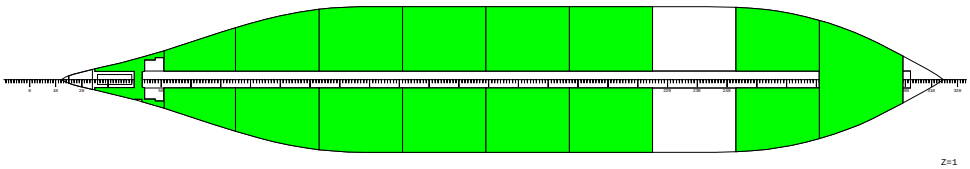












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00

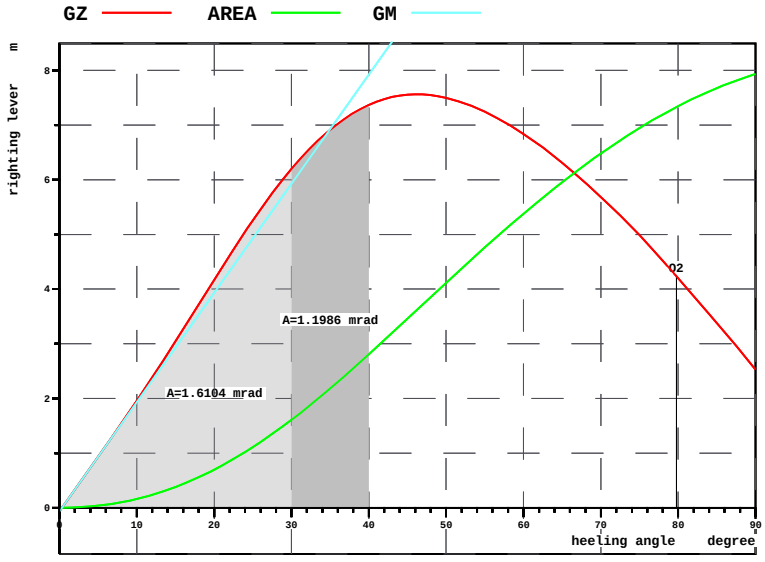
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		50839.2		149.70	-0.00	10.59	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	71438.0	162.77	-0.09 9.53
Total weight	96079.9	154.88	-0.06 10.59

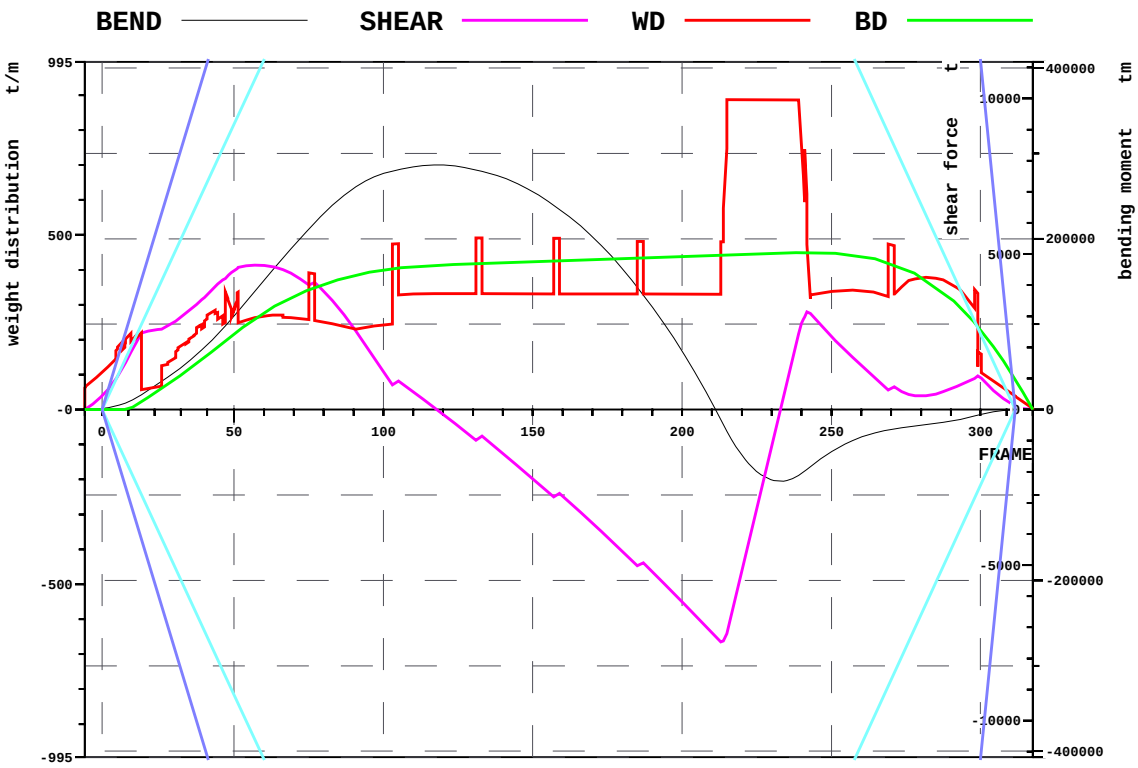
F L O A T I N G P O S I T I O N

Draught moulded	9.214	m	KM	22.03 m
Trim	-1.972	m	KG	10.59 m
Heel, PS=+	-0.3	deg		
TA	8.228	m	GM0	11.44 m
TF	10.200	m	GMCORR	0.00 m
Trimming moment	375174	tonm	GM	11.44 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.94	0.038	64.4	0.001
10.0	1.96	0.164	128.6	0.001
20.0	4.17	0.697	221.7	0.002
30.0	6.20	1.610	296.1	0.003
40.0	7.37	2.809	401.9	0.004
45.0	7.55	3.462	456.9	0.005
50.0	7.50	4.120	503.3	0.005
55.0	7.24	4.764	540.1	0.006
60.0	6.84	5.380	568.2	0.006
65.0	6.31	5.954	588.6	0.006
70.0	5.68	6.478	603.4	0.006
75.0	4.99	6.944	614.6	0.006
80.0	4.20	7.345	620.8	0.006
85.0	3.38	7.676	620.0	0.006
90.0	2.53	7.934	613.4	0.006



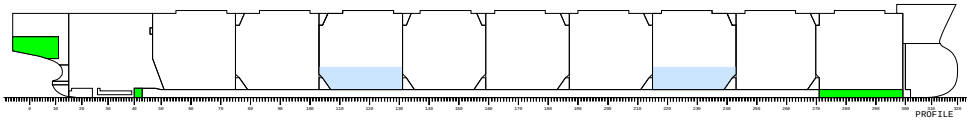
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.610	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.809	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.199	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.561	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	46.164	deg	OK
GM0.15	GM > 0.15 m	0.150	11.443	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.671		OK

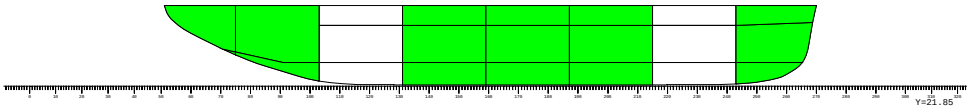


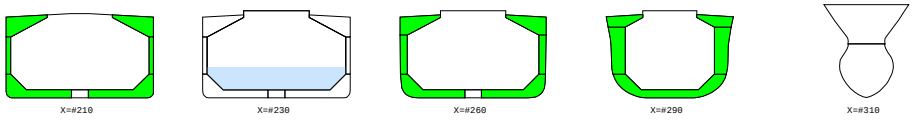
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1097	443	106.40	0.00
8.80	11.00	10355	1840	218.45	5.28
12.00	15.00	17235	2462	57.05	22.62
37.60	47.00	97841	4210	336.94	196.18
41.13	51.00	113339	4532	334.16	222.23
62.97	75.00	211326	4002	257.88	342.24
88.45	103.00	279378	794	473.39	402.35
113.93	131.00	280598	-985	490.82	417.04
139.41	159.00	234211	-2692	490.27	425.46
164.89	187.00	134309	-4932	481.11	433.80
190.37	215.00	-25421	-7198	816.92	442.11
215.85	243.00	-66583	3084	328.43	448.30
241.33	271.00	-22707	729	331.33	411.64
266.81	299.00	-6632	1088	332.70	236.96
278.01	313.00	-265	134	40.20	88.92

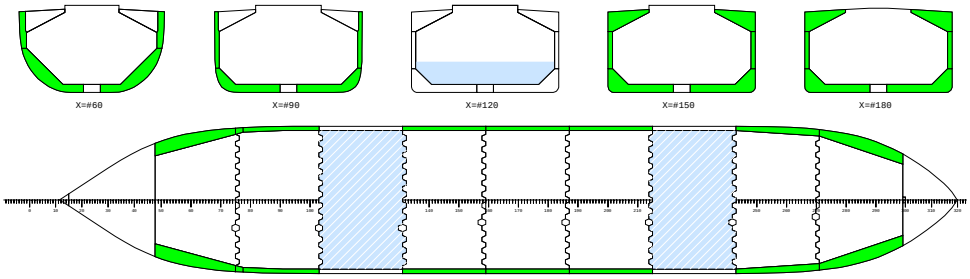
				X	FRAME
SHEAR FORCE (MIN)	-7467.6 t	(55.9%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	4642.6 t	(35.2%)		46.6 m	57
SAGGING MOMENT	-83755.3 tm	(14.4%)		207.6 m	234
HOGGING MOMENT	286370.0 tm	(39.0%)		101.2 m	117

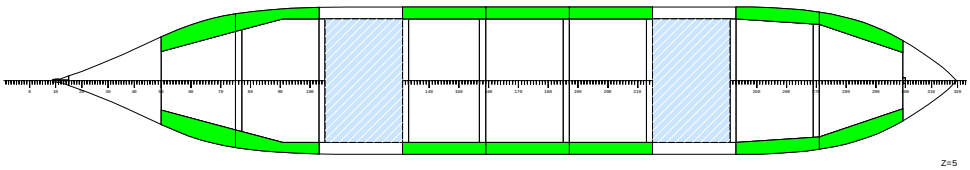
LOADING CONDITION - LSC4C23-02 : LOA. SEQU.LC4LC23 STEP2-GARGO 7-ARRIVAL BUNKERS

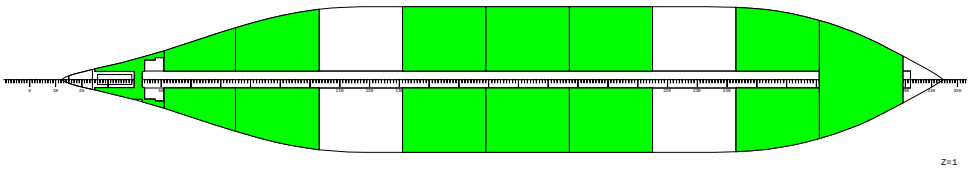












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		39478.8		152.11	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

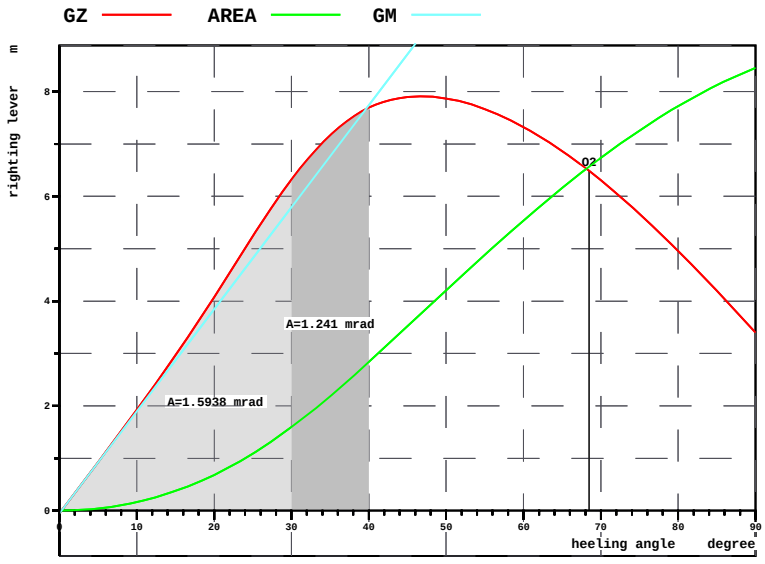
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		44290.4		156.87	-0.00	10.55	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	84637.2	153.35	-0.07 8.64
Total weight	109279.1	148.54	-0.05 9.77

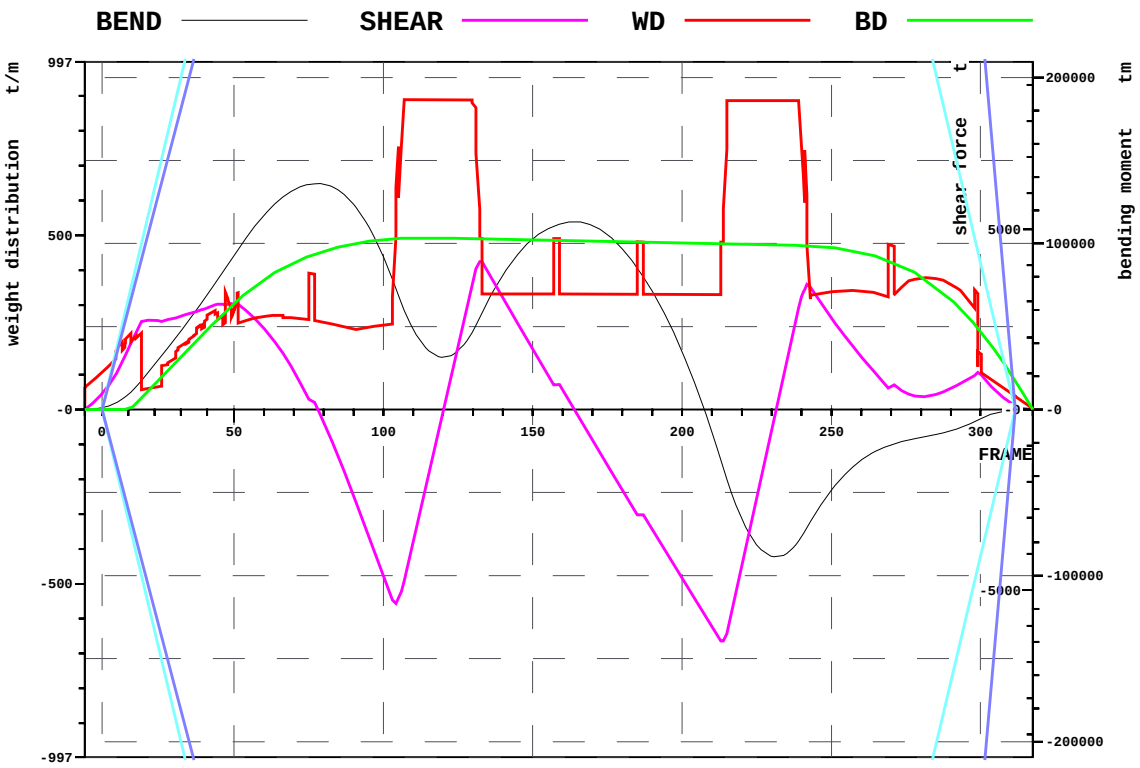
F L O A T I N G P O S I T I O N

Draught moulded	10.520	m	KM	20.94 m
Trim	1.184	m	KG	9.77 m
Heel, PS=+	-0.3	deg		
TA	11.112	m	GM0	11.17 m
TF	9.929	m	GMCORR	0.00 m
Trimming moment	-238885	tonm	GM	11.17 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.92	0.038	64.4	0.001
10.0	1.92	0.162	128.6	0.001
20.0	4.08	0.682	221.7	0.002
30.0	6.32	1.594	296.1	0.003
40.0	7.69	2.835	401.9	0.004
45.0	7.90	3.517	456.9	0.004
50.0	7.87	4.206	503.3	0.005
55.0	7.66	4.885	540.1	0.005
60.0	7.32	5.540	568.2	0.005
65.0	6.86	6.159	588.6	0.005
70.0	6.31	6.735	603.4	0.006
75.0	5.66	7.258	614.6	0.006
80.0	4.96	7.721	620.8	0.006
85.0	4.20	8.121	620.0	0.006
90.0	3.40	8.453	613.4	0.006



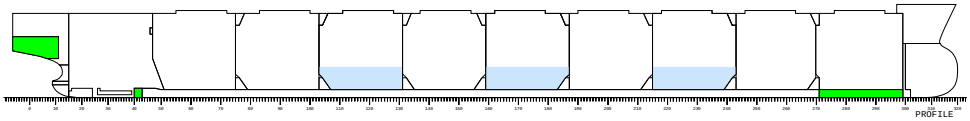
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.594	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.835	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.241	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.909	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	46.784	deg	OK
GM0.15	GM > 0.15 m	0.150	11.168	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.639		OK

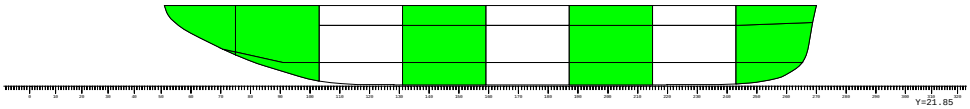


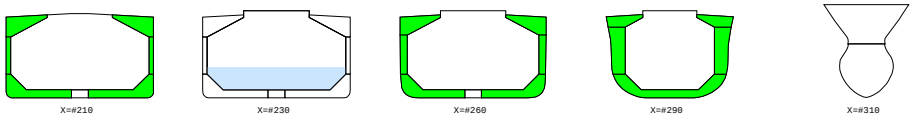
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1127	443	106.38	0.00
8.80	11.00	10435	1839	218.51	6.32
12.00	15.00	17313	2445	57.05	32.61
37.60	47.00	85100	2893	336.90	279.35
41.13	51.00	95438	2911	334.12	311.16
62.97	75.00	135586	288	257.85	439.61
88.45	103.00	78483	-5283	323.61	489.24
113.93	131.00	50785	3906	735.35	490.63
139.41	159.00	111622	689	490.20	485.70
164.89	187.00	79920	-2916	481.05	480.70
190.37	215.00	-42086	-6208	816.81	475.66
215.85	243.00	-67360	3386	328.39	468.48
241.33	271.00	-20728	689	331.29	418.23
266.81	299.00	-6330	1031	332.65	233.01
278.01	313.00	-274	123	40.20	85.53

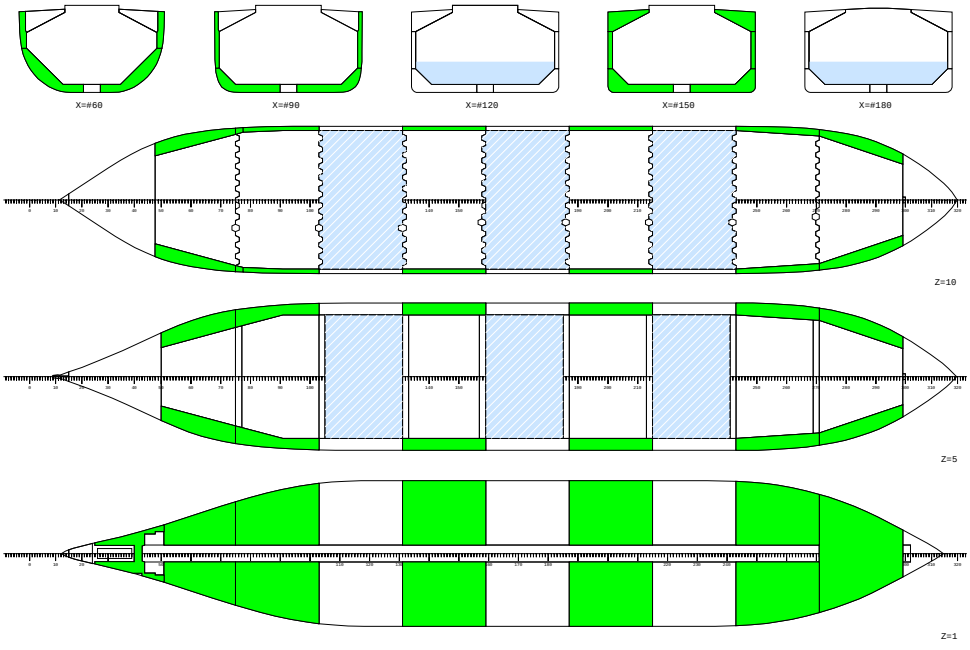
				X	FRAME
SHEAR FORCE (MIN)	-6415.9 t	(48.0%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	4103.3 t	(30.7%)		115.8 m	133
SAGGING MOMENT	-88692.8 tm	(14.6%)		204.7 m	231
HOGGING MOMENT	136079.1 tm	(24.6%)		66.6 m	79

LOADING CONDITION - LSC4C23-03 : LOA. SEQU.LC4LC23 STEP3-GARGO 5-ARRIVAL BUNKERS









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		59208.7		151.84	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

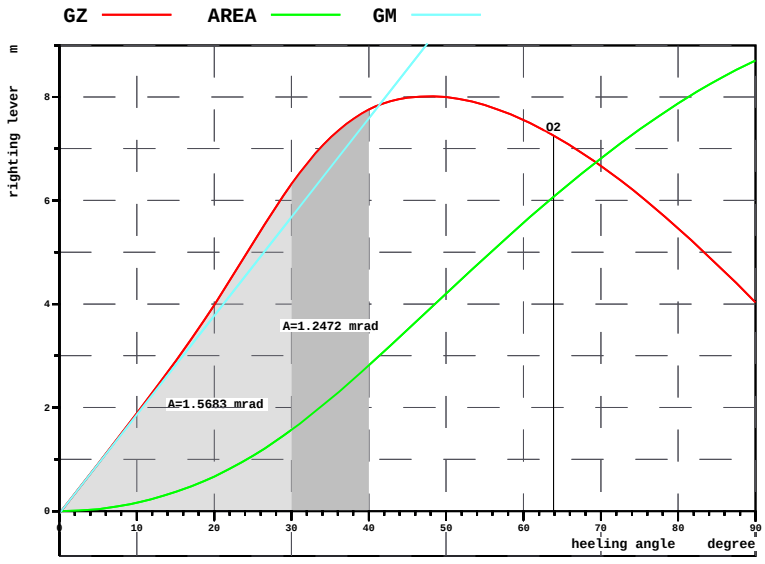
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		37712.2		157.69	-0.00	10.50	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	97789.0	153.02	-0.06 7.99
Total weight	122430.9	148.79	-0.05 9.14

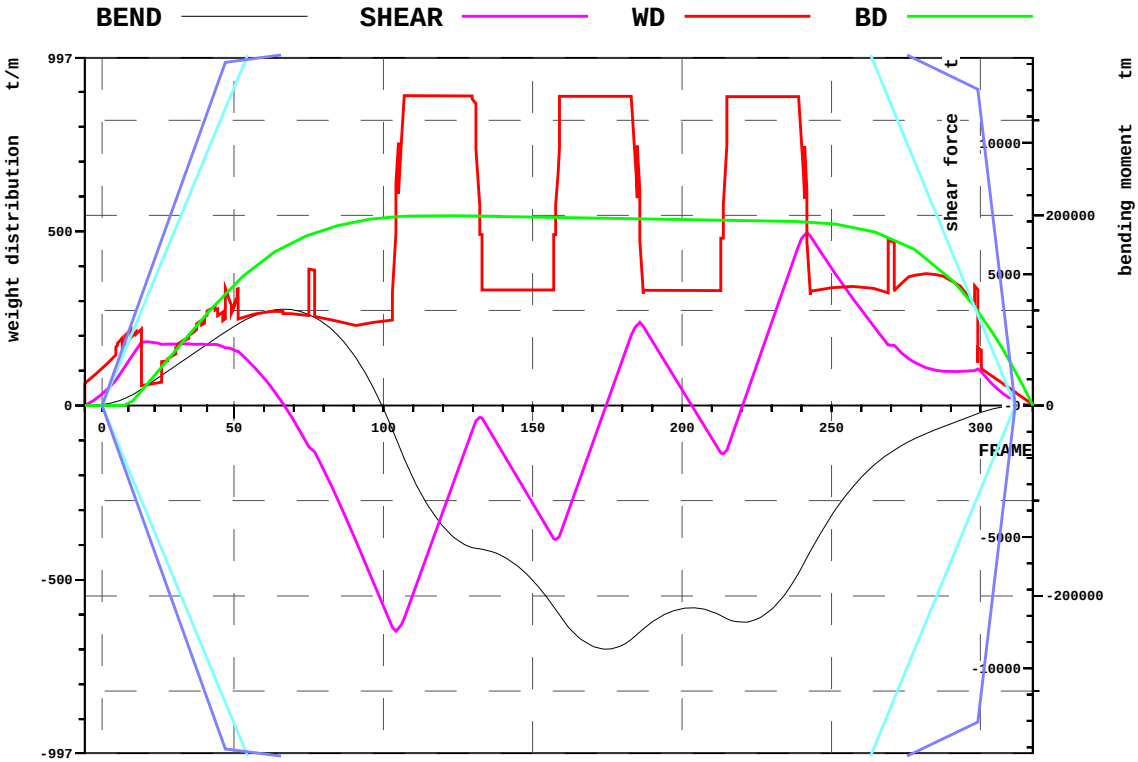
F L O A T I N G P O S I T I O N

Draught moulded	11.685	m	KM	20.07 m
Trim	0.907	m	KG	9.14 m
Heel, PS=+	-0.2	deg		
TA	12.138	m	GM0	10.93 m
TF	11.232	m	GMCORR	0.00 m
Trimming moment	-189653	tonm	GM	10.93 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.91	0.038	64.4	0.001
10.0	1.89	0.160	128.6	0.001
20.0	3.98	0.667	221.7	0.002
30.0	6.32	1.568	296.1	0.002
40.0	7.76	2.815	401.9	0.003
45.0	7.98	3.504	456.9	0.004
50.0	7.99	4.202	503.3	0.004
55.0	7.84	4.894	540.1	0.004
60.0	7.55	5.567	568.2	0.005
65.0	7.16	6.210	588.6	0.005
70.0	6.67	6.813	603.4	0.005
75.0	6.10	7.371	614.6	0.005
80.0	5.46	7.875	620.8	0.005
85.0	4.77	8.322	620.0	0.005
90.0	4.03	8.706	613.4	0.005



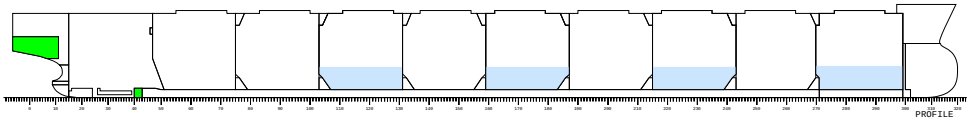
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.568	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.816	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.247	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	8.011	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.822	deg	OK
GM0.15	GM > 0.15 m	0.150	10.930	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.550		OK

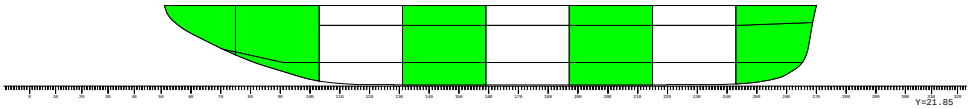


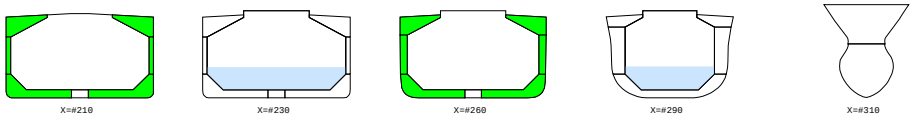
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1120	443	106.38	0.00
8.80	11.00	10406	1832	218.48	10.50
12.00	15.00	17227	2417	57.04	41.61
37.60	47.00	77357	2192	336.87	319.05
41.13	51.00	84963	2066	334.09	353.31
62.97	75.00	95781	-1574	257.83	489.02
88.45	103.00	-25168	-8438	323.58	540.80
113.93	131.00	-150173	-580	735.29	543.36
139.41	159.00	-220813	-4963	740.53	539.60
164.89	187.00	-234932	3011	321.96	535.77
190.37	215.00	-223963	-1700	816.74	531.90
215.85	243.00	-152803	6446	328.36	525.81
241.33	271.00	-46885	2292	331.26	473.79
266.81	299.00	-8493	1390	332.63	270.09
278.01	313.00	-372	172	40.20	101.99

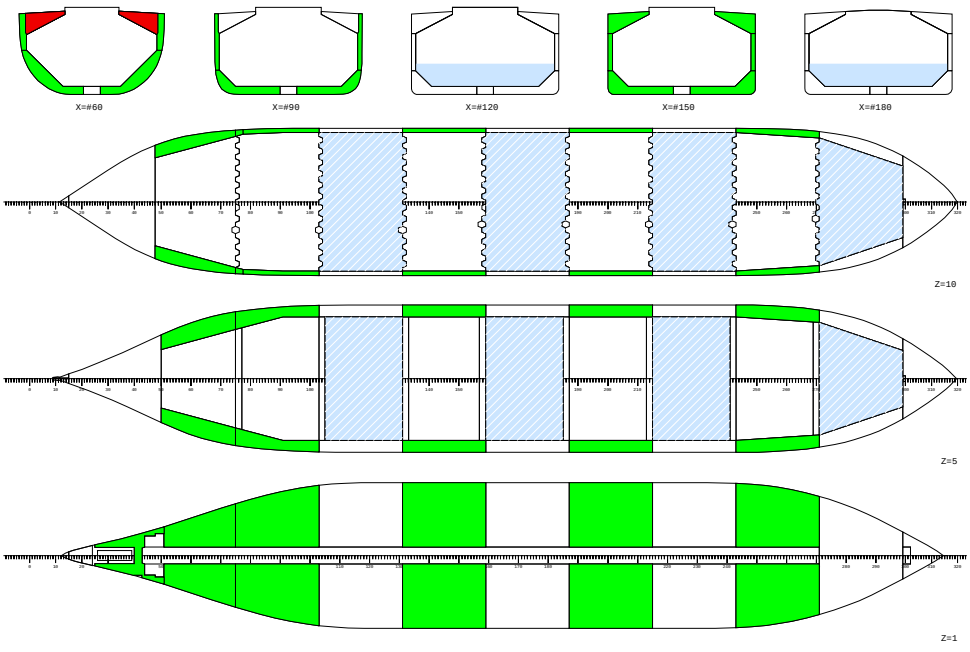
				X	FRAME
SHEAR FORCE (MIN)	-8583.4 t	(62.4%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	6590.1 t	(49.3%)		214.8 m	242
SAGGING MOMENT	-256065 tm	(35.3%)		153.7 m	175
HOGGING MOMENT	101428.1 tm	(22.2%)		55.1 m	66

LOADING CONDITION - LSC4C23-04 : LOA. SEQU.LC4LC23 STEP4-GARGO HOLD 1-50% BUNKER









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		74397.0		172.41	0.01	6.21	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

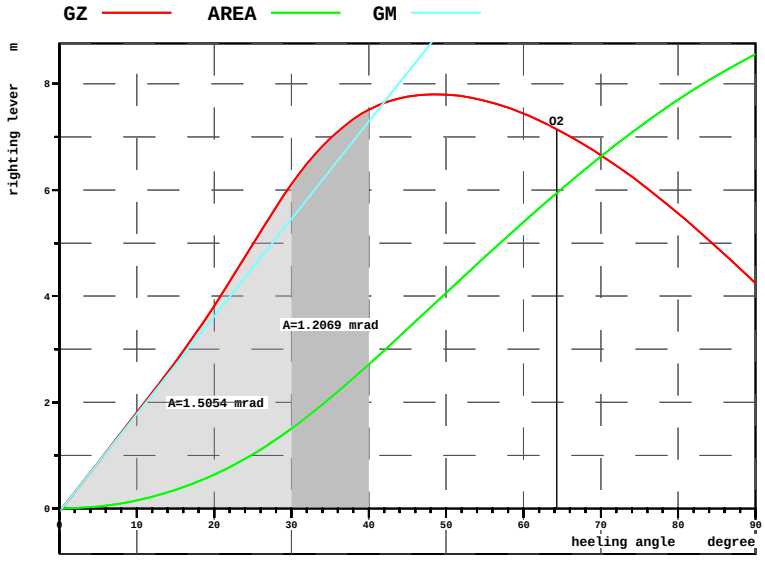
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		30452.5		134.80	-0.00	10.12	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	108086.4	157.81	-0.07 7.80
Total weight	132728.2	153.02	-0.06 8.89

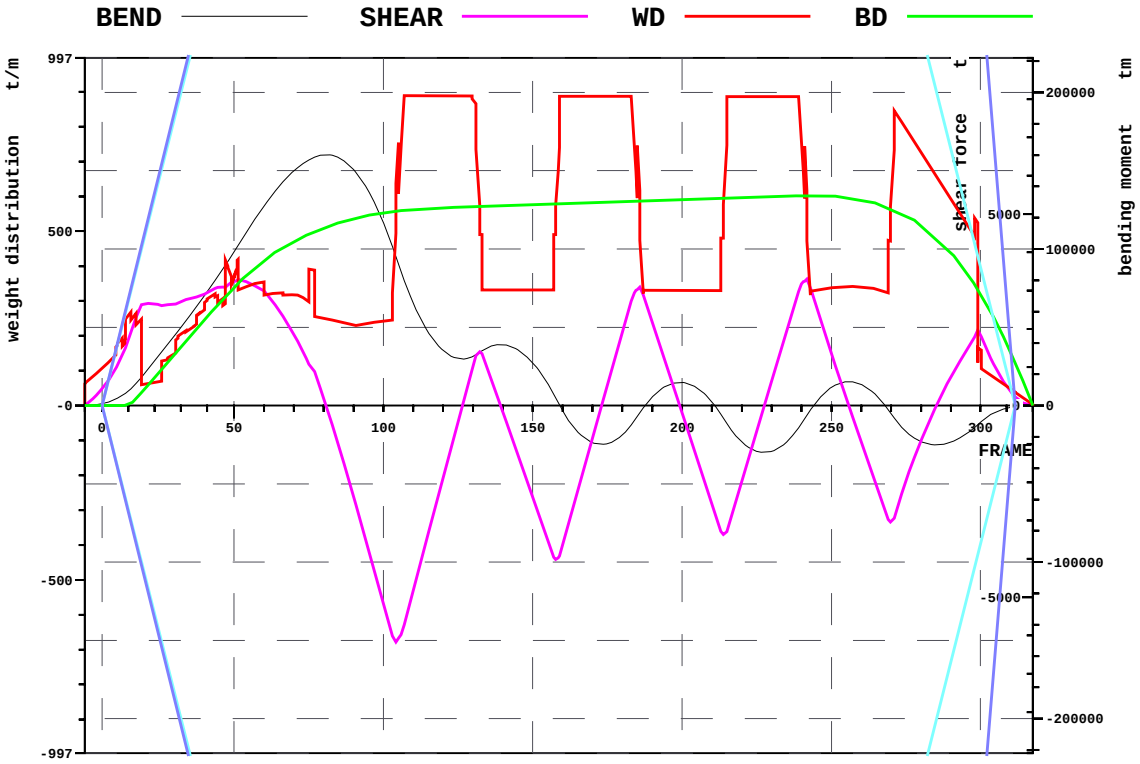
F L O A T I N G P O S I T I O N

Draught moulded	12.530	m	KM	19.42 m
Trim	-1.961	m	KG	8.89 m
Heel, PS=+	-0.3	deg		
TA	11.549	m	GM0	10.53 m
TF	13.510	m	GMCORR	0.00 m
Trimming moment	400842	tonm	GM	10.53 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.86	0.035	273.1	0.002
10.0	1.80	0.152	572.7	0.004
20.0	3.81	0.637	1197.1	0.009
30.0	6.11	1.505	1623.0	0.012
40.0	7.51	2.712	1920.4	0.014
45.0	7.76	3.380	2017.0	0.015
50.0	7.80	4.060	2076.3	0.016
55.0	7.68	4.736	2095.0	0.016
60.0	7.44	5.397	2072.0	0.016
65.0	7.09	6.032	2014.9	0.015
70.0	6.65	6.632	1930.4	0.015
75.0	6.14	7.191	1823.1	0.014
80.0	5.56	7.702	1696.6	0.013
85.0	4.92	8.159	1553.9	0.012
90.0	4.24	8.559	1399.3	0.011



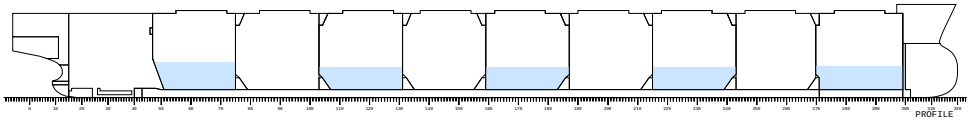
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.505	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.712	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.207	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.802	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	48.728	deg	OK
GM0.15	GM > 0.15 m	0.150	10.528	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	7.237		OK

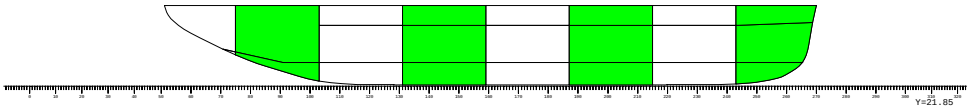


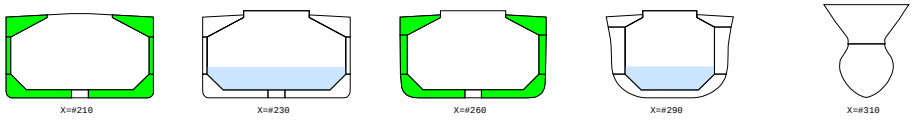
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1193	443	106.33	0.00
8.80	11.00	10667	1913	268.19	7.90
12.00	15.00	18042	2636	59.75	36.92
37.60	47.00	89935	3080	418.12	311.53
41.13	51.00	101279	3269	417.28	346.79
62.97	75.00	156808	1079	297.79	491.74
88.45	103.00	101592	-6011	323.45	555.71
113.93	131.00	32503	1303	734.99	570.42
139.41	159.00	46	-3928	740.22	578.80
164.89	187.00	-1507	2884	321.82	587.10
190.37	215.00	-11457	-3294	816.40	595.38
215.85	243.00	-2595	3071	328.23	601.34
241.33	271.00	-7914	-2936	845.23	557.10
266.81	299.00	-12184	1970	524.36	331.72
278.01	313.00	-604	245	40.18	127.36

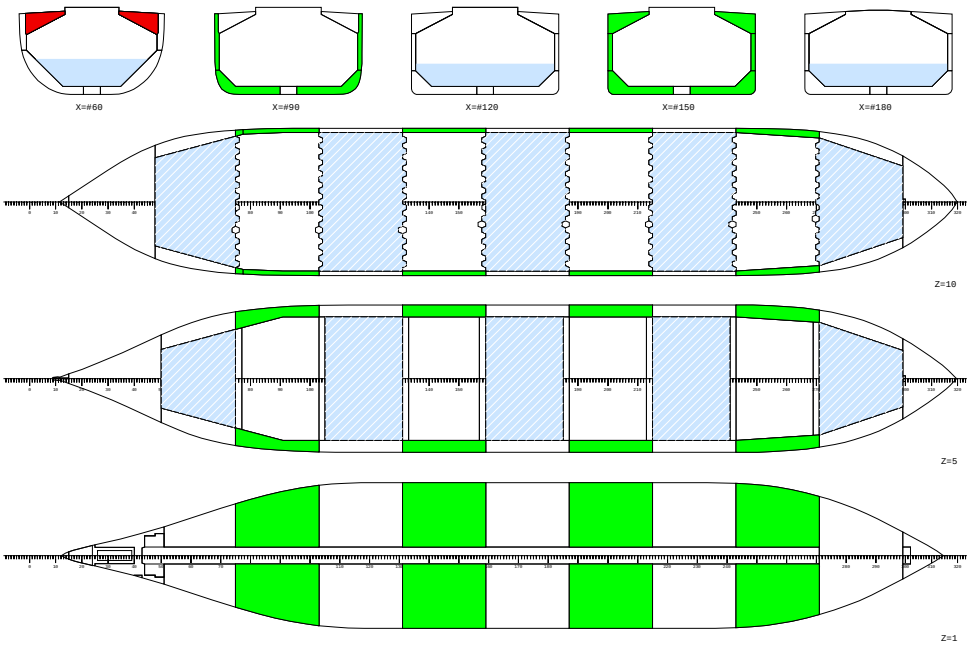
				X	FRAME
SHEAR FORCE (MIN)	-6173.7 t	(44.9%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	3298.1 t	(24.7%)		214.8 m	242
SAGGING MOMENT	-29881.8 tm	(4.7%)		200.8 m	227
HOGGING MOMENT	160103.5 tm	(27.8%)		69.5 m	82

LOADING CONDITION - LSC4C23-05 : LOA. SEQU.LC4LC23 STEP5-GARGO HOLD 9-50% BUNKER









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		90773.3		150.72	0.01	6.39	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

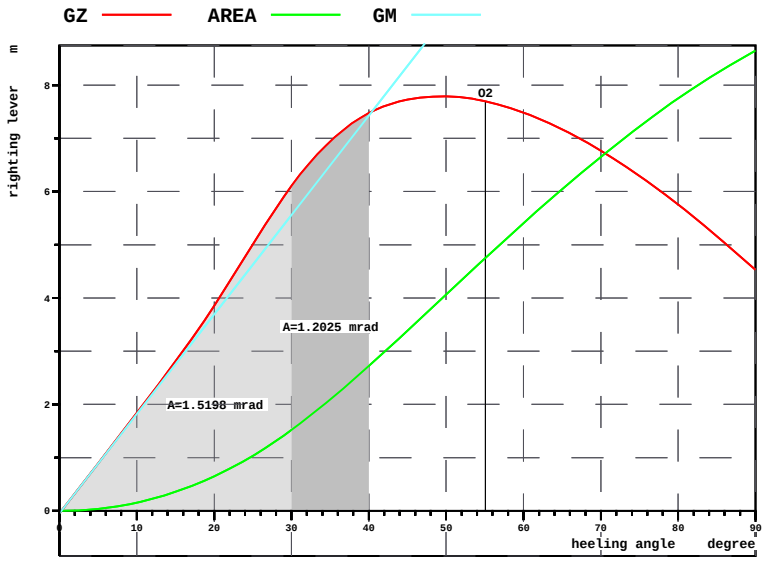
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
SUBTOTAL		24168.0		159.89	-0.00	10.08	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	118178.3	149.53	-0.06 7.58
Total weight	142820.1	146.51	-0.05 8.63

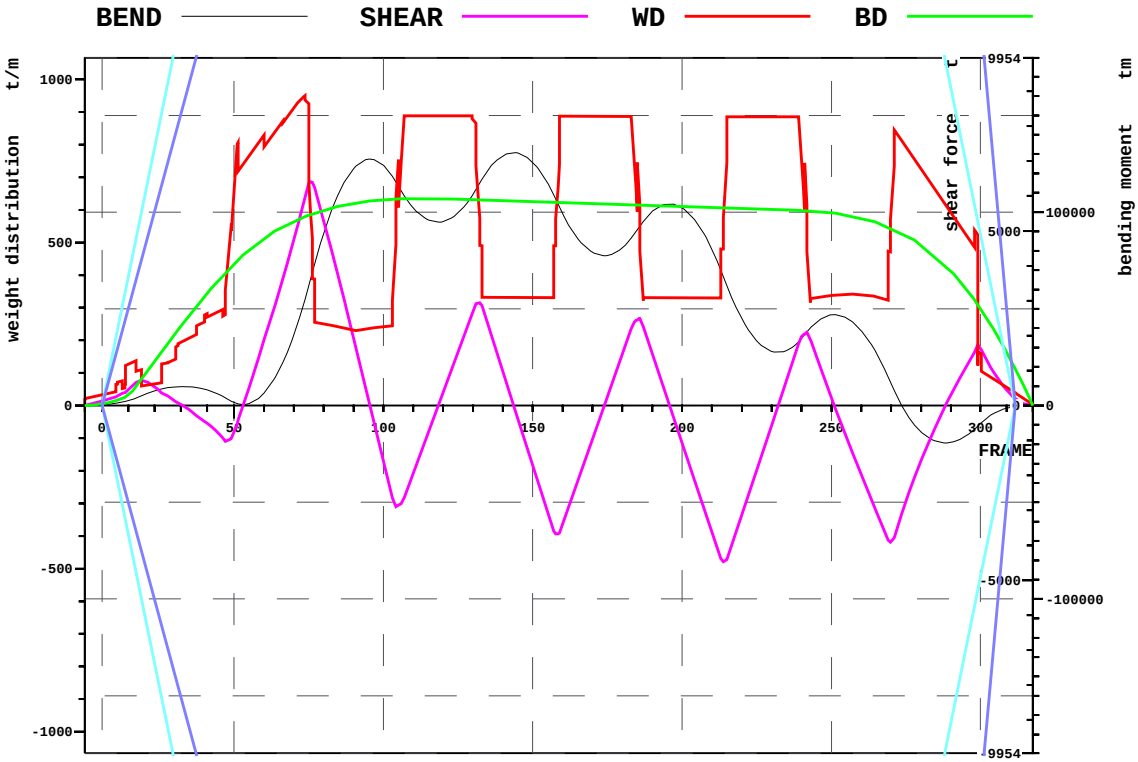
F L O A T I N G P O S I T I O N

Draught moulded	13.481	m	KM	19.34 m
Trim	1.926	m	KG	8.63 m
Heel, PS=+	-0.2	deg		
TA	14.444	m	GM0	10.71 m
TF	12.518	m	GMCORR	0.00 m
Trimming moment	-443566	tonm	GM	10.71 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.89	0.037	273.1	0.002
10.0	1.84	0.156	572.7	0.004
20.0	3.85	0.648	1197.1	0.008
30.0	6.11	1.520	1623.0	0.011
40.0	7.47	2.722	1920.4	0.013
45.0	7.73	3.387	2017.0	0.014
50.0	7.79	4.065	2076.3	0.015
55.0	7.70	4.742	2095.0	0.015
60.0	7.48	5.405	2072.0	0.015
65.0	7.17	6.045	2014.9	0.014
70.0	6.77	6.654	1930.4	0.014
75.0	6.29	7.224	1823.1	0.013
80.0	5.75	7.750	1696.6	0.012
85.0	5.16	8.227	1553.9	0.011
90.0	4.53	8.650	1399.3	0.010



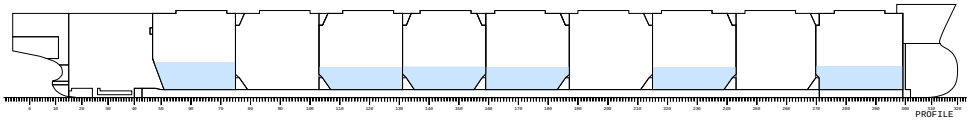
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.520	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.722	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.203	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.789	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	49.391	deg	OK
GM0.15	GM > 0.15 m	0.150	10.710	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.884		OK

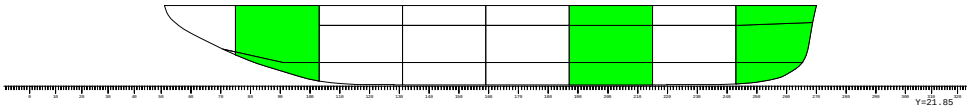


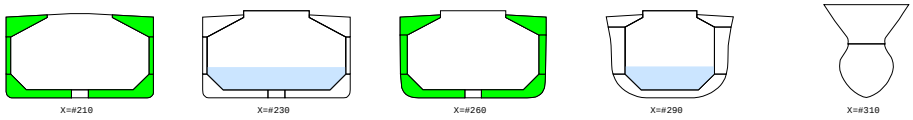
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	375	129	32.89	5.45
8.80	11.00	2910	530	130.44	40.42
12.00	15.00	5010	726	59.73	81.12
37.60	47.00	3989	-1021	354.47	404.85
41.13	51.00	950	-543	800.65	442.11
62.97	75.00	61615	6381	804.20	583.63
88.45	103.00	118463	-2645	323.34	632.41
113.93	131.00	111679	2917	734.74	630.67
139.41	159.00	102277	-3642	739.98	622.59
164.89	187.00	95217	2256	321.72	614.42
190.37	215.00	61901	-4410	816.13	606.23
215.85	243.00	40213	1880	328.12	595.74
241.33	271.00	7824	-3785	844.95	536.23
266.81	299.00	-10640	1741	524.19	308.12
278.01	313.00	-487	217	40.17	117.19

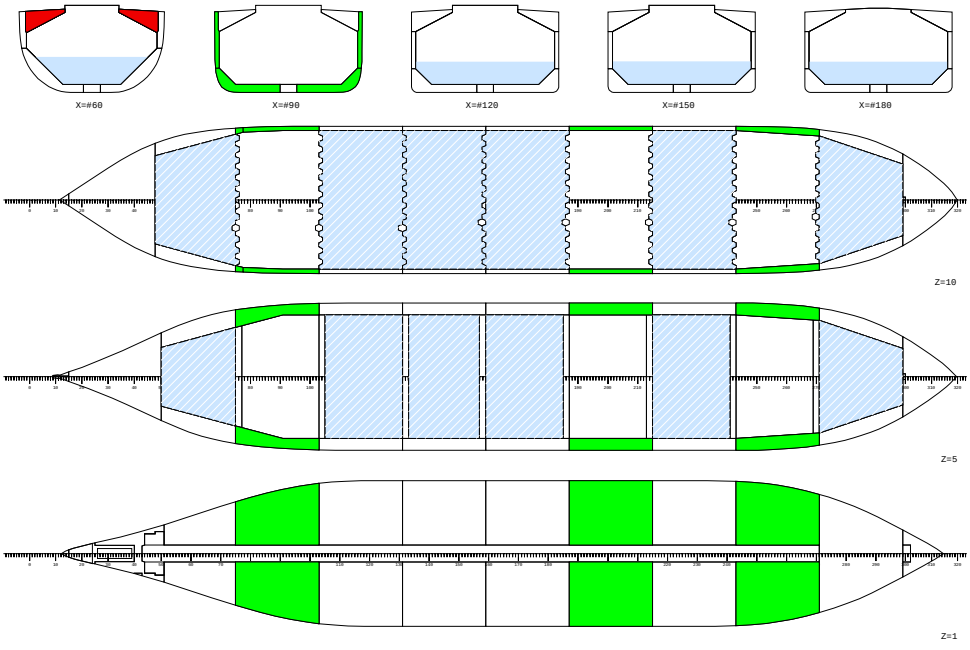
				X	FRAME
SHEAR FORCE (MIN)	-4470.7 t	(33.4%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	6407.7 t	(47.6%)		63.5 m	76
SAGGING MOMENT	-19269.1 tm	(10.8%)		256.5 m	288
HOGGING MOMENT	130756.1 tm	(17.8%)		126.0 m	144

LOADING CONDITION - LSC4C23-06 : LOA. SEQU.LC4LC23 STEP6-GARGO HOLD 6-50% BUNKER









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		109362.1		146.63	0.01	6.37	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

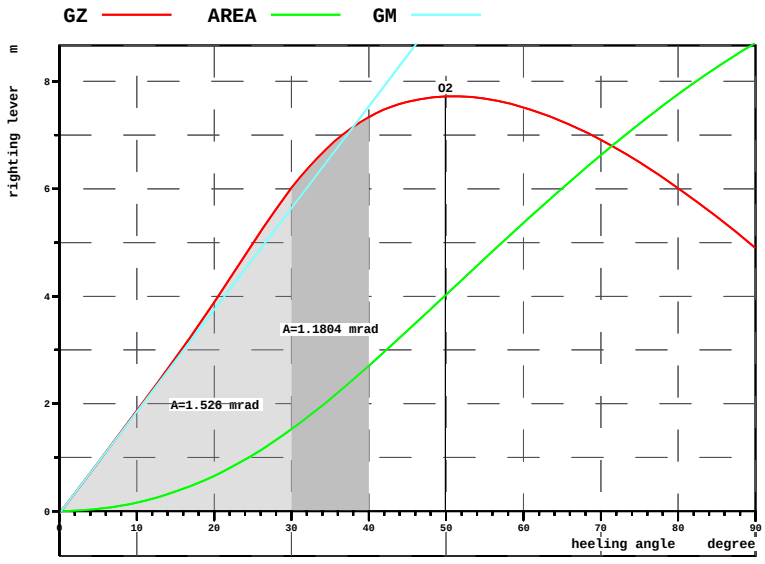
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
SUBTOTAL		17589.8		172.31	-0.00	9.79	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	130188.9	147.42	-0.05 7.24
Total weight	154830.8	144.97	-0.04 8.26

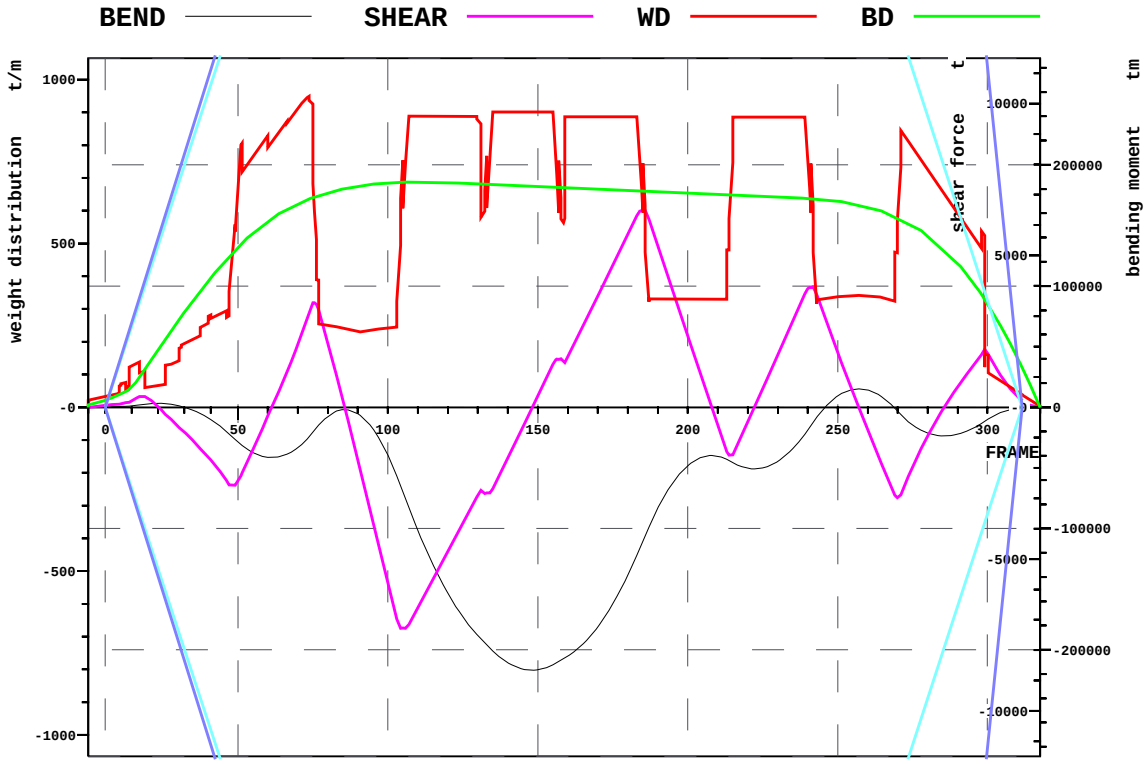
F L O A T I N G P O S I T I O N

Draught moulded	14.508	m	KM	19.11 m
Trim	2.629	m	KG	8.26 m
Heel, PS=+	-0.2	deg		
TA	15.823	m	GM0	10.85 m
TF	13.194	m	GMCORR	0.00 m
Trimming moment	-630876	tonm	GM	10.85 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.91	0.038	273.1	0.002
10.0	1.87	0.159	572.7	0.004
20.0	3.89	0.657	1197.1	0.008
30.0	6.02	1.526	1623.0	0.010
40.0	7.34	2.706	1920.4	0.012
45.0	7.62	3.360	2017.0	0.013
50.0	7.72	4.031	2076.3	0.013
55.0	7.68	4.704	2095.0	0.014
60.0	7.52	5.367	2072.0	0.013
65.0	7.26	6.013	2014.9	0.013
70.0	6.91	6.631	1930.4	0.012
75.0	6.50	7.217	1823.1	0.012
80.0	6.01	7.763	1696.6	0.011
85.0	5.48	8.265	1553.9	0.010
90.0	4.90	8.718	1399.3	0.009



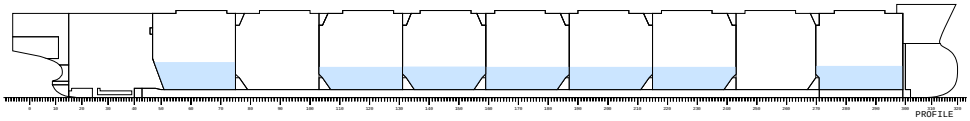
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.526	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.706	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.180	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.722	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	50.893	deg	OK
GM0.15	GM > 0.15 m	0.150	10.855	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.433		OK

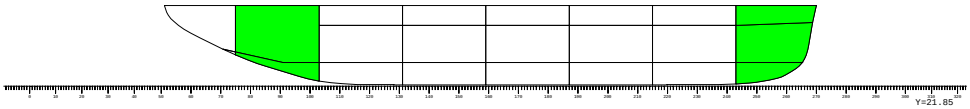


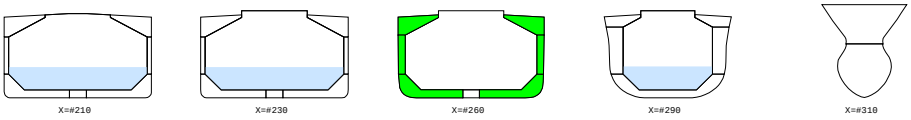
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	207	66	32.89	21.87
8.80	11.00	1354	263	130.46	70.11
12.00	15.00	2430	356	59.73	114.97
37.60	47.00	-21922	-2552	354.47	457.93
41.13	51.00	-30705	-2262	800.66	496.35
62.97	75.00	-20840	3448	804.06	639.43
88.45	103.00	-56681	-6972	323.34	685.79
113.93	131.00	-190786	-2732	865.53	681.08
139.41	159.00	-206804	1472	577.18	670.03
164.89	187.00	-98709	6199	321.72	658.89
190.37	215.00	-45734	-1563	816.14	647.73
215.85	243.00	-8075	3708	328.12	634.23
241.33	271.00	-6042	-2887	844.95	570.06
266.81	299.00	-11608	1917	524.19	328.21
278.01	313.00	-513	237	40.17	124.36

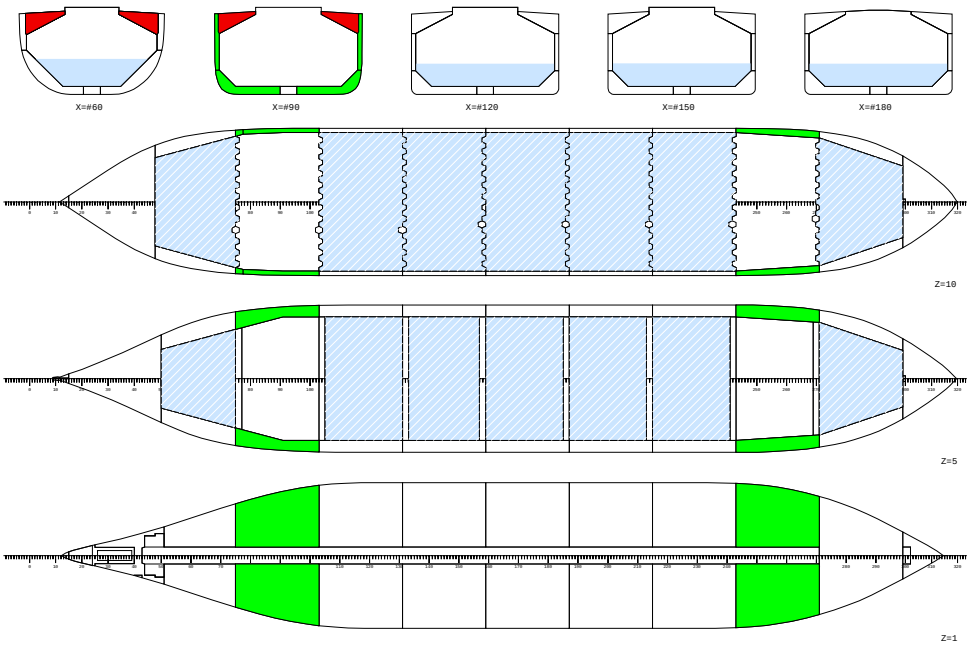
				X	FRAME
SHEAR FORCE (MIN)	-7283.0 t	(53.0%)	POSITION:	91.1 m	106
SHEAR FORCE (MAX)	6490.0 t	(48.6%)		163.7 m	186
SAGGING MOMENT	-216893 tm	(29.9%)		130.0 m	149
HOGGING MOMENT	15210.2 tm	(3.7%)		228.6 m	257

LOADING CONDITION - LSC4C23-07 : LOA. SEQU.LC4LC23 STEP7-GARGO 4-DEPARTURE BUNKE









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		129093.5		151.24	0.01	6.34	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00

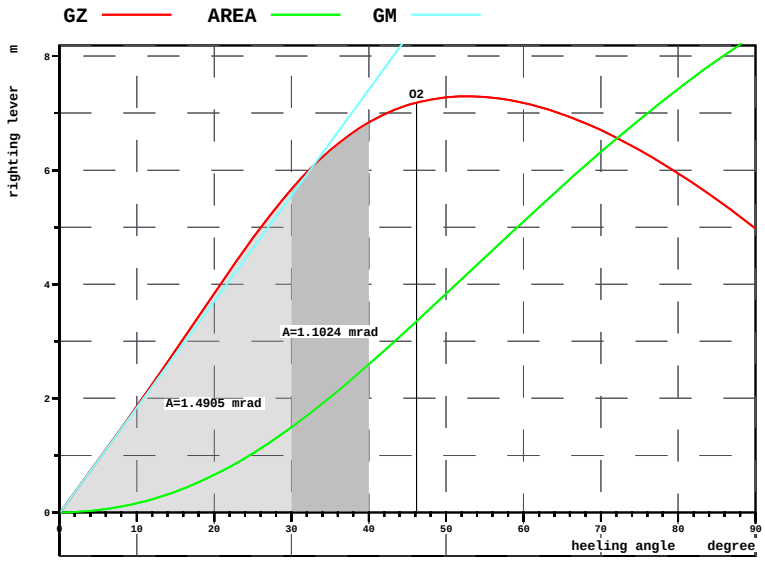
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
SUBTOTAL		11011.7		169.13	-0.00	9.16	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	146290.4	148.32	-0.02 7.25
Total weight	170932.3	145.97	-0.02 8.18

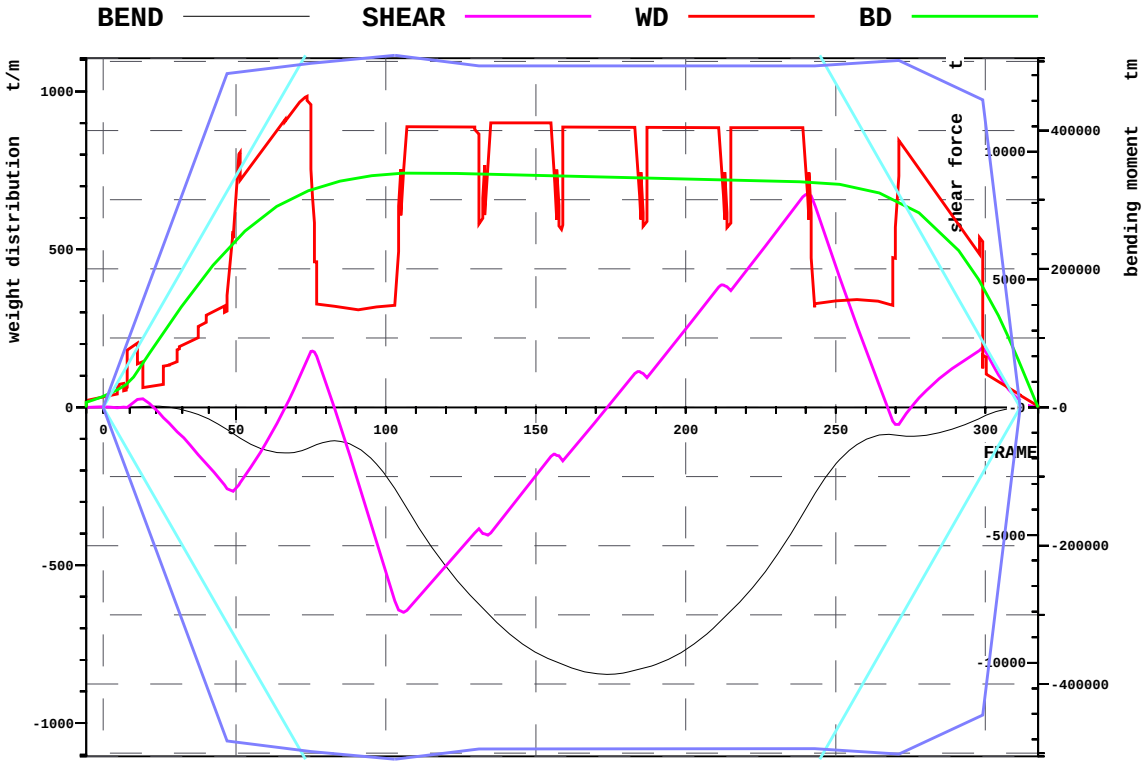
F L O A T I N G P O S I T I O N

Draught moulded	15.871	m	KM	18.83	m
Trim	1.500	m	KG	8.18	m
Heel, PS=+	-0.1	deg			
TA	16.621	m	GM0	10.65	m
TF	15.121	m	GMCORR	0.00	m
Trimming moment	-363239	tonm	GM	10.65	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.91	0.039	0.0	0.000
10.0	1.86	0.160	0.0	0.000
20.0	3.83	0.655	0.0	0.000
30.0	5.67	1.490	0.0	0.000
40.0	6.84	2.593	0.0	0.000
45.0	7.14	3.204	0.0	0.000
50.0	7.28	3.834	0.0	0.000
55.0	7.28	4.470	0.0	0.000
60.0	7.18	5.102	0.0	0.000
65.0	6.98	5.721	0.0	0.000
70.0	6.70	6.318	0.0	0.000
75.0	6.35	6.889	0.0	0.000
80.0	5.94	7.426	0.0	0.000
85.0	5.49	7.925	0.0	0.000
90.0	4.98	8.381	0.0	0.000



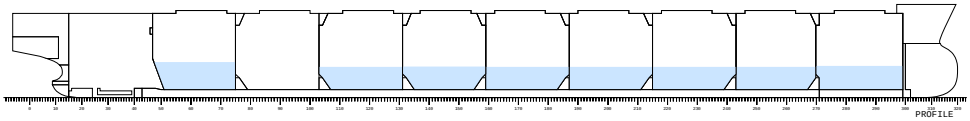
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.490	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.593	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.102	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.295	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	52.765	deg	OK
GM0.15	GM > 0.15 m	0.150	10.648	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.242		OK

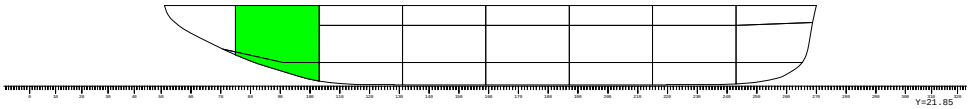


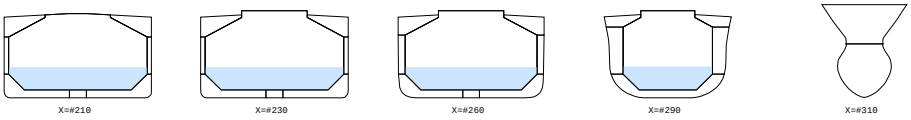
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	70	11	32.89	34.75
8.80	11.00	214	156	192.89	91.20
12.00	15.00	1124	334	62.46	138.83
37.60	47.00	-32288	-3190	354.79	498.08
41.13	51.00	-43577	-3044	800.98	538.07
62.97	75.00	-57802	2190	856.31	687.76
88.45	103.00	-117505	-7575	323.34	739.27
113.93	131.00	-284862	-4759	865.54	739.34
139.41	159.00	-371941	-2100	577.19	733.06
164.89	187.00	-375707	1163	588.04	726.70
190.37	215.00	-294529	4564	885.81	720.31
215.85	243.00	-124850	7925	328.12	711.50
241.33	271.00	-40825	-673	844.96	648.17
266.81	299.00	-13916	2353	524.20	381.12
278.01	313.00	-573	269	40.17	139.60

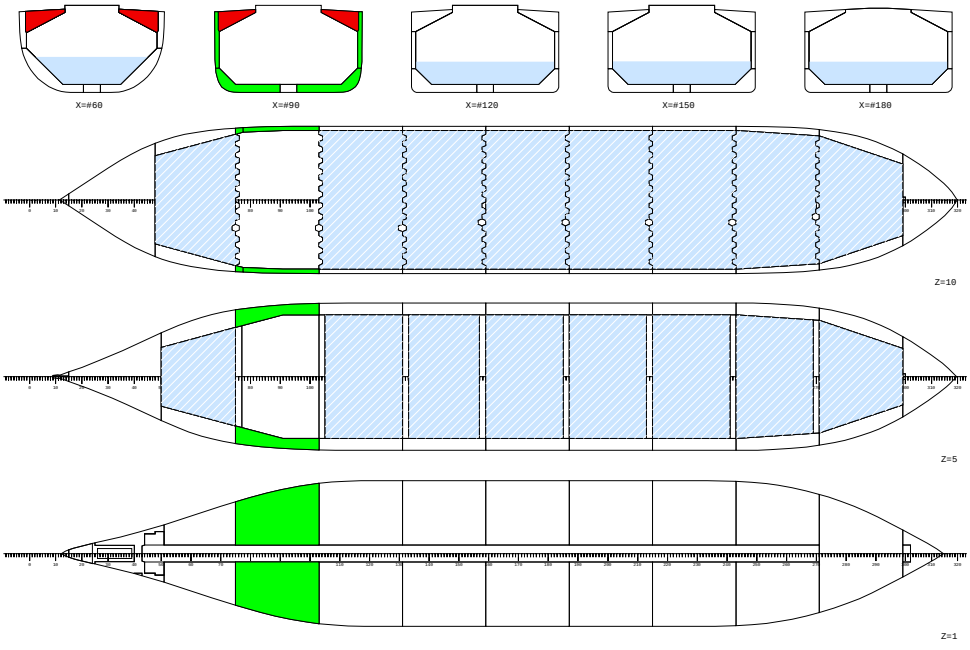
				X	FRAME
SHEAR FORCE (MIN)	-8027.2 t	(58.4%)	POSITION:	91.1 m	106
SHEAR FORCE (MAX)	8331.3 t	(62.3%)		213.1 m	240
SAGGING MOMENT	-386089 tm	(53.3%)		152.8 m	174
HOGGING MOMENT	1629.4 tm	(1.4%)		14.1 m	18

LOADING CONDITION - LSC4C23-08 : LOA. SEQU.LC4LC23 STEP8-GARGO 2-DEPARTURE BUNKE









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.2	NO.2 CARGO HOLD	19099.3	30.0	227.55	0.01	6.22	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		148192.8		161.07	0.01	6.33	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00

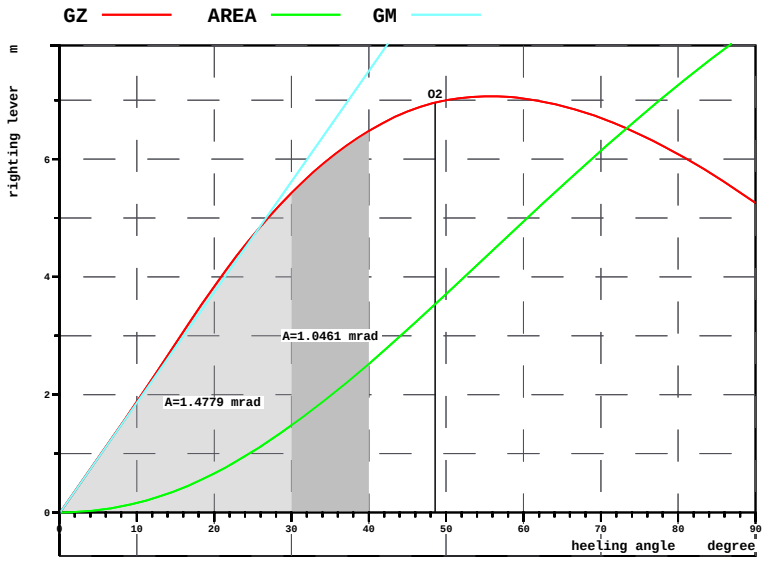
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
SUBTOTAL		4275.7		75.52	-0.00	6.08	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	158653.7	154.46	-0.02 6.96
Total weight	183295.6	151.44	-0.02 7.87

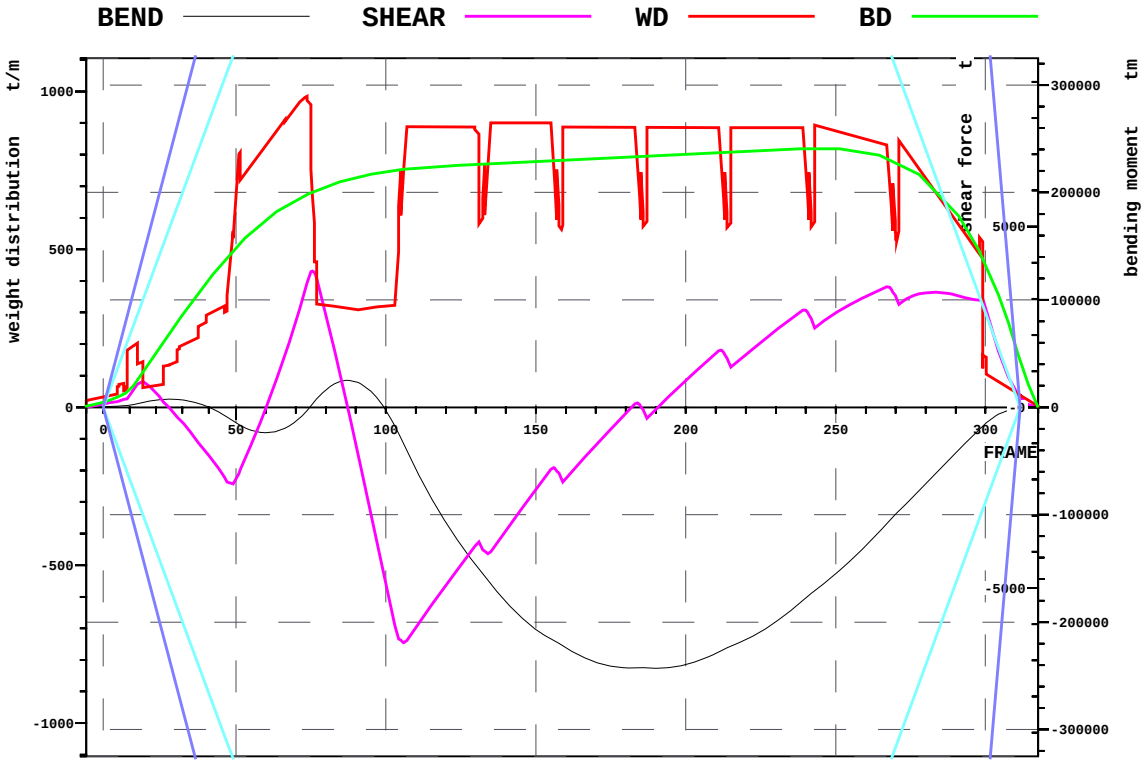
F L O A T I N G P O S I T I O N

Draught moulded	16.924	m	KM	18.62 m
Trim	-3.053	m	KG	7.87 m
Heel, PS=+	-0.1	deg		
TA	15.397	m	GM0	10.75 m
TF	18.451	m	GMCORR	0.00 m
Trimming moment	723484	tonm	GM	10.75 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.93	0.040	0.0	0.000
10.0	1.88	0.162	0.0	0.000
20.0	3.84	0.662	0.0	0.000
30.0	5.43	1.478	0.0	0.000
40.0	6.48	2.524	0.0	0.000
45.0	6.81	3.105	0.0	0.000
50.0	7.00	3.708	0.0	0.000
55.0	7.06	4.322	0.0	0.000
60.0	7.03	4.938	0.0	0.000
65.0	6.90	5.546	0.0	0.000
70.0	6.70	6.140	0.0	0.000
75.0	6.42	6.713	0.0	0.000
80.0	6.08	7.259	0.0	0.000
85.0	5.69	7.773	0.0	0.000
90.0	5.26	8.251	0.0	0.000



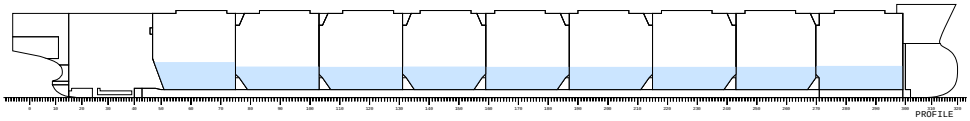
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.478	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.524	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.046	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.065	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	55.667	deg	OK
GM0.15	GM > 0.15 m	0.150	10.754	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.516		OK

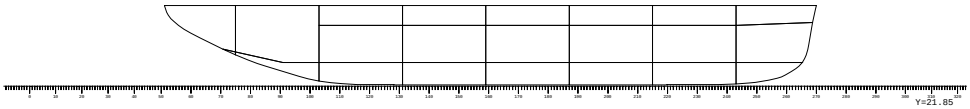


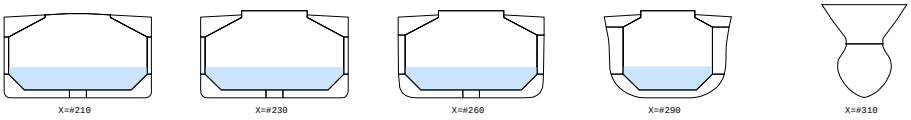
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	404	92	32.89	15.92
8.80	11.00	2320	441	192.87	64.29
12.00	15.00	4363	709	62.46	110.14
37.60	47.00	-9121	-2071	354.75	472.47
41.13	51.00	-16217	-1838	800.90	514.31
62.97	75.00	969	3756	856.22	678.93
88.45	103.00	-17416	-6029	323.31	749.67
113.93	131.00	-150240	-3725	865.45	769.03
139.41	159.00	-222122	-2071	577.13	782.04
164.89	187.00	-242551	-305	587.98	794.98
190.37	215.00	-222423	1108	885.72	807.88
215.85	243.00	-170593	2194	587.72	818.24
241.33	271.00	-96624	2846	558.23	768.31
266.81	299.00	-17402	3010	524.14	469.53
278.01	313.00	-735	287	40.16	153.16

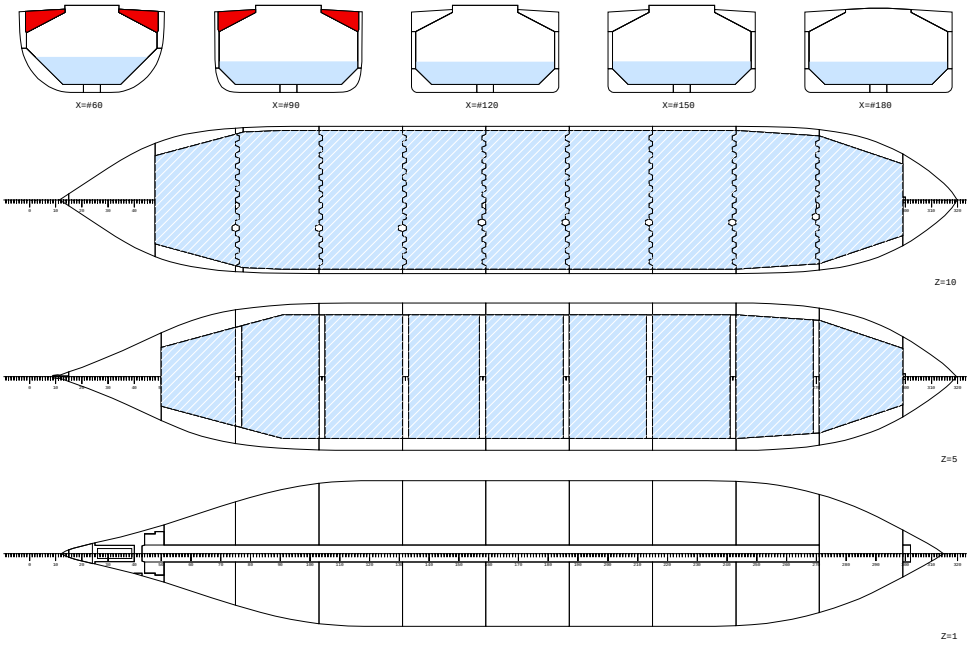
				X	FRAME
SHEAR FORCE (MIN)	-6511.8 t	(47.4%)	POSITION:	91.2 m	106
SHEAR FORCE (MAX)	3771.1 t	(28.0%)		63.5 m	76
SAGGING MOMENT	-242964 tm	(33.5%)		167.6 m	190
HOGGING MOMENT	25054.9 tm	(4.1%)		73.8 m	87

LOADING CONDITION - LSC4C23-09 : LOA. SEQU.LC4LC23 STEP9-GARGO 8-DEPARTURE BUNKE









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.2	NO.2 CARGO HOLD	19099.3	30.0	227.55	0.01	6.22	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.8	NO.8 CARGO HOLD	19407.5	30.0	76.84	0.02	6.27	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		167600.3		151.32	0.01	6.32	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

CONTENTS= (RHO=1)

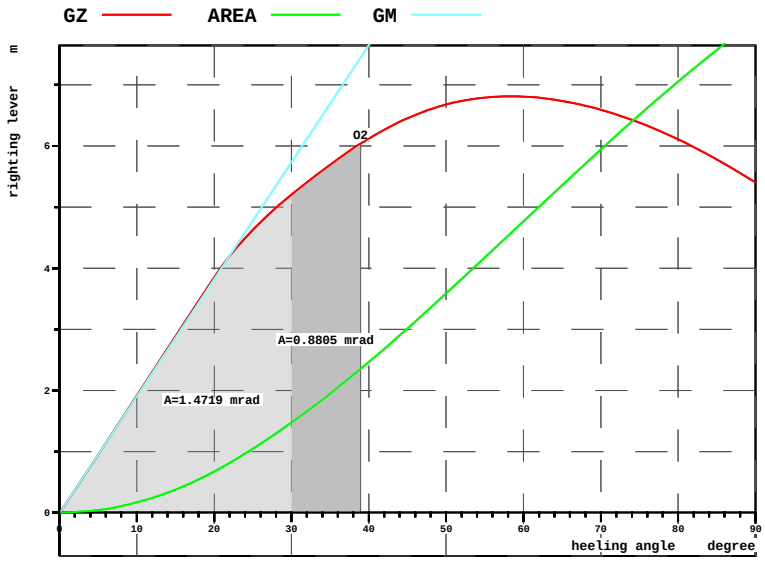
(OIL&WAT.	200.0	0.0	23.10	0.00	22.00	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	173785.5	147.73	-0.01 6.91
Total weight	198427.4	145.78	-0.01 7.75

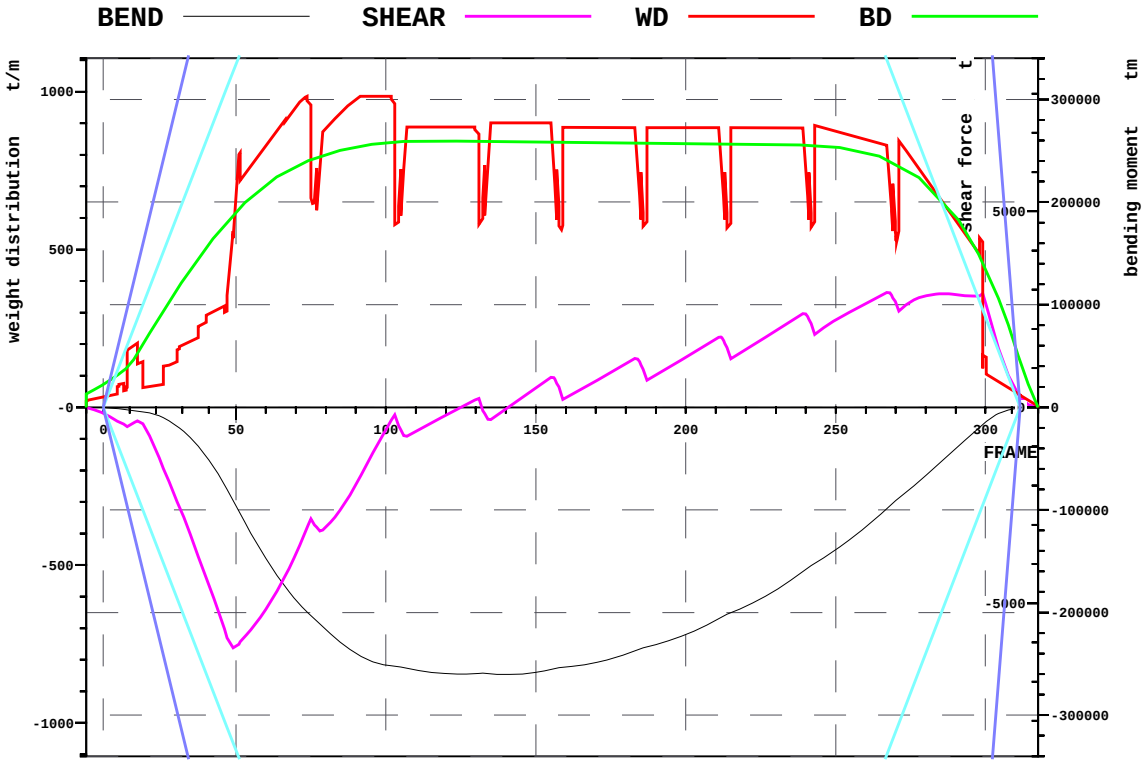
F L O A T I N G P O S I T I O N

Draught moulded	18.183	m	KM	18.73	m
Trim	0.710	m	KG	7.75	m
Heel, PS=+	-0.1	deg			
TA	18.538	m	GM0	10.98	m
TF	17.828	m	GMCORR	0.00	m
Trimming moment	-176768	tonm	GM	10.98	m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.01	0.000	0.0	0.000
5.0	0.95	0.041	0.0	0.000
10.0	1.92	0.166	0.0	0.000
20.0	3.85	0.670	0.0	0.000
30.0	5.20	1.472	0.0	0.000
40.0	6.12	2.464	0.0	0.000
45.0	6.45	3.013	0.0	0.000
50.0	6.68	3.587	0.0	0.000
55.0	6.79	4.175	0.0	0.000
60.0	6.81	4.769	0.0	0.000
65.0	6.74	5.361	0.0	0.000
70.0	6.59	5.943	0.0	0.000
75.0	6.38	6.510	0.0	0.000
80.0	6.11	7.055	0.0	0.000
85.0	5.78	7.574	0.0	0.000
90.0	5.40	8.062	0.0	0.000



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.472	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.352	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	0.880	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.812	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	58.416	deg	OK
GM0.15	GM > 0.15 m	0.150	10.983	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	3.507		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-299	-155	32.89	72.08
8.80	11.00	-3182	-416	192.87	145.70
12.00	15.00	-4357	-421	62.46	198.76
37.60	47.00	-80111	-5880	354.76	585.11
41.13	51.00	-101404	-6045	800.92	627.22
62.97	75.00	-202928	-2850	810.94	784.08
88.45	103.00	-252326	-188	578.50	839.52
113.93	131.00	-259342	221	865.47	842.94
139.41	159.00	-253278	196	577.14	840.01
164.89	187.00	-233378	689	588.00	837.00
190.37	215.00	-200188	1235	885.74	833.96
215.85	243.00	-152213	1860	587.73	828.37
241.33	271.00	-88667	2456	558.24	762.87
266.81	299.00	-16633	2900	524.16	456.20
278.01	313.00	-651	284	40.16	150.70

				X	FRAME
SHEAR FORCE (MIN)	-6133.3 t	(46.5%)	POSITION:	39.4 m	49
SHEAR FORCE (MAX)	2927.2 t	(21.6%)		237.7 m	267
SAGGING MOMENT	-260536 tm	(36.0%)		122.3 m	140
HOGGING MOMENT	2.7 tm			-5.0 m	-6

12. Loading/unloading sequence: From cond. No25 to cond. No2

LOADING/UNLOADING SEQUENCE SUMMARY FORM

Vessel name	Voyage No.	Condition	Yard	Id. Number

Port (specific or typical):	Condition at commencement of laoding/discharging:	
Total mass of cargo to be loaded/discharged:	Condition at end of loading/discharging:	
Dock water density (t/m^3):	Maximum loading/discharging rate:	Average Loading/discharging rate:
Number of loaders/dischargers:	Maximum Ballasting/Deballasting rate:	Average Ballasting/Deballasting rate:

Hold No.	9	8	7	6	5	4	3	2	1
volume of Hold, V(m^3)	18190	21557	21935	21647	21915	21917	21916	21214	16870
Height of hold, h (m)	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2	24.2

Note:During each pour it has to be controlled that allowable limits for hull girder shear force,bending moment and mass in holds are not exceeded.
Loading/discharging operations may have to be paused to allow for ballasting/deballasting in order to keep actual values within limits

Hold content at commencement of loading/discharging.

Ballast contents at commencement of loading/discharging.

Commencement of loading/discharging (sea).

Cargo mass (t)	16376	19408	19748	18589	19730	19731	197308	19099	15188
Density, (t/m^3)	2.99	2.99	2.99	2.99	2.99	2.99	2.99	2.99	2.99
Grade									

Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT
Top side	0	0	0	0	0	0	0	0	0	0	0
Double side	0	0	0	0	0	0	0	0	0	0	0
Double bottom	0	0	0	0	0	0	0	0	0	0	0

d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
18.14	0.20	17.94	49.60%	36.70%

CARGO OPERATIONS — BALLASTING OPERATIONS

Values at end of pour (from harbour to sea).

Pour No./ Grade	Hold 9	Hold 8	Hold 7	Hold 6	Hold 5	Hold 4	Hold 3	Hold 2	Hold 1	Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT	d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
1							19099.3t To 0% of Hold			Top side									100%			18.38	3.39	14.99	64.50%	63.00%
										Double side									100%							
										Double bottom									100%							
2		19408t To 0% of Hold								Top side			100%									15.22	-0.40	15.62	68.80%	60.50%
										Double side			100%													
										Double bottom			100%													
3						19731t To 0% of Hold				Top side							100%					15.13	1.65	13.48	56.70%	35.50%
										Double side							100%									
										Double bottom							100%									
4				18588.8t To 0% of Hold						Top side					100%							14.44	1.92	12.52	47.60%	17.80%
										Double side					100%											
										Double bottom					100%											
5	16376.3t To 0% of Hold									Top side	100%	100%										11.55	-1.96	13.51	44.90%	27.80%
										Double side	100%	100%														
										Double bottom	100%	100%														
6									15188.2t To 0% of Hold	Top side										100%		12.95	2.06	10.89	58.40%	29.00%
										Double side										100%						
										Double bottom										100%						
7					19730t To 0% of Hold					Top side						100%						12.84	3.57	9.27	54.60%	26.60%
										Double side						100%										
										Double bottom						100%										
8			19748t To 0% of Hold							Top side				100%								10.13	0.68	9.45	63.50%	50.90%
										Double side				100%												
										Double bottom				100%												
9								19730.8t To 0% of Hold		Top side								100%			100%	9.98	2.02	7.96	43.20%	63.10%
										Double side								100%			100%					
										Double bottom								100%			100%					

Total cargo onboard (t): 167600.3 **Remaining cargo to be loaded (t): 0**

Total amount of bunkers onboard (t): 5975.2 (100%)

Draft Survey n-1										Top side																
										Double side																
										Double bottom																
Draft Survey n										Top side																
										Double side																
										Double bottom																

Hold content at end of loading/discharging.

Ballast contents at end of loading/discharging.

Values at end of loading/discharging (sea).

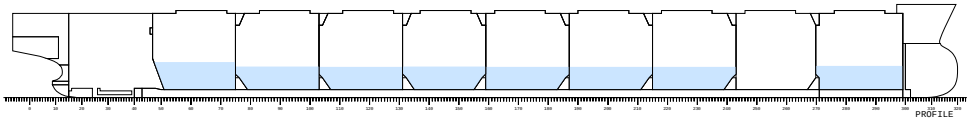
Cargo mass (t)											Wings or peaks:	APT+No.10	No.9 B.T.	No.8 B.T.	No.7 B.T.	No.6 B.T.	No.5 B.T.	No.4 B.T.	No.3 B.T.	No.2 B.T.	No.1 B.T.	FPT	d_{aft} (m)	Trim (m)	d_{fwd} (m)	Maximum S.F.(%)	Maximum B.M.(%)
											Top side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	9.98	2.02	7.96	53.80%	98.20%
											Double side	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%					
											Double bottom	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%					

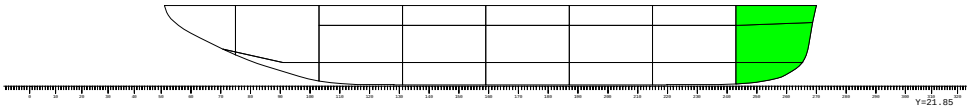
Total mass loaded/discharged (t): 167600.3

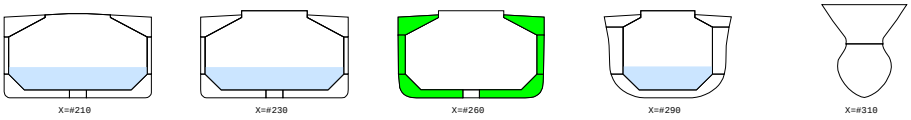
Net load on Double Bottom:	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	-8.96	tons/m^2
Net load in two adjacent holds:	0	0	0	0	0	0	0	0	0	tons

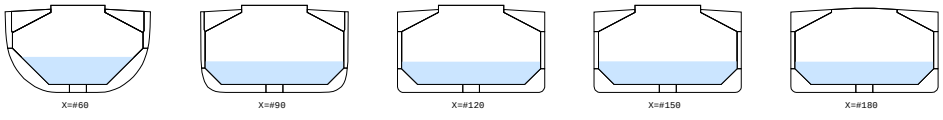
Net load on double bottom = (Mh/V)×h-T[t/m^2]
where: Mh = Mass in hold + mass in DB (t)
V = Total volume of hold (m^3) T = draught (m)
h = height of hold from inner bottom to top of coaming (m)

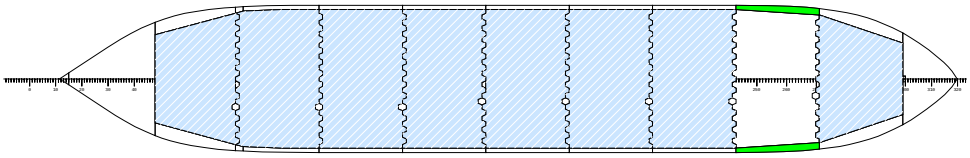
LOADING CONDITION - LSC25C2-01 : LOA. SEQU.LC25LC2 STEP1-GARGO 2 EMPTY-ARR. BUN.

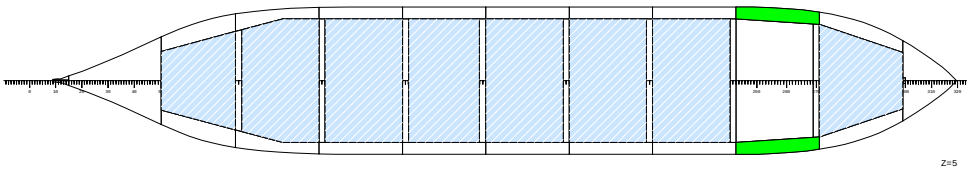


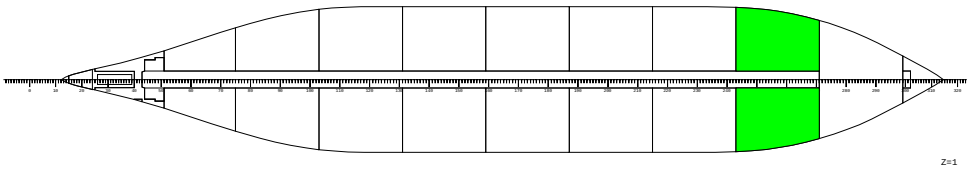












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.8	NO.8 CARGO HOLD	19407.5	30.0	76.84	0.02	6.27	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		148501.0		141.51	0.01	6.33	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61

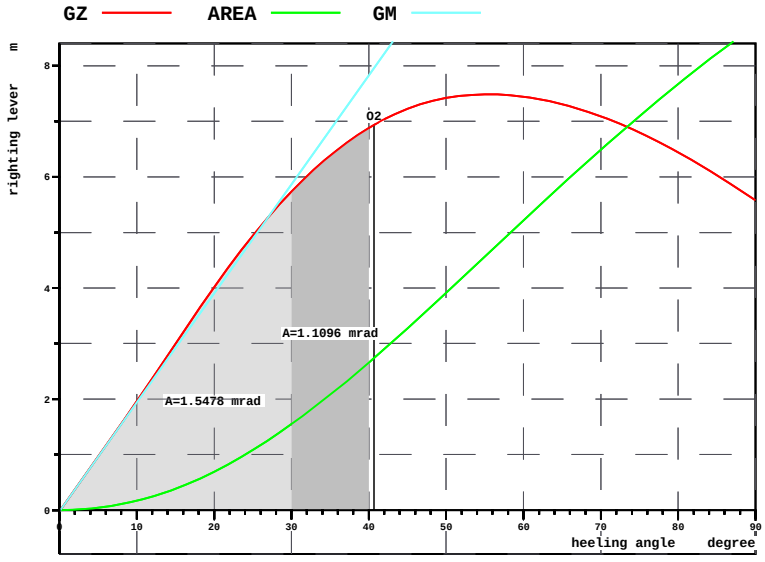
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
SUBTOTAL		6736.0		228.54	-0.00	11.11	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	156105.0	144.65	-0.03 6.63
Total weight	180746.9	142.93	-0.02 7.59

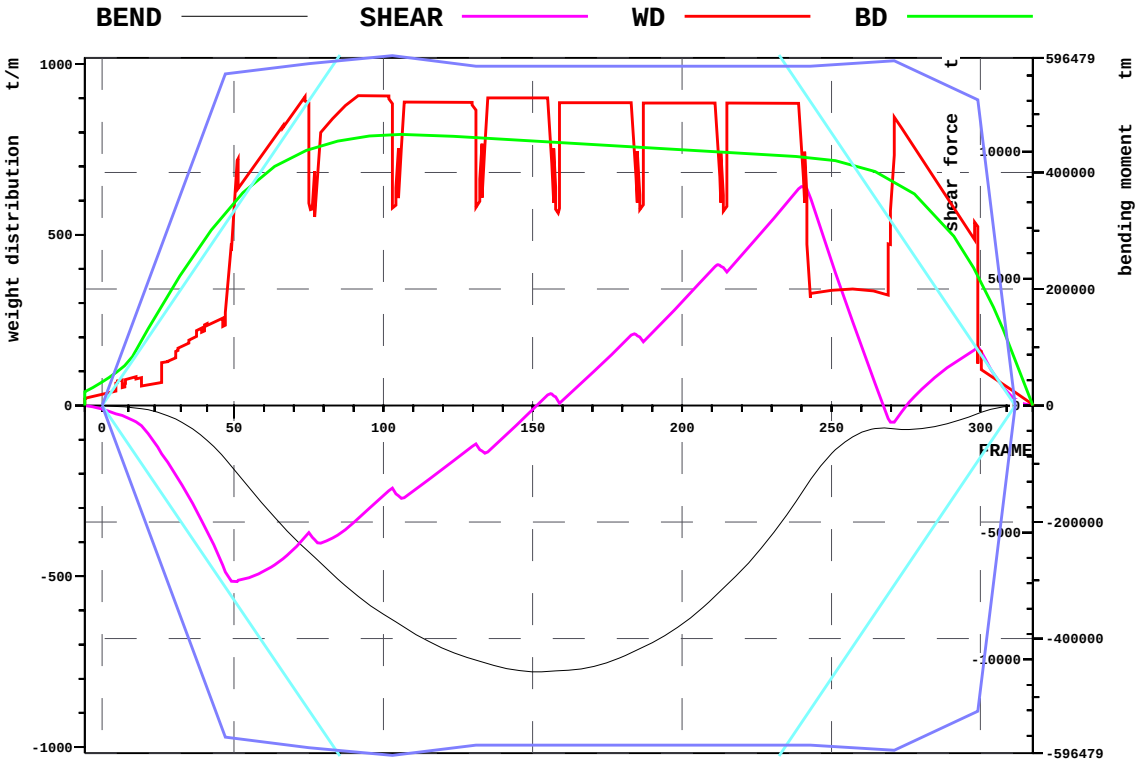
F L O A T I N G P O S I T I O N

Draught moulded	16.685	m	KM	18.84 m
Trim	3.398	m	KG	7.59 m
Heel, PS=+	-0.1	deg		
TA	18.384	m	GM0	11.25 m
TF	14.986	m	GMCORR	0.00 m
Trimming moment	-843685	tonm	GM	11.25 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.02	0.000	0.0	0.000
5.0	0.96	0.041	64.4	0.000
10.0	1.95	0.168	128.6	0.001
20.0	4.01	0.689	221.7	0.001
30.0	5.74	1.548	296.1	0.002
40.0	6.88	2.657	401.9	0.002
45.0	7.23	3.274	456.9	0.003
50.0	7.42	3.914	503.3	0.003
55.0	7.49	4.566	540.1	0.003
60.0	7.44	5.218	568.2	0.003
65.0	7.31	5.862	588.6	0.003
70.0	7.09	6.491	603.4	0.003
75.0	6.80	7.098	614.6	0.003
80.0	6.44	7.676	620.8	0.003
85.0	6.03	8.221	620.0	0.003
90.0	5.58	8.727	613.4	0.003



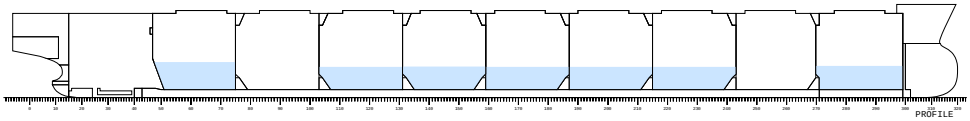
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.548	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.657	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.110	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.487	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	55.394	deg	OK
GM0.15	GM > 0.15 m	0.150	11.249	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.081		OK

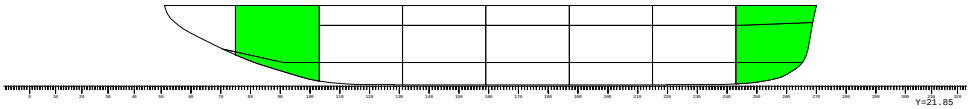


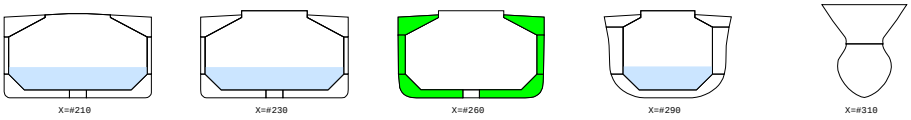
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	-366	-142	32.89	68.88
8.80	11.00	-3276	-534	80.62	138.74
12.00	15.00	-5370	-796	57.00	190.21
37.60	47.00	-91661	-6552	273.11	562.72
41.13	51.00	-115761	-6926	717.37	602.95
62.97	75.00	-251090	-5000	739.70	749.06
88.45	103.00	-367531	-3249	578.55	792.91
113.93	131.00	-436609	-1504	865.54	784.92
139.41	159.00	-454899	96	577.19	770.58
164.89	187.00	-414240	2505	588.05	756.17
190.37	215.00	-307773	5257	885.81	741.73
215.85	243.00	-126713	8175	328.12	724.85
241.33	271.00	-39952	-658	844.96	653.28
266.81	299.00	-13672	2334	524.20	379.52
278.01	313.00	-526	268	40.17	138.80

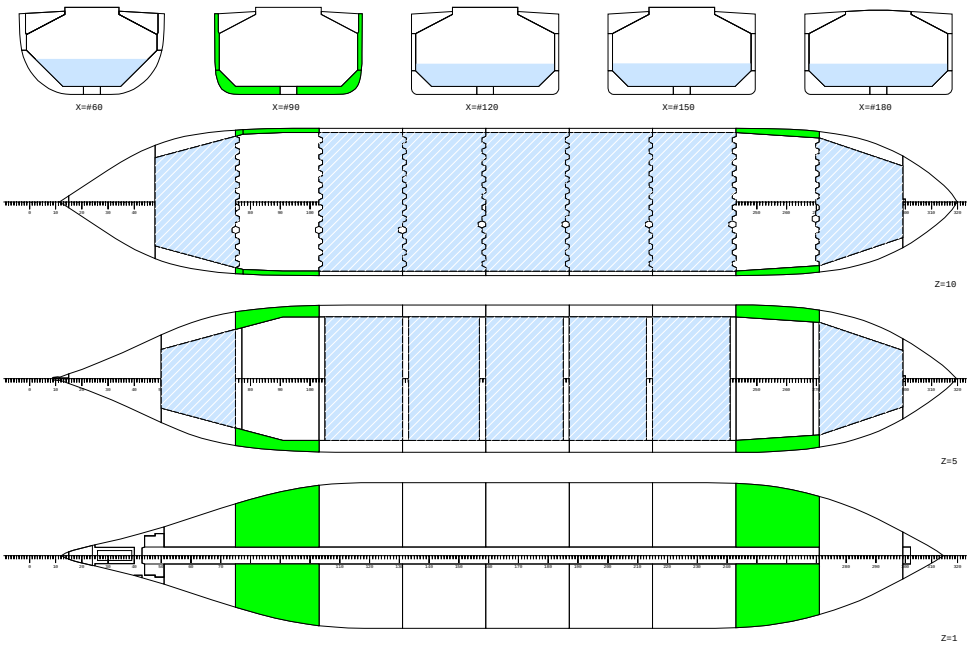
				X	FRAME
SHEAR FORCE (MIN)	-6926.1 t	(52.5%)	POSITION:	41.1 m	51
SHEAR FORCE (MAX)	8619.2 t	(64.5%)		213.1 m	240
SAGGING MOMENT	-456722 tm	(63.0%)		133.9 m	153
HOGGING MOMENT	0.4 tm			283.5 m	320

LOADING CONDITION - LSC25C2-02 : LOA. SEQU.LC25LC2 STEP2-GARGO 8 EMPTY-ARR. BUN.









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.4	NO.4 CARGO HOLD	19731.4	30.0	176.78	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		129093.5		151.24	0.01	6.34	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

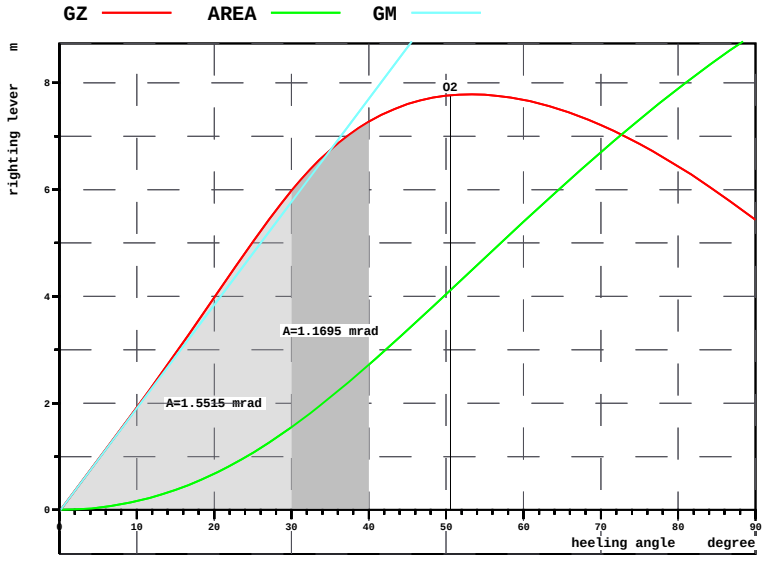
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
SUBTOTAL		11011.7		169.13	-0.00	9.16	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	140973.2	151.89	-0.03 6.67
Total weight	165615.1	148.94	-0.03 7.71

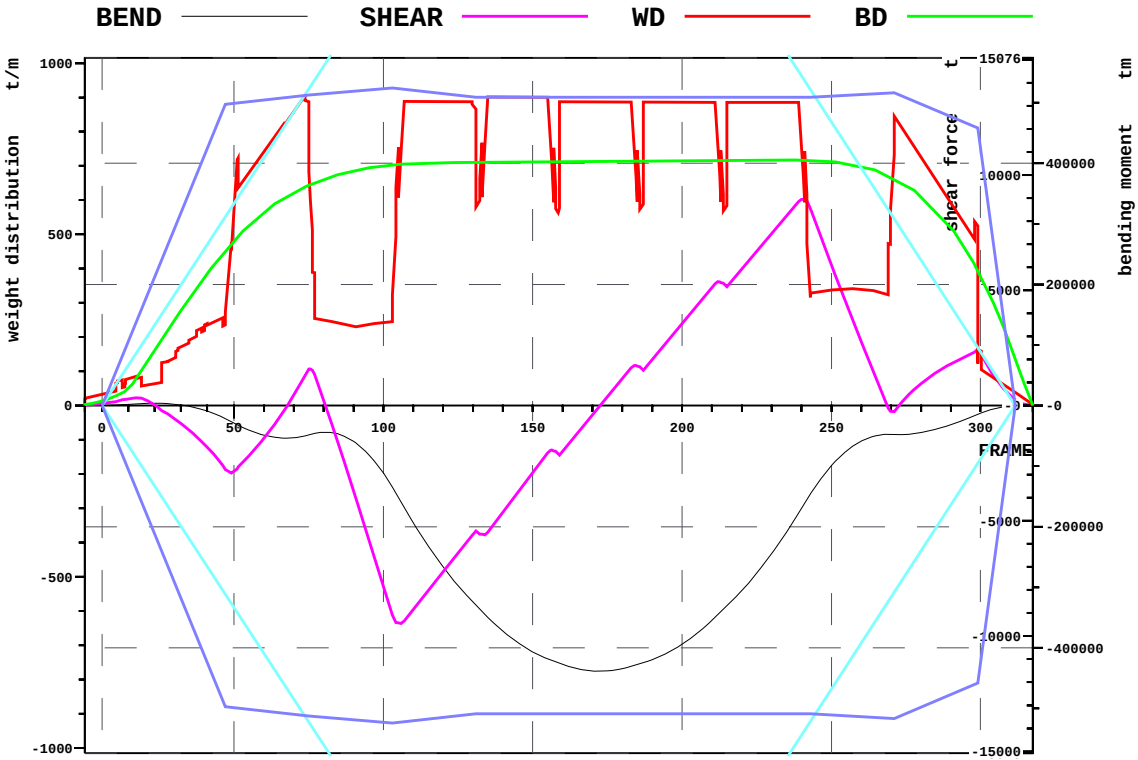
F L O A T I N G P O S I T I O N

Draught moulded	15.421	m	KM	18.78 m
Trim	-0.399	m	KG	7.71 m
Heel, PS=+	-0.1	deg		
TA	15.222	m	GM0	11.07 m
TF	15.620	m	GMCORR	0.00 m
Trimming moment	93376	tonm	GM	11.07 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.94	0.040	64.4	0.000
10.0	1.92	0.165	128.6	0.001
20.0	3.98	0.678	221.7	0.001
30.0	5.98	1.551	296.1	0.002
40.0	7.27	2.721	401.9	0.002
45.0	7.60	3.371	456.9	0.003
50.0	7.76	4.043	503.3	0.003
55.0	7.78	4.721	540.1	0.003
60.0	7.68	5.397	568.2	0.003
65.0	7.49	6.059	588.6	0.004
70.0	7.21	6.701	603.4	0.004
75.0	6.86	7.315	614.6	0.004
80.0	6.44	7.895	620.8	0.004
85.0	5.96	8.437	620.0	0.004
90.0	5.43	8.934	613.4	0.004



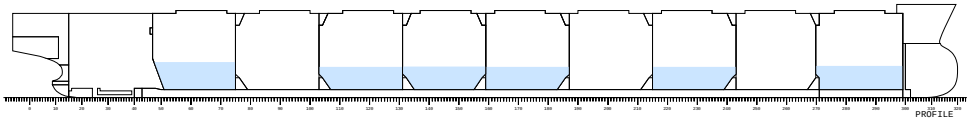
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.551	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.721	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.170	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.785	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	53.183	deg	OK
GM0.15	GM > 0.15 m	0.150	11.074	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.411		OK

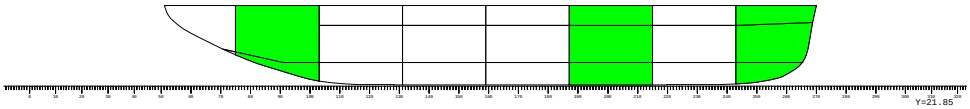


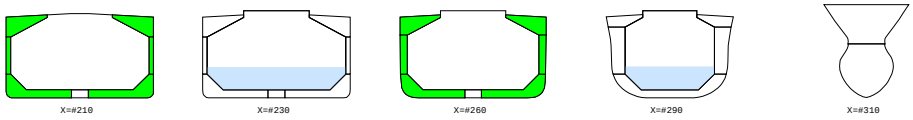
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	305	100	32.89	13.66
8.80	11.00	2082	310	80.51	58.30
12.00	15.00	3146	317	57.00	102.50
37.60	47.00	-21614	-2770	273.11	449.98
41.13	51.00	-31627	-2746	717.35	489.80
62.97	75.00	-48818	1590	784.99	643.29
88.45	103.00	-134581	-9092	323.34	702.51
113.93	131.00	-329462	-5441	865.52	710.62
139.41	159.00	-425427	-2153	577.18	712.39
164.89	187.00	-424667	1533	588.03	714.08
190.37	215.00	-330776	5153	885.79	715.73
215.85	243.00	-145420	8528	328.12	714.96
241.33	271.00	-47958	-258	844.95	659.09
266.81	299.00	-14420	2447	524.19	392.88
278.01	313.00	-590	273	40.17	142.29

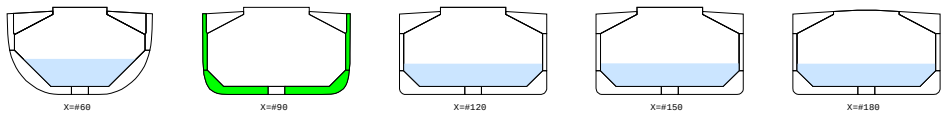
				X	FRAME
SHEAR FORCE (MIN)	-9448.4 t	(68.8%)	POSITION:	91.1 m	106
SHEAR FORCE (MAX)	8942.6 t	(66.9%)		213.1 m	240
SAGGING MOMENT	-438545 tm	(60.5%)		152.8 m	174
HOGGING MOMENT	3893.0 tm	(2.8%)		16.9 m	21

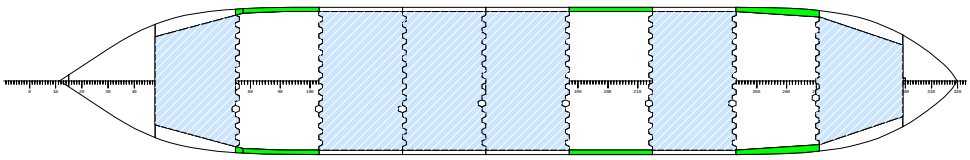
LOADING CONDITION - LSC25C2-03 : LOA. SEQU.LC25LC2 STEP3-GARGO 4 EMPTY-ARR. BUN.

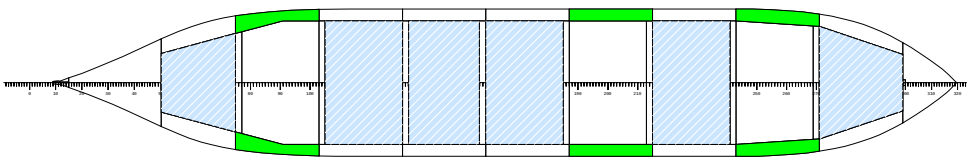


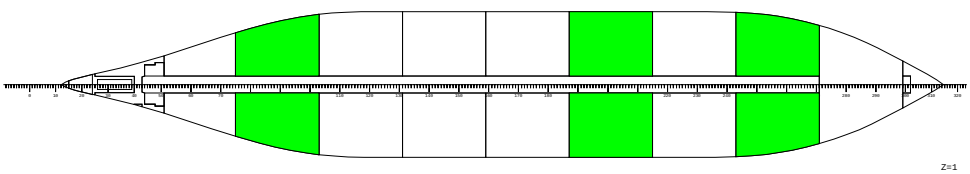












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.6	NO.6 CARGO HOLD	18588.8	30.0	126.65	0.01	6.30	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		109362.1		146.63	0.01	6.37	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		190.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	7.7	29.0	31.20	13.70	20.25	4.57
SUBTOTAL		34.1		29.96	13.73	21.83	4.57
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	30.8	10.0	9.68	10.50	19.02	305.51
R5.1S	F.W.TK.(S)	20.0	10.0	8.84	-10.34	19.02	182.21
SUBTOTAL		50.8		9.35	2.31	19.02	487.72
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.5	NO.5 H.F.O.TK	255.5	48.0	30.23	-15.97	20.10	247.61
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		493.1		31.23	-13.90	20.94	247.61
CONTENTS= (RHO=1)							
(OIL&WAT.		100.0	0.0	23.10	0.00	22.00	0.00

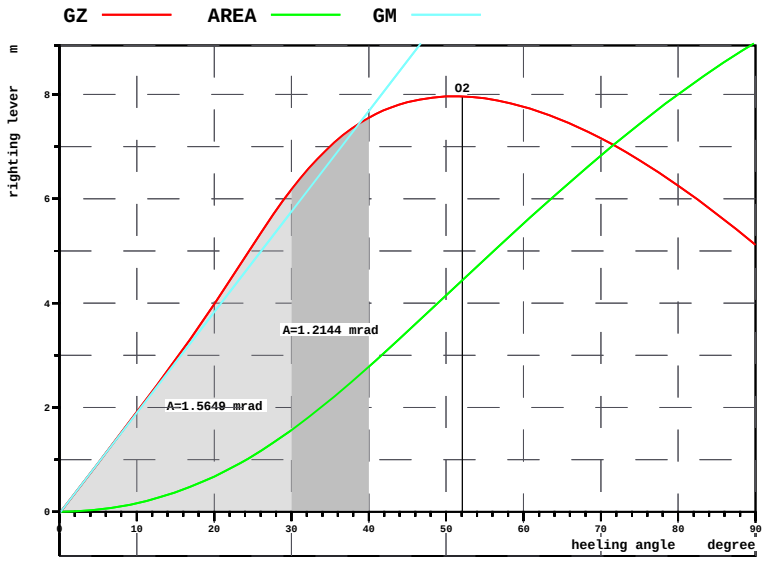
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
SUBTOTAL		17589.8		172.31	-0.00	9.79	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	127820.0	149.38	-0.04 6.96
Total weight	152461.9	146.57	-0.03 8.04

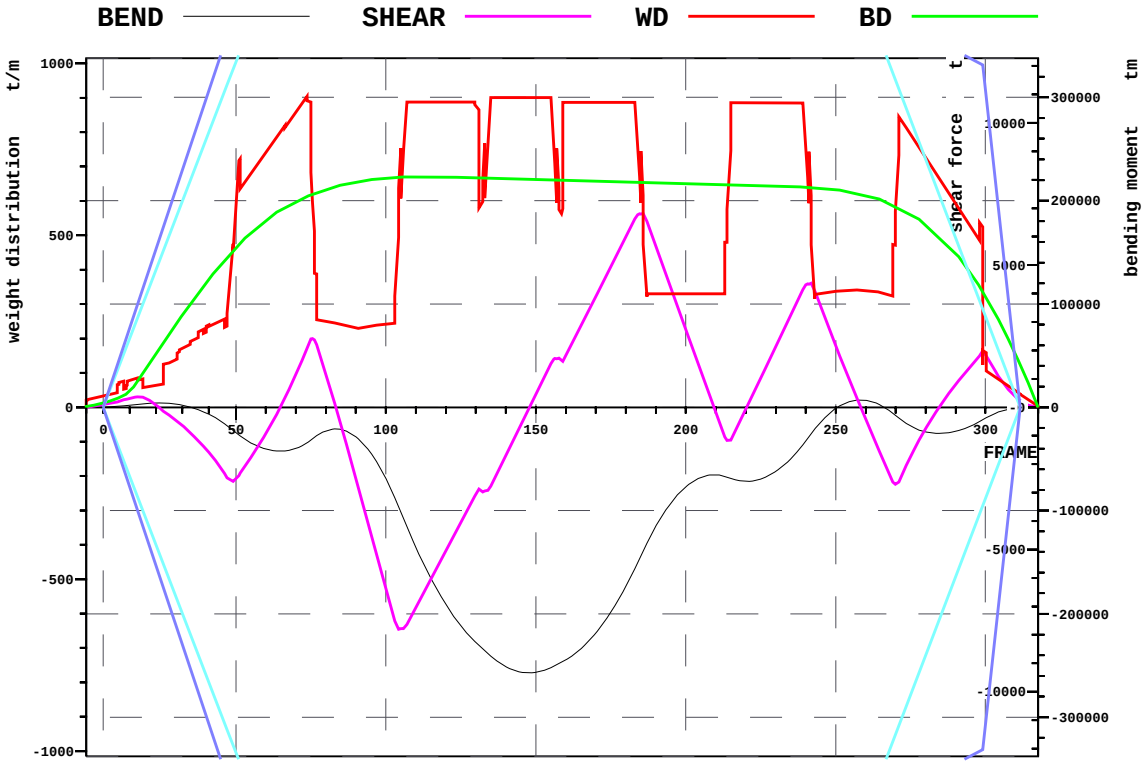
F L O A T I N G P O S I T I O N

Draught moulded	14.304	m	KM	19.09 m
Trim	1.649	m	KG	8.04 m
Heel, PS=+	-0.2	deg		
TA	15.129	m	GM0	11.05 m
TF	13.480	m	GMCORR	0.00 m
Trimming moment	-387696	tonm	GM	11.05 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.03	0.000	0.0	0.000
5.0	0.94	0.039	64.4	0.000
10.0	1.91	0.164	128.6	0.001
20.0	3.98	0.674	221.7	0.001
30.0	6.18	1.565	296.1	0.002
40.0	7.55	2.779	401.9	0.003
45.0	7.85	3.453	456.9	0.003
50.0	7.96	4.144	503.3	0.003
55.0	7.93	4.838	540.1	0.004
60.0	7.77	5.524	568.2	0.004
65.0	7.51	6.191	588.6	0.004
70.0	7.16	6.831	603.4	0.004
75.0	6.74	7.438	614.6	0.004
80.0	6.25	8.006	620.8	0.004
85.0	5.71	8.528	620.0	0.004
90.0	5.12	9.000	613.4	0.004



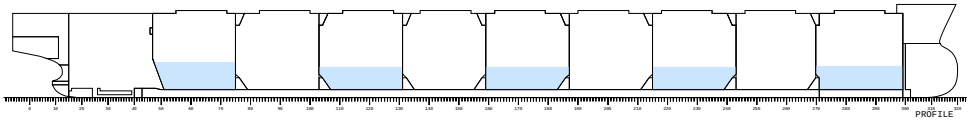
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.565	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.779	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.214	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.965	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	51.152	deg	OK
GM0.15	GM > 0.15 m	0.150	11.052	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.651		OK

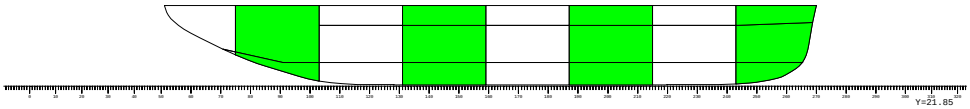


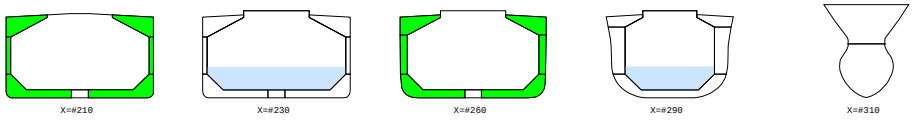
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	296	104	32.89	12.51
8.80	11.00	2149	333	80.57	54.77
12.00	15.00	3298	353	57.00	97.86
37.60	47.00	-17939	-2480	273.11	434.55
41.13	51.00	-26838	-2399	717.35	472.85
62.97	75.00	-31755	2404	784.99	617.80
88.45	103.00	-87615	-7514	323.34	668.11
113.93	131.00	-230275	-2877	865.52	667.55
139.41	159.00	-246066	1619	577.18	660.65
164.89	187.00	-131658	6532	321.71	653.67
190.37	215.00	-68988	-1150	816.13	646.66
215.85	243.00	-20939	4095	328.12	637.31
241.33	271.00	-10504	-2629	844.94	576.91
266.81	299.00	-11973	1983	524.19	335.16
278.01	313.00	-523	244	40.17	127.13

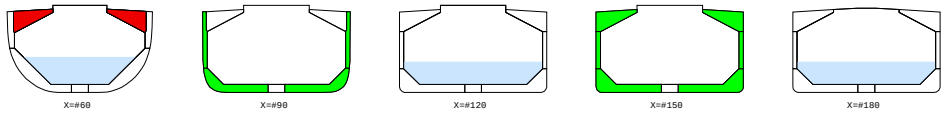
				X	FRAME
SHEAR FORCE (MIN)	-7800.1 t	(56.7%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	6816.7 t	(51.0%)		163.7 m	186
SAGGING MOMENT	-257091 tm	(35.5%)		130.0 m	149
HOGGING MOMENT	6986.0 tm	(1.7%)		228.7 m	257

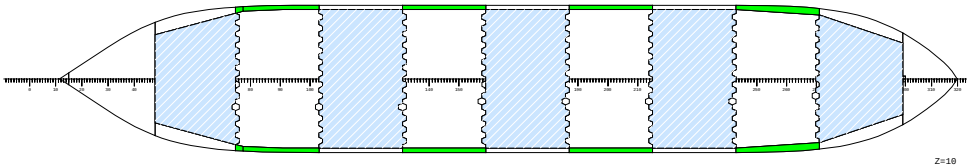
LOADING CONDITION - LSC25C2-04 : LOA. SEQU.LC25LC2 STEP4-GARGO 6 EMPTY-50% BUNK.

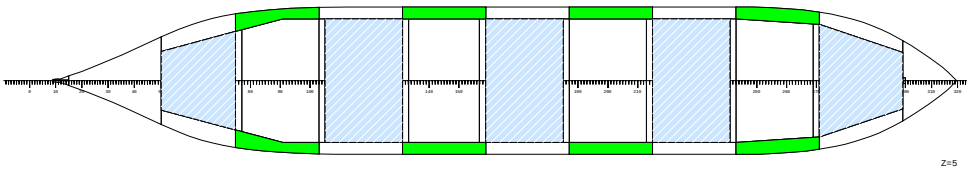


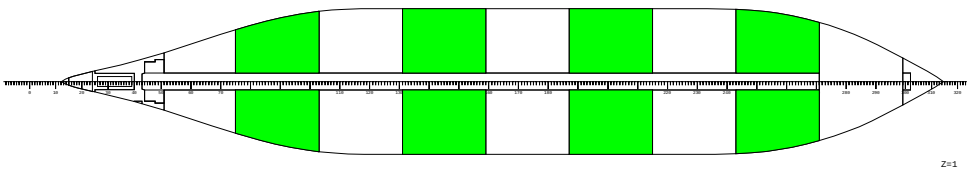












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
R1.9	NO.9 CARGO HOLD	16376.3	30.0	52.21	0.03	7.19	0.00
SUBTOTAL		90773.3		150.72	0.01	6.39	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

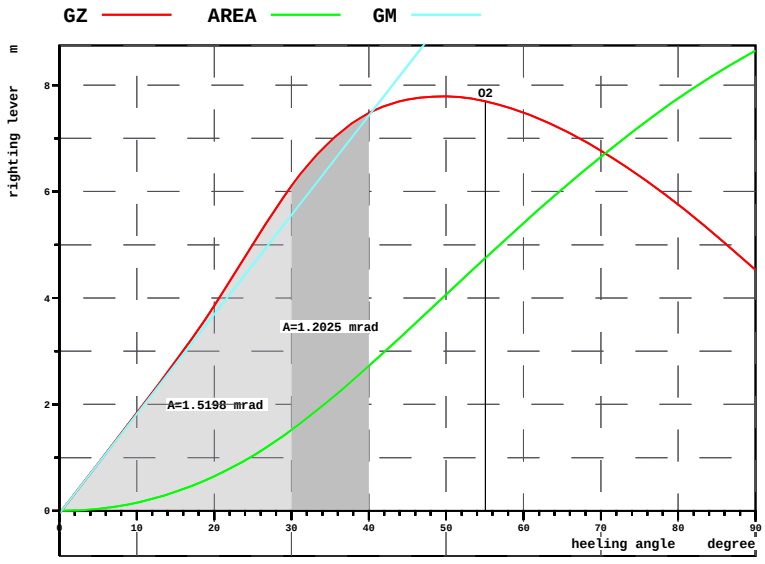
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
SUBTOTAL		24168.0		159.89	-0.00	10.08	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	118178.3	149.53	-0.06 7.58
Total weight	142820.1	146.51	-0.05 8.63

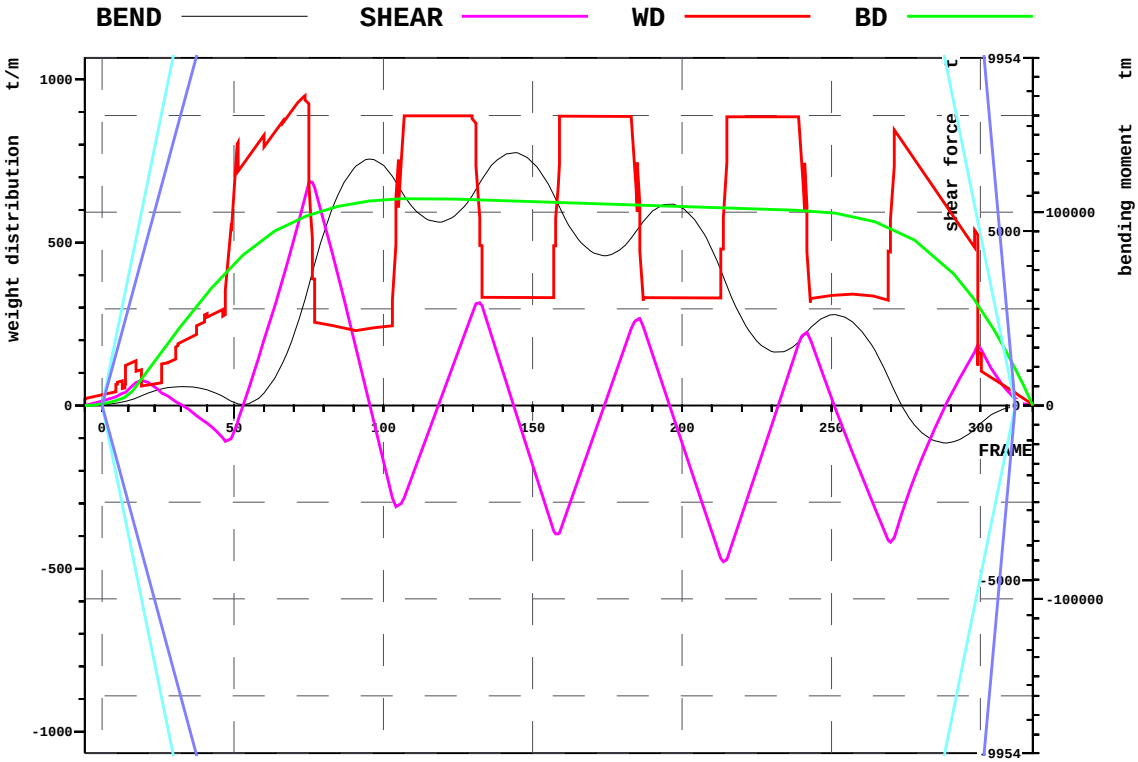
F L O A T I N G P O S I T I O N

Draught moulded	13.481	m	KM	19.34 m
Trim	1.926	m	KG	8.63 m
Heel, PS=+	-0.2	deg		
TA	14.444	m	GM0	10.71 m
TF	12.518	m	GMCORR	0.00 m
Trimming moment	-443566	tonm	GM	10.71 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.89	0.037	273.1	0.002
10.0	1.84	0.156	572.7	0.004
20.0	3.85	0.648	1197.1	0.008
30.0	6.11	1.520	1623.0	0.011
40.0	7.47	2.722	1920.4	0.013
45.0	7.73	3.387	2017.0	0.014
50.0	7.79	4.065	2076.3	0.015
55.0	7.70	4.742	2095.0	0.015
60.0	7.48	5.405	2072.0	0.015
65.0	7.17	6.045	2014.9	0.014
70.0	6.77	6.654	1930.4	0.014
75.0	6.29	7.224	1823.1	0.013
80.0	5.75	7.750	1696.6	0.012
85.0	5.16	8.227	1553.9	0.011
90.0	4.53	8.650	1399.3	0.010



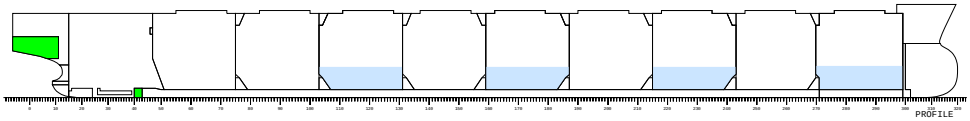
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.520	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.722	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.203	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.789	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	49.391	deg	OK
GM0.15	GM > 0.15 m	0.150	10.710	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.884		OK

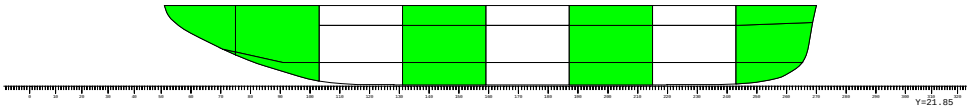


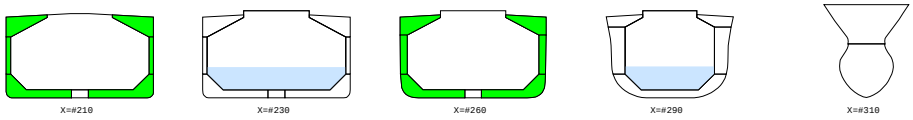
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	375	129	32.89	5.45
8.80	11.00	2910	530	130.44	40.42
12.00	15.00	5010	726	59.73	81.12
37.60	47.00	3989	-1021	354.47	404.85
41.13	51.00	950	-543	800.65	442.11
62.97	75.00	61615	6381	804.20	583.63
88.45	103.00	118463	-2645	323.34	632.41
113.93	131.00	111679	2917	734.74	630.67
139.41	159.00	102277	-3642	739.98	622.59
164.89	187.00	95217	2256	321.72	614.42
190.37	215.00	61901	-4410	816.13	606.23
215.85	243.00	40213	1880	328.12	595.74
241.33	271.00	7824	-3785	844.95	536.23
266.81	299.00	-10640	1741	524.19	308.12
278.01	313.00	-487	217	40.17	117.19

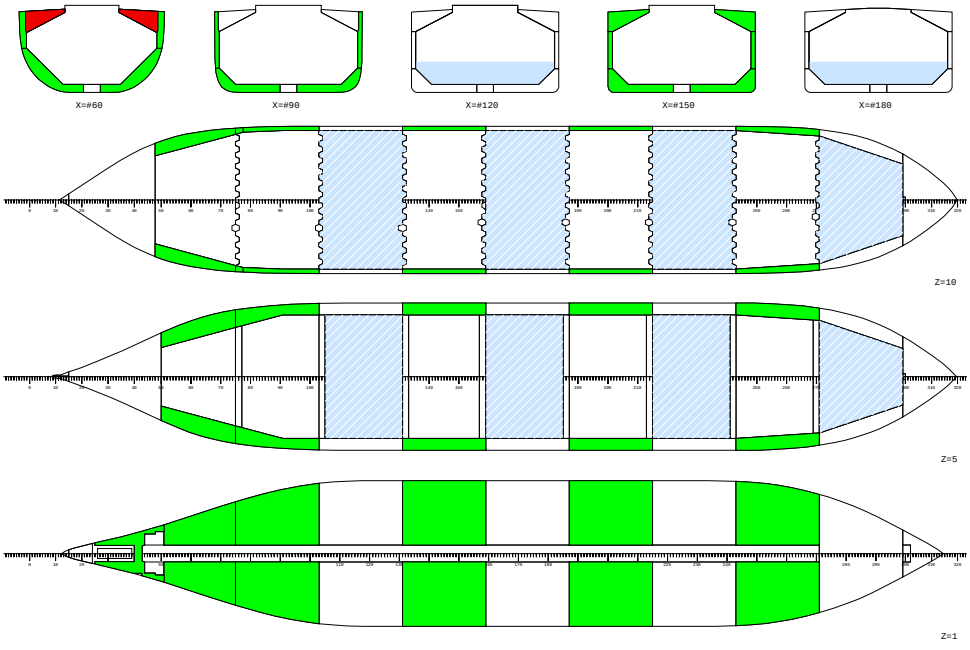
				X	FRAME
SHEAR FORCE (MIN)	-4470.7 t	(33.4%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	6407.7 t	(47.6%)		63.5 m	76
SAGGING MOMENT	-19269.1 tm	(10.8%)		256.5 m	288
HOGGING MOMENT	130756.1 tm	(17.8%)		126.0 m	144

LOADING CONDITION - LSC25C2-05 : LOA. SEQU.LC25LC2 STEP5-GARGO 9 EMPTY-50% BUNK.









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.1	NO.1 CARGO HOLD	15188.2	30.0	252.56	0.00	6.33	0.00
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		74397.0		172.41	0.01	6.21	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

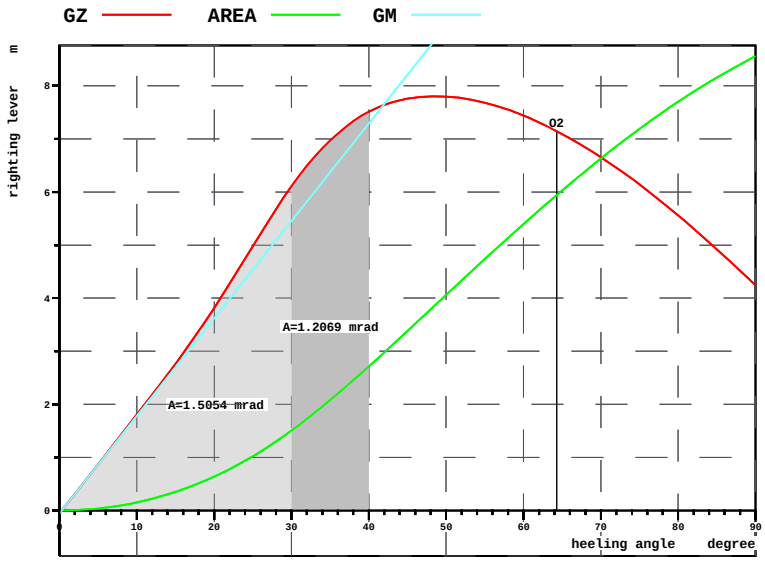
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		30452.5		134.80	-0.00	10.12	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	108086.4	157.81	-0.07 7.80
Total weight	132728.2	153.02	-0.06 8.89

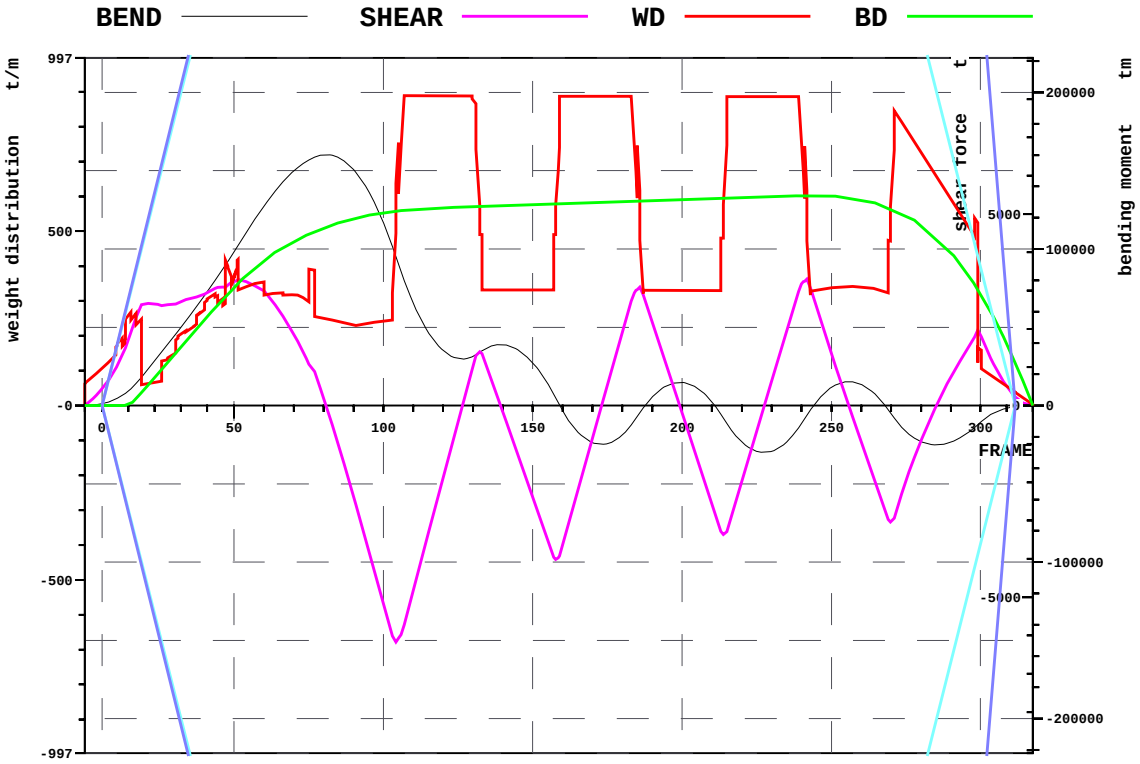
F L O A T I N G P O S I T I O N

Draught moulded	12.530	m	KM	19.42 m
Trim	-1.961	m	KG	8.89 m
Heel, PS=+	-0.3	deg		
TA	11.549	m	GM0	10.53 m
TF	13.510	m	GMCORR	0.00 m
Trimming moment	400842	tonm	GM	10.53 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.86	0.035	273.1	0.002
10.0	1.80	0.152	572.7	0.004
20.0	3.81	0.637	1197.1	0.009
30.0	6.11	1.505	1623.0	0.012
40.0	7.51	2.712	1920.4	0.014
45.0	7.76	3.380	2017.0	0.015
50.0	7.80	4.060	2076.3	0.016
55.0	7.68	4.736	2095.0	0.016
60.0	7.44	5.397	2072.0	0.016
65.0	7.09	6.032	2014.9	0.015
70.0	6.65	6.632	1930.4	0.015
75.0	6.14	7.191	1823.1	0.014
80.0	5.56	7.702	1696.6	0.013
85.0	4.92	8.159	1553.9	0.012
90.0	4.24	8.559	1399.3	0.011



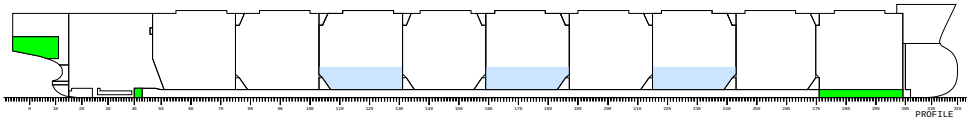
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.505	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.712	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.207	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.802	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	48.728	deg	OK
GM0.15	GM > 0.15 m	0.150	10.528	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	7.237		OK

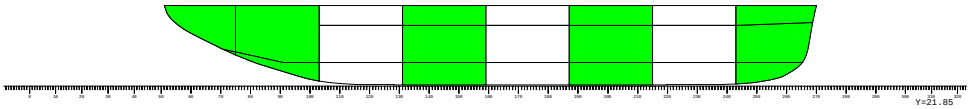


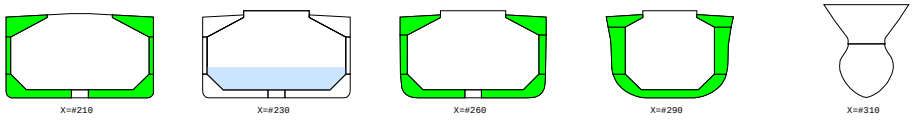
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1193	443	106.33	0.00
8.80	11.00	10667	1913	268.19	7.90
12.00	15.00	18042	2636	59.75	36.92
37.60	47.00	89935	3080	418.12	311.53
41.13	51.00	101279	3269	417.28	346.79
62.97	75.00	156808	1079	297.79	491.74
88.45	103.00	101592	-6011	323.45	555.71
113.93	131.00	32503	1303	734.99	570.42
139.41	159.00	46	-3928	740.22	578.80
164.89	187.00	-1507	2884	321.82	587.10
190.37	215.00	-11457	-3294	816.40	595.38
215.85	243.00	-2595	3071	328.23	601.34
241.33	271.00	-7914	-2936	845.23	557.10
266.81	299.00	-12184	1970	524.36	331.72
278.01	313.00	-604	245	40.18	127.36

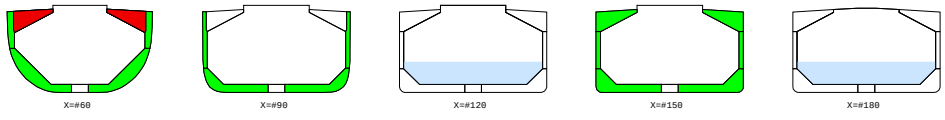
				X	FRAME
SHEAR FORCE (MIN)	-6173.7 t	(44.9%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	3298.1 t	(24.7%)		214.8 m	242
SAGGING MOMENT	-29881.8 tm	(4.7%)		200.8 m	227
HOGGING MOMENT	160103.5 tm	(27.8%)		69.5 m	82

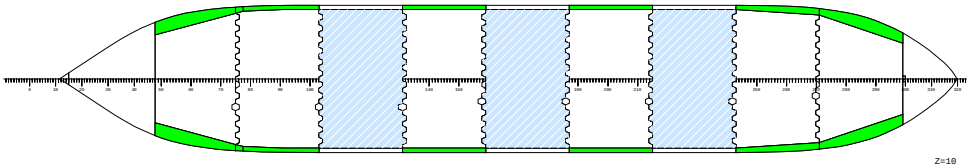
LOADING CONDITION - LSC25C2-06 : LOA. SEQU.LC25LC2 STEP6-GARGO 1 EMPTY-50% BUNK.

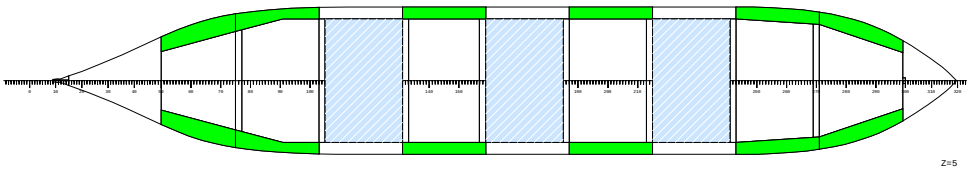


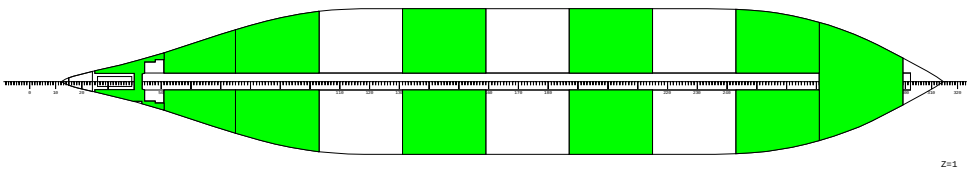












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.5	NO.5 CARGO HOLD	19730.0	30.0	151.31	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		59208.7		151.84	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		200.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	118.0	41.0	33.69	15.92	19.85	184.27
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		171.0		32.67	15.24	20.61	184.27
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	154.1	50.0	9.68	10.76	20.40	394.41
R5.1S	F.W.TK.(S)	99.8	50.0	8.84	-10.61	20.41	239.00
SUBTOTAL		254.0		9.35	2.36	20.41	633.41
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.3P	NO.3 H.F.O.TK(P)	352.1	58.0	56.08	17.20	21.68	1132.12
R3.3S	NO.3 H.F.O.TK(S)	352.1	58.0	56.08	-17.20	21.68	1132.12
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		2462.0		43.16	-4.57	22.21	2264.24

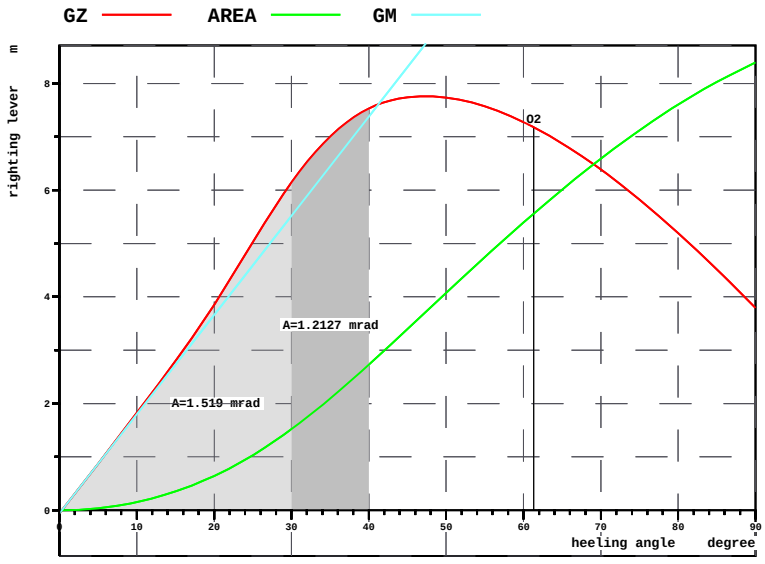
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		150.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		37712.2		157.69	-0.00	10.50	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	100157.9	150.39	-0.07 8.33
Total weight	124799.7	146.76	-0.06 9.39

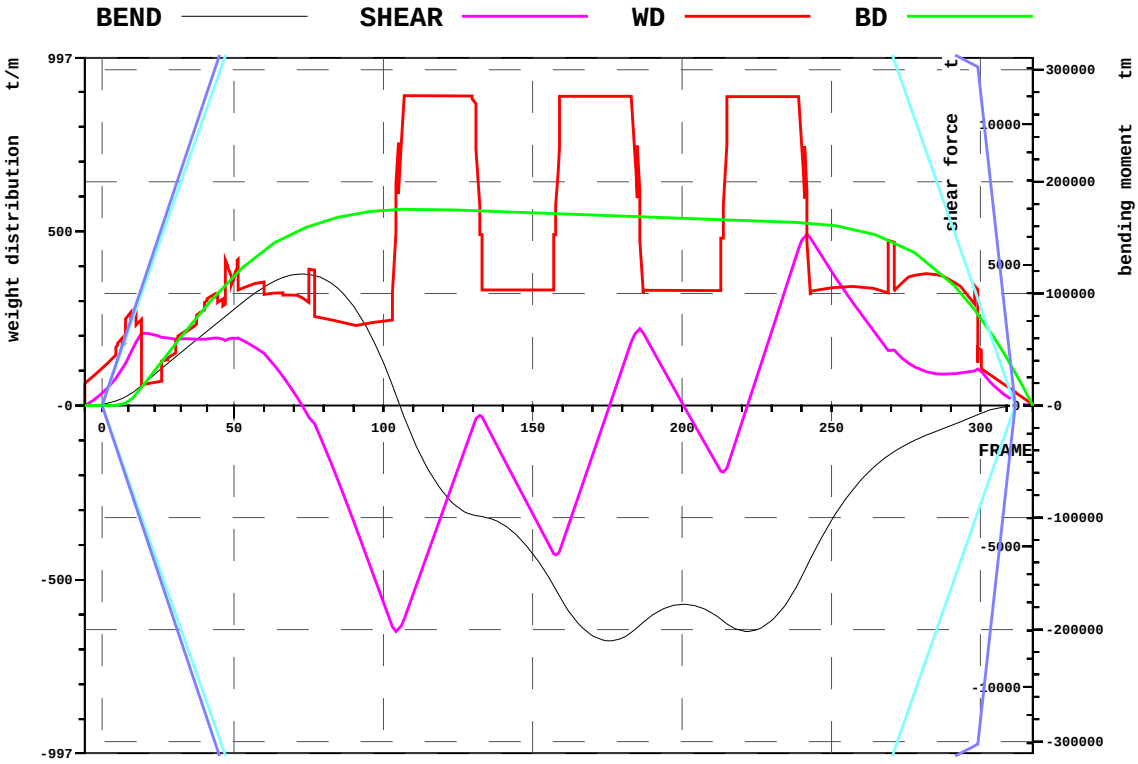
F L O A T I N G P O S I T I O N

Draught moulded	11.920	m	KM	20.04 m
Trim	2.069	m	KG	9.39 m
Heel, PS=+	-0.3	deg		
TA	12.954	m	GM0	10.65 m
TF	10.885	m	GMCORR	0.00 m
Trimming moment	-446971	tonm	GM	10.65 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.06	0.000	0.0	0.000
5.0	0.87	0.035	273.1	0.002
10.0	1.82	0.153	572.7	0.005
20.0	3.86	0.643	1197.1	0.010
30.0	6.15	1.519	1623.0	0.013
40.0	7.53	2.732	1920.4	0.015
45.0	7.74	3.400	2017.0	0.016
50.0	7.74	4.076	2076.3	0.017
55.0	7.57	4.745	2095.0	0.017
60.0	7.28	5.394	2072.0	0.017
65.0	6.88	6.013	2014.9	0.016
70.0	6.39	6.592	1930.4	0.015
75.0	5.82	7.125	1823.1	0.015
80.0	5.19	7.606	1696.6	0.014
85.0	4.51	8.030	1553.9	0.012
90.0	3.79	8.393	1399.3	0.011



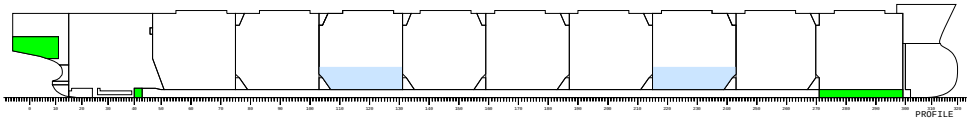
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.519	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.732	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.213	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.760	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	47.400	deg	OK
GM0.15	GM > 0.15 m	0.150	10.653	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	6.562		OK

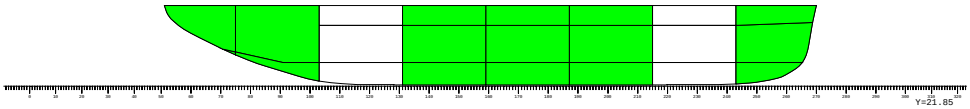


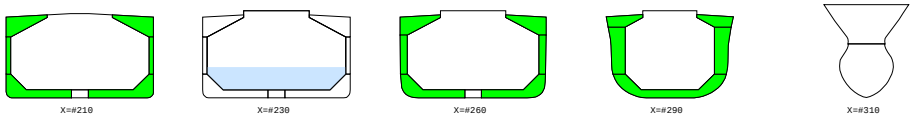
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1142	443	106.38	0.01
8.80	11.00	10486	1889	268.42	17.36
12.00	15.00	17695	2573	59.77	52.02
37.60	47.00	79690	2313	418.29	344.45
41.13	51.00	88089	2387	417.45	379.35
62.97	75.00	117094	-416	343.75	514.15
88.45	103.00	18086	-7864	323.58	561.42
113.93	131.00	-98353	-469	735.29	559.07
139.41	159.00	-170621	-5189	740.53	550.38
164.89	187.00	-193368	2573	321.96	541.62
190.37	215.00	-194826	-2224	816.74	532.82
215.85	243.00	-136696	5960	328.36	521.82
241.33	271.00	-41232	1968	331.26	465.35
266.81	299.00	-8011	1301	332.63	261.52
278.01	313.00	-369	159	40.20	97.68

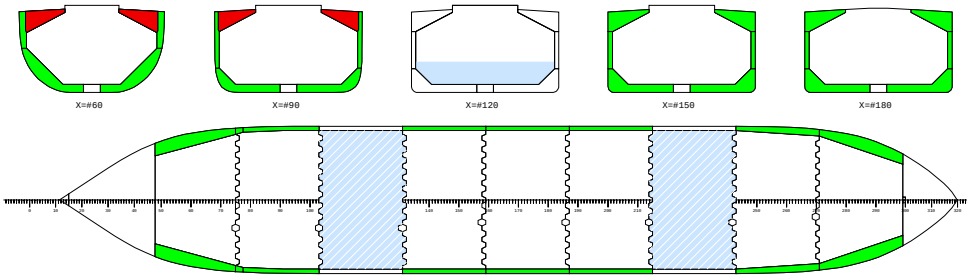
				X	FRAME
SHEAR FORCE (MIN)	-8032.2 t	(58.4%)	POSITION:	89.6 m	104
SHEAR FORCE (MAX)	6099.9 t	(45.6%)		214.8 m	242
SAGGING MOMENT	-210152 tm	(29.0%)		155.3 m	176
HOGGING MOMENT	117481.4 tm	(23.1%)		61.3 m	73

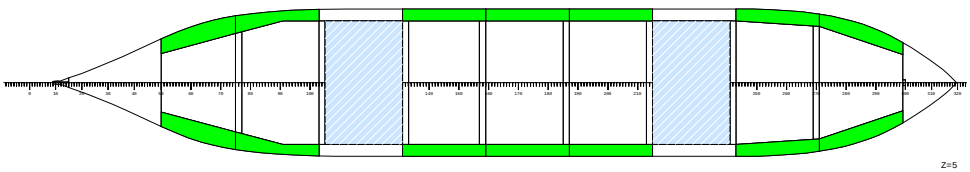
LOADING CONDITION - LSC25C2-07 : LOA. SEQU.LC25LC2 STEP7-GAR.5 EMPTY-DEP. BUNK.

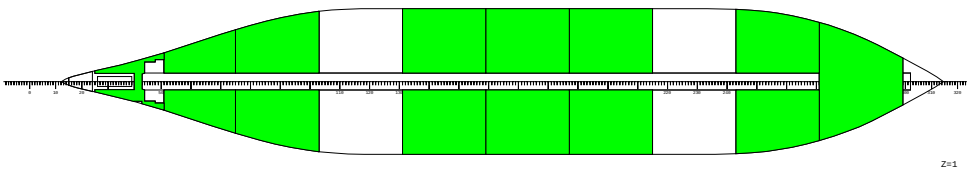












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
R1.7	NO.7 CARGO HOLD	19748.0	30.0	102.01	0.02	6.18	0.00
SUBTOTAL		39478.8		152.11	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

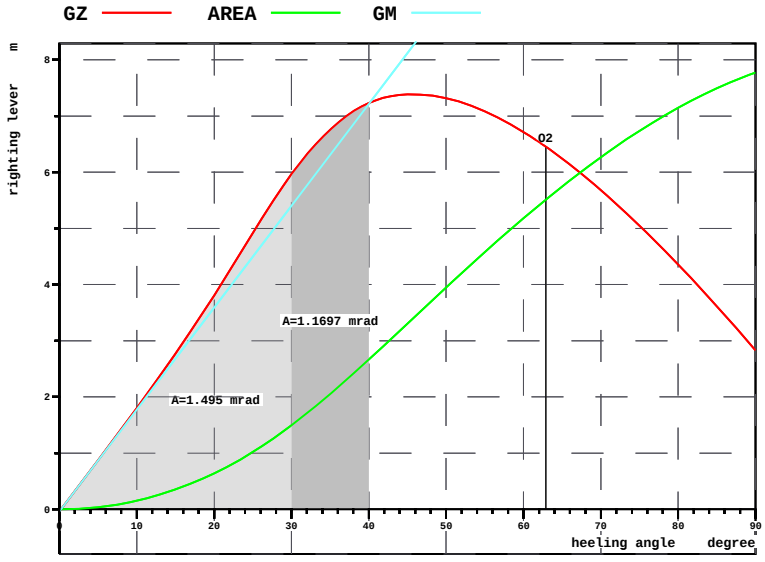
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		44290.4		156.87	-0.00	10.55	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	89954.4	147.46	-0.05 9.48
Total weight	114596.2	144.14	-0.04 10.38

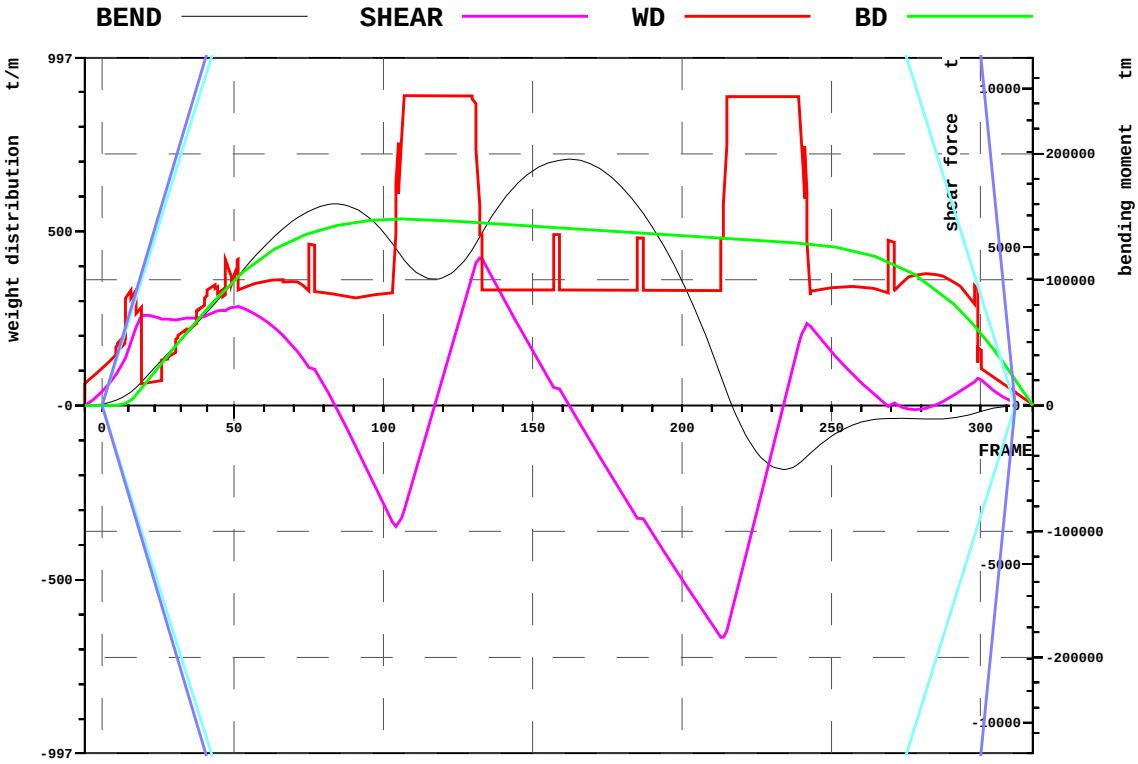
F L O A T I N G P O S I T I O N

Draught moulded	11.055	m	KM	20.76 m
Trim	3.574	m	KG	10.38 m
Heel, PS=+	-0.2	deg		
TA	12.841	m	GM0	10.39 m
TF	9.268	m	GMCORR	0.00 m
Trimming moment	-765935	tonm	GM	10.39 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.04	0.000	0.0	0.000
5.0	0.87	0.036	0.0	0.000
10.0	1.80	0.153	0.0	0.000
20.0	3.81	0.638	0.0	0.000
30.0	5.96	1.495	0.0	0.000
40.0	7.23	2.665	0.0	0.000
45.0	7.38	3.304	0.0	0.000
50.0	7.32	3.947	0.0	0.000
55.0	7.08	4.576	0.0	0.000
60.0	6.71	5.179	0.0	0.000
65.0	6.24	5.745	0.0	0.000
70.0	5.68	6.266	0.0	0.000
75.0	5.05	6.734	0.0	0.000
80.0	4.36	7.145	0.0	0.000
85.0	3.61	7.493	0.0	0.000
90.0	2.83	7.774	0.0	0.000



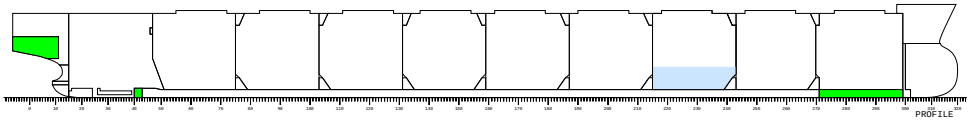
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.495	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.665	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.170	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.385	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	45.670	deg	OK
GM0.15	GM > 0.15 m	0.150	10.387	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	5.709		OK

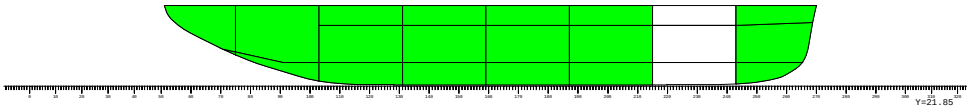


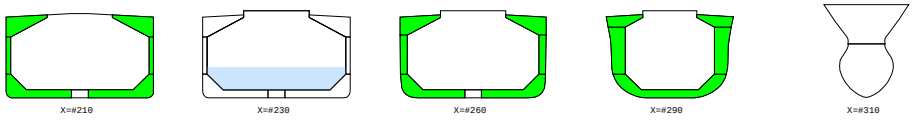
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1187	443	106.38	0.00
8.80	11.00	10697	1992	330.93	15.67
12.00	15.00	18561	2839	62.51	49.34
37.60	47.00	90884	2994	418.63	332.13
41.13	51.00	101801	3115	417.80	365.72
62.97	75.00	154298	1205	395.94	493.54
88.45	103.00	128725	-3675	323.60	534.09
113.93	131.00	128810	4499	735.34	525.34
139.41	159.00	194886	527	490.20	510.27
164.89	187.00	152479	-3574	481.04	495.12
190.37	215.00	10409	-7105	816.80	479.94
215.85	243.00	-37719	2509	328.39	462.62
241.33	271.00	-10176	85	331.28	403.13
266.81	299.00	-5438	860	332.65	216.89
278.01	313.00	-282	96	40.20	77.07

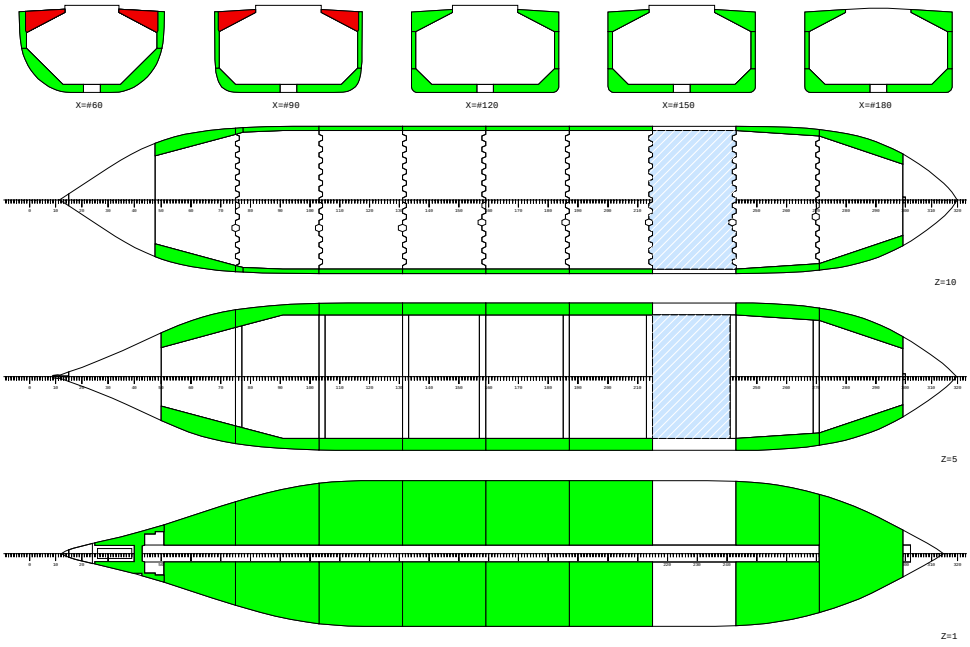
				X	FRAME
SHEAR FORCE (MIN)	-7304.4 t	(54.6%)	POSITION:	189.3 m	214
SHEAR FORCE (MAX)	4654.6 t	(34.8%)		115.1 m	132
SAGGING MOMENT	-50864.1 tm	(8.7%)		207.6 m	234
HOGGING MOMENT	195714.7 tm	(26.6%)		142.3 m	162

LOADING CONDITION - LSC25C2-08 : LOA. SEQU.LC25LC2 STEP8-GAR.7 EMPTY-DEP. BUNK.









NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Solid cargo (RHO=3.001)							
R1.3	NO.3 CARGO HOLD	19730.8	30.0	202.26	0.01	6.18	0.00
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00

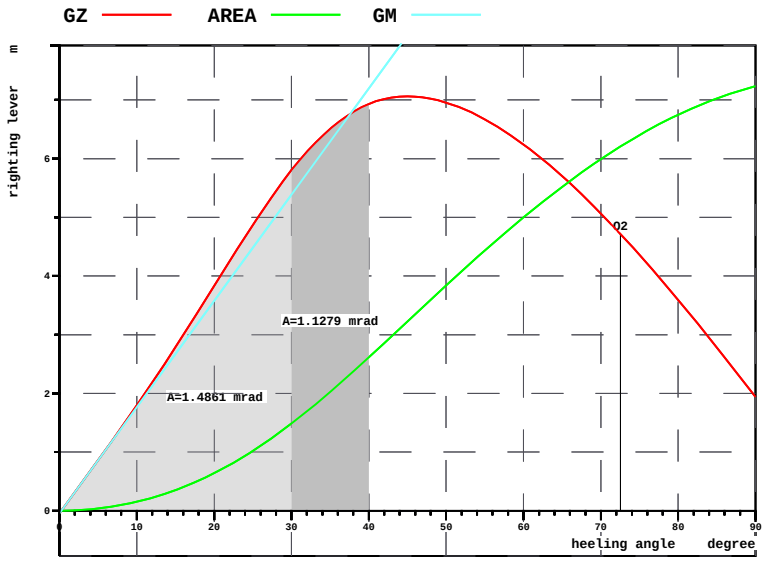
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		50839.2		149.70	-0.00	10.59	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	76755.2	155.21	-0.06 10.45
Total weight	101397.1	149.58	-0.05 11.23

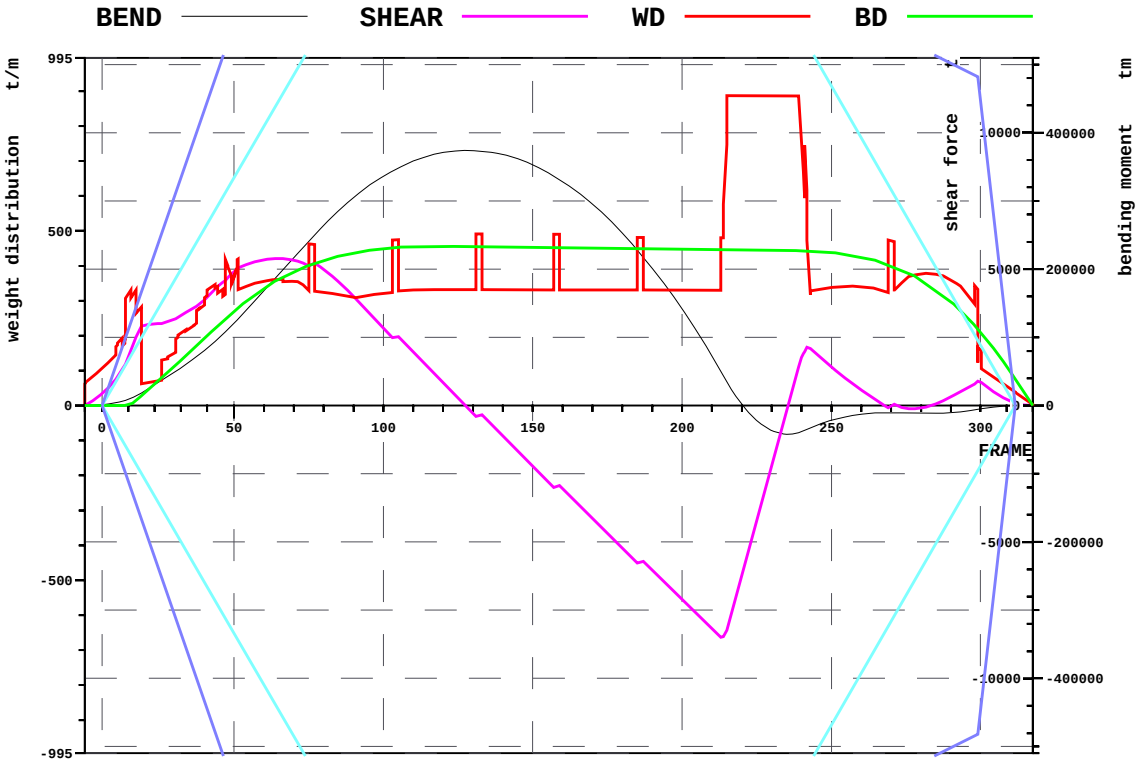
F L O A T I N G P O S I T I O N

Draught moulded	9.795	m	KM	21.60 m
Trim	0.682	m	KG	11.23 m
Heel, PS=+	-0.2	deg		
TA	10.136	m	GM0	10.37 m
TF	9.454	m	GMCORR	0.00 m
Trimming moment	-134095	tonm	GM	10.37 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.86	0.036	0.0	0.000
10.0	1.80	0.152	0.0	0.000
20.0	3.83	0.639	0.0	0.000
30.0	5.80	1.486	0.0	0.000
40.0	6.93	2.614	0.0	0.000
45.0	7.06	3.227	0.0	0.000
50.0	6.96	3.840	0.0	0.000
55.0	6.67	4.435	0.0	0.000
60.0	6.24	5.000	0.0	0.000
65.0	5.71	5.522	0.0	0.000
70.0	5.07	5.993	0.0	0.000
75.0	4.36	6.404	0.0	0.000
80.0	3.59	6.752	0.0	0.000
85.0	2.78	7.030	0.0	0.000
90.0	1.94	7.237	0.0	0.000



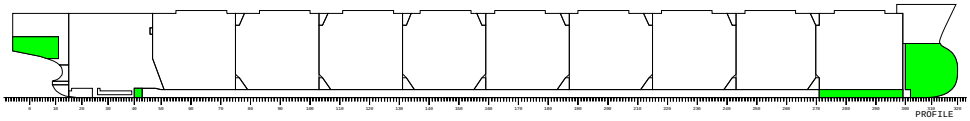
RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.486	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.614	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.128	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	7.063	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.994	deg	OK
GM0.15	GM > 0.15 m	0.150	10.374	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.834		OK

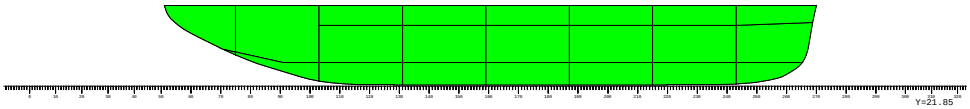


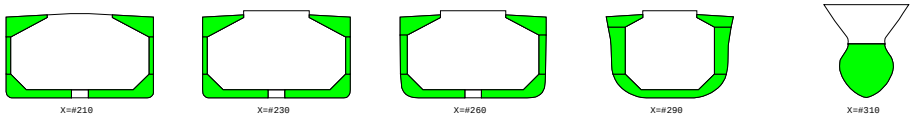
X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1137	443	106.40	0.00
8.80	11.00	10604	2014	330.97	5.32
12.00	15.00	18582	2912	62.52	27.85
37.60	47.00	108230	4570	418.68	247.10
41.13	51.00	125214	4998	417.85	276.78
62.97	75.00	240658	5116	395.99	400.64
88.45	103.00	343713	2485	473.38	451.43
113.93	131.00	373479	-404	490.81	454.93
139.41	159.00	331030	-2934	490.26	452.14
164.89	187.00	217741	-5710	481.10	449.26
190.37	215.00	34580	-8227	816.90	446.36
215.85	243.00	-32776	2089	328.43	441.34
241.33	271.00	-10620	48	331.32	394.48
266.81	299.00	-5547	897	332.69	218.82
278.01	313.00	-246	104	40.20	79.44

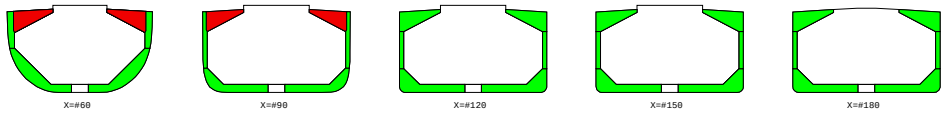
				X	FRAME
SHEAR FORCE (MIN)	-8488.6 t	(63.5%)	POSITION:	188.6 m	213
SHEAR FORCE (MAX)	5385.1 t	(40.5%)		52.7 m	64
SAGGING MOMENT	-41997.1 tm	(7.3%)		208.6 m	235
HOGGING MOMENT	374111.9 tm	(50.9%)		110.4 m	127

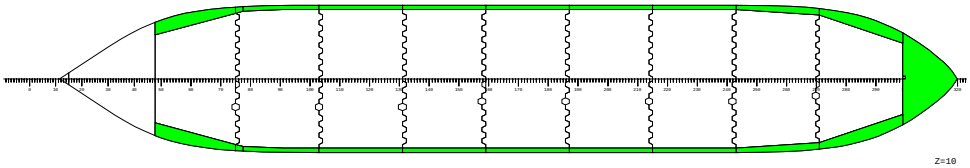
LOADING CONDITION - LSC25C2-09 : LOA. SEQU.LC25LC2 STEP9-GAR.3 EMPTY-DEP. BUNK.

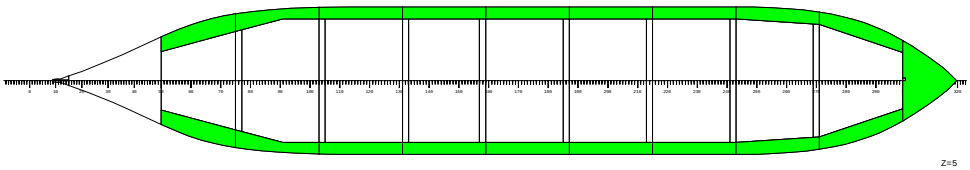


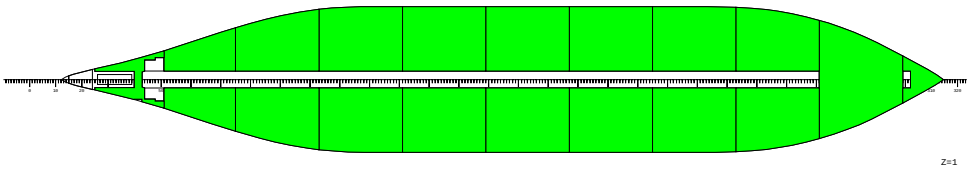












NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS= (RHO=1)							
(CREW&ST.		210.0	0.0	39.10	0.00	30.00	0.00
CONTENTS=Diesel Oil (RHO=0.85)							
R4.1	M.D.O.TK	287.9	100.0	33.83	15.96	21.76	0.00
R4.2	M.D.O.SERV.TK	26.5	100.0	29.60	13.73	22.28	0.00
R4.3	M.D.O.SETT.TK	26.5	100.0	31.20	13.73	22.28	0.00
SUBTOTAL		340.9		33.30	15.61	21.84	0.00
CONTENTS=Fresh Water (RHO=1)							
R5.1P	F.W.TK.(P)	308.3	100.0	9.67	10.85	22.05	0.00
R5.1S	F.W.TK.(S)	199.7	100.0	8.83	-10.71	22.06	0.00
SUBTOTAL		507.9		9.34	2.38	22.05	0.00
CONTENTS=Heavy Fuel Oil (RHO=0.99)							
R3.1P	NO.1 H.F.O.TK(P)	537.3	100.0	81.63	17.21	22.86	0.00
R3.1S	NO.1 H.F.O.TK(S)	537.3	100.0	81.63	-17.21	22.86	0.00
R3.2P	NO.2 H.F.O.TK(P)	439.9	100.0	68.97	16.97	22.93	0.00
R3.2S	NO.2 H.F.O.TK(S)	439.9	100.0	68.97	-16.97	22.93	0.00
R3.3P	NO.3 H.F.O.TK(P)	607.1	100.0	56.08	16.02	22.78	0.00
R3.3S	NO.3 H.F.O.TK(S)	607.1	100.0	56.08	-16.02	22.78	0.00
R3.4P	NO.4 H.F.O.TK(P)	493.9	100.0	43.54	15.60	22.86	0.00
R3.4S	NO.4 H.F.O.TK(S)	493.9	100.0	43.54	-15.60	22.86	0.00
R3.5	NO.5 H.F.O.TK	532.2	100.0	30.21	-15.92	21.86	0.00
R3.6	NO.1 H.F.O.SERV.TK	38.2	100.0	30.00	-13.73	21.78	0.00
R3.7	NO.2 H.F.O.SERV.TK	38.2	100.0	27.60	-13.73	21.78	0.00
R3.8	NO.1 H.F.O.SETT.TK	76.9	100.0	34.17	-9.20	21.92	0.00
R3.9	NO.2 H.F.O.SETT.TK	84.5	100.0	33.78	-12.09	21.82	0.00
SUBTOTAL		4926.4		57.50	-2.28	22.70	0.00
CONTENTS= (RHO=1)							
(OIL&WAT.		200.0	0.0	23.10	0.00	22.00	0.00

NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
CONTENTS=Water Ballast (RHO=1.025)							
R2.01	NO.1 W.B TK.	2833.2	100.0	253.04	0.00	2.67	0.00
R2.02P	NO.2 W.B TK(P)	1648.0	100.0	228.29	14.13	2.24	0.00
R2.02S	NO.2 W.B TK(S)	1648.0	100.0	228.29	-14.13	2.24	0.00
R2.03P	NO.3 W.B TK(P)	1706.6	100.0	203.10	14.56	2.09	0.00
R2.03S	NO.3 W.B TK(S)	1706.6	100.0	203.10	-14.56	2.09	0.00
R2.04P	NO.4 W.B TK(P)	1709.5	100.0	177.63	14.57	2.09	0.00
R2.04S	NO.4 W.B TK(S)	1709.5	100.0	177.63	-14.57	2.09	0.00
R2.05P	NO.5 W.B TK(P)	1709.5	100.0	152.15	14.57	2.09	0.00
R2.05S	NO.5 W.B TK(S)	1709.5	100.0	152.15	-14.57	2.09	0.00
R2.06P	NO.6 W.B TK(P)	1709.5	100.0	126.67	14.57	2.09	0.00
R2.06S	NO.6 W.B TK(S)	1709.5	100.0	126.67	-14.57	2.09	0.00
R2.07P	NO.7 W.B TK(P)	1694.9	100.0	101.26	14.51	2.10	0.00
R2.07S	NO.7 W.B TK(S)	1694.9	100.0	101.26	-14.51	2.10	0.00
R2.08P	NO.8 W.B TK(P)	1580.6	100.0	75.57	13.82	2.51	0.00
R2.08S	NO.8 W.B TK(S)	1580.6	100.0	75.57	-13.82	2.51	0.00
R2.09P	NO.9 W.B TK(P)	1650.4	100.0	51.30	13.52	5.34	0.00
R2.09S	NO.9 W.B TK(S)	1650.4	100.0	51.30	-13.52	5.34	0.00
R2.11P	NO.1 D.S.W.B.TK(P)	1306.4	100.0	255.57	16.61	14.53	0.00
R2.11S	NO.1 D.S.W.B.TK(S)	1306.4	100.0	255.57	-16.61	14.53	0.00
R2.12P	NO.2 D.S.W.B.TK(P)	575.3	100.0	229.72	21.23	13.27	0.00
R2.12S	NO.2 D.S.W.B.TK(S)	575.3	100.0	229.72	-21.23	13.27	0.00
R2.13P	NO.3 D.S.W.B.TK(P)	381.5	100.0	203.11	21.85	12.97	0.00
R2.13S	NO.3 D.S.W.B.TK(S)	381.5	100.0	203.11	-21.85	12.97	0.00
R2.14P	NO.4 D.S.W.B.TK(P)	381.5	100.0	177.63	21.85	12.97	0.00
R2.14S	NO.4 D.S.W.B.TK(S)	381.5	100.0	177.63	-21.85	12.97	0.00
R2.15P	NO.5 D.S.W.B.TK(P)	381.5	100.0	152.15	21.85	12.97	0.00
R2.15S	NO.5 D.S.W.B.TK(S)	381.5	100.0	152.15	-21.85	12.97	0.00
R2.16P	NO.6 D.S.W.B.TK(P)	381.5	100.0	126.67	21.85	12.97	0.00
R2.16S	NO.6 D.S.W.B.TK(S)	381.5	100.0	126.67	-21.85	12.97	0.00
R2.17P	NO.7 D.S.W.B.TK(P)	381.5	100.0	101.19	21.85	12.97	0.00
R2.17S	NO.7 D.S.W.B.TK(S)	381.5	100.0	101.19	-21.85	12.97	0.00
R2.18P	NO.8 D.S.W.B.TK(P)	557.2	100.0	75.39	21.77	16.21	0.00
R2.18S	NO.8 D.S.W.B.TK(S)	557.2	100.0	75.39	-21.77	16.21	0.00
R2.19P	NO.9 D.S.W.B.TK(P)	523.7	100.0	51.19	21.11	19.02	0.00
R2.19S	NO.9 D.S.W.B.TK(S)	523.7	100.0	51.19	-21.11	19.02	0.00
R2.21P	NO.1 T.S.W.B.TK(P)	906.8	100.0	252.05	15.40	23.27	0.00
R2.21S	NO.1 T.S.W.B.TK(S)	906.8	100.0	252.05	-15.40	23.27	0.00
R2.22P	NO.2 T.S.W.B.TK(P)	1144.6	100.0	228.32	17.66	22.78	0.00
R2.22S	NO.2 T.S.W.B.TK(S)	1144.6	100.0	228.32	-17.66	22.78	0.00
R2.23P	NO.3 T.S.W.B.TK(P)	1198.0	100.0	203.11	17.85	22.69	0.00
R2.23S	NO.3 T.S.W.B.TK(S)	1198.0	100.0	203.11	-17.85	22.69	0.00
R2.24P	NO.4 T.S.W.B.TK(P)	1198.0	100.0	177.63	17.85	22.69	0.00
R2.24S	NO.4 T.S.W.B.TK(S)	1198.0	100.0	177.63	-17.85	22.69	0.00
R2.25P	NO.5 T.S.W.B.TK(P)	1198.0	100.0	152.15	17.85	22.69	0.00
R2.25S	NO.5 T.S.W.B.TK(S)	1198.0	100.0	152.15	-17.85	22.69	0.00
R2.26P	NO.6 T.S.W.B.TK(P)	1198.0	100.0	126.67	17.85	22.69	0.00
R2.26S	NO.6 T.S.W.B.TK(S)	1198.0	100.0	126.67	-17.85	22.69	0.00

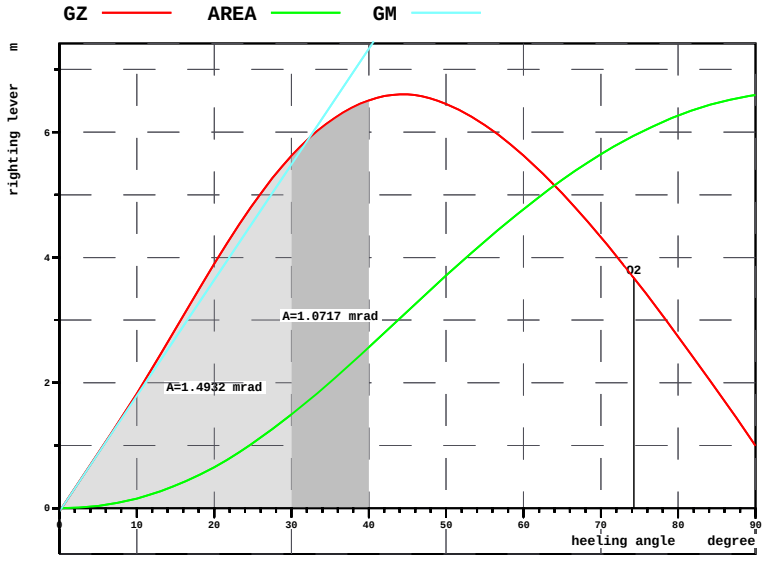
NAME	DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
R2.27P	NO.7 T.S.W.B.TK(P)	1198.0	100.0	101.19	17.85	22.69	0.00
R2.27S	NO.7 T.S.W.B.TK(S)	1198.0	100.0	101.19	-17.85	22.69	0.00
R2.31	F.P.TK	3562.9	100.0	272.45	-0.00	9.06	0.00
R2.32	A.P.TK.	1622.2	100.0	4.92	-0.08	16.27	0.00
R2.33	NO.10 D.B.W.B.TK.	313.9	100.0	31.53	0.06	1.68	0.00
SUBTOTAL		60974.2		162.63	-0.00	10.53	0.00

Lightweight	24641.9	132.01	0.00 13.67
Deadweight	67159.5	152.30	-0.07 11.67
Total weight	91801.3	146.85	-0.05 12.20

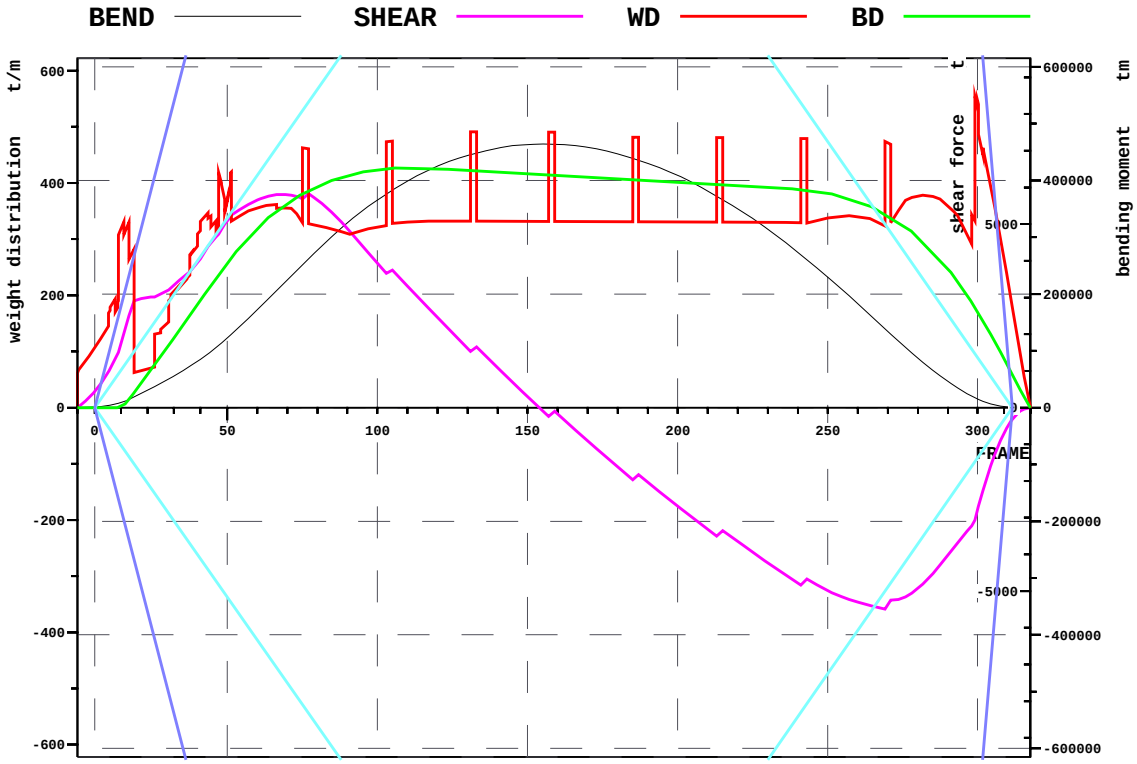
F L O A T I N G P O S I T I O N

Draught moulded	8.971	m	KM	22.79 m
Trim	2.027	m	KG	12.20 m
Heel, PS=+	-0.3	deg		
TA	9.984	m	GM0	10.58 m
TF	7.957	m	GMCORR	0.00 m
Trimming moment	-394576	tonm	GM	10.58 m

HEEL degree	GZ m	AREA rad*m	FSMOM tm	DGZ m
0.0	-0.05	0.000	0.0	0.000
5.0	0.88	0.036	0.0	0.000
10.0	1.83	0.153	0.0	0.000
20.0	3.90	0.652	0.0	0.000
30.0	5.62	1.493	0.0	0.000
40.0	6.51	2.565	0.0	0.000
45.0	6.60	3.139	0.0	0.000
50.0	6.45	3.710	0.0	0.000
55.0	6.11	4.259	0.0	0.000
60.0	5.63	4.772	0.0	0.000
65.0	5.02	5.238	0.0	0.000
70.0	4.33	5.646	0.0	0.000
75.0	3.56	5.991	0.0	0.000
80.0	2.74	6.266	0.0	0.000
85.0	1.88	6.467	0.0	0.000
90.0	0.99	6.593	0.0	0.000



RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve up to 30 deg.	0.055	1.493	mrad	OK
AREA40	Area under GZ curve up to 40 deg.	0.090	2.565	mrad	OK
AREA3040	Area under GZ curve btw. 30-40 deg.	0.030	1.072	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.604	m	OK
MAXGZ25	Max. GZ at an angle > 25 deg.	25.000	44.363	deg	OK
GM0.15	GM > 0.15 m	0.150	10.582	m	OK
V.IMOWEATHER	IMO weather criterion	1.000	4.008		OK



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	1157	443	106.41	0.00
8.80	11.00	10658	2015	331.01	5.28
12.00	15.00	18650	2914	62.53	27.06
37.60	47.00	109775	4723	418.74	235.58
41.13	51.00	127388	5195	417.90	263.71
62.97	75.00	250990	5686	396.05	380.13
88.45	103.00	376047	3661	473.44	424.62
113.93	131.00	445224	1530	490.87	422.41
139.41	159.00	463331	-98	490.32	413.91
164.89	187.00	435441	-1826	481.16	405.34
190.37	215.00	366115	-3350	405.42	396.74
215.85	243.00	260570	-4667	328.47	386.09
241.33	271.00	129115	-5250	331.37	336.83
266.81	299.00	17118	-3070	332.73	177.94
278.01	313.00	589	-335	182.47	60.50

				X	FRAME
SHEAR FORCE (MIN)	-5488.6 t	(40.5%)	POSITION:	239.5 m	269
SHEAR FORCE (MAX)	5830.2 t	(43.2%)		64.8 m	77
SAGGING MOMENT	-0.1 tm			283.6 m	320
HOGGING MOMENT	463872.7 tm	(63.1%)		135.9 m	155

APPENDIX

A1.Light ship weight distribution

	Weight	XMIN	XMAX	LCG	VCG
HULL	13993.50	18.19	266.83	141.00	11.86
AFT	670.51	-5.20	12.00	4.80	17.89
ENG	1800.88	12.00	41.49	28.89	13.39
ACCOMMO	435.40	20.00	37.60	31.66	34.22
FORT	942.54	266.83	283.62	272.54	15.38
MACHINERY	1486.40	4.70	266.83	135.77	11.61
EQUIPMENT	2134.50	6.98	278.20	136.91	24.26
ELECTRODE	374.00	6.98	286.61	127.28	13.14
PROPELLER	46.20	4.20	6.20	5.20	4.5
PAINT	350.00	-5.00	283.61	130.00	13.14
ENGTB	121.30	37.60	39.42	38.31	13.41
703TB	242.70	62.97	64.79	63.88	11.77
705TB	266.40	88.45	90.27	89.36	10.52
708TB	289.10	113.93	115.75	114.84	10.61
710TB	289.10	137.59	139.41	138.50	10.61
712TB	273.40	163.07	164.89	163.98	10.61
715TB	273.40	188.55	190.37	189.46	10.61
717TB	273.40	214.03	215.85	214.94	10.61
720TB	273.40	239.51	241.33	240.42	11.68
FORTB	105.80	265.86	267.86	266.86	12.58
TOTAL	24641.93			132.01	13.67

Addendum to Loading Manual
Maximum Allowable and Minimum Required Mass of Cargo for Double Bottom

Manager of Shanghai Office

NIPPON KAIJI KYOKAI

Instruction

The Allowable Mass Charts (Maximum Allowable and Minimum Required Mass of Cargo for Double Bottom) shown herein from page 4 through 20. The lateral axis of the chart is dx, the draft in meter at the longitudinal center of the hold (or holds) in question. The ordinate is cargo and double bottom ballast weight* in metric tons. Two solid lines show allowable maximum and required minimum values for the condition at sea. Two dotted lines show allowable maximum and required minimum values for the condition in harbor.

The master has to check every loading condition using this Allowable Mass Charts and Tables. The cargo and double bottom ballast weight for each hold is to be between required minimum and allowable maximum weight against each draft.

Special Attention

The Master has to consider double bottom ballast weight (to the double bottom height, 2.45 m) to check each loading condition according to the arrangements of double bottom ballast tanks. Double bottom fuel oil weight need not to be added, considering the consumption during navigation.

Arrangement of water ballast tank in double bottom as follows:

No.1 Cargo Hold	: No.1 W.B.T. is located below the hold.
No.2 Cargo Hold	: No.2 W.B.T.(P/S) are located below the hold.
No.3 Cargo Hold	: No.3 W.B.T.(P/S) are located below the hold.
No.4 Cargo Hold	: No.4 W.B.T.(P/S) are located below the hold.
No.5 Cargo Hold	: No.5 W.B.T.(P/S) are located below the hold.
No.6 Cargo Hold	: No.6 W.B.T.(P/S) are located below the hold.
No.7 Cargo Hold	: No.7 W.B.T.(P/S) are located below the hold.
No.8 Cargo Hold	: No.8 W.B.T.(P/S) are located below the hold.
No.9 Cargo Hold	: No.9 W.B.T.(P/S) are located below the hold.

Tank Name	Full Capacity (m^3)	Double Bottom Capacity (m^3)	Double Bottom Ballast Weight (MT)	
			Ballast height＜2.45m	Ballast height≧2.45m
No.1 W.B.T.	2,719	1,611	Actual Weight	1,651
No.2 W.B.T. (P/S)	3,175	2,178	Actual Weight	2,232
No.3 W.B.T. (P/S)	3,301	2,381	Actual Weight	2,441
No.4 W.B.T. (P/S)	3,309	2,493	Actual Weight	2,555
No.5 W.B.T. (P/S)	3,309	2,493	Actual Weight	2,555
No.6 W.B.T. (P/S)	3,309	2,493	Actual Weight	2,555
No.7 W.B.T. (P/S)	3,279	2,362	Actual Weight	2,421
No.8 W.B.T. (P/S)	3,040	1,961	Actual Weight	2,010
No.9 W.B.T. (P/S)	3,180	1,052	Actual Weight	1,078

Table.1 : capacity of each ballast tank

Sample Calculation

< Sample Loading Condition >

No. of C.H.	No.1	No.2	No.3	No.4	No.5	No.6	No.7	No.8	No.9
Cargo Weight (MT)	17,000	17,000	17,000	13,000	17,000	11,000	7,000	17,000	17,000

da=13.0m　df=13.0m

No.1, 2, 3, 5, 8, 9 ballast tank = empty　　No.4, 6, 7 ballast tank = full

< Data for judgments (Single Hold) >

No. of C.H.	No.1	No.2	No.3	No.4	No.5	No.6	No.7	No.8	No.9
Total Weight (MT)	17,000	17,000	17,000	15,555	17000	13,555	9,421	17,000	17,000
Draught (m)	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0

(Total Weight of No.4 C.H. = 13,000 + 2,555)

(Total Weight of No.6 C.H. = 11,000 + 2,555)

(Total Weight of No.7 C.H. = 7,000 + 2,421)

< Data for judgments (Adjacent Holds) >

No. of C.H.	No.1&2	No.2&3	No.3&4	No.4&5	No.5&6	No.6&7	No.7&8	No.8&9
Total Weight (MT)	34,000	34,000	32,555	32,555	30,555	22,976	26,421	34,000
Draught (m)	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0

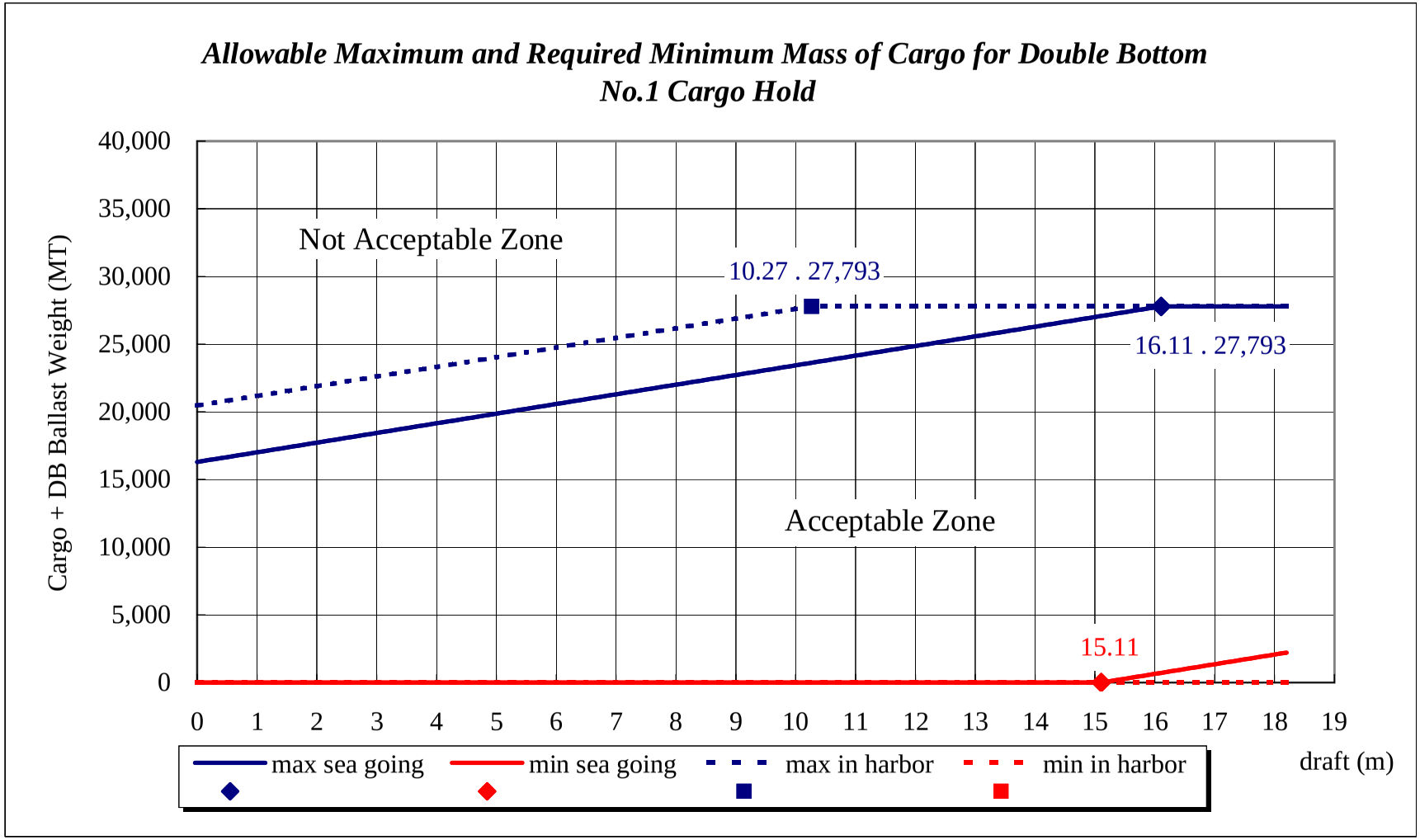
Judgment: ***OK !***

Hold No.

1

M_D	27,793
M_H	14,937
M_{HD}	27,793
M_{FULL}	16,452

$W_{max}(0)$	16,296.55
$W_{min}(0)$	-10,782.07



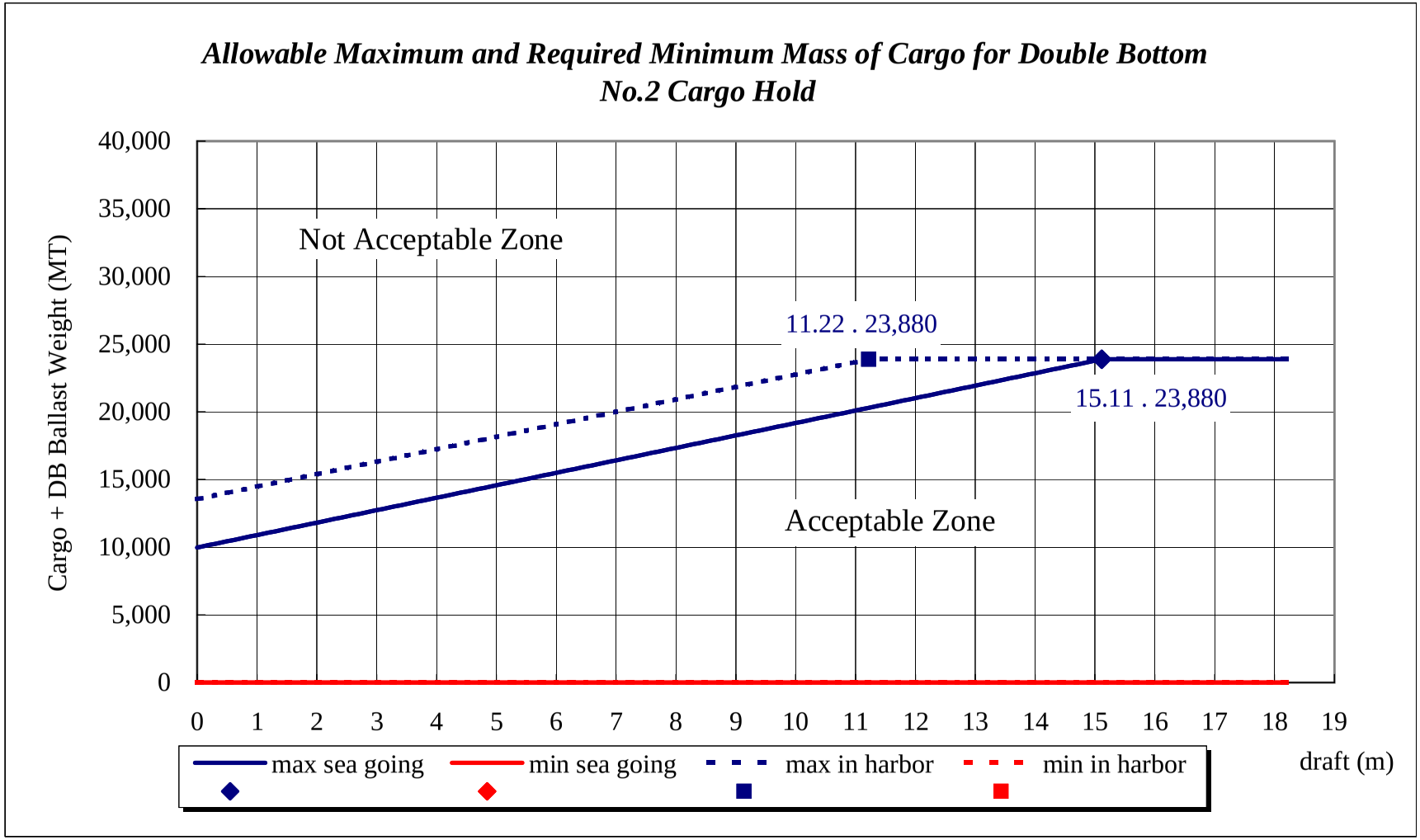
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	16,297	17,010	17,724	18,438	19,152	19,865	20,579	21,293	22,007	22,720	23,434	24,148	24,862	25,575	26,289	27,003	27,717	27,793	27,793	27,793
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	638	1,352	2,066	2,208
Harbor	W_{MAX}	20,466	21,179	21,893	22,607	23,321	24,034	24,748	25,462	26,176	26,889	27,603	27,793	27,793	27,793	27,793	27,793	27,793	27,793	27,793	27,793
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

2

M_D	23,880
M_H	19,243
M_{HD}	0
M_{FULL}	21,194

$W_{max}(0)$	9,972.93
$W_{min}(0)$	-16,747.87



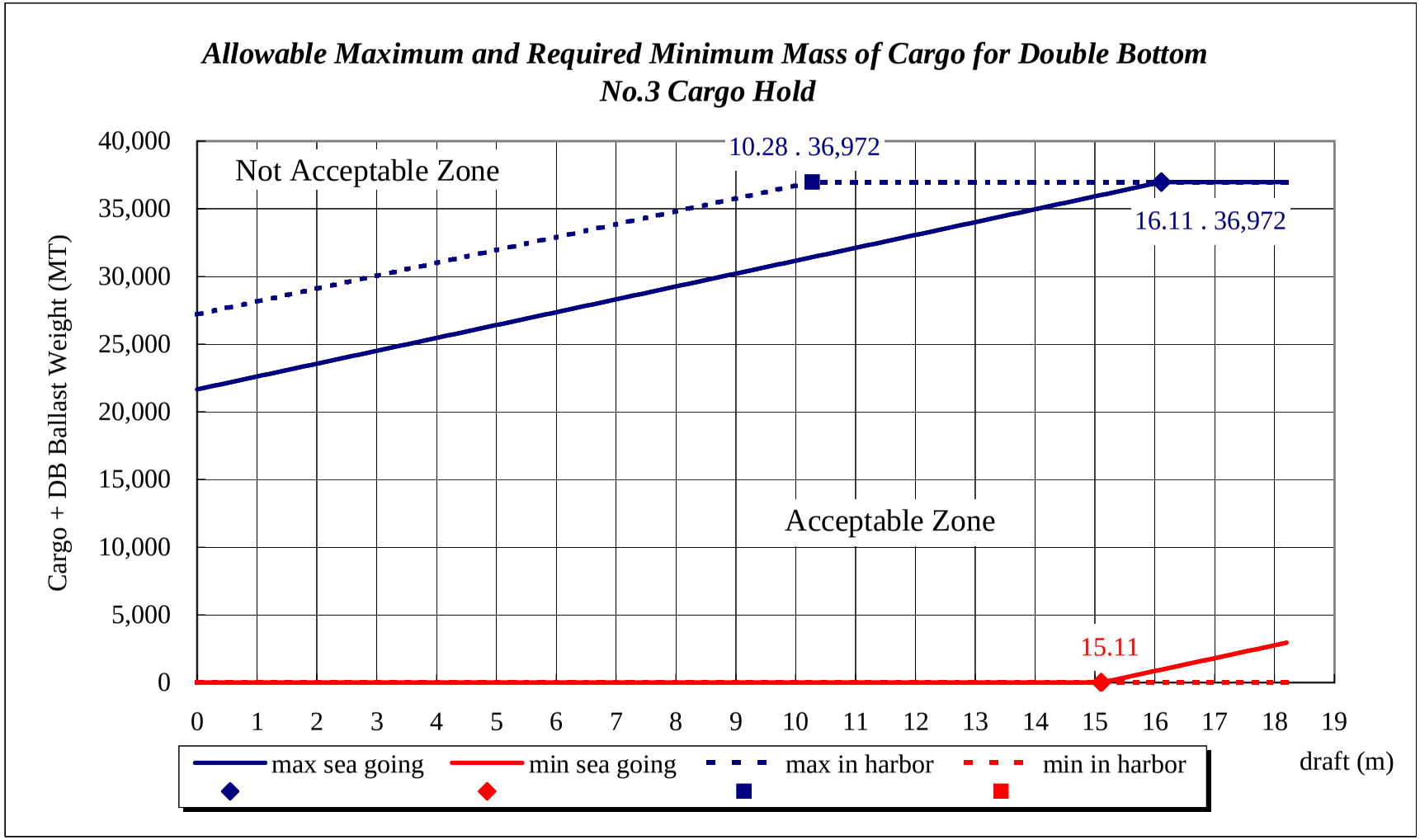
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	9,973	10,893	11,813	12,734	13,654	14,574	15,494	16,414	17,335	18,255	19,175	20,095	21,015	21,936	22,856	23,776	23,880	23,880	23,880	23,880
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Harbor	W_{MAX}	13,555	14,475	15,395	16,316	17,236	18,156	19,076	19,996	20,917	21,837	22,757	23,677	23,880	23,880	23,880	23,880	23,880	23,880	23,880	23,880
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

3

M_D	36,972
M_H	19,871
M_{HD}	36,972
M_{FULL}	21,886

$W_{max}(0)$	21,657.09
$W_{min}(0)$	-14,360.59



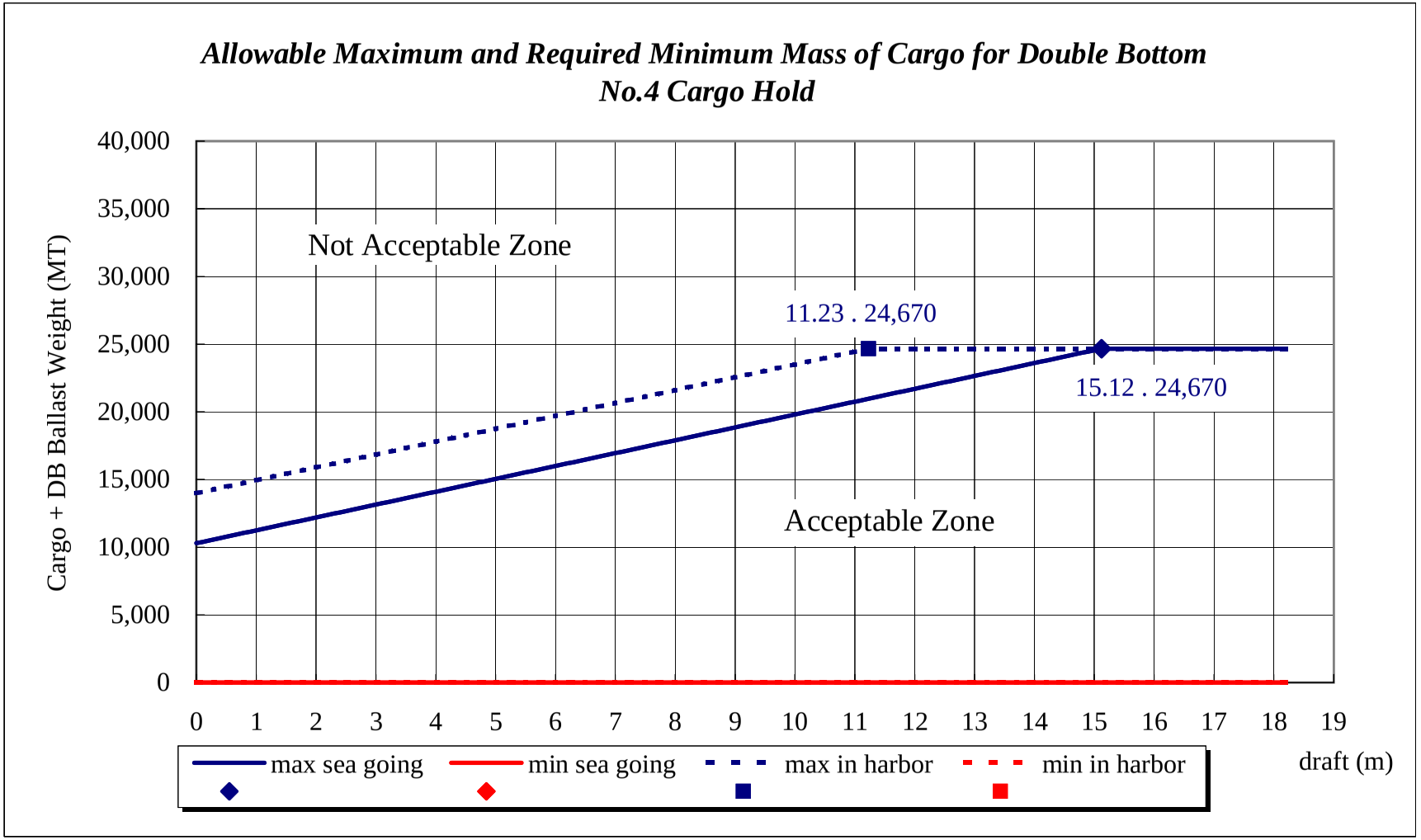
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	21,657	22,608	23,558	24,509	25,460	26,410	27,361	28,312	29,262	30,213	31,164	32,114	33,065	34,016	34,966	35,917	36,868	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	850	1,801	2,751	2,941
Harbor	W_{MAX}	27,203	28,154	29,104	30,055	31,006	31,956	32,907	33,857	34,808	35,759	36,709	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

4

M_D	24,670
M_H	19,871
M_{HD}	0
M_{FULL}	21,886

$W_{max}(0)$	10,293.72
$W_{min}(0)$	-17,301.91



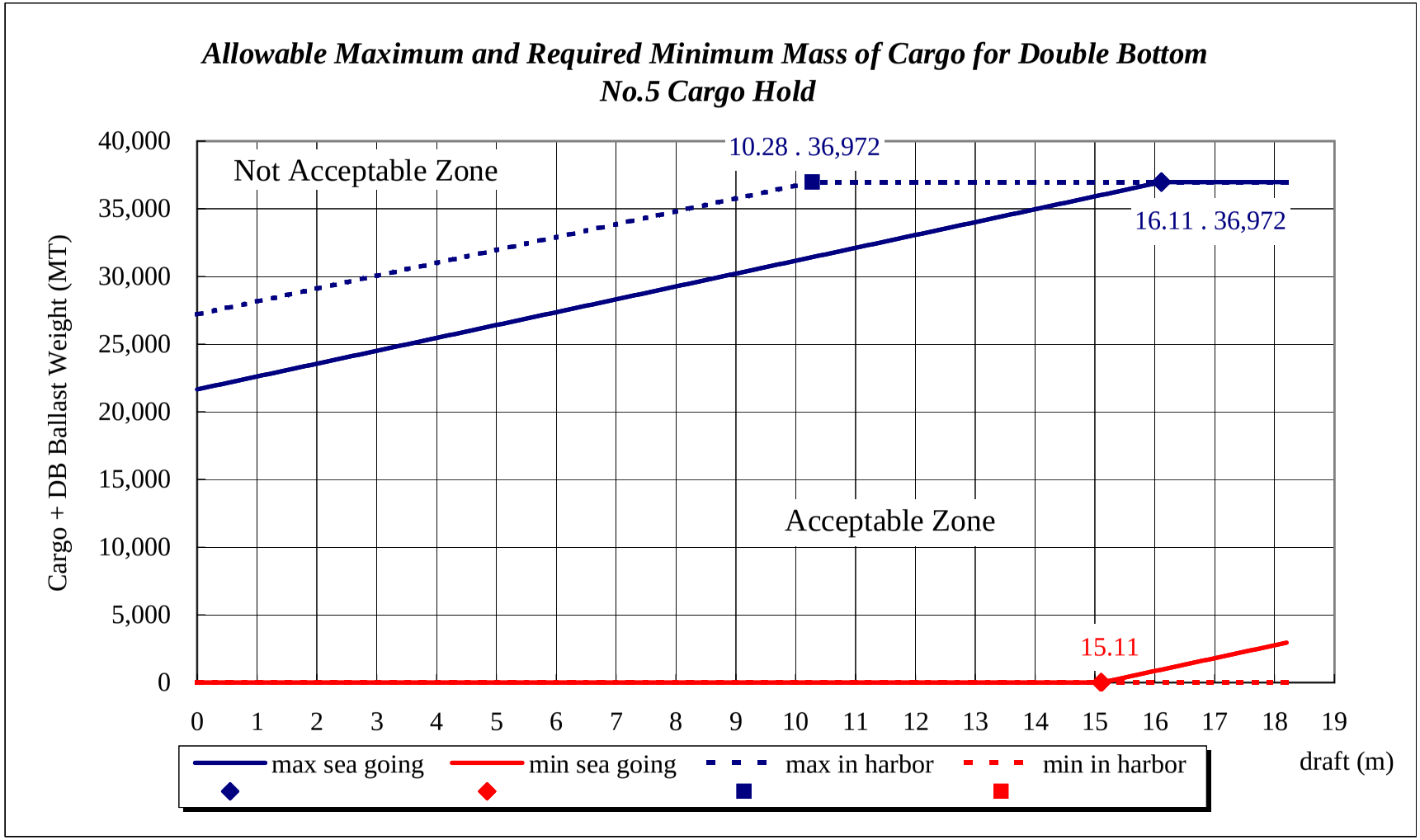
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	10,294	11,244	12,195	13,146	14,096	15,047	15,998	16,948	17,899	18,850	19,800	20,751	21,702	22,652	23,603	24,554	24,670	24,670	24,670	24,670
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Harbor	W_{MAX}	13,994	14,945	15,896	16,846	17,797	18,747	19,698	20,649	21,599	22,550	23,501	24,451	24,670	24,670	24,670	24,670	24,670	24,670	24,670	24,670
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

5

M_D	36,972
M_H	19,871
M_{HD}	36,972
M_{FULL}	21,886

$W_{max}(0)$	21,657.09
$W_{min}(0)$	-14,360.59



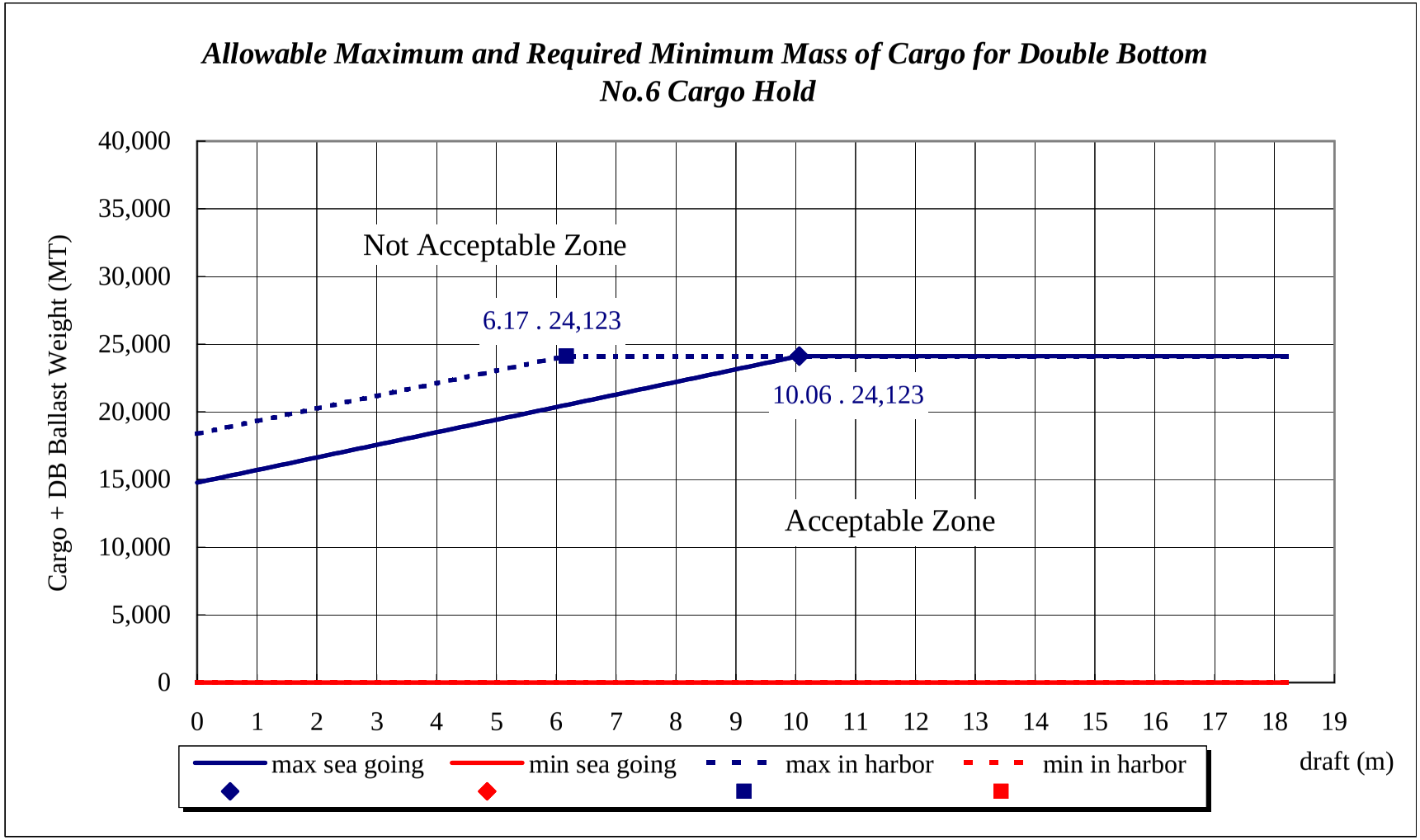
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	21,657	22,608	23,558	24,509	25,460	26,410	27,361	28,312	29,262	30,213	31,164	32,114	33,065	34,016	34,966	35,917	36,868	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	850	1,801	2,751	2,941
Harbor	W_{MAX}	27,203	28,154	29,104	30,055	31,006	31,956	32,907	33,857	34,808	35,759	36,709	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

6

M_D	24,123
M_H	19,437
M_{HD}	0
M_{FULL}	21,408

$W_{max}(0)$	14,770.52
$W_{min}(0)$	-16,918.40



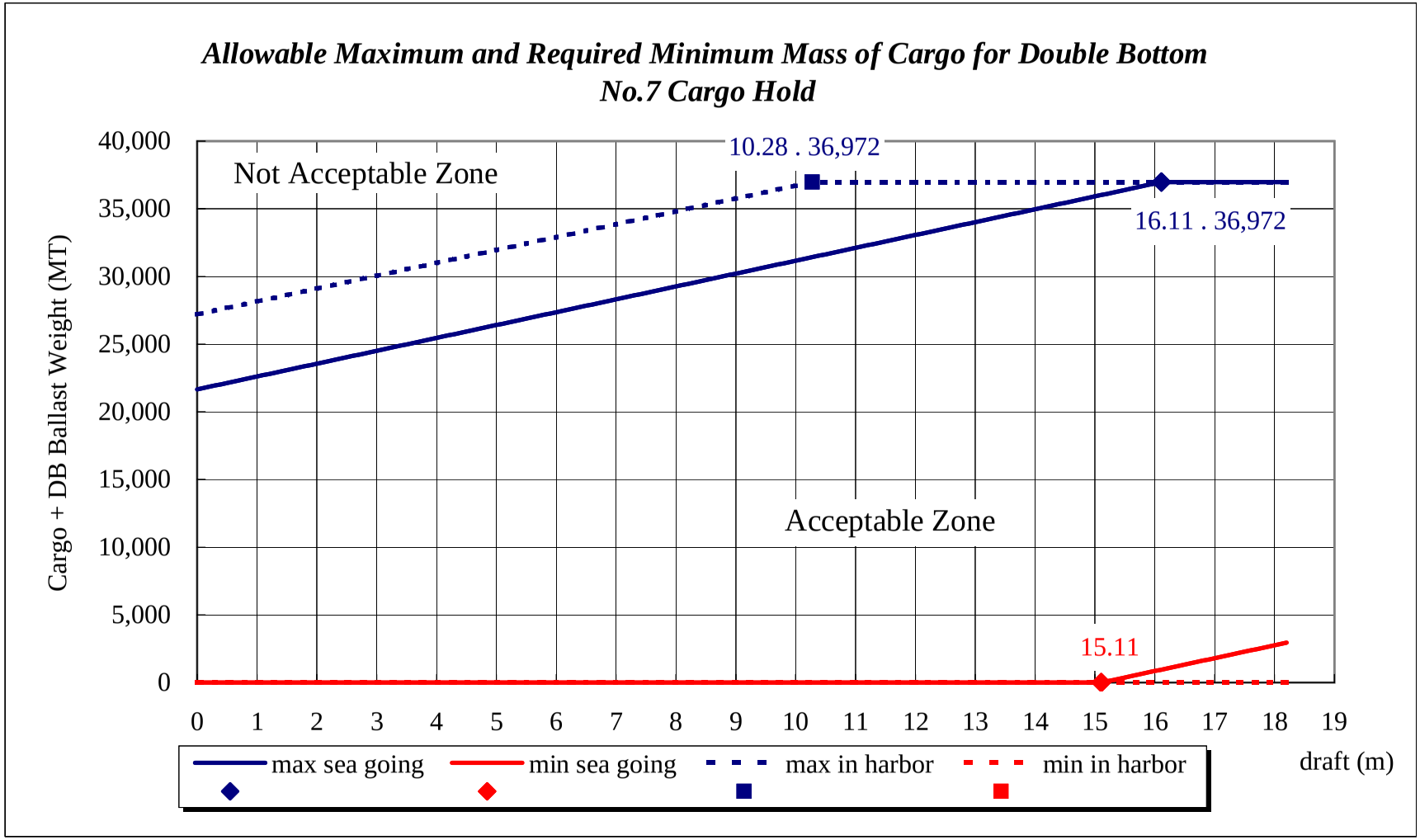
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	14,771	15,700	16,630	17,559	18,489	19,418	20,348	21,278	22,207	23,137	24,066	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Harbor	W_{MAX}	18,389	19,319	20,248	21,178	22,107	23,037	23,966	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123	24,123
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

7

M_D	36,972
M_H	19,871
M_{HD}	36,972
M_{FULL}	21,886

$W_{max}(0)$	21,657.09
$W_{min}(0)$	-14,360.59



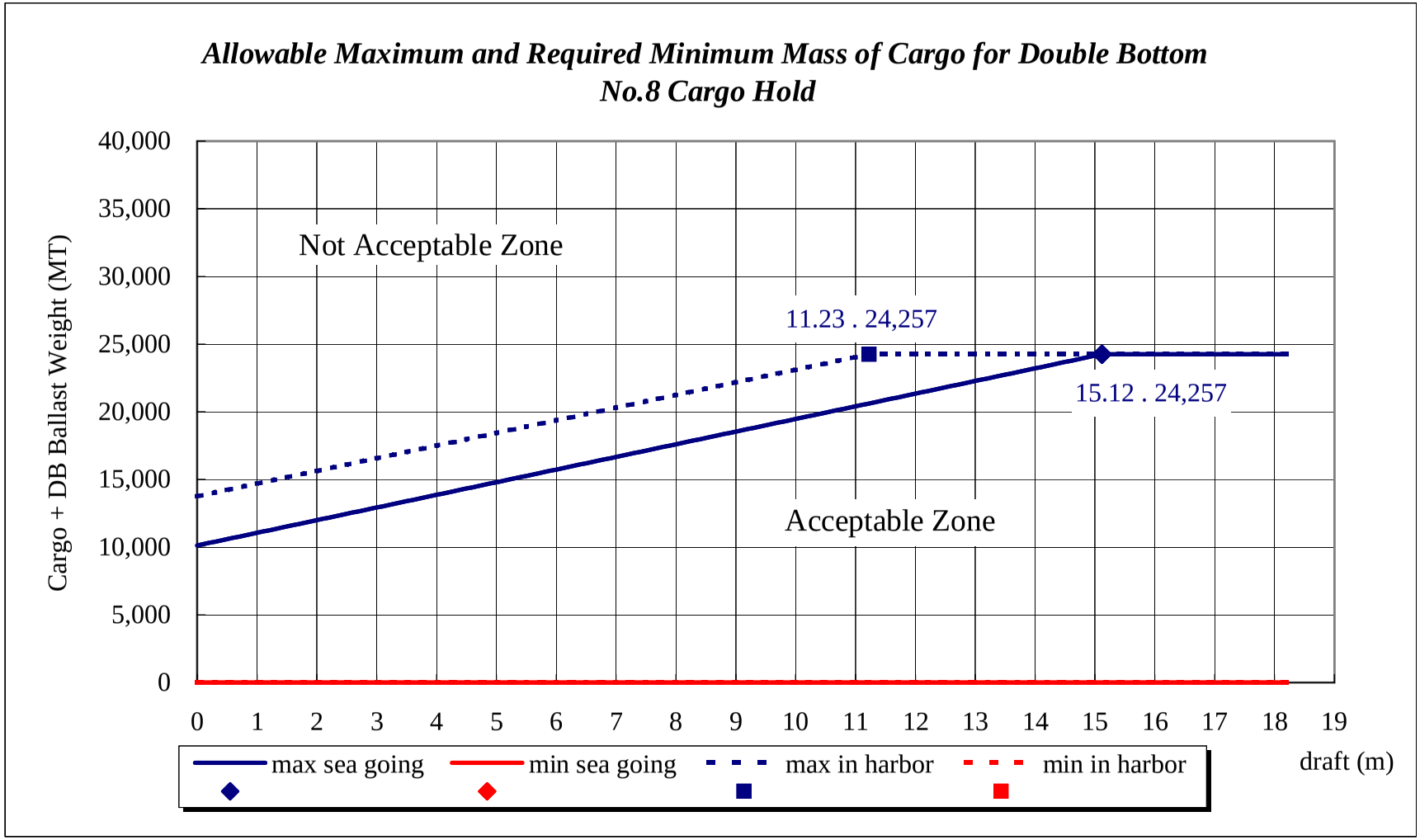
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	21,657	22,608	23,558	24,509	25,460	26,410	27,361	28,312	29,262	30,213	31,164	32,114	33,065	34,016	34,966	35,917	36,868	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	850	1,801	2,751	2,941
Harbor	W_{MAX}	27,203	28,154	29,104	30,055	31,006	31,956	32,907	33,857	34,808	35,759	36,709	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972	36,972
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

8

M_D	24,257
M_H	19,542
M_{HD}	0
M_{FULL}	21,524

$W_{max}(0)$	10,125.91
$W_{min}(0)$	-17,012.08



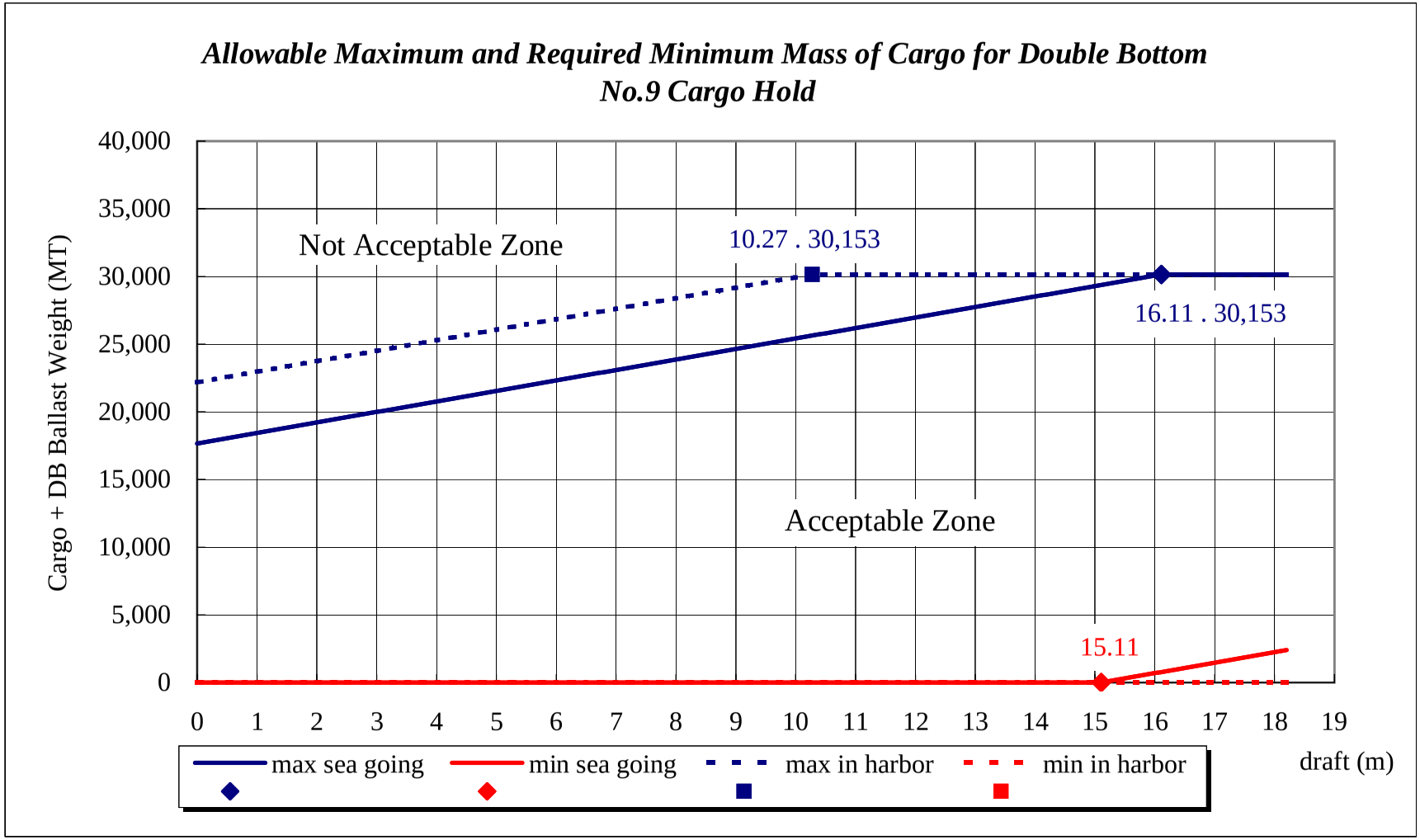
	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	10,126	11,061	11,995	12,930	13,865	14,800	15,734	16,669	17,604	18,538	19,473	20,408	21,343	22,277	23,212	24,147	24,257	24,257	24,257	24,257
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Harbor	W_{MAX}	13,764	14,699	15,634	16,569	17,503	18,438	19,373	20,308	21,242	22,177	23,112	24,046	24,257	24,257	24,257	24,257	24,257	24,257	24,257	24,257
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Hold No.

9

M_D	30,153
M_H	16,206
M_{HD}	30,153
M_{FULL}	17,849

$W_{max}(0)$	17,665.06
$W_{min}(0)$	-11,710.42



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	17,665	18,440	19,215	19,991	20,766	21,541	22,316	23,092	23,867	24,642	25,417	26,192	26,968	27,743	28,518	29,293	30,069	30,153	30,153	30,153
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	693	1,468	2,243	2,399
Harbor	W_{MAX}	22,188	22,963	23,738	24,514	25,289	26,064	26,839	27,615	28,390	29,165	29,940	30,153	30,153	30,153	30,153	30,153	30,153	30,153	30,153	30,153
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

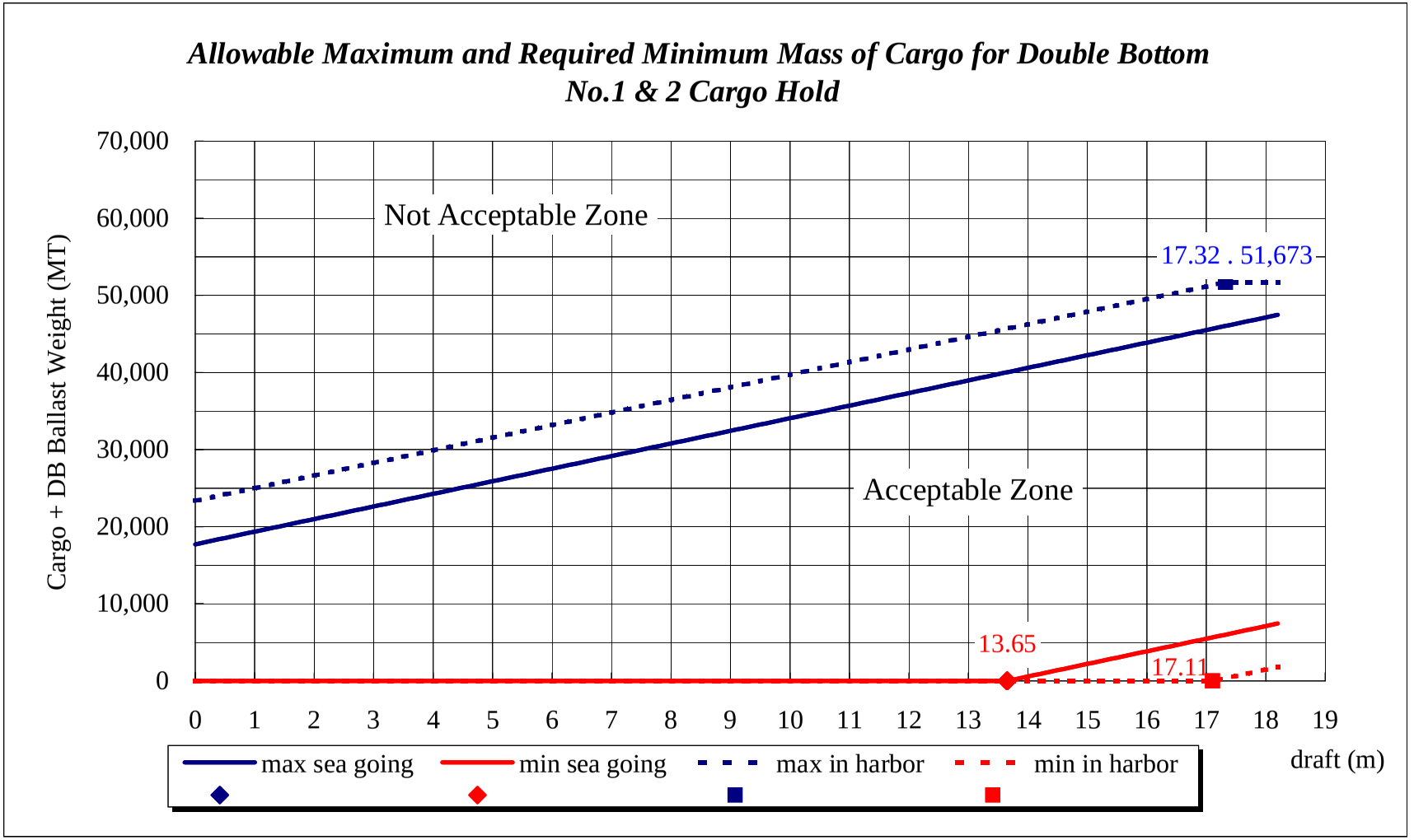
Hold No.

1 & 2

Hold No.	1
M_D	27,793
M_{FULL}	16,452

Hold No.	2
M_D	23,880
M_{FULL}	21,194

ds	18.200
$0.75d_s$	13.650
d_{hb}	9.727
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	17,721	19,355	20,989	22,623	24,257	25,891	27,525	29,159	30,793	32,427	34,061	35,695	37,329	38,963	40,597	42,231	43,865	45,499	47,133	47,460
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	572	2,206	3,840	5,474	7,108	7,435
Harbor	W_{MAX}	23,368	25,002	26,636	28,270	29,904	31,538	33,172	34,806	36,440	38,074	39,708	41,342	42,976	44,610	46,244	47,878	49,512	51,146	51,673	51,673
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,461	1,788

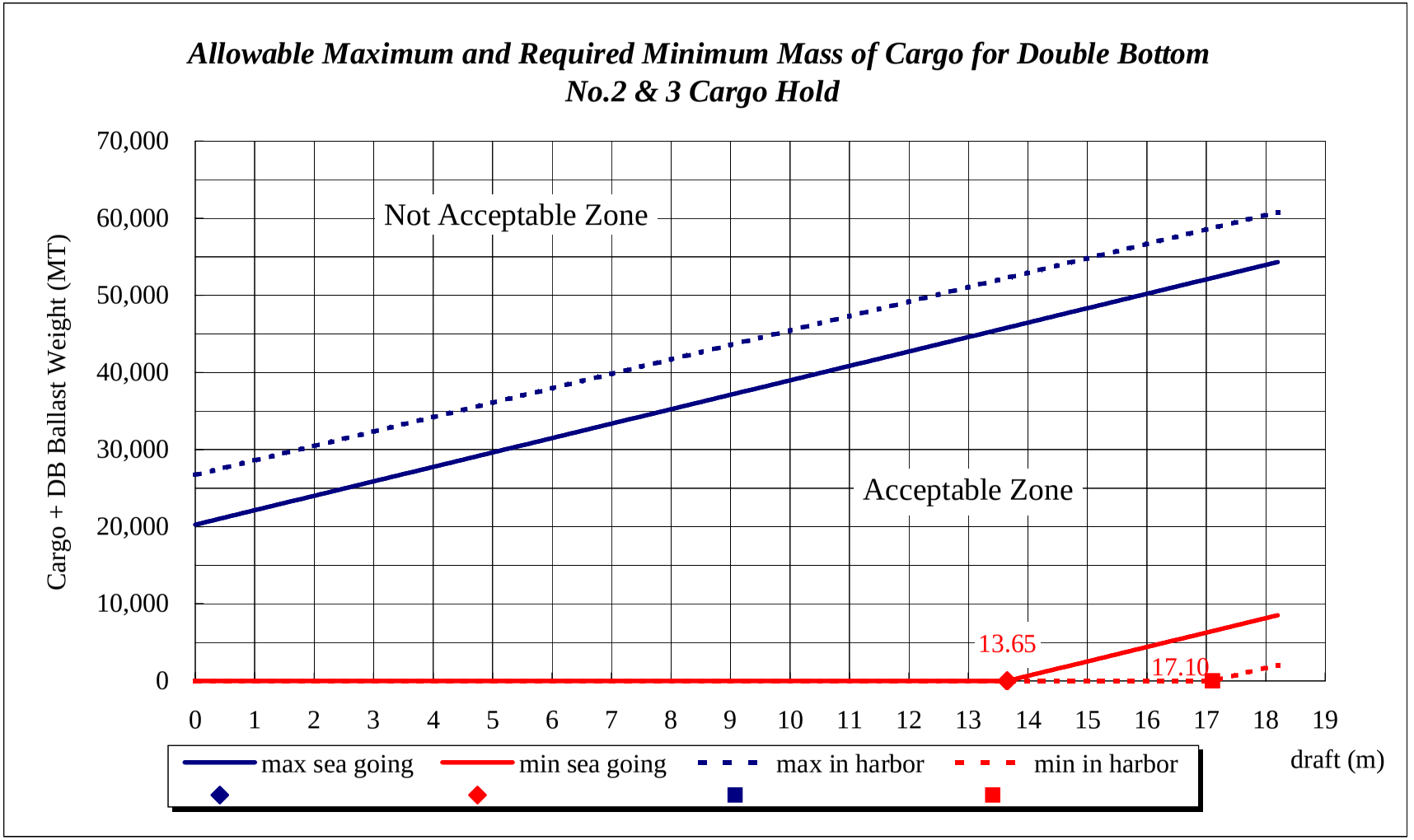
Hold No.

2 & 3

Hold No.	2
M_D	23,880
M_{FULL}	21,194

Hold No.	3
M_D	36,972
M_{FULL}	21,886

ds	18.200
$0.75d_s$	13.650
d_{hb}	9.891
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,267	22,138	24,008	25,879	27,750	29,621	31,492	33,363	35,234	37,104	38,975	40,846	42,717	44,588	46,459	48,330	50,201	52,071	53,942	54,316
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	655	2,526	4,397	6,267	8,138	8,512
Harbor	W_{MAX}	26,729	28,600	30,470	32,341	34,212	36,083	37,954	39,825	41,696	43,566	45,437	47,308	49,179	51,050	52,921	54,792	56,663	58,533	60,404	60,778
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,676	2,050

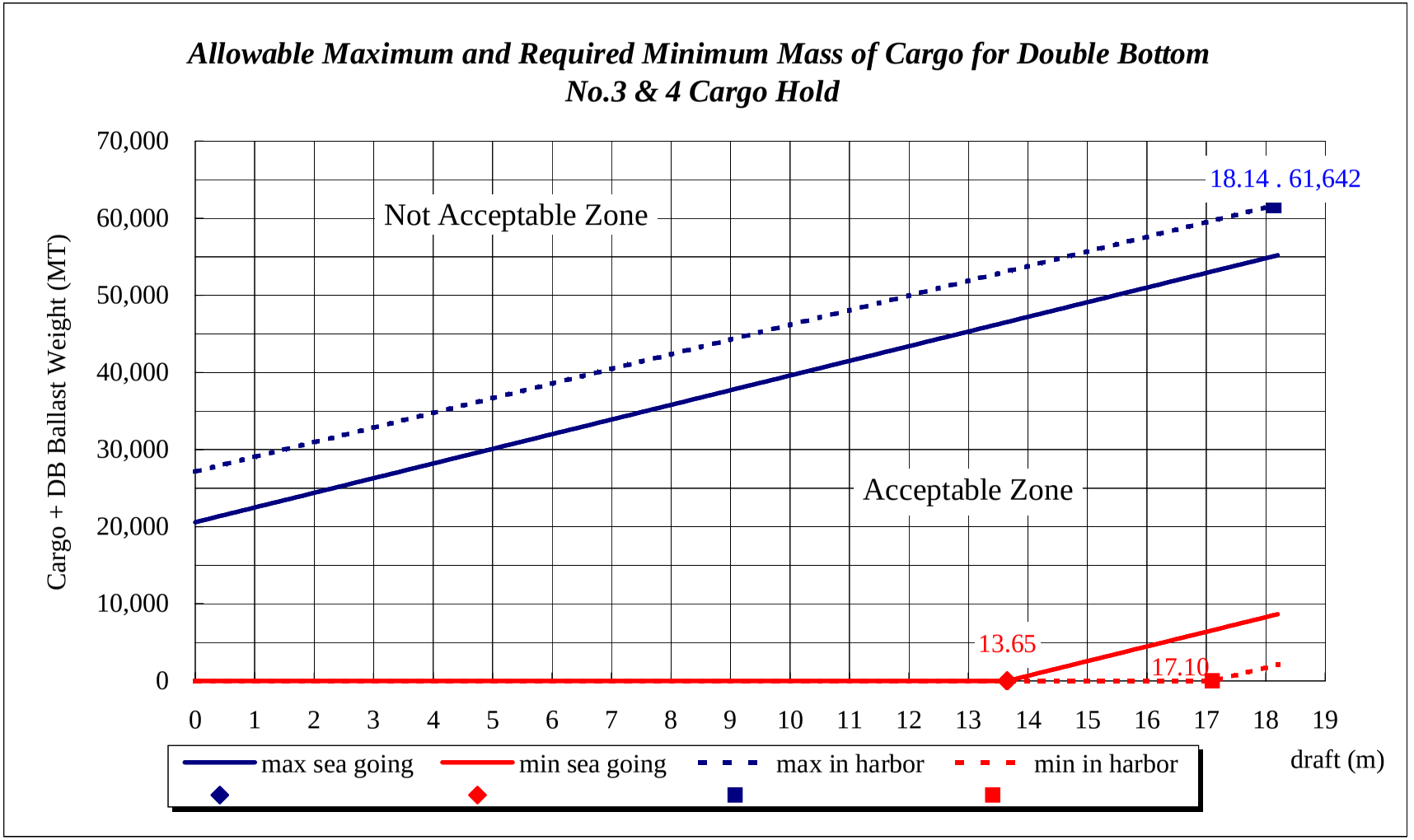
Hold No.

3 & 4

Hold No.	3
M_D	36,972
M_{FULL}	21,886

Hold No.	4
M_D	24,670
M_{FULL}	21,886

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.055
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,587	22,489	24,390	26,291	28,193	30,094	31,995	33,897	35,798	37,699	39,601	41,502	43,403	45,304	47,206	49,107	51,008	52,910	54,811	55,191
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	665	2,567	4,468	6,369	8,271	8,651
Harbor	W_{MAX}	27,153	29,055	30,956	32,857	34,758	36,660	38,561	40,462	42,364	44,265	46,166	48,068	49,969	51,870	53,772	55,673	57,574	59,475	61,377	61,642
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,705	2,085

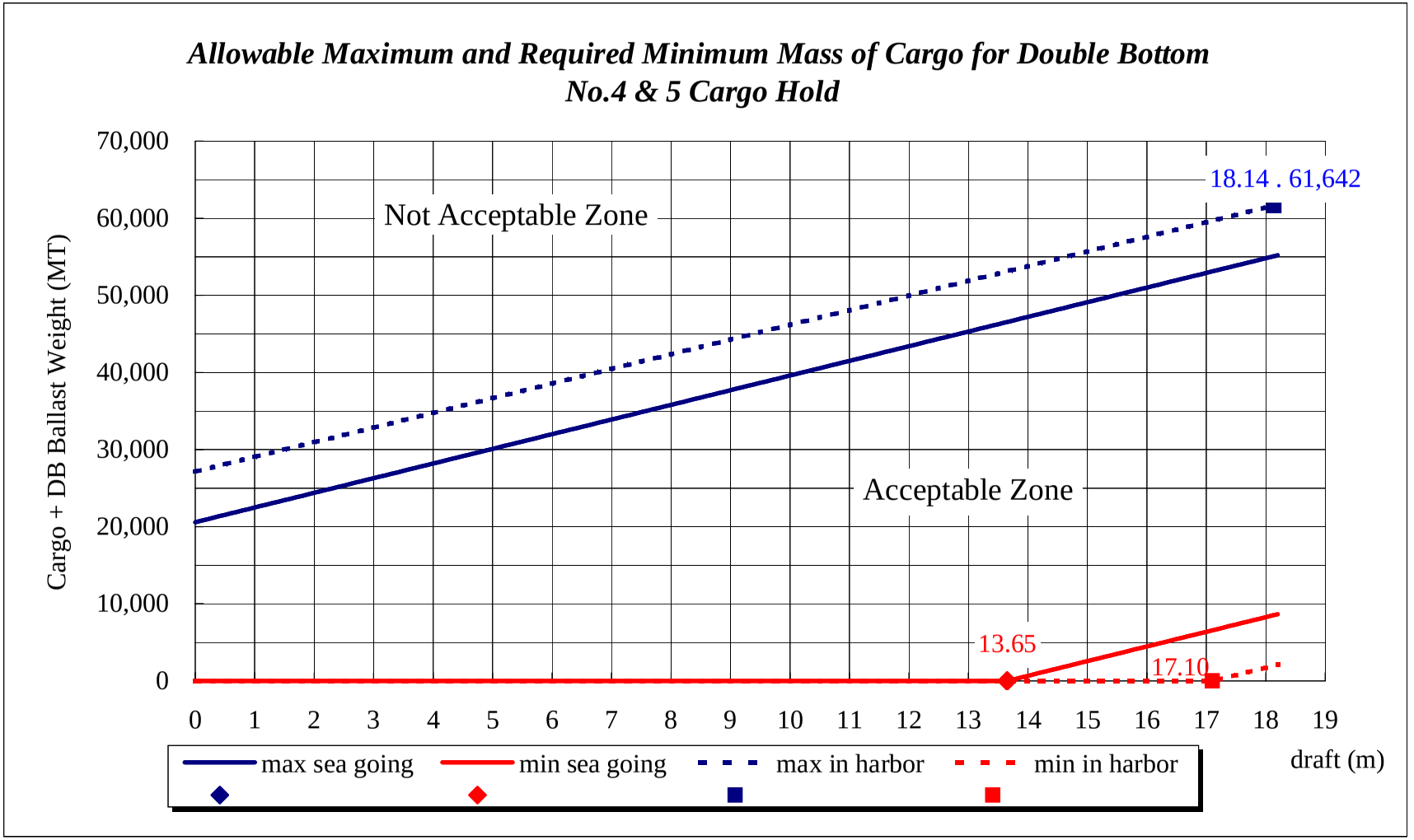
Hold No.

4 & 5

Hold No.	4
M_D	24,670
M_{FULL}	21,886

Hold No.	5
M_D	36,972
M_{FULL}	21,886

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.219
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,587	22,489	24,390	26,291	28,193	30,094	31,995	33,897	35,798	37,699	39,601	41,502	43,403	45,304	47,206	49,107	51,008	52,910	54,811	55,191
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	665	2,567	4,468	6,369	8,271	8,651
Harbor	W_{MAX}	27,153	29,055	30,956	32,857	34,758	36,660	38,561	40,462	42,364	44,265	46,166	48,068	49,969	51,870	53,772	55,673	57,574	59,475	61,377	61,642
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,705	2,085

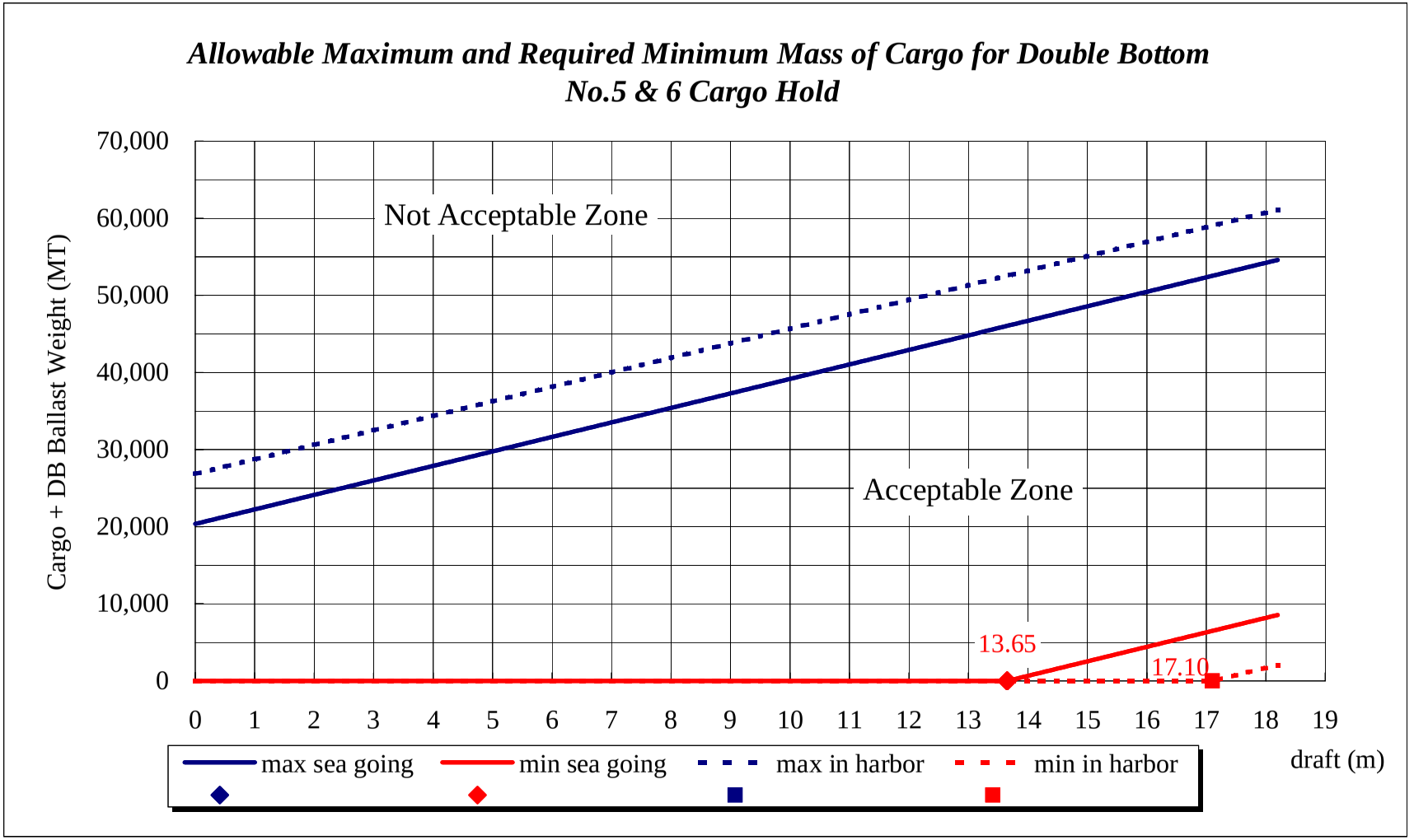
Hold No.

5 & 6

Hold No.	5
M_D	36,972
M_{FULL}	21,886

Hold No.	6
M_D	24,123
M_{FULL}	21,408

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.383
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,366	22,247	24,127	26,007	27,887	29,768	31,648	33,528	35,408	37,289	39,169	41,049	42,929	44,809	46,690	48,570	50,450	52,330	54,211	54,587
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	658	2,538	4,419	6,299	8,179	8,555
Harbor	W_{MAX}	26,860	28,741	30,621	32,501	34,381	36,262	38,142	40,022	41,902	43,783	45,663	47,543	49,423	51,304	53,184	55,064	56,944	58,825	60,705	61,081
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,685	2,061

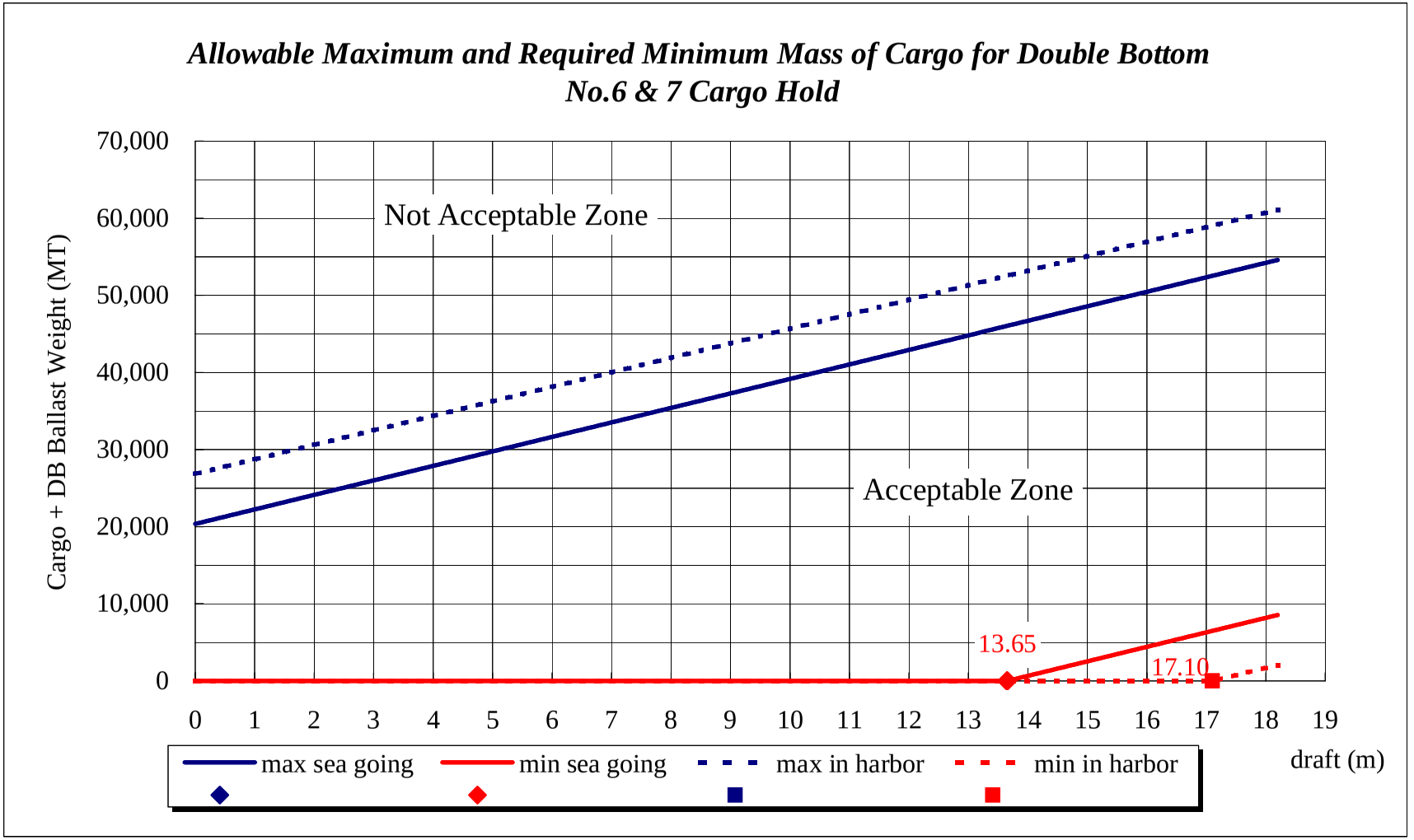
Hold No.

6 & 7

Hold No.	6
M_D	24,123
M_{FULL}	21,408

Hold No.	7
M_D	36,972
M_{FULL}	21,886

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.547
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,366	22,247	24,127	26,007	27,887	29,768	31,648	33,528	35,408	37,289	39,169	41,049	42,929	44,809	46,690	48,570	50,450	52,330	54,211	54,587
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	658	2,538	4,419	6,299	8,179	8,555
Harbor	W_{MAX}	26,860	28,741	30,621	32,501	34,381	36,262	38,142	40,022	41,902	43,783	45,663	47,543	49,423	51,304	53,184	55,064	56,944	58,825	60,705	61,081
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,685	2,061

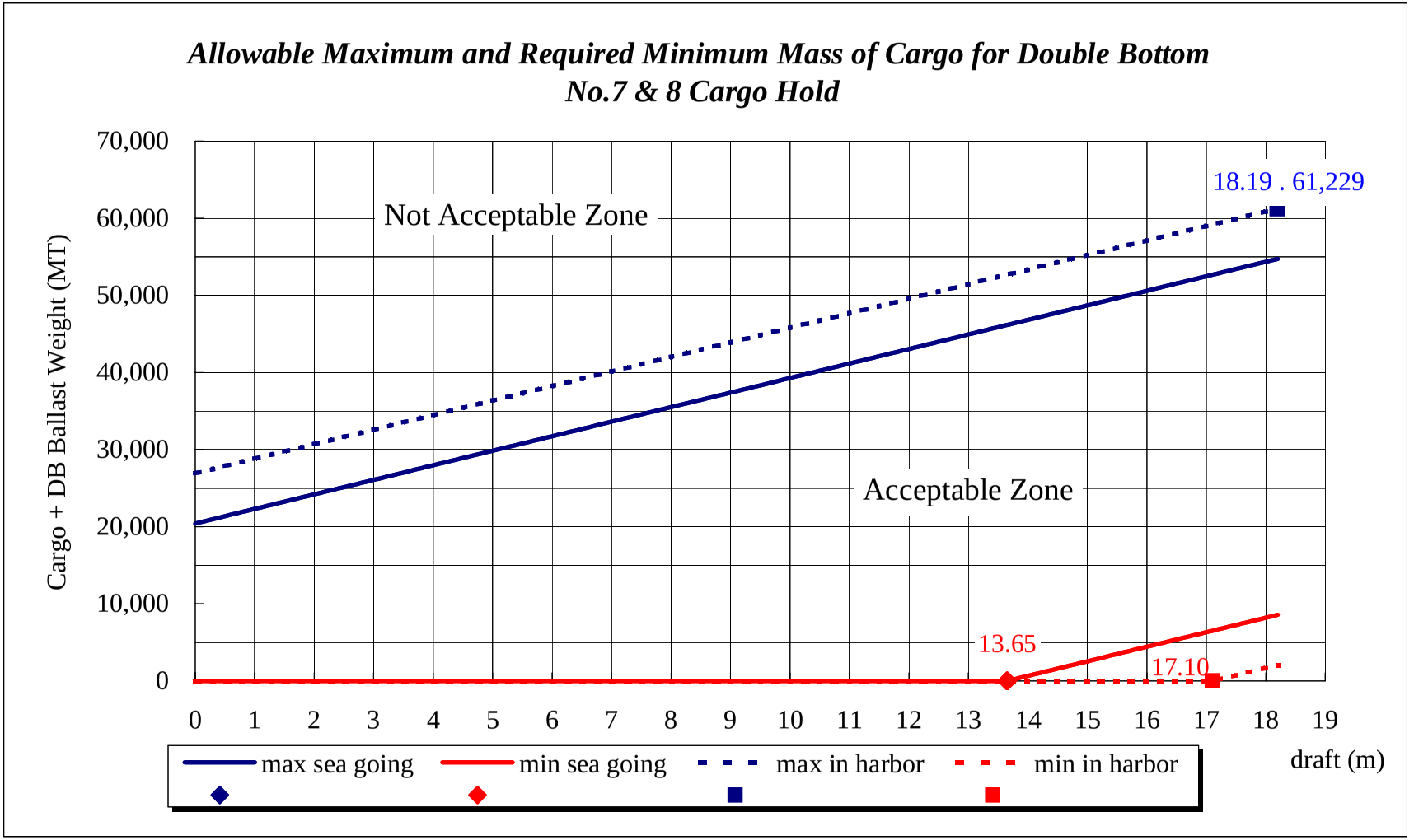
Hold No.

7 & 8

Hold No.	7
M_D	36,972
M_{FULL}	21,886

Hold No.	8
M_D	24,257
M_{FULL}	21,524

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.711
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	20,420	22,305	24,190	26,076	27,961	29,847	31,732	33,617	35,503	37,388	39,273	41,159	43,044	44,930	46,815	48,700	50,586	52,471	54,357	54,734
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	660	2,545	4,431	6,316	8,201	8,578
Harbor	W_{MAX}	26,931	28,817	30,702	32,587	34,473	36,358	38,243	40,129	42,014	43,900	45,785	47,670	49,556	51,441	53,327	55,212	57,097	58,983	60,868	61,229
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,690	2,067

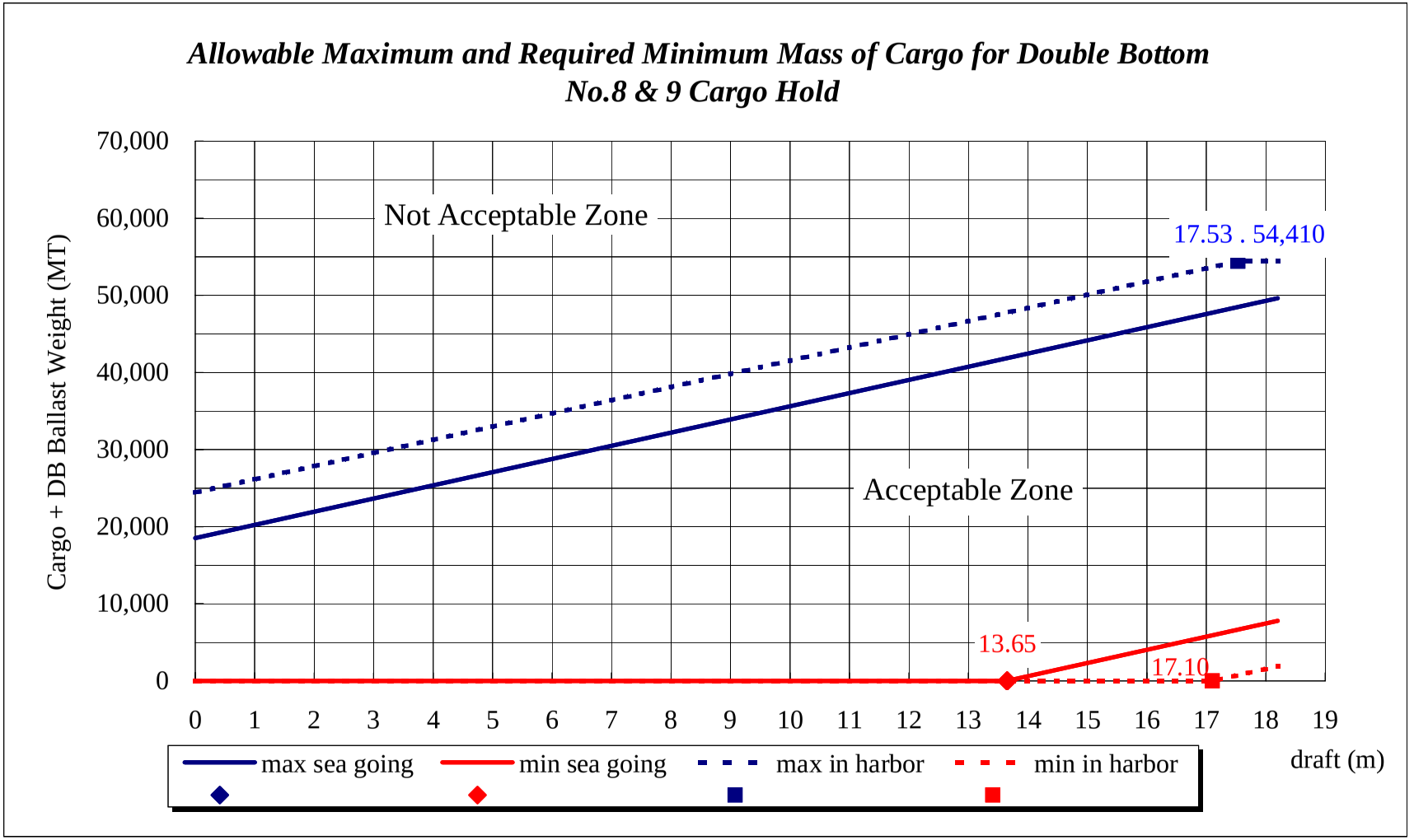
Hold No.

8 & 9

Hold No.	8
M_D	24,257
M_{FULL}	21,524

Hold No.	9
M_D	30,153
M_{FULL}	17,849

ds	18.200
$0.75d_s$	13.650
d_{hb}	10.873
d_{min}	13.650



	dx	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	18.2
Sea	W_{MAX}	18,522	20,232	21,942	23,652	25,362	27,072	28,782	30,492	32,201	33,911	35,621	37,331	39,041	40,751	42,461	44,171	45,881	47,591	49,301	49,643
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	598	2,308	4,018	5,728	7,438	7,780
Harbor	W_{MAX}	24,428	26,138	27,848	29,558	31,268	32,978	34,688	36,397	38,107	39,817	41,527	43,237	44,947	46,657	48,367	50,077	51,787	53,497	54,410	54,410
	W_{MIN}	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,532	1,874